

10

PREP JEE

MOCK TEST PAPER FOR JEE MAIN

SALIENT FEATURES

1. CONTAINS 10 TEST PAPER WITH DETAILED SOLUTION
2. 10 MOCK TEST PAPER
3. ALL TEST PAPERS ARE DESIGNED TO MEET THE NEED OF STUDENTS

THIS BOOK IS DESIGNED BY PREP JEE FOR ALL JEE MAIN ASPIRANTS

PREP JEE PUBLICATION, MIRGANJ INDIA

PREPJEE PUBLICATION

ALL RIGHTS RESERVED

© Authors

No Part of this publication may be re-produced or distributed in any means, electronic, mechanical, printing recording, scanning, web or otherwise without the written permission of the publisher. Prepjee Has obtained all the information in this book from the sources believed to be reliable and true. However, Prepjee don't take any responsibility for the absolute accuracy of any information published, and damages or loss suffered thereupon.

For further information about the products from PREPJEE,

Log on to www.prepjee.in or email to info@prepjee.in

PREPFACE

THIS BOOK HAS BEEN DEVELOPED AND DESIGNED TO OVERCOME WITH PRESSING NEED OF JEE MAIN ASPIRANTS. THE BOOK CONTAINS 10 MOCK TEST PAPERS WITH ANSWER KEY AND DETAILED SOLUTION. THIS WILL GUIDE THE COURSE OF PREPARATION OF AN ASPIRANT IN A DEFINITE AND REQUIRED DIRECTION, LIKE RADAR.

MOREOVER SOLUTION IS IN APPROPRIATE MANNER SO THAT AVERAGE STUDENT CAN UNDERSTAND AND SOLVE PROBLEMS EASILY. THIS BOOK ALSO HELPS STUDENTS IN CONCEPT BUILDING.

Any suggestion for the improvement of book is welcome

PREPJEE

RAJA BABU

MOCK TEST 1

Date : Depends On User

Time : 3 Hours

Max. Marks : 360

Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.

INSTRUCTIONS :

A. General :

1. This booklet is your Question Paper. Do not break the seals of this booklet before being instructed to do so by the invigilators.
2. Blank spaces and blank pages are provided in the question paper for your rough work. No additional sheets will be provided for rough work.
3. Blank papers, clipboards, log tables, slide rules, calculators, cameras, cellular phones, pagers, and electronic gadgets are NOT allowed inside the examination hall.
4. Write you name and roll number in the space provided on the back cover of this booklet.
5. **Using a black ball point pen, darken the bubbles on the upper original sheet.**
6. DO NOT TAMPER WITH/MUTILATE THE ORS OR THE BOOKLET.
7. On breaking the seals of the booklet check that it contains all the **90** questions and corresponding answer choices are legible. Read carefully the Instructions printed at the beginning of each section.

B. Filling the ORS

Use only Black ball point pen only for filling the ORS.

8. Write your Roll no. in the boxes given at the top left corner of your ORS with black ball point pen. Also, darken the corresponding bubbles with Black ball point pen only. Also fill your roll no on the back side of your ORS in the space provided (if the ORS is both side printed).
9. Fill your Paper Code as mentioned on the Test Paper and darken the corresponding bubble with Black ball point pen.
10. If student does not fill his/her roll no. and paper code correctly and properly, then his/her marks will not be displayed and 5 marks will be deducted (paper wise) from the total.
11. Since it is not possible to erase and correct pen filled bubble, you are advised to be extremely careful while darken the bubble corresponding to your answer.
12. Neither try to erase / rub / scratch the option nor make the Cross (X) mark on the option once filled. Do not scribble, smudge, cut, tear, or wrinkle the ORS. Do not put any stray marks or whitener anywhere on the ORS.
13. If there is any discrepancy between the written data and the bubbled data in your ORS, the bubbled data will be taken as final.

Please read the last page of this booklet for rest of the instructions

PART - A

SECTION - I

Straight Objective Type

This section contains 18 **multiple choice questions**. Each question has 5 choices (1), (2), (3), (4) and (5) out of which **ONLY ONE** is correct.

1. If a $\triangle ABC$, $\tan B = 2$, $\tan C = \frac{3}{2}$ and $a = \sqrt{65}$, then circumradius of the triangle is -

(1) 65 (2) $\frac{\sqrt{65}}{7}$

(3) $\frac{65}{14}$ (4) $\frac{65}{7}$

(5) None of these

2. The variable plane $(2\lambda + 1)x + (3 - \lambda)y + z = 4$ always passes through the line -

(1) $\frac{x}{1} = \frac{y}{2} = \frac{z}{-3}$ (2) $\frac{x}{0} = \frac{y}{0} = \frac{z+4}{1}$

(3) $\frac{x}{1} = \frac{y}{2} = \frac{z-4}{-7}$ (4) $\frac{x}{-1} = \frac{y}{2} = \frac{z-4}{7}$

(5) $\frac{x}{0} = \frac{y}{0} = \frac{z-4}{1}$

3. $\sum_{n=1}^{360} \left(\frac{1}{n\sqrt{n+1} + (n+1)\sqrt{n}} \right) =$

(1) $\frac{17}{19}$ (2) 2

(3) $\frac{18}{19}$ (4) $\frac{20}{19}$ (5) $\frac{1}{19}$

4. A straight line passing through $P(3, 1)$ meets the coordinate axes at A and B. It is given that distance of this line from the origin 'O' is maximum, then area of $\triangle OAB$ is -

(1) $\frac{50}{3}$ sq. units (2) $\frac{25}{3}$ sq. units

(3) $\frac{20}{3}$ sq. units (4) $\frac{100}{3}$ sq. units

(5) $\frac{200}{3}$ sq. units

5. Let P be any moving point on the circle $x^2 + y^2 - 2x = 1$. AB be the chord of contact of this point w.r.t the circle $x^2 + y^2 - 2x = 0$. The locus of the circumcentre of the triangle CAB (C being the centre of the circles) is

(1) $2x^2 + 2y^2 - 4x + 1 = 0$

(2) $x^2 + y^2 - 4x + 2 = 0$

(3) $x^2 + y^2 - 4x + 1 = 0$

(4) $2x^2 + 2y^2 - 4x + 3 = 0$

(5) $x^2 + y^2 - 4x + 3 = 0$

(SPACE FOR ROUGH WORK)

6. The length of latus rectum of the ellipse whose focus is $S(-1, 1)$, the corresponding directrix is $x - y + 3 = 0$ and the eccentricity is $\frac{1}{2}$, is equal to

- (1) $\frac{1}{2}$ (2) $\frac{1}{2\sqrt{2}}$
(3) $\frac{1}{\sqrt{2}}$ (4) $\frac{1}{4}$

(5) Not possible to calculate

7. Let z_1 and z_2 be n^{th} roots of unity which subtends a right angle at the origin then 'n' must be of the form -

- (1) $4K + 1, K \in I$ (2) $4K + 2, K \in I$
(3) $4K + 3, K \in I$ (4) $4K, K \in I$
(5) None of these

8. If $f(x)$ is an even function and satisfies the relation $x^2 f(x) - 2f\left(\frac{1}{x}\right) = g(x)$, where $g(x)$ is an odd function, then $f(5)$ is equals to -

- (1) 0 (2) $\frac{50}{75}$
(3) $\frac{49}{75}$ (4) $\frac{1}{75}$

(5) None of these

9. Let $f : \mathbb{R} \rightarrow \left[0, \frac{\pi}{2}\right)$ defined by

$f(x) = \tan^{-1}(x^2 + x + a)$, then set of value(s) of 'a' for which f is onto is

- (1) $[0, \infty)$ (2) $[2, 1]$
(3) $\left[\frac{1}{4}, \infty\right)$ (4) $\left\{\frac{1}{4}\right\}$

(5) None of these

10. If matrix A is given by $A = \begin{bmatrix} 6 & 11 \\ 2 & 4 \end{bmatrix}$, then the

determinant of $(A^{2005} - 6A^{2004})$ is equal to -

- (1) 2^{2006} (2) $(-11)2^{2005}$
(3) $7(-2^{4010})$ (4) $(-9)2^{2004}$
(5) -11.2^{4009}

11. Let p, q be statements. Which of the following is a negation of $p \rightarrow q$?

- (1) $p \wedge (\sim q)$ (2) $\sim q \rightarrow \sim p$
(3) $\sim p \vee q$ (4) $p \vee (\sim q)$
(5) none of these

(SPACE FOR ROUGH WORK)

12. Let $T = \int_0^{\ln 2} \frac{2e^{3x} + e^{2x} - 1}{e^{3x} + e^{2x} - e^x + 1} dx$, then e^T is equal to

- (1) $\frac{7}{4}$ (2) $\frac{7}{2}$
 (3) $\frac{11}{2}$ (4) $\frac{11}{4}$
 (5) None of these

13. If $\phi(x) = \cos x - \int_0^x (x-t)\phi(t) dt$, then

$\phi''(x) + \phi(x)$ is equal to

- (1) $-\cos x$ (2) 0
 (3) $\int_0^x (x-t)\phi(t) dt$ (4) $-\int_0^{-x} (x-t)\phi(t) dt$
 (5) $\cos x$

14. Let the function $g : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by

$g(t) = \frac{\pi}{2} - 2\cot^{-1}(3^{-t})$. Then g is -

- (1) even and is strictly increasing in $(-\infty, \infty)$
 (2) odd and is strictly decreasing in $(-\infty, \infty)$
 (3) even and is strictly decreasing in $(-\infty, \infty)$
 (4) odd and is strictly increasing in $(-\infty, \infty)$
 (5) $g(0) = \frac{\pi}{2}$

15. Let A and B are two independent events with $P(A) = \frac{1}{2}$ and $P(A \cup B) = \frac{2}{3}$, then the probabilities $P(B)$, $P(A/B)$ and $P(\bar{B}/A)$ are in

- (1) A.P. (2) G.P.
 (3) H.P. (4) A.G.P.
 (5) none of these

16. Three identical red balls, three identical green balls and three identical blue balls are placed in a row in a random order. Let A denotes the event that no two blue balls are adjacent. The odds against 'A' is

- (1) 7 : 4 (2) 5 : 3
 (3) 7 : 5 (4) 6 : 5
 (5) none of these

17. Let $f(x) = \frac{e^{\tan x} - e^x + \ln(\sec x + \tan x) - x}{\tan x - x}$,

$x \neq 0$ be a continuous function at $x = 0$. The value of $f(0)$ is equal to

- (1) $\frac{1}{2}$ (2) $\frac{2}{3}$
 (3) $\frac{3}{2}$ (4) 2
 (5) $\frac{4}{3}$

(SPACE FOR ROUGH WORK)

18. There are five different oranges and three different apples. Number of ways they can be divided into two groups of four fruits if each group must contain atleast one apple is

- (1) 95 (2) 65
(3) 60 (4) 35
(5) 30

SECTION – II

(One or more options correct Type)

This section contains 5 **multiple choice questions**. Each question has four choices (1), (2), (3) and (4) out of which **ONE** or **MORE** are correct.

19. If $\lim_{x \rightarrow 2} (\log_3(ax^2 + 3x + 1))^{\log_{(x-1)} 3} = \ell$, where ℓ is a finite real number then

- (1) $a = -1$
(2) 'a' can have more than one values
(3) $\ell = e^{-2/3}$
(4) $\ell = e^{-1/3}$

20. Let a function $f(x)$, $x \neq 0$ be such that

$$f(x) + f\left(\frac{1}{x}\right) = f(x) \cdot f\left(\frac{1}{x}\right) \text{ then } f(x) \text{ can be}$$

- (1) $1 - x^{2013}$
(2) $\sqrt{|x|} + 1$
(3) $\frac{\pi}{2 \tan^{-1} |x|}$
(4) $\frac{2}{1 + k \ln |x|}$, k being arbitrary constant

21. Let $f(x) = ax^3 + 2bx^2 + 3cx + 4d$ where a,b,c,d are non zero real numbers, be an increasing function and $4b^2 < 3c^2$. If $g(x) = af'(x) + 2bf''(x) + 9c^2$

$$\text{and } h(x) = \int_0^x g(t) dt, \text{ then}$$

- (1) $h(x)$ is increasing function
(2) $h(x)$ is decreasing function
(3) $h(x)$ is neither increasing non decreasing
(4) $h(x)$ is one -one function

22. If P is a point (x, y) on the line $y = -3x$ such that P and the point (3, 4) are on the opposite sides of the line $3x - 4y = 8$, then which of the following is/are **FALSE** ?

- (1) $y < -\frac{8}{5}$ (2) $y > -\frac{8}{5}$
(3) $y > -\frac{11}{5}$ (4) $y < -\frac{9}{5}$

23. A man wants to distribute 101 coins of a rupee each, among his 3 sons with the condition that no one receives more money than the combined total of other two. The number of ways of doing this is :

- (1) ${}^{103}C_2 - 3 \cdot {}^{52}C_2$ (2) $\frac{{}^{103}C_2}{3}$
(3) 1275 (4) $\frac{{}^{103}C_2}{6}$

(SPACE FOR ROUGH WORK)

SECTION – III

(Reasoning Type)

This section contains 3 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

24. Statement- 1 : A person purchase 1 kg of tomatoes from each of the 4 places at the rate of 1 kg, 2kg, 3kg, 4kg per rupee respectively. On the average he has purchased x kg of tomatoes per rupee, then the value of x is 2.5.

Statement- 2 : The average of $x_1, x_2, x_3, \dots, x_n$ is $\frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$.

- (1) Statement -1 is True, Statement -2 is True;
Statement -2 is a correct explanation for Statement -1
- (2) Statement-1 is True, Statement-2 is True ;
Statement-2 is **NOT** a correct explanation for Statement-1
- (3) Statement -1 is True, Statement -2 is False
- (4) Statement -1 is False, Statement -2 is True

25. Statement- 1 : Let L be the set of all lines in a plane and R be the relations in L defined as $R = \{(L_1, L_2) : L_1 \text{ is perpendicular to } L_2\}$. This relation is not equivalence relation.

Statement- 2 : Two lines perpendicular to given line are mutually perpendicular.

- (1) Statement -1 is True, Statement -2 is True;
Statement -2 is a correct explanation for Statement -1
- (2) Statement-1 is True, Statement-2 is True ;
Statement-2 is **NOT** a correct explanation for Statement-1
- (3) Statement -1 is True, Statement -2 is False
- (4) Statement -1 is False, Statement -2 is True

26. Let $0 < \alpha, \beta, \gamma < \frac{\pi}{2}$

Statement- 1 : If $\tan^3 \alpha, \tan^3 \beta, \tan^3 \gamma$ are the roots of the cubic equation $x^3 - 6x^2 + kx - 8 = 0$, then $\tan \alpha = \tan \beta = \tan \gamma$.

Statement- 2 : If $a^3 + b^3 + c^3 = 3abc$ and a,b,c are positive numbers, then $a = b = c$.

- (1) Statement-1 is True, Statement-2 is True ;
Statement-2 is **NOT** a correct explanation for Statement-1
- (2) Statement -1 is True, Statement -2 is True;
Statement -2 is a correct explanation for Statement -1
- (3) Statement -1 is True, Statement -2 is False
- (4) Statement -1 is False, Statement -2 is True

(SPACE FOR ROUGH WORK)

SECTION – IV

(Comprehension Type)

This section contains 2 paragraphs. Based on each paragraph, there are 2 questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 27 to 28

Consider lines $L_1 : \vec{r}_1 = 2\hat{i} + \hat{j} - \hat{k} + \lambda (\hat{i} + 2\hat{k})$, $\lambda \in \mathbb{R}$ and $L_2 : \vec{r}_2 = 3\hat{i} + \hat{j} + \mu (\hat{i} + \hat{j} - \hat{k})$, $\mu \in \mathbb{R}$. Let P be the plane which contains the line L_1 and is parallel to L_2 .

27. The shortest distance between the lines L_1 and L_2 , is

- (1) $\frac{1}{\sqrt{5}}$ (2) $\frac{1}{\sqrt{6}}$
(3) $\frac{1}{\sqrt{8}}$ (4) $\frac{1}{\sqrt{14}}$

28. Image of origin in the plane P, is -

- (1) $\left(\frac{2}{7}, \frac{-3}{7}, \frac{-1}{7}\right)$ (2) $\left(\frac{4}{7}, \frac{-6}{7}, \frac{-2}{7}\right)$
(3) $\left(\frac{-2}{7}, \frac{3}{7}, \frac{1}{7}\right)$ (4) $\left(\frac{-4}{7}, \frac{6}{7}, \frac{2}{7}\right)$

Paragraph for Question Nos. 29 to 30

A parabola whose focus is (3, 4) is touching the coordinate axes.

29. The equation of circle whose diameter is the portion of the tangent at the vertex of the parabola between the coordinate axes is

- (1) $x^2 + y^2 - 3x - 4y = 0$
(2) $x^2 + y^2 + 6x + 8y = 0$
(3) $x^2 + y^2 - 3x + 4y = 0$
(4) $x^2 + y^2 - 6x + 8y = 0$

30. If P and Q are ends of a focal chord of the

parabola, then $\frac{1}{SP} + \frac{1}{SQ}$ equal to

- (1) $\frac{12}{5}$
(2) $\frac{24}{5}$
(3) $\frac{5}{12}$
(4) $\frac{5}{24}$

(SPACE FOR ROUGH WORK)

PART-B

SECTION - I

Straight Objective Type

This section contains 18 multiple choice questions. Each question has 5 choices (1), (2), (3), (4) and (5) for its answer, out of which **ONLY ONE** is correct.

31. Ultraviolet light of wavelength λ_1 and $\lambda_2 (\lambda_2 > \lambda_1)$ when allowed to fall on hydrogen atoms in their ground state is found to liberate electrons with kinetic energies E_1 and E_2 respectively. The value of planck's constant can be found from the relation.

(1) $h = \frac{1}{c}(\lambda_2 - \lambda_1)(E_1 - E_2)$

(2) $h = \frac{1}{c}(\lambda_1 + \lambda_2)(E_1 + E_2)$

(3) $h = \frac{(E_1 - E_2)\lambda_1\lambda_2}{c(\lambda_2 - \lambda_1)}$

(4) $h = \frac{(E_1 + E_2)\lambda_1\lambda_2}{c(\lambda_1 + \lambda_2)}$

(5) $h = \frac{(E_1 + E_2)\lambda_1\lambda_2}{3c(\lambda_2 - \lambda_1)}$

32. A point source of light of power 'P' and wavelength ' λ ' is emitting light in all directions. The number of photons present in a spherical region of radius 'r' to radius $r + x$ with centre at the source is

(1) $\frac{P\lambda}{4\pi r^2 hc}$

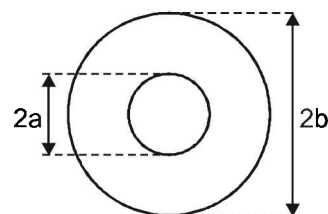
(2) $\frac{P\lambda x}{hc^2}$

(3) $\frac{P\lambda x}{4\pi r^2 hc}$

(4) $\frac{3P\lambda x}{4\pi r^2 hc}$

(5) None of these

33. A small circular wire loop of radius a is located at the centre of a much larger circular wire loop of radius b as shown above ($b \gg a$). Both loops are coaxial and coplanar. The larger loop carries a time (t) varying current $I = I_0 \cos \omega t$, where I_0 and ω are constants. The magnetic field generated by the current in the large loop induces in the small loop an emf that is approximately equal to which of the following.



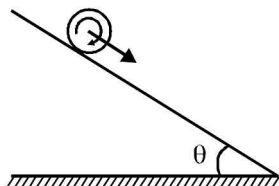
(1) $\frac{\pi \mu_0 I_0}{2} \frac{a^2}{b} \omega \cos \omega t$ (2) $\frac{\pi \mu_0 I_0}{2} \frac{a^2}{b} \omega \sin \omega t$

(3) $\frac{\pi \mu_0 I_0}{2} \frac{a}{b^2} \omega \sin \omega t$ (4) $\frac{\pi \mu_0 I_0}{2} \frac{a}{b^2} \omega \cos \omega t$

(5) $\frac{\pi \mu_0 I_0}{2} \frac{a}{b^2} \omega \cos^2 \omega t$

(SPACE FOR ROUGH WORK)

34. A uniform solid sphere of radius R and mass M purely rolls down an inclined plane. The coefficient of friction between the sphere and the inclined plane is μ . The maximum value of θ for this to be possible is : (Take $g = 10 \text{ m/s}^2$)



(1) $\tan^{-1} \mu$

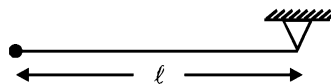
(2) $\tan^{-1} \left(\frac{7\mu}{2} \right)$

(3) $\tan^{-1} \left(\frac{5\mu}{3} \right)$

(4) $\tan^{-1} \left(\frac{3\mu}{2} \right)$

(5) $2\tan^{-1} \mu$

35. A simple pendulum is released from horizontal position as shown. Consider an angle θ which pendulum makes with downward vertical at any time. Find the value of $2\tan^2\theta$ when acceleration of the bob becomes horizontal. String is massless and inextensible.



(1) 2

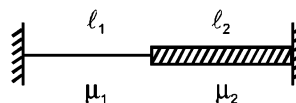
(2) 4

(3) 6

(4) 8

(5) 1

36. A composite wire is made by joining two uniform wires. If $\ell_1 = \ell_2 = \ell$ and $\mu_1 = \frac{\mu_2}{9} = \mu$. Tension in the strings is T , μ is mass per unit length. Then lowest frequency such that the junction is an antinode.



(1) $\frac{1}{4\ell} \sqrt{\frac{\ell}{\mu}}$

(2) $\frac{3}{4\ell} \sqrt{\frac{\ell}{\mu}}$

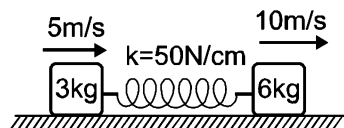
(3) $\frac{5}{4\ell} \sqrt{\frac{\ell}{\mu}}$

(4) $\frac{7}{4\ell} \sqrt{\frac{\ell}{\mu}}$

(5) $\frac{1}{\ell} \sqrt{\frac{\ell}{\mu}}$

(SPACE FOR ROUGH WORK)

37. A spring mass system is placed on a frictionless horizontal surface as shown in the figure. The spring is expanded by $\frac{1}{10}$ m and the blocks are given velocities as shown, then maximum extension of spring is :



- (1) $\frac{1}{10\sqrt{2}}$ m
 (2) $\frac{1}{5\sqrt{2}}$ m
 (3) $\frac{1}{20}$ m
 (4) $\frac{1}{10}$ m
 (5) $\frac{1}{\sqrt{2}}$ m

38. A particle is moving along straight line with initial velocity +7 m/sec and uniform acceleration -2 m/sec². The distance travelled by the particle in 4th second of its motion is :

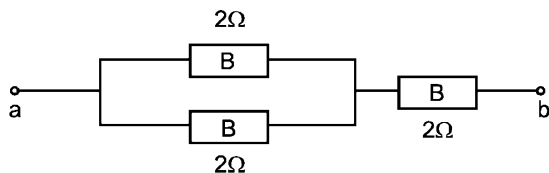
- (1) zero
 (2) 0.25 m
 (3) 0.5 m
 (4) 7 m
 (5) 0.75 m

39. When an small object is placed at a distance x_1 and x_2 from a lens on its principal axis, then real image and a virtual image are formed respectively having same magnitude of transverse magnification. Then the focal length of the lens is :

- (1) $x_1 - x_2$
 (2) $\frac{x_1 - x_2}{2}$
 (3) $\frac{x_1 + x_2}{2}$
 (4) $x_1 + x_2$
 (5) $\frac{x_1 + x_2}{4}$

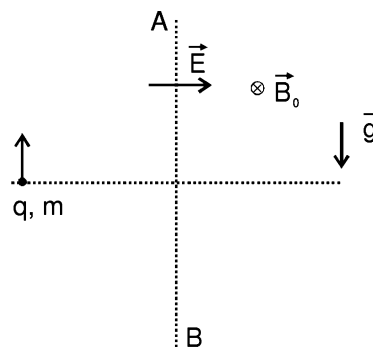
(SPACE FOR ROUGH WORK)

40. Three identical bulbs each of resistance 2Ω are connected as shown. The maximum power that can be consumed by individual bulb is 32W , then the maximum power consumed by the combination is :



- (1) 48 W
- (2) 96 W
- (3) 128 W
- (4) 160 W
- (5) 100 W

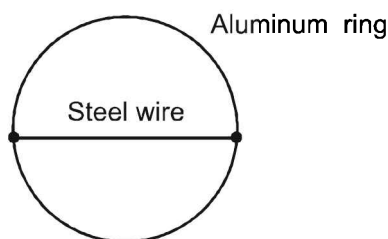
41. A uniform electric field E is present horizontally along the paper throughout the region but uniform magnetic field B_0 is present horizontally (perpendicular to plane of paper in inward direction) right to the line AB. A charge particle having charge q and mass m is projected vertically upward and it crosses the line AB after time t_0 . Find the speed of projection if particle moves with constant velocity after t_0 . (Given $qE = mg$)



- (1) gt_0
- (2) $2gt_0$
- (3) $\frac{gt_0}{2}$
- (4) $\frac{gt_0}{4}$
- (5) Particle can't move with constant velocity after crossing AB

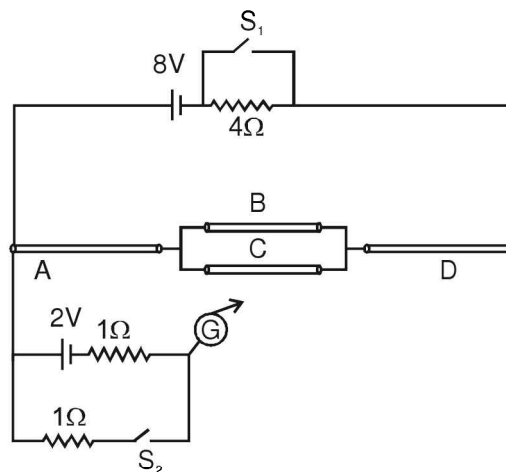
(SPACE FOR ROUGH WORK)

42. A steel wire is rigidly fixed along diameter of aluminium ring of radius R as shown. Linear expansion coefficient of steel is half of linear expansion coefficient for aluminium. If temperature of system is increased by $\Delta\theta$ then the thermal stress developed in steel wire is: (α_{Al} is linear expansion coefficient for aluminium and Young's modulus for steel is Y)



- (1) more than $R \alpha_{Al} \Delta\theta Y$
- (2) less than $R \alpha_{Al} \Delta\theta Y$
- (3) Equal to $R \alpha_{Al} \Delta\theta Y$
- (4) equal to $2R \alpha_{Al} \Delta\theta Y$
- (5) equal to $4R \alpha_{Al} \Delta\theta Y$

43. Four wire A, B, C and D each of length $\ell = 10\text{cm}$ and each of area of cross section is 0.1 m^2 are connected in the given circuit. Then, the position of null point is

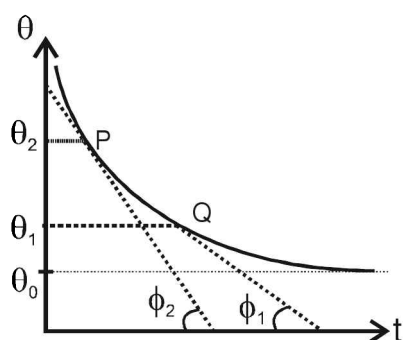


Given that resistivity $\rho_A = 1\ \Omega - \text{m}$
 $\rho_B = 3\ \Omega - \text{m}$
 $\rho_C = 6\ \Omega - \text{m}$
 $\rho_D = 1\ \Omega - \text{m}$

- (1) mid point of wire B or wire C when both the switches S_1 and S_2 open.
- (2) mid point of wire B when both the switches S_1 and S_2 are closed.
- (3) mid point of wire D when both the switches S_1 and S_2 are open.
- (4) 3 cm from the left end of B when both the switches S_1 and S_2 open.
- (5) None of these

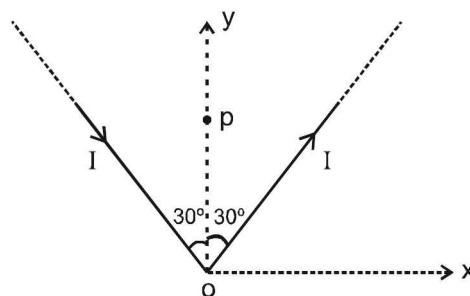
(SPACE FOR ROUGH WORK)

44. A body cools in a surrounding which is at a constant temperature of θ_0 . Assume that it obeys Newton's law of cooling. Its temperature θ is plotted against time t . Tangents are drawn to the curve at the points $P(\theta = \theta_2)$ and $Q(\theta = \theta_1)$. These tangents meet the time axis at angles of ϕ_2 and ϕ_1 , as shown. Then choose correct option :



- (1) $\frac{\tan \phi_2}{\tan \phi_1} = \frac{\theta_1 - \theta_0}{\theta_2 - \theta_0}$
- (2) $\frac{\tan \phi_2}{\tan \phi_1} = \frac{\theta_2 - \theta_0}{\theta_1 - \theta_0}$
- (3) $\frac{\tan \phi_1}{\tan \phi_2} = \frac{\theta_1}{\theta_2}$
- (4) $2 \frac{\tan \phi_2}{\tan \phi_1} = \frac{\theta_2 - \theta_0}{\theta_1 - \theta_0}$
- (5) None of these

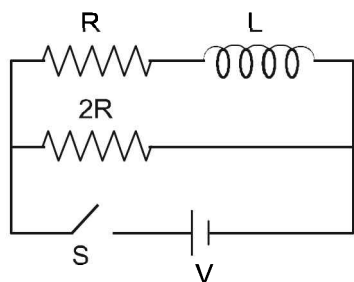
45. An infinite current carrying wire, carrying current I is bent in V shape, lying in x-y plane as shown in figure. Intensity of magnetic field at point P will be (take $OP = 2a$)



- (1) $\frac{\mu_0 I}{4\pi a} (2 + \sqrt{3}) \hat{k}$
- (2) $\frac{\mu_0 I}{4\pi a} (2 - \sqrt{3}) \hat{k}$
- (3) $\frac{\mu_0 I}{4\pi a} (2 + \sqrt{3}) (-\hat{k})$
- (4) $\frac{\mu_0 I}{4\pi a} (2 - \sqrt{3}) (-\hat{k})$
- (5) $7 \frac{\mu_0 I}{4\pi a} (2 + \sqrt{3}) \hat{k}$

(SPACE FOR ROUGH WORK)

46. Consider a L – R circuit shown in figure. There is no current in circuit. Switch S is closed at $t = 0$, time instant when current in inductor is equal to current in resistor $2R$ will be :



(1) $\frac{L}{R} \ln 2$

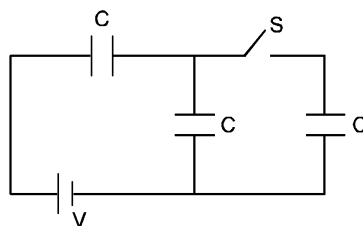
(2) $\frac{2L}{R} \ln 2$

(3) $\frac{L}{2R} \ln 2$

(4) $\frac{L}{2R}$

(5) $\frac{L}{R} \ln \frac{3}{2}$

47. Find the heat generated after the switch S is closed.



(1) $\frac{CV^2}{2}$

(2) $\frac{CV^2}{6}$

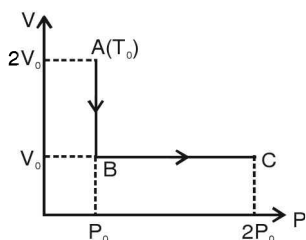
(3) $\frac{2CV^2}{3}$

(4) $\frac{CV^2}{12}$

(5) $\frac{5CV^2}{12}$

(SPACE FOR ROUGH WORK)

48. An ideal gas undergoes a thermodynamics process ABC shown in figure. Temperature at the state A is T_0 . Then which of the following options is **correct**.



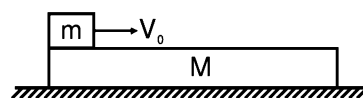
- (1) Change in internal energy in process from A to C must be negative
- (2) Change in internal energy in process from A to C must be positive
- (3) Total heat given to gas in process A to C must be positive
- (4) Total heat given to gas in process A to C must be negative
- (5) None of these

SECTION – II

One or more options correct Type

This section contains 5 **multiple choice questions**. Each question has four choices (1), (2), (3) and (4) out of which **ONE** or **MORE** are correct.

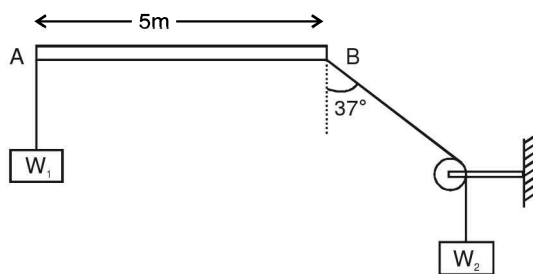
49. The coefficient of friction between the block of mass m and plank of mass M is μ and its value is such that block becomes stationary with respect to plank before it reaches the other end. Then :



- (1) the work done by friction on the block is negative
- (2) the work done by friction on the plank is positive
- (3) the net work done by friction is negative
- (4) net work done by the friction is zero

(SPACE FOR ROUGH WORK)

50. A single additional force F is to be applied to the rod AB to maintain it in horizontal equilibrium position as shown in figure. Two weights $W_1 = 10\text{N}$ and $W_2 = 50\text{N}$ are attached to the two ends of the rod at points A & B respectively using light inextensible strings. If the weight of the bar is to be neglected then choose correct options.



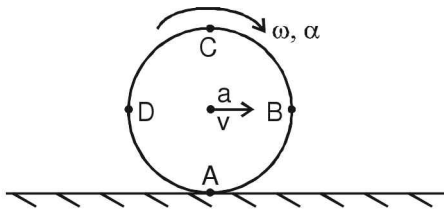
- (1) The horizontal component of required force F is 30N
- (2) The vertical component of required force F is 50N
- (3) Point of application of required force F is 3m to the right of point A .
- (4) Point of application of required force F is 1m to the left of point B .

51. Choose the right statement from the following :

- (1) The average KE of a molecule of any ideal gas is the same at the same temperature
- (2) The average translational KE of a molecule of any ideal gas is the same at the same temperature .
- (3) The average translational KE of a molecule of diatomic gas is 1.5 times the average rotational KE of a molecule at a given moderate temperature.
- (4) The average rotational KE of a molecule of a monoatomic gas increases linearly with increase in temperature.

(SPACE FOR ROUGH WORK)

52. A circular disc of radius R rolls without slipping on a rough horizontal surface. At the instant shown its linear velocity is V , linear acceleration a , angular velocity ω and angular acceleration α . Four points A, B, C and D lie on its circumference such that the diameter AC is vertical & BD horizontal then choose the correct options.



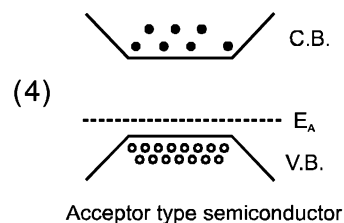
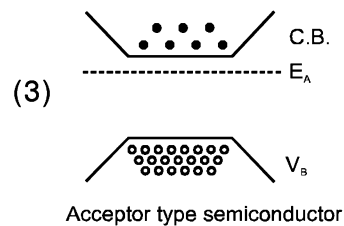
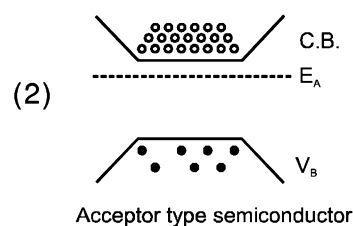
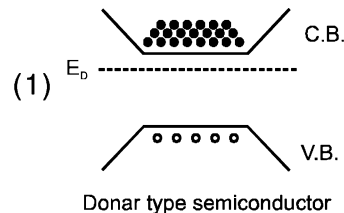
(1) $V_B = \sqrt{V^2 + (R\omega)^2}$

(2) $V_C = V + R\omega$

(3) $a_A = \sqrt{(a - R\alpha)^2 + (\omega^2 R)^2}$

(4) $a_D = \sqrt{(a + \omega^2 R)^2 + (R\alpha)^2}$

53. Identify correct diagram for donor and acceptor type semiconductor. If free electrons and holes are represented in diagram with $(\bullet)$ & $(\circ)$ respectively and other symbols have their usual meaning



(SPACE FOR ROUGH WORK)

SECTION - III

Reasoning Type

This section contains 3 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

54. STATEMENT-1 : Work done by a force on a body whose centre of mass does not move may be non-zero.

STATEMENT-2 : Work done by a force depends on the displacement of the centre of mass.

(1) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True.

55. STATEMENT-1: Drift velocity is the average velocity which is acquired by electron due to electric field applied by source.

STATEMENT-2 : Drift velocity is independent of temperature.

(1) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True.

56. STATEMENT-1 : A neutral conducting wire carrying no current, when placed in uniform magnetic field, experiences zero net force.

STATEMENT-2 : An electron in a conducting wire, carrying no current, doesn't experience any magnetic force, when placed in uniform magnetic field.

(1) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True.

(SPACE FOR ROUGH WORK)

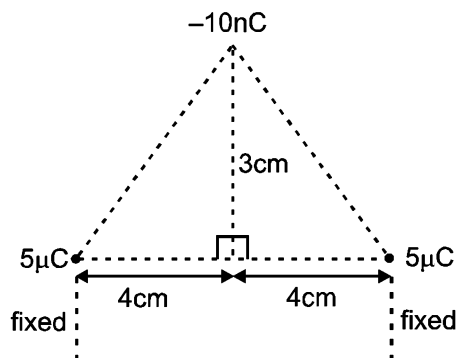
SECTION - IV

Comprehension Type

This section contains 2 paragraphs. Based on each paragraph, there are 2 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 57 to 58

Figure shows two fixed positive point charges of magnitude $5\mu\text{C}$. A negative point charge of -10nC and mass 1 kg is released as shown in figure. (Consider only electrostatic forces)



57. Initial acceleration of -10nC charge is :

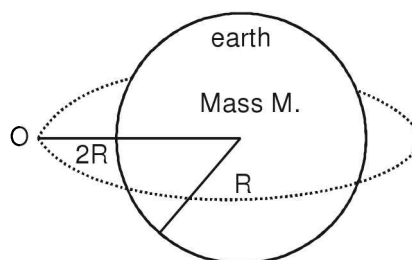
- (1) 0.216 m/s^2 (2) 3.6 m/s^2
(3) 2.16 m/s^2 (4) 360 m/s^2

58. Kinetic energy of -10nC charge when it crosses the line joining two fixed charges is :

- (1) 4.5 J (2) 4.5 mJ
(3) 3 J (4) 3 mJ

Paragraph for Question Nos. 59 to 60

A satellite is moving around earth (mass M , Radius R) in circular orbit of radius $2R$ in equatorial plane.



59. Find the total area on the earth, this satellite covers for communication purpose in its one complete revolution w.r.t earth.

- (1) $2\pi R^2$ (2) $\sqrt{3}\pi R^2$
(3) $2\sqrt{3}\pi R^2$ (4) $\frac{4}{\sqrt{3}}\pi R^2$

60. A particle is projected from this satellite radially outward relative to satellite with speed u . Find the minimum value of u so that particle escape from the gravitational force of earth. (Mass of earth $\gg \gg$ mass of satellite $\gg \gg$ mass of particle)

- (1) $\sqrt{\frac{GM}{R}}$ (2) $\sqrt{\frac{GM}{2R}}$
(3) $\sqrt{\frac{2GM}{R}}$ (4) $2\sqrt{\frac{GM}{R}}$

(SPACE FOR ROUGH WORK)

PART- C

Atomic masses : [H = 1, D = 2, Li = 7, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, Si = 28, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Cu = 63.5, Zn = 65, As = 75, Br = 80, Ag = 108, I = 127, Ba = 137, Hg = 200, Pb = 207]

SECTION – I

Straight Objective Type

This section contains 18 **multiple choice questions**. Each question has 5 choices (1), (2), (3), (4) and (5) out of which **ONLY ONE** is correct.

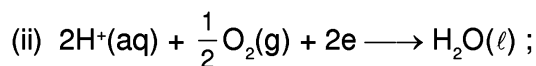
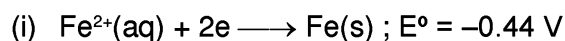
61. During the conversion of $\text{NH}_2\text{OH} \longrightarrow \text{N}_2\text{O}$, the equivalent weight of NH_2OH is (mol. wt. of NH_2OH is M)

- (1) M (2) M/2
(3) M/4 (4) M/5
(5) 2 M

62. Thermodynamically the most stable form of carbon is

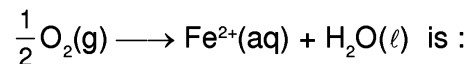
- (1) Diamond (2) Graphite
(3) Fullerenes (4) Coal
(5) Charcoal

63. If the half cell reactions are given as



$$E^\circ = +1.23 \text{ V}$$

the E° for the reaction, $\text{Fe}(\text{s}) + 2\text{H}^+ +$

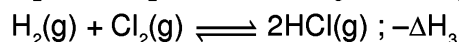
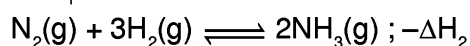
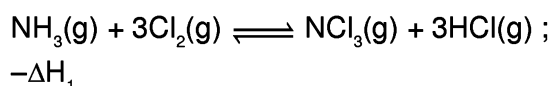


- (1) +1.67 V (2) 1.67 V
(3) +0.79 V (4) -0.79 V
(5) +1.58 V

64. Bromine water reacts with SO_2 to form :

- (1) H_2O and HBr (2) HBr and S
(3) H_2SO_4 and HBr (4) S and H_2O
(5) Br_2 and SO_3

65. Given,



The heat of formation of $\text{NCl}_3(\text{g})$ in terms of ΔH_1 , ΔH_2 and ΔH_3 is :

(1) $\Delta H_f = \Delta H_1 + \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$

(2) $\Delta H_f = -\Delta H_1 - \frac{\Delta H_2}{2} + \frac{3}{2}\Delta H_3$

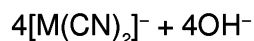
(3) $\Delta H_f = -\Delta H_1 + \frac{\Delta H_2}{2} - \frac{3}{2}\Delta H_3$

(4) $\Delta H_f = \Delta H_1 + \Delta H_2 + \Delta H_3$

(5) None of these

(SPACE FOR ROUGH WORK)

66. In the equation : $4M + 8CN^- + 2H_2O + O_2 \longrightarrow$



Identify the metal M :

- (1) Iron (2) Copper
(3) Zinc (4) Gold
(5) Sodium

67. The dissociation constant of acetic acid at a given temperature is 1.69×10^{-5} . The degree of dissociation of 0.01 M acetic acid in the presence of 0.01 M HCl is equal to

- (1) 0.41 (2) 0.13
(3) 1.69×10^{-3} (4) 0.013.
(5) 0.04

68. In an organic compound, phosphorus is estimated as :

- (1) H_3PO_4 (2) P_2O_5
(3) $Mg_2P_2O_7$ (4) $Mg_3(PO_4)_2$
(5) H_3PO_3

69. If 0.5 mole H_2 is reacted with 0.5 mole I_2 in a ten-litre container at $444^\circ C$ and at same temperature value of equilibrium constant K_c is 49, the ratio of $[HI]$ and $[I_2]$ will be :

- (1) 7 (2) $\frac{1}{7}$
(3) $\sqrt{\frac{1}{7}}$ (4) 49
(5) $\sqrt{7}$

70. Nitrolium is a mixture of

- (1) $Ca(CN)_2$ + graphite
(2) CaC_2 + graphite
(3) $CaCN_2$ + graphite
(4) $CaCN_2 + N_2$
(5) $CaC_2 + NH_3$

71. Which of the following complex species does not involve d^2sp^3 hybridisation ?

- (1) $[CoF_6]^{3-}$
(2) $[Co(NH_3)_6]^{3+}$
(3) $[Fe(CN)_6]^{3-}$
(4) $[Cr(NH_3)_6]^{3+}$
(5) $[Fe(CN)_6]^{4-}$

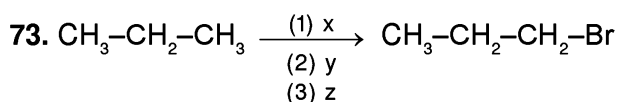
(SPACE FOR ROUGH WORK)

72. For a reaction $A + 2B \longrightarrow C + D$, the following data were obtained :

Expt. Initial Concentration (mol L ⁻¹)			Initial rate of formation of D (mol L ⁻¹ min ⁻¹)
	(A)	(B)	
1.	0.1	0.1	6.0×10^{-3}
2.	0.3	0.2	7.2×10^{-2}
3.	0.3	0.4	2.88×10^{-1}
4.	0.4	0.1	2.4×10^{-2}

The correct rate law expression will be

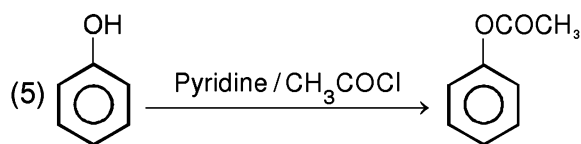
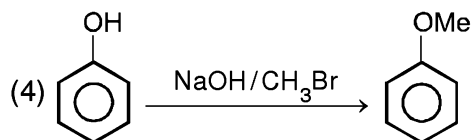
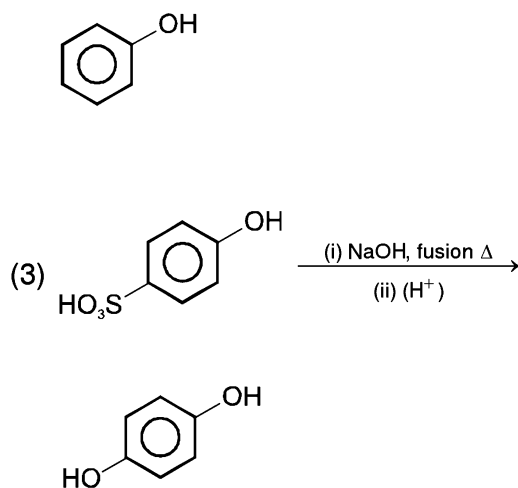
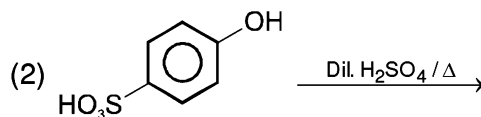
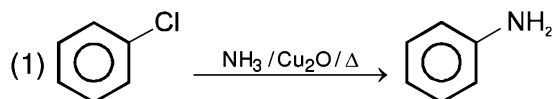
- (1) Rate = $k [A] [B]$ (2) Rate = $k [A] [B]^2$
 (3) Rate = $k [A]^2 [B]^2$ (4) Rate = $k [A]^2 [B]$
 (5) Rate = $k [A]^2 [B]^3$



x, y, z are respectively :

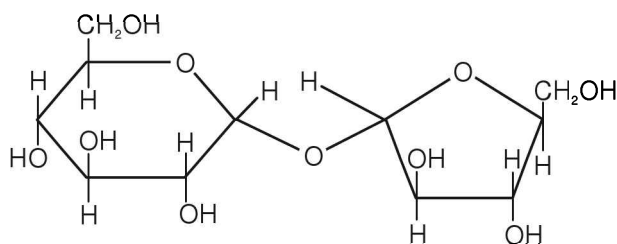
- (1) $\text{Br}_2 / h\nu$, conc. H_2SO_4 , ThO_2 / Δ
 (2) $\text{Cl}_2 / h\nu$, alc. KOH / Δ , $\text{HBr} / \text{R}_2\text{O}_2$
 (3) $\text{Cl}_2 / h\nu$, $\text{tBuO}^\bullet / \Delta$, HBr (dark)
 (4) $\text{F}_2 / h\nu$, $\text{Al}_2\text{O}_3 / \Delta$, NBS
 (5) $\text{Cl}_2 / h\nu$, alc. KOH / Δ , MnO_2

74. The aromatic electrophilic substitution reaction is :



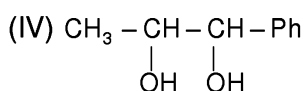
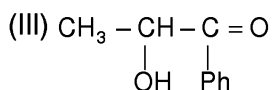
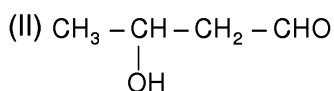
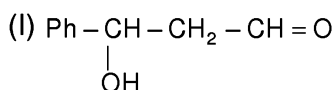
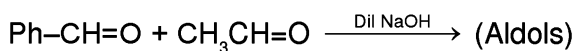
(SPACE FOR ROUGH WORK)

75. What is true about the following carbohydrate:



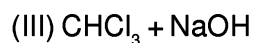
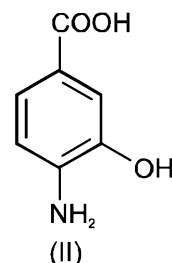
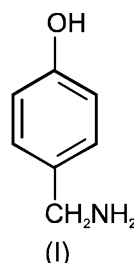
- (1) It anomerizes in solution phase.
- (2) It is a reducing sugar.
- (3) It shows the phenomenon of inversion of sugar in acidic medium.
- (4) It shows mutarotation.
- (5) It has 1,4 glycosidic linkage

76. Identify the option which represents the correct products of the following reaction,



- (1) I, II only
- (2) I, II, III only
- (3) III, IV only
- (4) I, III only
- (5) I, II, III, IV

77. The labels of two bottles containing compounds I and II were lost. The names of these compounds were available in a register. Following are structures of I and II. Which combination of reagents **cannot** help in distinguishing these compounds :



(1) I and II

(2) III and IV

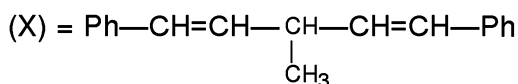
(3) II and III

(4) I and IV

(5) I, II and III

(SPACE FOR ROUGH WORK)

78. The correct statement about the following compound (X) is :



- (1) The total number of stereoisomers of (X) is six.
- (2) The total number of enantiomeric pairs of (X) is two.
- (3) If both the double bonds are trans then the number of enantiomers is two.
- (4) If the two double bonds are cis and trans respectively then the possible enantiomers are two.
- (5) If both the double bonds are cis then the number of enantiomers is four.

SECTION – II

One or more options correct Type

This section contains 5 **multiple choice questions**. Each question has four choices (1), (2), (3) and (4) out of which **ONE** or **MORE** are correct.

79. The correct statement(s) pertaining to the adsorption of a gas on a solid surface is (are)

- (1) Adsorption is always exothermic
- (2) Physisorption may transform into chemisorption at high temperature
- (3) Physisorption increases with increasing temperature but chemisorption decreases with increasing temperature
- (4) Chemisorption is more exothermic than physisorption, however it is very slow due to higher energy of activation.

80. The vapour pressure of the solution of two liquids A($p^\circ = 80$ mm) and B($p^\circ = 120$ mm) is found to be 100 mm when $x_A = 0.4$. The result shows that :

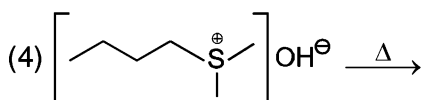
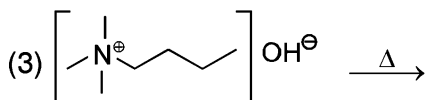
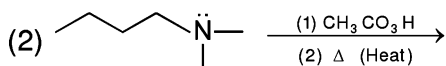
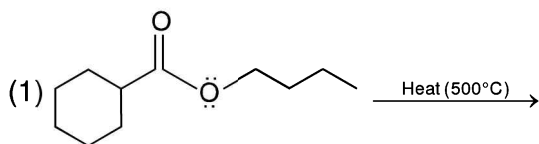
- (1) solution exhibits ideal behaviour
- (2) $\Delta H_{\text{solution}} < 0$
- (3) solution shows negative deviations
- (4) solution will show positive deviations for lower concentration and negative deviations for higher concentrations.

81. Which of the following order is/are correct :

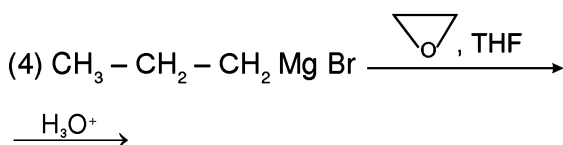
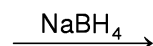
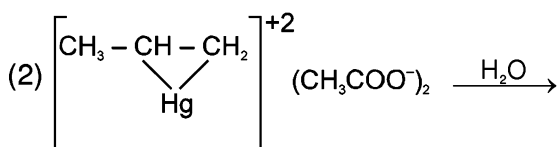
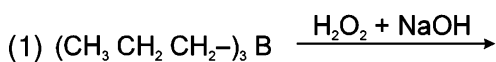
- (1) $\text{O}_2^+ < \text{O}_2 < \text{O}_2^-$
(Bond energy)
- (2) $\text{BF}_3 = \text{BCl}_3 = \text{BBr}_3$
(Bond angle)
- (3) $\text{N}-\text{H} < \text{P}-\text{H} < \text{As}-\text{H} < \text{Sb}-\text{H}$
(Bond length)
- (4) $\text{AlCl}_3 < \text{MgCl}_2 < \text{CaCl}_2$
(Ionic character)

(SPACE FOR ROUGH WORK)

82. But-1-ene is formed in reaction/s.



83. In which of the following reaction/s alcohol will be formed as the final product.



SECTION - III

Reasoning Type

This section contains 3 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

84. **Statement-1** : Cr^{+3} is a reducing agent and Mn^{+2} is oxidising agent.

Statement-2 : Mn^{+3} has d^5 configuration.

- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (3) Statement-1 is True, Statement-2 is False.
 (4) Statement-1 is False, Statement-2 is True.

85. **Statement-1** : For an element order of successive ionisation energy is $\text{IE}_1 < \text{IE}_2 < \text{IE}_3 < \text{IE}_4 \dots$

Statement-2 : Removal of an electron from cation is difficult than neutral atom because effective nuclear charge increases.

- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (3) Statement-1 is True, Statement-2 is False.
 (4) Statement-1 is False, Statement-2 is True.

(SPACE FOR ROUGH WORK)

86. **Statement-1** : Benzene and ethene both give reactions with electrophilic reagents.

Statement-2 : Benzene and ethene both have loosely bound π electrons, which can be donated to vacant orbital of the electrophile.

- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
- (3) Statement-1 is True, Statement-2 is False.
- (4) Statement-1 is False, Statement-2 is True.

SECTION - IV

Comprehension Type

This section contains 2 paragraphs. Based on each paragraph, there are 2 questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 87 to 88

Potassium crystallizes in body centred cubic lattice with a unit cell length $a = 5.2 \text{ \AA}$

87. What is the distance between nearest neighbours

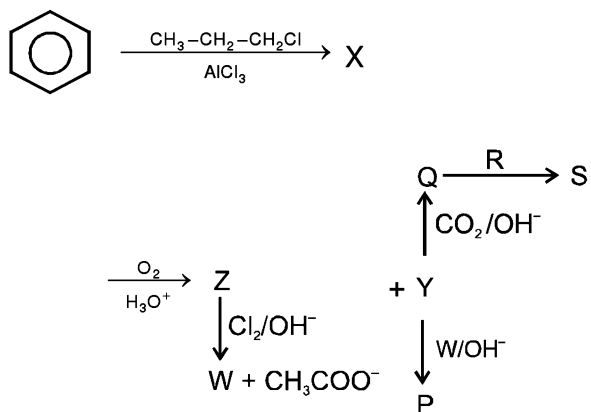
- (1) $4.5 \text{ \AA}$
- (2) $9.0 \text{ \AA}$
- (3) $5.2 \text{ \AA}$
- (4) None of these

88. How many nearest neighbours does each K atom have

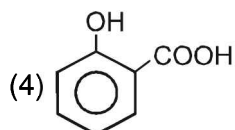
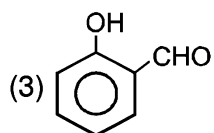
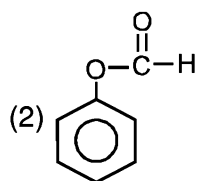
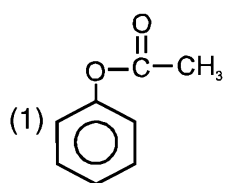
- (1) 6
- (2) 2
- (3) 8
- (4) 4

(SPACE FOR ROUGH WORK)

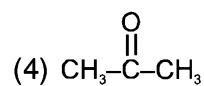
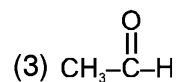
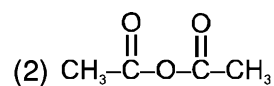
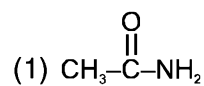
Paragraph for Question Nos. 89 to 90



89. Which of the following compound is 'P' ?



90. S is a well known pain killer which of the following is 'R' ?



(SPACE FOR ROUGH WORK)

MOCK TEST 2

CODE

Q

PREP JEE

PREPARE FOR JEE EXAMINATION WITH US

PREP JEE TEST SERIES (PTS)

JEE MAIN

TIME: 3 HOURS MAX. MARKS: 360

Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose

IMPORTANT INSTRUCTIONS

1. Immediately fill the particulars on this page of the Test Booklet with Blue / Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.
3. The test is of 3 hours duration.
4. The Test Booklet consists of 90 questions. The maximum marks are 360.
5. There are three parts in the question paper A, B, C consisting of Mathematics, Physics and Chemistry having 30 questions. Mathematics & Physics part consists of Two sections and Chemistry consists of One section.
6. For each question you will be awarded 4 marks if you darken all the bubble(s) corresponding to only the correct answer(s) and zero mark if no bubbles are darkened. In all other cases, $-1/4$ (one fourth (1)) marks will be deducted.
7. There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 6 above.

Filling the Top-half of the ORS :

Use only Black ball point pen only for filling the ORS. Do not use Gel / Ink / Felt pen as it might smudge the ORS.

8. Write your Roll no. in the boxes given at the top left corner of your ORS with black ball point pen. Also, darken the corresponding bubbles with Black ball point pen only. Also fill your roll no on the back side of your ORS in the space provided (if the ORS is both side printed).
9. Fill your Paper Code as mentioned on the Test Paper and darken the corresponding bubble with Black ball point pen.
10. If student does not fill his/her roll no. and paper code correctly and properly, then his/her marks will not be displayed and 5 marks will be deducted (paper wise) from the total.
11. Since it is not possible to erase and correct pen filled bubble, you are advised to be extremely careful while darken the bubble corresponding to your answer.
12. Neither try to erase / rub / scratch the option nor make the Cross (X) mark on the option once filled. Do not scribble, smudge, cut, tear, or wrinkle the ORS. Do not put any stray marks or whitener anywhere on the ORS.
13. If there is any discrepancy between the written data and the bubbled data in your ORS, the bubbled data will be taken as final.

VISIT : www.prepjee.in for more details consider buying more Test Paper and keep Practicing

PART-A

SECTION - I

Straight Objective Type

This section contains 27 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. Let P be a point on $y = x^2$. If normal at P meets parabola again at Q then minimum length of PQ is

- (1) $\frac{27}{4}$ (2) $\frac{3\sqrt{3}}{2}$
(3) $2\sqrt{3}$ (4) 18

2. A line L with slope 2 cuts parabola $y^2 = 8x$ to form a chord. If mid-point of chord lies on $x = 4$ then equation of line L is

- (1) $y = 2x - 6$ (2) $x = 2y - 6$
(3) $2x + y = 6$ (4) $x - 2y = 6$

3. If $f(x)$ is a non – constant differentiable function

such that $(f(x))^{1947} = \int_{1857}^x (f(t))^{1946} dt$, then

$f(2014) =$

- (1) $\frac{157}{1947}$ (2) $\frac{2013}{2014}$
(3) 0 (4) $\frac{1857}{1947}$

4. If x_1 and x_2 are solutions for x in quadratic

equation $x^2 + (\sin \theta - 1)x - \frac{1}{2}\cos^2 \theta = 0$, then

maximum value of $x_1^2 + x_2^2$ is equal to

- (1) 2 (2) 3
(3) $\frac{9}{4}$ (4) 4

5. The mean and variance of a binomial distribution are 30 and 5 respectively. The probability of two successes is

- (1) $\frac{875}{6^{34}}$
(2) $\frac{875}{2 \cdot 6^{34}}$
(3) $\frac{875}{6^{36}}$
(4) ${}^{36}C_2 \left(\frac{5}{6}\right)^{34} \left(\frac{1}{6}\right)^2$

6. The order of differential equation of family of circles passing through intersection of $L \equiv 3x + 4y - 7 = 0$ and $S \equiv x^2 + y^2 - 2x - 2y + 1 = 0$ is

- (1) 1
(2) 2
(3) 3
(4) 4

(SPACE FOR ROUGH WORK)

7. Let ABCD be a square with A(0, 0), C(2, 2). If M is mid – point of AB and P is a variable point on BC. The smallest value of DP + PM is
- (1) $\sqrt{13}$ (2) $\sqrt{2} + \sqrt{5}$
- (3) $2\sqrt{3}$ (4) $1 + 2\sqrt{2}$
8. An office employs thirty people. Five of them speak Sanskrit, Hindi, English. Nine speak Sanskrit, English. Twenty speak Hindi of which twelve also speak Sanskrit. Eighteen speak English. No one speak only Sanskrit. An employee selected at random is known to speak either Hindi or English, the probability that employee speaks only two of three languages is
- (1) 14/15 (2) 7/15
- (3) 1/6 (4) 5/6
9. ABC is vertical tower such that AB = BC = x. From the foot A of the tower, a horizontal line AQP is drawn. If the angle of elevation of the point B from P is α and that of C from Q is 2α , then AP, x, PQ are in
- (1) A.P.
- (2) G.P.
- (3) H.P.
- (4) None of these
10. The equation of plane through intersection of the planes $x + 2y + z - 1 = 0$ and $2x + y + 3z - 2 = 0$ and perpendicular to plane $x + y + z = 1$ is
- (1) $x + 4y - 3z + 1 = 0$
- (2) $x - 4y + 3z - 1 = 0$
- (3) $x - 4y + 3z = 0$
- (4) $x + 4y - 3z = 0$
11. A man running round a race course notes that the sum of the distance of two flag – posts from him is always 10 meters and distance between the flag – posts is 8 meters. The area of the path he encloses in square meters is
- (1) 15π
- (2) 12π
- (3) 18π
- (4) 8π
12. Let H be a set of hyperbolas. If a relation R on H be defined by
- $R = \{(h_1, h_2) : h_1, h_2 \text{ have same pair of asymptotes, } h_1, h_2 \in H\}$, then the relation R is
- (1) reflexive and symmetric but not transitive
- (2) symmetric and transitive but not reflexive
- (3) reflexive and transitive but not symmetric
- (4) Equivalence relation

(SPACE FOR ROUGH WORK)

13. Let $\omega = \frac{-1+i\sqrt{3}}{2}$, and $\omega, \omega, \alpha_1, \alpha_2, \dots, \alpha_8$ be

roots of equation $x^{10} + ax + b = 0$, where

$\alpha_i \neq \alpha_j, i \neq j$ value of $(\omega - \alpha_1)(\omega - \alpha_2) \dots (\omega - \alpha_8)$

=

(1) $90 \omega^2$

(2) $^{10}C_2 \omega^2$

(3) $10 \omega + a$

(4) 0

14. If $A = \{1, 2, 3, 4\}$ then which of the following are functions from A to itself ?

(1) $f_1 = \{(x, y) : y = x + 1\}$

(2) $f_2 = \{(x, y) : x + y > 4\}$

(3) $f_3 = \{(x, y) : x < y\}$

(4) $f_4 = \{(x, y) : x + y = 5\}$

15. Let p be the statement "Ram races", q be the statement "Ram wins". The statement $\sim(p \vee \sim q)$ is :

(1) Ram does not race and Ram does not win.

(2) It is not true that Ram races or Ram does not win.

(3) Ram does not race and Ram wins

(4) It is not true that Ram races and Ram does not win.

16. Let $I = \int_0^{\pi/6} \frac{\cos x}{x} dx$, $J = \int_{\pi/3}^{\pi/2} \frac{\cos x}{x} dx$. Which of

the following is **CORRECT** ?

(1) $I < \frac{\pi}{6}$, $J < \frac{\pi}{6}$

(2) $I > \frac{\pi}{6}$, $J < \frac{\pi}{6}$

(3) $I < \frac{\pi}{6}$, $J > \frac{\pi}{6}$

(4) $I > \frac{\pi}{6}$, $J > \frac{\pi}{6}$

17. If x_1 and x_2 are the real and distinct roots of $ax^2 + bx + c = 0$, then

$\lim_{x \rightarrow x_1} \left(1 + \sin(ax^2 + bx + c)\right)^{\frac{1}{x-x_1}}$ is equal to

(1) $e^{x_1-x_2}$

(2) $e^{x_2-x_1}$

(3) $e^{a(x_1-x_2)}$

(4) $e^{a(x_2-x_1)}$

(SPACE FOR ROUGH WORK)

18. Let p, q be real numbers such that $\frac{1}{p} - \frac{1}{q} = 1$

and $0 < p \leq \frac{1}{2}$ then

(1) $p + \frac{1}{2}p^2 + \frac{1}{3}p^3 + \dots \infty = q - \frac{1}{2}q^2 + \frac{1}{3}q^3$

$-\frac{1}{4}q^4 + \dots \infty$

(2) $p - \frac{1}{2}p^2 + \frac{1}{3}p^3 - \frac{1}{4}p^4 + \dots \infty = q + \frac{1}{2}q^2$

$+ \frac{1}{3}q^3 + \frac{1}{4}q^4 + \dots \infty$

(3) $p + \frac{p^2}{2!} + \frac{p^3}{3!} + \dots \infty = q - \frac{q^2}{2!} + \frac{q^3}{3!} - \frac{q^4}{4!}$

$+ \dots \infty$

(4) $p - \frac{1}{2}p^2 + \frac{1}{3}p^3 - \frac{1}{4}p^4 + \dots \infty = q$

$+ \frac{q^2}{2!} + \frac{q^3}{3!} + \frac{q^4}{4!} + \dots \infty$

19. $\int_0^{\pi/2} \frac{x dx}{1 + \sin x} =$

(1) $\ln 2 + \pi/4$

(2) $\ln 2 - \pi/4$

(3) $1 + \pi/2 + \ln 2$

(4) $\ln 2$

20. Number of acute angles α such that

$\cos \alpha \cos 2\alpha \cos 4\alpha = \frac{1}{8}$ are

(1) 1

(2) 2

(3) 3

(4) 4

21. The smallest positive integer n such that

$11^n - 1$ is divisible by 100 is :

(1) 4

(2) 5

(3) 10

(4) 11

22. The integral curve of equation $x dy - y dx = y dy$,

intersects the y -axis at $(0, 1)$ and the line $y = e$

at (a, b) . The value of $a + b =$

(1) 0

(2) 1

(3) $2e$

(4) e

23. A variable line $ax + by + c = 0$, where a, b, c

are in A.P., is normal to a circle

$(x - \alpha)^2 + (y - \beta)^2 = \gamma$, which is orthogonal to

circle $x^2 + y^2 - 4x - 4y - 1 = 0$. The value of

$\alpha + \beta + \gamma$ is equal to :

(1) 3

(2) 5

(3) 10

(4) 7

(SPACE FOR ROUGH WORK)

24. Let $A = [a_{ij}]_{3 \times 3}$, $B = [b_{ij}]_{3 \times 3}$ such that $a_{ij} = a_{ji}$, $b_{ij} = -b_{ji}$. Which of the following is **CORRECT** ?

- (1) AB is symmetric matrix
- (2) $|A| = 0$ and $|B| \neq 0$
- (3) If $a_{ij} = b_{ij}$ for $i < j$ then $C = A + B$ is necessarily an upper triangular matrix.
- (4) If $a_{ij} = b_{ij}$ for $i \leq j$ and $C = A + B$, then $|C| = |A|$.

25. If $\bar{a} = \hat{i} + \hat{j} + \hat{k}$, $\bar{b} = \hat{i} - \hat{j} + \hat{k}$, $\bar{c} = \hat{i} + 2\hat{j} - \hat{k}$, then

$$\text{value of } \begin{vmatrix} \bar{a} \cdot \bar{a} & \bar{a} \cdot \bar{b} & \bar{a} \cdot \bar{c} \\ \bar{b} \cdot \bar{a} & \bar{b} \cdot \bar{b} & \bar{b} \cdot \bar{c} \\ \bar{c} \cdot \bar{a} & \bar{c} \cdot \bar{b} & \bar{c} \cdot \bar{c} \end{vmatrix} =$$

- (1) 16
- (2) 24
- (3) 256
- (4) 144

26. Number of sequences of a 's and b 's of length 20 having all consecutive a 's or all consecutive b 's or both are :

- (1) ${}^{20}C_2$
- (2) $2 \times {}^{20}C_2$
- (3) ${}^{20}C_2 + 2$
- (4) 382

27. The equation of latus rectum of the rectangular hyperbola $xy = c^2$ is

- (1) $x - y = 2c + \sqrt{2}c$, $c > 0$
- (2) $x + y = 2c + 2\sqrt{2}c$, $c > 0$
- (3) $x + y = c + \sqrt{2}c$, $c > 0$
- (4) $x + y = 2\sqrt{2}c$

SECTION - II

Reasoning Type

This section contains 3 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

28. Let $n \geq 3$. A list of numbers $0 < x_1 < x_2 < \dots < x_n$ has mean μ and standard deviation σ . A new list of numbers is made as follows : $y_1 = 0$, $y_2 = x_2$, ..., $y_{n-1} = x_{n-1}$, $y_n = x_1 + x_n$. The mean and the standard deviation of the new list are $\hat{\mu}$ and $\hat{\sigma}$.

Statement-1 : $\sigma < \hat{\sigma}$ and $\mu = \hat{\mu}$

Statement-2 : Variance is independent of choice of origin while mean dependent on choice of origin.

- (1) Statement -1 is True, Statement -2 is True; Statement -2 is a correct explanation for Statement -1
- (2) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1
- (3) Statement -1 is True, Statement -2 is False
- (4) Statement -1 is False, Statement -2 is True

(SPACE FOR ROUGH WORK)

29. Statement- 1 : A vector in xy plane through origin and perpendicular to vector $4\hat{i} + 3\hat{j} + \hat{k}$ is parallel to $(3\hat{i} - 4\hat{j})$

Statement- 2 : A vector in plane of non-collinear vectors $\vec{a}$, $\vec{b}$ and perpendicular to vector $\vec{c}$ is parallel to $(\vec{a} \times \vec{b}) \times \vec{c}$.

(1) Statement -1 is True, Statement -2 is True;
Statement -2 is a correct explanation for Statement -1

(2) Statement-1 is True, Statement-2 is True ;
Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement -1 is True, Statement -2 is False

(4) Statement -1 is False, Statement -2 is True

30. Let $f(x) = \sin x + \cos x + \tan x + \sin^{-1}x + \tan^{-1}x$

Statement- 1 : If p and q are maximum and minimum values of $f(x)$ then

$$\frac{p+q}{2} = \sin 1 + \tan 1 + \frac{3\pi}{4}$$

Statement- 2 : If $y = g(x)$ is strictly monotonic increasing on $x \in [a, b]$ then minimum value of y is $g(a)$ and maximum value of y is $g(b)$.

(1) Statement -1 is True, Statement -2 is True;
Statement -2 is a correct explanation for Statement -1

(2) Statement-1 is True, Statement-2 is True;
Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement -1 is True, Statement -2 is False

(4) Statement -1 is False, Statement -2 is True

(SPACE FOR ROUGH WORK)

PART-B

SECTION - I

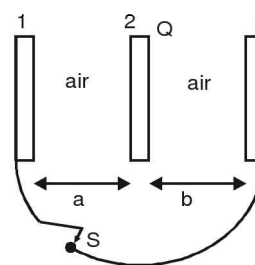
Straight Objective Type

This section contains **28 multiple choice questions**. Each question has choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

31. A solid sphere of mass 2 kg and density 10^5 kg/m^3 hanging from a string is lowered into a vessel of uniform cross-section area 10 m^2 containing a liquid of density $0.5 \times 10^5 \text{ kg/m}^3$, until it is fully immersed. The increase in pressure of liquid at the base of the vessel is (Assume liquid does not spill out of the vessel and take $g = 10 \text{ m/s}^2$)

- (1) 2 N/m^2
- (2) 1 N/m^2
- (3) 4 N/m^2
- (4) $\frac{1}{2} \text{ N/m}^2$

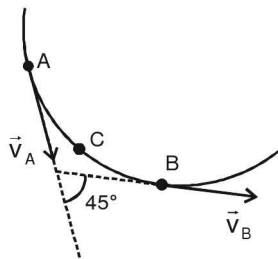
32. A system of three large parallel conducting plates 1, 2 and 3 are shown in figure, charge on middle plate is Q . Plate 1 and plate 3 are connected with conducting wire and switch S and initially plate 1 and plate 3 are neutral. After closing switch S which of the following option is **correct** -



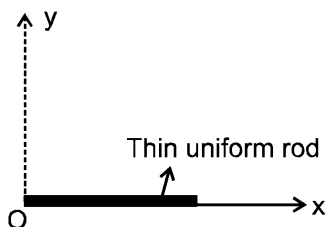
- (1) Charge at right surface of plate 2 is $\frac{Qa}{a+b}$
- (2) Magnitude of charge at right surface of plate 1 is $\frac{Qb}{a+b}$
- (3) Total charge of the system is Q
- (4) All the above options are correct

(SPACE FOR ROUGH WORK)

33. A particle moves along a constant curvature path between A and B (length of curve wire AB is 100m) with a constant speed of 72 km/hr. The acceleration of particle at mid point 'C' of curve AB is :

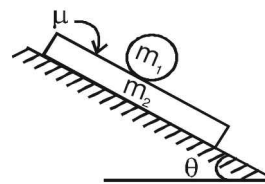


- (1) Zero
 (2) 3.14 m/s^2
 (3) 50.84 m/s^2
 (4) 50.84 km/hr^2
34. The locus of all the points on the x - y plane about which the moment of inertia of the uniform rod (axes perpendicular to xy-plane) shown in figure, is same as that about axis perpendicular to rod and passing through point 'O' is :



- (1) circle (2) parabola
 (3) straight line (4) ellipse

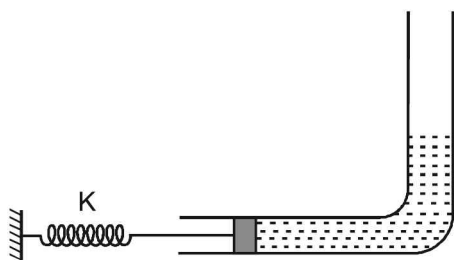
35. A uniform solid sphere of mass m_1 is placed on the plank of mass m_2 and the coefficient of friction between the plank and sphere is μ . If the inclined plane is frictionless then frictional force on m_1 is (Assume that the plank and uniform solid sphere are released on the inclined plane from rest)



- (1) parallel to the incline plane upwards.
 (2) parallel to the incline plane downwards.
 (3) Zero
 (4) $0 < f < \mu m_1 g \cos \theta$

(SPACE FOR ROUGH WORK)

36. A massless piston connected to a spring of spring constant k fits smoothly inside an L shaped fixed glass tube as shown in the figure. The piston can slide without friction and without leakage of water. The cross sectional area of the tube is s and the density of the liquid is ρ . If the system is slightly disturbed from its equilibrium position then the angular frequency of its small oscillations is: (m is the total mass of liquid and the liquid is ideal)

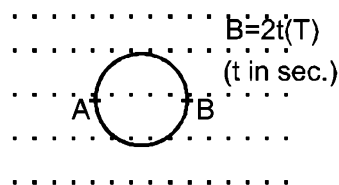


- (1) $\sqrt{\frac{2\rho s g}{m}} + \sqrt{\frac{k}{m}}$
- (2) $\sqrt{\frac{\rho s g}{m}} + \sqrt{\frac{k}{m}}$
- (3) $\sqrt{\frac{k + 2\rho s g}{m}}$
- (4) $\sqrt{\frac{k + \rho s g}{m}}$

37. In a damped oscillator the amplitude of vibrations of mass $m = 150$ grams falls by $\frac{1}{e}$ times of its initial value in time t_0 due to viscous forces. The time t_0 and the percentage loss in mechanical energy during the above time interval t_0 respectively are (Let damping constant be 50 grams/s)

- (1) $6s, \frac{e^2 - 1}{e^2} \times 100$ (2) $3s, \frac{e^2 - 1}{e^2} \times 100$
- (3) $6s, \frac{e - 1}{e} \times 100$ (4) $3s, \frac{e - 1}{e} \times 100$

38. A circular coil is placed in a uniform but time varying magnetic field as shown in figure. Then which of the following is correct statement :



- (1) A variable current flows from 'A' to 'B' in the lower half of ring
- (2) A variable current flows from 'B' to 'A' in the lower half of ring
- (3) A constant current flows from 'A' to 'B' in the lower half of ring
- (4) A constant current flows from 'B' to 'A' in the lower half of ring

(SPACE FOR ROUGH WORK)

39. The distance moved by the screw of a screw gauge is 2 mm in four rotations and there are 50 divisions on its cap. When nothing is put between its jaws, 30th division of circular scale coincides with reference line, with zero of circular scale lying above the reference line. When a plate is placed between the jaws, main scale reads 2 division and circular scale reads 20 division. Thickness of plate is :
- (1) 1.5 mm (2) 1.2 mm
(3) 1.4 mm (4) 1.6 mm
40. Two stars at a distance 20 light years are viewed through a telescope having a lens of 20 cm diameter. Then the minimum separation of these stars for which they appears distinguishable is ($\lambda = 6000 \text{ \AA}$) :
- (1) 7.32×10^{-5} light years
(2) 3.66×10^{-5} light years
(3) 14.64×10^{-5} light years
(4) None of these
41. A car is moving towards an approaching bike. Bike is moving with velocity 30 m/s. Frequency of horn sounded by car is 100 Hz, while it is observed by bike rider as 120 Hz. The actual duration of horn is 6 sec. Find the time interval for which the bike rider hears it. (Take velocity of sound in air as 330 m/s)
- (1) 5 sec
(2) 6 sec
(3) 7 sec
(4) Cannot be calculate with given information
42. A carnot refrigeration cycle absorbs heat at 270 K and rejects heat at 300 K. If the cycle is absorbing 1260 KJ/min at 270 K, then work required per second is :
- (1) 2.33 KJ/sec.
(2) 4.66 KJ/sec.
(3) 1 KJ/sec.
(4) 4 KJ/sec.
43. The radius of the orbit of an electron in a Hydrogen-like atom is $3 a_0$, where a_0 is the Bohr radius. Its orbital angular momentum is $\frac{3h}{2\pi}$. It is given that h is Planck's constant and R is Rydberg constant. The possible wavelength, when the atom de-excites is :
- (1) $\frac{4}{5R}$
(2) $\frac{4}{9R}$
(3) $\frac{1}{2R}$
(4) $\frac{9}{32R}$

(SPACE FOR ROUGH WORK)

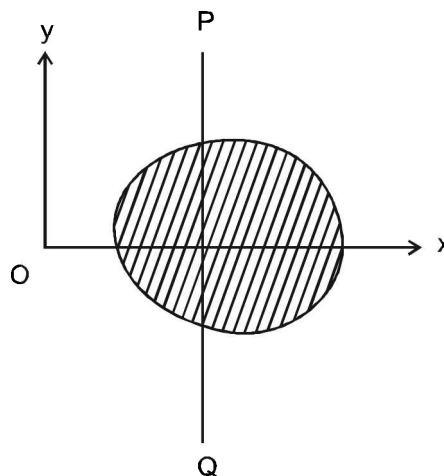
44. A particle leaves the origin at $t = 0$ with an initial velocity $\vec{v} = 3v_0\hat{i}$. It experiences a constant acceleration $\vec{a} = -2a_1\hat{i} - 5a_2\hat{j}$. The time at which the particle reaches its maximum x coordinate is (a_1 , a_2 & v_0 are positive number)

- (1) $\frac{3V_0}{10a_2}$
 (2) $\frac{3V_0}{9a_1 - 5a_2}$
 (3) $\frac{3V_0}{2a_1}$
 (4) $\frac{3V_0}{2a_1 + 5a_2}$

45. A man beats drum at a certain distance from a large mountain. He slowly increases the rate of beating and finds that the echo is not heard distinctly, when the drum beating is at the rate of 40 per minute. He moves by 80 meter towards the mountain and finds that the echo is again not heard distinctly, when the rate of beating of the drum is 1 per second. What is the initial distance of the man from the mountain ?

- (1) 120 m
 (2) 240 m
 (3) 270 m
 (4) 340 m

46. Line PQ is parallel to y-axis and moment of inertia of a rigid body about PQ line is given by $I = 2x^2 - 12x + 27$, where x is in meter and I is in kg-m^2 . The minimum value of I is :



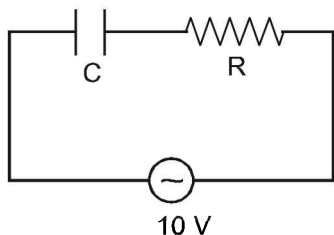
- (1) 27 kg m^2
 (2) 11 kg m^2
 (3) 17 kg m^2
 (4) 9 kg m^2

47. A ring of mass m and radius R is rolling without sliding on a flat fixed horizontal surface, with a constant angular velocity. The angle between the velocity and acceleration vectors of a point P on the rim of the ring at the same horizontal level as the centre of the ring may be -

- (1) zero
 (2) 90°
 (3) 135°
 (4) $\tan^{-1}1/2$

(SPACE FOR ROUGH WORK)

48. In a series RC circuit shown in figure applied voltage is 10 V and voltage across capacitor is found to be 8 V. Then voltage across R and phase difference between current and applied voltage respectively will be



- (1) 6V, $\tan^{-1}\left(\frac{4}{3}\right)$
 (2) 3V, $\tan^{-1}\left(\frac{3}{4}\right)$
 (3) 6V, $\tan^{-1}\left(\frac{5}{3}\right)$
 (4) 2V, $\tan^{-1}(1)$

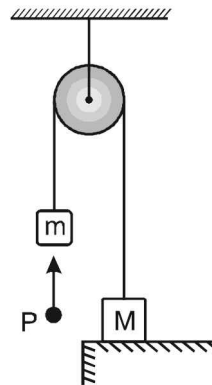
49. The electric field of a plane electromagnetic wave in vacuum is

$$E_y = 0.5 \cos \left[2\pi \times 10^8 \left\{ t - \frac{x}{c} \right\} \right] \text{ V/m; } E_z = E_x = 0.$$

Then select correct alternative.

- (1) Direction of propagation of wave is along +ve x-axis
 (2) Plane of polarization is x-y plane
 (3) Amplitude of magnetic field is $1.66 \times 10^{-9} \text{ T}$
 (4) All of above

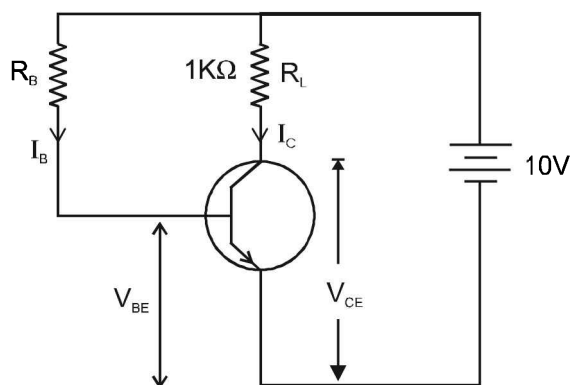
50. A light inelastic thread passes over a small frictionless pulley. Two blocks of masses $m = 1 \text{ kg}$ and $M = 3 \text{ kg}$ are attached with the thread and heavy block rests on a surface. A particle P of mass 1 kg moving upward with a velocity 10 m/s collides with the lighter block elastically at $t = 0$. ($g = 10 \text{ m/s}^2$). Choose the **incorrect** option :



- (1) Velocity with which heavy block is jerked into motion is 2.5 m/s
 (2) Maximum height ascended by heavy block after it is jerked into motion is 0.625 m
 (3) Impulse of tension in jerk is 7.5 Ns
 (4) Time after which block again strikes the surface is 2.5 s

(SPACE FOR ROUGH WORK)

51. In the circuit shown in figure, the current gain $\beta = 100$ for a npn transistor. The bias resistance R_B so that $V_{CE} = 5V$ is ($V_{BE} \ll 10V$)



- (1) $2 \times 10^3 \Omega$
 (2) $10^5 \Omega$
 (3) $2 \times 10^5 \Omega$
 (4) $5 \times 10^5 \Omega$

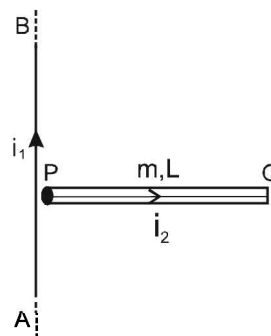
52. A point mass M is placed at a distance 'd' from infinite sheet. The magnitude of gravitational flux through the sheet is :

- (1) $2GM\pi$ (2) $4GM\pi$
 (3) $3GM\pi$ (4) $GM\pi$

53. The activity of radioactive sample is measured as 9750 counts per minute at $t = 0$, and 975 counts per minute at $t = 5$ minutes. The decay constant approximately is :

- (1) 0.230 per minute
 (2) 0.461 per minute
 (3) 0.691 per minute
 (4) 0.922 per minute

54. A metallic uniform rod PQ is hinged at point P and it can rotate about point P in vertical plane and an infinite long wire AB having current i_1 is placed as shown in the figure. If the mass of rod is m and length L , then the current i_2 in rod PQ such that it remains in equilibrium as shown is (separation between P and current carrying wire AB is very small) (consider gravity)



- (1) $\frac{2mg\pi}{\mu_0 i_1}$ (2) $\frac{mg\pi}{2\mu_0 i_1}$
 (3) $\frac{mg\pi}{4\mu_0 i_1}$ (4) $\frac{mg\pi}{\mu_0 i_1}$

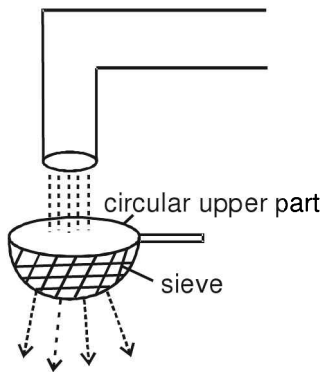
55. Maximum deviation occurs in a prism of refracting angle A_1 and having critical angle of its material as A_2 , when the angle of incidence is -

- (1) $\sin^{-1}\left[\frac{\sin(A_1 - A_2)}{\sin A_1}\right]$ (2) $\sin^{-1}\left[\frac{\sin A_2}{\sin A_1}\right]$
 (3) $\sin^{-1}\left(\frac{\sin A_1}{\sin A_2}\right)$ (4) $\sin^{-1}\left[\frac{\sin(A_1 - A_2)}{\sin A_2}\right]$

(SPACE FOR ROUGH WORK)

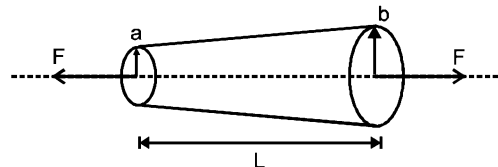
56. Water, having positive ions of charge '+e' dissolved in it with concentration c (Number of ions/volume), falls from a tube of cross sectional area 'a' with a speed 'v' in a sieve such that water is not retained in the sieve. The value of $\int \vec{B} \cdot d\vec{\ell}$ integrated over the upper

circular part of the sieve will be : [$\vec{B}$ indicates the magnetic field produced by Na^+ ions and $d\vec{\ell}$ is along the tangent on the periphery of circular part of the sieve].



- (1) $\mu_0 avce$
- (2) $2\mu_0 avce$
- (3) $\mu_0 ace/v$
- (4) $\mu_0 / avce$

57. A slightly conical wire of length L and radii a and b is stretched by two forces each of magnitude F , applied parallel to its length in opposite directions and normal to end faces. If Y denotes the Young's modulus, then the extension produced is :



(1) $\frac{FL}{\pi(a^2 + b^2)Y}$

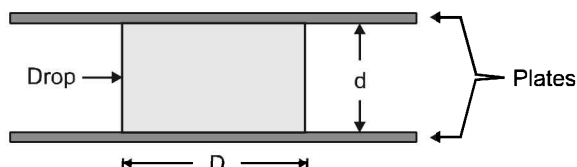
(2) $\frac{FL}{\pi(a^2 - b^2)Y}$

(3) $\frac{FL}{\pi abY}$

(4) None of these

(SPACE FOR ROUGH WORK)

58. A drop of liquid of surface tension σ is in between the two smooth parallel glass plates held at a distance d apart from each other in zero gravity. The liquid wets the plate so that the drop is a cylinder of diameter D with its curved surface at right angles to both the plates. Determine the force acting on each of the plates from drops under the following considerations.



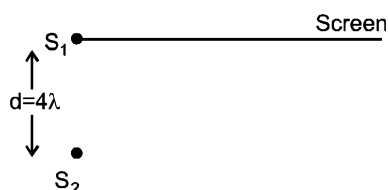
- (1) $\frac{\sigma\pi D}{2}$
- (2) $\frac{\sigma^2\pi D}{2}$
- (3) zero
- (4) None of these

SECTION - II

Reasoning Type

This section contains 2 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

59. **Statement-1:** For the situation shown in the figure two monochromatic identical coherent light sources produce interference pattern on the screen. The intensity of minima nearest to S_1 is not exactly zero.



Statement-2 : Minimum intensity is zero when interfering waves have same intensity at the location of super position.

- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
 - (3) Statement-1 is True, Statement-2 is False
 - (4) Statement-1 is False, Statement-2 is True
60. **Statement-1 :** Centre of gravity and centre of mass have same physical meaning.
- Statement-2 :** In uniform gravitational field, centre of mass and centre of gravity are at the same point.
- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 - (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
 - (3) Statement-1 is True, Statement-2 is False
 - (4) Statement-1 is False, Statement-2 is True

(SPACE FOR ROUGH WORK)

PART- C

Atomic masses : [H = 1, D = 2, Li = 7, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, Si = 28, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Cu = 63.5, Zn = 65, As = 75, Br = 80, Ag = 108, I = 127, Ba = 137, Hg = 200, Pb = 207]

SECTION

Straight Objective Type

This section contains 30 questions. Each question has four choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

61. A solution is prepared by urea and water. If mole fraction of water is 0.8 in the solution. Find the ratio of mass of urea & water.

(1) $\frac{5}{6}$

(2) $\frac{6}{5}$

(3) $\frac{1}{1}$

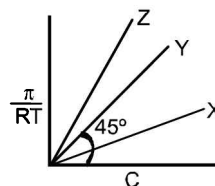
(4) $\frac{4}{1}$

62. Following graph shows the relation between

$\frac{\pi}{RT}$ and C where π is the osmotic pressure,

T is temperature in K, R is gas constant and C is molarity of solute. Given that solute Y neither dissociates nor polymerises in solution. Identify the incorrect statement

(i - Van't Hoff factor : M – Molar mass).



(1) $i_x < 1$ and $i_z > 1$

(2) $i_x > 1$ and $i_z < 1$

(3) $M_{\text{ABNORMAL}} \geq M_{\text{NORMAL}}$ for Y & X

(4) $M_{\text{ABNORMAL}} \leq M_{\text{NORMAL}}$ for Y & Z

(SPACE FOR ROUGH WORK)

63. For the electrons of oxygen atom, which of the following statements is correct ?

- (1) Z_{eff} for an electron in a 2s orbital is the same as Z_{eff} for an electron in a 2p orbital.
- (2) An electron in the 2s orbital has the same energy as an electron in the 2p orbital.
- (3) Z_{eff} for an electron in a 1s orbital is the same as Z_{eff} for an electron in a 2s orbital.
- (4) The two electrons present in the 2s orbital have different spin quantum numbers (m_s).

64. Decomposition of A follows first order kinetics by the following equation.

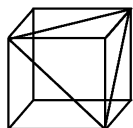


If initially, total pressure was 800 mm of Hg and after 10 minutes it is found to be 650 mm of Hg. What is half-life of A ?

(Assume only A is present initially)

- (1) 10 mins
- (2) 5 mins
- (3) 7.5 mins
- (4) 15 mins

65. Consider the plane shown in the figure:



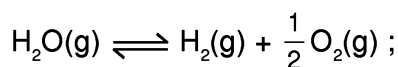
. A rock salt (NaCl) lattice is cut across this plane in which anions occupy fcc positions. It passes through centers of how many cations?

- (1) zero
- (2) 4
- (3) 6
- (4) 12

66. Which of the following statement is correct?

- (1) S—S bond is present in $H_2S_2O_6$.
- (2) In peroxosulphuric acid (H_2SO_5) sulphur is in +8 oxidation state.
- (3) Copper powder along with Al_2O_3 and K_2O is used as a catalyst in the preparation of NH_3 by Haber's process.
- (4) Change in enthalpy is positive for the preparation of SO_3 by catalytic oxidation of SO_2 .

67. Steam undergoes decomposition at high temperature as per the reaction



$$\Delta H^\circ = 200 \text{ kJ mol}^{-1}, \Delta S^\circ = 40 \text{ kJ mol}^{-1}.$$

The temperature at which equilibrium constant is unity is :

- (1) 3000 Kelvin
- (2) 5000 Kelvin
- (3) 5333 Kelvin
- (4) 5 Kelvin

68. Pure $PCl_5(g)$ was placed in a flask at 2 atm and left for some time at a certain temp. Where the following equilibrium was established



Under the given condition, pure oxygen was found to effuse 2.084 time faster than the equilibrium mixture then K_p of the above reaction is :

- (1) $\frac{2}{3}$
- (2) $\frac{3}{2}$
- (3) $\frac{4}{5}$
- (4) $\frac{5}{2}$

(SPACE FOR ROUGH WORK)

69. $M(OH)_x$ has $K_{sp} = 4 \times 10^{-12}$ and solubility $10^{-4} M$. Hence x is :

- (1) 1 (2) 2
(3) 3 (4) 4

70. Borax on heating strongly above its melting point melts to a liquid, which then solidifies to a transparent mass commonly known as borax-bead. The transparent glassy mass consists of :

- (1) sodium pyroborate
(2) boric anhydride
(3) sodium meta-borate
(4) boric anhydride and sodium metaborate

71. Which of the following is incorrect statement regarding $A = [Co(en)_2Cl_2]^+$ and $B = [Co(NH_3)_4Cl_2]^+$.

- (1) A is more thermodynamically stable than B.
(2) A shows optical isomerism.
(3) A and B both are diamagnetic.
(4) A is outer orbital complex while B is inner orbital complex.

72. How many P–OH bond in pyrophosphoric acid:

- (1) Two (2) Three
(3) Four (4) One

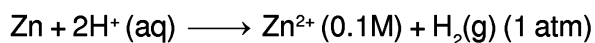
73. In order to refine 'blister copper' it is melted in furnace and is stirred with green logs of wood. The purpose is:

- (1) to expel the dissolved gases in blister copper.
(2) to reduce the metallic oxide impurities with hydrocarbon gases liberated from the wood.
(3) to bring the impurities to surface and oxidize them.
(4) to increase the carbon content in the copper.

74. The incorrect statement among the following is :

- (1) beryllium oxide is amphoteric in nature
(2) solubility of sulphates of second group elements decreases down the group
(3) reducing power of hydride of alkali metal decreases down the group
(4) Beryllium has diagonal relationship with aluminium

75. The e.m.f. of a cell corresponding to the reaction



is 0.169 volt at 25°C. Calculate the pH of the solution at the hydrogen electrode.

$$E^\circ_{Zn^{2+}/Zn} = -0.76 \text{ volt and } E^\circ_{H^+/H_2} = 0$$

- (1) 9 (2) 11.5
(3) 10 (4) 10.5

(SPACE FOR ROUGH WORK)

76. (I) V_2O_5 , Cr_2O_3 are amphoteric oxides.
 (II) Interstitial compounds are very reactive
 (III) In its higher oxidation states, manganese forms stable compounds with oxygen and fluorine.

Correct statements amongs the following are-

- (1) I, II (2) II, III
 (3) I, III (4) I, II

77. Which of the following statements does not correct from the point of view of molecular orbital?

- (1) Be_2 is not a stable molecule
 (2) He_2 is not stable but He^+ is expected to exist
 (3) Bond strength of N_2 is maximum amongst the homonuclear diatomic molecules
 (4) The order of energies of molecular orbitals in O_2 molecule is
 $E(\sigma 2s) < E(\sigma^* 2s) < E(\pi 2p_x) = E(\pi 2p_y) < E(\sigma 2p_z) < E(\pi^* 2p_x) = E(\pi^* 2p_y) < E(\sigma^* 2p_z)$

78. The equivalent conductance of 0.01 M solution of weak acid HA is $19.6 \text{ S cm}^2 \text{ eq}^{-1}$. The equivalent conductance of the electrolyte at infinite dilution 392 then pH of the solution will be :

- (1) 2.3 (2) 5
 (3) 3.3 (4) 4.7

79. In which of the following pairs the second one is the stronger base than first ?

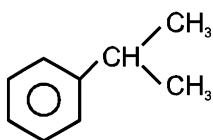
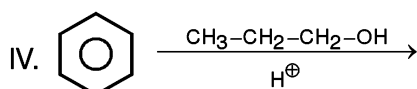
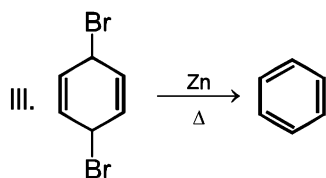
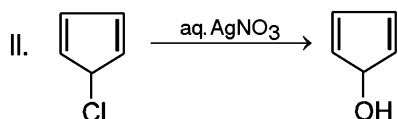
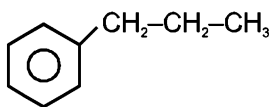
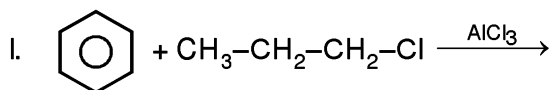
- (1) $CH_3COO^\ominus$, $HCOO^\ominus$
 (2) $HO^\ominus$, $NH_2^\ominus$
 (3) $CH_2 = \overset{\ominus}{C}H$, $H - C \equiv \overset{\ominus}{C}$
 (4) CH_3NH_2 , CH_3OH

80. When grignard reagent is treated with isopropyl formate followed by acid hydrolysis we get :

- (1) Aldehyde
 (2) 2° alcohol
 (3) 3° alcohol
 (4) 1° alcohol

(SPACE FOR ROUGH WORK)

81. Which of the following reactions report the correct major products ?

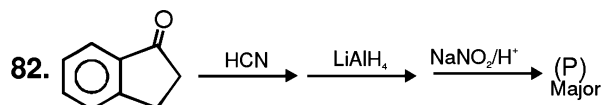


(1) I, III

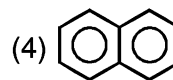
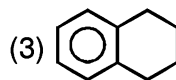
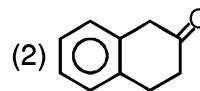
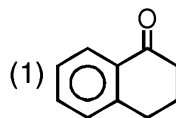
(2) III, IV

(3) II, III, IV

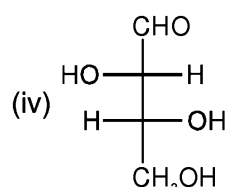
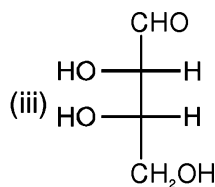
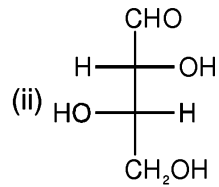
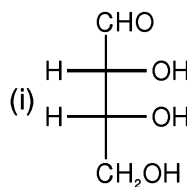
(4) I, III, II



Product P will be :



83. An optically active aldotetrose which retains its optical activity with aqueous Br_2 but losses it with conc. HNO_3 has the following structure :



(1) i,ii

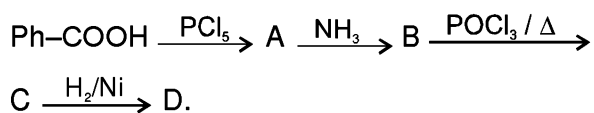
(2) i,iii

(3) ii, iii

(4) ii,iv

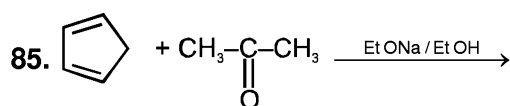
(SPACE FOR ROUGH WORK)

84. Consider the following sequence of reaction :

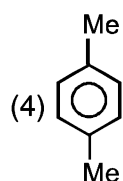
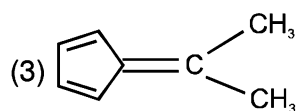
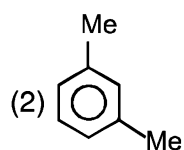
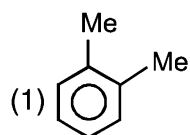


The final product D is :

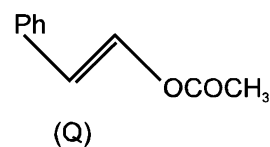
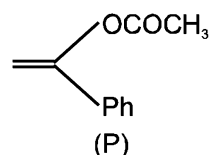
- (1) Benzonitrile
- (2) Benzylamine
- (3) Aniline
- (4) Benzamide



The product [X] is :

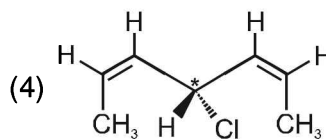
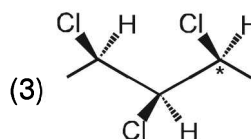
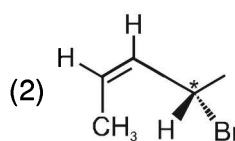
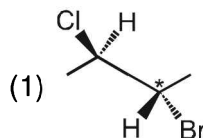


86. The product of acid hydrolysis of P and Q can be distinguished by :



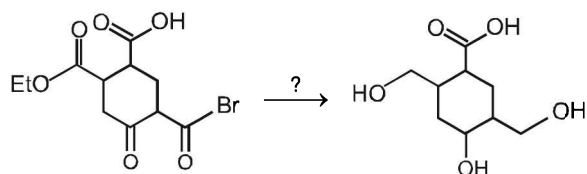
- (1) Lucas reagent
- (2) 2, 4-DNP
- (3) Fehling solution
- (4) NaHCO_3

87. In which of the following species, the configuration at the starred atom is changed if any change in configuration of neighbour atom is carried out.

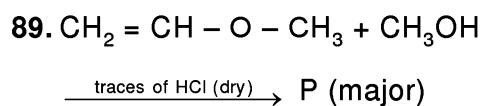


(SPACE FOR ROUGH WORK)

88. Which of the following reagent can be used for the given conversion :



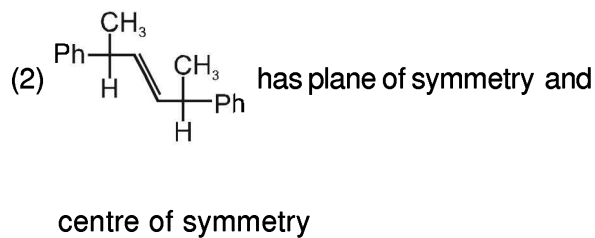
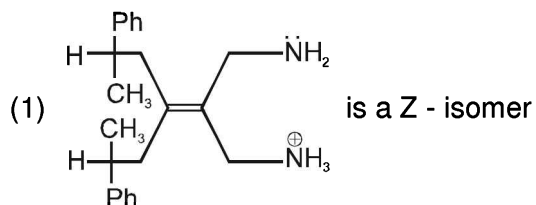
- (1) LiAlH_4
- (2) $\text{Na} / \text{C}_2\text{H}_5\text{OH}$
- (3) DIBAL-H
- (4) NaBH_4



The product "P" is/are :

- (1) $\text{CH}_3 - \text{O} - \text{CH}_2\text{CH}_2 - \text{O} - \text{CH}_3$
- (2) $\text{CH}_3 - \text{CH} = \text{O} + \text{CH}_3 - \text{O} - \text{CH}_3$
- (3) $\text{CH}_3 - \text{CH} \begin{cases} \text{OCH}_3 \\ \text{OCH}_3 \end{cases}$
- (4) $\text{CH}_3 - \overset{\text{O}}{\underset{\text{O}}{\parallel}} \text{C} - \text{CH}_3 + \text{CH}_3\text{OH}$

90. Which of the following statement is incorrect :



- (3) Only one conformation of ethylene glycol ($\text{HOCH}_2\text{CH}_2\text{OH}$) has centre of symmetry
- (4) The gauche form of meso tartaric acid is chiral

(SPACE FOR ROUGH WORK)

VISIT: WWW.PREPJEE.IN

MOCK TEST 3

CODE

Q

PREPJEE

PREPARE FOR JEE EXAMINATION WITH US

PREPJEE TEST SERIES (PTS)

JEE MAIN

TIME: 3 HOURS MAX.MARKS: 360

Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose

IMPORTANT INSTRUCTIONS

1. Immediately fill the particulars on this page of the Test Booklet with Blue / Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.
3. The test is of 3 hours duration.
4. The Test Booklet consists of 90 questions. The maximum marks are 360.
5. There are three parts in the question paper A, B, C consisting of Mathematics, Physics and Chemistry having 30 questions. Mathematics & Physics part consists of Two sections and Chemistry consists of One section.
6. For each question you will be awarded 4 marks if you darken all the bubble(s) corresponding to only the correct answer(s) and zero mark if no bubbles are darkened. In all other cases, $-1/4$ (one fourth (1)) marks will be deducted.
7. There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 6 above.

Filling the Top-half of the ORS :

Use only Black ball point pen only for filling the ORS. Do not use Gel / Ink / Felt pen as it might smudge the ORS.

8. Write your Roll no. in the boxes given at the top left corner of your ORS with black ball point pen. Also, darken the corresponding bubbles with Black ball point pen only. Also fill your roll no on the back side of your ORS in the space provided (if the ORS is both side printed).
9. Fill your Paper Code as mentioned on the Test Paper and darken the corresponding bubble with Black ball point pen.
10. If student does not fill his/her roll no. and paper code correctly and properly, then his/her marks will not be displayed and 5 marks will be deducted (paper wise) from the total.
11. Since it is not possible to erase and correct pen filled bubble, you are advised to be extremely careful while darken the bubble corresponding to your answer.
12. Neither try to erase / rub / scratch the option nor make the Cross (X) mark on the option once filled. Do not scribble, smudge, cut, tear, or wrinkle the ORS. Do not put any stray marks or whitener anywhere on the ORS.
13. If there is any discrepancy between the written data and the bubbled data in your ORS, the bubbled data will be taken as final.

VISIT : www.prepjee.in for more details consider buying more Test Paper and keep Practicing

PART-A

SECTION - I

Straight Objective Type

This section contains 27 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. Equation of a variable circle is $x^2 + y^2 - 2(t^2 - 3t + 1)x - 2(t^2 + 2t)y + 2 = 0$.

Locus of centre of circle is

- (1) $x^2 - 2xy + y^2 + 12x - 13y + 10 = 0$
(2) $x^2 - 2xy + y^2 - 12x - 13y + 10 = 0$
(3) $x^2 - 2xy + y^2 - 12x + 13y + 10 = 0$
(4) $x^2 - 2xy + y^2 - 12x - 13y + 11 = 0$

2. If a_r is the coefficient of x^r in the expression of $(1 + x + x^2)^n$, then the value of

$a_1 + 4a_4 + 7a_7 + 10a_{10} + \dots$ is equal to

- (1) 3^{n-1} (2) $n 3^{n-1}$
(3) 2^n (4) $\frac{2^n}{3}$

3. If the curve $y = 1 + \sqrt{4 - x^2}$ and the line $y = (x - 2)k + 4$ has two distinct points of intersection then the range of k is

- (1) $[1, 3]$ (2) $\left[\frac{5}{12}, \infty\right)$
(3) $\left(\frac{5}{12}, \frac{3}{4}\right]$ (4) $\left[\frac{5}{12}, \frac{3}{4}\right)$

4. If x_1 and x_2 are solutions for x in quadratic equation $x^2 + (\sin \theta - 1)x - \frac{1}{2} \cos^2 \theta = 0$, then

maximum value of $x_1^2 + x_2^2$ is equal to

- (1) 2 (2) 3
(3) $\frac{9}{4}$ (4) 4

5. The mean and variance of a binomial distribution are 30 and 5 respectively. The probability of two successes is

- (1) $\frac{875}{6^{34}}$
(2) $\frac{875}{2 \cdot 6^{34}}$
(3) $\frac{875}{6^{36}}$
(4) ${}^{36}C_2 \left(\frac{5}{6}\right)^{34} \left(\frac{1}{6}\right)^2$

6. A piece of string is cut into two pieces. The point at which the string is cut was chosen at random. What is the probability that the longer piece is at least three times as long as the shorter piece?

- (1) $\frac{1}{4}$ (2) $\frac{1}{3}$
(3) $\frac{2}{5}$ (4) $\frac{1}{2}$

(SPACE FOR ROUGH WORK)

7. Let ABCD be a square with A(0, 0), C(2, 2). If M is mid – point of AB and P is a variable point on BC. The smallest value of DP + PM is
- (1) $\sqrt{13}$
 (2) $\sqrt{2} + \sqrt{5}$
 (3) $2\sqrt{3}$
 (4) $1 + 2\sqrt{2}$
8. An office employs thirty people. Five of them speak Sanskrit, Hindi, English. Nine speak Sanskrit, English. Twenty speak Hindi of which twelve also speak Sanskrit. Eighteen speak English. No one speak only Sanskrit. An employee selected at random is known to speak either Hindi or English, the probability that employee speaks only two of three languages is
- (1) 14/15 (2) 7/15
 (3) 1/6 (4) 5/6
9. ABC is vertical tower such that AB = BC = x. Form the foot A of the tower, a horizontal line AQP is drawn. If the angle of elevation of the point B from P is α and that of C from Q is 2α , then AP, x, PQ are in
- (1) A.P. (2) G.P.
 (3) H.P. (4) None of these
10. If $\cos P = \frac{1}{7}$ and $\cos Q = \frac{13}{14}$, where P and Q are acute angles. Then $P - Q =$
- (1) 30° (2) 60°
 (3) 45° (4) 75°
11. A man running round a race course notes that the sum of the distance of two flag – posts from him is always 10 meters and distance between the flag – posts is 8 meters. The area of the path he encloses in square meters is
- (1) 15π (2) 12π
 (3) 18π (4) 8π
12. Set of equations
- $$x^2 - b_n x + \underbrace{111\dots 1}_{n\text{-times}} = n, \quad n = 2, 3, 4, \dots, 9 \text{ and } b_n \in \mathbb{N}, \text{ have integer roots. Which of the following are CORRECT ?}$$
- (1) Common root of these equations is 9.
 (2) No three of these equations have a common root.
 (3) A root of an equation with $n = 4$ can be 123.
 (4) A root of an equation with $n = 4$ is 1234.
13. If $S = \sum_{r=1}^7 \tan^2 \frac{r\pi}{16}$, then S is equal to
- (1) 44 (2) 40
 (3) 34 (4) 35

(SPACE FOR ROUGH WORK)

14. If $A = \{1, 2, 3, 4\}$ then which of the following are functions from A to itself ?

(1) $f_1 = \{(x, y) : y = x + 1\}$

(2) $f_2 = \{(x, y) : x + y > 4\}$

(3) $f_3 = \{(x, y) : x < y\}$

(4) $f_4 = \{(x, y) : x + y = 5\}$

15. Let p be the statement "Ram races", q be the statement "Ram wins". The statement $\sim (p \vee \sim q)$ is :

(1) Ram does not race and Ram does not win.

(2) It is not true that Ram races or Ram does not win.

(3) Ram does not race and Ram wins

(4) It is not true that Ram races and Ram does not win.

16. If $A = \{z : |z - 2 - 2i| = |z - 6 - 6i|\}$,

$$B = \{z : \arg(z - 4 - 4i) = \pi/4\},$$

$\operatorname{Re}(z) = a, \operatorname{Im}(z) = b, a \in \mathbb{N}, b \in \mathbb{N}, a \leq 6, b \leq 6$ then

(1) $n(A \cup B) = 7$

(2) $n(A \cap B) = 1$

(3) $A \subseteq B$

(4) $B \subseteq A$

17. If a the first term, d the common difference, ℓ the last term and s the sum of n terms of an A.P. are connected by the equation $(d + 2\ell)^2 = 8ds$ then

(1) $d = a$

(2) $d = 2a$

(3) $d = 3a$

(4) $d = 4a$

18. Let p, q be real numbers such that $\frac{1}{p} - \frac{1}{q} = 1$

and $0 < p \leq \frac{1}{2}$ then

(1) $p + \frac{1}{2}p^2 + \frac{1}{3}p^3 + \dots \infty = q - \frac{1}{2}q^2 + \frac{1}{3}q^3$

$-\frac{1}{4}q^4 + \dots \infty$

(2) $p - \frac{1}{2}p^2 + \frac{1}{3}p^3 - \frac{1}{4}p^4 + \dots \infty = q + \frac{1}{2}q^2$

$+\frac{1}{3}q^3 + \frac{1}{4}q^4 + \dots \infty$

(3) $p + \frac{p^2}{2!} + \frac{p^3}{3!} + \dots \infty = q - \frac{q^2}{2!} + \frac{q^3}{3!} - \frac{q^4}{4!}$

$+ \dots \infty$

(4) $p - \frac{1}{2}p^2 + \frac{1}{3}p^3 - \frac{1}{4}p^4 + \dots \infty = q$

$+\frac{q^2}{2!} + \frac{q^3}{3!} + \frac{q^4}{4!} + \dots \infty$

(SPACE FOR ROUGH WORK)

19. A box has 2 red, 3 white and 6 black balls. In how many ways can 3 balls be taken out from the box if at least one of the balls is white. All given balls are considered to be distinct.

- (1) 135
- (2) 61
- (3) 84
- (4) 109

20. Number of acute angles α such that

$$\cos \alpha \cos 2\alpha \cos 4\alpha = \frac{1}{8} \text{ are}$$

- (1) 1
- (2) 2
- (3) 3
- (4) 4

21. The smallest positive integer n such that $11^n - 1$ is divisible by 100 is

- (1) 4
- (2) 5
- (3) 10
- (4) 11

22. If P is the positive integer for which $1 + x^4 + x^8 + \dots + x^{4(p-1)}$ is divisible $(1 + x + \dots + x^{p-1})$ and both the roots of $f(x) = x^2 - (p+6)x - 2q = 0$ are prime numbers, then the value of $p + q + 2$ is

- (1) -1
- (2) 2
- (3) -2
- (4) 0

23. A variable line $ax + by + c = 0$, where a, b, c are in A.P., is normal to a circle $(x-\alpha)^2 + (y-\beta)^2 = \gamma$, which is orthogonal to circle $x^2 + y^2 - 4x - 4y - 1 = 0$. The value of $\alpha + \beta + \gamma$ is equal to :

- (1) 3
- (2) 5
- (3) 10
- (4) 7

24. Number of solutions of equation

$$(\sin \theta + \cos \theta) \tan \theta = \sqrt{\sec^2 \theta - 1} \text{ in } [0, 2\pi],$$

are

- (1) 1
- (2) 2
- (3) 3
- (4) 4

(SPACE FOR ROUGH WORK)

25. The area of an equilateral triangle inscribed in circle $x^2 + y^2 + 2gx + 2fy + c = 0$, is

(1) $\frac{3\sqrt{3}}{2}(g^2 + f^2 - c)$

(2) $\frac{3\sqrt{3}}{4}(g^2 + f^2 - c)$

(3) $\frac{3\sqrt{3}}{4}(g^2 + f^2 + c)$

(4) $\frac{3\sqrt{3}}{2}(g^2 + f^2 + c)$

26. Number of sequences of a's and b's of length 20 having all consecutive a's or all consecutive b's or both are :

(1) ${}^{20}C_2$

(2) $2 \times {}^{20}C_2$

(3) ${}^{20}C_2 + 2$

(4) 382

27. With usual notations in triangle ABC, if

$A = \frac{\pi}{2}$ and $a + b + c = \Delta$, then $b + c - a$ is

equal to

(1) 1

(2) 2

(3) 3

(4) 4

SECTION - II

Reasoning Type

This section contains 3 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

28. Let $n \geq 3$. A list of numbers $0 < x_1 < x_2 < \dots < x_n$ has mean μ and standard deviation σ . A new list of numbers is made as follows : $y_1=0$, $y_2= x_2, \dots, y_{n-1} = x_{n-1}, y_n = x_1 + x_n$. The mean and the standard deviation of the new list are $\hat{\mu}$ and $\hat{\sigma}$.

STATEMENT-1 : $\sigma < \hat{\sigma}$ and $\mu = \hat{\mu}$

STATEMENT-2 : Variance is independent of choice of origin while mean dependent on choice of origin.

- (1) Statement -1 is True, Statement -2 is True;
Statement -2 is a correct explanation for Statement -1
- (2) Statement-1 is True, Statement-2 is True;
Statement-2 is **NOT** a correct explanation for Statement-1
- (3) Statement -1 is True, Statement -2 is False
- (4) Statement -1 is False, Statement -2 is True

(SPACE FOR ROUGH WORK)

29. STATEMENT- 1 : For any real number θ , $\sin^6\theta + \cos^6\theta \leq 1$.

STATEMENT- 2 : If x and y are non-zero real numbers such that $x^2 + y^2 = 1$, then $x^4 \leq 1/2$ and $y^4 \leq 1/2$.

(1) Statement -1 is True, Statement -2 is True;

Statement -2 is a correct explanation for Statement -1

(2) Statement-1 is True, Statement-2 is True ;

Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement -1 is True, Statement -2 is False

(4) Statement -1 is False, Statement -2 is True

30. STATEMENT-1 : If $A(1, 2)$ and $B(7, 10)$ are two points. If $P(x, y)$ is such that $\angle APB = 60^\circ$, then area of $\triangle APB$ will be maximum if P lies on $L \equiv 3x + 4y = 16$

STATEMENT-2 : In statement-1 the area of triangle APB is maximum if P lies on the perpendicular bisector of AB .

(1) Statement -1 is True, Statement -2 is True;

Statement -2 is a correct explanation for Statement -1

(2) Statement-1 is True, Statement-2 is True ;

Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement -1 is True, Statement -2 is False

(4) Statement -1 is False, Statement -2 is True

(SPACE FOR ROUGH WORK)

PART-B

SECTION - I

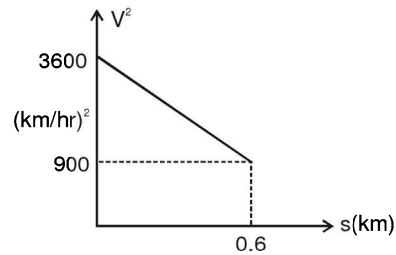
Straight Objective Type

This section contains **28 questions**. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

31. A solid sphere of mass 2 kg and density 10^5 kg/m^3 hanging from a string is lowered into a vessel of uniform cross-section area 10 m^2 containing a liquid of density $0.5 \times 10^5 \text{ kg/m}^3$, until it is fully immersed. The increase in pressure of liquid at the base of the vessel is (Assume liquid does not spill out of the vessel and take $g = 10 \text{ m/s}^2$)

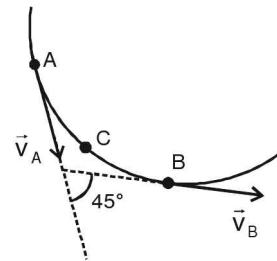
- (1) 2 N/m^2
- (2) 1 N/m^2
- (3) 4 N/m^2
- (4) $\frac{1}{2} \text{ N/m}^2$

32. A graph between the square of speed of a particle moving along a straight line and the distance 'S' moved is shown in the figure. The acceleration of the particle in km/hr^2 is -



- (1) 2250
- (2) 225
- (3) -2250
- (4) -225

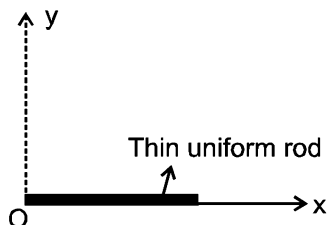
33. A particle moves along a constant curvature path between A and B (length of curve wire AB is 100m) with a constant speed of 72 km/hr. The acceleration of particle at mid point 'C' of curve AB is



- (1) Zero
- (2) 3.14 m/s^2
- (3) 50.84 m/s^2
- (4) 50.84 km/hr^2

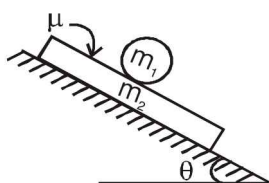
(SPACE FOR ROUGH WORK)

34. The locus of all the points on the $x - y$ plane about which the moment of inertia of the uniform rod (axes perpendicular to xy -plane) shown in figure, is same as that about axis perpendicular to rod and passing through point 'O' is :



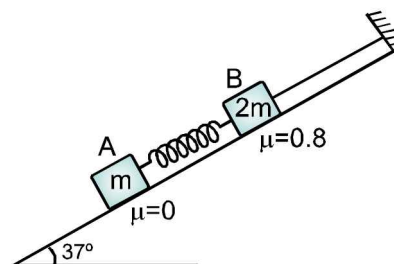
- (1) circle (2) parabola
(3) straight line (4) ellipse

35. A uniform solid sphere of mass m_1 is placed on the plank of mass m_2 and the coefficient of friction between the plank and sphere is μ . If the inclined plane is frictionless then frictional force on m_1 is (Assume that the plank and uniform solid sphere are released on the inclined plane from rest)



- (1) parallel to the incline plane upwards.
(2) parallel to the incline plane downwards.
(3) Zero
(4) $0 < f < \mu m_1 g \cos \theta$

36. Initially both the blocks are at static equilibrium. Then the acceleration of blocks A & B just after cutting the string is respectively : ($g = 10 \text{ m/s}^2$)



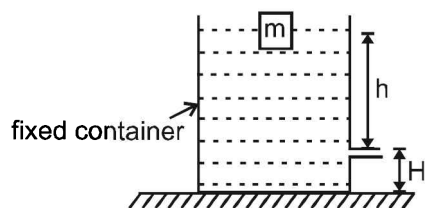
- (1) $7.5 \text{ m/s}^2, 7.5 \text{ m/s}^2$
(2) $2.6 \text{ m/s}^2, \text{zero}$
(3) $\text{zero}, 2.6 \text{ m/s}^2$
(4) zero, zero

37. In a damped oscillator the amplitude of vibrations of mass $m = 150$ grams falls by $\frac{1}{e}$ times of its initial value in time t_0 due to viscous forces. The time t_0 and the percentage loss in mechanical energy during the above time interval t_0 respectively are (Let damping constant be 50 grams/s)

- (1) $6s, \frac{e^2 - 1}{e^2} \times 100$
(2) $3s, \frac{e^2 - 1}{e^2} \times 100$
(3) $6s, \frac{e - 1}{e} \times 100$
(4) $3s, \frac{e - 1}{e} \times 100$

(SPACE FOR ROUGH WORK)

38. If the pin hole in the container is very small compared to the area of the base of container and a block floats in the ideal liquid then the speed of efflux is.

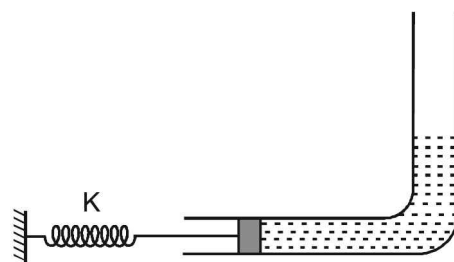


- (1) less than $\sqrt{2gh}$
 (2) greater than $\sqrt{2gh}$
 (3) equal to $\sqrt{2gh}$
 (4) equal to $\sqrt{2gH}$

39. The distance moved by the screw of a screw gauge is 2 mm in four rotations and there are 50 divisions on its cap. When nothing is put between its jaws, 30th division of circular scale coincides with reference line, with zero of circular scale lying above the reference line. When a plate is placed between the jaws, main scale reads 2 division and circular scale reads 20 division. Thickness of plate is :

- (1) 1.5 mm
 (2) 1.2 mm
 (3) 1.4 mm
 (4) 1.6 mm

40. A massless piston connected to a spring of spring constant k fits smoothly inside an L shaped fixed glass tube as shown in the figure. The piston can slide without friction and without leakage of water. The cross sectional area of the tube is s and the density of the liquid is ρ . If the system is slightly disturbed from its equilibrium position then the angular frequency of its small oscillations is: (m is the total mass of liquid and the liquid is ideal)

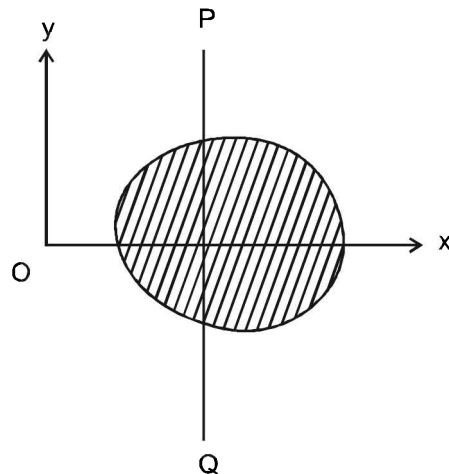


- (1) $\sqrt{\frac{2\rho s g}{m}} + \sqrt{\frac{k}{m}}$ (2) $\sqrt{\frac{\rho s g}{m}} + \sqrt{\frac{k}{m}}$
 (3) $\sqrt{\frac{k + 2\rho s g}{m}}$ (4) $\sqrt{\frac{k + \rho s g}{m}}$

41. A car is moving towards an approaching bike. Bike is moving with velocity 30 m/s. Frequency of horn sounded by car is 100 Hz, while it is observed by bike rider as 120 Hz. The actual duration of horn is 6 sec. Find the time interval for which the bike rider hears it. (Take velocity of sound in air as 330 m/s)
- (1) 5 sec (2) 6 sec (3) 7 sec
 (4) Cannot be calculate with given information

(SPACE FOR ROUGH WORK)

42. A carnot refrigeration cycle absorbs heat at 270 K and rejects heat at 300 K. If the cycle is absorbing 1260 KJ/min at 270 K, then work required per second is :
- 2.33 KJ/sec.
 - 4.66 KJ/sec.
 - 1 KJ/sec.
 - 4 KJ/sec.
43. A particle moves so that its position as a function of time in MKS unit is $\vec{r} = \hat{i} + 4t^2\hat{j} + t\hat{k}$.
- The trajectory of the particle is -
- parabola
 - ellipse
 - circle
 - None of these
44. A particle leaves the origin at $t = 0$ with an initial velocity $\vec{v} = 3v_0\hat{i}$. It experiences a constant acceleration $\vec{a} = -2a_1\hat{i} - 5a_2\hat{j}$. The time at which the particle reaches its maximum x coordinate is (a_1 , a_2 & v_0 are constant positive number)
- $\frac{3V_0}{10a_2}$
 - $\frac{3V_0}{9a_1 - 5a_2}$
 - $\frac{3V_0}{2a_1}$
 - $\frac{3V_0}{2a_1 + 5a_2}$
45. A man beats drum at a certain distance from a large mountain. He slowly increases the rate of beating and finds that the echo is not heard distinctly, when the drum beating is at the rate of 40 per minute. He moves by 80 meter towards the mountain and finds that the echo is again not heard distinctly, when the rate of beating of the drum is 1 per second. What is the initial distance of the man from the mountain ?
- 120 m
 - 240 m
 - 270 m
 - 340 m
46. Line PQ is parallel to y-axis and moment of inertia of a rigid body about PQ line is given by $I = 2x^2 - 12x + 27$, where x is in meter and I is in kg-m². The minimum value of I is :



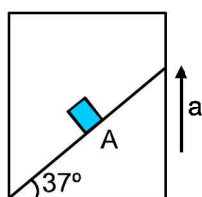
- 27 kg m²
- 11 kg m²
- 17 kg m²
- 9 kg m²

(SPACE FOR ROUGH WORK)

47. A ring of mass m and radius R is rolling without sliding on a flat fixed horizontal surface, with a constant angular velocity. The angle between the velocity and acceleration vectors of a point P on the rim of the ring at the same horizontal level as the centre of the ring may be -

- (1) zero
- (2) 90°
- (3) 135°
- (4) $\tan^{-1}1/2$

48. A board is fixed to the floor of an elevator such that board forms angle $\theta = 37^\circ$ with the horizontal floor of the elevator accelerating upwards. A block is placed at point A on the board as shown in figure. When velocity $4\sqrt{2}$ m/s (relative to board) is given to the block up the incline of the board, it comes to rest after moving a distance 1.6 m relative to the board. When it returns to point A, its velocity is 4 m/s (relative to board) down the board. Acceleration of elevator is constant and assume no collision. Choose the correct option :



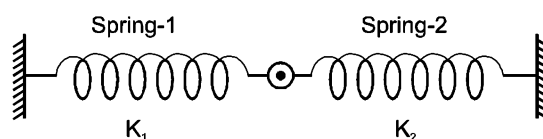
(1) the constant acceleration a of the elevator is 5 m/s^2

(2) the coefficient of friction between board and block is 0.25

(3) friction is absent between block and board

(4) acceleration of the block with respect to ground during descent is $5\sqrt{5} \text{ m/s}^2$ in magnitude

49. Spring-1 has a natural length of 0.5 meter and force constant $K_1 = 25 \text{ N/m}$ and spring -2 has a natural length of 1 meter and a force constant $K_2 = 10 \text{ N/m}$. They are joined together and their free ends are stretched so that the ends are fixed to the two walls 2 meter apart as shown in figure. The new lengths of the spring 1 and 2 at equilibrium are respectively : (springs are massless)



(1) $\frac{9}{14} \text{ m}, \frac{19}{14} \text{ m}$

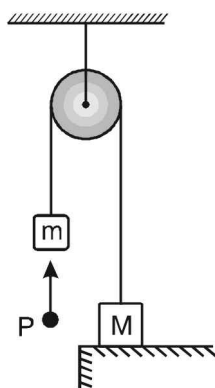
(2) $\frac{8}{14} \text{ m}, \frac{20}{14} \text{ m}$

(3) $\frac{7}{14} \text{ m}, \frac{21}{14} \text{ m}$

(4) $\frac{16}{14} \text{ m}, \frac{12}{14} \text{ m}$

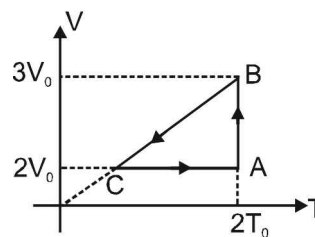
(SPACE FOR ROUGH WORK)

50. A light inelastic thread passes over a small frictionless pulley. Two blocks of masses $m = 1 \text{ kg}$ and $M = 3 \text{ kg}$ are attached with the thread and heavy block rests on a surface. A particle P of mass 1 kg moving upward with a velocity 10 m/s collides with the lighter block elastically at $t = 0$. Choose the **incorrect** option. ($g = 10 \text{ m/s}^2$)



- (1) Velocity with which heavy block is jerked into motion is 2.5 m/s
- (2) Maximum height ascended by heavy block after it is jerked into motion is 0.625 m
- (3) Impulse of tension in jerk is 7.5 Ns
- (4) Time after which block again strikes the surface is 2.5 s

51. One mole of an ideal gas in initial state A undergoes a cyclic process ABCA, as shown in the figure. Its pressure at A is P_0 . Choose the correct option from the following :



- (1) Internal energies at A and B are different
- (2) Work done by the gas in process AB is

$$2P_0V_0 \log_{10} \frac{3}{2}$$

- (3) Pressure at C is $\frac{2P_0}{3}$

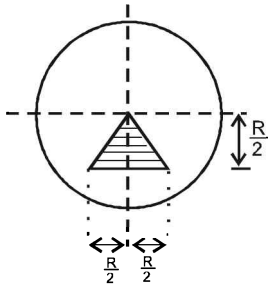
- (4) Temperature at C is $\frac{T_0}{3}$

52. The rate of change of kinetic energy is “n” times (n is constant with appropriate dimension) the velocity at any moment of a particle of mass m, which starts moving in a straight line. The constant acceleration with which particle is moving is :

- | | |
|-------------------|--------------------|
| (1) mn | (2) $\frac{m}{n}$ |
| (3) $\frac{n}{m}$ | (4) $\frac{1}{mn}$ |

(SPACE FOR ROUGH WORK)

53. In the figure as shown, a triangular portion is cut from a circular disc of radius R . The distance of centre of mass of the remaining part from the centre of disc is



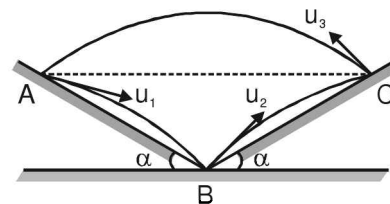
(1) $\frac{R}{3(4\pi - 1)}$

(2) $\frac{R}{4(4\pi - 1)}$

(3) $\frac{2R}{3(4\pi - 1)}$

(4) $\frac{3R}{4(4\pi - 1)}$

54. Three particles are projected in air with the minimum possible speeds, such that the first goes from A to B, the second goes from B to C and the third goes from C to A. Points A and C are at the same level. The two inclines make the same angle α with the horizontal as shown. Then the relation among the projection speeds of the three particles is :



(1) $u_3 = u_1 + u_2$

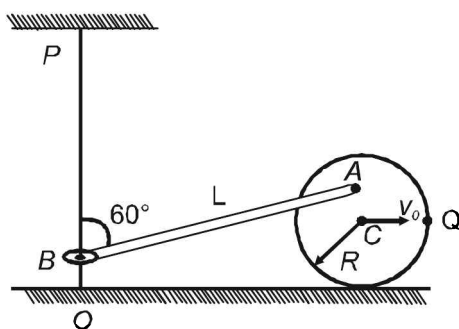
(2) $u_3^2 = 2u_1u_2$

(3) $\frac{1}{u_3} = \frac{1}{u_1} + \frac{1}{u_2}$

(4) $u_3^2 = u_1^2 + u_2^2$

(SPACE FOR ROUGH WORK)

55. A smooth rod OP is fixed vertically. A disc of mass m and radius R is rolling without slipping at the given instant on a rough horizontal surface as shown. Velocity of centre of mass C of disc is v_0 . A rod AB of length L is connected with disc at A, $R/2$ distance vertically up from centre of disc with pin joint (about which it can rotate freely), other end of the rod is attached to a small smooth ring B, which can move freely over rod OP. At an instant rod AB makes an angle 60° with the vertical. The velocity of ring on the vertical rod at this instant is –



- (1) $\frac{3\sqrt{3}}{2} v_0$
- (2) $\frac{3}{2} v_0$
- (3) v_0
- (4) perpendicular to velocity of point Q (Q and C are at same horizontal level)

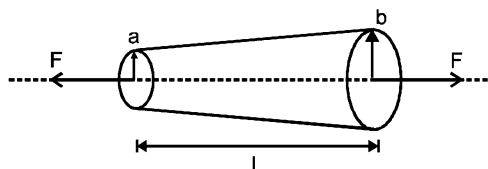
56. A point object moves on a circular path such that distance covered by it is given by function

$$S = \left(\frac{t^2}{2} + 2t \right) \text{ meter (t in second). The ratio of}$$

the magnitude of acceleration at $t = 2$ sec. and $t = 5$ sec. is 1 : 2 then radius of the circle is

- (1) $1m$
- (2) $3\sqrt{51} m$
- (3) $\sqrt{51} m$
- (4) $3m$

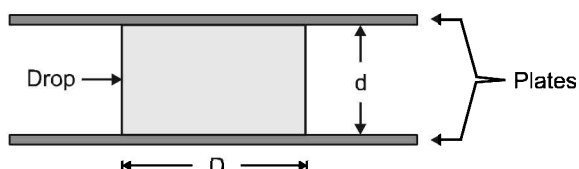
57. A slightly conical wire of length L and radii a and b is stretched by two forces each of magnitude F , applied parallel to its length in opposite directions and normal to end faces. If Y denotes the Young's modulus, then the extension produced is :



- (1) $\frac{FL}{\pi(a^2 + b^2)Y}$
- (2) $\frac{FL}{\pi(a^2 - b^2)Y}$
- (3) $\frac{FL}{\pi abY}$
- (4) None of these

(SPACE FOR ROUGH WORK)

58. A drop of liquid of surface tension σ is in between the two smooth parallel glass plates held at a distance d apart from each other in zero gravity. The liquid wets the plate so that the drop is a cylinder of diameter D with its curved surface at right angles to both the plates. Determine the force acting on each of the plates from drops under the following considerations.



- (1) $\frac{\sigma\pi D}{2}$
- (2) $\frac{\sigma^2\pi D}{2}$
- (3) zero
- (4) None of these

SECTION - II

Reasoning Type

This section contains **2 reasoning type questions**. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

59. **Statement-1** : If kinetic energy of a body is constant, the net force on it is zero.

Statement-2 : According to work energy theorem work done by all forces on a system converts into kinetic energy of system.

- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
- (3) Statement-1 is True, Statement-2 is False
- (4) Statement-1 is False, Statement-2 is True

60. **Statement-1** : Centre of gravity and centre of mass have same physical meaning.

Statement-2 : In uniform gravitational field, centre of mass and centre of gravity are at the same point.

- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
- (3) Statement-1 is True, Statement-2 is False
- (4) Statement-1 is False, Statement-2 is True

(SPACE FOR ROUGH WORK)

PART- C

Atomic masses : [H = 1, D = 2, Li = 7, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, Si = 28, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Cu = 63.5, Zn = 65, As = 75, Br = 80, Ag = 108, I = 127, Ba = 137, Hg = 200, Pb = 207]

SECTION

Straight Objective Type

This section contains 30 questions. Each question has four choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

61. A solution is prepared by urea and water. If mole fraction of water is 0.8 in the solution. Find the ratio of mass of urea & water.

(1) $\frac{5}{6}$

(2) $\frac{6}{5}$

(3) $\frac{1}{1}$

(4) $\frac{4}{1}$

62. For the electrons of oxygen atom, which of the following statements is correct ?

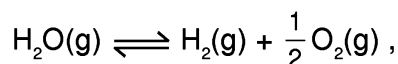
(1) Z_{eff} for an electron in a 2s orbital is the same as Z_{eff} for an electron in a 2p orbital.

(2) An electron in the 2s orbital has the same energy as an electron in the 2p orbital.

(3) Z_{eff} for an electron in a 1s orbital is the same as Z_{eff} for an electron in a 2s orbital.

(4) The two electrons present in the 2s orbital have different spin quantum numbers (m_s).

63. Steam undergoes decomposition at high temperature as per the reaction :



$$\Delta H^\circ = 200 \text{ kJ mol}^{-1} \quad \Delta S^\circ = 40 \text{ kJ mol}^{-1}.$$

The temperature at which equilibrium constant is unity is :

(1) 3000 Kelvin

(2) 5000 Kelvin

(3) 5333 Kelvin

(4) 5 Kelvin

64. Pure $\text{PCl}_5(\text{g})$ was placed in a flask at 2 atm and left for some time at a certain temp. Where the following equilibrium was established $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ Under the given condition, pure oxygen was found to effuse 2.084 time faster than the equilibrium mixture then K_p of the above reaction is :

(1) $\frac{2}{3}$

(2) $\frac{3}{2}$

(3) $\frac{4}{5}$

(4) $\frac{5}{2}$

(SPACE FOR ROUGH WORK)

65. $M(OH)_x$ has $K_{sp} = 4 \times 10^{-12}$ and solubility $10^{-4} M$. Hence x is

- (1) 1 (2) 2
(3) 3 (4) 4

66. Borax on heating strongly above its melting point melts to a liquid, which then solidifies to a transparent mass commonly known as borax-bead. The transparent glassy mass consists of :

- (1) sodium pyroborate
(2) boric anhydride
(3) sodium meta-borate
(4) boric anhydride and sodium metaborate

67. How many P–OH bond in pyrophosphoric acid

- (1) Two (2) Three
(3) Four (4) One

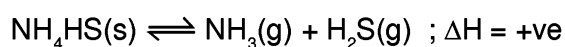
68. The incorrect statement among the following is :

- (1) beryllium oxide is amphoteric in nature
(2) solubility of sulphates of second group elements decreases down the group
(3) reducing power of hydride of alkali metal decreases down the group
(4) Beryllium has diagonal relationship with aluminium

69. Which of the following statements is not correct from the point of view of molecular orbital?

- (1) Be_2 is not a stable molecule
(2) He_2 not stable but He^+ is expected to exist
(3) Bond strength of N_2 is maximum amongst the homonuclear diatomic molecules
(4) The order of energies of molecular orbitals in O_2 molecule is
 $E(\sigma 2s) < E(\sigma^* 2s) < E(\pi 2p_x) = E(\pi 2p_y) < E(\sigma 2p_z) < E(\pi^* 2p_x) = E(\pi^* 2p_y) < E(\sigma^* 2p_z)$

70. Some statements are given for the following equilibrium.



S_1 : On increase in temperature, equilibrium pressure of ammonia increases.

S_2 : On increase in volume of container at constant temperature, equilibrium pressure of ammonia increases.

S_3 : On increase in mass of $NH_4HS(s)$ in the container at constant temperature, equilibrium pressure of ammonia increases.

- (1) T T T (2) F F F
(3) T T F (4) T F F

71. Arrange the following compounds in the increasing order of F–Xe–F bond angle :

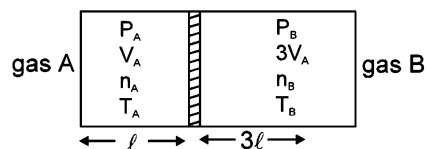


- (1) $XeF_2 < XeF_4 < XeF_5^-$
(2) $XeF_5^- < XeF_4 < XeF_2$
(3) $XeF_2 < XeF_5^- < XeF_4$
(4) $XeF_2 = XeF_4 = XeF_5^-$

(SPACE FOR ROUGH WORK)

72. Two gas samples are separated by a conducting frictionless piston as shown in the figure :

If the piston is in equilibrium in the position shown. Then which of the following statement is incorrect :

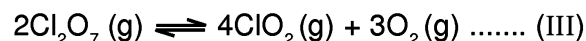
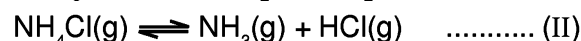
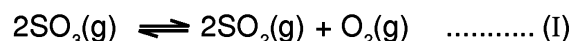


- (1) $T_A = T_B$
- (2) $P_A = P_B$
- (3) $n_B = 3n_A$
- (4) Total K.E. of Gas A = $3 \times$ Total K.E. of gas B

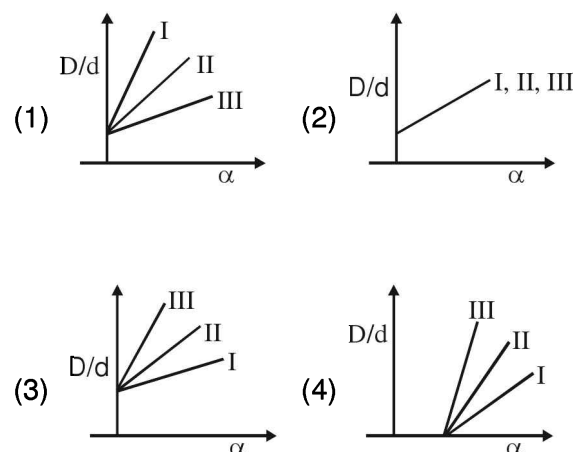
73. Given that : $\Delta G_f^\circ (\text{CuO}) = -30.4 \text{ kcal/mole}$
 $\Delta G_f^\circ (\text{Cu}_2\text{O}) = -34.98 \text{ kcal/mole}$ $T = 298 \text{ K}$
 Now on the basis of above data which of the following predictions will be most appropriate under the standard conditions and reversible reaction.

- (1) Finely divided form of CuO kept in excess O_2 would be completely converted to Cu_2O
- (2) Finely divided form of Cu_2O kept in excess O_2 would be completely converted to CuO
- (3) Finely divided form of CuO kept in excess O_2 would be converted to a mixture of CuO and Cu_2O (having more of CuO)
- (4) Finely divided form of CuO kept in excess O_2 would be converted to a mixture of CuO and Cu_2O (having more of Cu_2O)

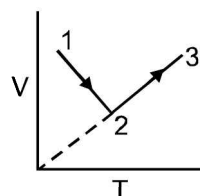
74. If the plot of " D/d " vs ' α ' is made for three different dissociation reactions



then which sketch may represent the correct variation



75. Following graph is constructed for the fixed amount of the gas.



Incorrect statment is :

- (1) From 1 - 2 pressure will increase
- (2) From 2 - 3 pressure remains constant
- (3) Gas pressure at 3 is greater than state 1
- (4) From 1 - 2 pressure will decrease

(SPACE FOR ROUGH WORK)

76. 40 mL of $\frac{M}{4}$ solution of a weak monobasic acid

($K_a = 2 \times 10^{-12}$ at 25°C) is treated with $M/12$ Barium hydroxide solution. The concentration of H^+ ions at the equivalence point is :

- (1) $5 \times 10^{-13} \text{ M}$
- (2) $\sqrt{5} \times 10^{-2} \text{ M}$
- (3) $4\sqrt{2} \times 10^{-13} \text{ M}$
- (4) $2\sqrt{5} \times 10^{-13} \text{ M}$

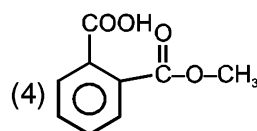
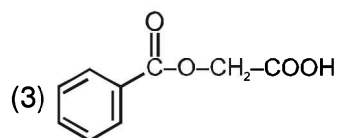
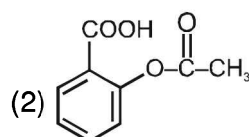
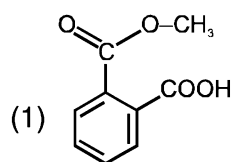
77. Select the correct statement :

- (1) Double chain silicates are known as amphiboles.
- (2) In cyclic silicates two oxygen atoms per tetrahedron are shared.
- (3) Orthosilicates contain discrete $(\text{SiO}_4)^{4-}$ units.
- (4) All of these

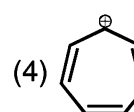
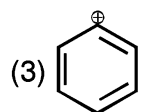
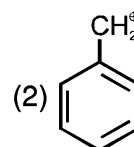
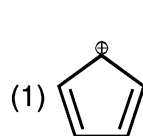
78. Select the correct statements

- (1) Oxides of boron (B_2O_3) and silicon (SiO_2) are acidic in nature.
- (2) Oxides of aluminium (Al_2O_3) and gallium (Ga_2O_3) are amphoteric in nature.
- (3) Oxides of germanium (GeO_2) and tin (SnO_2) are acidic in nature.
- (4) both (1) and (2)

79. Aspirin is a pain relieving drug. It is an 2-Alkanoyloxybenzoic acid derivative having molecular formula ($\text{C}_9\text{H}_8\text{O}_4$). Its structure is :

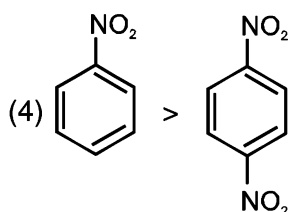
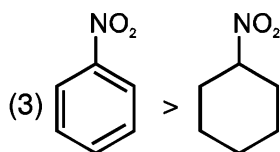
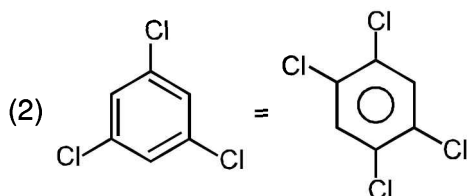
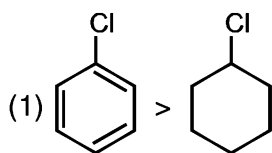


80. The most stable carbocation is :

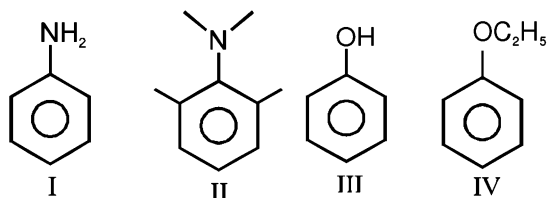


(SPACE FOR ROUGH WORK)

81. Which of the following is not correct for dipole moment ?



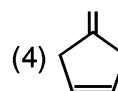
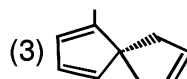
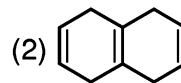
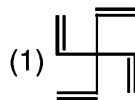
82. The order of electron density in the benzene ring in the following compounds :



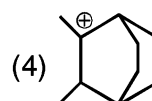
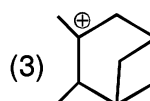
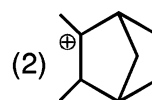
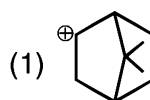
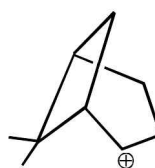
(1) I > II > III > IV (2) II > IV > I > III

(3) IV > III > II > I (4) I > III > IV > II

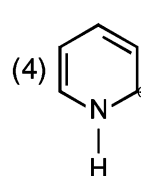
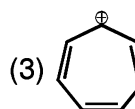
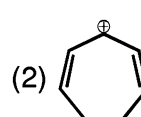
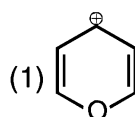
83. Catalytic hydrogenation of which of the following, gives a product that gives two monophotochlorination products (considering structural isomers only)?



84. The most probable rearranged carbocation of the given carbocation

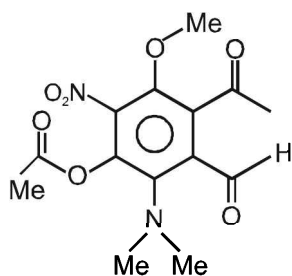


85. Which of the following is not aromatic cation ?



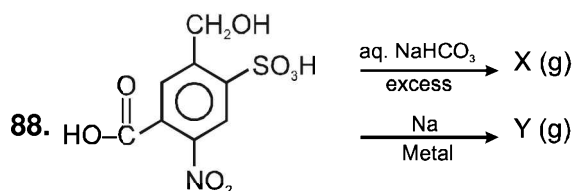
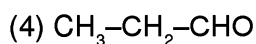
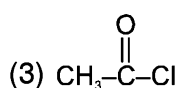
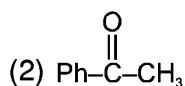
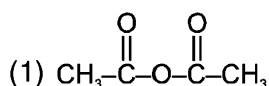
(SPACE FOR ROUGH WORK)

86. Find the total + m groups attached to the benzene ring in the given compound?



- (1) 2 (2) 4
(3) 5 (4) 3

87. Which of the following compounds will give yellow precipitate with NaOH/I₂?



In the above reaction find out the number of moles of the gases formed in the both reactions respectively?

	H ₂ (g)	SO ₂ (g)	CO ₂ (g)	NO ₂ (g)
(1)	1.5	1	1	0
(2)	3	1	1	1
(3)	1.5	0	3	0
(4)	1.5	0	2	0

89. How many different types of products will be obtained after reductive ozonolysis of o-Xylene (Number of moles of same compound will not be included).

- (1) 2 (2) 3
(3) 4 (4) 5

90. In which of the following all groups show positive inductive effect?

- (1) -ONa, -Cl, CCl₃
(2) -C(=O)Cl, -COONa, CH₃
(3) -CMe₃, -SO₃Na, -COONa
(4) -COONa, -ONa, NH_3^+

(SPACE FOR ROUGH WORK)

VISIT: WWW.PREPJEE.IN
FOR MORE DEATILS ABOUT
TEST PAPERS

MOCK TEST 4

CODE

Q

PREPJEE

PREPARE FOR JEE EXAMINATION WITH US

PREPJEE TEST SERIES (PTS)
JEE MAIN

TIME: 3 HOURS MAX.MARKS: 360

Duration : 3 Hours

VISIT : WWW.PREPJEE.IN

Max. Marks : 309

Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.

IMPORTANT INSTRUCTIONS

1. Immediately fill the particulars on this page of the Test Booklet with Blue / Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **360**.
5. There are three parts in the question paper A, B, C consisting of **Mathematics, Physics and Chemistry** having 30 questions. Each part consists of **Three** sections.
6. For each question in **Section 1**, you will be awarded **3 marks** if you darken all the bubble(s) corresponding to only the correct answer(s) and **zero mark** if no bubbles are darkened. In all other cases, **(-1) marks will be awarded**.
For each question in **Section 2 & 3**, you will be awarded **4 marks** if you darken the bubble corresponding to only the correct answer and **zero mark** if no bubbles are darkened. **No negative** marks will be awarded for incorrect answers in this section.
7. There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 6 above.

Filling the Top-half of the ORS :

Use only Black ball point pen only for filling the ORS. Do not use Gel / Ink / Felt pen as it might smudge the ORS.

8. Write your Roll no. in the boxes given at the top left corner of your ORS with black ball point pen. Also, darken the corresponding bubbles with Black ball point pen only. Also fill your roll no on the back side of your ORS in the space provided (if the ORS is both side printed).
9. Fill your Paper Code as mentioned on the Test Paper and darken the corresponding bubble with Black ball point pen.
10. If student does not fill his/her roll no. and paper code correctly and properly, then his/her marks will not be displayed and 5 marks will be deducted (paper wise) from the total.
11. Since it is not possible to erase and correct pen filled bubble, you are advised to be extremely careful while darken the bubble corresponding to your answer.
12. Neither try to erase / rub / scratch the option nor make the Cross (X) mark on the option once filled. Do not scribble, smudge, cut, tear, or wrinkle the ORS. Do not put any stray marks or whitener anywhere on the ORS.
13. If there is any discrepancy between the written data and the bubbled data in your ORS, the bubbled data will be taken as final.

DO NOT BREAK THE SEALS WITHOUT BEING
INSTRUCTED TO DO SO BY THE INVIGILATOR

PART- A

SECTION - I

Straight Objective Type

This section contains 17 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. For three square matrix A, B & C if $ABC = O$ & $|A| \neq 0$, B is non-zero singular matrix then
 - (1) C must be zero matrix
 - (2) C must be non-singular matrix
 - (3) C must be singular matrix
 - (4) Data insufficient
2. If $\vec{a} = 2x^2\hat{i} + 4x\hat{j} + \hat{k}$ and $\vec{b} = 7\hat{i} - 2\hat{j} + x\hat{k}$ are inclined at an obtuse angle and $\vec{a}$ is parallel to $\vec{c} = \hat{i} + 20\hat{j} + 50\hat{k}$ then
 - (1) $x = \frac{1}{10}$
 - (2) $x = 1$
 - (3) $x = -\frac{1}{10}$
 - (4) no such value of x exist
3. The median of set of nine distinct observations is 20.5. If each of the observations of the set is increased by 2, then the median of the new set
 - (1) is increased by 2
 - (2) is decreased by 2
 - (3) is two times the original median
 - (4) remains the same as that of the original set
4. The coefficient of x^{48} in the product $(x - 2)(x - 4)(x - 6)\dots(x - 98)$ is
 - (1) -99^2
 - (2) -2450
 - (3) -2500
 - (4) -2352
5. A closet has 5 pairs of different types of shoes. The number of ways in which 4 shoes can be drawn from it such that there will be no complete pair is
 - (1) 200
 - (2) 160
 - (3) 40
 - (4) 80

(SPACE FOR ROUGH WORK)

6. If A is a square matrix of order 3 such that $|A| = 2$, then $|(\text{adj}A^{-1})^{-1}|$ is
- (1) $\frac{1}{4}$
 (2) 2
 (3) 4
 (4) 8
7. Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{c} = \hat{i} + \hat{j} - 2\hat{k}$.
 A vector coplanar with $\vec{b}$ and $\vec{c}$, whose projection on $\vec{a}$ is of magnitude $\sqrt{\frac{2}{3}}$ is
- (1) $2\hat{i} + 2\hat{j} - 3\hat{k}$
 (2) $-2\hat{i} - \hat{j} + 5\hat{k}$
 (3) $2\hat{i} + 3\hat{j} + 3\hat{k}$
 (4) $2\hat{i} + \hat{j} + 5\hat{k}$
8. In the expansion of $(3^{-x/4} + 3^{5x/4})^n$ the sum of binomial coefficient is 64 and term with the greatest binomial coefficient exceed the third term by $(n - 1)$, the value of x must be
- (1) 0
 (2) 1
 (3) 2
 (4) 3
9. Number of divisors of the number $N = 2^3 \cdot 3^5 \cdot 5^7 \cdot 7^9$ which are perfect square is
- (1) 24
 (2) 60
 (3) 119
 (4) 120
10. The lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{4-z}{k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar if
- (1) $k = 0$ or -1
 (2) $k = 0$ or 1
 (3) $k = 0$ or -3
 (4) $k = 3$ or -3
11. If the system of linear equations $x + 2ay + az = 0$, $x + 3by + bz = 0$, and $x + 4cy + cz = 0$ has a non-zero solution, then a, b, c
- (1) are in AP
 (2) are in GP
 (3) are in HP
 (4) satisfy $a + 2b + 3c = 0$

(SPACE FOR ROUGH WORK)

12. Let $\vec{a} = \hat{j} - \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} - \hat{k}$. The vector $\vec{b}$

satisfying $\vec{a} \times \vec{b} + \vec{c} = \vec{0}$ and $\vec{a} \cdot \vec{b} = 3$ is

- (1) $2\hat{i} - \hat{j} + 2\hat{k}$ (2) $\hat{i} - \hat{j} - 2\hat{k}$
(3) $\hat{i} + \hat{j} - 2\hat{k}$ (4) $-\hat{i} + \hat{j} - 2\hat{k}$

13. If A and B be two events such that

$$P(\bar{A} \cap \bar{B}) = \frac{1}{6}, P(A \cap B) = \frac{1}{4} \text{ and } P(\bar{A}) = \frac{1}{4},$$

then

- (1) $P(B) = \frac{1}{3}$
(2) A and B are dependent events
(3) $P(\bar{A} \cup \bar{B}) = \frac{1}{4}$
(4) $\bar{A}$ and $\bar{B}$ are exhaustive events

14. There are six periods in each working day in a school in which 5 subjects can be arranged if each subject is allotted at least one period and no period remains vacant is

- (1) 210
(2) 1800
(3) 360
(4) 120

15. Let ABC be a triangle where $A = (2, 3, 5)$, $B = (-1, 3, 2)$ and $C = (\lambda, 5, \mu)$. If the median through A is equally inclined to the axes, then

- (1) $\lambda = \mu = 5$
(2) $\lambda = 5, \mu = 7$
(3) $\lambda = 7, \mu = 10$
(4) $\lambda = 0, \mu = 0$

16. If the mean and variance of a binomial variate X are $\frac{7}{3}$ and $\frac{14}{9}$ respectively. Then probability that X takes value 6 or 7 is equal to

- (1) $\frac{1}{729}$ (2) $\frac{5}{729}$
(3) $\frac{7}{729}$ (4) $\frac{13}{729}$

17. The angle between $\hat{i}$ and the line of intersection of the plane $\vec{r} \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 0$ and $\vec{r} \cdot (3\hat{i} + 3\hat{j} + \hat{k}) = 0$ is

- (1) $\cos^{-1} \left(\frac{1}{3} \right)$ (2) $\cos^{-1} \left(\frac{1}{\sqrt{3}} \right)$
(3) $\cos^{-1} \left(\frac{2}{\sqrt{3}} \right)$ (4) $\cos^{-1} \left(\frac{7}{\sqrt{122}} \right)$

(SPACE FOR ROUGH WORK)

SECTION - II**Straight Objective Type**

This section contains 10 questions. Each question has 5 choices (1), (2), (3), (4) and (5) for its answer, out of which **ONLY ONE** is correct.

18. Image of line $L \equiv \frac{x}{1} = \frac{1-y}{1} = \frac{z}{0}$ in the $x-z$ plane is

(1) $x-1=y=\frac{z}{0}$ (2) $x-y=1, z=2$

(3) $\frac{x-1}{0} = \frac{y}{1} = \frac{z}{1}$ (4) $\frac{x-1}{1} = \frac{y}{1} = \frac{z}{1}$

(5) $\frac{x-1}{1} = \frac{y}{0} = \frac{z}{1}$

19. The value of $({}^7C_0 + {}^7C_1) + ({}^7C_1 + {}^7C_2) + \dots + ({}^7C_6 + {}^7C_7)$ is

(1) $2^8 - 2$ (2) $2^8 - 1$

(3) $2^8 + 1$ (4) 2^8

(5) $2^7 - 2$

20. The remainder obtained when $1! + 2! + 3! + \dots + 95!$ is divided by 15 is

(1) 3 (2) 14

(3) 1 (4) 4

(5) none of these

21. If $\vec{a}, \vec{b}, \vec{c}$ are three vectors mutually perpendicular

to each other and $|\vec{a}| = 1, |\vec{b}| = 3$ and $|\vec{c}| = 5$,

then $[\vec{a} - 2\vec{b} \quad \vec{b} - 3\vec{c} \quad \vec{c} - 4\vec{a}] =$

(1) 0 (2) -24

(3) 3600 (4) -345

(5) 15

22. If A is a square matrix and $A^3 + 3A^2 + 3A = O$

then $A^2 + 2A + I$ is equal to

(1) $(A + I)^{-1}$ (2) O

(3) I (4) $3A + I$

(5) $-(A + I)$

23. One card is drawn from each of two ordinary packs of 52 cards. The probability that at least one of them is an ace of heart is

(1) $\frac{103}{2704}$ (2) $\frac{1}{2704}$

(3) $\frac{2}{52}$ (4) $\frac{2601}{2704}$

(5) $\frac{51}{52}$

(SPACE FOR ROUGH WORK)

24. A unit vector coplanar with $\hat{i} + \hat{j} + 2\hat{k}$ and

$\hat{i} + 2\hat{j} + \hat{k}$ and perpendicular to $\hat{i} + \hat{j} + \hat{k}$ is

(1) $\frac{\hat{j} - \hat{k}}{\sqrt{2}}$

(2) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$

(3) $\frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}}$

(4) $\frac{\hat{j} + \hat{k}}{\sqrt{2}}$

(5) $\frac{-\hat{j} + 2\hat{k}}{\sqrt{5}}$

25. Let mean of 100 data points be 55 and variance be 16. Now if every data point is increased by 2 units, then the sum of mean and standard deviation of new data points will be

(1) 71

(2) 59

(3) 63

(4) 61

(5) 55

26. The total number of ways of selecting 10 balls out of an unlimited number of identical white, red and blue balls is equal to

(1) ${}^{12}C_2$

(2) ${}^{13}C_3$

(3) ${}^{10}C_2$

(4) ${}^{10}C_3$

(5) ${}^{12}C_3$

27. A variable plane passes through a fixed point (a, b, c) and meets the coordinate axes in A, B, C. Locus of the point common to the planes through A, B, C and parallel to coordinate planes, is

(1) $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$

(2) $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

(3) $ax + by + cz = 1$

(4) $ax + by + cz = 2$

(5) $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 2$

SECTION - III

Reasoning Type

This section contains 3 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

28. **Statement- 1** : If a,b,c, d are real number and

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $A^3 = O$, then $A^2 = O$.

Statement- 2 : : For matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ we have $A^2 - (a + d)A + (ad - bc)I = O$.

(1) Statement -1 is True, Statement -2 is True ; Statement -2 is a correct explanation for Statement -1

(2) Statement-1 is True, Statement-2 is True ; Statement-2 is **NOT** a correct explanation for Statement-1

(3) Statement -1 is True, Statement -2 is False

(4) Statement -1 is False, Statement -2 is True

(SPACE FOR ROUGH WORK)

29. Consider the system of the equations

$$kx + y + z = 1, x + ky + z = k \text{ and } x + y + kz = k^2$$

Statement- 1 : System of equations has infinite solutions for $k = 1$.

Statement- 2 : The determinant $\begin{vmatrix} 1 & 1 & 1 \\ k & k & 1 \\ k^2 & 1 & k \end{vmatrix} = 0$

then $k = 1$ or -1

- (1) Statement -1 is True, Statement -2 is True
; Statement -2 is a correct explanation for Statement -1
- (2) Statement-1 is True, Statement-2 is True ;
Statement-2 is **NOT** a correct explanation for Statement-1
- (3) Statement -1 is True, Statement -2 is False
- (4) Statement -1 is False, Statement -2 is True

30. STATEMENT-1 : A card is drawn from a well shuffled ordinary deck of 52-playing cards. Let A be the event that 'card drawn is an Ace' and B be the event that 'card drawn is a spade'. Then the events A and B are independent

STATEMENT-2 : Let A and B be two non-empty events. If $P(A/B) = P(A)$, then the events A and B are independent.

- (1) Statement-1 is True, Statement-2 is True;
Statement-2 is a correct explanation for Statement-1.
- (2) Statement-1 is True, Statement-2 is True;
Statement-2 is NOT a correct explanation for Statement-1
- (3) Statement-1 is True, Statement-2 is False
- (4) Statement-1 is False, Statement-2 is True

(SPACE FOR ROUGH WORK)

SECTION - I

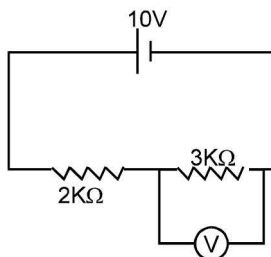
Straight Objective Type

This section contains 17 multiple choice questions. Each question has choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

31. An infinitely long cylindrical object with radius R has a charge distribution that depends upon distance r from the axis like this : $\rho = ar + br^2$ ($r \leq R$, a and b are non zero constant, ρ is volume charge density). If electric field outside the cylinder is zero then value of $\frac{a}{b}$ is :

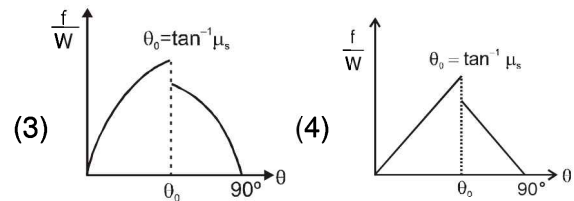
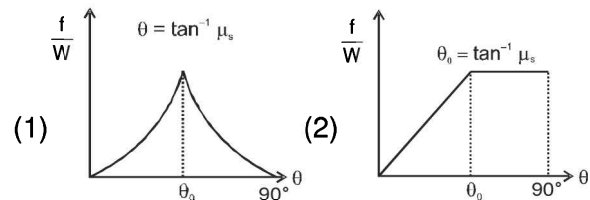
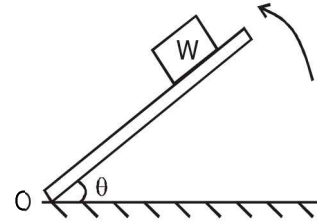
- (1) $3R/4$
- (2) $-3R/4$
- (3) $-4R/3$
- (4) $4R/3$

32. In the circuit shown in figure, the resistance of voltmeter is $6K\Omega$. The voltmeter reading will be:



- (1) 6V
- (2) 5V
- (3) 4V
- (4) 3V

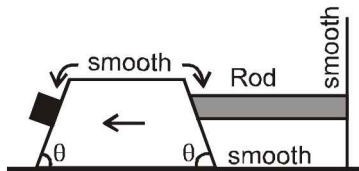
33. A block of weight W rests on a rough horizontal plank hinged at point 'O'. The plank starting from horizontal position rotates such that the angle θ it makes with horizontal increases slowly upto 90° . If f is the friction between the block and the plank, identify the graph of $\frac{f}{W}$ versus θ . (μ_s and μ_k are coefficients of static and kinetic friction between the plank and block) ?



(SPACE FOR ROUGH WORK)

34. In the diagram shown, no relative motion takes place between the wedge and the block placed on it. The rod slides downwards over the wedge and pushes the wedge to move in horizontal direction. The mass M of wedge is equal to 1 kg and is same as that of the block. If $\tan \theta = \frac{1}{\sqrt{3}}$, find the mass of rod.

(Neglect rotation of the rod)

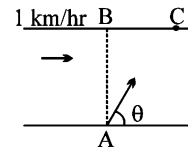


- (1) 1 kg
(2) 2 kg
(3) 3 kg
(4) 4 kg

35. A ball is dropped from a bridge 125 m above a river. 2 s later a second ball is thrown straight down. What must be the initial velocity of second ball so that both hit the water surface simultaneously?

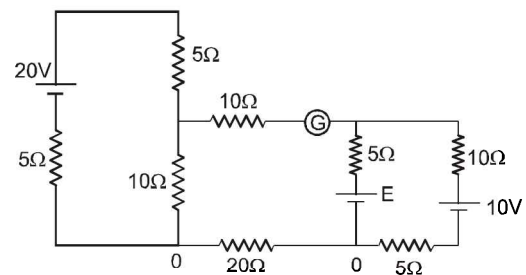
- (1) $\frac{80}{3}\text{ m/s}$ (2) $\frac{160}{3}\text{ m/s}$
(3) $\frac{40}{3}\text{ m/s}$ (4) $\frac{20}{3}\text{ m/s}$

36. A river is flowing with a speed of 1 km/hr . A swimmer wants to go to point 'C' starting from 'A'. He swims with a speed of 5 km/hr , at an angle θ w.r.t. the river. If $AB = BC = 400\text{ m}$. Then the value of θ is:



- (1) 37° (2) 30°
(3) 60° (4) 53°

37. What should be value of E for which galvanometer shows no deflection :

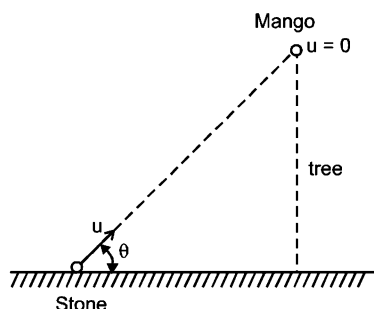


- (1) 10 V
(2) 5 V
(3) 15 V
(4) 20 V

38. The magnifying power of the telescope is found to be 9 and the separation between the lenses is 20 cm for relaxed eye. Then the focal lengths of component lenses are :
- (1) $18\text{ cm}, 2\text{ cm}$ (2) $10\text{ cm}, 10\text{ cm}$
(3) $9\text{ cm}, 11\text{ cm}$ (4) None of these

(SPACE FOR ROUGH WORK)

39. A stone is projected with speed u aiming the mango at some height from ground as shown in figure. At the instant of projection of stone the mango released (dropped) from the tree. Then which of the following is correct. (Distance between foot of tree and point of projection is less than R (Range))



- (1) The stone will hit the mango for all values of u .
 (2) The stone will hit the mango for a special value of u .
 (3) The stone will never hit the mango
 (4) None of these
40. Equation of trajectory of a projectile is given by $y = -x^2 + 10x$ where x and y are in meters and x is along horizontal and y is vertically upward and particle is projected from origin. Then which of the following options is **incorrect**: ($g = 10 \text{ m/s}^2$)
- (1) initial velocity of particle is $\sqrt{505} \text{ m/s}$
 (2) horizontal range is 10 m
 (3) maximum height is 25 m
 (4) angle of projection with horizontal is $\tan^{-1}(5)$

41. At an instant particle-A is at origin and moving with constant velocity $(3\hat{i} + 4\hat{j}) \text{ m/s}$ and particle-B is at $(4,4)\text{m}$ and moving with constant velocity $(4\hat{i} - 3\hat{j}) \text{ m/s}$. Then at this instant which of the following options is **incorrect** :

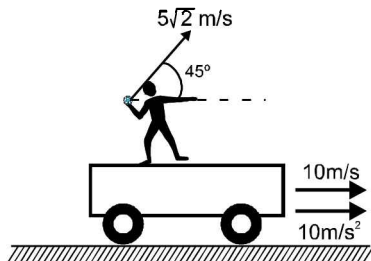
- (1) relative velocity of B w.r.t. A is $(\hat{i} - 7\hat{j}) \text{ m/s}$
 (2) approach velocity of A and B is $3\sqrt{2} \text{ m/s}$
 (3) relative velocity of B w.r.t. A remains constant
 (4) approach velocity of A and B remains constant

42. An astronaut is on the surface of a planet whose air resistance is negligible. To measure the acceleration due to gravity (g), he throws a stone upwards. He observes that the stone reaches to a maximum height of $h = 10\text{m}$ (which is negligible compared to radius of planet) and reaches the surface 4 seconds after it was thrown. Find the acceleration due to gravity (g) on the surface of that planet :

- (1) 5 m/s^2
 (2) 10 m/s^2
 (3) 7.5 m/s^2
 (4) 2.5 m/s^2

(SPACE FOR ROUGH WORK)

43. A man standing on a truck which moves with a constant horizontal acceleration $a (= 10 \text{ m/s}^2)$ when speed of the truck is 10 m/s . The man throws a ball with velocity $5\sqrt{2} \text{ m/s}$ with respect to truck. In the direction shown in the diagram. Find the displacement of ball in meters in one second observed as observed by the man. ($g = 10 \text{ m/s}^2$)



- (1) 20 m
(2) 10 m
(3) 5 m
(4) zero
44. When an small object is placed at a distance x_1 and x_2 from a lens on its principal axis, then real image and a virtual image are formed respectively having same magnitude of transverse magnification. Then the focal length of the lens is :

- (1) $x_1 - x_2$ (2) $\frac{x_1 - x_2}{2}$
(3) $\frac{x_1 + x_2}{2}$ (4) $x_1 + x_2$

45. A prism produces same deviation 40° for two different angles of incidence 37° and 42° in same surrounding medium on both sides. Then prism angle is :
- (1) 30°
(2) 31°
(3) 39°
(4) 35°

46. Plane surface of a thin plano-convex lens is silvered. If a point object is placed on principal axis at a distance 60 cm from the lens and final image is formed at a distance 30 cm from the lens on same side. What will be the distance of final image from the lens if the plane surface is not silvered :
- (1) 120 cm
(2) 180 cm
(3) 30 cm
(4) 90 cm

47. In a compound microscope
- (1) the objective has a shorter focal length
(2) the objective has a shorter aperture
(3) (1) and (2) both are correct
(4) the aperture of objective and eyepiece are same.

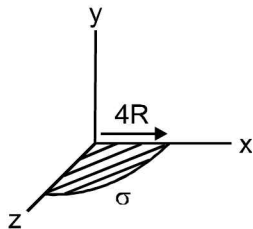
(SPACE FOR ROUGH WORK)

SECTION - II

Straight Objective Type

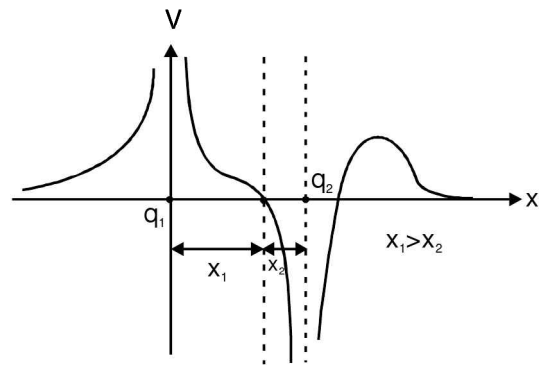
This section contains 10 multiple choice questions. Each question has choices (1), (2), (3), (4) and (5) out of which **ONLY ONE** is correct.

48. Quarter non-conducting disc of radius $4R$ having uniform surface charge density σ is placed in xz -plane then which of the following is **incorrect** :



- (1) electric potential at $(0, 3R, 0)$ is $\frac{\sigma R}{4 \epsilon_0}$
- (2) electric potential at $(0, 0, 0)$ is $\frac{\sigma R}{2 \epsilon_0}$
- (3) electric field at $(0, 3R, 0)$ is symmetric with x and z axis
- (4) electric field intensity at $(-4R, 0, -4R)$ is equally inclined with x and z axis
- (5) electric potential at $(0, 2R, 0)$ is $\frac{\sigma R}{4 \epsilon_0}$

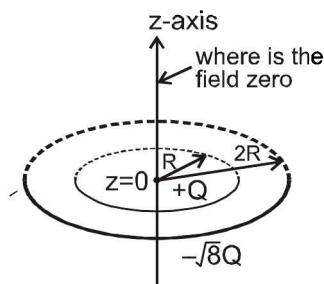
49. Two point charges (q_1 and q_2) are placed on x -axis. Figure shows graph of potential (V) on x -axis with x -co-ordinate distance x_1 and x_2 shown in figure then which of the following option is **true** :



- (1) $q_1 = \frac{x_1}{x_2} q_2$
- (2) $q_2 > 0$
- (3) $q_1 = -\frac{x_1^2}{x_2^2} q_2$
- (4) $q_1 = -\frac{x_1}{x_2} q_2$
- (5) $q_1 < 0$

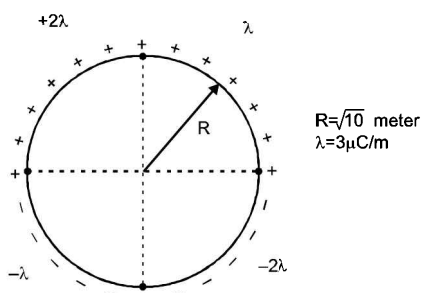
(SPACE FOR ROUGH WORK)

50. Two concentric rings, one of radius R and total charge $+Q$ and the second of radius $2R$ and total charge $-\sqrt{8}Q$, lie in x - y plane (i.e., $z = 0$ plane). The common centre of rings lies at origin and the common axis coincides with z -axis. The charge is uniformly distributed on both rings. The net electric field on z -axis is zero at a distance $\frac{8R}{(\sqrt{2})^x}$ from origin. Then find x .



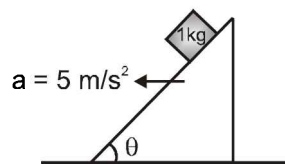
- (1) 2 (2) 4 (3) 6
(4) 8 (5) 5

51. A non-conducting circular ring is having charge distribution as shown in figure. Find electric field intensity at the centre of ring :



- (1) 9×10^3 N/C (2) 5.4×10^3 N/C
(3) 5.4×10^4 N/C (4) 2.7×10^4 N/C
(5) Zero

52. A block of mass 1 kg is at rest relative to a smooth wedge being moved leftwards with constant acceleration 5 m/s^2 . Let N be the normal reaction between the block and the wedge. Then N and $\tan \theta$ are :



- (1) $N = 5\sqrt{5} \text{ N}$ and $\tan \theta = \frac{1}{2}$
(2) $N = 15 \text{ N}$ and $\tan \theta = \frac{1}{2}$
(3) $N = 5\sqrt{5} \text{ N}$ and $\tan \theta = 2$
(4) $N = 15 \text{ N}$ and $\tan \theta = 2$
(5) $N = 20 \text{ N}$ and $\tan \theta = 1$

53. Two coaxial rings are separated by a distance x . R is radius of larger ring having charge Q uniformly distributed on it. The other ring is having very small radius r ($\ll R$) then magnitude of electric flux through smaller ring due to larger ring is maximum if x is equal to :

- (1) 0 (2) $\frac{R}{3}$
(3) $\frac{R}{\sqrt{3}}$ (4) $\frac{R}{\sqrt{2}}$
(5) R

(SPACE FOR ROUGH WORK)

54. The capacitance of a parallel plate air capacitor is $1\mu\text{F}$. This region is now filled with a dielectric slab of dielectric constant 4 and capacitor is connected with battery of EMF 5 volt and zero internal resistance. Now slab is taken out, then during the removal of slab then which of the following options is **incorrect** :

- (1) $-15\mu\text{C}$ charge flows through the cell
- (2) $75\mu\text{J}$ energy is absorbed by the cell
- (3) energy stored in the capacitor is reduced by $75\mu\text{J}$
- (4) the external agent has to do $37.5\mu\text{J}$ amount of work to take out the slab
- (5) charge on capacitor reduced by $15\mu\text{C}$

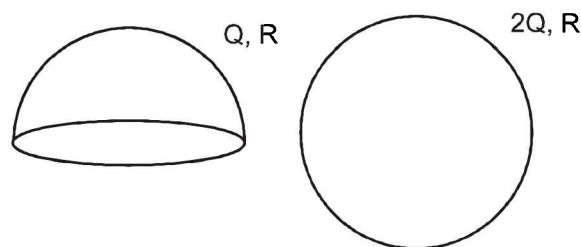
55. The spherical planets have the same mass but densities in the ratio 1: 8. For these planets, the :

- (1) acceleration due to gravity will be in the ratio 4 : 1
- (2) acceleration due to gravity will be in the ratio 1 : 4
- (3) escape velocities from their surfaces will be in the ratio $\sqrt{2} : 1$
- (4) escape velocities from their surfaces will be in the ratio 1 : 2
- (5) ratio of radius of two planet is 1 : 2

56. In the capacitor discharge formula $q = q_0 e^{-t/\tau}$ the symbol τ represents :

- (1) the time, it takes for capacitor to loose q_0/e charge.
- (2) the time it takes for capacitor to loose charge $q_0\left(1 - \frac{1}{e}\right)$
- (3) the time it takes for capacitor to loose essentially all of its initial charge.
- (4) time in which charge of capacitor becomes half of initial value
- (5) none of the above.

57. Figure shows a hemisphere of charge Q and radius R and a sphere of charge $2Q$ and radius R . The total potential energy of hemisphere is U_H and that the sphere is U_S . Then.



- (1) $2U_H = U_S$
- (2) $2U_H < U_S$
- (3) $2U_H > U_S$
- (4) $U_H = U_S$
- (5) Data insufficient

(SPACE FOR ROUGH WORK)

SECTION - III

Reasoning Type

This section contains 3 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

58. **Statement-1:** Work and kinetic energy are frame dependent physical quantities.

Statement -2: Magnitude of a real force is depends on frame of reference.

(1) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True.

59. **Statement-1:** When light ray enter in a rectangular glass slab, at angle of incidence $0 < \theta < 90$ then dispersion of light is not observer.

Statement-2: In dispersion, a beam of white light refracts at different angles of refraction when it enter oblique in a medium.

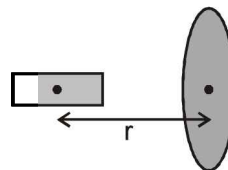
(1) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True.

60. **Statement-1:** The gravitational force between a thin rod of finite length and a disc of infinite radius is inversely proportional to the square of distance between their centre of mass. Mass is uniformly distributed on both.



Statement-2: Gravitational field intensity due to a unifrom disc of infinite radius is independent of distance from disc.

(1) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True.

(SPACE FOR ROUGH WORK)

PART- C

Atomic masses : [H = 1, D = 2, Li = 7, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, Si = 28, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Cu = 63.5, Zn = 65, As = 75, Br = 80, Ag = 108, I = 127, Ba = 137, Hg = 200, Pb = 207]

SECTION - I

Straight Objective Type

This section contains 17 questions. Each question has five choices (1), (2), (3), and (4) for its answer, out of which **ONLY ONE** is correct.

61. CsBr has b.c.c. structure with edge length 4.3 Å. The shortest inter ionic distance in between Cs^+ and Br^- is :

- (1) 3.72 (2) 1.86
(3) 7.44 (4) 4.3

62. In hcp (ABAB...) and ccp (ABCABC...) structures made up of spheres of equal size, the volume occupied per sphere (including the empty spaces) is (a = radius of sphere) :

- (1) $5.66 a^3$ (2) $1.33 a^3$
(3) $2.66 a^3$ (4) $7.40 a^3$

63. Two 1st order reactions have half-lives in the ratio 3 : 2. Then the ratio of time intervals t_1 : t_2 , will be? Where t_1 is the time period for 25% completion of the first reaction and t_2 is time required for 75% completion of the second reaction. [$\log 2 = 0.3$, $\log 3 = 0.48$]

- (1) 0.2 : 1 (2) 0.42 : 1
(3) 0.28 : 1 (4) 0.3 : 1

64. For a certain reaction the variation of the rate constant with temperature is given by the equation

$$\ln k_t = \ln k_0 + \left(\frac{\ln 3}{10} \right) t \quad (t \geq 0^\circ\text{C})$$

The value of the temperature coefficient of the reaction rate is therefore

- (1) 4 (2) 3
(3) 10 (4) 2

(SPACE FOR ROUGH WORK)

65. Which of the following is wrong :

- (1) Enthalpy (numerical value) of physisorption is greater than that of chemisorption
- (2) Physisorption is not very specific but chemisorption is highly specific
- (3) Chemisorption takes place at relatively high temperatures
- (4) In physisorption generally multi-molecular layers are formed on the adsorbent

66. Which of the following ions will be most effective in coagulating the As_2S_3 sol :

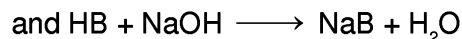
- (1) Fe^{3+} (2) Ba^{2+}
- (3) Cl^- (4) PO_4^{3-}

67. Electrolysis of a solution of HSO_4^- ions produces $\text{S}_2\text{O}_8^{--}$. Assuming 75% current efficiency, what current should be employed to achieve a production rate of 1 mole of $\text{S}_2\text{O}_8^{--}$ per hour ?

- (1) +71.5 amp (2) 35.7 amp
- (3) 142.96 amp (4) 285.93 amp

68. If $\text{HA} + \text{NaOH} \longrightarrow \text{NaA} + \text{H}_2\text{O}$

$$\Delta H = -12 \text{ kcal}$$



$$\Delta H = -11 \text{ kcal}$$

then equimolar solution of which acid has higher pH -

- (1) HA
- (2) HB
- (3) both have same pH
- (4) information insufficient

69. Barium ions, CN^- and Co^{2+} form an ionic complex. If that is supposed to be 75% ionised in water with vant Hoff factor (i) equal to four, then the coordination numbers of Co^{2+} in the complex can be:

- (1) 6
- (2) 4
- (3) 6 & 4 both possible
- (4) 5

70. A solution of 0.2 mole KI ($\alpha = 100\%$) in 1000 g water freezes at $T_1^\circ\text{C}$. Now to this solution 0.1 mole HgI_2 is added and the resulting solution freezes at $T_2^\circ\text{C}$. Which of the following is correct :

- (1) $T_1 = T_2$ (2) $T_1 > T_2$
- (3) $T_1 < T_2$ (4) Can't be predicted

(SPACE FOR ROUGH WORK)

71. In the reaction $\text{CrO}_5 + \text{H}_2\text{SO}_4 \rightarrow \text{Cr}_2(\text{SO}_4)_3 + \text{H}_2\text{O} + \text{O}_2$ one mole of CrO_5 will liberate how many moles of O_2
- (1) $5/2$ (2) $5/4$
(3) $9/2$ (4) none of these

72. In the following reaction $2\text{MnO}_4^- + 5\text{H}_2\text{O}_2^{18} + 6\text{H}^+ \rightarrow 2\text{Mn}^{2+} + 8\text{H}_2\text{O} + 5\text{O}_2$
The radioactive oxygen will appear in :
- (1) H_2O
(2) O_2
(3) both
(4) above reaction does not take place

73. 20 ml of H_2O_2 after acidification with dil H_2SO_4 required 30 ml of $\frac{N}{12}$ KMnO_4 for complete oxidation. The strength of H_2O_2 solution is [Molar mass of $\text{H}_2\text{O}_2 = 34$]
- (1) 2 g/L (2) 4 g/L
(3) 8 g/L (4) 6 g/L

74. How many ml water should be added to 100ml HCl solution ($d = 1.5 \text{ g/ml}$) 80% by wt. to make it a solution of 40% by wt. of density = 1 g/ml.
- (1) 100 ml (2) 300 ml
(3) 200 ml (4) none of these

75. 16 gm of an ideal gas SO_x occupies 5.6 lt. at STP. The value of x is
- (1) $x = 3$ (2) $x = 2$
(3) $x = 4$ (4) none

76. The volume occupied by 2.0 mole of N_2 at 200 K and 8.21 atm pressure, if $\frac{P_C V_C}{RT_C} = \frac{3}{8}$

and $\frac{P_r V_r}{T_r} = 2.4$, is

- (1) 1.8 L (2) $\left(\frac{5}{8}\right) \text{ L}$
(3) 12.8 L (4) 3.6 L

77. The increase in volume of air, when temperature of 600 ml of it, is increased from 27°C to 47°C under constant pressure, is :
- (1) 20 mL (2) 80 mL
(3) 40 mL (4) 500 mL

(SPACE FOR ROUGH WORK)

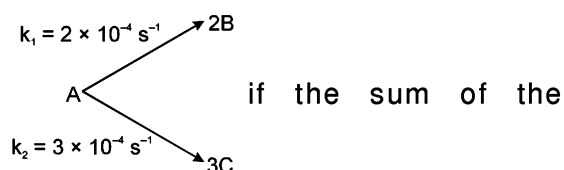
SECTION - II**Straight Objective Type**

This section contains 10 questions. Each question has five choices (1), (2), (3), (4) and (5) for its answer, out of which **ONLY ONE** is correct.

78. A crystal of formula AB_3 has A ions at the cube corners and B ions at the edge centres. The coordination numbers of A and B are respectively

(1) 6 and 6 (2) 2 and 6
(3) 6 and 2 (4) 8 and 8
(5) 8 and 6

79. For the following parallel chain reaction



concentration of B and C at any time is 2M then what will be $[B]_t$ and $[C]_t$ respectively?

(1) $\frac{11}{12}M, \frac{13}{12}M$ (2) $\frac{3}{4}M, \frac{5}{4}M$
(3) $\frac{4}{5}M, \frac{6}{5}M$ (4) $\frac{8}{13}M, \frac{18}{13}M$
(5) $\frac{5}{13}M, \frac{15}{13}M$

80. When a graph is plotted between $\log x/m$ and $\log p$, it is straight line with an angle 45° and intercept 0.3010 on y-axis. If initial pressure is 0.3 atm, what will be the amount of gas adsorbed per gm of adsorbent :

(1) 0.4 (2) 0.6
(3) 0.8 (4) 0.1
(5) 0.5

81. The reduction potential of hydrogen half cell will be negative if ($T = 298 \text{ K}$)

(1) $P_{H_2} = 1 \text{ atm}$ and $[H^+] = 1.0 \text{ M}$
(2) $P_{H_2} = 2 \text{ atm}$ and $[H^+] = 2.0 \text{ M}$
(3) $P_{H_2} = 2 \text{ atm}$ and $[H^+] = 1.0 \text{ M}$
(4) $P_{H_2} = 1 \text{ atm}$ and $[H^+] = 2.0 \text{ M}$
(5) $P_{H_2} = 1 \text{ atm}$ and $[H^+] = 1.5 \text{ M}$

82. 106.2 g 1 molal aqueous solution of ethylene glycol is cooled to -3.72°C . Mass of ice separated during cooling is (K_f water = 1.86, freezing point of water = 0°C)

(1) 25 g (2) 50 g
(3) 60 g (4) 40 g
(5) 30 g

(SPACE FOR ROUGH WORK)

83. 3 mole of a mixture of FeSO_4 and $\text{Fe}_2(\text{SO}_4)_3$ required 100 ml of 2M KMnO_4 solution in acidic medium. Hence mole fraction of FeSO_4 in the mixture is

- (1) $\frac{3}{5}$ (2) $\frac{2}{3}$
(3) $\frac{2}{5}$ (4) $\frac{1}{3}$
(5) $\frac{5}{2}$

84. Common salt obtained from sea - water contains 96% NaCl by mass. The approximate number of molecules present in 10.0 g of the salt is (At. wt. $\text{Na} = 23$)

- (1) 10^{21} (2) 10^{22}
(3) 10^{23} (4) 10^{24}
(5) 10^{25}

85. A mixture of $\text{HCl}(\text{g})$, $\text{PCl}_3(\text{g})$ and $\text{PCl}_5(\text{s})$, each have equal number of Cl -atom then their molar ratio in the mixture is :

- (1) 1 : 3 : 5 (2) 1 : 1 : 1
(3) 15 : 2 : 3 (4) 15 : 5 : 3
(5) 15 : 3 : 2

86. At 27°C the ratio of rms velocities of ozone to oxygen is

- (1) $\sqrt{5/5}$ (2) $\sqrt{4/3}$
(3) $\sqrt{2/3}$ (4) 0.25
(5) $\sqrt{3/4}$

87. Equal amount (mass) of methane and ethane have their total translational kinetic energy energy in the ratio 3 : 1 then their temperatures are in the ratio.

- (1) 5 : 8 (2) 45 : 8
(3) 15 : 8 (4) 8 : 5
(5) 8 : 15

SECTION - III

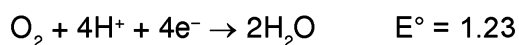
Comprehension Type

This section contains 1 paragraphs. Based open each paragraph, there are 3 questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

(SPACE FOR ROUGH WORK)

Paragraph for Question Nos. 88 to 90

Redox reactions play a pivotal role in chemistry and biology. The values of standard redox potential (E°) of two half-cell reactions decide which way the reaction is expected to proceed. A simple example is a Daniel cell in which zinc goes into solution and copper gets deposited. Given below are a set of half-cell reactions (acidic medium) along with their E° (V with respect to normal hydrogen electrode) values.



88. Among the following, identify the correct statement :

- (1) Chloride ion is oxidised by O_2
- (2) Fe^{2+} is oxidised by iodine
- (3) Iodide ion is oxidised by chlorine
- (4) Mn^{2+} is oxidised by chlorine

89. While Fe^{3+} is stable, Mn^{3+} is not stable in acid solution because :

- (1) O_2 oxidises Mn^{2+} to Mn^{3+}
- (2) O_2 oxidises both Mn^{2+} to Mn^{3+} and Fe^{2+} to Fe^{3+}
- (3) Fe^{3+} oxidises H_2O to O_2
- (4) Mn^{3+} oxidises H_2O to O_2

90. Sodium fusion extract, obtained from aniline, on treatment with iron (II) sulphate and H_2SO_4 in presence of air gives a Prussian blue precipitate. The blue colour is due to the formation of :

- (1) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
- (2) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$
- (3) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_2$
- (4) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_3$

(SPACE FOR ROUGH WORK)

MOCK TEST 5

CODE

Q

PREP JEE

PREPARE FOR JEE EXAMINATION WITH US

PREP JEE TEST SERIES (PTS)
JEE MAIN

TIME: 3 HOURS MAX.MARKS: **288**

Duration : 3 Hours

WWW.PREPJEE.IN

Max. Marks : 288

Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.

IMPORTANT INSTRUCTIONS

1. Immediately fill the particulars on this page of the Test Booklet with Blue / Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **288**.
5. There are three parts in the question paper A, B, C consisting of **Mathematics, Physics** and **Chemistry** having 30 questions. Each part consists of **Two** sections.
6. For each question in **Section 1**, you will be awarded **3 marks** if you darken all the bubble(s) corresponding to only the correct answer(s) and **zero mark** if no bubbles are darkened. **No negative** marks will be awarded for incorrect answers in this section.
For each question in **Section 2**, you will be awarded **4 marks** if you darken the bubble corresponding to only the correct answer and **zero mark** if no bubbles are darkened. In all other cases, **(-1) marks will be awarded**.
7. There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 6 above.

Filling the Top-half of the ORS :

Use only Black ball point pen only for filling the ORS. Do not use Gel / Ink / Felt pen as it might smudge the ORS.

8. Write your Roll no. in the boxes given at the top left corner of your ORS with black ball point pen. Also, darken the corresponding bubbles with Black ball point pen only. Also fill your roll no on the back side of your ORS in the space provided (if the ORS is both side printed).
9. Fill your Paper Code as mentioned on the Test Paper and darken the corresponding bubble with Black ball point pen.
10. If student does not fill his/her roll no. and paper code correctly and properly, then his/her marks will not be displayed and 5 marks will be deducted (paper wise) from the total.
11. Since it is not possible to erase and correct pen filled bubble, you are advised to be extremely careful while darken the bubble corresponding to your answer.
12. Neither try to erase / rub / scratch the option nor make the Cross (X) mark on the option once filled. Do not scribble, smudge, cut, tear, or wrinkle the ORS. Do not put any stray marks or whitener anywhere on the ORS.
13. If there is any discrepancy between the written data and the bubbled data in your ORS, the bubbled data will be taken as final.

DO NOT BREAK THE SEALS WITHOUT BEING
INSTRUCTED TO DO SO BY THE INVIGILATOR

PART-A

SECTION - I

Straight Objective Type

This section contains 24 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. If $A = \sin^{-1} \left\{ \cot \left(\sin^{-1} \sqrt{\frac{13}{17}} \right) \right\}$ then

- (1) $A = \tan^{-1} \frac{2}{3}$ (2) $A = \tan^{-1} \frac{2}{\sqrt{13}}$
(3) $A = \operatorname{cosec}^{-1} \frac{3}{2}$ (4) $A = \cot^{-1} \frac{2}{3}$

2. If $\lim_{h \rightarrow 0} \left[\frac{f(a+h)}{f(a)} \right] = c$ where $f(x)$ is a continuous

function such that $f(x) > 0$ for all $x \in \mathbb{R}$ and $[.]$ denotes greatest integer function then which of the following statements is always true ?

- (1) If $x = a$ is a point of local minima then $c \in \mathbb{N}$
(2) If $x = a$ is a point of local maxima then $c \in \mathbb{N}$
(3) If $x = a$ is a point of local minima then $c \in \mathbb{I}^-$
(4) If $x = a$ is a point of local maxima then $c \in \mathbb{I}^-$

3. If A and B be positive acute angles satisfying $3 \sin^2 A + 2 \sin^2 B = 1$ and $3 \sin 2A - 2 \sin 2B = 0$, then

- (1) $B = \pi/4 - A/2$
(2) $A = \pi/4 - 2B$
(3) $B = \pi/2 - A/4$
(4) $A = \pi/4 - B/2$

4. If $\lim_{\theta \rightarrow 0} \left(\frac{1 + a \cos \theta}{\theta^2} - \frac{b \sin \theta}{\theta^3} \right) = 1$, then the value

of $a + b$ is $-\lambda$, find the value of λ .

- (1) 4
(2) 1
(3) 2
(4) 0

5. The maximum values of $\frac{\log_e x}{x}$ is

- (1) 1
(2) $\frac{2}{e}$
(3) e
(4) $\frac{1}{e}$

(SPACE FOR ROUGH WORK)

6. If $f(x) = \frac{\sin(e^{x-2}-1)}{\ln(x-1)}$, $x \neq 2$ is continuous at

$x = 2$, then $f(2) =$

- (1) 0
- (2) 2
- (3) 1
- (4) $f(2)$ is not possible

7. Let $f(x) = x^x$; $x \in (1, \infty)$ and $g(x)$ be inverse function of $f(x)$ then $g'(x)$ must be

- (1) $\frac{1}{x(1+\ln x)}$
- (2) $\frac{1}{x(1+\ln(g(x)))}$
- (3) $\frac{1}{g(x).(1+\ln(g(x)))}$
- (4) non existent

8. Rolle's Theorem holds for the function

$f(x) = x^3 + bx^2 + cx$, $1 \leq x \leq 2$ at $x = \frac{4}{3}$, then the

value of $b + c$ is

- (1) 2
- (2) 3
- (3) 4
- (4) 5

9. A curve passes through the point $(2, 0)$ and the slope of the tangent at any point (x, y) is $x^2 - 2x$ for all values of x . The point of local maximum on the curve is

- (1) $\left(0, \frac{4}{3}\right)$
- (2) $\left(0, \frac{2}{3}\right)$
- (3) $\left(1, \frac{2}{3}\right)$
- (4) $\left(2, \frac{4}{3}\right)$

10. If $f(x) = ||x|^2 - 2|x| - 3|$ then $f(x)$ is non differentiable at x equal to

- (1) $-1, 0, 1$
- (2) $1, 2$
- (3) $-3, -1, 0, 1, 3$
- (4) $-3, 0, 3$

11. If in a $\triangle ABC$, $a \cos^2 \left(\frac{C}{2}\right) + c \cos^2 \left(\frac{A}{2}\right) = \frac{3b}{2}$,

then the sides a, c, b

- (1) are in A.P.
- (2) are in G.P.
- (3) are in H.P.
- (4) satisfy $a + c = 2b$

(SPACE FOR ROUGH WORK)

12. If $f(\theta) = (\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2$,
then minimum value of $f(\theta)$ is

- (1) 7
- (2) 8
- (3) 9
- (4) 4

13. The set $(A \cup B \cup C) \cap (A \cap B' \cap C')' \cap C'$ is
equal to

- (1) $B \cap C'$
- (2) $A \cap C$
- (3) $B' \cap C'$
- (4) $A \cap (B \cup C)'$

14. The value of

$$\cos^{-1} \left[\cot \left\{ \sin^{-1} \left(\sqrt{\frac{2-\sqrt{3}}{4}} \right) + \cos^{-1} \left(\frac{\sqrt{12}}{4} \right) + \sec^{-1} (\sqrt{2}) \right\} \right]$$

is

- (1) 0
- (2) $\frac{\pi}{4}$
- (3) $\frac{\pi}{6}$
- (4) $\frac{\pi}{2}$

15. If $x^{3/4(\log_3 x)^2 + \log_3 x - 5/4} = \sqrt{3}$ then x can not
assume

- (1) one positive integral value
- (2) one irrational value
- (3) two positive rational values
- (4) two distinct positive integral values

16. $\lim_{x \rightarrow \infty} \frac{2 \cdot x^{1/2} + 3 \cdot x^{1/3} + 4 \cdot x^{1/4} + \dots + n \cdot x^{1/n}}{(3x-4)^{1/2} + (3x-4)^{1/3} + \dots + (3x-4)^{1/n}}$ (here

$n \in \mathbb{N}, n \geq 2$) is equal to

- (1) $\frac{2}{\sqrt{3}}$
- (2) $\frac{\sqrt{3}}{2}$
- (3) $\frac{1}{2}$
- (4) $\frac{1}{\sqrt{3}}$

17. A solution (x, y) of the system of equation
 $x - y = 1/3$ and $\cos^2(\pi x) - \sin^2(\pi y) = 1/2$ is
given by

- (1) $(2/3, 1/3)$
- (2) $(5/3, 4/3)$
- (3) $(13/6, 11/6)$
- (4) $(5/12, 1/12)$

(SPACE FOR ROUGH WORK)

18. Let $R = \{(3, 3), (6, 6), (9, 9), (12, 12), (6, 12), (3, 9), (3, 12), (3, 6)\}$ be a relation on the set $A = \{3, 6, 9, 12\}$, the relation is

- (1) Reflexive and Transitive only
- (2) Reflexive only
- (3) an Equivalence relation
- (4) Reflexive and Symmetric

19. If $f(x)$ is a differentiable function satisfying $f'(x) < 2$ for all $x \in \mathbb{R}$ and $f(1) = 2$, then greatest possible integral value of $f(3)$ is

- (1) 5
- (2) 6
- (3) 4
- (4) Data insufficient

20. Let R be the relation on the set of real numbers defined by $a R b$. If $|a - b| \leq 1$. Then R is

- (1) Reflexive and symmetric
- (2) Symmetric only
- (3) Transitive only
- (4) Anti-symmetric only

21. If $f(x) = \begin{cases} x^2 & , x < 2 \\ x^3 + 3x & , x > 2 \\ a & , x = 2 \end{cases}$. Find range of values of 'a' for which $f(x)$ is strictly monotonically increasing at $x = 2$.

- (1) $[0, 14]$
- (2) $[4, 14]$
- (3) $\mathbb{R}$
- (4) $(4, 14)$

22. If the thrice repeated roots of equation $x^4 + ax^3 + bx^2 + cx - 1 = 0$ is 1, then $a + b + 2c$ is equal to

- (1) 0
- (2) 1
- (3) -1
- (4) 2

23. The curves $x^3 - 3xy^2 = a$ and $3x^2y - y^3 = b$, where a and b are constants, cut each other

- (1) at an angle $\frac{\pi}{3}$
- (2) at an angle $\frac{\pi}{4}$
- (3) orthogonally
- (4) at an angle $\frac{\pi}{6}$

(SPACE FOR ROUGH WORK)

24. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 2$, then $\lim_{x \rightarrow 0} [f(x)]$ equals,

(where $[.]$ denotes the greatest integer function)

- (1) 2
- (2) 1
- (3) 0
- (4) limit does not exist

SECTION - II

Straight Objective Type

This section contains 6 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

25. If $\sin A + \sin B = \sqrt{3} (\cos B - \cos A)$, then value of $\sin 3A + \sin 3B$ is equal to

- (1) 0
- (2) 1
- (3) 2
- (4) -1

26. Let $f : \mathbb{R} - \{1, 2, 3\} \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \frac{1}{x-1} + \frac{2}{x-2} + \frac{3}{x-3}. \text{ Then } f(x) \text{ is}$$

- (1) one-one
- (2) many one
- (3) monotonically increasing function
- (4) monotonically decreasing function

27. If $f(x) = 1 - x - x^3$, then all real values of x satisfying the inequality

$$1 - f(x) - f^3(x) > f(1 - 5x) \text{ are}$$

- (1) $(-2, 0) \cup (2, \infty)$
- (2) $(-\infty, -2)$
- (3) $(1, 2)$
- (4) $(0, 2)$

28. If $f(x) = \begin{cases} ax+b & , \quad x \leq 2 \\ x^2-5x+6 & , \quad 2 < x < 3 \\ px^2+qx+1 & , \quad x \geq 3 \end{cases}$ is differentiable

every where, then

- (1) $a = -1, p = -\frac{4}{9}$
- (2) $b = 2, q = \frac{5}{3}$
- (3) $a = 1, b = 2$
- (4) $a = -1, q = -\frac{5}{3}$

(SPACE FOR ROUGH WORK)

29. If $\alpha, \beta, \gamma \in \left(0, \frac{\pi}{2}\right)$ then the value of

$$\frac{\sin(\alpha + \beta + \gamma)}{\sin \alpha + \sin \beta + \sin \gamma} \text{ is}$$

- (1) less than 1
- (2) greater than 1
- (3) equal to 2
- (4) greater than 2

30. If $f(x)$ is continuous, increasing function

such that domain of $g(x) = \sqrt{f(x) - x}$ is $\mathbb{R}$

and $h(x) = \frac{1}{1-x}$, then domain of

$$\phi(x) = \sqrt{f(f(f(x))) - h(h(h(x)))} \text{ is}$$

- (1) $\mathbb{R}$
- (2) $\{0, 1\}$
- (3) $\mathbb{R} - \{0, 1\}$
- (4) $\mathbb{R}^+ - \{1\}$

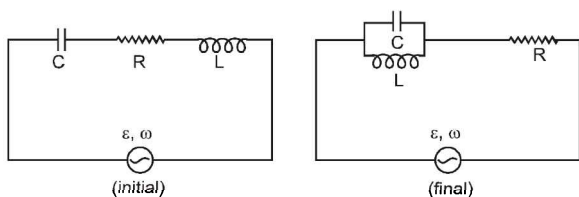
(SPACE FOR ROUGH WORK)

SECTION - I

Straight Objective Type

This section contains 24 multiple choice questions. Each question has choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

31. The r.m.s. value of current flowing through the resistor in a series L–C–R a.c. circuit is $I = \varepsilon / R$. Now the inductor and capacitor are connected in parallel and joined in series with the resistor as shown in figure. The current in the circuit is now. (Symbols have their usual meaning)



- (1) equal to I
- (2) more than I
- (3) less than I
- (4) zero

32. A conductor of length ℓ is placed horizontally perpendicular to a horizontal uniform magnetic field B . Suddenly a charge Q is passed through it and it is found to jump to height h . Choose the correct relation : (Take acceleration due to gravity g)

(1) $h = \frac{B^2 \ell^2 Q^2}{m^2 g}$

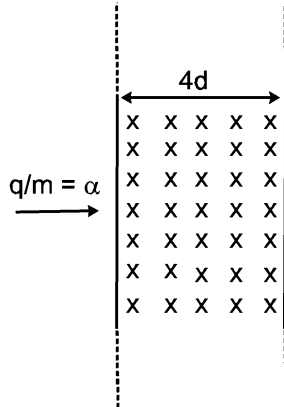
(2) $h = \frac{2B^2 \ell^2 Q^2}{m^2 g}$

(3) $h = \frac{B^2 \ell^2 Q^2}{2m^2 g}$

(4) $h = \frac{2B^2 \ell^2 Q^2}{mg}$

(SPACE FOR ROUGH WORK)

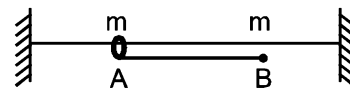
33. If a charged particle of charge to mass ratio $\frac{q}{m} = \alpha$ is entering in a magnetic field of strength B at a speed $v = (2\alpha d)(B)$, then which of the following is correct :



- (1) angle subtended by charged particle at the centre of circular path is 2π .
 (2) the charge will move on a circular path and will come out from magnetic field at a distance $4d$ from the point of insertion.
 (3) the time for which particle will be in the magnetic field is $\frac{2\pi}{\alpha B}$.
 (4) the charged particle will subtend an angle of 90° at the centre of circular path

34. A point charge is moving in clockwise direction in a circle with constant speed. Consider the magnetic field produced by the charge at a point P (not centre of the circle) on the axis of the circle.
 (1) only the magnitude of the magnetic field remains constant
 (2) only the direction of the magnetic field remains constant
 (3) magnetic field is constant in both direction and magnitude
 (4) both magnitude and direction of magnetic field change

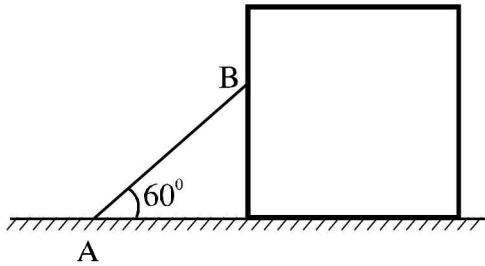
35. A small ring of mass m is connected with a particle of same mass by an ideal string & the whole system is released as shown in figure. Coefficient of friction between ring A and wire is $3/5$, ring A will start sliding when connecting string will make an angle θ with the vertical, then θ will be : (particle is free to move and ring can slide only)



- (1) 30° (2) 45°
 (3) 60° (4) None of these

(SPACE FOR ROUGH WORK)

36. Rod AB is placed against a block which is moving towards right with a speed of 1m/s. If at an instant when the rod makes an angle 60° with the horizontal and end A is sliding towards left with a speed of 1m/s. Then the speed of the point of contact 'B' of the rod is :



- (1) 1m/s
 (2) $\frac{\sqrt{3}}{2}$ m/s
 (3) $\frac{1}{2}$ m/s
 (4) $\frac{\sqrt{7}}{\sqrt{3}}$ m/s

37. A particle starts from rest from the top of an inclined plane and again comes to rest on reaching the bottom most point of incline plane. If the coefficient of friction on some part of inclined plane is $3 \tan \theta$ and rest portion is smooth then the ratio of the rough length of the inclined plane to smooth length would be: (' θ ' is the plane angle)

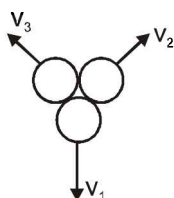
- (1) 3:1
 (2) $\frac{\sin \theta}{\cos^2 \theta}$
 (3) 1 : 2
 (4) $\sqrt{2}:1$

38. If particle move under the effect of superposition of two SHM simultaneously and it is given $x = 10(1 - \cos \pi t)$ m and $y = 10 \sin \pi t$ m : ($\pi = 22/7$) then distance travelled by a particle in $\frac{7}{22}$ sec.

- (1) 10 m
 (2) 20 m
 (3) 70 m
 (4) zero

(SPACE FOR ROUGH WORK)

39. A rocket is projected vertically upwards. It explodes at the topmost point of its trajectory into three identical fragments. One of the fragments comes straight vertical downward in time t_1 while the other two fragments land at a time t_2 after explosion. Then the height at which the explosion occurred in terms of t_1 and t_2 .



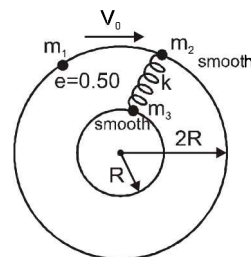
(1) $\frac{gt_1t_2}{2} \left(\frac{2t_1+t_2}{t_1+2t_2} \right)$

(2) $\frac{3gt_1t_2}{2} \left(\frac{2t_1+t_2}{t_1+2t_2} \right)$

(3) $\frac{gt_1t_2}{2} \left(\frac{t_1-2t_2}{2t_1+t_2} \right)$

(4) $\frac{gt_1t_2}{2} \left(\frac{t_1+2t_2}{2t_1+t_2} \right)$

40. Three particles each of mass m can slide on fixed frictionless horizontal circular tracks in the same horizontal plane as shown in the figure. The coefficient of restitution being $e = 0.5$. Assuming that m_2 and m_3 are at rest initially and lie along a radial line before impact and the spring is initially unstretched, then maximum extension in spring in subsequent motion.



(1) $\frac{3}{4} v_0 \sqrt{\frac{m}{k}}$

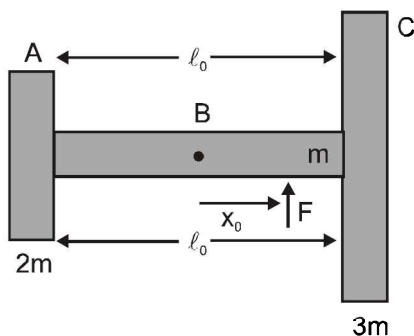
(2) $\frac{3}{4} v_0 \sqrt{\frac{m}{5k}}$

(3) $\frac{3}{4} v_0 \sqrt{\frac{2m}{5k}}$

(4) $\frac{3}{5} v_0 \sqrt{\frac{m}{k}}$

(SPACE FOR ROUGH WORK)

41. A frame made of three uniform rod placed on smooth Horizontal surface. Which is free to move any where, shape of frame is H shown in figure A, B and C part are uniform and point B is mid point of the rod, middle rod is rigidly attached at the midpoint of rod A and rod C. Now a horizontal force F (perpendicular rod B) is applied such as whole system move with pure translational motion. Then x_0 in the term of ℓ_0 is.

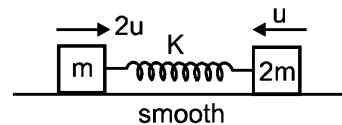


- (1) $\frac{-\ell_0}{6}$
- (2) $\frac{-\ell_0}{12}$
- (3) $\frac{\ell_0}{12}$
- (4) $\frac{7}{12}\ell_0$

42. In a binary star system mass $2m$ and $3m$ revolve about their centre of mass, which is at rest. If radius of star of mass $3m$ is r_0 then the time period of revolution of binary star system is:

- (1) $\frac{5}{2}\pi r_0 \sqrt{\frac{r_0}{2GM}}$ (2) $5\pi r_0 \sqrt{\frac{r_0}{GM}}$
- (3) $5\pi r_0 \sqrt{\frac{r_0}{2GM}}$ (4) None of these

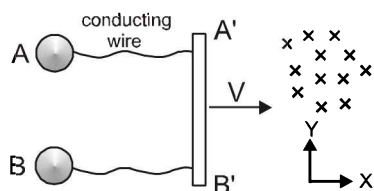
43. At $t = 0$ spring is in its natural length and velocity of two block shown in figure then which of the following option is **incorrect**



- (1) length of spring is maximum at a time $t = 7\pi \sqrt{\frac{3m}{8K}}$
- (2) length of spring is minimum at a time $t = \frac{\pi}{2} \sqrt{\frac{2m}{3K}}$
- (3) Acceleration of the two block is zero at a time $t = \pi \sqrt{\frac{2m}{3K}}$
- (4) Acceleration of the two block never be equal to zero, while acceleration of centre of mass is always zero.

(SPACE FOR ROUGH WORK)

44. Initially A' and B' (ends of the conducting rod) are moving with velocity 'V' and are connected to two stationary balls A and B with conducting wires as shown, after some time (when the rod has reached into the magnetic field) wires are cut:



- (1) A will become positively charged & B will become negatively charged
- (2) A will become negatively charged & B will become positively charged
- (3) If rod is moving in y – direction then A become negatively charged and B positively charged.
- (4) If rod is moving in y – direction then A is positively charged & B is negatively charged.

45. An electromagnetic wave travels in vacuum represented by,

$$\vec{E} = (E_1\hat{i} + E_2\hat{j})\cos(kz - \omega t)$$

then select incorrect option :

- (1) wave travels along positive z-direction
- (2) the associated magnetic field is

$$\vec{B} = \frac{1}{c}(E_1\hat{i} + E_2\hat{j})\cos(kz - \omega t)$$

- (3) The associated magnetic field is

$$\vec{B} = \frac{1}{c}(E_1\hat{i} - E_2\hat{j})\cos(kz - \omega t)$$

- (4) The given electromagnetic wave is plane polarized light

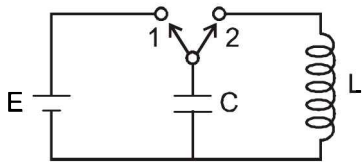
46. In the fraunhofer diffraction from a single slit illuminated by polychromatic light, the first minimum with wavelength λ_1 is found to be coincident with the third minimum at λ_2 . Then the relationship between these two wavelengths is
- (1) $3\lambda_1 = 2\lambda_2$
 - (2) $3\lambda_1 = \lambda_2$
 - (3) $\lambda_1 = 3\lambda_2$
 - (4) $2\lambda_1 = 3\lambda_2$

(SPACE FOR ROUGH WORK)

47. A polaroid is placed at 45° to an incoming light of intensity I_0 . Now the intensity of light passing through polaroid after polarisation would be :

- (1) I_0 (2) $I_0/2$
 (3) $I_0/4$ (4) zero

48. In the given figure switch (1) is connected for long time, then at $t = 0$, switch '1' is opened and switch '2' is connected. Then after time $t = 0$, when inductor coil and capacitor both has same energy first time will be :

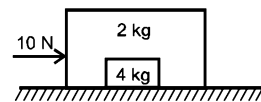


- (1) $\frac{5\pi}{4}\sqrt{LC}$
 (2) $\frac{3\pi}{4}\sqrt{LC}$
 (3) $\frac{\pi}{4}\sqrt{LC}$
 (4) $\frac{\pi}{6}\sqrt{LC}$

49. A car of mass 500 kg is accelerating up an inclined plane 1 in 50 with an acceleration 1 m/s^2 . The power delivered by the engine at an instant is 600 Watts. The speed of the car at this instant is (neglect air resistance and rolling friction) (take $g = 10 \text{ m/s}^2$)

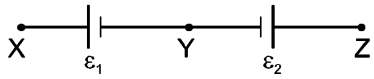
- (1) 2 m/s
 (2) 1 m/s
 (3) 6 m/s
 (4) 10 m/s

50. A small block of mass 4 kg is placed inside a cart of mass 2 kg placed on horizontal surface. A force 10 N is applied on the cart as shown. Neglect friction everywhere. The acceleration of the cart with respect to the ground is :



- (1) $5/3 \text{ m/s}^2$
 (2) 2.5 m/s^2
 (3) 5 m/s^2
 (4) 1 m/s^2

(SPACE FOR ROUGH WORK)

51. A man carries a weight tied at the end of a stick $\frac{3}{2}$ m long, which is placed on his shoulder. He holds the other end of the stick with his hand. In order that the force on his shoulder may be three times the weight hanging at the end, the distance between his hand and the shoulder should be (neglect weight of the stick)
- (1) 0.25 m
 - (2) 0.3 m
 - (3) 0.4 m
 - (4) 0.5 m
52. A capacitor of capacitance C is fully charged by connecting it across a battery of emf V , and then disconnected and reconnected to the same battery but polarity is reversed. Long time after changing polarity, the total heat dissipated in the connecting wires in the process is :
- (1) CV^2
 - (2) $\frac{3}{2}CV^2$
 - (3) $2CV^2$
 - (4) $\frac{5}{2}CV^2$
53. An electric field given by $\vec{E} = 4\hat{i} - 3(y^2 + 2)\hat{j}$ passes Gaussian cube of side 1m placed with one corner at origin such that its sides represents x , y and z axes. The magnitude of net charge enclosed within the cube is
- (1) $4\epsilon_0$
 - (2) $3\epsilon_0$
 - (3) $5\epsilon_0$
 - (4) zero
54. Two cells of emf ϵ_1 and ϵ_2 ($\epsilon_2 < \epsilon_1$) are joined as shown in figure :
- 
- When a potentiometer is connected between X and Y it balances for 300 cm length against ϵ_1 . On connecting the same potentiometer between X and Z it balances for 100 cm length against ϵ_1 and ϵ_2 . Then the ratio $\frac{\epsilon_2}{\epsilon_1}$ is :
- (1) $\frac{1}{3}$
 - (2) $\frac{3}{4}$
 - (3) $\frac{1}{4}$
 - (4) $\frac{2}{3}$

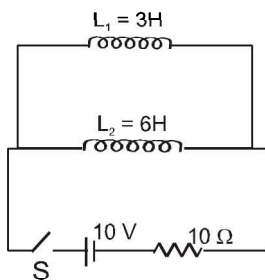
(SPACE FOR ROUGH WORK)

SECTION - II

Straight Objective Type

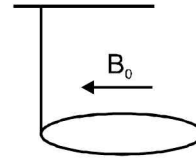
This section contains 6 multiple choice questions. Each question has choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

55. Two inductor coils of self inductance $3H$ and $6H$ respectively are connected with a resistance 10Ω and a battery $10V$ as shown in figure. At $t = 0$ the switch 'S' is closed then the ratio of the magnetic flux at steady state in the inductors L_1 and L_2 is :



- (1) 1 : 2
- (2) 1 : 1
- (3) 2 : 1
- (4) 1 : 3

56. A uniform current carrying ring of mass m and radius R is connected to a massless string as shown. A uniform magnetic field B_0 exist in the region to keep the ring in equilibrium in horizontal position. The current in the ring is: (Take acceleration due to gravity g)

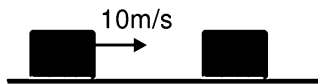


- (1) $\frac{mg}{\pi R B_0}$ clockwise as seen from above the ring
- (2) $\frac{mg}{\pi R B_0}$ anticlockwise as seen from above the ring
- (3) $\frac{mg}{2\pi R B_0}$ clockwise as seen from above the ring
- (4) $\frac{mg}{2\pi R B_0}$ anticlockwise as seen from above the ring

(SPACE FOR ROUGH WORK)

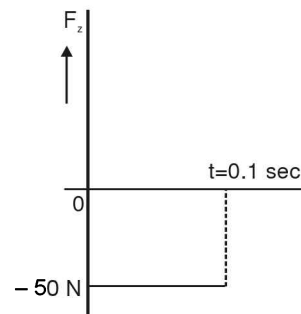
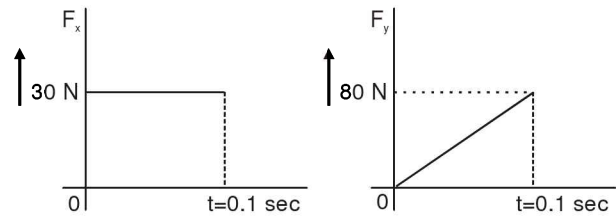
57. In the figure shown a block A moving with velocity 10m/s on a horizontal surface collides with another block B (of same mass) at rest initially. The coefficient of restitution is $\frac{1}{2}$. Neglect friction every

where. The distance between the blocks at 5s after the collision takes place is :



- (1) 20 m
- (2) 10 m
- (3) 25 m
- (4) 30 m

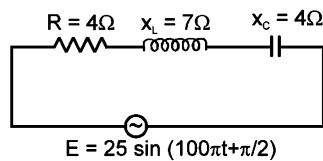
58. A particle is acted upon by a force whose component's variations with time are shown in diagrams. Then the magnitude of change in momentum of the particle in 0.1 sec will be



- (1) $2 \text{ kg } \frac{\text{m}}{\text{sec}}$
- (2) $10 \text{ kg } \frac{\text{m}}{\text{sec}}$
- (3) $12 \text{ kg } \frac{\text{m}}{\text{sec}}$
- (4) $5\sqrt{2} \text{ kg } \frac{\text{m}}{\text{sec}}$

(SPACE FOR ROUGH WORK)

59. In the series LCR circuit as shown in figure, the heat developed in 80 seconds and amplitude of wattless current is :



- (1) 4000 J, 3A
 (2) 8000 J, 3A
 (3) 4000 J, 4A
 (4) 8000 J, 5A
60. Moment of inertia of a uniform solid cone about an axis passing through its centre of gravity and parallel to its base is (m is mass of cone, its height is h and its radius $R = h$)

- (1) $\frac{10}{3} MR^2$
 (2) $\frac{3}{5} MR^2$
 (3) $\frac{3}{10} MR^2$
 (4) $\frac{3}{16} MR^2$

(SPACE FOR ROUGH WORK)

PART- C

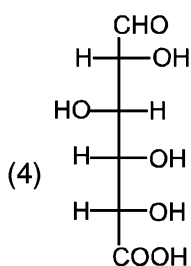
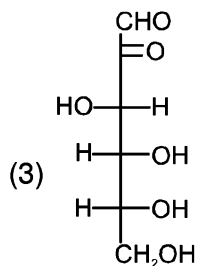
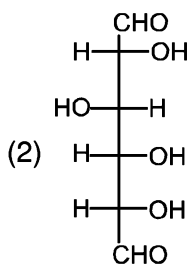
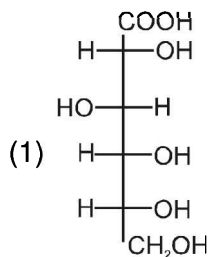
Atomic masses : [H = 1, D = 2, Li = 7, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, Si = 28, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Cu = 63.5, Zn = 65, As = 75, Br = 80, Ag = 108, I = 127, Ba = 137, Hg = 200, Pb = 207]

SECTION - I

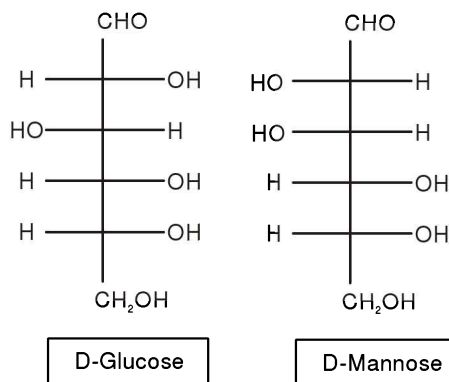
Straight Objective Type

This section contains 24 questions. Each question has five choices (1), (2), (3), and (4) for its answer, out of which **ONLY ONE** is correct.

61. D-Glucose upon treatment with bromine-water gives :



62.



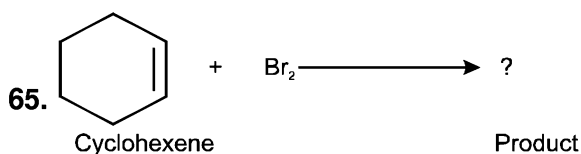
The above structures are related to each other as :

- (1) identical substance
- (2) diastereomers
- (3) enantiomers
- (4) epimers

(SPACE FOR ROUGH WORK)

63. RNA forms loop structure because
- (1) It always contain uracyl instead of thymine
 - (2) of presence of nearby complementary bases
 - (3) all RNAs have to form loop structure to function
 - (4) they are always single stranded

64. A protein attached to carbohydrate moiety is called as :
- (1) Lipoprotein
 - (2) Nucleoprotein
 - (3) Apoprotein
 - (4) Glycoprotein



The correct name of the product obtained is:

- (1) cis-1,2-dibromocyclohexane
- (2) cis-1,4-dibromocyclohexane
- (3) trans-1,2-dibromocyclohexane
- (4) trans-1,4-dibromocyclohexane

66. Which of the following statements is correct?

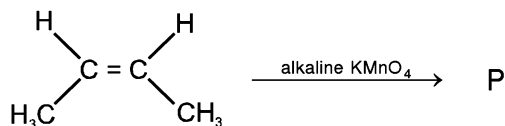
- (1) $-\text{NO}_2$ group activates the benzene ring for attack of electrophile at ortho and para position.
- (2) $-\text{NH}_2$ group activates the benzene ring for attack of electrophile at ortho and para position.
- (3) Both $-\text{NO}_2$ group as well as $-\text{NH}_2$ group activate the benzene ring for attack of electrophile at ortho and para position.
- (4) Neither $-\text{NO}_2$ group nor $-\text{NH}_2$ group activate the benzene ring for attack of electrophile at ortho and para position.

67. The isoelectric point of an amino acid is :

- (1) The pH at which it exists in the acidic form
- (2) The pH at which exists in the basic form
- (3) The pH at which it exists in the Zwitterion form
- (4) The pH which is equal to its pK_a value

(SPACE FOR ROUGH WORK)

68. Which of the following statements is true for the reaction given below ?

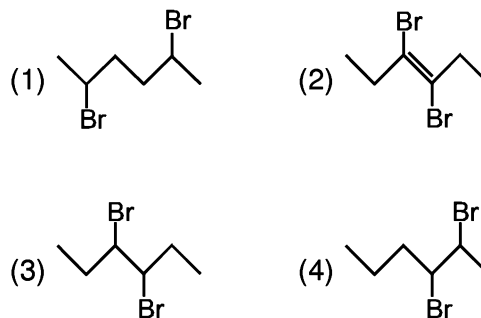


- (1) P is a meso compound 2, 3-butanediol formed by syn addition.
- (2) P is a meso compound 2, 3-butanediol formed by anti addition.
- (3) P is a racemic mixture of d- and l- 2, 3-butanediol formed by anti addition.
- (4) P is a racemic mixture of d- and l- 2, 3-butanediol formed by syn addition.

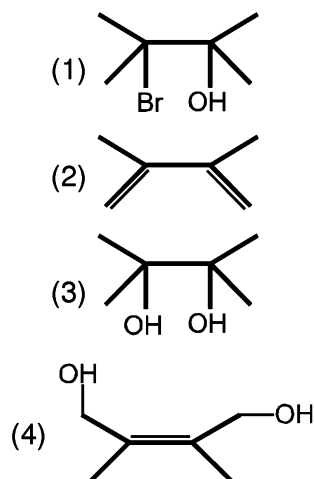
69. Which one of the following combinations is found in DNA?

- (1) Guanine and guanidine
- (2) Guanidine and cytosine
- (3) Guanine and cytosine
- (4) Adenine and guanidine

70. Compound 'X' on heating with Zn dust gives compound 'Y' which on treatment with O_3 followed by reaction with Zn dust gives propionaldehyde. The structure of 'X' is :



71. 2, 3-Dimethylbut-2-ene when reacted with bromine forms a compound which upon heating with alcoholic KOH produce the following major product :



(SPACE FOR ROUGH WORK)

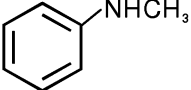
72. The compound which reacts with excess bromine to produce 2, 4, 6-tribromophenol, is :

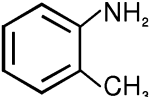
- (1) 1, 3-cyclohexadiene
- (2) 1, 3-cyclohexanedione
- (3) salicylic acid
- (4) cyclohexanone

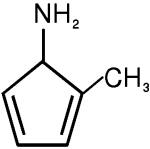
73. Ethyl acetate reacts with $\text{NH}_2\text{NHCONH}_2$ to form

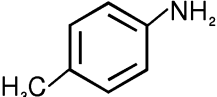
- (1) $\text{CH}_3\text{CONHCONHNH}_2$
- (2) $\text{CH}_3\text{CON}(\text{NH}_2)\text{CONH}_2$
- (3) $\text{CH}_3\text{CONHNHCONH}_2$
- (4) $\text{CH}_3\text{CH}_2\text{NHNHCONH}_2$

74. The compound X ($\text{C}_7\text{H}_9\text{N}$) reacts with benzenesulfonyl chloride to give Y ($\text{C}_{13}\text{H}_{13}\text{NO}_2\text{S}$) which is insoluble in alkali. The compound X is :

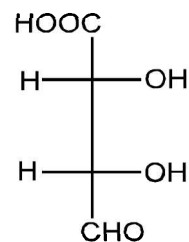
- (1) 

(3) 

(2) 

(4) 

75. The configurations of the carbon atoms C_2 and C_3 in the following compound are respectively



(1) 2R, 3R

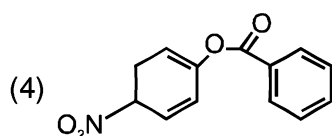
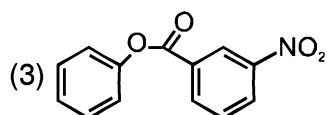
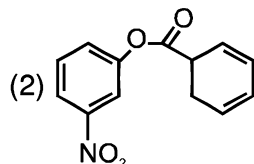
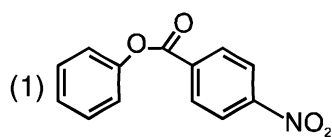
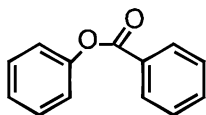
(2) 2S, 3S

(3) 2R, 3S

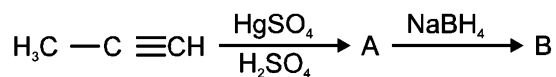
(4) 2S, 3R

(SPACE FOR ROUGH WORK)

76. Major product of mononitration of the following compound is :



77. The product obtained from the following sequence of reactions is



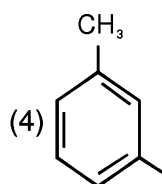
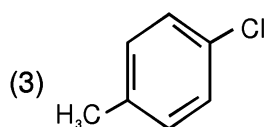
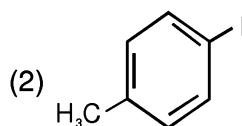
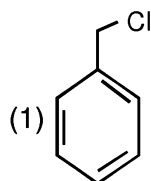
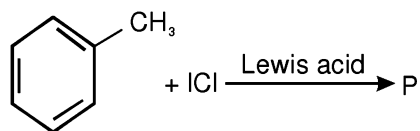
(1) propanal

(2) 2-propanol

(3) 1-propanol

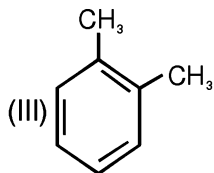
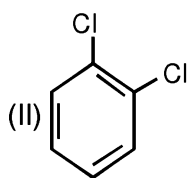
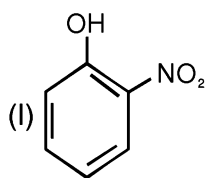
(4) propane

78. The product (P) of the following reaction is :



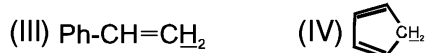
(SPACE FOR ROUGH WORK)

79. The correct order of dipole moment for the following molecules is :



- (1) $I = II = III$ (2) $I < II < III$
 (3) $I > II > III$ (4) $II < III < I$

80. The order of acidities of the H-atoms underlined in the following compounds is in the order :



- (1) $\text{IV} > \text{II} > \text{I} > \text{III}$
 (2) $\text{II} > \text{IV} > \text{III} > \text{I}$
 (3) $\text{III} > \text{IV} > \text{I} > \text{II}$
 (4) $\text{I} > \text{III} > \text{II} > \text{IV}$

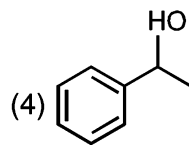
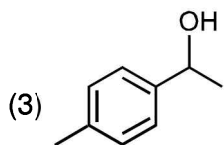
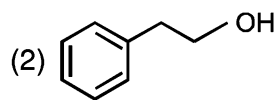
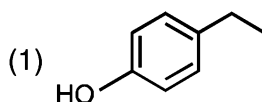
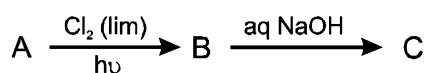
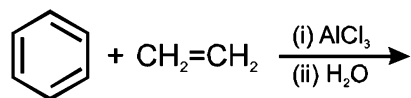
81. The compound that is chiral is :

- (1) 3-methyl-3-hexene
 (2) 4-chloro-1-methylcyclohexane
 (3) 2-phenylpentane
 (4) 1,3-diisopropylbenzene

82. The monomer/s of the following polymer is/are :

- $(-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}(\text{CH}_3)-)_n$
 (1) ethylene
 (2) propylene
 (3) 2-butene
 (4) ethylene + propylene

83. The product (C) of the following sequence of reactions is :



(SPACE FOR ROUGH WORK)

84. The order of reactivity of ammonia with the following compound is :

- (I) $\text{CH}_2=\text{CHBr}$
- (II) $\text{CH}_3-\text{CH}_2-\text{COCl}$
- (III) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{Cl}$
- (IV) $\text{C}(\text{CH}_3)_3\text{C-Br}$

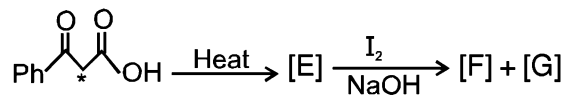
- (1) $\text{IV} > \text{II} > \text{I} > \text{III}$
- (2) $\text{II} > \text{IV} > \text{III} > \text{I}$
- (3) $\text{III} > \text{IV} > \text{II} > \text{I}$
- (4) $\text{I} > \text{IV} > \text{II} > \text{III}$

SECTION - II

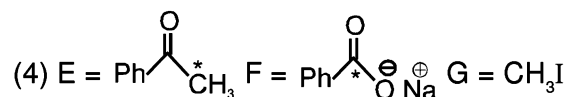
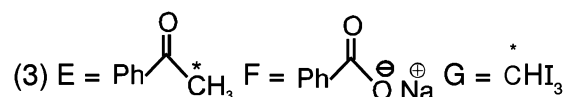
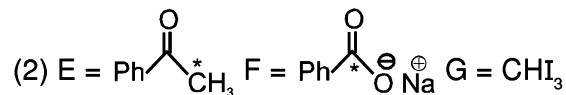
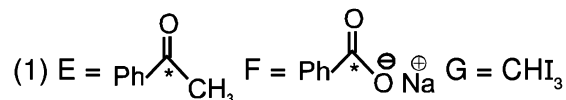
Straight Objective Type

This section contains 6 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

85. In the following reaction sequence, the correct structures of E, F and G are :

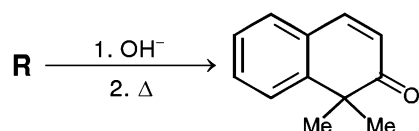
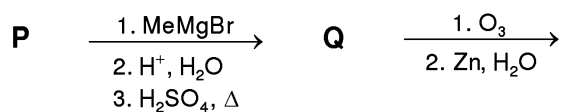


(*implies ^{13}C labelled carbon)

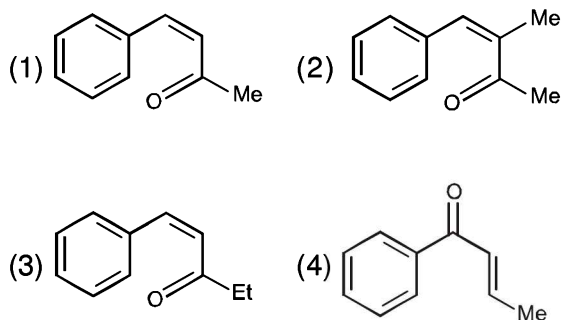


(SPACE FOR ROUGH WORK)

86. A carbonyl compound **P**, which gives positive iodoform test, undergoes reaction with MeMgBr followed by dehydration to give an olefin **Q**. Ozonolysis of **Q** leads to a dicarbonyl compound **R**, which undergoes intramolecular aldol reaction to give predominantly **S**.



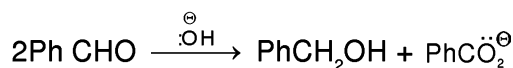
The structure of the carbonyl compound **P** is :



87. Reaction of cyclohexanone with dimethylamine in the presence of catalytic amount of an acid forms a compound if water during the reaction is continuously removed. The compound formed is generally known as

- (1) an amine
- (2) an imine
- (3) an enamine
- (4) a Schiff's base

88. In Cannizzaro reaction given below

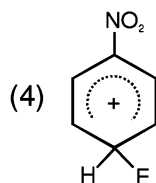
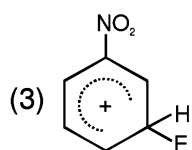
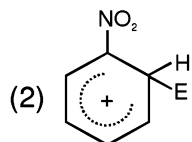
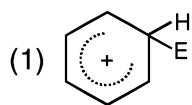


the slowest step is :

- (1) the transfer of hydride to the carbonyl group
- (2) the abstraction of proton from the carboxylic group
- (3) the deprotonation of PhCH_2OH
- (4) the attack of $:\text{OH}^-$ at the carboxyl group

(SPACE FOR ROUGH WORK)

89. The electrophile, E^+ , attacks the benzene ring to generate the intermediate σ -complex. Of the following, which σ -complex is of lowest energy ?



90. Phenol is heated with a solution of mixture of KBr and $KBrO_3$. The major product obtained in the above reaction is :

- (1) 2-Bromophenol
- (2) 3-Bromophenol
- (3) 4-Bromophenol
- (4) 2, 4, 6 -Tribromophenol

(SPACE FOR ROUGH WORK)

MOCK TEST 6

CODE

Q

PREP JEE

PREPARE FOR JEE EXAMINATION WITH US

PREP JEE TEST SERIES (PTS)

JEE MAIN

TIME: 3 HOURS MAX. MARKS: **288**

Duration : 3 Hours

www.prepjee.in

Max. Marks : 270

Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.

IMPORTANT INSTRUCTIONS

1. Immediately fill the particulars on this page of the Test Booklet with Blue / Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **270**.
5. There are three parts in the question paper A, B, C consisting of **Mathematics, Physics** and **Chemistry** having 30 questions. Mathematics & Physics part consists of **Two** sections and Chemistry consists **One Section**.
6. For each question in **Section 1**, you will be awarded **3 marks** if you darken all the bubble(s) corresponding to only the correct answer(s) and **zero mark** if no bubbles are darkened. In all other cases, **(-1) marks will be awarded**.
For each question in **Section 2**, you will be awarded **3 marks** if you darken the bubble corresponding to only the correct answer and **zero mark** if no bubbles are darkened. **No negative** marks will be awarded for incorrect answers in this section.
7. There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 6 above.

Filling the Top-half of the ORS :

- Use only Black ball point pen only for filling the ORS. Do not use Gel / Ink / Felt pen as it might smudge the ORS.**
8. Write your Roll no. in the boxes given at the top left corner of your ORS with black ball point pen. Also, darken the corresponding bubbles with Black ball point pen only. Also fill your roll no on the back side of your ORS in the space provided (if the ORS is both side printed).
 9. Fill your Paper Code as mentioned on the Test Paper and darken the corresponding bubble with Black ball point pen.
 10. If student does not fill his/her roll no. and paper code correctly and properly, then his/her marks will not be displayed and 5 marks will be deducted (paper wise) from the total.
 11. Since it is not possible to erase and correct pen filled bubble, you are advised to be extremely careful while darken the bubble corresponding to your answer.
 12. Neither try to erase / rub / scratch the option nor make the Cross (X) mark on the option once filled. Do not scribble, smudge, cut, tear, or wrinkle the ORS. Do not put any stray marks or whitener anywhere on the ORS.
 13. If there is any discrepancy between the written data and the bubbled data in your ORS, the bubbled data will be taken as final.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

PART-A

SECTION - I

Straight Objective Type

This section contains 24 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. The area bounded by the curves

$$\sqrt{x} + \sqrt{y} = 1 \text{ and } x + y = 1 \text{ is}$$

(1) $\frac{1}{3}$ (2) $\frac{1}{6}$

(3) $\frac{1}{2}$ (4) $\frac{5}{6}$

2. If $a, b, c, d \in \mathbb{R}$ then the equation

$$(x^2 + ax - 3b)(x^2 - cx + b)(x^2 - dx + 2b) = 0 \text{ has}$$

(1) 6 real roots (2) At least 2 real roots

(3) 4 real roots (4) 3 real roots

3. The value of $\int_0^{100\pi} ([\cot^{-1} x] + [\tan^{-1} x]) dx$ is (where $[.]$ denotes greatest integer function)

(1) $100\pi + 1 \cot 1$ (2) $100\pi + 2 \cot 2$

(3) $2 \cot 2$ (4) $100\pi - 2 \cot 2$

4. Let $f(x) = \int_{-2}^x e^{(1+t)^2} dt$ and $g(x) = f(h(x))$ where $h(x)$ is defined for all $x \in \mathbb{R}$. If $g'(2) = e^4$ and $h'(2) = 1$ then absolute value of sum of all possible values of $h(2)$ is

(1) 2

(2) 3

(3) 4

(4) 0

5. $\sum_{j=1}^{21} a_j = 693$, where $a_1, a_2, \dots, a_{21}$ are in A.P.,

then $\sum_{i=0}^{10} a_{2i+1}$ is

(1) 330

(2) 363

(3) 1386

(4) 333

6. $\int \frac{\sin 2x}{\sin 5x \sin 3x} dx =$

(1) $\ln \sin 3x - \ln \sin 5x + c$

(2) $\frac{1}{3} \ln \sin 3x + \frac{1}{5} \ln \sin 5x + c$

(3) $\frac{1}{3} \ln \sin 3x - \frac{1}{5} \ln \sin 5x = c$

(4) $3 \ln \sin 3x - 5 \ln \sin 5x + c$

(SPACE FOR ROUGH WORK)

7. The general solution of differential equation

$$(x^6 y^4 + x^2) dy = (1 - x^5 y^5 - xy) dx, \text{ is}$$

(1) $\ln |x| = xy + \frac{x^4 y^4}{4} + C$

(2) $\ln |y| = xy + \frac{x^4 y^4}{4} + C$

(3) $\ln |x| = xy + \frac{x^5 y^5}{5} + C$

(4) $\ln |y| = xy + \frac{x^5 y^5}{5} + C$

8. A spherical acetone drop evaporates at a rate proportional to its surface area at that instant. The radius of this drop initially is 3mm and after one hour it is found to be 2 mm. If $r(t)$ represents radius of the drop at time 't', Then

(1) $r(t) = 3 - t$

(2) $r(t) = t - 3$

(3) $r(t) = 3 + t^2 - 2t$

(4) $r(t) = 3 - t^3$

9. If $x_1, x_2, x_3, \dots, x_{2008}$ are in HP and

$$\sum_{i=1}^{2007} x_i x_{i+1} = \lambda x_1 x_{2008}, \text{ then } \lambda \text{ is}$$

(1) 2008

(2) 1998

(3) 1863

(4) 2007

10. $\lim_{h \rightarrow 0} \frac{1}{h} \left[\int_a^{x+h} \sin^4 t \, dt - \int_a^x \sin^4 t \, dt \right] =$

(1) $\sin^4 x$

(2) $4 \sin^3 x \cos x$

(3) 0

(4) $\frac{\sin^5 x}{5}$

11. If α, β be the roots of $x^2 - x - 1 = 0$ and

$A_n = \alpha^n + \beta^n$, then A.M. of A_{n-1} and A_n is

(1) $2A_{n+1}$

(2) $(1/2)A_{n+1}$

(3) $2A_{n-2}$

(4) $\frac{1}{2}(A_{n-2})$

12. Solution of the differential equation

$$(2x - 10y^3) \frac{dy}{dx} + y = 0 \text{ is}$$

(1) $(x - 2y^3) \cdot y^2 = c$

(2) $\left(x - \frac{2y^3}{5} \right) y^2 = c$

(3) $(x + 2y^3) \cdot y^2 = c$

(4) $\left(x + \frac{2y^3}{5} \right) y^2 = c$

(SPACE FOR ROUGH WORK)

13. If 2nd, 5th, and 9th terms of an A.P. are in G.P., then the sum of all possible ratios of the first term to the common difference of A.P.

(1) $\frac{9}{8}$

(2) 9

(3) 8

(4) 1

14. If α, β are the roots of the equation

$$ax^2 + bx + c = 0, \text{ then } \frac{\alpha}{a\beta + b} + \frac{\beta}{a\alpha + b} =$$

(1) $\frac{2}{c}$

(2) $-\frac{2}{a}$

(3) $\frac{2}{a}$

(4) $\frac{2}{b}$

15. Let $S_1, S_2, \dots, S_n$ be squares such that for each $n \geq 1$, the length of a side of S_n equals the length of a diagonal of S_{n+1} . If the length of a side of S_1 is 10 cm, then for which of the following values of n is the area of S_n less than 1 sq. cm ?

(1) 7

(2) 8

(3) 6

(4) 5

16. The sum to n terms of the series

$$11 + 103 + 1005 + \dots \text{ is}$$

(1) $\frac{1}{9}(10n - 1) + n^2$ (2) $\frac{1}{9}(10^n - 1) + 2n$

(3) $\frac{10}{9}(10^n - 1) + n^2$ (4) $\frac{10}{9}(10^n - 1) + 2n$

17. The value of the integral $\int_{-\pi/2}^{\pi/2} \log\left(\frac{a - \sin\theta}{a + \sin\theta}\right) d\theta$,

$a > 1$ is -

(1) 0

(2) 1

(3) 2

(4) -2

18. The set of values of p for which both the roots of the quadratic equation,

$$4x^2 - 20px + (25p^2 + 15p - 66) = 0$$
 are less than 2 will be

(1) $(4/5, 2)$

(2) $(2, \infty)$

(3) $(-1, 4/5)$

(4) $(-\infty, -1)$

(SPACE FOR ROUGH WORK)

19. Find the area covered by the curve

$y = \max. \{2 - x, 2, 1 + x\}$ with x-axis from $x = -1$ to $x = 1$ is

(1) $\frac{1}{2}$

(2) $\frac{5}{2}$

(3) $\frac{7}{2}$

(4) $\frac{9}{2}$

20. The value of $\int_0^{\pi/2} x \cot x \, dx$ is

(1) $\pi \ln 2$

(2) $-\pi \ln 2$

(3) $(\pi \ln 2)/2$

(4) $2\pi \ln 2$

21. The degree and order of differential equation of family of all parabolas whose axis is x-axis are respectively

(1) 1, 2

(2) 2, 1

(3) 1, 1

(4) 2, 2

22. The area bounded by parabola $y^2 = x$, straight line $y = 4$ and y-axis is

(1) $\frac{16}{3}$

(2) $\frac{64}{3}$

(3) $7\sqrt{2}$

(4) $\frac{32}{3}$

23. The ratio in which the area bounded by the curves $y^2 = 12x$ and $x^2 = 12y$ is divided by the line $x = 3$ is

(1) 15 : 49

(2) 13 : 480

(3) 13 : 37

(4) 1 : 1

24. The value of $\int_0^{\infty} f(x^n + x^{-n}) \log x \frac{dx}{x}$ is

(1) ∞

(2) 1

(3) -1

(4) 0

(SPACE FOR ROUGH WORK)

SECTION - II
Comprehension Type

This section contains 3 paragraphs. Based on each paragraph, there are 2 questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 25 to 26

If $S = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=0}^{kn} f\left(\frac{r}{n}\right)$, (for $k \in \mathbb{N}$), then

$S = \int_0^k f(x) dx$. In this process $\frac{r}{n}$ transforms to x , $\sum$ transforms to $\int$ and $\frac{1}{n}$ transforms to dx .

25. $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\sin^2 \frac{\pi}{2n} + \sin^2 \frac{2\pi}{2n} + \dots + \sin^2 \frac{n\pi}{2n} \right)$ is

equal to

(1) $\frac{2}{\pi}$ (2) $\frac{\pi}{2}$

(3) $\frac{1}{2}$ (4) 1

26. $\lim_{n \rightarrow \infty} \frac{1}{n^4} \left((n^2 + 1^2)(n^2 + 2^2)(n^2 + 3^2) \dots (n^2 + n^2) \right)^{1/n}$

is equal to

(1) $2 \ln 5 + 2(\tan^{-1} 2) - 4$ (2) $\frac{25e^{2\tan^{-1} 2}}{e^4}$

(3) $\frac{\pi}{2} + (\ln 2) - 2$ (4) $\frac{2e^{\pi/2}}{e^2}$

Paragraph for Question Nos. 27 to 28

A differential equation of the form

$$\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)} \text{ where } f \text{ and } g \text{ are homogeneous}$$

function of x and y , and of the same degree, is called homogeneous differential equation and can be solved easily by putting $y = vx$.

27. The solution of the differential equation

$$\frac{dy}{dx} = \frac{x^2 + xy + y^2}{x^2} \text{ is}$$

(1) $\tan^{-1} \left(\frac{x}{y} \right) = \ln y + c$

(2) $\tan^{-1} \left(\frac{y}{x} \right) = \ln y + c$

(3) $\tan^{-1} \left(\frac{y}{x} \right) = \ln x + c$

(4) $\sin^{-1} \left(\frac{y}{x} \right) = \ln x + c$

28. The solution of the differential equation,

$$\frac{dy}{dx} + \frac{x(x^2 + 3y^2)}{y(y^2 + 3x^2)} = 0 \text{ is}$$

(1) $x^4 + y^4 + x^2y^2 = c$

(2) $x^4 + y^4 + 3x^2y^2 = c$

(3) $x^4 + y^4 + 6x^2y^2 = c$

(4) $x^4 + y^4 + 9x^2y^2 = c$

(SPACE FOR ROUGH WORK)

Paragraph for Question Nos. 29 to 30

We are giving the concept of arithmetic mean of m^{th} power. Let $a, b > 0$ and $a \neq b$ and let m be a real number. Then

$$\frac{a^m + b^m}{2} > \left(\frac{a+b}{2} \right)^m, \text{ if } m \in \mathbb{R} \sim [0, 1]$$

However if $m \in (0, 1)$, then

$$\frac{a^m + b^m}{2} < \left(\frac{a+b}{2} \right)^m$$

Obviously if $m \in \{0, 1\}$, then

$$\frac{a^m + b^m}{2} = \left(\frac{a+b}{2} \right)^m$$

If $a = b$ then equality comes in all cases.

On the basis of above information, answer the following questions :

29. If a, b be positive and $a + b = 1$ ($a \neq b$)

and If $A = \sqrt[3]{a} + \sqrt[3]{b}$ then the correct statement is

(1) $A > 2^{2/3}$

(2) $A = \frac{2^{2/3}}{3}$

(3) $A < 2^{2/3}$

(4) $A = 2^{2/3}$

30. If x, y be positive real numbers such that

$x^2 + y^2 = 8$, then the maximum value of $x + y$ is

(1) 2

(2) 4

(3) 6

(4) 8

(SPACE FOR ROUGH WORK)

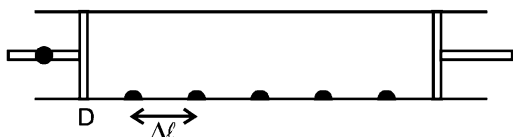
PART-B

SECTION - I

Straight Objective Type

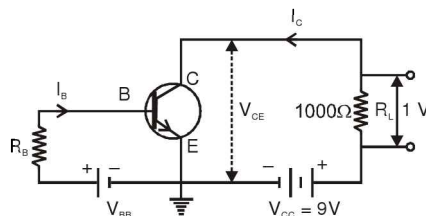
This section contains 24 multiple choice questions. Each question has choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

31. In a Kundt's tube distance between two consecutive heap is $\Delta \ell$ for air, while it is increased by 50% for a gas in the same tube with same resonite if speed of sound in air is $\frac{1000}{3} \text{ m/s}^{-1}$ then speed of sound in gas at same temperature :



- (1) 1500 m/s
- (2) 500 m/s
- (3) 1000 m/s
- (4) To calculate speed in gas degree of freedom and molecular mass of the gas is required

32. An N-P-N transistor is connected in common emitter configuration in which collector supply is 9V and the voltage drop across the load resistance of 1000Ω connected in the collector circuit is 1 V. If current amplification factor is $(25/26)$, If the internal resistance of the transistor is 200Ω , then which of the following options is **incorrect** :



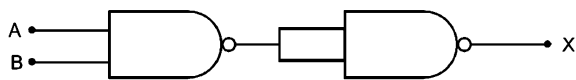
- (1) $V_{CE} = 8 \text{ V}$
 - (2) collector current is 1.0 mA
 - (3) voltage gain $\frac{50}{23}$, and power gain is 4.6
 - (4) emitter current is 2.04 mA
33. A piece of iron is heated in a flame. It first becomes dull red then becomes reddish yellow and finally turns to white hot. The correct explanation for the above observation is possible by using :
- (1) Wein's displacement Law
 - (2) Kirchoff's Law
 - (3) Newton's Law of cooling
 - (4) Stefan's Law

(SPACE FOR ROUGH WORK)

34. In a common emitter (CE) amplifier having a voltage gain G , the transistor used has transconductance 0.03 mho and current gain 25 . If the above transistor is replaced with another one with transconductance 0.02 mho and current gain 20 , the voltage gain will be :
- (1) $1.5 G$ (2) $\frac{1}{3} G$
- (3) $\frac{5}{4} G$ (4) $\frac{2}{3} G$
35. If we study the vibration of a pipe open at both ends, then the following statement is **not** true:
- (1) Fundamental frequency will be inversely proportional to the length of pipe
- (2) All harmonics of the fundamental frequency will be generated
- (3) Pressure change will be maximum at both ends
- (4) Open end will be displacement antinode
36. If photon of energy E and an electron have same energy E (kinetic energy), and De-Broglie wavelength of an electron is λ_e and De-Broglie wavelength of photon is λ_p . Then correct relation between λ_e and λ_p is :
- (1) $\lambda_p \propto \lambda_e$ (2) $\lambda_p \propto \sqrt{\lambda_e}$
- (3) $\lambda_p \propto \frac{1}{\sqrt{\lambda_e}}$ (4) $\lambda_p \propto \lambda_e^2$
37. The half life of a radioactive isotope 'X' is 20 years. It decays into another stable element 'Y'. The two elements 'X' and 'Y' were found to be in the ratio $1 : 7$ in a sample of a given rock. Then age of the rock is estimated to be :
- (1) 60 years
- (2) 80 years
- (3) 100 years
- (4) 40 years
38. During an adiabatic process, the pressure of a gas is found to be proportional to the cube of its temperature. The ratio of $\frac{C_p}{C_v}$ for the gas is :
- (1) 2
- (2) $\frac{5}{3}$
- (3) $\frac{3}{2}$
- (4) $\frac{4}{3}$
39. Vernier constant of the Vernier callipers is :
- (1) least count of Vernier callipers
- (2) value of 1 division of main scale
- (3) value of 1 division of vernier scale
- (4) None of these

(SPACE FOR ROUGH WORK)

40. The output(X) of the logic circuit shown in figure will be :

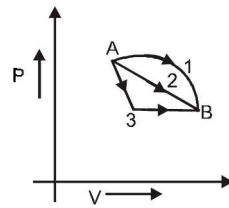


- (1) $X = \overline{A \cdot B}$
 (2) $X = A \cdot B$
 (3) $X = \overline{A + B}$
 (4) $2X = \overline{A \cdot B}$

41. A slab of stone of area 0.36 m^2 and thickness 0.1 m is exposed on the lower surface to steam at 100°C . A block of ice at 0°C rests on the upper surface of the slab. In one hour 4.8 kg of ice is melted. The thermal conductivity of slab is approximately : (Given latent heat of fusion of ice $= 3.36 \times 10^5 \text{ J kg}^{-1}$)

- (1) $1.24 \text{ J/m/s/}^\circ\text{C}$
 (2) $1.29 \text{ J/m/s/}^\circ\text{C}$
 (3) $2.05 \text{ J/m/s/}^\circ\text{C}$
 (4) $1.02 \text{ J/m/s/}^\circ\text{C}$

42. An ideal gas goes from state A to state B via three different processes as indicated in the P-V diagram :



If Q_1, Q_2, Q_3 indicate the heat absorbed by the gas along the three processes and $\Delta U_1, \Delta U_2, \Delta U_3$ indicate the change in internal energy along the three processes respectively, then

- (1) $Q_1 > Q_2 > Q_3$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$
 (2) $Q_3 > Q_2 > Q_1$ and $\Delta U_1 = \Delta U_2 = \Delta U_3$
 (3) $Q_1 = Q_2 = Q_3$ and $\Delta U_1 > \Delta U_2 > \Delta U_3$
 (4) $Q_3 > Q_2 > Q_1$ and $\Delta U_1 > \Delta U_2 > \Delta U_3$

43. The equation of a simple harmonic wave is given by

$$y = 3 \sin \frac{\pi}{2} (50t - x)$$

Where x and y are in meters and t is in seconds. The ratio of maximum particle velocity to the wave velocity is

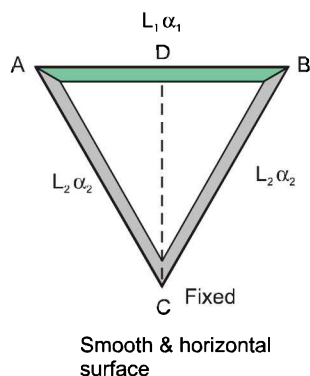
- (1) 2π
 (2) $\frac{3}{2}\pi$
 (3) 3π
 (4) $\frac{2}{3}\pi$

(SPACE FOR ROUGH WORK)

44. The input resistance of a silicon transistor is $100\ \Omega$. Base current is changed by $40\ \mu\text{A}$ which results in a change in collector current by 2mA . This transistor is used as a common emitter amplifier with a load resistance of $4\ \text{K}\Omega$. The voltage gain of the amplifier is :

- (1) 2000 (2) 3000
(3) 4000 (4) 1000

45. Figure shows isosceles triangle frame ABC of two different material shown in figure. Thermal expansion coefficient of the rod ADB is α_1 and for rod ACB is α_2 . End C is fixed and whole system is placed on smooth horizontal surface and D is midpoint of rod AB and CD is perpendicular to the AB. If temperature of the system is increase such as it is found that distance CD remain fixed then.



- (1) $\frac{\ell_1}{\ell_2} = 2\sqrt{\frac{\alpha_2}{\alpha_1}}$ (2) $\frac{\ell_1}{\ell_2} = 2\sqrt{\frac{\alpha_1}{\alpha_2}}$
(3) $\frac{\ell_1}{\ell_2} = \sqrt{\frac{\alpha_1}{\alpha_2}}$ (4) $\frac{\ell_1}{\ell_2} = \sqrt{\frac{\alpha_2}{\alpha_1}}$

46. The positron is the anti-matter counterpart of the electron. It has the same mass as an electron but the opposite charge. A photon is a particle of light. It has no charge, but has non-zero energy and momentum, proportional to the lights frequency. Away from all other matter an electron and positron moving towards each other with equal and opposite velocities.

(1) can annihilate into one photon, conserving both energy and momentum.

(2) can annihilate into one photon because energy and momentum are not conserved in quantum mechanics.

(3) cannot annihilate into one photon because energy cannot be conserved.

(4) cannot annihilate into one photon because momentum cannot be conserved.

47. If refractive index and dielectric constant of the material of a glass slab is $\frac{3}{2}$ and 3 respectively then magnetic permeability of the glass is (In SI system)

- (1) 5.32×10^{-7} (2) 3.94×10^{-7}
(3) 10^{-7} (4) 9.42×10^{-7}

48. Choose the correct statements from the following

(1) Good reflectors are good emitters of thermal radiations

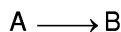
(2) Burns caused by water at 100°C are more severe than those caused by steam at 100°C .

(3) All bodies emit thermal radiations at all temperatures greater than $0\ \text{K}$.

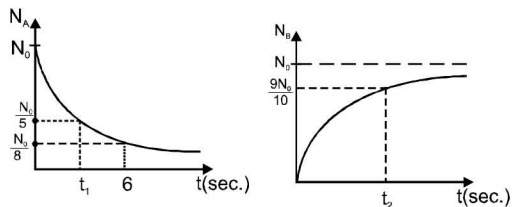
(4) It is possible to construct a heat engine of 100% efficiency.

(SPACE FOR ROUGH WORK)

49. In a decay process A decays to B,



Two graphs of number of nuclei of A and B versus time is given then, which of the following options is **incorrect** :



- (1) $t_2 - t_1 = 4$
- (2) $t_2 - t_1 = 2$
- (3) $t_1 = 2 \log_2 5$
- (4) $t_2 = \log_2 100$

50. Select correct statement regarding waves on a string [all symbols have their usual meanings] :

- (1) Power transfer through any point in standing wave is $\mu v V_p^2$
- (2) Energy is not conserve between consecutive node and antinode
- (3) Two travelling waves of same frequency which are moving in opposite direction must form standing wave.
- (4) Speed of particle in travelling wave is maximum where slope is maximum

51. A source emit sound waves of frequency 1000 Hz. The source moves to the right with a speed of 32 m/s relative to ground. On the right a reflecting surface moves towards left with a speed of 64 m/s relative to ground. The speed of sound in air is 332 m/s, then which of the following options is **incorrect** :

- (1) wavelength of sound in ahead of source is 0.3 m
- (2) number of waves arriving per second which meets the reflected surface is 1320
- (3) speed of reflected wave is 268 m/s
- (4) wavelength of reflected waves is nearly 0.2 m

52. $p = \frac{a - t^2}{bx}$:

p = pressure

x = distance

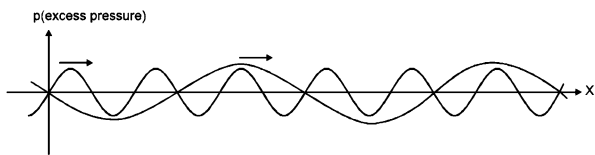
t = time

find the dimensions of $\frac{a}{b}$:

- (1) $[M^2 L T^{-3}]$
- (2) $[M T^{-2}]$
- (3) $[M L^3 T^{-1}]$
- (4) $[L T^{-3}]$

(SPACE FOR ROUGH WORK)

53. Two longitudinal sinusoidal pressure waves one having lower frequency of 2 Hz & both travelling in same direction through the same medium as shown in figure are superimposed. Then the difference in frequency of the two waves is -



- (1) 1 Hz
(2) 2 Hz
(3) 3 Hz
(4) 4 Hz
54. Three samples of a radioactive substance have activity in a ratio 2:5:7, then after two half lives the ratio of their activities will be :
- (1) 2:5:7
(2) 1:3:5
(3) 7:5:2
(4) data insufficient

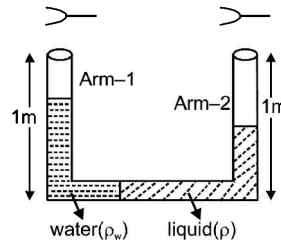
SECTION - II

Comprehension Type

This section contains 3 paragraphs. Based open each paragraph, there are 2 questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Paragraph for Question Nos. 55 to 56

A U-tube, open from both the ends, contains two arms, arm-1 and arm-2 each of having equal cross-section and height of each arm is 1 m. Water of density ρ_w and an unknown liquid of density ρ is filled as shown.



A tuning fork of frequency 300 Hz is vibrated on arm-1, then a loud sound of fundamental tone is produced. If the same tuning fork is vibrated on arm-2, again a loud sound of 1st overtone is produced. ($V_{\text{sound}} = 300 \text{ m/sec.}$, $g = 10 \text{ m/sec}^2$, density of water $\rho_w = 10^3 \text{ kg/m}^3$, atmospheric pressure = 10^5 Pa). Neglect the effect of surface tension and end correction.

55. Density of the unknown liquid (ρ) is :

- (1) $2\rho_w$ (2) $2.5 \rho_w$
(3) $3\rho_w$ (4) $3.5 \rho_w$

56. Now we use a tuning fork of frequency 302 Hz, instead of 300 Hz, with how much velocity should we move the tuning fork, so that resonance is created with the air column in any arm ?

- (1) 2 m/sec. towards the tube
(2) 2 m/sec. away from the tube
(3) 4 m/sec. towards the tube
(4) 4 m/sec. away from the tube

(SPACE FOR ROUGH WORK)

Paragraph for Question Nos. 57 to 58

A message signal of 12 kHz and peak voltage 20 V is used to modulate a carrier wave of frequency 12 MHz and peak voltage 30 V. Then using the above information answer the following questions :

57. The modulation index in the given amplitude modulation is :

(1) $1/5$ (2) $2/3$
(3) $3/2$ (4) 1

58. The lower side band (LSB) frequency of modulated wave is :

(1) 11 MHz to 12 MHz
(2) 10 MHz to 12 MHz
(3) 12.012 MHz to 12 MHz
(4) 11.988 MHz to 12 kHz

59. How many times maximum sound can be observed on line AB :

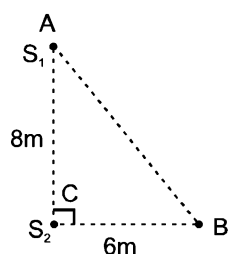
(1) 7
(2) 6
(3) 5
(4) None of these

60. If power of sources are same and equal to 7200π watt. Resultant intensity (in watt/m²) at point B will be :

(1) $(\sqrt{50} - \sqrt{18})^2$
(2) $(\sqrt{30} - \sqrt{18})^2$
(3) $(\sqrt{50} + \sqrt{18})^2$
(4) None of these

Paragraph for Question Nos. 59 to 60

Two coherent point sound sources S_1 and S_2 are placed as shown in the figure. Both are emitting sound of frequency 165 Hz. S_1 is ahead of S_2 in phase by π -radian. (Speed of sound is 330 m/s)



(SPACE FOR ROUGH WORK)

PART- C

Atomic masses : [H = 1, D = 2, Li = 7, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, Si = 28, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Cu = 63.5, Zn = 65, As = 75, Br = 80, Ag = 108, I = 127, Ba = 137, Hg = 200, Pb = 207]

SECTION - I

Straight Objective Type

This section contains 30 questions. Each question has five choices (1), (2), (3), and (4) for its answer, out of which **ONLY ONE** is correct.

- 61.** Which of the following MO has lowest energy for B_2 molecule ?
(1) $\sigma 2p_x$ (2) $\sigma^* 2p_x$
(3) $\pi 2p_y$ (4) $\pi^* 2p_y$
- 62.** Unequal bond lengths are present in :
(1) BF_3
(2) CO_3^{2-} (present in $CaCO_3$)
(3) HNO_3
(4) SO_4^{2-}
- 63.** In which of the following species, each atom carries same number of lone pair of electrons on it?
(1) XeO_4^{2-} (2) XeF_2
(3) XeO_6^{4-} (4) O_3
- 64.** Which of the following has been named correctly ?
(1) $S_2O_3^{2-}$ thiosulphite ion
(2) N_3^- nitride ion
(3) $HAsO_3^{-2}$ monohydrogenarsenite ion
(4) $H_2PO_4^-$ dihydrogenphosphite ion
- 65.** The oxide of an element which is stored under water and which exhibits allotropy is a powerful
(1) dehydrating agent
(2) reducing agent
(3) Fumigant
(4) Oxidizing agent

(SPACE FOR ROUGH WORK)

66. Which of the following statements is INCORRECT ?

- (1) Complex $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Br}_2$ can show both hydrate as well as ionization isomerism.
- (2) Complex $[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})](\text{NO}_3)_3$ can show hydrate isomerism.
- (3) Complex $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_6]$ cannot show coordination isomerism.
- (4) $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)\text{Cl}]\text{Cl}$ can show both ionization as well as linkage isomerism.

67. Usually a disilicate share only one oxygen of silicate unit. But if in the disilicate, two O atoms are shared, then formula of its salt with potassium is :

- (1) $\text{K}_6\text{Si}_2\text{O}_7$
- (2) $\text{K}_4\text{Si}_2\text{O}_6$
- (3) $\text{K}_2\text{Si}_2\text{O}_6$
- (4) $\text{K}_8\text{Si}_2\text{O}_8$

68. Bubbling CO_2 through which of the following will produce a white precipitate :

- (1) NaAlO_2
- (2) Na_2CO_3
- (3) NaOH
- (4) MgSO_4

69. When conc. HNO_3 is treated with P_4O_{10} it forms

- (1) N_2O
- (2) NO
- (3) NO_2
- (4) N_2O_5

70. Which is not correctly matched ?

- (1) Basic strength of oxides
 $\text{Cs}_2\text{O} < \text{Rb}_2\text{O} < \text{K}_2\text{O} < \text{Na}_2\text{O} < \text{Li}_2\text{O}$
 - (2) Stability of peroxides
 $\text{Na}_2\text{O}_2 < \text{K}_2\text{O}_2 < \text{Rb}_2\text{O}_2 < \text{Cs}_2\text{O}_2$
 - (3) Stability of bicarbonates
 $\text{LiHCO}_3 < \text{NaHCO}_3 < \text{KHCO}_3 < \text{RbHCO}_3 < \text{CsHCO}_3$
 - (4) Melting point
 $\text{NaF} < \text{NaCl} < \text{NaBr} < \text{NaI}$
- (1) 1 and 4
 - (2) 1 and 3
 - (3) 1 and 2
 - (4) 2 and 3

71. A substance X when heated with Y produces residue and odourless, gas which turns lime water milky. The silver nitrate is added to residue yellow precipitate is formed. X and Y could be

- | | X | Y | | X | Y |
|-----|----------------|---|-----|--------------------------|---|
| (1) | MnO_2 | S | (2) | NaBrO_3 | C |
| (3) | KI | S | (4) | Na_2SO_4 | C |

72. An aqueous solution contains Al^{3+} & Zn^{2+} both. To this solution NH_4OH is added in excess.

- (1) only $\text{Al}(\text{OH})_3$ will be precipitated
- (2) only $\text{Zn}(\text{OH})_2$ will be precipitated
- (3) both will be precipitated
- (4) no precipitate will appear

(SPACE FOR ROUGH WORK)

73. (I) When copper ore is mixed with silica, in a reverberatory furnace copper matte is produced. The copper matte contains sulphides of copper (II) and iron (II).

(II) Zone refining is based on the principle that impurities are more soluble in molten metal than in solid metal.

(III) In the metallurgy of aluminium, graphite anode is oxidised to carbon monoxide and carbon dioxide.

Correct statements amongs the following are-

- (1) I, II (2) II, III
(3) I, III (4) I, II

74. Among the following statements which is INCORRECT :

(1) In the preparation of compounds of Xe, Bartlett had taken O_2PtF_6 as a base compound because both O_2 and Xe have almost same ionisation enthalpy.

(2) Nitrogen does not show allotropy.

(3) A brown ring is formed in the ring test for NO_3^- ion. It is due to the formation of $[\text{Fe}(\text{H}_2\text{O})_5(\text{NO})]^{2+}$

(4) On heating with concentrated NaOH solution in an inert atmosphere of CO_2 , red phosphorus gives PH_3 gas.

75. Which of the following statement is incorrect.

(1) Be dissolves in alkali forming



(2) LiCl is soluble in pyridine.

(3) Alkaline earth metal ion, because of their much larger charge to size ratio exert a much stronger electrostatic attraction on the oxygen of water molecule surrounding them.

(4) BeF_2 form complex ion with NaF in which Be goes with cation.

76. (l) V_2O_5 , Cr_2O_3 are amphoteric oxides.

(II) Interstitial compounds are very reactive

(III) In its higher oxidation states, manganese forms stable compounds with oxygen and fluorine.

Correct statements amongs the following are-

- (1) I, II (2) II, III
(3) I, III (4) I, II

77. $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ Sodium thiosulphate is used in photography to :

(1) remove reduced silver

(2) remove undecomposed AgBr as soluble silver thiosulphate complex

(3) convert the metallic silver to silver salt

(4) reduce the silver bromide grains to metallic silver

(SPACE FOR ROUGH WORK)

78. H_2S reacts with lead acetate forming a black compound which reacts with H_2O_2 to form another compound. The colour of the compound is :

- (1) pink (2) black
(3) yellow (4) white

79. (I) $[\text{MnCl}_6]^{3-}$, $[\text{FeF}_6]^{3-}$ and $[\text{CoF}_6]^{3-}$ are paramagnetic having four, five and four unpaired electrons respectively.

(II) Valence bond theory gives a quantitative interpretation of the thermodynamic stabilities of coordination compounds.

(III) The crystal field splitting Δ_o , depends upon the field produced by the ligand and charge on the metal ion.

Amongst the following correct statements are :

- (1) I, II (2) I, III
(3) I, II, III (4) II, III

80. When $[\text{K}]^+ [\text{AgF}_4]^-$ is reacted with BF_3 a red solid is formed. This red solid on reaction with Xe forms a brown solid and a fluoride of Xenon. This fluoride is

- (1) XeF_2 (2) XeF_4
(3) XeF_6 (4) XeF_8

81. Which of the following ions does not have S–S linkage ?

- (1) $\text{S}_2\text{O}_8^{2-}$ (2) $\text{S}_2\text{O}_6^{2-}$
(3) $\text{S}_2\text{O}_5^{2-}$ (4) $\text{S}_2\text{O}_3^{2-}$

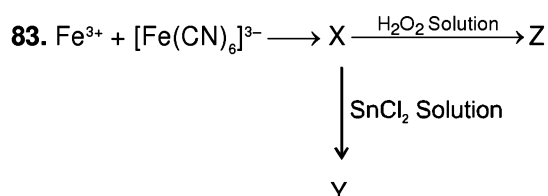
82. Which of the following statement(s) is/are correct ?

(1) Chlorine dioxide (ClO_2) is powerful oxidising agent but bleaching action is lower than Cl_2

(2) ClO_2 in alkaline solution undergoes disproportionation.

(3) SF_4 has a square planar shape with S having two lone pair of electrons.

(4) Sulphur tetrafluoride hydrolysed by water to give SO_2 and HF



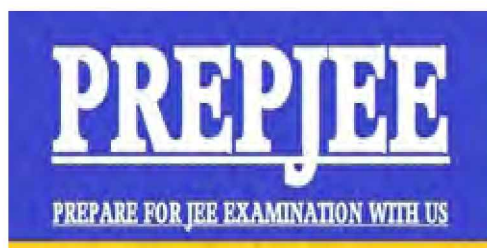
Correct observation is

- (1) X : White Y and Z are same (blue)
(2) X : Brown Y and Z are same (blue)
(3) X : Brown Y is blue, Z white
(4) X : White Y is brown, Z white

(SPACE FOR ROUGH WORK)

84. Which one is incorrect statement among the following ?
- (1) PH_5 , SCl_6 and FeCl_3 do not exist.
 - (2) $p\pi - d\pi$ bond is present in SO_2 molecule.
 - (3) 12 P–O bonds are present in P_4O_6 molecule.
 - (4) Bond angle in SiH_4 less than that in CH_4 .
85. (I) $[\text{Co}(\text{EDTA})]^-$ has two optical isomers.
 (II) $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)_2]^+$ show linkage isomers.
 (III) For $[\text{Pt}(\text{py})(\text{NH}_3)(\text{NO}_2)\text{ClBrI}]$, theoretically fifteen different geometrical isomers are possible.
 (IV) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2] \cdot 2\text{H}_2\text{O}$ can show hydrate as well as ionisation isomerism.
 Amongst the following correct statements are :
- (1) II, III
 - (2) III
 - (3) I, III
 - (4) I, II & III
86. Identify the least stable ion amongst the following :
- (1) Ne^-
 - (2) F^-
 - (3) B^-
 - (4) C^-
87. When haematite ore is burnt in air with coke along with lime at 200°C , the process not only produces steel but also produces an important compound (A), which is useful in making building materials. The compound (A) is -
- (1) SiO_2
 - (2) CaSiO_3
 - (3) FeO
 - (4) Fe_2O_3
88. Which of the following is the **correct** order of ionisation energy :
- (1) $\text{Be}^+ > \text{Be}$
 - (2) $\text{Be} > \text{Be}^+$
 - (3) $\text{C} > \text{Be}$
 - (4) $\text{B} > \text{Be}$
 - (1) 2, 3
 - (2) 3, 4
 - (3) 1, 3
 - (4) None of these
89. In which of the following complex ion, the metal ion will have t_{2g}^6, e_g^0 configuration according to CFT:
- (1) $[\text{FeF}_6]^{3-}$
 - (2) $[\text{Fe}(\text{CN})_6]^{3-}$
 - (3) $[\text{Fe}(\text{CN})_6]^{4-}$
 - (4) None of these
90. Ammonia, on reaction with hypochlorite anion can form :
- (1) NO
 - (2) NH_4Cl
 - (3) N_2H_4
 - (4) HNO_2

(SPACE FOR ROUGH WORK)



JEE MAIN TEST SERIES
JMTS
TARGET : JEE (MAIN)

PAPER

P

COURSE : PREP JEE JMTS

MOCK TEST 7

IMPORTANT INSTRUCTIONS

1. Immediately fill the particulars on this page of the Test Booklet with Blue / Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **360**.
5. There are three parts in the question paper A, B, C consisting of **Mathematics, Physics and Chemistry** having 30 questions. Mathematics & Physics part consists of **Two** sections and Chemistry consists of **One** section.
6. For each question you will be awarded **4 marks** if you darken all the bubble(s) corresponding to only the correct answer(s) and **zero mark** if no bubbles are darkened. In all other cases, **$\frac{1}{4}$ (one fourth (1)) marks will be deducted.**
7. There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 6 above.

Filling the Top-half of the ORS :

Use only Black ball point pen only for filling the ORS. Do not use Gel / Ink / Felt pen as it might smudge the ORS.

8. Write your Roll no. in the boxes given at the top left corner of your ORS with black ball point pen. Also, darken the corresponding bubbles with Black ball point pen only. Also fill your roll no on the back side of your ORS in the space provided (if the ORS is both side printed).
9. Fill your Paper Code as mentioned on the Test Paper and darken the corresponding bubble with Black ball point pen.
10. If student does not fill his/her roll no. and paper code correctly and properly, then his/her marks will not be displayed and 5 marks will be deducted (paper wise) from the total.
11. Since it is not possible to erase and correct pen filled bubble, you are advised to be extremely careful while darken the bubble corresponding to your answer.
12. Neither try to erase / rub / scratch the option nor make the Cross (X) mark on the option once filled. Do not scribble, smudge, cut, tear, or wrinkle the ORS. Do not put any stray marks or whitener anywhere on the ORS.
13. If there is any discrepancy between the written data and the bubbled data in your ORS, the bubbled data will be taken as final.

PART - A

SECTION - I

Straight Objective Type

This section contains 27 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. Let a, b, c are three Arithmetic means between two numbers such that $a+b+c=15$ and p, q, r be Harmonic means between same numbers such that $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = \frac{5}{3}$, then the numbers are

- (1) 3, 10 (2) 3, 12
(3) 3, 15 (4) 9, 1

2. If $(1-p)(1+3x+9x^2+27x^3+81x^4+243x^5) = 1-p^6$, $p \neq 1$, then the value of $\frac{p}{x}$ may be equal to

- (1) $\frac{1}{3}$ (2) 3
(3) $\frac{1}{2}$ (4) 2

3. $\int \frac{\sec x}{a+b \tan x} dx$ is equal to

- (1) $\frac{1}{\sqrt{a^2+b^2}} \ln \tan \frac{x}{2} + C$
(2) $\frac{1}{\sqrt{a^2+b^2}} \ln \left(\tan \left(\frac{x + \tan^{-1}\left(\frac{a}{b}\right)}{2} \right) \right) + C$
(3) $\frac{1}{\sqrt{a^2+b^2}} \ln \left(\tan \left(\frac{x + \tan^{-1}\left(\frac{b}{a}\right)}{2} \right) \right) + C$
(4) $\frac{1}{\sqrt{a^2+b^2}} \ln \sin \frac{x}{2} + C$

4. If the roots of the equation $x^2 + px + c = 0$ are 2, -2 and the roots of the equation $x^2 + bx + q = 0$ are -1, -2, then the roots of the equation $x^2 + bx + c = 0$ are
(1) -3, -2 (2) -1, 4
(3) 1, -4 (4) -2, 3

5. Let $R = \{(x, y) : x^2 + y^2 = 1, x, y, \in R\}$ be a relation in R then the relation R
(1) reflexive
(2) symmetric
(3) transitive
(4) equivalence

Space for Rough Work

6. At the foot of a mountain the elevation of its peak is found to be $\frac{\pi}{4}$. After ascending 'h' m toward the mountain up a slope of $\frac{\pi}{6}$ inclination, the elevation is found to be $\frac{\pi}{3}$. Height of the mountain is

- (1) $\frac{h}{2}(\sqrt{3}+1)m$ (2) $h(\sqrt{3}+1)m$
 (3) $\frac{h}{2}(\sqrt{3}-1)m$ (4) $h(\sqrt{3}-1)m$

7. In how many ways can five different examination papers of maths, physics, chemistry, computer and physical education be arranged so that physics and chemistry papers never come together

- (1) 31 (2) 48
 (3) 60 (4) 72

8. Let P and Q be two statements, then $\sim (\sim P \wedge Q) \wedge (P \vee Q)$ is logically equivalent to

- (1) Q (2) P
 (3) $P \vee Q$ (4) $P \wedge Q$

9. The probabilities of different faces of biased dice to appear are as follows

Face number

1 2 3 4 5 6

Probability

0.1 0.32 0.21 0.15 0.05 0.17

The dice is thrown and it is known that either the face number 1 or 2 will appear. Then, the probability of the face number 1 to appear is

- (1) $\frac{5}{21}$ (2) $\frac{5}{12}$
 (3) $\frac{7}{23}$ (4) $\frac{3}{10}$

10. Set of all values of m for which two points P & Q lie on the line $y = mx + 8$ so that

$\angle APB = \angle AQB = \frac{\pi}{2}$ where $A \equiv (-4, 0)$,

$B \equiv (4, 0)$ is

- (1) $(-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty) - \{-2, 2\}$
 (2) $[-\sqrt{3}, \sqrt{3}] - \{-2, 2\}$
 (3) $(-\infty, -1) \cup (1, \infty) - \{-2, 2\}$
 (4) none of these

11. The set of all values of a for which equation

$\frac{x^2}{8-a} + \frac{y^2}{a-2} = 1$ will represent an ellipse, is

- (1) (1, 4) (2) $(-\infty, 2) \cup (8, \infty)$
 (3) $(2, 8) - \{5\}$ (4) (2, 8)

Space for Rough Work

12. Consider the following statements :

S_1 : For an odd function $f(x)$, graph of $y = f(x)$ always passes through origin.

S_2 : If f and g are two bijective function then $f(g(x))$ is also bijective.

S_3 : All points of intersection of $y = f(x)$ and $y = f^{-1}(x)$ lies on $y = x$ only.

State, in order, whether S_1, S_2, S_3, S_4 are true or false

- (1) T F T (2) T T F
(3) F T T (4) F F F

13. The equation of the plane containing the line $\vec{r} = \vec{a} + \lambda \vec{b}$ and perpendicular to the plane $\vec{r} \cdot \vec{n} = d$ is

- (1) $(\vec{r} - \vec{b}) \cdot (\vec{n} \times \vec{a}) = 0$
(2) $(\vec{r} - \vec{a}) \cdot (\vec{n} \times (\vec{a} \times \vec{b})) = 0$
(3) $(\vec{r} - \vec{a}) \cdot (\vec{n} \times \vec{b}) = 0$
(4) $(\vec{r} - \vec{b}) \cdot (\vec{n} \times (\vec{a} \times \vec{b})) = 0$

14. The distance between the lines $\vec{r} = \hat{i} + 2\hat{j} - 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$, $\lambda \in \mathbb{R}$ and $\vec{r} = 3\hat{i} + 3\hat{j} - 5\hat{k} + \mu(2\hat{i} + 3\hat{j} + 6\hat{k})$, $\mu \in \mathbb{R}$ is

- (1) $\frac{\sqrt{293}}{14}$ (2) $\frac{\sqrt{293}}{7}$
(3) $\frac{\sqrt{193}}{7}$ (4) $\frac{\sqrt{193}}{14}$

15. The value of 'a' for which all extremums of function $f(x) = x^3 - 3ax^2 + 3(a^2 - 1)x + 1$, lie in the interval $(-2, 4)$ is

- (1) (3, 4) (2) (-1, 3)
(3) (-3, -1) (4) none of these

16. If line $2x - 3y - k = 0$ is normal to parabola $y^2 = 4x$, then k is equal to

- (1) $\frac{22}{9}$ (2) $\frac{2}{3}$
(3) $\frac{44}{9}$ (4) $\frac{4}{27}$

17. The value of $\lim_{x \rightarrow \infty} \frac{\cot^{-1}(x^{-a} \log_a x)}{\sec^{-1}(a^x \log_x a)}$, ($a > 1$) is equal to

- (1) 0 (2) $\pi/2$
(3) 1 (4) does not exist

18. Orthocentre of the triangle whose sides are given by the equations $x - 2y + 3 = 0$, $2x - 3y + 5 = 0$ and $4x + 2y + 7 = 0$ is

- (1) $\left(-2, \frac{1}{2}\right)$ (2) (2, 3)
(3) $\left(-2, -\frac{1}{2}\right)$ (4) (1, 1)

Space for Rough Work

19. The value of ' λ ' so that the system of equations

$$x + 2y + 3 = 0,$$

$$3x + 4y + 4 = 0,$$

$$7x - 10y + \lambda = 0,$$

is consistent, is

- (1) -39 (2) 20
(3) -20 (4) none of these

20. Range of $y = \ln x + \sin^{-1} x$ is

- (1) $(-\infty, \infty)$ (2) $\left(0, \frac{\pi}{2}\right]$
(3) $(0, \infty)$ (4) $\left(-\infty, \frac{\pi}{2}\right]$

21. Complete solution set of the inequation

$$\sec^{-1} x \geq \frac{1}{2} \text{ is}$$

- (1) $\left[\sec \frac{1}{2}, \infty\right)$
(2) $(-\infty, -1] \cup \left[\sec \frac{1}{2}, \infty\right)$
(3) $[0, \pi]$
(4) $\left[\frac{\pi}{3}, \pi\right]$

22. The slope of the tangent to curve

$$y = \int_x^{x^2} \cos^{-1} t^2 dt \text{ at } x = \frac{1}{\sqrt[4]{2}} \text{ is}$$

- (1) $\left(\frac{\sqrt[4]{8}}{2} - \frac{3}{4}\right)\pi$
(2) $\left(\frac{\sqrt[4]{8}}{3} - \frac{1}{4}\right)\pi$
(3) $\left(\frac{\sqrt[5]{8}}{4} - \frac{1}{3}\right)\pi$

(4) none of these

23. The mean of 10 numbers is 12.5 ; the mean of the first six numbers is 15 and the last five numbers is 10. The sixth number is :

- (1) 15 (2) 12
(3) 18 (4) None of these

24. A solution of the differential equation

$$\left(\frac{dy}{dx}\right)^2 - \frac{dy}{dx} (e^x + e^{-x}) + 1 = 0 \text{ is given by}$$

- (1) $y + e^{-x} = C$
(2) $y - e^{-x} = C$
(3) $y + e^x = C$
(4) $y - e^{2x} = C$

Space for Rough Work

25. Let $f(x) = \begin{cases} \sin 2x & , 0 < x \leq \pi/6 \\ ax + b & , \frac{\pi}{6} < x < 1 \end{cases}$ continuous

and derivable then

(1) $a = 1, b = \frac{1}{\sqrt{2}} + \frac{\pi}{6}$

(2) $a = b = \frac{1}{\sqrt{2}}$

(3) $a = 1, b = \frac{\sqrt{3}}{2} - \frac{\pi}{6}$

(4) none of these

26. The area of the region bounded by the curves $y = |x-1|$ and $y = 3 - |x|$ is

(1) 2 sq. unit

(2) 3 sq. unit

(3) 4 sq. unit

(4) 6 sq. unit

27. If ${}^nC_0 - {}^nC_1 + {}^nC_2 - \dots + (-1)^r {}^nC_r = 28$, then n may be equal to-

(1) 9

(2) 8

(3) 5

(4) 6

SECTION - II

Reasoning Type

This section contains 3 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

28. STATEMENT-1 :

$\text{Arg}(2 + 3i) + \text{Arg}(2 - 3i) = 0$ ($\text{Arg } z$ stands for principal argument of z)

STATEMENT-2 :

$\text{Arg } z + \text{Arg } \bar{z} = 0, z = x + iy, \forall x, y \in \mathbb{R}$

($\text{Arg } z$ stands for principal argument of z)

(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True

Space for Rough Work

29. Statement-1 : $f(x) = x + \cos x$ is strictly increasing.

Statement-2 : If $f(x)$ is strictly increasing, then $f'(x)$ may vanish at some finite number of points.

(1) Statement -1 is True, Statement -2 is True ; Statement -2 is a correct explanation for Statement -1

(2) Statement-1 is True, Statement-2 is True ; Statement-2 is **NOT** a correct explanation for Statement-1

(3) Statement -1 is True, Statement -2 is False

(4) Statement -1 is False, Statement -2 is True

30. STATEMENT - 1 : If the roots of equation $ax^4 + bx^3 + cx^2 + dx + e = 0$ are in H.P then the roots of the equation $ex^4 + dx^3 + cx^2 + bx + a = 0$ are in A.P.

STATEMENT- 2 : The equation having reciprocals of the roots of the equation $f(x) = 0$, as roots is $f\left(\frac{1}{x}\right) = 0$

(1) Statement -1 is True, Statement -2 is True ; Statement -2 is a correct explanation for Statement -1

(2) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1

(3) Statement -1 is True, Statement -2 is False

(4) Statement -1 is False, Statement -2 is True

Space for Rough Work

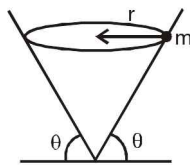
PART – B

SECTION - I

Straight Objective Type

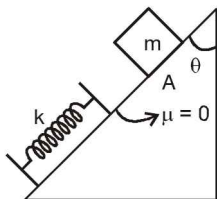
This section contains 28 multiple choice questions. Each question has choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

31. A ball of mass 'm' is rotating in a circle of radius r with speed v inside a smooth cone as shown in figure. Let N be the normal reaction on the ball by the cone, then choose the **incorrect** option



- (1) $N = mg \cos \theta$ (2) $g \sin \theta = \frac{v^2}{r} \cos \theta$
 (3) $N \sin \theta - \frac{mv^2}{r} = 0$ (4) none of these

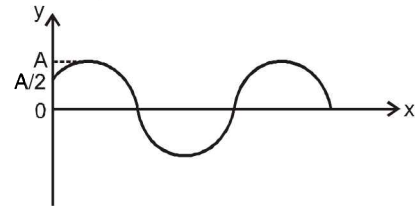
32. A block of mass 'm' is released from rest at point A. The compression in spring, when the speed of block is maximum



- (1) $\frac{mg \sin \theta}{k}$ (2) $\frac{2mg \sin \theta}{k}$
 (3) $\frac{mg \cos \theta}{k}$ (4) $\frac{mg}{k}$

33. Here given snap shot of a progressive wave at $t = \frac{T}{6}$ with time period = T. Then the equation of the wave if wave is going in +ve x-direction and if wave is going in -ve x-direction will be respectively.

(Here, $T = \frac{2\pi}{\omega}$)



- (1) $y = A \sin \left(kx - \omega t + \frac{\pi}{2} \right)$, $y = A \sin \left(kx + \omega t - \frac{\pi}{6} \right)$
 (2) $y = A \cos (kx - \omega t)$, $y = A \sin (kx + \omega t)$
 (3) $y = A \sin (kx - \omega t)$, $y = A \sin \left(kx + \omega t - \frac{\pi}{6} \right)$
 (4) $y = A \sin \left(kx - \omega t + \frac{\pi}{2} \right)$, $y = A \sin \left(kx + \omega t + \frac{\pi}{6} \right)$

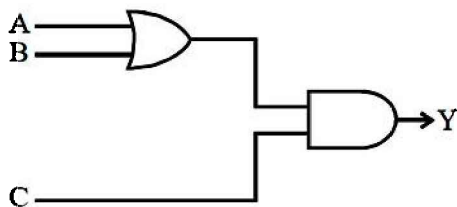
34. A ring of radius R is having two charges q and 2q distributed on its two half parts. The electric potential at a point on its axis at a distance $2\sqrt{2}R$ from its centre is :

- (1) $\frac{3kq}{R}$
 (2) $\frac{kq}{3R}$
 (3) $\frac{kq}{R}$
 (4) $\frac{kq}{\sqrt{3}R}$

Space for Rough Work

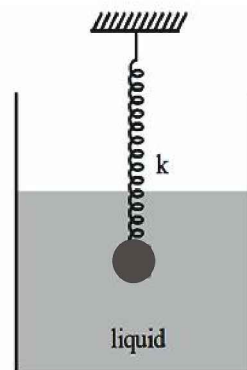
35. In frequency modulated wave :
- (1) frequency varies with time
 - (2) amplitude varies with time
 - (3) both frequency and amplitude vary with time
 - (4) both frequency and amplitude are constant

36. To get output $Y = F$ (False) from circuit of figure. The input must not be :

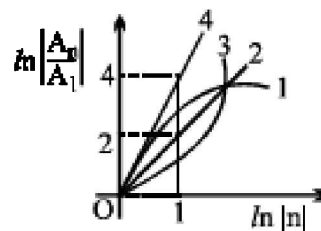


- (1) 110
 - (2) 100
 - (3) 101
 - (4) 110
37. Light of wavelength $6000\text{\AA}$ is incident normally on a slit of width $24 \times 10^{-5} \text{ cm}$. Find out the angular position of second minimum from central maximum ?
- (1) 15°
 - (2) 30°
 - (3) 45°
 - (4) 60°

38. Viscous force acting on a sphere falling in a liquid of coefficient of viscosity η is given by $F_{\text{viscous}} = 6\pi\eta rv$. If the spring constant of the system shown in the figure is $300\pi \text{ N/m}$ and mass of the sphere is 2 kg , the ratio of natural frequency and damping ratio is given by :



- (1) ηr
 - (2) $100 \eta r$
 - (3) $\frac{100}{\eta r}$
 - (4) $\frac{100}{\pi \eta r}$
39. The figure shows a graph between $\ln \left| \frac{A_n}{A_1} \right|$ and $\ln |n|$, where A_n is the area enclosed by the n^{th} orbit in a hydrogen like atom. The correct curve is :



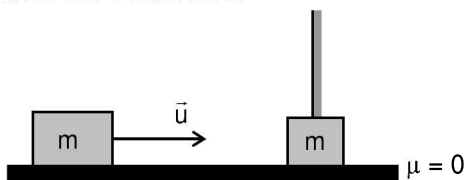
- (1) 4
- (2) 3
- (3) 2
- (4) 1

Space for Rough Work

40. A plane mirror is placed in y-z plane facing towards negative x-axis. The mirror is moving parallel to y-axis with a speed of 2 cm/s. A point object p move is moving in front of the mirror with a velocity $(1 \text{ cm/s}) \hat{i} + (1 \text{ cm/s}) \hat{j}$. Then which of the following statements is **incorrect** ?

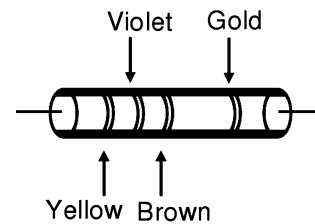
- (1) The velocity of image is $(-\hat{i} + \hat{j}) \text{ cm/s}$
- (2) The velocity of image with respect to mirror is $-(\hat{i} + \hat{j}) \text{ cm/s}$
- (3) The velocity of image with respect to object is $-(2\hat{i}) \text{ cm/s}$
- (4) The unit vector in the direction of reflected ray is $\frac{(\hat{i} + \hat{j})}{\sqrt{2}}$

41. An object of mass m is moving with velocity $\vec{u}$ towards a plane mirror kept on a stand as shown in the figure. The mass of the mirror and stand system is m . A head-on elastic collision takes place between the object and the mirror stand, the velocity of image before and after the collision is



- (1) $\vec{u}, 2\vec{u}$
- (2) $-\vec{u}, -2\vec{u}$
- (3) $-\vec{u}, 2\vec{u}$
- (4) $\vec{u}, -2\vec{u}$

42. What is resistance of following resistor ?



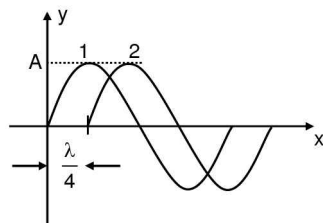
- (1) $47 \pm 5\%$
- (2) $470 \pm 5\%$
- (3) $57 \pm 5\%$
- (4) $47 \pm 10\%$

43. Magnetic moments of two identical magnets are M and $2M$ respectively. Both are combined in such a way that their similar poles are same side. The time period in this is case ' T_1 '. If polarity of one of the magnets is reversed its period becomes ' T_2 ' then find out ratio of their time periods T_1/T_2 .

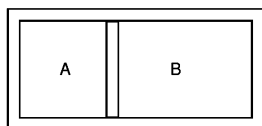
- (1) $\sqrt{3}$
- (2) $\frac{1}{\sqrt{3}}$
- (3) 1
- (4) $\frac{1}{3}$

Space for Rough Work

44. In the given figure two identical waves each of intensity I_0 ; 1 and 2 are superimposed. The resulting intensity is :



- (1) I_0 (2) $2\sqrt{2}I_0$
 (3) $4I_0$ (4) $2I_0$
45. A piston can slide without friction inside a horizontal cylindrical vessel, which contains an ideal monoatomic gas. The piston and the cylinder both are made of a perfect heat insulating material. Initially, the piston is in equilibrium and divides the cylindrical into two part A and B, which are not necessarily equal. The temperatures of the gases in both the parts are equal. Now, the piston is held in its initial position and the gas in part A is supplied some amount of heat, then the piston is released. What will the piston do in its subsequent motion.



- (1) It will execute oscillatory motion of definite amplitude.
 (2) It will stop exactly at middle of the cylinder after several oscillations of decreasing amplitude.
 (3) It will stop exactly where it has started after several oscillations of decreasing amplitude.
 (4) Certainly, it will execute oscillations of decreasing amplitude but where it will finally stop depends on the amount of heat supplied.

46. A radioactive isotope is being formed at a constant rate K . At $t = 0$, the number of active nuclei is N_0 . The decay constant of isotope is λ . The number of active nuclei.
- (1) first increase then decreases
 (2) goes on increasing
 (3) goes on decreasing
 (4) is $\frac{K}{\lambda}$ after a time $t \gg \frac{1}{\lambda}$

47. Two geometrically identical cycle wheels have different number of spokes connecting centre to rim. One is having 20 spokes and other having only 10 (the rim and the spokes are resistanceless). A resistance of value R is connected between centre and rim in both the wheels. If same magnetic field is applied perpendicular to the plane of these wheels and both wheels are rotating with the same angular speed, then the current in R will be :
- (1) double in first wheel than in the second wheel
 (2) four times in first wheel than in the second wheel
 (3) will be double in second wheel than that of the first wheel
 (4) will be equal in both these wheels

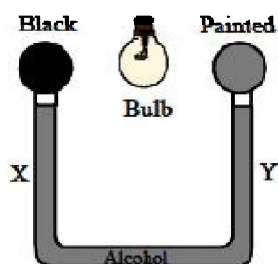
48. In Searle's apparatus, when experimental wire is loaded and unloaded, the air bubble in spirit level gets shifted :
- (1) towards reference wire while loading & experimental wire while unloading
 (2) towards experimental wire while loading & towards reference wire while unloading
 (3) towards experimental wire, both the times, during loading & unloading
 (4) towards reference wire, both the times during loading & unloading

Space for Rough Work

49. A column of liquid of length L is in a uniform capillary tube. The temperature of the tube and column of liquid is raised by ΔT . If γ be the coefficient of volume expansion of the liquid and α be the coefficient of linear expansion of the material of the tube, then the increase ΔL in the length of the column will be.

(1) $\Delta L = L(\gamma - \alpha)\Delta T$ (2) $\Delta L = L(\gamma - 2\alpha)\Delta T$
 (3) $\Delta L = L(\gamma - 3\alpha)\Delta T$ (4) $\Delta L = L\gamma\Delta T$

50. The following figure shows two air-filled bulb connected by a U-tube partly filled with alcohol. What happened to the level of alcohol in the limbs X and Y when an electric bulb plated midway between the bulb is lighted ?



- (1) The level of alcohol in limb X falls while that in limb Y rises
 (2) The level of alcohol in limb X rises while that in limb Y falls
 (3) The level of alcohol falls in both limbs
 (4) The level of alcohol rises in both limbs

51. Consider what happens when you jump up in the air. Which of the following is the most accurate statement ?

- (1) You are able to spring up because the earth exerts a force upward on you which is stronger than downward force you exert on the earth
 (2) Since the ground is stationary, it cannot exert the upward force necessary to propel you into the air.
 (3) When you push down on the earth with a force greater than your weight, the earth will push back with the same magnitude force and thus propel you into the air
 (4) When you jump up the earth exerts a force F_1 on you and you exert a force F_2 on the earth. You go up because $F_1 > F_2$, and this is so because ratio of F_1 to F_2 is earth's mass ratio to your mass

52. The temperature at the bottom of a pond of depth L is 4°C . The temperature of the air, just about the layer of ice frozen at the pond's surface is -2°C for the past many days. The thermal conductivity of ice is three times that of water. The thickness of the frozen layer of ice must be :

- (1) $2L/3$ (2) $L/2$
 (3) $L/3$ (4) $3L/5$

53. A telescope consisting of an objective of focal length 60 cm and a single-lens eyepiece of focal length 5 cm is focussed at a distant object in such a way that parallel rays emerge from the eye piece. If the object subtends an angle of 2 at the objective, then find the angular width of the image.

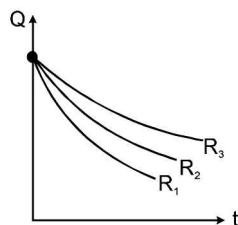
- (1) 6 (2) 12
 (3) 24 (4) $1/6$

Space for Rough Work

54. For a particle moving along x-axis, the acceleration a of the particle in terms of its x-coordinate x is given by $a = -9x$, where x is in meters and a is in m/s^2 . Take acceleration, velocity and displacement in positive x-direction as positive. The initial velocity of particle at $x = 0$ is $u = +6 \text{ m/s}$. The velocity of particle at $x = 2 \text{ m}$ will be :

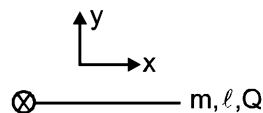
- (1) $+6\sqrt{2} \text{ m/s}$
 (2) $-6\sqrt{2} \text{ m/s}$
 (3) 72 m/s
 (4) 0

55. Three identical capacitors are given a charge Q each and they are then allowed to discharge through resistance R_1 , R_2 and R_3 separately. Their charges, as a function of time are shown in the graph below. The smallest of the three resistances is



- (1) R_3
 (2) R_2
 (3) R_1
 (4) cannot be predicted

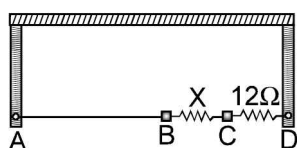
56. A rod of mass ' m ', length ' ℓ ' and charge ' Q ' is released from horizontal position. It can rotate about a fixed smooth horizontal axis passing through its one end. The mass and charge is uniformly distributed. A uniform electric field $-E\hat{i}$ exists in the space. When the rod becomes vertical its kinetic energy is



- (1) $mg \frac{\ell}{2}$
 (2) $\frac{mg\ell}{2} + QE \frac{\ell}{2}$
 (3) $mg \frac{\ell}{2} - QE \frac{\ell}{2}$
 (4) $\frac{QE\ell}{2} - mg \frac{\ell}{2}$

Space for Rough Work

57. A thin uniform wire AB of length 1 m, an unknown resistance X and a resistance of $12\ \Omega$ are connected by thick conducting strips, as shown in the figure. A battery and a galvanometer (with a sliding jockey connected to it) are also available. To obtain the value of X appropriate connections are made. Which one represents the correct combination if no deflection takes place in galvanometer.



- (1)

- 58.** White light is incident normally on a glass plate (in air) of thickness 500 nm and refractive index of 1.5. The wavelength (in nm) in the visible region (400 nm - 700nm) that is strongly reflected by the plate is:
- (1) 450 (2) 600
(3) 400 (4) 500

SECTION - II

Reasoning Type

This section contains 2 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

59. **STATEMENT-1** : An electron beam from a discharge tube is incident on a sample of atomic H-gas, it is observed that the number of spectral lines emitted from gas increases with increase in potential difference of discharge tube.
- STATEMENT-2** : de Broglie wavelength of electron is inversely proportional to the square root of voltage applied across discharge tube :
- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (2) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1.
- (3) Statement-1 is True, Statement-2 is False.
- (4) Statement-1 is False, Statement-2 is True.

Space for Rough Work

60. **STATEMENT-1** : A cyclist is cycling on a rough horizontal circular track with increasing speed. Then the net frictional force on cycle is always directed towards centre of the circular track.

STATEMENT-2 : For a particle moving in a circle, component of its acceleration towards centre, that is, centripetal acceleration should exist (except when speed is zero instantaneously).

(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True

Space for Rough Work

PART- C

Atomic masses : [H = 1, D = 2, Li = 7, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, Si = 28, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Cu = 63.5, Zn = 65, As = 75, Br = 80, Ag = 108, I = 127, Ba = 137, Hg = 200, Pb = 207]

Straight Objective Type

This section contains 30 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

61. A colloidal solution is subjected to an electrical field. The particles move towards anode. The coagulation of same sol is studied using NaCl, BaCl₂ and AlCl₃ solutions. Their coagulating power should be :

- (1) NaCl > BaCl₂ > AlCl₃
- (2) BaCl₂ > AlCl₃ > NaCl
- (3) AlCl₃ > BaCl₂ > NaCl
- (4) BaCl₂ > NaCl > AlCl₃

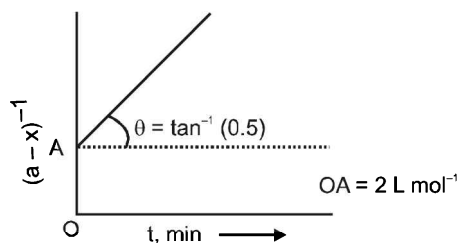
62. KMnO₄ can be prepared from K₂MnO₄ as per the reaction :



The reaction can go to completion by removing OH⁻ ions by adding :

- | | |
|---------------------|---------------------|
| (1) KOH | (2) CO ₂ |
| (3) SO ₂ | (4) HCl |

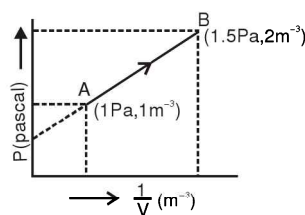
63. Given is the graph between $(a - x)^{-1}$ and time.



Hence, rate at the start of the reaction is

- (1) 1.25 mol L⁻¹ min⁻¹
- (2) 0.125 mol L⁻¹ min⁻¹
- (3) 0.5 mol L⁻¹ min⁻¹
- (4) 1.25 mol min⁻¹

64. 1 mole of an ideal gas is subjected to a reversible process as shown in the figure.

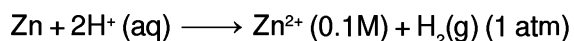


Calculate the work done (in joule) in the process. [ln 2 = 0.7]

- | | |
|---------------|---------------|
| (1) 0.6 Joule | (2) 6 Joule |
| (3) 8 Joule | (4) 0.8 Joule |

(SPACE FOR ROUGH WORK)

65. The e.m.f. of a cell corresponding to the reaction



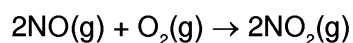
is 0.26 volt at 25°C. Calculate the pH of the solution at the hydrogen electrode.

$$E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ volt and } E^\circ_{\text{H}^+/\text{H}_2} = 0$$

- (1) 8 (2) 9
(3) 1 (4) 2

66. Calculate the change in pressure (in atm) when 2 mole of NO and 16 g O₂ in a 6.25 litre originally at 27°C react to produce the maximum quantity of NO₂ possible according to the equation.

$$(\text{Take } R = \frac{1}{12} \text{ atm L/mol K})$$



- (1) 1 (2) 2
(3) 3 (4) 4

67. Which of the following combinations of quantum numbers is allowed ?

	n	l	m	m _s
(1)	3	2	1	0
(2)	2	0	0	$-\frac{1}{2}$
(3)	3	-3	-2	$+\frac{1}{2}$
(4)	1	0	1	$+\frac{1}{2}$

68. At a particular temperature, the vapour pressures of two liquids A and B are respectively 120 and 180 mm of mercury. If 2 moles of A and 3 moles of B are mixed to form an ideal solution, the vapour pressure of the solution at the same temperature will be (in mm of mercury)

- (1) 156 (2) 145
(3) 150 (4) 108

69. Chemical A is used for water softening to remove temporary hardness. A reacts with Na₂CO₃ to generate caustic soda. When CO₂ is bubbled through A, it turns cloudy. What is the chemical formula of A.

- (1) CaCO₃ (2) CaO
(3) Ca(OH)₂ (4) Ca(HCO₃)₂

70. Time taken to produce 1 mol of photon by a 100 W (Js⁻¹) yellow lamp is (Given λ of yellow light is 500 nm) :

- (1) 340 min (2) 12 min
(3) 36 min (4) 40 min

71. Which equilibrium can be described as an acid base reaction using the Lewis acid – base definition but not using the Bronsted– Lowry definition ?

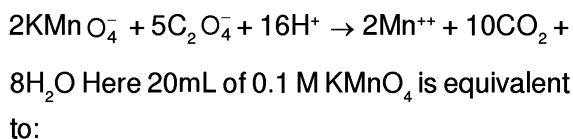
- (1) $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightleftharpoons 2\text{NH}_4^+ + \text{SO}_4^{2-}$
(2) $\text{NH}_3 + \text{CH}_3\text{COOH} \rightleftharpoons \text{NH}_4^+ + \text{CH}_3\text{COO}^-$
(3) $\text{H}_2\text{O} + \text{CH}_3\text{COOH} \rightleftharpoons \text{H}_3\text{O}^+ + \text{CH}_3\text{COO}^-$
(4) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cu}(\text{NH}_3)_4]^{2+} + 4\text{H}_2\text{O}$

(SPACE FOR ROUGH WORK)

72. N_2 and O_2 are converted into monoanions N_2^- and O_2^- respectively. Which of the following statements is wrong ?

- (1) In N_2^- , N–N bond weakens
- (2) In O_2^- , O–O bond order increases
- (3) In O_2^- , O–O bond order decreases
- (4) N_2^- becomes paramagnetic

73. $KMnO_4$ reacts with oxalic acid according to the equation



- (1) 20 mL of 0.5 M $H_2C_2O_4$
- (2) 50 mL of 0.5 M $H_2C_2O_4$
- (3) 50 mL of 0.1 M $H_2C_2O_4$
- (4) 20 mL of 0.1 M $H_2C_2O_4$

74. 6.02×10^{20} molecules of urea are present in 100 mL of its solution. The concentration of solution is :

- | | |
|------------|-------------|
| (1) 0.01 M | (2) 0.001 M |
| (3) 0.1 M | (4) 0.02 M |

75. Which of the following lanthanoid ions is diamagnetic ?

(At nos. Ce = 58, Sm = 62, Eu = 63, Yb = 70)

- | | |
|---------------|---------------|
| (1) Sm^{2+} | (2) Eu^{2+} |
| (3) Yb^{2+} | (4) Ce^{2+} |

76. Among the following statements which is INCORRECT :

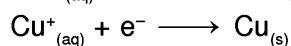
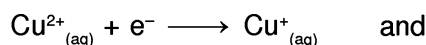
- (1) In the preparation of compounds of Xe, Bartlett had taken O_2PtF_6 as a base compound because both O_2 and Xe have almost same ionisation enthalpy.
- (2) Nitrogen does not show allotropy.
- (3) A brown ring is formed in the ring test for NO_3^- ion. It is due to the formation of $[Fe(H_2O)_5(NO)]^{2+}$
- (4) On heating with concentrated NaOH solution in an inert atmosphere of CO_2 , red phosphorus gives PH_3 gas.

77. The entropy change can be calculated by using the expression $\Delta S = \frac{q_{rev}}{T}$. When water freezes in a glass beaker, choose the correct statement among the following :

- (1) $\Delta S_{(system)}$ decreases but $\Delta S_{(surroundings)}$ remains the same.
- (2) $\Delta S_{(system)}$ increases but $\Delta S_{(surroundings)}$ decreases.
- (3) $\Delta S_{(system)}$ decreases but $\Delta S_{(surroundings)}$ increases.
- (4) $\Delta S_{(system)}$ decreases and $\Delta S_{(surroundings)}$ also decreases.

(SPACE FOR ROUGH WORK)

78. The electrode potentials for

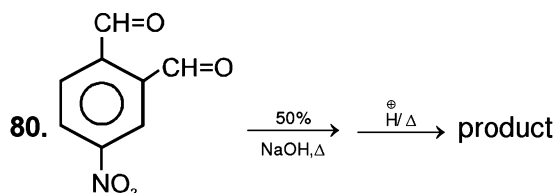
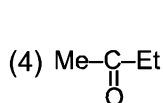
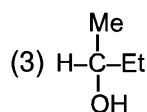
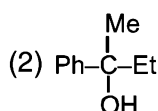
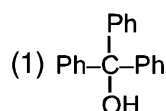


are +0.15 V and +0.50V respectively. The value

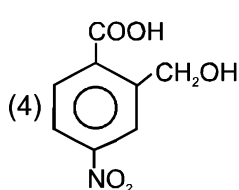
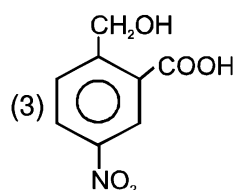
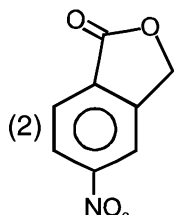
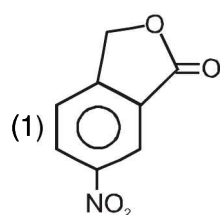
of $E^{\circ}_{\text{Cu}^{2+}/\text{Cu}}$ will be :

- (1) 0.500 V (2) 0.325 V
(3) 0.650 V (4) 0.150 V

79. An acid halide (X) reacts with excess of Grignard reagent (Y), the product can be :

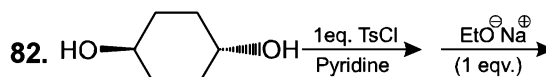


Product is :



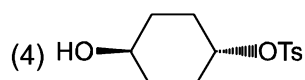
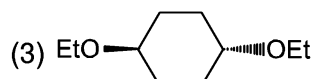
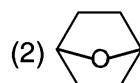
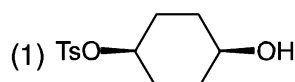
81. Acetaldehyde and Propyne can be distinguish by :

- (i) Tollen's reagent (ii) I_2/NaOH
(iii) Lucas reagent (iv) neutral FeCl_3
(1) (i) , (ii) & (iii) (2) (ii) & (iii)
(3) (i) & (ii) (4) (iii) & (iv)

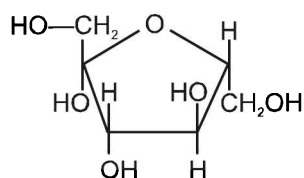


Product

The product is : :

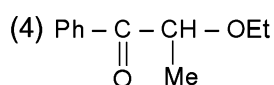
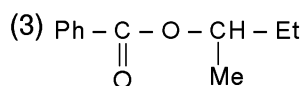
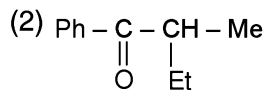
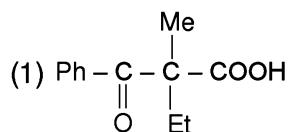
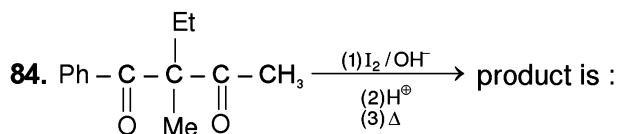


83. The following carbohydrate is :

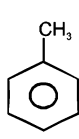
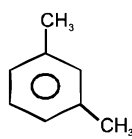
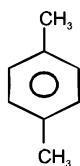
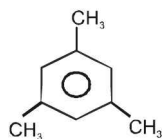


- (1) a ketohexose (2) an aldohexose
(3) an α -furanose (4) a ketopentose

(SPACE FOR ROUGH WORK)



85. Arrange the following compound in decreasing order of rate of electrophilic aromatic substitution reaction :



(I)

(II)

(III)

(IV)

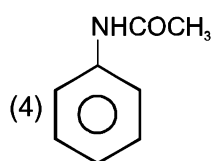
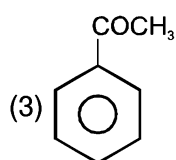
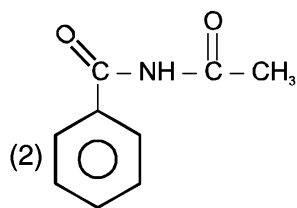
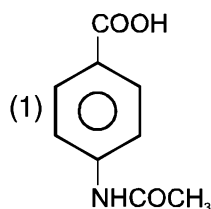
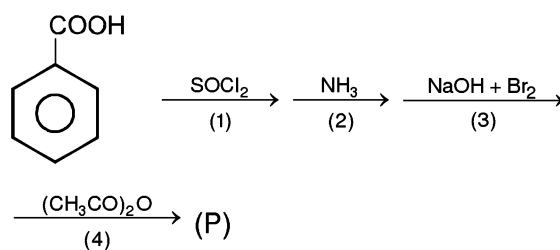
(1) (I) > (III) > (II) > (IV)

(2) (III) > (I) > (II) > (IV)

(3) (I) > (II) > (III) > (IV)

(4) (I) > (IV) > (II) > (III)

86. The structure of product (P) is :



87. D-Glucose and D-Mannose are :

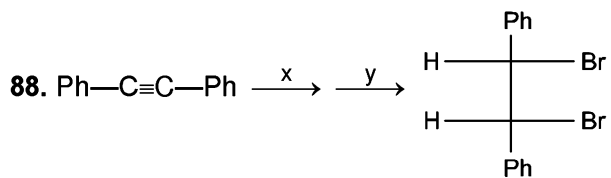
(1) Anomers

(2) Enantiomers

(3) Geometrical Isomers

(4) Epimers

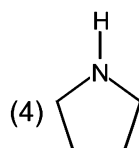
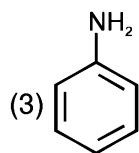
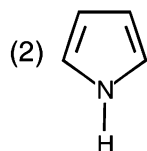
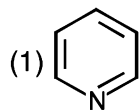
(SPACE FOR ROUGH WORK)



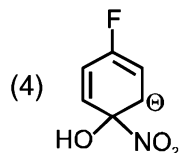
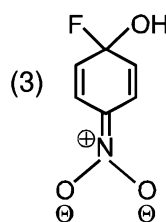
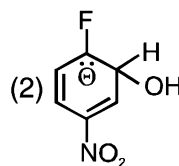
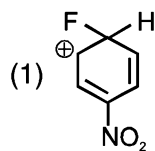
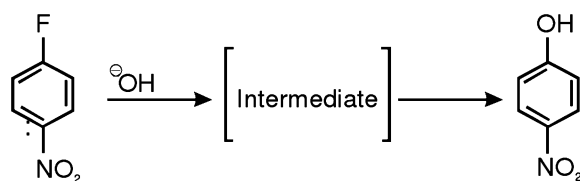
The reagents x and y are respectively :

- (1) $\text{Br}_2 / \text{CCl}_4, \text{H}_2 / \text{Ni}$
- (2) HBr, HBr
- (3) $\text{Na} / \text{NH}_3 (\ell), \text{Br}_2 / \text{CCl}_4$
- (4) $\text{Br}_2 / \text{CCl}_4, \text{Lindlar's hydrogenation}$

89. Which molecule have highest K_b value?



90. The intermediate of the following reaction is



(SPACE FOR ROUGH WORK)

Name of the Candidate

--

Roll Number

--	--	--	--	--	--	--	--

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified all the information filled in by the Candidate.

Signature of the Invigilator

PREPJEE EDUSHOP

**NOW GET ALL DOCUMENTS IN
PDF FORMAT**

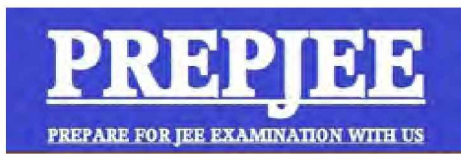
**PREPJEE EDUSHOP HAS
FOLLOWING TYPES OF PRODUCTS**

- 1. JEE MAIN SAMPLE PAPER**
- 2. JEE ADVANCE SAMPLE PAPER**
- 3. FORMULA BOOKS FOR JEE AND
OTHER COMPETITIVE EXAMINATION**
- 4. JEE ADVANCE 13 YEARS QUESTION PAPERS
WITH SOLUTION**
- 5. JEE MAIN 13 YEARS QUESTION PAPERS
WITH SOLUTION**

www.prepjee.in

To know more: sms PJ at 9852954321 | e-mail: support@prepjee.in

MOCK TEST 8



JEE MAIN TEST SERIES JMTS TARGET : JEE (MAIN)

PAPER



COURSE : PREP JEE JMTS

WWW.PREPJEE.IN

Time : 3 Hours

Max. Marks : 360

Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.

IMPORTANT INSTRUCTIONS

1. Immediately fill the particulars on this page of the Test Booklet with Blue / Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **360**.
5. There are three parts in the question paper A, B, C consisting of **Mathematics, Physics** and **Chemistry** having 30 questions. Mathematics & Physics part consists of **Two** sections and Chemistry consists of **One** section.
6. For each question you will be awarded **4 marks** if you darken all the bubble(s) corresponding to only the correct answer(s) and **zero mark** if no bubbles are darkened. In all other cases, **¼ (one fourth (1)) marks will be deducted**.
7. There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 6 above.

Filling the Top-half of the ORS :

Use only Black ball point pen only for filling the ORS. Do not use Gel / Ink / Felt pen as it might smudge the ORS.

8. Write your Roll no. in the boxes given at the top left corner of your ORS with black ball point pen. Also, darken the corresponding bubbles with Black ball point pen only. Also fill your roll no on the back side of your ORS in the space provided (if the ORS is both side printed).
9. Fill your Paper Code as mentioned on the Test Paper and darken the corresponding bubble with Black ball point pen.
10. If student does not fill his/her roll no. and paper code correctly and properly, then his/her marks will not be displayed and 5 marks will be deducted (paper wise) from the total.
11. Since it is not possible to erase and correct pen filled bubble, you are advised to be extremely careful while darken the bubble corresponding to your answer.
12. Neither try to erase / rub / scratch the option nor make the Cross (X) mark on the option once filled. Do not scribble, smudge, cut, tear, or wrinkle the ORS. Do not put any stray marks or whitener anywhere on the ORS.
13. If there is any discrepancy between the written data and the bubbled data in your ORS, the bubbled data will be taken as final.

PART - A

SECTION - I

Straight Objective Type

This section contains 27 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. The sum of three numbers in GP is 42. If the first two numbers increased by 2 and third term decreased by 4 the resulting numbers form an AP, then the middle term of the G.P. is

(1) 6
(2) 10
(3) 12
(4) 24

2. If $x = 5$ and $y = -2$ then $x - 2y = 9$. The contrapositive of this statement is

(1) If $x - 2y = 9$ then $x = 5$ and $y = -2$
(2) If $x - 2y \neq 9$ then $x \neq 5$ and $y \neq -2$
(3) If $x - 2y \neq 9$ then $x \neq 5$ or $y \neq -2$
(4) If $x - 2y \neq 9$ then either $x \neq 5$ or $y = -2$

3. If $A = \begin{bmatrix} a & x & y \\ x & b & z \\ y & z & c \end{bmatrix}$, $a, b, c, x, y, z \in \{1, 2, 3, 4, 5, 6\}$, a, b, c, x, y, z are distinct then number of matrices A with trace equal to 10 are

Note: The trace of matrix is the sum of its diagonal entries.]

(1) $3 \cdot (3!)^2$ (2) $2 \cdot (3!)^2$
(3) $(3!)^2$ (4) $(3!)^3$

4. If roots of the quadratic equation $x^2 + 2(a+1)x + a^2 - 6a + 8 = 0$, $a \in \mathbb{R}$ are of opposite in sign then complete set of values of 'a' is

(1) (1, 4) (2) (2, 4)
(3) $(-2, -4)$ (4) $(2, -4)$

5. In a city 20 percent of the population travels by car, 50 percent travels by bus and 10 percent travels by both car and bus. Then persons travelling neither by car nor by bus is

(1) 80 percent
(2) 40 percent
(3) 60 percent
(4) 70 percent

6. Number of divisors of 240 in the form of $4n + 2, n \in W$ is equal to

- (1) 4 (2) 8
(3) 10 (4) 3

7. The modulus of $\frac{\sqrt{3+4i} \cdot \sqrt[3]{12-5i}}{\sqrt[3]{(12+5i)}}$

(where $i = \sqrt{-1}$), is

- (1) 5 (2) $\sqrt{5} \cdot \sqrt[3]{13}$
(3) $\sqrt[3]{65}$ (4) $\sqrt{5}$

8. If a coin be tossed n times then probability that the head comes odd number of times

- (1) $\frac{1}{2}$ (2) $\frac{1}{2^n}$
(3) $\frac{1}{2^{n-1}}$ (4) $\frac{1}{n^2}$

9. Equation of the straight line meeting the circle with centre at origin and radius equal to 5 in two points at equal distances of 3 units from the point $A(3, 4)$, is

- (1) $6x + 8y - 41 = 0$
(2) $6x - 8y + 41 = 0$
(3) $8x + 6y + 41 = 0$
(4) none of these

10. $\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x \sin x \cdot \cos x}$ is equal to

- (1) $\frac{2}{3}$ (2) $\frac{3}{5}$
(3) $\frac{3}{2}$ (4) $\frac{3}{4}$

11. If length of the latus rectum of the parabola $25[(x-2)^2 + (y-3)^2] = (3x-4y+7)^2$ is λ , then 100λ is

- (1) 10 (2) 20
(3) 30 (4) 40

12. If $f(x) = \begin{cases} x & , 0 \leq x \leq 1 \\ 2 - e^{x-1} & , 1 < x \leq 2 \\ x - e & , 2 < x \leq 3 \end{cases}$ and

$g'(x) = f(x), x \in [1, 3]$, then

- (1) $g(x)$ has no local maxima
(2) $g(x)$ has no local minima
(3) $g(x)$ has local maxima at $x = 1 + \ln 2$ and local minima at $x = e$.
(4) $g(x)$ has local minima at $x = 1 + \ln 2$ and local maxima at $x = e$.

13. If the planes $x - cy - bz = 0, cx - y + az = 0$ and $bx + ay - z = 0$ pass through a straight line, then the value of $a^2 + b^2 + c^2 + 2abc$ is

- (1) 0 (2) 1
(3) -1 (4) 2

14. The function $f(x) = 2\ln(x - 2) - x^2 + 4x + 1$ increases in the interval

- (1) $(-\infty, 1) \cup (2, 3)$ (2) $(1, 2)$
(3) $(2, 3)$ (4) $(3, \infty)$

15. If $|\vec{a}| = 3$, $|\vec{b}| = 1$, $|\vec{c}| = 4$ and $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, then value of $\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a} =$

- (1) 13 (2) -13
(3) 26 (4) -26

16. If function $f : (1, 3) \rightarrow \mathbb{R}$ satisfies the inequality

$$\frac{\cos(2x - 6) - 37}{2} < f(x) < \frac{x^2 |2x - 6|}{x - 3},$$

$\forall x \in (1, 3)$, then

- (1) $\lim_{x \rightarrow 3^-} f(x) = 18$
(2) $\lim_{x \rightarrow 3^-} f(x) = -18$
(3) $\lim_{x \rightarrow 3^-} f(x)$ can not be determined from the

given inequality

- (4) $\lim_{x \rightarrow 3^-} f(x)$ does not exist

17. $f(x) = |x| + |\sin x|$, $0 < x \leq 2\pi$, is

- (1) continuous everywhere except at $x = \pi, 2\pi$.
(2) continuous and differentiable everywhere
(3) differentiable everywhere except at $x = \pi$.
(4) none of these

18. Which of the following is correct ?

- (1) $\tan^{-1}\left(\frac{\pi}{4}\right) + \cot^{-1}\left(\frac{\pi}{4}\right) = 2$
(2) $\sec^{-1}\left(\frac{1}{3}\right) + \operatorname{cosec}^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{2}$
(3) $\sin^{-1}(\pi) + \cos^{-1}(\pi) = \frac{\pi}{2}$
(4) $\tan^{-1}(2\pi) + \tan^{-1}\left(\frac{1}{2\pi}\right) = \frac{\pi}{2}$

19. $\int_{\pi}^{5\pi/4} \frac{\sin 2x}{\cos^4 x + \sin^4 x} dx$ is equal to

- (1) $\frac{\pi}{2}$ (2) $\frac{\pi}{4}$
(3) $\frac{\pi}{3}$ (4) $\frac{\pi}{6}$

20. Let $f(x) : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is an invertible function such that $f'(x) > 0$ and $f''(x) > 0 \forall x \in [1, 5]$. If $f(1) = 1$ and $f(5) = 5$ and area under the curve $y = f(x)$ above x -axis from $x = 1$ to $x = 5$ is 8 sq. units, then area bounded by $y = f^{-1}(x)$ above x -axis from $x = 1$ to $x = 5$ is
- (1) 8 (2) 12
(3) 16 (4) 20

21. If $\sum_{i=1}^{18} (x_i - 8) = 9$ and $\sum_{i=1}^{18} (x_i - 8)^2 = 45$, then standard deviation of $x_1, x_2, \dots, x_{18}$ is
- (1) $\frac{4}{9}$ (2) $\frac{9}{4}$
(3) $\frac{3}{2}$ (4) None of these

22. Solution of the differential equation

$$\frac{dy}{dx} + \frac{1+y^2}{\sqrt{1-x^2}} = 0 \text{ is}$$

- (1) $\tan^{-1} y + \sin^{-1} x = c$
(2) $\tan^{-1} x + \sin^{-1} y = c$
(3) $\tan^{-1} y \cdot \sin^{-1} x = c$
(4) $\tan^{-1} y - \sin^{-1} x = c$

23. Acute angle between the curves $y = x^2 - 2$ and $y = x^3 - 3x$ at their point of intersection (2, 2) is

- (1) $\tan^{-1}\left(\frac{1}{21}\right)$ (2) $\frac{\pi}{4}$
(3) $\tan^{-1}\left(\frac{5}{37}\right)$ (4) $\tan^{-1}\left(\frac{4}{41}\right)$

24. If a, b, c are non-zero real constant and the number of terms in the expansion of $(ax + by + c)^n$ is 36, then n ($n \in \mathbb{N}$) is equal to

- (1) 7 (2) 8
(3) 9 (4) 10

25. The number of values of x in the interval

$$\left[0, \frac{11\pi}{2}\right] \text{ satisfying the equation } 6\sin^2 x + \sin$$

$$x - 2 = 0 \text{ is}$$

- (1) 9 (2) 10
(3) 11 (4) 12

26. $\int \frac{x^2+2}{x^4+4} dx$ is equal to

(1) $\frac{1}{2} \tan^{-1} \left(\frac{x^2+2}{2x} \right) + c$

(2) $\frac{1}{2} \tan^{-1} \left(\frac{x^2-2}{2x} \right) + c$

(3) $\frac{1}{2} \tan^{-1} \left(\frac{2x}{x^2-2} \right) + c$

(4) $\frac{1}{2} \tan^{-1} (x^2+2) + c$

27. If $y = \sqrt{\frac{1+\cos 2\theta}{1-\cos 2\theta}}$; then $\frac{dy}{d\theta}$ at $\theta = \frac{3\pi}{4}$ is

(1) -2

(2) 2

(3) ± 2

(4) none of these

SECTION - II
Reasoning Type

This section contains 3 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

28. **STATEMENT-1** : Total number of tangents

of the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ that are perpendicular to the line $2x + 5y - 3 = 0$ is 2.

STATEMENT-2 : The equations of tangents

to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ in terms of its

slope m are $y = mx \pm \sqrt{a^2m^2 - b^2}$, if $a^2m^2 - b^2 > 0$.

(1) Statement -1 is True, Statement -2 is True ; Statement -2 is a correct explanation for Statement -1

(2) Statement-1 is True, Statement-2 is True ; Statement-2 is **NOT** a correct explanation for Statement-1

(3) Statement -1 is True, Statement -2 is False

(4) Statement -1 is False, Statement -2 is True

Space for Rough Work

29. STATEMENT - 1 : Number of rectangles on chessboard is ${}^8C_2 \cdot {}^8C_2$.

STATEMENT - 2 : To form a rectangle we have to select any two of the horizontal lines and any two of the vertical lines.

(1) Statement -1 is True, Statement -2 is True ; Statement -2 is a correct explanation for Statement -1

(2) Statement-1 is True, Statement-2 is True ; Statement-2 is **NOT** a correct explanation for Statement-1

(3) Statement -1 is True, Statement -2 is False

(4) Statement -1 is False, Statement -2 is True

30. STATEMENT - 1 If $y = f(x)$ is increasing in $[\alpha, \beta]$, then its range is $[f(\alpha), f(\beta)]$

STATEMENT- 2 Every increasing function need not to be continuous.

(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(2) Statement-1 is True, Statement-2 is True; Statement-2 is **NOT** a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True

Space for Rough Work

PART – B

SECTION - I

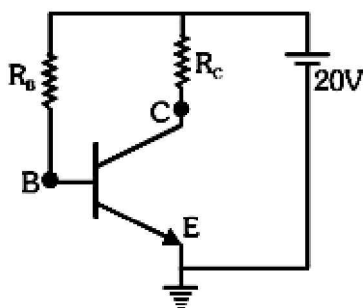
Straight Objective Type

This section contains 28 multiple choice questions. Each question has choices (1), (2), (3) and (4) out of which **ONLY ONE** is correct.

31. A particle is projected with velocity u horizontally from the top of a smooth sphere of radius ' a ' so that it slides down the outside of the sphere. If the particle leaves the sphere when it has fallen a vertical distance $\left(\frac{a}{4}\right)$, the value of u is :

- (1) $\sqrt{ag}$ (2) $\sqrt{2ag}$
(3) $\frac{\sqrt{ag}}{2}$ (4) $\frac{3\sqrt{ag}}{2}$

32. For given CE biasing circuit, if voltage across collector-emitter is 12V and current gain is 100 and base current is 0.04 mA then determine the value collector resistance R_C .



- (1) 1200Ω (2) 200Ω
(3) 400Ω (4) 2000Ω

33. If light beam is incident at polarising angle (56.3°) on air-glass interface, then what is the angle of refraction in glass ?

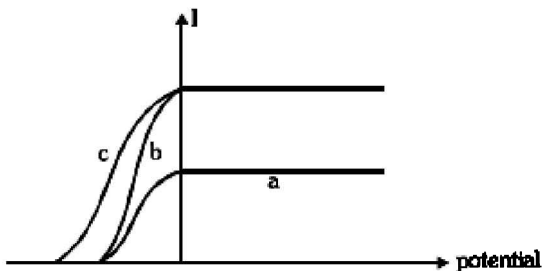
- (1) 56.3°
(2) 28.15°
(3) 33.7°
(4) Refraction will not take place

34. The 660 V rails on a subway can kill a person upon contact. A 10,000 V Van de Graff generator, however, will only give a mild shock. Which of the following best explains this seeming paradox ?

- (1) The generator provides more energy per charge, but since it has few charges it transfers a lesser amount of energy.
(2) The generator provides more energy, but since there is little energy per charge the current is small.
(3) Most of the energy provided by the generator is dissipated in the air because air presents a smaller resistance than the human body
(4) Most of the energy flows directly to the ground without going through the human body since the generator is grounded.

Space for Rough Work

35. In photoelectric experiment the plot between anode potential and photoelectric current is shown in figure. Which of the following is correct ?

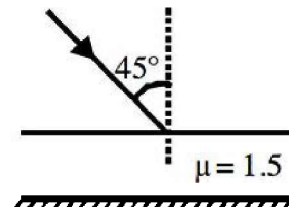


- (1) Frequency of light corresponding to 'a' is same as that 'b' and is different for 'c'
- (2) Frequency of light corresponding to 'a' is different from 'b' but the intensities are same
- (3) Frequency of light corresponding to 'b' is same as that 'c' but intensities are different.
- (4) None of these.

36. Choose the INCORRECT statement.

- (1) Cut-off wavelength of X-rays is independent of filament voltage.
- (2) Presence of K_{α} -line in X-ray spectrum means that L-series will also be present.
- (3) Increase in filament current increases intensity of X-ray.
- (4) Presence of L-series in X-ray spectrum means that K-series will also be present.

37. One side of a glass slab is silvered as shown. A ray of light is incident on the other side at angle of incidence $i = 45^\circ$. Refractive index of glass is given as 1.5. The deviation of the ray of light from its initial path when it comes out of the slab is :

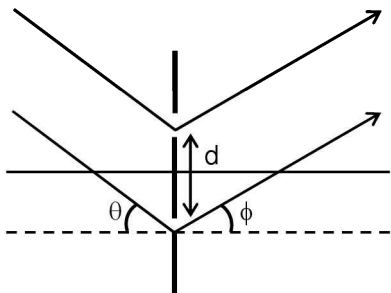


- (1) 90°
- (2) 180°
- (3) 120°
- (4) 45°

38. Find the value of earth's magnetization (magnetic moment per unit volume). Assume that the earth's field can be approximated by a giant bar magnet of magnetic moment $8.0 \times 10^{22} \text{ Am}^2$. The earth's radius is 6400 km.

- (1) $24 \times 10^4 \text{ Am}^{-1}$
- (2) 240 Am^{-1}
- (3) 729 Am^{-1}
- (4) 72.9 Am^{-1}

39. In the given figure light is incident at an angle θ with the normal to a plane containing two slits of separation d . Select the expression that correctly describes the positions of the interference maxima in terms of the incoming angle θ and outgoing angle ϕ .



(1) $\sin \phi + \sin \theta = \left(m + \frac{1}{2}\right) \frac{\lambda}{d}$

(2) $d \sin \theta = m\lambda$

(3) $\sin \phi - \sin \theta = (m + 1) \frac{\lambda}{d}$

(4) $\sin \phi + \sin \theta = m \frac{\lambda}{d}$

40. A container has an ideal gas at pressure P in thermal equilibrium. Assuming the mass of a molecule is m and all the molecules are moving with same speed v in random directions. The expression for number of collision per second which the molecules make with unit area of container wall is :

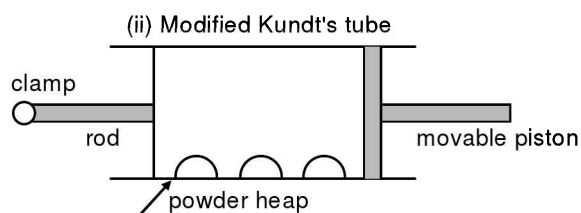
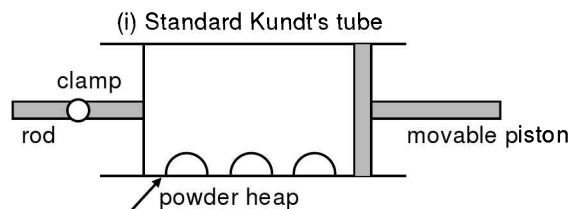
(1) $\frac{P}{mv}$

(2) $\frac{3P}{2mv}$

(3) $\frac{\sqrt{3}P}{2mv}$

(4) $\frac{3P}{\sqrt{2}mv}$

41. In the Kundts tube experiment [Shown in figure (i)], the rod is clamped at the end instead of clamping it at the center as shown in [figure (ii)]. It is known that speed of sound in air is 330 m/s, powder piles up at successive distance of 0.6 m and length of rod used is 1 m, speed of sound in rod is :



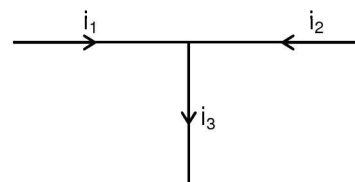
(1) 550 m/s

(2) 1100 m/s

(3) 1200 m/s

(4) 600 m/s

42. In the given figure if $i_1 = 3 \sin \omega t$ and $i_2 = 4 \cos \omega t$, then i_3 is :



(1) $5 \sin (\omega t + 53^\circ)$

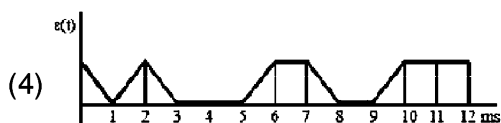
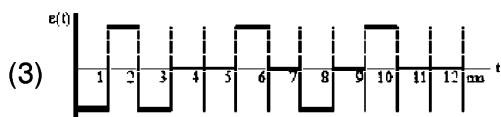
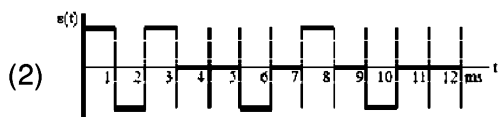
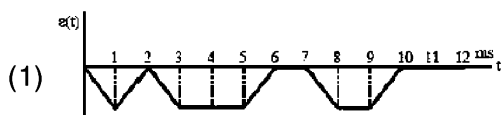
(2) $5 \sin (\omega t + 37^\circ)$

(3) $5 \sin (\omega t + 45^\circ)$

(4) $5 \cos (\omega t + 53^\circ)$

Space for Rough Work

43. A spy taps the series link between two computers by wrapping a small coil around the current-carrying wire connecting the computer. The current versus time for the transmission between the two computers is a signal as shown in figure. Then the sketch of the induced emf in the coil versus time as detected by the spy will be :



44. The vernier of a circular scale is divided into 50 divisions which coincides with 49 main scale division. Each main scale division is 0.5° . The least count of the instrument is :

- (1) $0.6'$ (2) $1'$
(3) $6'$ (4) $30'$

45. A metal wire of length L_1 and area of cross section A is attached to a rigid support. Another metal wire of length L_2 and of the same cross sectional area is attached to the free end of the first wire. A body of mass M is then suspended from the free end of the second wire. If Y_1 and Y_2 are the Young's moduli of the wires respectively, the effective force constant of the system of two wires is :

(1) $\frac{[(Y_1 Y_2)A]}{[2(Y_1 L_2 + Y_2 L_1)]}$

(2) $\frac{[(Y_1 Y_2)A]}{(L_1 + L_2)^{1/2}}$

(3) $\frac{[(Y_1 Y_2)A]}{(Y_1 L_2 + Y_2 L_1)}$

(4) $\frac{(Y_1 Y_2)^{1/2} A}{(L_1 + L_2)^{1/2}}$

46. You do a very precise experiment to test the properties of a piece of wire. You stretch the wire by applying an outward force to each end of the wire and measure its change in length. If you then precisely double this force and the wire still remains elastic but is beyond proportionality limit, the change in length of the wire.

- (1) will exactly double
(2) will just slightly more than double
(3) will just slightly less than double
(4) will decrease by exactly a factor of two

Space for Rough Work

47. Two bodies of different materials and same surface area, coated by similar material, are heated to same temperature. They are allowed to cool down in same surroundings.

If $\frac{dQ}{dt}$ represents rate of loss of heat and

$\frac{d\theta}{dt}$ represents rate of fall of temperature, then for both :

(1) $\frac{dQ}{dt}$ is same but $\frac{d\theta}{dt}$ is different

(2) $\frac{dQ}{dt}$ is different but $\frac{d\theta}{dt}$ is same

(3) $\frac{d\theta}{dt}$ as well as $\frac{dQ}{dt}$ is same

(4) $\frac{d\theta}{dt}$ as well as $\frac{dQ}{dt}$ is different

48. A spring of force constant k is cut in two parts at its one third length. When both the parts are stretched by same amount, the work done in the two part, will be :

(1) equal in both

(2) greater for the longer part

(3) greater for the shorter part

(4) data insufficient

49. A person can not see clearly an object kept at a distance beyond of 100 cm. Find the nature and the power of lens to be used for seeing clearly the object at infinity.

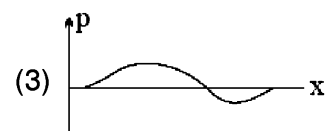
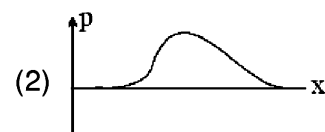
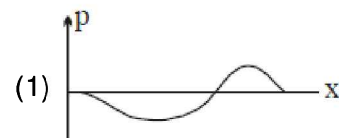
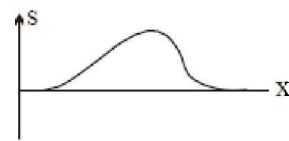
(1) $-1D$

(2) $+1D$

(3) $-3D$

(4) $3D$

50. For displacement(s)-x graph shown for a sound wave, select appropriate excess pressure (p)-x graph.



Space for Rough Work

51. A projectile is thrown at an angle 30° from the horizontal ground with velocity 10 m/s. Find angle between displacement vector and velocity vector at time $t = 1$ sec .

- (1) 20°
- (2) 30°
- (3) 40°
- (4) 50°

52. At time $t = 0$, a 2 kg particle has position vector $\vec{r} = (4\hat{i} - 2\hat{j})$ m relative to the origin. Its velocity is given by $\vec{v} = 2t^2\hat{i}$ (m/s) for $t \geq 0$. The torque acting on the particle about the origin at $t = 2$ s, is :

- (1) 32 N-m ($\hat{k}$) (2) 16 N-m ($\hat{k}$)
- (3) 16 N-m ($\hat{k}$) (4) 12 N-m ($\hat{k}$)

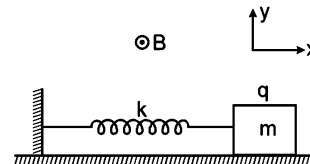
53. A soap bubble (surface tension = T) is charged to a maximum surface density of charge = σ , when it is just going to burst. Its radius R is given by:

- (1) $R = \frac{\sigma^2}{8 \epsilon_0 T}$
- (2) $R = 8 \epsilon_0 \frac{T}{\sigma^2}$
- (3) $R = \frac{\sigma}{\sqrt{8 \epsilon_0 T}}$
- (4) $R = \frac{\sqrt{8 \epsilon_0 T}}{\sigma}$

54. Select the correct choice(s):

- (1) The gravitational field inside a spherical cavity, within a spherical planet (mass is uniformly distributed) must be non zero and uniform.
- (2) When a body is projected horizontally at an appreciable large height above the earth, with a velocity less than for a circular orbit, it will fall to the earth along a parabolic path.
- (3) A body of zero total mechanical energy placed in a gravitational field will escape the field
- (4) Earth's satellite must be in equatorial plane.

55. A spring of spring constant 'K' is fixed at one end has a small block of mass m and charge q is attached at the other end. The block rests over a smooth horizontal surface. A uniform and constant magnetic field B exists normal to the plane of paper as shown in figure. An electric field $\vec{E} = E_0 \hat{i}$ (E_0 is a positive constant) is switched on at $t = 0$ sec. The block moves on horizontal surface without ever lifting off the surface. Then the normal reaction acting on the block is :



- (1) Maximum at extreme position and minimum at mean position.
- (2) Maximum at mean position and minimum at extreme position.
- (3) Is uniform throughout the motion.
- (4) Is both maximum and minimum at mean position.

Space for Rough Work

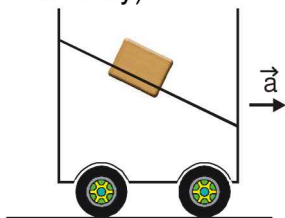
56. A man of mass 60 kg standing on a platform executing S.H.M. in the vertical plane. The displacement from the mean position varies as $y = 0.5 \sin (2\pi ft)$. The value of f , for which the man will feel weightlessness at the highest point is : (y is in metres)

- (1) $\frac{g}{4\pi}$ (2) $4\pi g$
 (3) $\frac{\sqrt{2g}}{2\pi}$ (4) $2\pi\sqrt{2g}$

57. Current is flowing in a circular conductor PQRS of centre O with angle POR = 60° . If B_1 and B_2 are the magnitudes of the magnetic fields at O due to the currents in PQR and PSR respectively, then the ratio of $\frac{B_1}{B_2}$ will be :

- (1) 5 (2) 2
 (3) 10 (4) 0.5

58. An open water tanker moving on a horizontal straight road has a cubical block of cork floating over its surface. If the tanker has an acceleration of a as shown, the acceleration of the cork w.r.t. container is (ignore viscosity) -



- (1) Zero (2) $\frac{a^2}{g}$
 (3) $\frac{a}{y}\sqrt{g^2 - a^2}$ (4) a

SECTION - II

Reasoning Type

This section contains 2 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

59. **STATEMENT-1** : A parallel beam of light traveling in air can be displaced laterally by a parallel transparent slab by distance more than the thickness of the plate.

And

STATEMENT-2 : The lateral displacement of light traveling in air increased with rise in value of refractive index of slab.

- (1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1
 (2) Statement-1 is true, statement-2 is true and statement-2 is NOT correct explanation for statement-1
 (3) Statement-1 is true, statement-2 is false
 (4) Statement-1 is false, statement-2 is true

60. **STATEMENT-1** : When source of sound moves towards a stationary observer the wave length of sound wave received by the observer is different from the wave length of the wave emitted by the source.

STATEMENT-2 : The wavelength of the sound wave propagating in the direction of motion of source is smaller than when the source does not move. It is because the source moves by some distance.

- (1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1
 (2) Statement-1 is true, statement-2 is true and statement-2 is NOT correct explanation for statement-1
 (3) Statement-1 is true, statement-2 is false
 (4) Statement-1 is false, statement-2 is true

Space for Rough Work

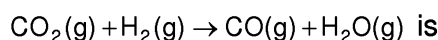
PART- C

Atomic masses : [H = 1, D = 2, Li = 7, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, Si = 28, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Cu = 63.5, Zn = 65, As = 75, Br = 80, Ag = 108, I = 127, Ba = 137, Hg = 200, Pb = 207]

Straight Objective Type

This section contains 30 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

61. The ΔH_f° for CO_2 (g), CO (g) and H_2O (g), are -393.5 , -110.5 and -241.8 KJ mol^{-1} respectively. The standard enthalpy change (in KJ) for the reaction



- (1) 524.1 (2) 41.2
(3) -262.5 (4) -41.2

62. Match list I with list II and select the correct answer using the codes given below the lists :

List I

- I. Cyanide Process
II. Floatation process
III. Electrolytic reduction
IV. Zone refining

List II

- A. Ultra pure Ge
B. Pine oil
C. Extraction of Al
D. Extraction of Au

Codes:

- (1) I-C, II-A, III-D, IV-B
(2) I-D, II-B, III-C, IV-A
(3) I-C, II-B, III-D, IV-A
(4) I-D, II-A, III-C, IV-B

63. For the equilibrium, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s}) \rightleftharpoons \text{CuSO}_4 \cdot 3\text{H}_2\text{O}(\text{s}) + 2\text{H}_2\text{O}(\text{g})$, $K_p = 1.086 \times 10^{-4} \text{ atm}^{-2}$ at 25°C . The efflorescent nature of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}(\text{s})$ can be noticed when the vapour pressure of water in the atmosphere is (in mm Hg) :

- (1) 10.92 (2) 7.90
(3) < 7.90 (4) > 7.90

64. The reaction of XeF_6 with silica (SiO_2) gives:

- (1) XeO_3 and SiF_4
(2) XeF_4 and SiF_4
(3) XeOF_2 and SiF_4
(4) XeOF_4 and SiF_4

65. In FCC lattice A, B, C, D atoms are arranged at corners, face centres, octahedral voids and tetrahedral voids respectively, then the body diagonal contains :

- (1) 2A, 2D (2) 2A, C, 2D
(3) 2A, 2B, D (4) 2A, 2B, 2C

66. (I) $[\text{Co}(\text{EDTA})]^-$ has two optical isomers.

(II) $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)_2]^+$ show linkage isomers.

(III) For $[\text{Pt}(\text{py})(\text{NH}_3)(\text{NO}_2)\text{ClBr}]$, theoretically fifteen different geometrical isomers are possible.

(IV) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl}_2 \cdot 2\text{H}_2\text{O}$ can show hydrate as well as ionisation isomerism.

Amongs the following correct statements are:

- (1) II, III (2) III
(3) I, III (4) I, II & III

67. The vander Waal constant 'a' for the gases CH_4 , N_2 , NH_3 and O_2 are 2.25, 1.39, 4.17 and $1.3 \text{ L}^2 \text{ atm-mol}^{-2}$ respectively. The gas which shows highest critical temperature is :
- (1) CH_4 (2) N_2
(3) NH_3 (4) O_2
68. Zn Amalgam is prepared by electrolysis of aqueous ZnCl_2 using Hg cathode (9gm). How much current is to be passed through ZnCl_2 solution for 1000 seconds to prepare a Zn Amalgam with 25% Zn by wt. (Zn = 65.4)
- (1) 5.6 amp (2) 7.2 amp
(3) 8.85 amp (4) 11.2 amp
69. A certain acid-base indicator is red in acidic solution and blue in basic solution. At pH = 5, 75% of the indicator is present in the solution in its blue form. Calculate the pH at which the indicator shows 90% red form?
(Given $10^{-4.523} = 3 \times 10^{-5}$)
- (1) 3.56 (2) 5.47
(3) 2.5 (4) 7.4
70. A mixture of (a) and (b), which are two miscible liquids, is distilled under equilibrium conditions at atmospheric pressure. The mole fraction of (a) in solution is 0.3 and in vapour phase is 0.6. If the solution behaves ideally the ratio of p_a° to p_b° is :
- (1) 1.5 (2) 3.5
(3) 2.5 (4) 4.0
71. The non-stoichiometric compound $\text{Fe}_{0.94}\text{O}$ is formed when x% of Fe^{2+} ions are replaced by Fe^{3+} ions, then x is :
- (1) 18 (2) 15
(3) 12 (4) 6
72. Which of the following statement is false about the alkali metals ?
- (1) All alkali-metal salts impart a characteristic colour to the Bunsen flame.
(2) Among LiOH , CsOH , KOH , and RbOH , CsOH has the highest solubility in water.
(3) Among the alkali metals, cesium is the most reactive.
(4) The reducing character of the alkali metal hydrides follow the order :
 $\text{LiH} > \text{NaH} > \text{KH} > \text{RbH} > \text{CsH}$.
73. Among the following select the correct the statements :
- (I) PH_5 donot exist.
(II) $p\pi-d\pi$ bond is present in SO_2 .
(III) SeF_4 and CH_4 have the same shape.
(IV) I_3^+ has a bent shape.
- (1) II, III (2) I, II
(3) I, IV (4) I, II, IV
74. If r_0 be the radius of first Bohr's orbit of H-atom, the de-Broglie's wavelength of an electron revolving in the third Bohr's orbit will be :
- (1) $2\pi r_0$ (2) $4\pi r_0$
(3) $6\pi r_0$ (4) πr_0

(SPACE FOR ROUGH WORK)

75. An aqueous solution of 6.3 g of oxalic acid dihydrate is made up to 250 ml. The volume of 0.1 N NaOH required to completely neutralise 10 ml of this solution is

- (1) 20 ml (2) 40 ml
(3) 10 ml (4) 4 ml

76. A buffer solution is prepared in which the concentration of NH_3 is 0.30 M and the concentration of NH_4^+ is 0.20 M. If the equilibrium constant, K_b for NH_3 equals 1.8×10^{-5} , what is the pH of this solution ? ($\log 2.7 = 0.433$).

- (1) 9.08 (2) 9.43
(3) 11.72 (4) 8.73

77. A reaction which is of first order w.r.t. reactant A, has a rate constant 6 min^{-1} . If we start with $[\text{A}] = 0.5 \text{ mol L}^{-1}$, when would $[\text{A}]$ reach the value of 0.05 mol L^{-1}

- (1) 0.384 min (2) 0.15 min
(3) 3 min (4) 3.84 min

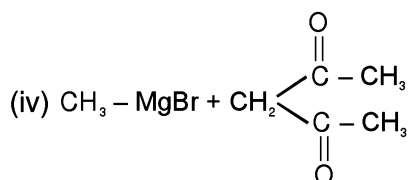
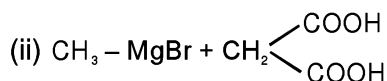
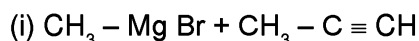
78. Pyrolusite is MnO_2 used to prepare KMnO_4 .

Steps are, $\text{MnO}_2 \xrightarrow{\text{I}} \text{MnO}_4^{2-} \xrightarrow{\text{II}} \text{MnO}_4^-$.

Steps I and II are respectively :

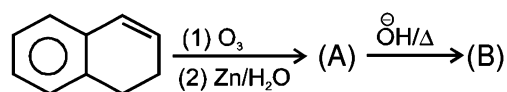
- (1) fuse with KOH / air and electrolytic oxidation.
(2) fuse with KOH / KNO_3 and electrolytic reduction.
(3) fuse with conc. HNO_3 / air and electrolytic reduction.
(4) dissolve in H_2O and oxidation.

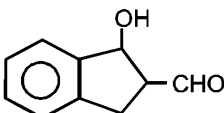
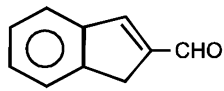
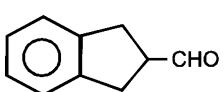
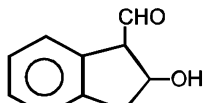
79. In which of the following reaction CH_4 will be obtained ?



- (1) (i), (ii) & (iii)
(2) (i), (ii), (iii) & (iv)
(3) (iii) & (iv)
(4) (iii) & (i), (iv)

80. In the given reaction sequence B is :



- (1)  (2) 
(3)  (4) 

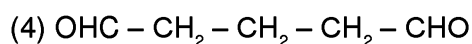
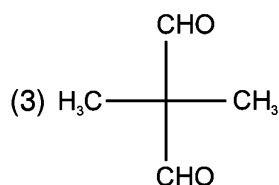
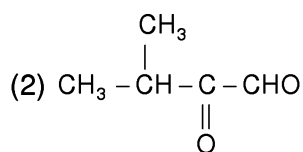
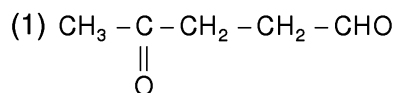
81. Which of the following is used in relieving pain?

- (1) Acetyl salicylic acid
- (2) Methyl salicylate
- (3) Methylacetate
- (4) Phenylacrylate

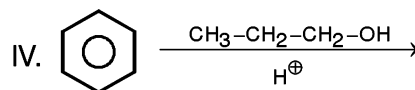
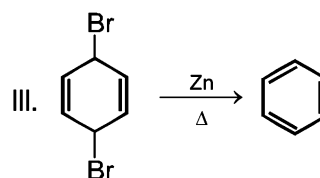
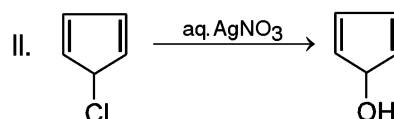
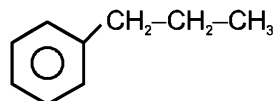
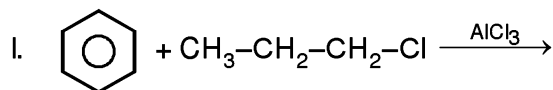
82. The compound responds to Tollen's reagent is :

- (1) CH_3COCH_3
- (2) CH_3CHO
- (3) CH_3CONH_2
- (4) CH_3COOH

83. Compound X ($\text{C}_5\text{H}_8\text{O}_2$) forms a dioxime with NH_2OH . On Clemmensen reduction it forms a compound which forms only one monochloro derivative. 'X' can be :



84. Which of the following reactions report the correct major products.



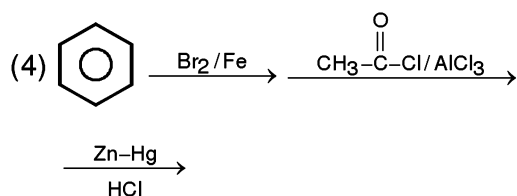
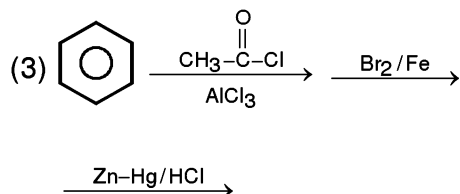
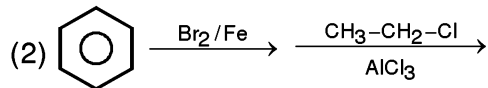
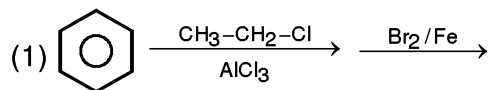
(1) I, III

(3) II, III, IV

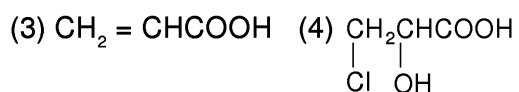
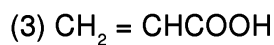
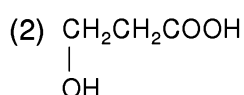
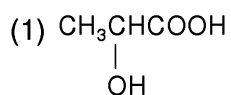
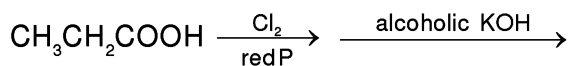
(2) III, IV

(4) I, III, II

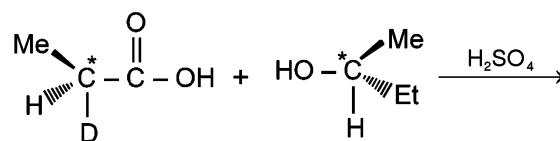
85. Which of the following method would be better for the synthesis of 1-bromo-3-ethylbenzene.



86. End product of the following reaction is :



87. The major product (ester) of the following reaction is :

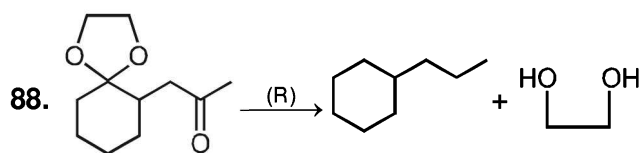


(1) A single stereoisomer (optically inactive)

(2) A mixture of diastereomers (both optically active)

(3) A racemic mixture (optically inactive)

(4) A single stereoisomer (optically active)



The reagent (R) is :

(1) $\text{N}_2\text{H}_4 / \text{KOH}$

(2) LiAlH_4

(3) $\text{Zn-Hg} / \text{Conc. HCl}$

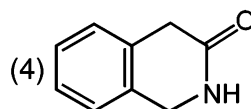
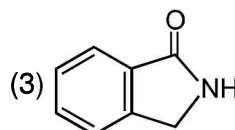
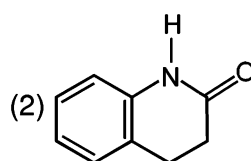
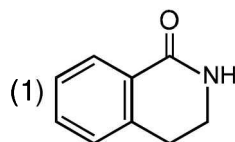
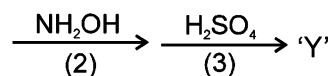
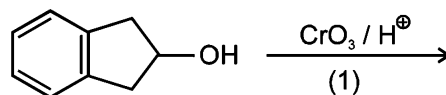
(4) H_2 / Ni

(SPACE FOR ROUGH WORK)

89. Choose the correct statement :

- (1) Salicylic acid (o-Hydroxybenzoic acid) is stronger than its m-, p-isomers and benzoic acid itself.
- (2) Acidity of salicylic acid is only due to steric inhibition of resonance, as – OH group forces – COOH out of the plane of ring.
- (3) The orbitals which are in the same plane must take part in resonance.
- (4) All the resonating structures have real existence.

90. The most probable product of the following sequence of reactions would be :



(SPACE FOR ROUGH WORK)

Name of the Candidate

--

Roll Number

--	--	--	--	--	--	--	--

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified all the information filled in by the Candidate.

Signature of the Invigilator

PREPJEE EDUSHOP

**NOW GET ALL DOCUMENTS IN
PDF FORMAT**

**PREPJEE EDUSHOP HAS
FOLLOWING TYPES OF PRODUCTS**

- 1. JEE MAIN SAMPLE PAPER**
- 2. JEE ADVANCE SAMPLE PAPER**
- 3. FORMULA BOOKS FOR JEE AND
OTHER COMPETITIVE EXAMINATION**
- 4. JEE ADVANCE 13 YEARS QUESTION PAPERS
WITH SOLUTION**
- 5. JEE MAIN 13 YEARS QUESTION PAPERS
WITH SOLUTION**

www.prepjee.in

To know more: sms PJ at 9852954321 | e-mail: support@prepjee.in

COURSE : PREP JEE JMTS

WWW.PREPJEE.IN

Time : 3 Hours

Max. Marks : 360

Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.

MOCK TEST 9

IMPORTANT INSTRUCTIONS

1. Immediately fill the particulars on this page of the Test Booklet with Blue / Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **360**.
5. There are three parts in the question paper A, B, C consisting of **Mathematics, Physics** and **Chemistry** having 30 questions. Mathematics & Physics part consists of **Two** sections and Chemistry consists of **One** section.
6. For each question you will be awarded **4 marks** if you darken all the bubble(s) corresponding to only the correct answer(s) and **zero mark** if no bubbles are darkened. In all other cases, **$\frac{1}{4}$ (one fourth (1)) marks will be deducted.**
7. There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 6 above.

Filling the Top-half of the ORS :

Use only Black ball point pen only for filling the ORS. Do not use Gel / Ink / Felt pen as it might smudge the ORS.

8. Write your Roll no. in the boxes given at the top left corner of your ORS with black ball point pen. Also, darken the corresponding bubbles with Black ball point pen only. Also fill your roll no on the back side of your ORS in the space provided (if the ORS is both side printed).
9. Fill your Paper Code as mentioned on the Test Paper and darken the corresponding bubble with Black ball point pen.
10. If student does not fill his/her roll no. and paper code correctly and properly, then his/her marks will not be displayed and 5 marks will be deducted (paper wise) from the total.
11. Since it is not possible to erase and correct pen filled bubble, you are advised to be extremely careful while darken the bubble corresponding to your answer.
12. Neither try to erase / rub / scratch the option nor make the Cross (X) mark on the option once filled. Do not scribble, smudge, cut, tear, or wrinkle the ORS. Do not put any stray marks or whitener anywhere on the ORS.
13. If there is any discrepancy between the written data and the bubbled data in your ORS, the bubbled data will be taken as final.

PART- A

SECTION - I

Straight Objective Type

This section contains 27 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. The set $(A \cup B \cup C) \cap (A \cap B' \cap C')' \cap C'$ equals

- (1) $A \cap C$
- (2) $B \cap C'$
- (3) $(B \cup C)'$
- (4) none of these

2. The mean of 10 numbers is 12.5 ; the mean of the first six numbers is 15 and the last five numbers is 10. The sixth number is :

- (1) 15
- (2) 12
- (3) 18
- (4) None of these

3. Consider the following argument :

"If it is a good watch then it is a Titan watch.
It is a Titan watch therefore it is a good watch". This argument is

- (1) Valid
- (2) invalid
- (3) may be valid or invalid
- (4) invalid if conditional connective is replaced by biconditional connective

4. Area bounded between the curves $xy = -2$, $x + y + 1 = 0$, $y = 2$ and y -axis is equal to -

- (1) $1 + 2\ln 2$
- (2) $2 + 2\ln 2$
- (3) $2 - 2\ln 2$
- (4) $1 - 2\ln 2$

5. The total number of integral values of a for which the equation $a^2 - 2a + 3 = |\cos x - 2|$ have real solutions for x is

- (1) 4
- (2) 6
- (3) 2
- (4) 3

(SPACE FOR ROUGH WORK)

6. If the coefficients of second, third and fourth terms in the expansion of $(1+x)^{2n}$ are in A.P., then which of the following is **TRUE**.

- (1) $n^2 - 9n + 7 = 0$
- (2) $3n^2 - 9n + 7 = 0$
- (3) $3n^2 + 9n + 7 = 0$
- (4) $2n^2 - 9n + 7 = 0$

7. Length of the common chord of the circles $(x-1)^2 + (y+1)^2 = c^2$ and $(x+1)^2 + (y-1)^2 = c^2$ is

- (1) $\frac{1}{2} \sqrt{c^2 - 2}$
- (2) $\sqrt{c^2 - 2}$
- (3) $c + 2$
- (4) $2\sqrt{c^2 - 2}$

8. If $z = \frac{1}{2}(i\sqrt{3} - 1)$, then the value of $(z - z^2 + 2z^3)(2 - z + z^2)$ is

- (1) 3
- (2) 7
- (3) -1
- (4) 5

9. The function

$$f(x) = \begin{cases} 1 - 2x + 3x^2 - 4x^3 + \dots \text{to } \infty, & x \neq -1 \\ 1 & , x = -1 \end{cases} \text{ is}$$

- (1) continuous and derivable at $x = -1$
- (2) neither continuous nor derivable at $x = -1$
- (3) continuous but not derivable at $x = -1$
- (4) none of these

10. If $I = \int_0^2 \frac{dx}{\sqrt{x + \sqrt{x + \sqrt{x + \dots \infty}}}}$, then e^{2-I} equal

- to
- (1) 10
 - (2) 1
 - (3) 2
 - (4) 5

11. If by dropping a stone in a quiet lake a wave moves in circle at a speed of 3.5 cm/sec, then the rate of increase of the enclosed circular region when the radius of the circular wave is

10 cm, is $\left(\pi = \frac{22}{7}\right)$

- (1) 220 sq. cm/sec
- (2) 110 sq. cm/sec
- (3) 35 sq. cm/sec
- (4) 350 sq. cm/sec

(SPACE FOR ROUGH WORK)

12. The solution of the equation $y^2x dx + ydx - xdy = 0$ is

(1) $x^2 + \frac{x}{y} + c = 0$ (2) $\frac{x^2}{2} - \frac{x}{y} + c = 0$

(3) $\frac{x^2}{2} + \frac{x}{y} + c = 0$ (4) $\frac{x^2}{2} - xy + c = 0$

13. Equation of circle centered at extremity of positive minor axis and passing through the focus (on positive x-axis) of the ellipse

$$\frac{x^2}{16} + \frac{y^2}{9} = 1 \text{ is}$$

(1) $x^2 + y^2 - 6y - 7 = 0$

(2) $x^2 + y^2 - 6y = 0$

(3) $x^2 + y^2 - 6y - 16 = 0$

(4) $x^2 + y^2 - 3x - 3y = 0$

14. If function $f(x) = \log_2 \left(\frac{\sin x - \cos x + 3\sqrt{2}}{\sqrt{2}} \right)$, then

(1) Domain of $f(x)$ is $\mathbb{R}$

(2) Domain of $f(x)$ is $\mathbb{R} - \{n\pi\}, n \in \mathbb{I}$

(3) Range of $f(x)$ is $[0, 1]$

(4) Range of $f(x)$ is $\mathbb{R} - [1, 2]$

15. The angle of elevation of the top of a tower standing on a horizontal plane from a point A is α . After walking a distance d towards the foot of the tower the angle of elevation is found to be β . The height of the tower is-

(1) $\frac{d}{\cot \alpha + \cot \beta}$

(2) $\frac{d}{\cot \alpha - \cot \beta}$

(3) $\frac{d}{\tan \beta - \tan \alpha}$

(4) $\frac{d}{\tan \beta + \tan \alpha}$

16. The number of real points at which line

$$3y - 2x = 14 \text{ cuts hyperbola } \frac{(x-1)^2}{9} - \frac{(y-2)^2}{4} = 5$$

is

(1) 0

(2) 1

(3) 2

(4) 4

(SPACE FOR ROUGH WORK)

17. If $\int (x^9 + x^6 + x^3)(2x^6 + 3x^3 + 6)^{1/3} dx = \frac{1}{a}(2x^9 + 3x^6 + 6x^3)^{4/3} + c$, then 'a' equal to

(1) $\frac{1}{6}$ (2) $\frac{1}{24}$

(3) 24 (4) 6

18. If $\sum_{k=1}^{100} \tan^{-1} \frac{1}{2k^2} = \tan^{-1} \left(\frac{p}{q} \right)$ where p and q are

coprime number, then $2p - q$ is

(1) 101 (2) 99

(3) 201 (4) 98

19. The value of $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{\sin x}{x - \sin x}}$ is

(1) 1 (2) -1

(3) e (4) $\frac{1}{e}$

20. If A, B and C are three square matrices of the same size such that $B = CAC^{-1}$ and $CA^3C^{-1} = B^k$. then find the value of k.

(1) 3 (2) 4

(3) 2 (4) None of these

21. From the point (15, 12) three normals are drawn to the parabola $y^2 = 4x$, then centroid of triangle formed by three co-normal points is

(1) $\left(\frac{16}{3}, 4 \right)$ (2) $\left(4, \frac{26}{3} \right)$

(3) $\left(\frac{26}{3}, 0 \right)$ (4) (6, 4)

22. A missile is destroyed before hitting the target by a probability of $\frac{1}{3}$. If the missile is not destroyed probability of hitting the target is $\frac{3}{4}$. Three missiles are fired. The probability of exactly two hits the target is

(1) $\frac{1}{16}$ (2) $\frac{3}{16}$

(3) $\frac{3}{8}$ (4) $\frac{1}{8}$

23. If $\log_2(a + b) + \log_2(c + d) \geq 4$. Then the minimum value of the expression $a + b + c + d$ is

(1) 2 (2) 4

(3) 8 (4) none of these

(SPACE FOR ROUGH WORK)

24. In $\triangle ABC$, the angles A, C, B are in A.P. Then

$\frac{a+b}{\sqrt{a^2-ab+b^2}}$ equals to

(1) $\sin \frac{A}{2}$ (2) $\frac{1}{2} \cos \frac{A+B}{2}$

(3) $\cos \frac{B}{2}$ (4) $2 \cos \left(\frac{A-B}{2} \right)$

25. Equation $3x^2 - 8xy - 3y^2 = 0$ and $x - 2y = 3$ represents the sides of a triangle which is
(1) right angled but not isosceles

(2) isosceles

(3) equilateral

(4) right angled and isosceles

26. Number of solutions to $x + y + z = 10$ where $1 \leq x, y, z \leq 6$

(1) 35 (2) 36

(3) 27 (4) 66

27. The shortest distance between the z -axis and the line, $x + y + 2z - 3 = 0$, $2x + 3y + 4z - 4 = 0$ is

(1) 1 (2) 2

(3) 3 (4) 0

SECTION - II

Reasoning Type

This section contains 3 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

28. STATEMENT -1: $\frac{d}{dx} \{ \tan^{-1}(\sec x + \tan x) \} \neq \frac{d}{dx}$

$\{ \cot^{-1}(\operatorname{cosec} x + \cot x) \}$, $x \in \left(0, \frac{\pi}{4} \right)$.

STATEMENT-2: $\sec^2 x - \tan^2 x = 1 = \operatorname{cosec}^2 x - \cot^2 x$

x if $x \in \left(0, \frac{\pi}{2} \right)$

(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True

(SPACE FOR ROUGH WORK)

29. **Statement 1** : For real x , $\frac{x^2 + 34x - 71}{x^2 + 2x - 7}$ cannot

lie in the interval (5, 9)

Statement 2 : For real constant a, b, c ; $a \neq 0$, the solutions of equation $ax^2 + bx + c = 0$ are real if $b^2 - 4ac \geq 0$ otherwise imaginary

(1) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation of Statement-1.

(2) Statement-1 is true, Statement-2 is true; Statement-2 is not the correct explanation of Statement-1.

(3) Statement-1 is false, Statement-2 is true.

(4) Statement-1 is true, Statement-2 is false.

30. **Statement 1** : $\cot \theta - \tan \theta = 2$, then $\theta = (4n + 1) \pi/8$

Statement 2 : $\sin 2x + \cos 2x + \sin x + \cos x + 1 = 0$ has no solution for 'x' in the 1st quadrant.

(1) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation of Statement-1.

(2) Statement-1 is true, Statement-2 is true; Statement-2 is not the correct explanation of Statement-1.

(3) Statement-1 is false, Statement-2 is true.

(4) Statement-1 is true, Statement-2 is false.

(SPACE FOR ROUGH WORK)

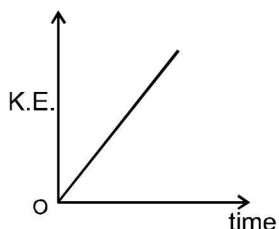
PART-B

SECTION - I

Straight Objective Type

This section contains 28 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

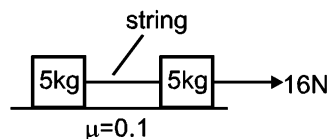
31. A body moves along a straight line and the variation of its kinetic energy with time is linear as depicted in the figure below :



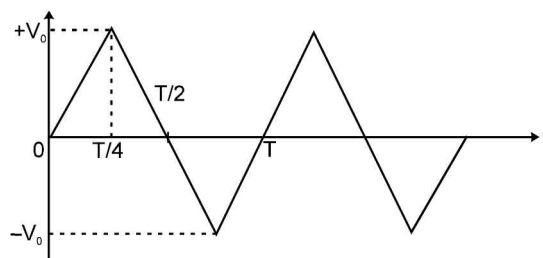
Then force acting on the body is :

- (1) directly proportional to velocity
- (2) inversely proportional to velocity
- (3) zero
- (4) constant

32. Two blocks of same mass are kept on a horizontal rough surface as shown in the figure. Coefficient of friction between the blocks and surface is 0.1. A force of 16 N is applied on one block as shown in figure. Find the tension (in Newton) in the string.



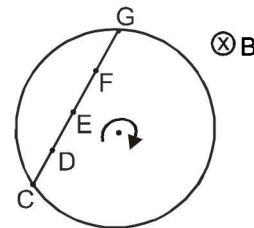
- (1) 4
 - (2) 6
 - (3) 8
 - (4) 10
33. The voltage time (V - t) graph for triangular wave having peak value. V_0 is as shown in figure. The average value of voltage V in time interval from $t = 0$ to T is :



- (1) 0
- (2) $\frac{V_0}{2}$
- (3) $\frac{V_0}{\sqrt{2}}$
- (4) None of these

(SPACE FOR ROUGH WORK)

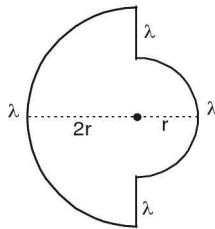
34. If the ratio of the intensity of two coherent sources is 4 then the visibility $[(I_{\max} - I_{\min}) / (I_{\max} + I_{\min})]$ of the fringes is -
 (1) 4
 (2) 4/5
 (3) 3/5
 (4) 9
35. In a hydrogen atom following the Bohr's postulates the product of linear momentum and angular momentum is proportional to $(n)^x$ where 'n' is the orbit number. Then 'x' is :
 (1) 0
 (2) 2
 (3) -2
 (4) 1
36. Consider atoms H, He^+ , Li^{++} in their ground states. Suppose E_1 , E_2 and E_3 are minimum energies required so that the atoms H, He^+ , Li^{++} can achieve their first excited states respectively, then -
 (1) $E_1 = E_2 = E_3$
 (2) $E_1 > E_2 > E_3$
 (3) $E_1 < E_2 < E_3$
 (4) $E_1 = E_2 = E_3$
37. A charged particle moves in a circular path in a uniform magnetic field. If its velocity is reduced then its time period will:
 (1) increase (2) decrease
 (3) remain same (4) none
38. A conducting disc rotates about its axis (passing through centre and perpendicular to the plane of disc) in a uniform magnetic field B and magnetic field is perpendicular to plane of disc. There are certain points marked on a chord joining two points C and G on the periphery of the disc. Point E is the midpoint of the chord. Points D and F are the midpoints of segments CE and EG respectively. The emf induced is maximum between the points: (Do not consider the emf due to centrifugal forces)



- (1) C and D
 (2) C and E
 (3) C and G
 (4) D and F

(SPACE FOR ROUGH WORK)

39. Two semicircular rings lying in same plane, of uniform linear charge density λ have radius r and $2r$. They are joined using two straight uniformly charged wires of linear charge density λ and length r as shown in figure. The magnitude of electric field at common centre of semi circular rings is -

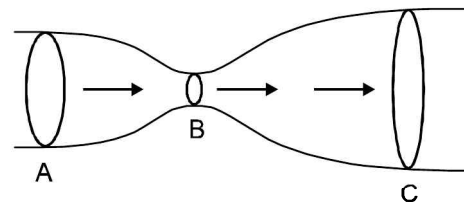


- (1) $\frac{1}{4\pi\epsilon_0} \frac{3\lambda}{2r}$
- (2) $\frac{1}{4\pi\epsilon_0} \frac{\lambda}{2r}$
- (3) $\frac{1}{4\pi\epsilon_0} \frac{2\lambda}{r}$
- (4) $\frac{1}{4\pi\epsilon_0} \frac{\lambda}{r}$

40. An object is placed at a distance of $R/2$ from the centre of earth. Knowing mass is distributed uniformly, acceleration of that object due to gravity at that point is: (g = acceleration due to gravity on the surface of earth and R is the radius of earth)

- (1) g
- (2) $2g$
- (3) $g/2$
- (4) none of these

41. Figure shows a solid resistance wire of uniform material but non-uniform cross-section area. A, B and C are the three sections. Current flowing is i , electric field at a section is E , potential is V and flux of electric field cut by a cross-section is ϕ then :



- (1) $i_B > i_A > i_C$
- (2) $E_B = E_A > E_C$
- (3) $\phi_A = \phi_B = \phi_C$
- (4) $V_B > V_A > V_C$

(SPACE FOR ROUGH WORK)

42. Moment of inertia of a uniform quarter disc of radius R and mass M about an axis through its centre of mass and perpendicular to its plane is :

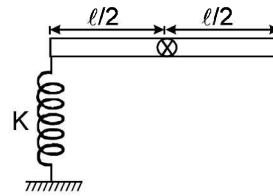
(1) $\frac{MR^2}{2} - M \left(\frac{4R}{3\pi} \right)^2$

(2) $\frac{MR^2}{2} - M \left(\sqrt{2} \frac{4R}{3\pi} \right)^2$

(3) $\frac{MR^2}{2} + M \left(\frac{4R}{3\pi} \right)^2$

(4) $\frac{MR^2}{2} + M \left(\sqrt{2} \frac{4R}{3\pi} \right)^2$

43. A uniform rod of mass m and length l is hinged at its mid point in such a way that it can rotate in the vertical plane about a horizontal axis passing through the hinge. One of its ends is attached to a spring of spring constant k which is unstretched when the rod is horizontal. If this end is now given a small displacement and released angular frequency of the resulting motion is



(1) $\sqrt{\frac{k}{m}}$

(2) $\sqrt{\frac{2k}{m}}$

(3) $\sqrt{\frac{3k}{m}}$

(4) $\sqrt{\frac{g}{l}}$

(SPACE FOR ROUGH WORK)

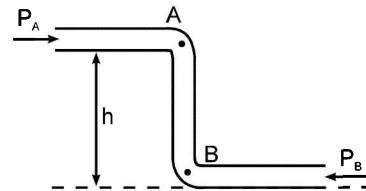
44. A uniform metal rod (fixed at both ends) of 2 mm^2 cross-section is cooled from 40°C to 20°C . The co-efficient of the linear expansion of the rod is 12×10^{-6} per degree & it's young modulus of elasticity is 10^{11} N/m^2 . The energy stored per unit volume of the rod is:

- (1) 2880 J/m^3
- (2) 1500 J/m^3
- (3) 5760 J/m^3
- (4) 1440 J/m^3

45. Two identical spherical drops of water are falling (vvertically downwards) through air with a steady velocity of 5 cm/sec . If both the drops coalesce (combine) to form a new spherical drop, the terminal velocity of the new drop will be- (neglect bouyant force on the drops.)

- (1) $5 \times 2 \text{ cm/sec}$
- (2) $5 \times \sqrt{2} \text{ cm/sec}$
- (3) $5 \times (4)^{1/3} \text{ cm/sec}$
- (4) $\frac{5}{\sqrt{2}} \text{ cm/sec.}$

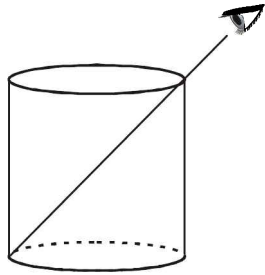
46. Figure shows an ideal fluid flowing through a uniform cross-sectional tube in the vertical tube with liquid velocities v_A & v_B and pressure P_A & P_B . Knowing that tube offers no resistance to fluid flow then which of the following is true.



- (1) $P_B > P_A$
- (2) $P_B < P_A$
- (3) $P_A = P_B$
- (4) none of these

(SPACE FOR ROUGH WORK)

47. A glass beaker has diameter 4cm wide at the bottom. An observer observes the edge of bottom when beaker is empty as shown in figure. When the beaker is completely filled with liquid of refractive index $n = \sqrt{5/2}$, he can just see the centre of bottom, then the height of glass beaker is:



- (1) 4 cm
(2) $\sqrt{5/2}$ cm
(3) 16 cm
(4) None of these

48. Two point objects are placed on principal axis of a thin converging lens. One is 20 cm from the lens and other is on the other side of lens at a distance of 40 cm from the lens. The images of both objects coincide. The magnitude of focal length of lens is

- (1) $\frac{80}{3}$ cm
(2) $\frac{40}{3}$ cm
(3) 40 cm
(4) $\frac{20}{3}$ cm

49. A force of 6.4N stretches a vertical spring by 0.1m. The mass that must be suspended from the spring so that it oscillates with a time period of $\pi/4$ second.

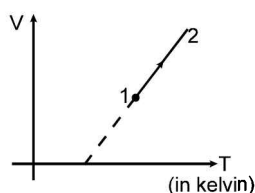
- (1) $\frac{\pi}{4}$ kg
(2) $\frac{4}{\pi}$ kg
(3) 1 kg
(4) 10 kg

(SPACE FOR ROUGH WORK)

50. Identify the correct statement :

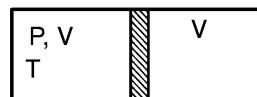
- (1) The fractional change in the time period of a pendulum on changing the temperature is independent of length of pendulum.
- (2) As the length of the simple pendulum is increased, the maximum velocity of its bob during the oscillation will also increase.
- (3) The greater the mass of pendulum bob the greater is the time period of oscillation.
- (4) When the acceleration of the lift in which spring is oscillating is increased, the time period of oscillation also increases.

51. V–T diagram for a process of a given mass of ideal gas is as shown in the figure. During the process pressure of gas.



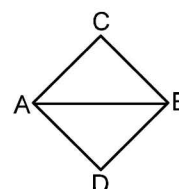
- (1) first increases then decreases
- (2) continuously decreases
- (3) continuously increases
- (4) first decreases then increases.

52. An insulated container is divided into two equal portions. One portion contains one mole of an ideal monoatomic gas at pressure P and temperature T, while the other portion is a perfect vacuum. If the intermediate partition is removed and the gas is allowed to expand, the change in the internal energy of gas is



- (1) $\frac{3}{2} RT$
- (2) PV
- (3) $\frac{RT}{2}$
- (4) Zero

53. Two identical rectangular rods of metal are welded end to end in series between temperature 0°C and 100°C and 10 J of heat is conducted (in steady state process) through the rod in 2.00 min. If 5 such rods are taken and joined as shown in figure maintaining the same temperature difference between A and B, then the time in which 20 J heat will flow through the rods is :



- (1) 30 sec.
- (2) 2 min.
- (3) 1 min.
- (4) 20 sec.

(SPACE FOR ROUGH WORK)

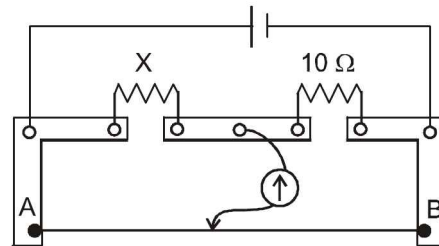
54. A string of length 1.5 m with its two ends clamped is vibrating in fundamental mode. Amplitude at the centre of the string is 4 mm. Minimum distance between the two points having amplitude 2 mm is :

- (1) 1 m
- (2) 75 cm
- (3) 60 cm
- (4) 50 cm

55. A particle is revolving in a circle of radius R with initial speed u. It starts retarding with constant retardation $\frac{v^2}{4\pi R}$. The number of revolutions it makes in time $\frac{8\pi R}{u}$ is :

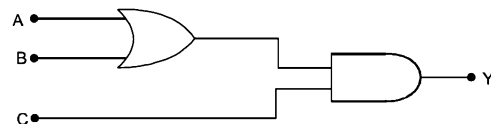
- (1) 3
- (2) 4
- (3) 2
- (4) none of these

56. A meter bridge is set-up as shown, to determine an unknown resistance 'X' using a standard 10 ohm resistor. The galvanometer shows null point when tapping-key is at 52 cm mark. The end-corrections are 1 cm and 2 cm respectively for the ends A and B. The determined value of 'X' is



- (1) 10.2 ohm
- (2) 10.6 ohm
- (3) 10.8 ohm
- (4) 11.1 ohm

57. To get an output Y = 1 in given circuit which of the following input will be correct:



- | | A | B | C |
|-----|---|---|---|
| (1) | 1 | 0 | 0 |
| (2) | 1 | 0 | 1 |
| (3) | 1 | 1 | 0 |
| (4) | 0 | 1 | 0 |

(SPACE FOR ROUGH WORK)

58. Half angular width of the central maxima in the fraunhofer diffraction due to slit width $\frac{1200}{\sqrt{2}} \mu\text{m}$ is 45° . Then wavelength of the light is:

- (1) $600 \mu\text{m}$
- (2) $1200 \mu\text{m}$
- (3) $\frac{600}{\sqrt{2}} \mu\text{m}$
- (4) $\frac{1200}{\sqrt{2}} \mu\text{m}$

SECTION - II

Reasoning Type

This section contains 2 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

59. **STATEMENT-1** : Two uniform proton beams going in the same direction repel each other.

STATEMENT-2 : Magnitude of electric force between two protons is always higher than magnetic force between them.

- (1) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1
- (2) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1
- (3) Statement-1 is True, Statement-2 is False
- (4) Statement-1 is False, Statement-2 is True.

60. **STATEMENT-1** : An ideal gas is enclosed within a container fitted with a piston. When volume of this enclosed gas is increased at constant temperature, the pressure exerted by the gas on the piston decreases.

STATEMENT-2 : In the above situation the rate of molecules striking the piston decreases. If the rate at which molecules of a gas having same average speed striking a given area of the wall decreases, the pressure exerted by gas on the wall decreases.

- (1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
- (3) Statement-1 is True, Statement-2 is False
- (4) Statement-1 is False, Statement-2 is True

(SPACE FOR ROUGH WORK)

PART- C

Atomic masses : [H = 1, D = 2, Li = 7, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, Si = 28, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Cu = 63.5, Zn = 65, As = 75, Br = 80, Ag = 108, I = 127, Ba = 137, Hg = 200, Pb = 207]

Straight Objective Type

This section contains 30 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

61. What is the activation energy for a reaction if its rate doubles when the temperature is raised from 20°C to 35°C ?

($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$, $\log 2 = 0.3010$)

- (1) 269 kJ mol⁻¹ (2) 34.7 kJ mol⁻¹
(3) 15.1 kJ mol⁻¹ (4) 342 kJ mol⁻¹

62. A hydrogen gas electrode is made by dipping platinum wire in a solution of HCl of pH = 10 and by passing hydrogen gas around the platinum wire at one atm pressure. The oxidation potential of electrode would be?

- (1) 0.59 V (2) 0.118 V
(3) 1.18 V (4) 0.059 V

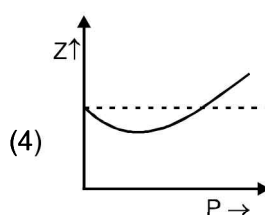
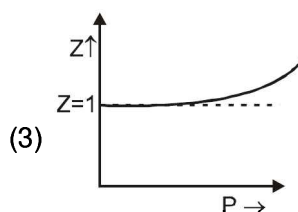
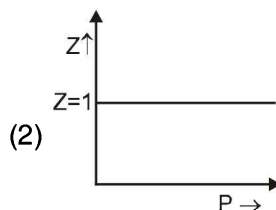
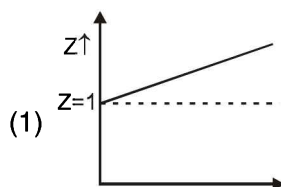
63. Which of the following structure is similar to graphite ?

- (1) B (2) B₄C
(3) B₂H₆ (4) BN

64. Select the **correct** statement :

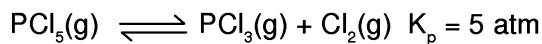
- (1) Perxenate ion is $[\text{XeO}_6]^{4-}$ with octahedral geometry.
(2) XeF₂ is bent molecule with 3 lone pairs (l.p)
(3) XeOF₄, XeF₄, XeO₂F₂ all contains one lone pair only
(4) None of these

65. At Boyle's temperature, the graph of Z v/s P is drawn select correct graph.



(SPACE FOR ROUGH WORK)

66. An equimolar mixture of PCl_3 and Cl_2 is taken. Calculate the equilibrium pressure, if 75% of PCl_3 is converted into PCl_5 ?



- (1) 60 atm (2) 75 atm
(3) 45 atm (4) 120 atm

67. Which of the following combinations of quantum numbers is allowed ?

n	l	m	m_s
(1) 3	2	1	0

(2) 2	0	0	$-\frac{1}{2}$
-------	---	---	----------------

(3) 3	-3	-2	$+\frac{1}{2}$
-------	----	----	----------------

(4) 1	0	1	$+\frac{1}{2}$
-------	---	---	----------------

68. Manganese oxide (MnO_2) is heated in stream of hydrogen to give water and new oxide Mn_xO_y . If 17.4 g MnO_2 produces 12.6 g of Mn_xO_y then y : x will be [at. wt. of Mn = 55]

(1) 1 (2) $\frac{1}{2}$

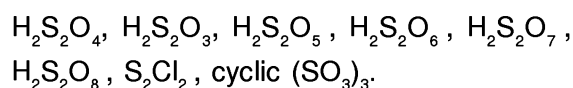
(3) $\frac{3}{2}$ (4) $\frac{5}{2}$

69. Equal volumes of 0.02 M weak base (BOH) is mixed with 0.02 M HCl. What will be $[\text{H}^+]$ concentration at equivalence point ?

$$K_{b(\text{BOH})} = 10^{-6}$$

- (1) 10^{-5} M (2) 10^{-9} M
(3) 10^{-10} M (4) $1.414 \times 10^{-5} \text{ M}$

70. In how many of the following compounds of sulphur, there is S-S bond (only single bond not double bond between two sulphur atoms).



(1) 7 (2) 6

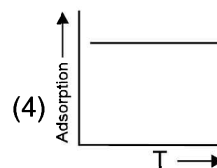
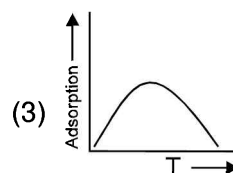
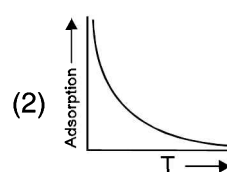
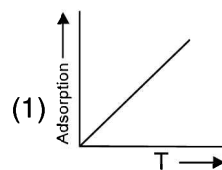
(3) 5 (4) 4

71. The edge lengths of the unit cells in terms of the radius of spheres constituting fcc, bcc and simple cubic unit cell are respectively -

(1) $2r, \frac{4r}{\sqrt{3}}, 2\sqrt{2}r$ (2) $\frac{4r}{\sqrt{3}}, 2\sqrt{2}r, 2r$

(3) $2r, 2\sqrt{2}r, \frac{4r}{\sqrt{3}}$ (4) $2\sqrt{2}r, \frac{4r}{\sqrt{3}}, 2r$

72. Following is the variation of physical adsorption with temperature :



(SPACE FOR ROUGH WORK)

73. Potassium dichromate in alkaline solution, with 30% H_2O_2 produces :

- (1) CrO_3 (2) $\text{CrO}(\text{O}_2)_2$
(3) K_3CrO_8 (4) K_2CrO_4

74. K_f for water is $1.86 \text{ K kg mol}^{-1}$. If your automobile radiator holds 0.5 Kg of water, how many grams of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) must you add to get the freezing point of the solution lowered to -2.8°C ?

- (1) 72 g (2) 46.6 g
(3) 93 g (4) 36 g

75. The incorrect expression among the following is :

- (1) $\frac{\Delta G_{\text{system}}}{\Delta S_{\text{total}}} = T$
(2) In isothermal process, $w_{\text{reversible}} = -nRT \ln \frac{V_f}{V_i}$
(3) $\ln K = -\frac{(\Delta H^\circ - T\Delta S^\circ)}{RT}$
(4) $K = e^{-\Delta G^\circ/RT}$

76. Which of the following is not correctly matched?

- (1) Sodium (ethylenediaminetetraacetato)chromate(II) – $\text{Na}_2[\text{Cr}(\text{CH}_3\text{COO})_4(\text{en})]$
(2) Dichloridobis(ethane-1, 2-diamine)cobalt(III) ion – $[\text{Co}(\text{Cl})_2(\text{en})_2]^+$
(3) Tris(bipyridyl)iron(II)ion – $[\text{Fe}(\text{NH}_4\text{C}_5\text{H}_4\text{N})_3]^{2+}$
(4) Ammineaquadibromidocopper(II) – $[\text{Cu}(\text{NH}_3)(\text{H}_2\text{O})\text{Br}_2]$

77. The entropy change can be calculated by using

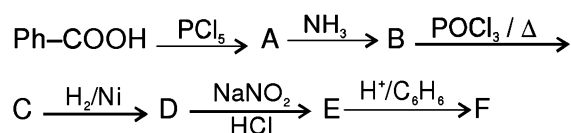
the expression $\Delta S = \frac{q_{\text{rev}}}{T}$. When water freezes in a glass beaker, choose the correct statement among the following :

- (1) $\Delta S_{(\text{system})}$ decreases but $\Delta S_{(\text{surroundings})}$ remains the same.
(2) $\Delta S_{(\text{system})}$ increases but $\Delta S_{(\text{surroundings})}$ decreases.
(3) $\Delta S_{(\text{system})}$ decreases but $\Delta S_{(\text{surroundings})}$ increases.
(4) $\Delta S_{(\text{system})}$ decreases and $\Delta S_{(\text{surroundings})}$ also decreases.

78. Van arkel process is represented by following:

- (1) $\text{Ni} + 4\text{CO} \xrightarrow{330-350 \text{ K}} [\text{Ni}(\text{CO})_4] \xrightarrow{450-470 \text{ K}} \text{Ni} + 4\text{CO}$
(2) $2\text{Al}_2\text{O}_3 + 3\text{C} \xrightarrow{\text{Electrolysis}} 4\text{Al} + 3\text{CO}_2$
(3) $\text{Ti}(\text{s}) + 2\text{I}_2 \xrightarrow{523 \text{ K}} \text{TiI}_4(\text{g}) \xrightarrow{1700 \text{ K}} \text{Ti}(\text{s}) + 2\text{I}_2(\text{g})$
(4) $2[\text{M}(\text{CN})_2]^- (\text{aq}) + \text{Zn}(\text{s}) \longrightarrow [\text{Zn}(\text{CN})_4]^{2-} (\text{aq}) + 2\text{M}(\text{s})$

79. Consider the following sequence of reaction.

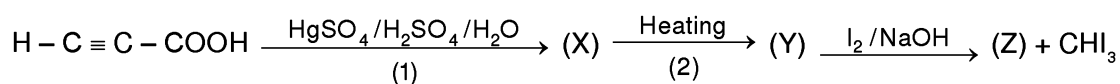


The final product E is :

- (1) Benzonitrile (2) Benzylamine
(3) Diphenylmethane (4) Benzylalcohol

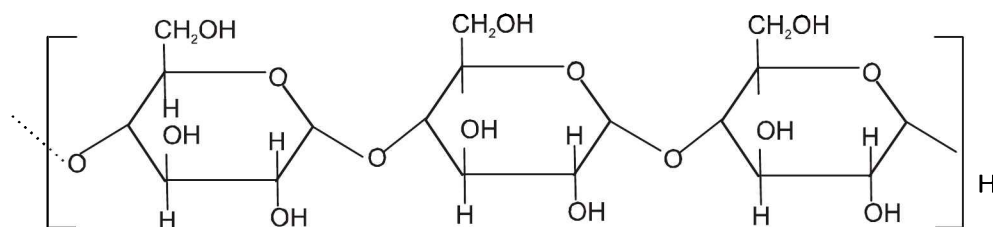
(SPACE FOR ROUGH WORK)

80. In the following reaction sequence, the correct structure of X, Y, and Z are :



	(X)	(Y)	(Z)
(1)	$\text{CH}_2 \begin{cases} \text{COOH} \\ \text{CHO} \end{cases}$	$\text{CH}_2 \begin{cases} \text{COOH} \\ \text{COOH} \end{cases}$	$\text{CH}_3 \text{COONa}$
(2)	$\text{H}_2\text{C} = \text{CH} - \text{COOH}$	$\text{CH}_3 - \underset{\text{OH}}{\text{CH}} - \text{COOH}$	$\begin{matrix} \text{COONa} \\ \\ \text{COONa} \end{matrix}$
(3)	$\text{CH}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{COOH}$	$\text{CH}_3 - \text{CH} = \text{O}$	HCOONa
(4)	$\text{O} = \text{CH} - \text{CH}_2 - \text{COOH}$	$\text{CH}_3 - \text{CH} = \text{O}$	HCOONa

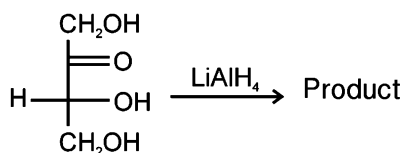
81. The following poly-saccharide has :



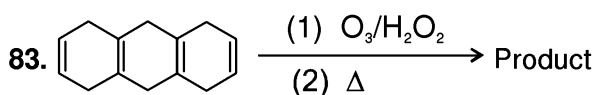
- (1) Glucose monomer and α - glycosidic linkage (2) Fructose monomer and β - glycosidic linkage
 (3) Glucose monomer and β - glycosidic linkage (4) Fructose monomer and α - glycosidic linkage

(SPACE FOR ROUGH WORK)

82. In the given reaction the product is.



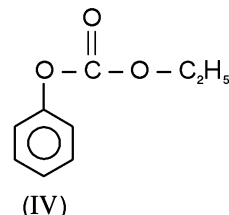
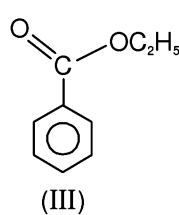
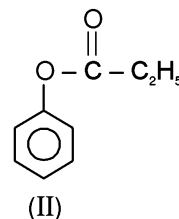
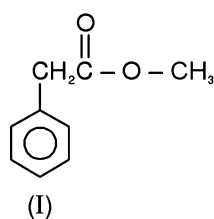
- (1) An optically inactive mixture of two compounds
- (2) one optical inactive compound and two optical active compounds in mixture
- (3) An optically inactive mixture of three compounds
- (4) An optically active mixture of three compounds



Select incorrect statement about product out of following :

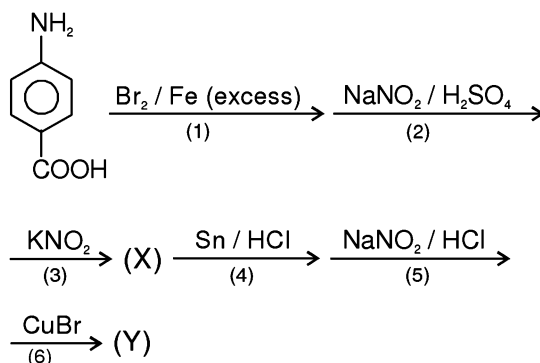
- (1) Product release four mole CO_2 gas
- (2) Product is 3,4-Dioxoheptanedioic acid
- (3) Only one product is formed having significant enol content
- (4) Product can show 2,4-DNP test positive

84. The correct reactivity order of following compounds towards nitration in aromatic ring is :



- (1) III > I > II > IV
- (2) II > IV > I > III
- (3) IV > I > II > III
- (4) IV > II > I > III

85. Observe the following synthesis.

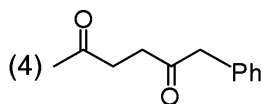
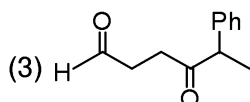
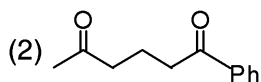
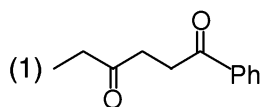
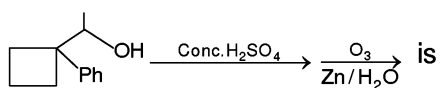


Find out sum of total Bromine atoms in (X) and (Y) :

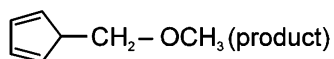
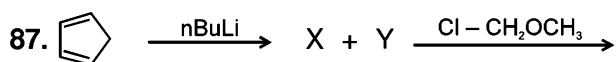
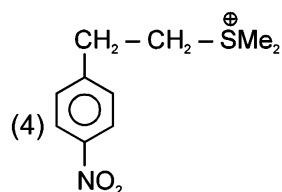
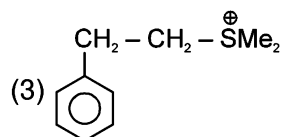
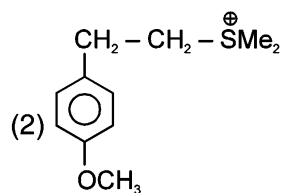
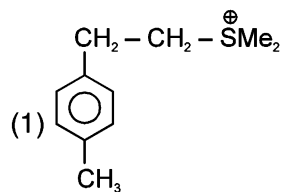
- (1) 7
- (2) 8
- (3) 5
- (4) 6

(SPACE FOR ROUGH WORK)

86. The correct product of following reaction is :



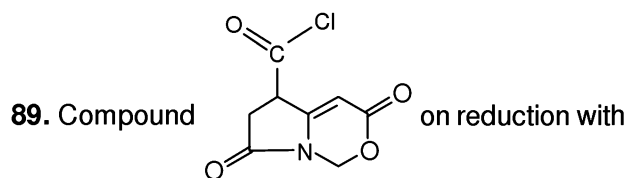
88. Which of the following substrates will undergo fastest reaction through E1cb route ?



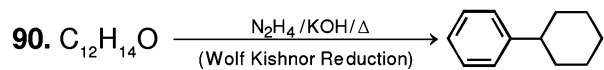
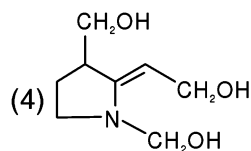
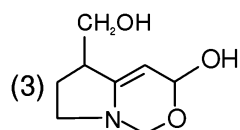
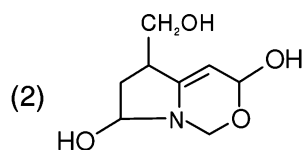
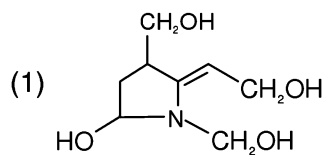
In the above reaction which of the following are incorrect.

- (1) step-1 is an acid-base reaction
- (2) step-2 is an S_N^2 reaction
- (3) X = n-Butane ; Y = aromatic compound
- (4) the nucleophile in 2nd reaction is $\text{:Bu}^\ominus$

(SPACE FOR ROUGH WORK)



LiAlH_4 produces :



How many total isomeric reactants can be converted in to given product by Wolff Kishner reduction ?

(1) 5

(2) 3

(3) 7

(4) 4

(SPACE FOR ROUGH WORK)

Name of the Candidate

--

Roll Number

--	--	--	--	--	--	--	--

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified all the information filled in by the Candidate.

Signature of the Invigilator

PREP JEE EDUSHOP

**NOW GET ALL DOCUMENTS IN
PDF FORMAT**

**PREP JEE EDUSHOP HAS
FOLLOWING TYPES OF PRODUCTS**

- 1. JEE MAIN SAMPLE PAPER**
- 2. JEE ADVANCE SAMPLE PAPER**
- 3. FORMULA BOOKS FOR JEE AND
OTHER COMPETITIVE EXAMINATION**
- 4. JEE ADVANCE 13 YEARS QUESTION PAPERS
WITH SOLUTION**
- 5. JEE MAIN 13 YEARS QUESTION PAPERS
WITH SOLUTION**

www.prepjee.in

To know more: sms PJ at 9852954321 | e-mail: support@prepjee.in

MOCK TEST 10



JEE MAIN TEST SERIES JMTS-T-10 TARGET : JEE (MAIN+ADVANCED)

PAPER



COURSE : PREP JEE JMTS

WWW.PREPJEE.IN

Time : 3 Hours

Max. Marks : 360

Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.

IMPORTANT INSTRUCTIONS

1. Immediately fill the particulars on this page of the Test Booklet with Blue / Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The Answer Sheet is kept inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars carefully.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **90** questions. The maximum marks are **360**.
5. There are three parts in the question paper A, B, C consisting of **Mathematics, Physics** and **Chemistry** having 30 questions. Mathematics & Physics part consists of **Two** sections and Chemistry consists of **One** section.
6. For each question you will be awarded **4 marks** if you darken all the bubble(s) corresponding to only the correct answer(s) and **zero mark** if no bubbles are darkened. In all other cases, **$\frac{1}{4}$ (one fourth (1)) marks will be deducted.**
7. There is only one correct response for each question. Filling up more than one response in any question will be treated as wrong response and marks for wrong response will be deducted accordingly as per instructions 6 above.

Filling the Top-half of the ORS :

- Use only Black ball point pen only for filling the ORS. Do not use Gel / Ink / Felt pen as it might smudge the ORS.**
8. Write your Roll no. in the boxes given at the top left corner of your ORS with black ball point pen. Also, darken the corresponding bubbles with Black ball point pen only. Also fill your roll no on the back side of your ORS in the space provided (if the ORS is both side printed).
 9. Fill your Paper Code as mentioned on the Test Paper and darken the corresponding bubble with Black ball point pen.
 10. If student does not fill his/her roll no. and paper code correctly and properly, then his/her marks will not be displayed and 5 marks will be deducted (paper wise) from the total.
 11. Since it is not possible to erase and correct pen filled bubble, you are advised to be extremely careful while darken the bubble corresponding to your answer.
 12. Neither try to erase / rub / scratch the option nor make the Cross (X) mark on the option once filled. Do not scribble, smudge, cut, tear, or wrinkle the ORS. Do not put any stray marks or whitener anywhere on the ORS.
 13. If there is any discrepancy between the written data and the bubbled data in your ORS, the bubbled data will be taken as final.

PART- A

SECTION - I

Straight Objective Type

This section contains 27 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

1. $\int_0^{\pi/2} \frac{1+2\cos x}{(2+\cos x)^2} dx$ is equal to

- (1) $\frac{1}{2}$
- (2) 0
- (3) -1
- (4) 3

2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be two one-one and onto functions such that they are the mirror images of each other about the line $y = a$. If $h(x) = f(x) + g(x)$, then $h(x)$ is
- (1) one-one and onto
 - (2) one-one and into
 - (3) many-one onto
 - (4) many-one and into

3. The fundamental period of $\sin \frac{2x}{3} + \cos 4x +$

$|\tan 3x| + \operatorname{sgn}(x^2 + 4x + 15)$ is $k\pi$, then the value of k is-

- (1) π
- (2) 2π
- (3) 3π
- (4) $\pi/2$

4. If $\omega (\neq 1)$ is a cube root of unity then the value of $(1 + \omega)(1 + \omega^2)(1 + \omega^3)(1 + \omega^4)(1 + \omega^5)(1 + \omega^6) \dots (1 + \omega^{30})$ equals to

- (1) 243
- (2) 1
- (3) 1024
- (4) 625

5. The graph of $f(x) = \frac{x^5}{20} - \frac{x^4}{12} + 5$ has

- (1) no local extreme, one point of inflection
- (2) Two local maximum one local minimum, two point of inflection
- (3) One local maximum, one local minimum, one point of inflection
- (4) One local maximum, one local minimum, two point of inflection

(SPACE FOR ROUGH WORK)

6. The function $f(x) = |x + 2| + |x - 1|$ is

- (1) decreasing in $(1, \infty)$
- (2) increasing in $(0, \infty)$
- (3) strictly increasing in $(-2, 1)$
- (4) increasing in $(1, \infty)$

7. The point of intersection of tangent to the curve $y = x^4$ at $(1, 1)$ with the normal to the same curve at $(-1, 1)$ is

(1) $\left(\frac{17}{15}, \frac{13}{15}\right)$

(2) $\left(\frac{7}{5}, \frac{3}{5}\right)$

(3) $\left(\frac{17}{15}, \frac{23}{15}\right)$

(4) does not exist

8. The general solution of differential equation $x(1 + y^2) dx + y(1 + x^2) dy = 0$ is

(1) $\frac{1}{2} \ln(1 + x^2)(1 + y^2) = k$

(2) $(1 + x^2)(1 + y^2) = c$

(3) $(1 + y^4) = c(1 + x^2)$

(4) $\ln[(1 + x^2)/(1 + y^2)] = k$

9. The term independent of x in the expansion of

$$(1 + x + 2x^2) \left(3x^2 - \frac{1}{3x^2}\right)^4 \text{ is}$$

(1) 10

(2) 2

(3) 0

(4) 6

10. Five distinct letters are to be transmitted through a communication channel. A total number of 15 blanks is to be inserted between the two letters with at least three between every two. the number of ways in which this can be done is

(1) 1200

(2) 1800

(3) 2400

(4) 3000

11. A problem in mathematics is given to 3 students whose chances of solving it

individually are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. The probability that

the problem will be solved at least by one is

(1) $\frac{1}{4}$

(2) $\frac{1}{24}$

(3) $\frac{23}{24}$

(4) $\frac{3}{4}$

(SPACE FOR ROUGH WORK)

12. Find the sum of 'n' terms of the series

$$\frac{3}{1^2} + \frac{5}{1^2 + 2^2} + \frac{7}{1^2 + 2^2 + 3^2} + \frac{9}{1^2 + 2^2 + 3^2 + 4^2} + \dots$$

- (1) $\frac{6}{n+1}$ (2) $\frac{6n}{n+1}$
(3) 6 (4) $\frac{6(n+1)}{n}$

13. If range of $\ell n(x^2 + x + k)$ is all real values then

- (1) $k \geq \frac{1}{4}$ (2) $k \leq \frac{1}{4}$
(3) $k = \frac{1}{4}$ only (4) $k > \frac{1}{4}$

14. The coordinates of the centroid of the triangle whose sides are $12x^2 - 20xy + 7y^2 = 0$ and $2x - 3y + 4 = 0$ are

- (1) $\left(\frac{8}{3}, \frac{-8}{3}\right)$
(2) $\left(-\frac{8}{3}, \frac{8}{3}\right)$
(3) $\left(\frac{8}{3}, \frac{8}{3}\right)$
(4) $\left(-\frac{8}{3}, -\frac{8}{3}\right)$

15. Area of the triangle formed by tangents and chord of contact of the circle $x^2 + y^2 + 6x + 8y = 0$ if tangents are drawn from the point (1, 1) is—

- (1) $\frac{340}{11}$ sq. units
(2) $\frac{360}{41}$ sq. units
(3) $\frac{320}{41}$ sq. units
(4) none of these

16. Let P, Q, R be points with position vectors

$$\vec{r}_1 = 3\hat{i} - 2\hat{j} - \hat{k}, \vec{r}_2 = \hat{i} + 3\hat{j} + 4\hat{k} \text{ and } \vec{r}_3 = 2\hat{i} + \hat{j} - 2\hat{k}$$

relative to an origin O. The distance of P from the plane OQR is (Magnitude)

- (1) 2 (2) 3
(3) 1 (4) $11/\sqrt{3}$

17. The locus of the mid point of the focal radii of a variable point moving on the parabola $y^2 = 4ax$ is a parabola whose

- (1) Latus rectum is half the latus rectum of the original parabola
(2) Vertex is $\left(\frac{a}{3}, 0\right)$
(3) Directrix is x-axis
(4) Focus has the co-ordinates (a, a)

(SPACE FOR ROUGH WORK)

18. If the mean of a set of observations $x_1, x_2, \dots, x_n$ is $\bar{x}$, then the mean of the observations $x_i + 2i$; $i = 1, 2, \dots, n$ is
- (1) $\bar{x} + 2$
 - (2) $\bar{x} + 2n$
 - (3) $\bar{x} + (n + 1)$
 - (4) $\bar{x} + n$

19. Negation of the conditional : "If it rains. I shall go to school" is
- (1) It rains and I shall go to school
 - (2) It rains and I shall not go to school
 - (3) It does not rain and I shall go to school
 - (4) None of these

20. In an equilateral triangle, inradius(r), circumradius (R) and exradius (r_1) are in
- (1) A.P.
 - (2) G.P.
 - (3) H.P.
 - (4) none of these

21. If $\cos^6 \alpha + \sin^6 \alpha + k \sin^2 2\alpha = 1$ ($0 < \alpha, < \pi/2$), then 'k' is-
- (1) $3/4$
 - (2) $1/4$
 - (3) $1/3$
 - (4) $1/8$

22. If R and R' are symmetric relations (not disjoint) on a set A , then the relation $R \cap R'$ is-
- (1) reflexive
 - (2) symmetric
 - (3) transitive
 - (4) None of these

23. $\lim_{\theta \rightarrow 0} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right)$, where $[.]$ is the greatest integer function and $n \in \mathbb{N}$, is-
- (1) $2n$
 - (2) $2n + 1$
 - (3) $2n - 1$
 - (4) does not exist

24. The function $f(x) = \begin{cases} |x - 3| & , x \geq 1 \\ \frac{x^2}{4} - \frac{3x}{2} + \frac{13}{4} & , x < 1 \end{cases}$ is -
- (1) continuous and differentiable at $x = 3$
 - (2) continuous at $x = 3$, but not differentiable at $x = 3$
 - (3) not continuous at $x = 3$, but differentiable at $x = 3$
 - (4) None of these

(SPACE FOR ROUGH WORK)

25. If $A = \sin^{-1} \left\{ \cot \left(\sin^{-1} \sqrt{\frac{13}{17}} \right) \right\}$ then

(1) $A = \tan^{-1} \frac{2}{3}$

(2) $A > \tan^{-1} \frac{2}{3}$

(3) $A < \tan^{-1} \frac{2}{3}$

(4) None of these

26. If f be a twice differentiable function such that

$f''(x) = -f(x)$ and $f'(x) = g(x)$, If $h(x) = (f(x))^2 + (g(x))^2$ & $h(5) = 9$ then $h(10)$ is equal to :

(1) -2

(2) -1

(3) 0

(4) 9

27. The line $y = x + 5$ touches

(1) the parabola $y^2 = 25x$

(2) the ellipse $4x^2 + 9y^2 = 36$

(3) the hyperbola $\frac{x^2}{29} - \frac{y^2}{4} = 1$

(4) the circle $x^2 + y^2 = 25$

SECTION - II

Reasoning Type

This section contains 3 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

28. **STATEMENT-1** : Equation $\frac{x^2}{10-a} + \frac{y^2}{4-a} = 1$

represents an ellipse if $a < 4$

and

STATEMENT-2 : Eccentricity of an ellipse is less than 1.

(1) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is a correct explanation for STATEMENT-1

(2) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is **NOT** a correct explanation for STATEMENT-1

(3) STATEMENT-1 is True, STATEMENT-2 is False

(4) STATEMENT-1 is False, STATEMENT-2 is True

(SPACE FOR ROUGH WORK)

29. STATEMENT-1 : The system of equation $x - 9y + bz = 0$, $9x + y - cz = 0$, $-bx + cy + z = 0$ does not have non-trivial solution.

and

STATEMENT-2 : If A is skew symmetric matrix, then $I - A$ is non-singular.

(1) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is a correct explanation for STATEMENT-1

(2) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 **is NOT** a correct explanation for STATEMENT-1

(3) STATEMENT-1 is True, STATEMENT-2 is False

(4) STATEMENT-1 is False, STATEMENT-2 is True

30. STATEMENT-1 : Let $\vec{r} = (\hat{i} + \hat{j}) + \lambda (2\hat{i} + 3\hat{j} - 6\hat{k})$

be a straight line, where λ is an arbitrary real number. If position vector of a point P is $(2\hat{i} + \hat{j} - \hat{k})$, then length of perpendicular from P on the given straight line is 5.

STATEMENT-2 : If a non-zero vector $\vec{a}$ is perpendicular to another non-zero vector $\vec{b}$ and $\vec{c}$ is a vector such that $\vec{c} = \vec{a} + \vec{b}$, then

$$|\vec{b}| = \frac{|\vec{c} \times \vec{a}|}{|\vec{a}|}.$$

(1) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True

(SPACE FOR ROUGH WORK)

PART-B

SECTION - I

Straight Objective Type

This section contains 28 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

31. A perfectly straight portion of a uniform rope has mass M and length L . At end A of the segment, the tension in the rope is T_A and at end B it is T_B ($T_B > T_A$). Neglect effect of gravity and no contact force acts on the rope in between points A and B. The tension in the rope at a distance $L/5$ from end A is

- (1) $T_B - T_A$
- (2) $(T_A + T_B)/5$
- (3) $(4T_A + T_B)/5$
- (4) $(T_A - T_B)/5$

32. For a LCR series circuit with an A.C. source of angular frequency ω .

(1) circuit will be capacitive if $\omega > \frac{1}{\sqrt{LC}}$

(2) circuit will be inductive if $\omega = \frac{1}{\sqrt{LC}}$

(3) power factor of circuit will be unity if capacitive reactance equals inductive reactance

(4) current will be leading voltage if

$\omega > \frac{1}{\sqrt{LC}}$

33. The distance of n^{th} bright fringe to the $(n+1)^{\text{th}}$ dark fringe in Young's experiment is equal to:

(1) $\frac{n\lambda D}{d}$

(2) $\frac{n\lambda D}{2d}$

(3) $\frac{\lambda D}{2d}$

(4) $\frac{\lambda D}{d}$

(SPACE FOR ROUGH WORK)

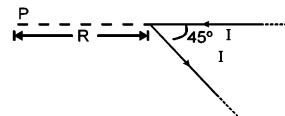
34. A nucleus with mass number 220 initially at rest emits an α particle. If the Q value of the reaction is 6.50 MeV and only one photon of energy 1.00 MeV is subsequently emitted after the emission of the α particle, the kinetic energy of the α particle is:

- (1) 5.50 MeV
- (2) 5.40 MeV
- (3) 4.50 MeV
- (4) 4.00 MeV

35. Which of the following statements is/are correct for decay processes :

- (1) In k-capture process a proton in nucleus combines with an electron and converts itself into a neutron. Antineutrino is also emitted in the process
- (2) Production of antineutrino along with electron in β^- -decay helps to explain the line spectrum
- (3) Production of neutrino along with the positron in β^+ -decay helps to explain the continuous spectrum
- (4) None of these

36. A long straight wire, carrying current I , is bent at its midpoint to form an angle of 45° . Magnetic field at point P, distance R from point of bending is equal to :



- (1) $\frac{(\sqrt{2}-1)\mu_0 I}{4\pi R}$
- (2) $\frac{(\sqrt{2}+1)\mu_0 I}{4\pi R}$
- (3) $\frac{(\sqrt{2}+1)\mu_0 I}{4\sqrt{2}\pi R}$
- (4) $\frac{(\sqrt{2}-1)\mu_0 I}{2\sqrt{2}\pi R}$

37. A wire of fixed length is wound in such a way that it forms a solenoid of length ' ℓ ' and radius ' r '. Its self inductance is found to be L . Now if same wire is wound in such

a way that it forms a solenoid of length $\frac{\ell}{2}$

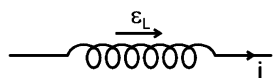
and radius $\frac{r}{2}$, then the self inductance will

be :

- (1) $2L$
- (2) L
- (3) $4L$
- (4) $8L$

(SPACE FOR ROUGH WORK)

38. At a given instant the current and self induced emf in an inductor are directed as shown in figure. If the induced emf is 17 volt and rate of change of current is 25 kA/s the correct statement is :



- (1) Current is increasing and inductance of coil is $3.4 \mu\text{H}$
 (2) Current is decreasing and inductance of coil is $680 \mu\text{H}$
 (3) Current is decreasing and inductance of coil is $3.4 \mu\text{H}$
 (4) Current is increasing and inductance of will is $6.8 \mu\text{H}$

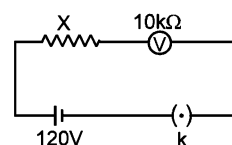
39. Two concentric conducting spheres of radii R and $2R$ carrying charges Q and $-3Q$ respectively. If the charge on inner sphere is doubled, the potential difference between two spheres will :

- (1) become two times
 (2) become four times
 (3) remain same
 (4) be halved

40. Altitude at which acceleration due to gravity decreases by 0.1% approximately: (Radius of earth = 6400 km)

- (1) 3.2 km
 (2) 6.4 km
 (3) 2.4 km
 (4) 1.6 km

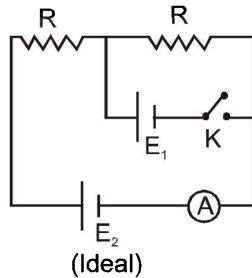
41. A D.C. supply of 120 V is connected to a large resistance X . A voltmeter of resistance $10\text{k}\Omega$ placed in series in the circuit reads 4V. This is an unusual use of voltmeter for measuring very high resistance. The value of X is :



- (1) $390 \text{ k}\Omega$
 (2) $290 \text{ k}\Omega$
 (3) $190 \text{ k}\Omega$
 (4) $300 \text{ k}\Omega$

(SPACE FOR ROUGH WORK)

42. In the given circuit, when key K is open, reading of ideal ammeter is I. Now key K is closed, then choose correct statement(s).

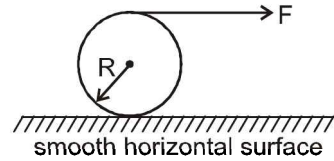


- (1) If $E_1 < IR$ reading of ammeter is less than I.
 (2) If $IR < E_1 < 2IR$ reading of ammeter is greater than I
 (3) If $E_1 = 2IR$ reading of ammeter is zero
 (4) If $E_1 = 2IR$ reading of ammeter is I

43. A uniform disc of radius $R = 0.2$ is given velocity 10m/s over a rough surface. After some time its kinetic energy becomes zero. Then initial angular velocity is :

- (1) $\frac{100}{3}$ rad/sec.
 (2) 100 rad/sec.
 (3) 50 rad/sec.
 (4) cannot be determined

44. Thread is tightly wound over a cylinder of mass M which lies on smooth horizontal surface as shown. Free end of thread is pulled by horizontal force F as shown. Then acceleration of centre of cylinder at the shown instant is



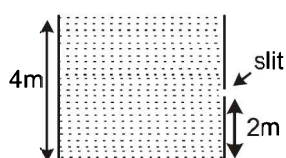
- (1) $\frac{F}{2M}$ (2) $\frac{F}{M}$
 (3) $\frac{3F}{2M}$ (4) $\frac{2F}{M}$

45. A steel rod of length 1m is heated from 25°C to 75°C keeping its length constant. The longitudinal strain developed in the rod is (Given : Coefficient of linear expansion of steel = $12 \times 10^{-6}/^\circ\text{C}$)

- (1) 6×10^{-6}
 (2) -6×10^{-5}
 (3) -6×10^{-4}
 (4) zero

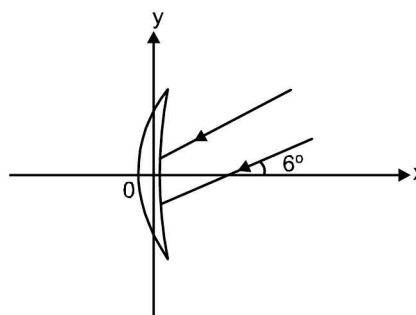
(SPACE FOR ROUGH WORK)

46. A rain drop of radius 0.3 mm falling vertically downwards in air has a terminal velocity of 1m/s. The viscosity of air is 18×10^{-5} poise. The viscous force on the drop is -
- 101.73×10^{-4} dyne
 - 101.73×10^{-5} dyne
 - 16.95×10^{-5} dyne
 - 16.95×10^{-4} dyne
47. A fixed container is filled with a liquid of density ρ up to a height 4m. A horizontal slit of small width but of area = 0.5 m^2 is made at a height of 2m from bottom. The speed of top surface of the water level is (area of top surface of container is 4 m^2 and $g = 10 \text{ m/s}^2$) :



- $\sqrt{\frac{20}{63}} \text{ m/sec}$
- $\sqrt{\frac{40}{63}} \text{ m/sec}$
- $\sqrt{\frac{80}{63}} \text{ m/sec}$
- None of these

48. For a prism kept in air it is found that for an angle of incidence 60° , the angle of refraction 'A', angle of deviation ' δ ' and angle of emergence 'e' become equal. Then the refractive index of the prism is
- 1.73
 - 1.15
 - 1.5
 - 1.33
49. Two parallel rays are incident on a thin lens of refractive index 1.5 and radius of curvature of its surfaces are 30 cm & 60 cm as shown in figure. Origin of coordinate axes is at the optical centre of the lens. Then y-coordinate of the point of focus for the two incident rays is approximately :



- $-4\pi \text{ cm}$
- zero
- 720 cm
- $-\frac{4\pi}{3} \text{ cm}$

(SPACE FOR ROUGH WORK)

50. If the length of a simple pendulum is doubled then the % change in the time period is :
- (1) 50 (2) 41.4
(3) 25 (4) 100
51. The average rotational kinetic energy of hydrogen molecule at a temperature T is E . The average translational kinetic energy of helium at same temperature will be :
- (1) $\frac{2E}{3}$
(2) $\frac{5E}{3}$
(3) E
(4) $\frac{3E}{2}$
52. In a process the pressure of an ideal gas is proportional to square of the volume of the gas. If the temperature of the gas increases in this process, then work done by this gas :
- (1) is positive
(2) is negative
(3) is zero
(4) may be positive
53. A monatomic & a diatomic gas both at the N.T.P. having same no. of moles are compressed adiabatically to half of initial volume :
- (1) Increase in temp. will be more for the diatomic gas
(2) Increase in temp. will be more for the monatomic gas
(3) Increase in temp. will be same for both the gases.
(4) None of these
54. The wave-function for a certain standing wave on a string fixed at both ends is $y(x,t) = 0.5 \sin(0.025\pi x) \cos 500t$ where x and y are in centimeters and t is in seconds. The shortest possible length of the string is :
- (1) 126 cm
(2) 160 cm
(3) 40 cm
(4) 80 cm

(SPACE FOR ROUGH WORK)

55. A particle begins to move with a tangential acceleration of constant magnitude 0.6 m/s^2 in a circular path. If it slips when its total acceleration becomes 1 m/s^2 , then the angle through which it would have turned before it starts to slip is :
- (1) $1/3 \text{ rad}$
 (2) $2/3 \text{ rad}$
 (3) $4/3 \text{ rad}$
 (4) 2 rad
56. A vernier calipers has 1 mm marks on the main scale. It has 20 equal division on the Vernier scale which match with 16 main scale divisions. For this Vernier calipers, the least count is :
- (1) 0.02 mm
 (2) 0.05 mm
 (3) 0.1 mm
 (4) 0.2 mm
57. The input resistance of a silicon transistor is 100Ω . Base current is changed by $40 \mu\text{A}$ which results in a change in collector current by 2mA. This transistor is used as a common emitter amplifier with a load resistance of $4 \text{ K}\Omega$. The voltage gain of the amplifier is :
- (1) 2000
 (2) 3000
 (3) 4000
 (4) 1000
58. A compass needle which is allowed to move in a horizontal plane is taken to a geomagnetic pole. It :
- (1) will become rigid showing no movement
 (2) will stay in any position
 (3) will stay in north-south direction only
 (4) will stay in east-west direction only

(SPACE FOR ROUGH WORK)

SECTION - II

Reasoning Type

This section contains 2 reasoning type questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

59. STATEMENT-1 : Heavy nuclides tend to have more number of neutrons than protons.

STATEMENT-2 : In heavy nuclei, as there is coulombic repulsion between protons, so excess of neutrons are preferable.

(1) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True.

60. STATEMENT-1 : Steady flow can be maintained in compressible liquids.

STATEMENT-2 : Steady flow means all physical parameters related to the flow being constant with time though they may vary with position.

(1) Statement-1 is True, Statement-2 is True; Statement-2 **is** a correct explanation for Statement-1

(2) Statement-1 is True, Statement-2 is True; Statement-2 **is NOT** a correct explanation for Statement-1

(3) Statement-1 is True, Statement-2 is False

(4) Statement-1 is False, Statement-2 is True.

(SPACE FOR ROUGH WORK)

PART- C

Atomic masses : [H = 1, D = 2, Li = 7, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, Si = 28, P = 31, S = 32, Cl = 35.5, K = 39, Ca = 40, Cr = 52, Mn = 55, Fe = 56, Cu = 63.5, Zn = 65, As = 75, Br = 80, Ag = 108, I = 127, Ba = 137, Hg = 200, Pb = 207]

Straight Objective Type

This section contains 30 questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

61. A solution of 0.2 mole KI ($\alpha = 100\%$) in 1000 g water freezes at T_1 °C. Now to this solution 0.1 mole HgI_2 is added and the resulting solution freezes at T_2 °C. Which of the following is correct :

- (1) $T_1 = T_2$
- (2) $T_1 > T_2$
- (3) $T_1 < T_2$
- (4) Can't be predicted

62. 20 ml of H_2O_2 after acidification with dil H_2SO_4

required 30 ml of $\frac{N}{12}$ KMnO_4 for complete

oxidation. The strength of H_2O_2 solution is [Molar mass of $\text{H}_2\text{O}_2 = 34$]

- (1) 2 g/L
- (2) 4 g/L
- (3) 8 g/L
- (4) 6 g/L

63. Which of the following is the **correct** order of ionisation energy :

- (1) $\text{Be}^+ > \text{Be}$
- (2) $\text{Be} > \text{Be}^+$
- (3) $\text{C} > \text{Be}$
- (4) $\text{B} > \text{Be}$
- (1) 2, 3
- (2) 3, 4
- (3) 1, 3
- (4) None of these

64. In the synthesis of sodium carbonate by Solvay (or ammonia soda) process the recovery of ammonia is done by treating NH_4Cl with Ca(OH)_2 . The by-product obtained in this process is -

- (1) CaCl_2
- (2) NaCl
- (3) NaOH
- (4) NaHCO_3

65. When conc. HNO_3 is treated with P_4O_{10} it forms

- (1) N_2O
- (2) NO
- (3) NO_2
- (4) N_2O_5

66. $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ Sodium thiosulphate is used in photography to :

- (1) remove reduced silver
- (2) remove undecomposed AgBr as soluble silver thiosulphate complex
- (3) convert the metallic silver to silver salt
- (4) reduce the silver bromide grains to metallic silver

67. Barium ions, CN^- and Co^{2+} form an ionic complex. If that is supposed to be 75% ionised in water with vant Hoff factor (i) equal to four, then the coordination numbers of Co^{2+} in the complex can be :

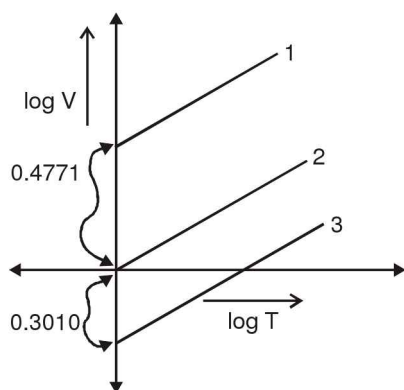
- (1) 6
- (2) 4
- (3) 6 & 4 both possible
- (4) 5

(SPACE FOR ROUGH WORK)

68. Zn Amalgam is prepared by electrolysis of aqueous ZnCl_2 using Hg cathode (9gm). How much current is to be passed through ZnCl_2 solution for 1000 seconds to prepare a Zn Amalgam with 25% Zn by wt. ($\text{Zn} = 65.4$)

- (1) 5.6 amp (2) 7.2 amp
(3) 8.85 amp (4) 11.2 amp

69. At constant P of 0.0821 atm $\log V$ v/s $\log T$ graph is plotted for 3 samples of ideal gas as shown. Value of moles of gas in these three samples is respectively :



- (1) 3, 1, $\frac{1}{2}$ (2) $\frac{1}{2}$, 1, 3
(3) 1, $\frac{1}{2}$, 3 (4) $\frac{1}{2}$, 3, 1

70. Which is false for MOT ?

- (1) HOMO is not responsible for basic character and nucleophilicity of the molecule.
(2) After passing through node sign of MO changes.
(3) In N_2 molecule electronic configuration energy levels are filled considering sp mixing.
(4) If no. of nodal planes for any MO in O_2 is zero or even then it is gerade.

71. At 1400 K, $K_c = 2.5 \times 10^{-3}$ for the reaction $\text{CH}_4(\text{g}) + 2\text{H}_2\text{S}(\text{g}) \rightleftharpoons \text{CS}_2(\text{g}) + 4\text{H}_2(\text{g})$. A 10.0L reaction vessel at 1400 K contains 2.00 mole of CH_4 , 3.0 mol of CS_2 , 3.0 mole of H_2 and 4.0 mole of H_2S . Then

- (1) The reaction will proceed in backward direction to reach equilibrium
(2) The reaction will proceed in forward direction to reach equilibrium
(3) This reaction is at equilibrium with above concentrations.
(4) The information is insufficient to decide the direction of progress of reaction

(SPACE FOR ROUGH WORK)

72. Which of the following statement is correct?

- (1) S-S bond is present in $\text{H}_2\text{S}_2\text{O}_6$.
- (2) In peroxosulphuric acid (H_2SO_5) sulphur is in +8 oxidation state.
- (3) Copper powder along with Al_2O_3 and K_2O is used as a catalyst in the preparation of NH_3 by Haber's process.
- (4) Change in enthalpy is positive for the preparation of SO_3 by catalytic oxidation of SO_2 .

73. A buffer solution is prepared in which the concentration of NH_3 is 0.30 M and the concentration of NH_4^+ is 0.20 M. If the equilibrium constant, K_b for NH_3 equals 1.8×10^{-5} , what is the pH of this solution? ($\log 2.7 = 0.433$).

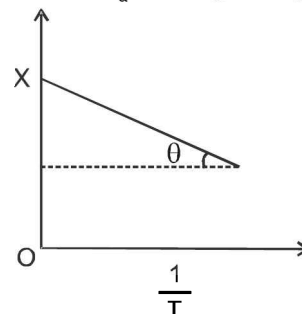
- (1) 9.08 (2) 9.43
- (3) 11.72 (4) 8.73

74. In hcp (ABAB...) and ccp (ABCABC...) structures made up of spheres of equal size, the volume occupied per sphere (including the empty spaces) is (a = radius of sphere):

- (1) $5.66 a^3$ (2) $1.33 a^3$
- (3) $2.66 a^3$ (4) $7.40 a^3$

75. Graph between $\log K$ and $\frac{1}{T}$ [Where K is rate constant (s^{-1}) and T is temperature (K)] is a straight line with $\text{OX} = 5$, $\theta = \tan^{-1} \left[-\frac{1}{2.303} \right]$

Hence E_a and $\log A$ respectively will be :



- (1) $2.303 \times 2 \text{ cal}, 5$
- (2) $\frac{2}{3.303} \text{ cal}, e^5$
- (3) 2 cal, 5
- (4) None of these

76. Pyrolusite is MnO_2 used to prepare KMnO_4 .

Steps are, $\text{MnO}_2 \xrightarrow{\text{I}} \text{MnO}_4^{2-} \xrightarrow{\text{II}} \text{MnO}_4^-$.
Steps I and II are respectively :

- (1) fuse with KOH / air and electrolytic oxidation.
- (2) fuse with KOH / KNO_3 and electrolytic reduction.
- (3) fuse with conc. HNO_3 / air and electrolytic reduction.
- (4) dissolve in H_2O and oxidation.

(SPACE FOR ROUGH WORK)

77. Which of the following ions will be most effective in coagulating the As_2S_3 sol :

- (1) Fe^{3+} (2) Ba^{2+}
(3) Cl^- (4) PO_4^{3-}

78. If $\text{HA} + \text{NaOH} \longrightarrow \text{NaA} + \text{H}_2\text{O}$

$$\Delta H = -12 \text{ kcal}$$

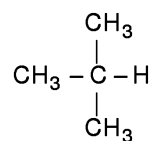
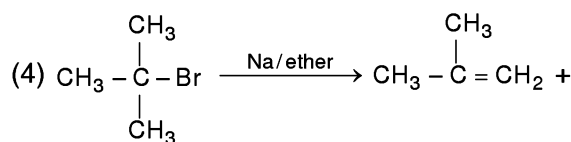
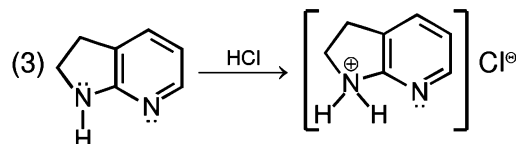
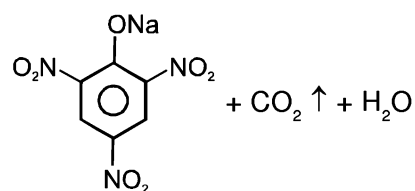
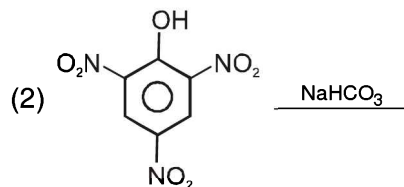
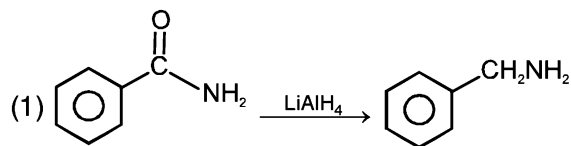
and $\text{HB} + \text{NaOH} \longrightarrow \text{NaB} + \text{H}_2\text{O}$

$$\Delta H = -11 \text{ kcal}$$

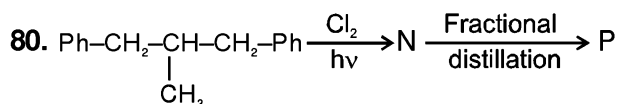
then equimolar solution of which acid has higher pH -

- (1) HA
(2) HB
(3) both have same pH
(4) information insufficient

79. Which of the following reaction is not feasible?

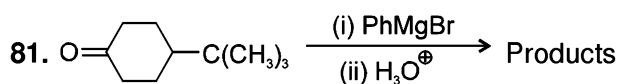


(SPACE FOR ROUGH WORK)



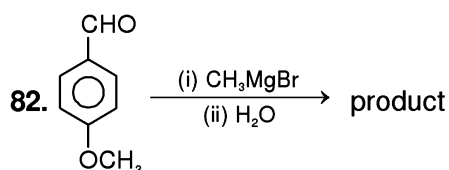
The number of possible isomers (N) & number of fractions (P) are :

- (1) 7,4 (2) 6,5
(3) 7,5 (4) 6,4



Product/s are :

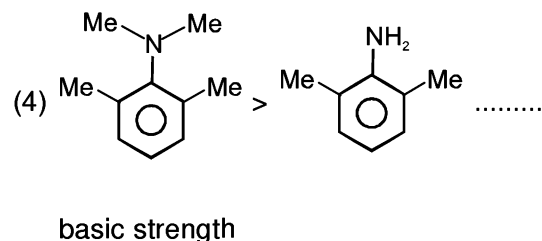
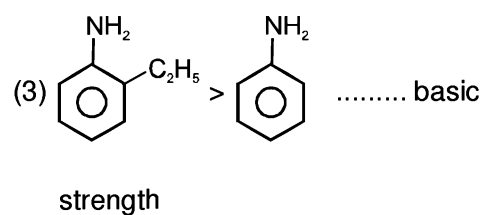
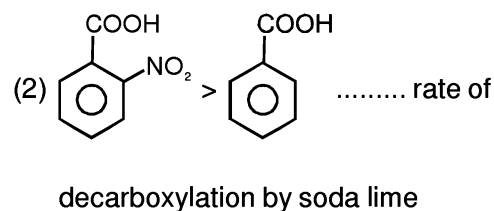
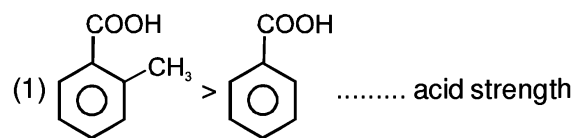
- (1) Diastereomeric alcohols and resolvable
(2) Racemic mixture of two alcohols
(3) Diastereomeric alcohols and both are tertiary alcohols
(4) Two tertiary alcohols and one of them is optically active



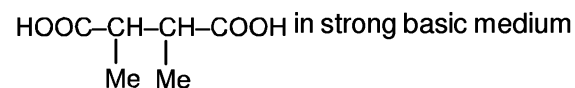
The nature of reaction is :

- (1) Chemoselective
(2) Regioselective
(3) Stereoselective
(4) A and B both

83. The incorrect order is :



84. The most stable form of meso

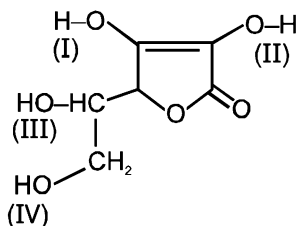


is :

- (1) Anti form
(2) Gauche form
(3) Fully eclipsed form
(4) Partially eclipsed form

(SPACE FOR ROUGH WORK)

85. In the ascorbic acid (vitamin-C),



The decreasing order of acidic strength of H will be :

(1) I > II > III > IV

(2) I > II > IV > III

(3) IV > III > I > II

(4) I > III > IV > II

86. Isobutyl alcohol and secondary butyl alcohol can be distinguished by.

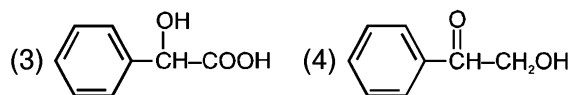
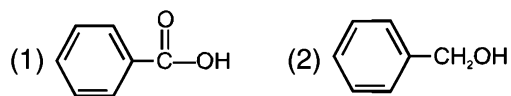
(1) Oxidation with alkaline KMnO_4 & tollen's reagent.

(2) Oxidation with acidic dichromate & tollen's reagent

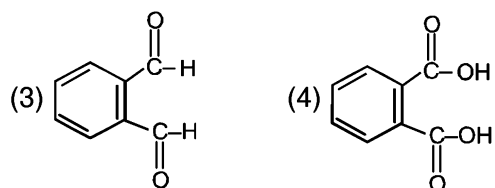
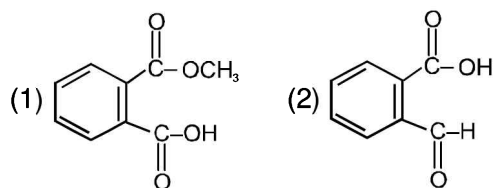
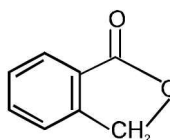
(3) Oxidation by heating with copper followed by reaction with $\text{I}_2 / \text{OH}^\ominus$.

(4) Oxidation by concentrated H_2SO_4 followed by reaction with fehling solution.

87. In the reaction, the major product is -

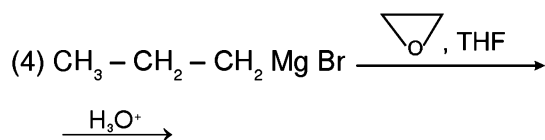
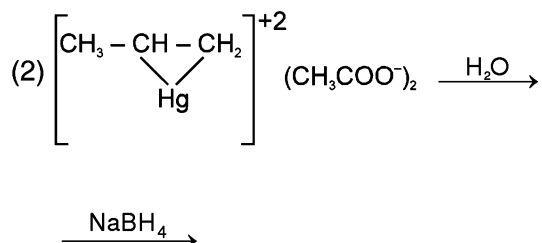
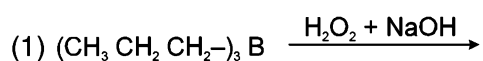


88. Which of the following reagent on reaction with conc. NaOH followed by acidification gives following lactone as the :

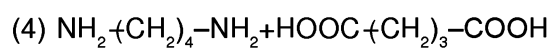
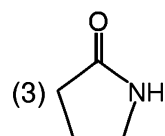
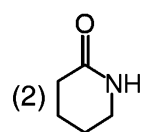
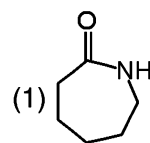
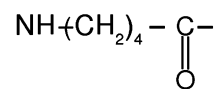
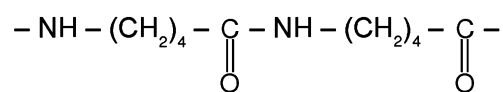


(SPACE FOR ROUGH WORK)

89. In which of the following reaction alcohol will not formed as the final product.



90. Which of the following is monomeric unit of the polymer Nylon-5 ?



(SPACE FOR ROUGH WORK)

Name of the Candidate

--

Roll Number

--	--	--	--	--	--	--	--

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified all the information filled in by the Candidate.

Signature of the Invigilator

PREPJEE EDUSHOP

**NOW GET ALL DOCUMENTS IN
PDF FORMAT**

**PREPJEE EDUSHOP HAS
FOLLOWING TYPES OF PRODUCTS**

- 1. JEE MAIN SAMPLE PAPER**
- 2. JEE ADVANCE SAMPLE PAPER**
- 3. FORMULA BOOKS FOR JEE AND
OTHER COMPETITIVE EXAMINATION**
- 4. JEE ADVANCE 13 YEARS QUESTION PAPERS
WITH SOLUTION**
- 5. JEE MAIN 13 YEARS QUESTION PAPERS
WITH SOLUTION**

www.prepjee.in

To know more: sms PJ at 9852954321 | e-mail: support@prepjee.in

DATE : Depends on User

COURSE : PREP JEE JMTS

MOCK TEST 1 - DETAILED SOLUTION

PART- A MATHEMATICS

1. If a $\triangle ABC$, $\tan B = 2$

Sol. In a $\triangle ABC$ $\sum \tan A = \prod \tan A$

$$\therefore \tan A = \frac{7}{4} \Rightarrow \sin A = \frac{7}{\sqrt{65}}$$

$$\therefore \frac{a}{\sin A} = 2R \Rightarrow R = \frac{65}{14}$$

2. The variable plane

Sol. In the give plane can be written as

$$(x + 3y + z - 4) + \lambda(2x - y) = 0$$

$$\text{i.e., } P_1 + \lambda P_2 = 0$$

$\therefore$ we have to find the equation of their line of intersection

$$\therefore P_1 : x + 3y + z - 4 = 0 \quad \therefore \ell + 3m + n = 0$$

$$P_2 : 2x - y = 0 \quad 2\ell - m + 0.n = 0$$

$$\Rightarrow \frac{\ell}{0+1} = \frac{m}{2-0} = \frac{n}{-1-6}$$

$$\Rightarrow \frac{\ell}{1} = \frac{m}{2} = \frac{n}{-7}$$

and for the point on the line let $y = 0$, we get $x = 0$ and $z = 4$

$$\therefore \text{required line is } \frac{x-0}{1} = \frac{y-0}{2} = \frac{z-4}{-7}$$

3. $\sum_{n=1}^{360} \left(\frac{1}{n\sqrt{n+1} + (n+1)\sqrt{n}} \right)$

Sol. $\therefore T_n = \frac{1}{\sqrt{n}\sqrt{n+1}(\sqrt{n+1} + \sqrt{n})} = \frac{\sqrt{n+1} - \sqrt{n}}{\sqrt{n+1}\sqrt{n}}$

$$= \frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}}$$

$$\therefore S = \sum_{n=1}^{360} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right) = 1 - \frac{1}{\sqrt{361}} = 1 - \frac{1}{19} = \frac{18}{19}$$

4. A straight line passing

Sol. Line will be perpendicular to OP

$$\therefore \text{Equation line is } y - 1 = -3(x - 3)$$

$$\Rightarrow y = -3x + 10 \quad \therefore A\left(\frac{10}{3}, 0\right) \text{ and } B(0, 10)$$

$$\therefore \text{Required area} = \frac{1}{2} \times \frac{10}{3} \times 10 = \frac{50}{3}$$

5. Let P be any

Sol. Let $P(1 + \sqrt{2} \cos \theta, \sqrt{2} \sin \theta)$ and $\therefore C(1, 0)$

circumcenter of triangle ABC is midpoint of PC

$\therefore$ Let circumcenter be (h, k)

$$\therefore 2h = 1 + \sqrt{2} \cos \theta + 1, k = \frac{\sqrt{2} \sin \theta + 0}{2}$$

$$\Rightarrow (2(h-1))^2 + (2k)^2 = 2$$

$$\Rightarrow 2(x-1)^2 + 2(y)^2 = 1$$

$$= 2x^2 + 2y^2 - 4x + 1 = 0$$

6. The length of latus

Sol. $\therefore \lambda(L.R.) = 2e(\text{distance between focus and corresponding directrix})$

$$= 2 \times \frac{1}{2} \left| \frac{-1-1+3}{\sqrt{1+1}} \right| = \frac{1}{\sqrt{2}}$$

7. Let z_1 and z_2 be n^{th}

Sol. Let $z = \left(1\right)^{\frac{1}{n}} = e^{i\left(\frac{2m\pi}{n}\right)}$, $m = 0, 1, 2, \dots, (n-1)$

$$\text{Let } z_1 = e^{i\left(\frac{2m_1\pi}{n}\right)} \text{ and } z_2 = e^{i\left(\frac{2m_2\pi}{n}\right)}$$

$\therefore$ two roots subtend a right angle at $(0, 0)$

$$\therefore \frac{2m_1\pi}{n} - \frac{2m_2\pi}{n} = \pm \frac{\pi}{2} \Rightarrow 4(m_1 - m_2) = \pm n$$

$$\Rightarrow n = 4K, K \in \mathbb{I}$$

8. If $f(x)$ is an even function

Sol. $x^2 f(x) - 2f\left(\frac{1}{x}\right) = g(x)$ (1)

replacing x by $\frac{1}{x}$, we get

$$\frac{1}{x^2} f\left(\frac{1}{x}\right) - 2f(x) = g\left(\frac{1}{x}\right)$$

$$\Rightarrow f\left(\frac{1}{x}\right) - 2x^2 f(x) = x^2 g\left(\frac{1}{x}\right)$$
 (2)

by eliminating $f\left(\frac{1}{x}\right)$ using (1) and (2) we get

$$f(x) = - \left(\frac{g(x) + 2x^2 g\left(\frac{1}{x}\right)}{3x^2} \right)$$

$\therefore g(x)$ is odd

$$\therefore f(-x) = - \left(\frac{g(x) - 2x^2 g\left(\frac{1}{x}\right)}{3x^2} \right) = -f(x)$$

$\Rightarrow f(x)$ is odd but $f(x)$ is even

$\Rightarrow f(x) = 0 \forall x \in \mathbb{R}$

$\therefore f(5) = 0$

9. Let $f: \mathbb{R} \rightarrow \left[0, \frac{\pi}{2}\right)$

Sol. codomain is $\left[0, \frac{\pi}{2}\right)$

$\therefore$ we have to find possible value (s) 'a' for which range is $\left[0, \frac{\pi}{2}\right)$

$$\therefore x^2 + x \geq -\frac{1}{4}$$

$\therefore$ if $a = \frac{1}{4}$, then only $x^2 + x + a \geq 0$

then only range will be $\left[0, \frac{\pi}{2}\right)$

10. If matrix A is given by

Sol. $\therefore |A^{2005} - 6A^{2004}| = |A|^{2004} |A - 6I|$

$$= 2^{2004} \begin{vmatrix} 0 & 11 \\ 2 & -2 \end{vmatrix} = (-22)2^{2004} = (-11)2^{2005}$$

12. Let $T = \int_0^{\ln 2} \frac{2e^{3x} + e^{2x} - 1}{e^{3x} + e^{2x} - e^x + 1} dx$

$$\text{Sol. } \therefore T = \int_0^{\ln 2} \frac{(3x^{3x} + 2e^{2x} - e^x) - (e^{3x} + e^{2x} - e^x + 1)}{(e^{3x} + 3e^{2x} - e^x + 1)} dx$$

$$[\ln(e^{3x} + e^{2x} - e^x + 1) - x]_0^{\ln 2}$$

$$= (\ln(8 + 4 - 2 + 1) - \ln 2) - (\ln 2 - 0)$$

$$= \ln \frac{11}{2} - \ln 2 = \ln \frac{11}{4} \therefore e^T = \frac{11}{4}$$

13. If $\phi(x) = \cos x - \int_0^x (x-t)\phi(t) dt$

$$\text{Sol. } \phi(x) = \cos x - \int_0^x (x-t)\phi(t) dt$$

$$= \cos x - x \int_0^x \phi(t) dt + \int_0^x t \phi(t) dt$$

$$\Rightarrow \phi'(x) = -\sin x - \int_0^x \phi(t) dt - x \phi(x) + x \phi(x)$$

$$\Rightarrow \phi''(x) = -\cos x - \phi(x)$$

$$\Rightarrow \phi(x) + \phi''(x) = -\cos x$$

14. Let the function

$$\text{Sol. } g(t) = \frac{\pi}{2} - 2\cot^{-1}(3^{-t})$$

$$\therefore g(-t) = \frac{\pi}{2} - 2\cot^{-1}(3^t) = \frac{\pi}{2} - 2\tan^{-1}(3^{-t})$$

$$= \frac{\pi}{2} - 2\left(\frac{\pi}{2} - \cot^{-1}(3^{-t})\right)$$

$$= -\frac{\pi}{2} + 2\cot^{-1}(3^{-t})$$

$$= -\left(\frac{\pi}{2} - 2\cot^{-1}(3^{-t})\right) = -g(t)$$

$\therefore g(t)$ is odd

$$\therefore g'(t) = \frac{-2(3^{-t})\ln 3}{1 + (3^{-t})^2}$$

$\therefore g'(t) < 0 \forall t \in \mathbb{R}$

$\therefore g$ is strictly decreasing in $(-\infty, \infty)$

15. Let A and B are two

Sol. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\Rightarrow \frac{2}{3} = \frac{1}{2} + P(B) - P(A) \cdot P(B) \Rightarrow P(B) = \frac{1}{3}$$

$$\therefore P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)P(B)}{P(B)} \Rightarrow P(A) = \frac{1}{2}$$

$$P(\bar{B}/A) = \frac{P(\bar{B} \cap A)}{P(A)} = \frac{P(\bar{B})P(A)}{P(A)} = 1 - \frac{1}{3} = \frac{2}{3}$$

∴ P(B), P(A/B), P($\bar{B}/A$) are in A.P.

16. Three identical red balls.....

Sol. $n(S) = \frac{9!}{3!3!3!}$

$$n(A) = {}^7C_3 \times 1 \times \frac{6!}{3!3!}$$

$$\therefore P(A) = \frac{{}^7C_3 \times \frac{6!}{3!3!}}{\frac{9!}{3!3!3!}} = \frac{5}{12}$$

$$\therefore P(\bar{A}) = \frac{7}{12}$$

$$\therefore \text{odds against A is } = \frac{P(\bar{A})}{P(A)} = 7 : 5$$

17. Let $f(x) = \dots\dots\dots$

Sol. $\therefore f(0) = \lim_{x \rightarrow 0} \frac{e^{\tan x} - e^x + \ln(\sec x + \tan x) - x}{\tan x - x} \quad \left(\frac{0}{0} \right)$

$$= \lim_{x \rightarrow 0} \frac{e^{\tan x} - e^x}{\tan x - x} + \lim_{x \rightarrow 0} \frac{\ln(\sec x + \tan x) - x}{\tan x - x}$$

$$= \lim_{x \rightarrow 0} \frac{e^x(e^{\tan x - x} - 1)}{(\tan x - x)} + \lim_{x \rightarrow 0} \frac{\ln(\sec x + \tan x) - x}{\left(\frac{\tan x - x}{x^3} \right)(x^3)}$$

$$= 1 + \lim_{x \rightarrow 0} \frac{3\ln(\sec x + \tan x) - x}{x^3}$$

$$= 1 + \lim_{x \rightarrow 0} \frac{\sec x - 1}{x^2} \quad (\text{by L.H. rule})$$

$$= 1 + \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = 1 + \frac{1}{2} = \frac{3}{2}$$

18. There are five different

Sol. Required number of ways = $\frac{5!}{3!2!} \times \frac{3!}{1!2!} = 10 \times 3 = 30$

19. If $\lim_{x \rightarrow 2} (\log_3(ax^2 + 3x + 1))^{\log_{(x-1)} 3} = \ell$

Sol. $\lim_{x \rightarrow 2} (\log_3(ax^2 + 3x + 1))$ must be equal to 1

$$\text{So, } 4a + 7 = 3 \Rightarrow a = -1$$

$$\text{Also } \ell = e^{\lim_{x \rightarrow 2} \left(\frac{\log_3(-x^2 + 3x + 1) - 1}{\log_3(x-1)} \right)} = e^{\lim_{x \rightarrow 2} \left(\frac{(x-1)(3-2x)}{(-x^2 + 3x + 1)} \right)} = e^{-1/3}$$

20. Let a function $f(x)$, $x \neq 0$

Sol. A & B are obvious

$$\left(f\left(\frac{1}{x}\right) - 1 \right) = \frac{1}{(f(x) - 1)} \text{ i.e., } \left(f\left(\frac{1}{x}\right) - 1 \right) \text{ is reciprocal of } (f(x) - 1)$$

$$\text{Now, for } f(x) = \frac{\pi/2}{\tan^{-1}|x|}, f(x) - 1 = \frac{\cot^{-1}|x|}{\tan^{-1}|x|}, f\left(\frac{1}{x}\right) - 1 = \frac{\tan^{-1}|x|}{\cot^{-1}|x|}$$

$$\text{Also for } f(x) = \frac{2}{1 + k \ln|x|}, f(x) - 1 = \frac{1 - k \ln|x|}{1 + k \ln|x|}, f\left(\frac{1}{x}\right) - 1 = \frac{1 + k \ln|x|}{1 - k \ln|x|}$$

21. Let $f(x) = ax^3 + 2bx^2 + 3cx + 4d$

Sol. $f(x) = ax^3 + 2bx^2 + 3cx + 4d$

$$f'(x) = 3ax^2 + 4bx + 3c > 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow 16b^2 - 36ac < 0 \quad \text{and} \quad a > 0$$

$$g(x) = 3a^2x^2 + 16abx + 3ac + 8b^2 + 9c^2$$

$$h'(x) = g(x)$$

$$\text{Discriminant of } g(x) = 4a^2 [40b^2 - 9ac - 27c^2]$$

$$= 4a^2 [(4b^2 - 9ac) + (36b^2 - 27c^2)] < 0$$

$$\Rightarrow h(x) \text{ is increasing and one-one function}$$

22. If P is a point (x, y) on the

Sol. Since $3.3 - 4.4 - 8 = -15 < 0$

$$\Rightarrow 3x - 4y - 8 > 0 \Rightarrow 3\left(-\frac{y}{3}\right) - 4y - 8 > 0$$

$$\Rightarrow 5y < -8 \Rightarrow y < -\frac{8}{5}$$

23. A man wants to distribute.....

Sol. Let the amount received by the sons be Rs. x, Rs. y and Rs. z respectively, then

$$x \leq y + z = 101 - x$$

$$\text{i.e., } 2x \leq 101 \therefore x \leq 50, y \leq 50, z \leq 50$$

$$x + y + z = 101$$

$$\text{The corresponding multinomial is } (1 + x + \dots + x^{50})^3$$

$$\therefore \text{coefficient of } x^{101} \text{ in the expansion of } (1 + x + \dots + x^{50})^3$$

$$= 1 \times {}^{103}C_{101} - 3 \cdot {}^{52}C_{50}$$

$$= {}^{103}C_2 - 3 \cdot {}^{52}C_2$$

24. **Statement- 1 :** A person

Sol. $x = \frac{4}{1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4}} = \frac{48}{25} = 1.92$

26. Let $0 < \alpha, \beta, \gamma < \frac{\pi}{2}$

Statement- 1 : If $\tan^3 \alpha, \tan^3 \beta, \dots$

Sol. $\tan^3 \alpha + \tan^3 \beta + \tan^3 \gamma = 6$ and $\tan^3 \alpha \tan^3 \beta \tan^3 \gamma = 8$

$$\Rightarrow \tan \alpha \tan \beta \tan \gamma = 2$$

$$\therefore \tan^3 \alpha + \tan^3 \beta + \tan^3 \gamma = 3 \tan \alpha \tan \beta \tan \gamma$$

$$\Rightarrow (\sum \tan \alpha) (\sum \tan^2 \alpha - \sum \tan \alpha \tan \beta) = 0$$

$$\therefore \sum \tan \alpha \neq 0$$

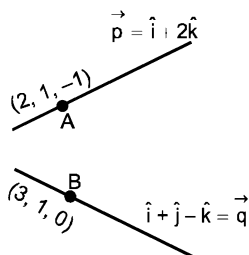
$$\therefore \sum \tan^2 \alpha - \sum \tan \alpha \tan \beta = 0$$

$$\Rightarrow \tan \alpha = \tan \beta = \tan \gamma$$

27. The shortest distance

28. Image of origin in

Sol. (27 & 28)



$$\overrightarrow{AB} = (3\hat{i} + \hat{j} - 2\hat{i} - \hat{j} + \hat{k}) =$$

$$\therefore \vec{p} \times \vec{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 2 \\ 1 & 1 & -1 \end{vmatrix} = \hat{i}(-2) - \hat{j}(-1-2) + \hat{k}(1)$$

$$= -2\hat{i} + 3\hat{j} + \hat{k}$$

$$\therefore \text{required shortest distance} = \frac{|\overrightarrow{AB} \cdot (\vec{p} \times \vec{q})|}{|\vec{p} \times \vec{q}|} = \frac{|-2+3+1|}{\sqrt{4+9+1}}$$

$$= \frac{1}{\sqrt{14}}$$

$\therefore$ equation of plane containing L_1 is

$$a(x-2) + b(y-1) + c(z+1) = 0 \quad \dots (1)$$

and $a+2c=0$ and $\therefore$ plane is parallel to L_2

$$\therefore a+b-c=0$$

$$\therefore \frac{a}{-2} = \frac{b}{3} = \frac{c}{1} = \lambda \text{ (let)}$$

$\therefore$ put $a = -2\lambda, b = 3\lambda, c = \lambda$ in (1), we get

$$-2\lambda(x-2) + 2\lambda(y-1) + \lambda(z+1) = 0$$

$$\Rightarrow -2(x-2) + 3(y-1) + z+1 = 0$$

$$\Rightarrow 2x - 3y - z - 2 = 0$$

$$\therefore \frac{x-0}{2} = \frac{y-0}{-3} = \frac{z-0}{-1} = -2 \left(\frac{0-0-0-2}{4+9+1} \right)$$

$$\therefore \text{mirror image is } \left(\frac{4}{7}, \frac{-6}{7}, \frac{-2}{7} \right)$$

29. The equation of circle

30. If P and Q are ends of a

Sol. (29 & 30)

$\therefore$ feet of $\perp r$ from focus upon any tangent to parabola lies on the tangent at the vertex

$\therefore$ required equation of circle is

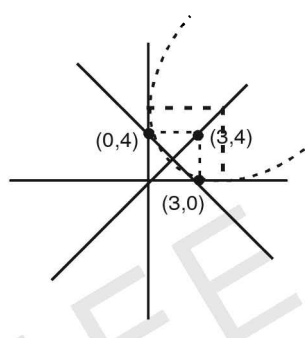
$$(x-3)(x-0) + (y-4)(y-0) = 0$$

$$x^2 - 3x + y^2 - 4y = 0$$

$$x^2 + y^2 - 3x - 4y = 0$$

$\therefore$ Equation of tangent at the vertex is

$$\frac{x}{3} + \frac{y}{4} = 1 \Rightarrow 4x + 3y - 12 = 0$$



$$\therefore l(L,R) = 4 \left| \frac{12+12-12}{\sqrt{16+9}} \right| = \left(\frac{12}{5} \right) \times 4 = \frac{48}{5} = \frac{24}{5}$$

$$\therefore \frac{2}{\frac{1}{SP} + \frac{1}{SQ}} = \frac{24}{5} \Rightarrow \frac{1}{SP} + \frac{1}{SQ} = \frac{5}{12}$$

PART- B

PHYSICS

31. Ultraviolet light of wavelength

$$\text{Sol. } E_1 = \frac{hc}{\lambda_1} - 13.6\text{eV} \quad ; \quad E_2 = \frac{hc}{\lambda_2} - 13.6\text{eV}$$

$$E_1 - E_2 = hc \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right) \quad ; \quad E_1 - E_2 = hc \left(\frac{\lambda_2 - \lambda_1}{\lambda_1 \lambda_2} \right)$$

$$h = \frac{(E_1 - E_2) \lambda_1 \lambda_2}{c(\lambda_2 - \lambda_1)}$$

32. A point source of light of power 'P' and

Sol. The energy stored in the spherical region between r and r + x is

$$= \text{energy density} \times 4\pi r^2 dx = \frac{I}{c} \times 4\pi r^2 dx = \frac{P}{4\pi r^2 c} 4\pi r^2 dx = \frac{P dx}{c}$$

$$\therefore \text{Number of photons in the given volume is} = \frac{Pdx}{h\nu c} = \frac{P\lambda dx}{hc^2}$$

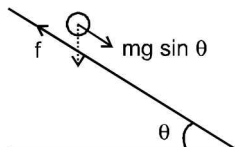
33. A small circular wire loop of radius a

Sol. The magnetic flux in inner loop due to current in outer loop is

$$\phi = BA = \left(\frac{\mu_0 2\pi i}{4\pi b} \right) \cdot \pi a^2 = \left(\mu_0 \frac{\pi a^2}{2b} \right) i_0 \cos \omega t$$

$$\therefore e = \frac{d\phi}{dt} = \mu_0 \frac{\pi a^2 \omega i_0}{2b} \sin \omega t$$

34. A uniform solid sphere of radius R



Sol.

$$mg \sin \theta - f = ma \quad \dots\dots\dots(1)$$

$$\& \quad fR = \frac{2}{5}mR^2 \frac{a}{R}$$

$$\therefore f = \frac{2}{5}ma$$

put in (1)

$$\therefore mg \sin \theta = \frac{7}{5}ma$$

$$a = \frac{5}{7}g \sin \theta$$

$$\therefore f = \frac{2}{7}mg \sin \theta$$

$$\text{Now } f \leq f_{s_{\max}}$$

$$\frac{2}{7}mg \sin \theta \leq \mu mg \cos \theta \Rightarrow \theta \leq \tan^{-1} \left(\frac{7\mu}{2} \right)$$

35. A simple pendulum is released from horizontal

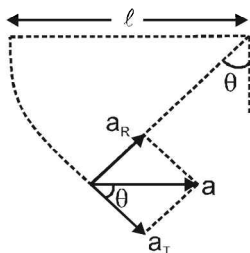
$$\text{Sol. } \tan \theta = \frac{a_R}{a_T} \quad \dots(i)$$

$$a_T = g \sin \theta \dots(ii)$$

M.E.C.

$$\frac{1}{2}mv^2 = mg\ell \cos \theta$$

$$\Rightarrow a_R = \frac{v^2}{\ell} = 2g \cos \theta \quad \dots(iii)$$



(i), (ii) and (iii)

$$\tan^2 \theta = 2$$

36. A composite wire is made by joining two.....

$$\text{Sol. For antinode } f_1 = \frac{(2n_1 - 1)}{4\ell} \sqrt{\frac{\ell}{\mu}}$$

$$f_2 = \frac{(2n_2 - 1)}{4\ell} \sqrt{\frac{\ell}{9\mu}} = \frac{(2n_2 - 1)}{12\ell} \sqrt{\frac{\ell}{\mu}}$$

$$f = \frac{1}{4\ell} \sqrt{\frac{\ell}{\mu}} \quad \frac{(2n_1 - 1)}{4\ell} = \frac{2n_2 - 1}{12\ell}$$

$$6n_1 - 3 = 2n_2 - 1$$

$$6n_1 - 2n_2 = 2$$

$$3n_1 - n_2 = 1,$$

$$n_1 = 1, n_2 = 2$$

37. A spring mass system is placed on a.....

$$\text{Sol. } \frac{1}{2}kx_1^2 + \frac{1}{2}\mu v_{\text{rel}}^2 = \frac{1}{2}kx_{\text{max}}^2$$

$$\frac{1}{2} \times 5000 \cdot \frac{1}{100} + \frac{1}{2} \times 2(5)^2 = \frac{1}{2} \times 5000 x_{\text{max}}^2$$

$$25 + 25 = 2500 x_{\text{max}}^2$$

$$\Rightarrow \frac{50}{2500} = x_{\text{max}}^2 \Rightarrow x_{\text{max}} = \frac{1}{\sqrt{50}} = \frac{1}{5\sqrt{2}} \text{ m.}$$

38. A particle is moving along straight.....

$$\text{Sol. Distance travelled} = |S_{3 \rightarrow 3.5}| + |S_{3.5 \rightarrow 4}|$$

$$= \left| \frac{1+0}{2} \times 0.5 \right| + \left| \frac{0+(-1)}{2} \times 0.5 \right| = 0.25 + 0.25 = 0.5 \text{ m.}$$

39. When an small object is placed at a distance.....

$$\text{Sol. } m = \frac{f}{f + u}$$

$$m_1 = \frac{f}{f - x_1}$$

$$m_2 = \frac{f}{f - x_2} \quad m_1 = -m_2$$

$$2f = (x_1 + x_2)$$

$$f = \frac{x_1 + x_2}{2}$$

40. Three identical bulbs each of resistance.....

$$\text{Sol. } P = i^2 R$$

$$32 = i^2 (2)$$

$$i = 4A$$

$$P_{\text{max}} = (2)^2 (2) + (2^2)2 + (4)^2 (2)$$

$$= 8 + 8 + 32 = 48 \text{ W}$$

41. A uniform electric field E is present horizontally.....

Sol. If particle move with constant velocity

$$\text{then } qvB_0 \cos\theta = mg$$

$$qvB_0 \sin\theta = qE$$

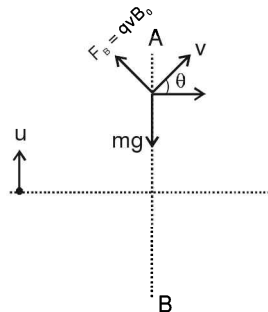
$$\tan\theta = 1 \Rightarrow \theta = 45^\circ$$

$$\text{and } v \cos\theta = \frac{qE}{m} t_0 = gt_0$$

$$v \sin\theta = u - gt_0$$

$$\tan\theta = \frac{u - gt_0}{gt_0}$$

$$u = 2gt_0$$



42. A steel wire is rigidly fixed along diameter.....

Sol. Stress = $2R [\alpha_{Al} - \alpha_{st}] \Delta\theta Y$

If aluminium ring is allowed to expand freely

so stress < $2R (\alpha_{Al}) \Delta\theta Y$

43. Four wire A, B, C and D each of length

Sol. Both open.

$$R_A = 1\Omega, R_B = 3\Omega, R_C = 6\Omega, R_D = 1\Omega$$

Current in upper circuit when S_1 is open is $I_1 = 1A$

Null point is mid point of B or C when S_2 and S_1 both open

When both closed, current in upper circuit

$$I_1 = 2A$$

Current in lower circuit $I_2 = 1A$

$$\Delta V = 2 - I_2 (1) = 1V$$

null point is mid point of A.

$\therefore$ **Ans.** (1)

44. A body cools in a surrounding which.....

Sol. For $\theta - t$ plot, rate of cooling = $\frac{d\theta}{dt}$ = slope of the curve.

$$\text{At P, } \frac{d\theta}{dt} = |\tan(180 - \phi_2)| = \tan\phi_2 = k(\theta_2 - \theta_0),$$

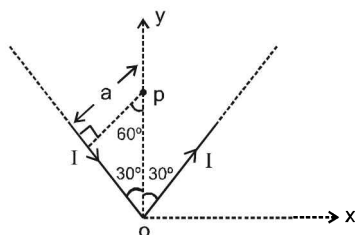
where k = constant.

$$\text{At Q, } \frac{d\theta}{dt} = |\tan(180 - \phi_1)| = \tan\phi_1 = k(\theta_1 - \theta_0).$$

$$\therefore \frac{\tan\phi_2}{\tan\phi_1} = \frac{\theta_2 - \theta_0}{\theta_1 - \theta_0}$$

45. An infinite current carrying wire, carrying

$$\text{Sol. } \vec{B} = 2 \frac{\mu_0 I}{4\pi a} [\sin 60^\circ + \sin 90^\circ] \hat{k}$$



$$\vec{B} = \frac{\mu_0 I}{4\pi a} [2 + \sqrt{3}] \hat{k}$$

46. Consider a L - R circuit shown in figure.

$$\text{Sol. } i_1 = \frac{V}{R} (1 - e^{-t/\tau})$$

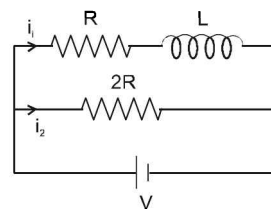
$$i_2 = \frac{V}{2R}$$

$$i_2 = i_2$$

$$\Rightarrow \frac{V}{R} (1 - e^{-t/\tau}) = \frac{V}{2R}$$

$$\Rightarrow 1 - e^{-t/\tau} = \frac{1}{2} \Rightarrow e^{-t/\tau} = \frac{1}{2}$$

$$\Rightarrow e^{t/\tau} = 2 \Rightarrow \frac{t}{\tau} = \ln 2 \Rightarrow t = \tau \ln 2 = \frac{L}{R} \ln 2$$



47. Find the heat generated after the

$$\text{Sol. } V_1 = \frac{1}{2} \left(\frac{C}{2} \right) v^2 = \frac{CV^2}{4}$$

$$Q_1 = \frac{CV}{2} \quad Q_1 = \frac{2CV}{3}$$

$$W_B = V(Q_1 - Q_1) = \frac{CV^2}{6}$$

$$V_1 = \frac{1}{2} \left(\frac{2C}{3} \right) v^2 = \frac{CV^2}{3}$$

$$V_1 + \omega_B = V_1 + H \Rightarrow \frac{CV^2}{4} + \frac{CV^2}{6} - \frac{CV^2}{3} = \frac{CV^2}{12}$$

48. An ideal gas undergoes a thermodynamics.....

Sol. Temperature at B and C is $\frac{T_0}{2}$ and T_0 respectively so change

in internal energy must be zero.

A to C

$$\Delta U = 0$$

So total work done is equal to heat given to system = $W_{A \text{ to } C}$

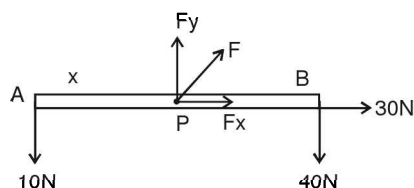
$$= -nR \frac{T_0}{2} = -P_0 V_0$$

49. The coefficient of friction between the

Sol. Friction on the block is left hand side while friction on plank its right hand side. Displacement of the both block and plank is right hand side so work done by the friction on the block is negative while on the plank is positive respectively and friction is kinetic friction so work done by internal kinetic friction must be negative.

50. A single additional force F is to be applied.....

Sol. Let P be point of application of force.



$$T_A = F_y x - 40 \times 5 = 0$$

$$F_y x = 200 \quad \dots(1)$$

$$F_y = 10 + 40 \quad \dots(2)$$

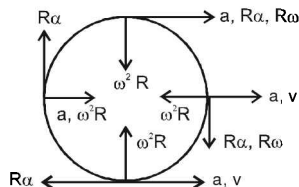
$$F_y = 50 \text{ N}, F_x = 30 \text{ N}, x = 4 \text{ m}.$$

51. Choose the right statement from.....

Sol. Options 4 is false.

52. A circular disc of radius R rolls without.....

Sol.



53. Identify correct diagram for donor and acceptor

Sol. In donor type semiconductor fermi level of donor atom is close to conduction band while in acceptor type semiconductor fermi level of acceptor atom is close to valence band.

54. STATEMENT-1 : Work done by a force

Sol. Work done depends on displacement of point of application of force and not on displacement of centre of mass. Hence statement-2 is false.

55. STATEMENT-1: Drift velocity is the average.....

Sol. Statement-1 is True, Statement-2 is False. It is because of drift velocity of electron depends on temperature.

56. Statement-1 : A neutral conducting wire.....

Sol. Statement-1 is true, Statement-2 is false

57. Initial acceleration of -10nC charge is :

$$\text{Sol. } a = \frac{2Kq_1q_2 \cos 53^\circ}{r^2 m}$$

$$= \frac{2 \times 9 \times 10^9 \times 10 \times 10^{-9} \times 5 \times 10^{-6} \times \frac{3}{5}}{(5 \times 10^{-2})^2 \times 1} = 0.216 \text{ m/s}^2.$$

58. Kinetic energy of -10nC charge

$$\text{Sol. K.E.} = 2 \times 9 \times 10^9 \times 5 \times 10^{-6} \times 10 \times 10^{-9} \left[\frac{1}{4 \times 10^{-2}} - \frac{1}{5 \times 10^{-2}} \right]$$

$$= 2 \times 9 \times 5 \times 10 \times 10^{9-6-9+2} \left[\frac{5-4}{20} \right]$$

$$= 2 \times 9 \times 5 \times \frac{10}{20} \times 10^{-4} = 4.5 \text{ mJ}.$$

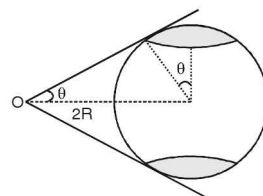
59. Find the total area on the earth, this

Sol. All area of earth will be covered other than shaded area.

$$\text{shaded area} = 2 \left[2\pi R^2 (1 - \cos \theta) \right]$$

$$= 4\pi R^2 \left[1 - \frac{\sqrt{3}}{2} \right]$$

$$= 2\pi R^2 [2 - \sqrt{3}]$$



$$\text{area covered } 4\pi R^2 - 2\pi R^2 [2 - \sqrt{3}]$$

$$= 2\pi R^2 \sqrt{3}$$

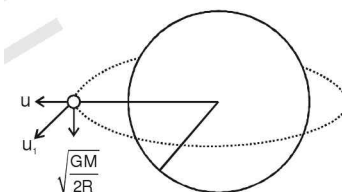
60. A particle is projected from this satellite.....

$$\text{Sol. } \frac{1}{2} m u_1^2 - \frac{GMm}{2R} = 0$$

$$v_1 = \sqrt{\frac{GM}{R}}$$

$$\frac{GM}{2R} + u^2 = \frac{GM}{R}$$

$$v = \sqrt{\frac{GM}{2R}}$$



PART- C

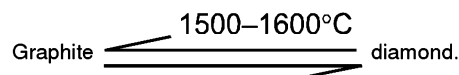
CHEMISTRY

61. During the conversion of $\text{NH}_2\text{OH} \longrightarrow \text{N}_2\text{O}$

$$\text{Sol. } E = \frac{M}{n - \text{factor}}, \text{N}^{-1} \rightarrow \text{N}^{+1} \therefore n\text{-factor} = 2.$$

62. Thermodynamically the most

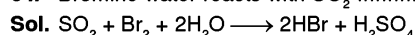
Sol. It is stable thermodynamically at low temperature.



63. If the half cell reactions are given

$$\text{Sol. } E^\circ_{\text{Cell}} = E^\circ_{\text{OP}} + E^\circ_{\text{RP}} = 0.44 + 1.23 = 1.67 \text{ V}$$

64. Bromine water reacts with SO_2



65. Given $\text{NH}_3(\text{g}) + 3\text{Cl}_2(\text{g}) \rightleftharpoons \text{NCl}_3(\text{g}) + 3\text{HCl}(\text{g})$

Sol. $-\Delta H_1 = \Delta H_{\text{f NCl}_3} + 3 \times \left(-\frac{\Delta H_3}{2} \right) - \left(-\frac{\Delta H_2}{2} \right)$

66. In the equation : $4\text{M} + 8\text{CN}^- + 2\text{H}_2\text{O} + \text{O}_2$

Sol. Metal M is Ag or Au

67. The dissociation constant of acetic

Sol. $\text{CH}_3\text{COOH}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$
 $t = 0 \quad 0.01$
 $t = \text{eq} \quad 0.01 - x \quad x \quad x$
 $[\text{H}^+] = x + 0.01 \approx 0.01 \text{ M} \quad \text{also } 0.01 - x \approx 0.01$

$\therefore K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]}$

$\Rightarrow 1.69 \times 10^{-5} = \frac{0.01 \times [\text{CH}_3\text{COO}^-]}{0.01}$

$\therefore [\text{CH}_3\text{COO}^-] = 1.69 \times 10^{-5} \text{ M}$

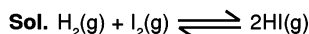
So, degree of dissociation of CH_3COOH

$= \frac{1.69 \times 10^{-5}}{0.01} = 1.69 \times 10^{-3}$

68. In an organic compound, phosphorus

Sol. Organic compound + Conc. HNO_3 + Magnesia mixture ($\text{MgSO}_4 + \text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$) $\rightarrow$ Precipitate is dried and ignited to give magnesium pyrophosphate, $\text{Mg}_2\text{P}_2\text{O}_7$.

69. If 0.5 mole H_2 is reacted with 0.5 mole



$K_c = \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} \quad \text{if } [\text{H}_2] = [\text{I}_2]$

$K_c = \frac{[\text{HI}]^2}{[\text{I}_2]^2} \quad \text{or} \quad \frac{[\text{HI}]}{[\text{I}_2]} = \sqrt{K_c} = \sqrt{49} = 7$

70. Nitrolum is a mixture

Sol. ($\text{CaCN}_2 + \text{C}$)
Nitrolum

71. Which of the following complex

Sol. F^- is WFE ligand.

72. For a reaction $\text{A} + 2\text{B} \rightarrow \text{C} + \text{D}$

Sol. $r = k [\text{A}]^x [\text{B}]^y$

$6.0 \times 10^{-3} = K (0.1)^x (0.1)^y$ (1)

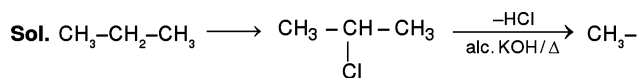
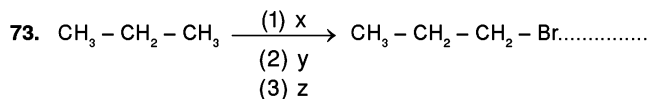
$2.4 \times 10^{-2} = K (0.1)^x (0.1)^y$ (2)

(2) / (1) gives $x = 1$

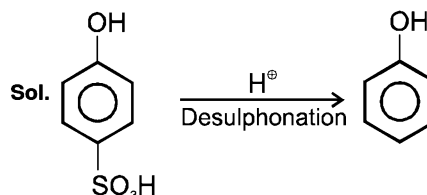
$7.2 \times 10^{-2} = K (0.3)^x (0.2)^y$ (3)

$2.88 \times 10^{-1} = K (0.3)^x (0.4)^y$ (4)

(4) / (3) gives $y = 2$.

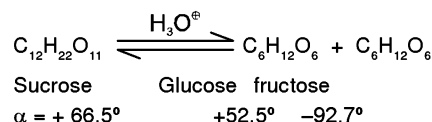


74. The aromatic electrophilic substitution

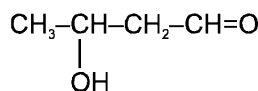
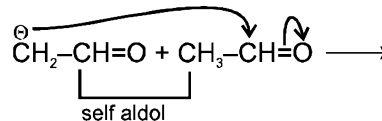
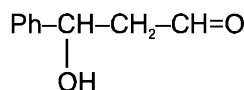
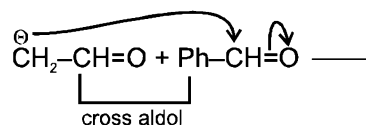
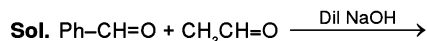


75. What is true about the following carbohydrate

Sol. It shows the phenomenon of inversion of sugar in acidic medium.



76. Identify the option which represents



77. The labels of two bottles containing compounds

Sol. Neutral FeCl_3 gives positive test with I and II so I and II cannot distinguish with FeCl_3 solution.

$\text{CHCl}_3 + \text{NaOH}$ react with both compound so they cannot distinguish.

78. The correct statement about the following

Sol. The compound is symmetrical and contains three stereocentre so total isomer is $(2)^{3-1} = 4$.

cis - R - trans and trans-S-cis are mirror image so they are enantiomers of each other.

79. The correct statement(s) pertaining to.....

Sol. (1) $\Delta H = -ve$ for adsorption

(2) fact

(4) chemical bonds are stronger than vander waal's forces so chemical adsorption is more exothermic.

80. The vapour pressure of the solution

Sol. $P_{\text{Total}} = 0.4 \times 80 + 0.6 \times 120 = 104 > 100$ mm of Hg.

81. Which of the following order is/are

Sol. All have sp^2 hybridisation i.e. same angle.

82. But-1-ene is formed in reaction

Sol. (1) Pyrolysis of esters, syn 1, 2 Elimination

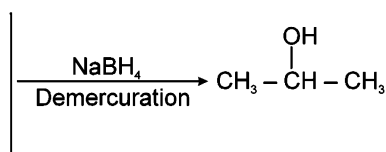
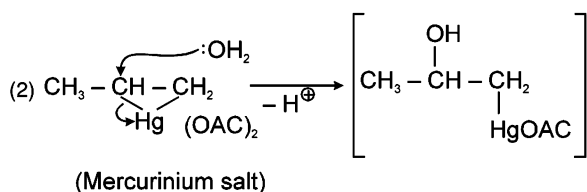
(2) Cope elimination of Trialkyl amine oxides,

(3) Hofmann elimination of quaternary ammonium hydroxide

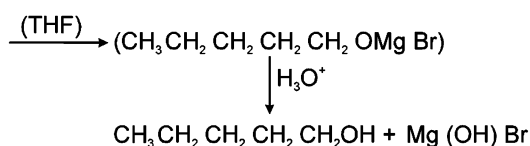
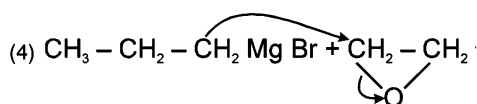
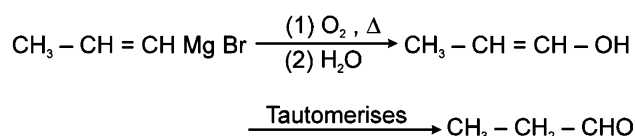
(4) Hofmann elimination of sulphonium hydroxide

83. In which of the following reaction

Sol. (1) $(\text{CH}_3\text{CH}_2\text{CH}_2)_3\text{B} \xrightarrow{\text{HO}^\ominus} 3 \text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + \text{H}_3\text{BO}_3$



(3)



84. **Statement-1** : Cr^{+3} is a reducing agent

Sol. $E^\circ_{\text{Cr}^{+3}/\text{Cr}^{+2}} = -0.41 \text{ V}$; $E^\circ_{\text{Mn}^{+3}/\text{Mn}^{+2}} = 1.57 \text{ V}$

85. **Statement-1** : For an element order of

Sol. Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

86. **Statement-1** : Benzene and ethene both give

Sol. Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

87. What is the distance between

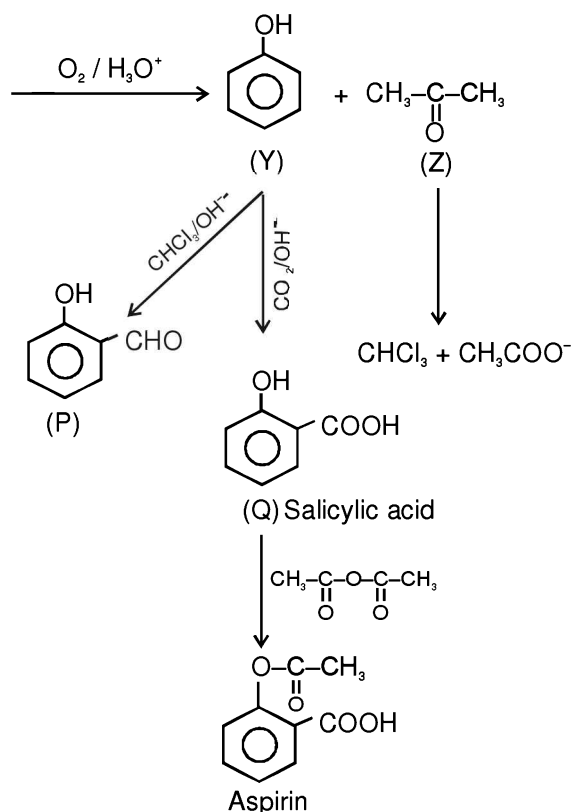
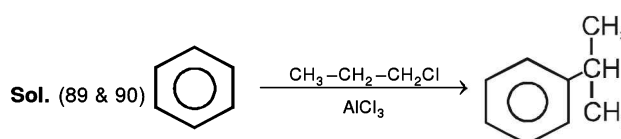
Sol. distance between nearest neighbours $= 2R = \frac{5.2 \times \sqrt{3}}{2} = 4.5 \text{ \AA}$

88. How many nearest neighbours

Sol. Coordination number (BCC) $\longrightarrow 8$.

89. Which of the following compound

90. S is a well known pain killer



PART - A MATHEMATICS

- Let $P(t, t^2)$, $Q(u, u^2)$
slope of normal at P = slope of line joining PQ

$$-\frac{1}{2t} = \frac{u^2 - t^2}{u - t} \Rightarrow u = -t - \frac{1}{2t}$$

$$PQ^2 = (t - u)^2 + (t^2 - u^2)^2 = (t - u)^2 (1 + 1(t + u)^2)$$

$$= \left(2t + \frac{1}{2t}\right)^2 \left(1 + \frac{1}{4t^2}\right)$$
Let $z = 2t$

$$PQ^2 = \left(z^2 + 2 + \frac{1}{z^2}\right) \left(1 + \frac{1}{z^2}\right) = z^2 + \frac{3}{z^2} + 3 + \frac{1}{z^4}$$

$$\frac{d(PQ^2)}{dz} = 2z - \frac{6}{z^3} - \frac{4}{z^5} = \frac{2z^6 - 6z^2 - 4}{z^5}$$

$$= \frac{2(z^6 - 3z^2 - 2)}{z^5} = \frac{2(z^2 + 1)(z^4 - z^2 - 2)}{z^5}$$

$$= \frac{2(z^2 + 1)^2(z^2 - 2)}{z^5} = 0 \Rightarrow z^2 = 2 \Rightarrow z = \pm\sqrt{2}$$

$$2t = \pm\sqrt{2}$$

$$t = \pm \frac{1}{\sqrt{2}}$$

$$PQ^2 = \left(\sqrt{2} + \frac{1}{\sqrt{2}}\right)^2 \left(1 + \frac{1}{2}\right) = \frac{9}{2} \cdot \frac{3}{2} = \frac{27}{4}$$

$$PQ = \frac{3\sqrt{3}}{2}$$
- Let $M(4, y_1)$ be mid - point
 $T = S_1$
 $yy_1 - 4(x + 4) = y_1^2 - 8 \times 4$
 $yy_1 - 4x = y_1^2 - 16$
slope = $\frac{4}{y_1} = 2 \Rightarrow y_1 = 2$

DETAILED SOLUTION MOCK TEST 2

- $$\begin{aligned} 2y - 4x &= -12 \\ y &= 2x - 6 \end{aligned}$$
- $$1947 (f(x))^{1946} f'(x) = (f(x))^{1946} \text{ (by Liebnits theorem)}$$

$$f(x) = \frac{x}{1947} + c$$

$$f(x) = \frac{x - 1857}{1947}$$

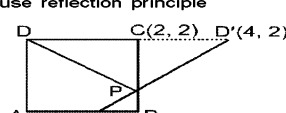
$$f(2014) = \frac{157}{1947}$$
- $$x_1 + x_2 = (1 - \sin\theta), x_1 x_2 = -\frac{1}{2} \cos^2\theta$$
Now, $x_1^2 + x_2^2 = (x_1 + x_2)^2 - 2x_1 x_2$

$$= (1 - \sin\theta)^2 + \cos^2\theta$$

$$= 2 - 2\sin\theta$$

$$\Rightarrow x_1^2 + x_2^2 \Big|_{\max} = 2 + 2 = 4.$$
- $$np = 30, npq = 5$$

$$q = \frac{1}{6}, p = \frac{5}{6} \Rightarrow n = 36$$
Required probability = ${}^{36}C_2 \left(\frac{1}{6}\right)^{34} \left(\frac{5}{6}\right)^2$

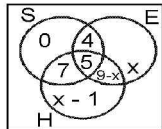
$$= \frac{36.35}{2} \frac{25}{6^{34} \times 36} = \frac{875}{2 \times 6^{34}}$$
- Equation of circle $(x^2 + y^2 - 2x - 2y + 1) + \lambda(3x + 4y - 7) = 0$
Arbitrary constant is 1
So order is 1
- We use reflection principle

Let $D'(4, 2)$ be image of D in BC
 $\Rightarrow DP + PM = D'P + PM$
Now $D'P + PM$ will be least if P is collinear with M and D'

Equation of MD' is $3y = 2x - 2$
Equation of BC is $x = 2$

$$\therefore P\left(2, \frac{2}{3}\right) \therefore \text{minimum DP + PM} = \sqrt{1 + \frac{4}{9}} + \sqrt{4 + \frac{16}{9}}$$

$$= \frac{\sqrt{13}}{3} + \frac{2\sqrt{13}}{3} = \sqrt{13}$$

8. Let x be number of employs speaking only English



$\Rightarrow$ numbers of employs speaking only English and Hindi will be $18 - 4 - 5 - x = 9 - x$
number of employs speaking only Hindi will be $20 - 7 - 5 - (9 - x) = x - 1$
 $\Rightarrow 30 = 0 + 7 + 5 + 4 + x + (9 - x) + (x - 1)$
 $6 = x$

$$\text{required probability} = \frac{14}{30} = \frac{7}{15}$$

9. In $\triangle PAB$ $\tan \alpha = \frac{x}{AP}$

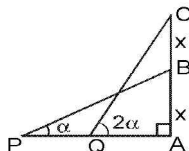
$$\text{In } \triangle QAC \tan 2\alpha = \frac{2x}{AQ}$$

$$PQ = AP - AQ$$

$$= \frac{x}{\tan \alpha} - \frac{2x}{\tan 2\alpha} = x \tan \alpha$$

$$\Rightarrow AP, PQ = \frac{x}{\tan \alpha} \cdot x \tan \alpha = x^2$$

$$\Rightarrow AP, x, PQ \text{ are in G.P.}$$



10. $(x + 2y + z - 1) + \lambda(2x + y + 3z - 2) = 0$
 $(1 + 2\lambda)x + (2 + \lambda)y + (1 + 3\lambda)z + (-1 - 2\lambda) = 0$
perpendicular to $x + y + z = 1$
 $1(1 + 2\lambda) + 2(2 + \lambda) + 1(1 + 3\lambda) = 0$
 $6\lambda + 4 = 0$

$$\lambda = -\frac{2}{3}$$

$$x - 4y + 3z - 1 = 0$$

11. Path of man is an ellipse. Let $a > b$

$$2a = 10, 2ae = 8, b^2 = a^2(1 - e^2)$$

$$a = 5, e = 4/5, b = 3$$

$$\text{area} = \pi ab = 15\pi$$

12. Reflexive h_1Rh_1 as have same pair of asymptotes
symmetric $h_1Rh_2 \Rightarrow h_2Rh_1$
transitive h_1Rh_2 and h_2Rh_3
 $\Rightarrow h_1, h_2$ have same pair of asymptotes
and h_2, h_3 have same pair of asymptotes
 $\Rightarrow h_1, h_3$ have same pair of asymptotes

13. $x^{10} + ax + b = (x - \omega)(x - \omega)(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_8)$

$$\Rightarrow (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_8) = \frac{x^{10} + ax + b}{(x - \omega)^2}$$

$$\Rightarrow (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_8)$$

$$= \lim_{x \rightarrow \omega} \frac{x^{10} + ax + b}{(x - \omega)^2} \text{ (Apply L.H rule)}$$

14. $f_1 = \{(1, 2), (2, 3), (3, 4)\}$

4 has no image under f_1 hence not function

$f_2 = \{(1, 4), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4), (4, 1), (4, 2), (4, 3)\}$

2 has more than one image under f_2 hence not a function

$f_3 = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$

1 has more than one image under f_3 hence not a function

$f_4 = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$ it is function

15. $\sim(p \vee \sim q) \equiv \sim p \wedge \sim(\sim q) \equiv \sim p \wedge q$

16. $0 < x < \frac{\pi}{6}$

$$\cos x > x$$

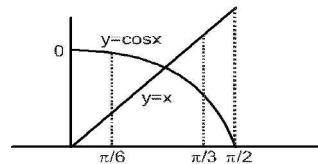
$$\Rightarrow \frac{\cos x}{x} > 1$$

$$\int_0^{\pi/6} \frac{\cos x}{x} dx > \int_0^{\pi/6} 1 dx$$

$$I > \frac{\pi}{6} ; \quad \frac{\pi}{3} < x < \frac{\pi}{2}$$

$$\cos x < x$$

$$\frac{\cos x}{x} < 1 \Rightarrow \int_{\pi/3}^{\pi/2} \frac{\cos x}{x} dx < \int_{\pi/3}^{\pi/2} 1 dx \Rightarrow J < \frac{\pi}{6}$$



17. $ax^2 + bx + c = a(x - x_1)(x - x_2)$

$$\Rightarrow \lim_{x \rightarrow x_1} (1 + \sin(ax^2 + bx + c))^{\frac{1}{x - x_1}}$$

$$= e^{\lim_{x \rightarrow x_1} \frac{\sin(a(x - x_1)(x - x_2))}{(x - x_1)}}$$

$$= e^{\lim_{x \rightarrow x_1} \frac{\sin(a(x - x_1)(x - x_2))}{a(x - x_1)(x - x_2)} a(x - x_2)} = e^{a(x_1 - x_2)}$$

18. $q - p = pq$

$$q - p - pq + 1 = 1$$

$$(q + 1) - p(1 + q) = 1$$

$$(1 + q)(1 - p) = 1$$

$$\ln(1 + q) = -\ln(1 - p)$$

$$q - \frac{q^2}{2} + \frac{q^3}{3} - \frac{q^4}{4} + \dots \infty = p + \frac{p^2}{2} + \frac{p^3}{3} + \frac{p^4}{4} + \dots \infty$$

19. $x \cdot \int \frac{dx}{1 + \sin x} - \int 1 \cdot \left(\int \frac{dx}{1 + \sin x} \right) dx$

$$\tan\left(\frac{x}{2}\right) = t \quad dx = \frac{2dt}{1 + t^2}$$

$$= x \cdot \int \frac{2dt}{(1 + t)^2} - \int 1 \cdot \left(\int \frac{2dt}{(1 + t)^2} \right) dx$$

$$= \left[x \cdot \frac{(-2)}{(1 + t)} \right] - \int \frac{-2}{1 + t} dx$$

$$= \frac{-2 \cos(x/2)}{\cos(x/2) + \sin(x/2)} \Bigg|_0^{\pi/2} + 2 \int_0^{\pi/2} \frac{\cos(x/2)}{\cos(x/2) + \sin(x/2)} dx$$

$$= \left(\frac{-2 \cdot \frac{1}{\sqrt{2}}}{\frac{\sqrt{2}}{1}} \right) - \left(\frac{-2}{1} \right) + \int_0^{\pi/2} \left(\frac{\sin \frac{x}{2} + \cos \frac{x}{2} + \cos \frac{x}{2} - \sin \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2}} \right) dx$$

$$= (-1 + 2) + \pi/2 + 2\ell n |\cos x/2 + \sin x/2|_0^{\pi/2}$$

$$= 1 + \frac{\pi}{2} + 2\ell n(\sqrt{2}) - 2\ell n(1) = 1 + \frac{\pi}{2} + \ell n 2$$

$$20. \Rightarrow \frac{\sin 8\alpha}{8\sin \alpha} = \frac{1}{8}$$

$$8\alpha = n\pi + (-1)^n \alpha$$

$$n = 2m \quad 7\alpha = 2m\pi$$

$$\Rightarrow \alpha = \frac{2m\pi}{7} : m = 1, \Rightarrow \alpha = \frac{2\pi}{7}$$

$$n = (2m + 1) \quad 8\alpha = (2m + 1)\pi - \alpha \Rightarrow \alpha = (2m + 1) \frac{\pi}{9}$$

$$m = 0, 1 \quad \alpha = \frac{\pi}{9}, \frac{2\pi}{9}$$

$\therefore$ Number of acute angles are 3

$$21. 11^n - 1 = (10 + 1)^n - 1$$

$$= {}^nC_0 10^n + {}^nC_1 10^{n-1} + \dots + {}^nC_{n-2} 10 + 1 - 1$$

$$= 10^2 (\text{some integer}) + 10n \Rightarrow n = 10$$

$$22. (x - y)dy - ydx = 0 \quad \frac{dy}{dx} = \frac{y}{x - y}$$

$$y = vx \Rightarrow v + x \frac{dv}{dx} = \frac{y}{x - y}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{v}{1 - v} ; x \frac{dv}{dx} = \frac{v - v(1 - v)}{1 - v}$$

$$\frac{1 - v}{v^2} dv = \frac{dx}{x} \Rightarrow \frac{-1}{v} = \ell n v + \ell n x + \ell n c$$

$$-\frac{x}{y} = \ell n(yc) \Rightarrow yc = e^{-x/y}$$

$$c = 1$$

$$y = e^{-x/y} \Rightarrow \frac{e}{x} = e^{-x/e}$$

$$(a, b) = (-e, e)$$

$$a + b = 0$$

$$23. ax + by + c = 0$$

$$a + c = 2b \Rightarrow a - 2b + c = 0$$

$$x = 1, y = -2$$

$$(1, -2) = (\alpha, \beta)$$

$$(x - 1)^2 + (y + 2)^2 = \gamma$$

$$\Rightarrow x^2 + y^2 - 2x + 4y + 5 - \gamma = 0$$

$$\text{it is orthogonal to } x^2 + y^2 - 4x - 4y - 1 = 0$$

$$\Rightarrow 4 - 8 = 5 - \gamma - 1$$

$$\gamma = 8$$

$$\alpha + \beta + \gamma = 1 - 2 + 8 = 7$$

$$24. (1) (AB)^T = B^T A^T$$

$$= -BA$$

$$\neq AB$$

$$(2) B^{-1} \text{ does not exist}$$

$$(3) a_{12} = b_{12}, a_{13} = b_{13}, a_{23} = b_{23}$$

$$A = \begin{bmatrix} a_{11} & b_{12} & b_{13} \\ a_{21} & a_{22} & b_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} a_{11} & b_{12} & b_{13} \\ b_{12} & a_{22} & b_{23} \\ b_{13} & b_{23} & a_{33} \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & b_{12} & b_{13} \\ -b_{12} & 0 & b_{23} \\ -b_{31} & -b_{23} & 0 \end{bmatrix}$$

$$A + B = \begin{bmatrix} a_{11} & 2b_{12} & 2b_{13} \\ 0 & a_{22} & 2b_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

$$25. \begin{vmatrix} \bar{a} \cdot \bar{a} & \bar{a} \cdot \bar{b} & \bar{a} \cdot \bar{c} \\ \bar{b} \cdot \bar{a} & \bar{b} \cdot \bar{b} & \bar{b} \cdot \bar{c} \\ \bar{c} \cdot \bar{a} & \bar{c} \cdot \bar{b} & \bar{c} \cdot \bar{c} \end{vmatrix} = \begin{vmatrix} 3 & 1 & 2 \\ 1 & 3 & -2 \\ 2 & -2 & 6 \end{vmatrix}$$

$$= ((54 - 4 - 4) - (12 + 12 + 6)) = (16)$$

$$26. 20a^s, 0b^s$$

$$a a \dots a \quad 1 \text{ ways}$$

$$19 a^s, 1 b^s \quad 20 \text{ ways}$$

$$18 a^s, 2 b^s \quad 19 \text{ ways} + 3 - 2 = 20$$

$$\vdots$$

$$11 a^s, 9 b^s \quad 12 \text{ ways} + 10 - 2 = 20$$

$$10 a^s, 10 b^s \quad 11 \text{ ways} + 11 - 2 = 20$$

$$9 a^s, 11 b^s \quad 10 \text{ ways} + 12 - 2 = 20$$

$$\vdots$$

$$1 a^s, 19 b^s \quad 20 \text{ ways}$$

$$0 a^s, 20 b^s \quad 1 \text{ ways}$$

$$\text{required} = 1 + 19 \times 20 + 1 = 382$$

$$27. x = y, xy = c^2$$

$$\Rightarrow A(c, c)$$

$$OA = c\sqrt{2}, e = \sqrt{2}$$

$$As = eOA$$

$$= 2c$$

$$S(\alpha, \alpha)$$

$$\sqrt{(\alpha - c)^2 + (\alpha - c)^2} = 2c$$

$$|\alpha - c| \sqrt{2} = 2c$$

$$\alpha - c = \sqrt{2} c \Rightarrow \alpha = c + \sqrt{2} c$$

$$S = (c + \sqrt{2} c, c + \sqrt{2} c)$$

$$x + y = 2c + 2\sqrt{2} c$$

$$28. \hat{\mu} = \frac{0 + x_2 + x_3 + \dots + x_{n-1} + (x_1 + x_n)}{n} = \mu$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum y_i^2 - \hat{\mu}^2$$

$$= \frac{1}{n} [0^2 + x_2^2 + x_3^2 + \dots + (x_{n-1})^2 + (x_1 + x_n)^2] - \mu^2$$

$$\hat{\sigma}^2 - \sigma^2 = \frac{2x_1 x_n}{n} > 0$$

$$\text{so } \hat{\sigma} > \sigma$$

$$29. \vec{r} = x\hat{i} + y\hat{j} \text{ is perpendicular to } 4\hat{i} + 3\hat{j} + \hat{k}$$

$$\text{then } 4x + 3y = 0$$

$$y = -\frac{4x}{3}$$

$$\vec{r} = x\hat{i} - \frac{4x}{3}\hat{j} = \frac{x}{3} (3\hat{i} - 4\hat{j})$$

$$30. f'(x) = \cos x - \sin x + \sec^2 x + \frac{1}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1+x^2}} > 0$$

$$\forall x \in [-1, 1]$$

PART- B

PHYSICS

31. The increase in pressure in liquid is given by buoyant force divided by the area of bottom surface.

$$\Delta P = \frac{\rho_L V g}{A} = 1 \text{ N/m}^2$$

32. As $E_p = 0$
 $\therefore q' = Q - q$
 $q' = Q/2$
 and $Q - q' = Q/2$

$$\text{Now } V_1 = V_3$$

$$V_2 - V_1 = V_2 - V_3$$

$$\frac{qa}{A\epsilon_0} = \frac{(Q-q)b}{A\epsilon_0} \Rightarrow q = \frac{Qb}{a+b}$$

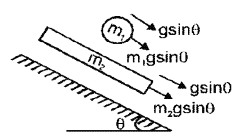
$$\therefore Q/2 = \frac{-Qb}{a+b} = \frac{Qb}{a+b} = \frac{Qa}{a+b} = Q/2$$

33. $a_c = \frac{v^2}{R} = \frac{(20)^2}{\frac{400}{\pi}} = \pi \text{ m/s}^2$

34. $\frac{m\ell^2}{3} = \frac{m\ell^2}{12} + m\left(x - \frac{\ell}{2}\right)^2 + y^2$

$$\left(x - \frac{\ell}{2}\right)^2 + y^2 = \frac{\ell^2}{3} - \frac{\ell^2}{12} = \frac{3\ell^2}{12} = \frac{\ell^2}{4}$$

35.



Neither there is relative motion nor there is tendency of relative motion.

36. Let the gravitational potential energy be zero when spring is at its natural length. Further let H be the height of the water column at natural length of spring. Now if x is the small extension in the spring.

$$\frac{1}{2} kx^2 + (\rho s x) g \left(\frac{H}{2} + \frac{x}{2} \right) + \frac{1}{2} mv^2 = C$$

$$\Rightarrow \frac{kx dx}{dt} + \rho s g x \frac{dx}{dt} + \rho s g \frac{H}{2} \frac{dx}{dt} + mv \frac{d^2 x}{dt^2} = 0$$

$$\Rightarrow \omega = \sqrt{\frac{k + \rho s g}{m}}$$

37. $A' = Ae^{-bt/2m}$

$$\frac{A}{e} = Ae^{-bt_0/2m}$$

$$\frac{bt_0}{2m} = 1$$

$$t_0 = \frac{2m}{b} = \frac{2 \times 150}{50} = 6s$$

$$E' = \frac{1}{2} KA^2 e^{-bt/m} = \frac{1}{2} KA^2 e^{-2}$$

$$\% \text{Loss} = \frac{E - E'}{E} \times 100 = \frac{1 - e^{-2}}{1} \times 100 = \frac{e^2 - 1}{e^2} \times 100$$

38. Magnetic field is increasing with constant rate and its direction is upwards the plane of paper. So induced current will be constant from B to A clockwise

39. Pitch = $\frac{1}{2} \text{ mm}$

$$\text{least count} = \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}$$

$$\text{Reading} = \left[2 \times \frac{1}{2} + 20 \times 0.01 - (-30 \times 0.01) \right] = 1.5 \text{ mm}$$

40. $\Delta \theta = \frac{1.22 \lambda}{a} = \frac{1.22 \times 6 \times 10^{-5}}{20 \text{ cm}} \text{ cm} = 0.366 \times 10^{-5} \text{ rad}$

so minimum separation between the stars
 $= D \Delta \theta = 7.32 \times 10^{-5} \text{ light years.}$

41. Velocity of bike $V_b = 30 \text{ m/s}$
 Let velocity of car be V_c

$$f_{\text{obs}} = f_{\text{actual}} \times \frac{V_{\text{so}} + V_b}{V_{\text{so}} - V_c}$$

$$120 \text{ Hz} = 100 \text{ Hz} \times \frac{330 + 30}{330 - V_c} \Rightarrow V_c = 30 \text{ m/s}$$

Distance between initial and final pulse of horn

$$= V_{\text{so}} \times \Delta t - V_c \Delta t$$

(Δt is duration of emission of sound)

$$= (330 - 30) \times 6 = 1800 \text{ m}$$

This wave train travels with 330 m/s and cross bike rider moving with velocity of 30 m/s towards it.

$$\text{time taken} = \frac{\text{length of wave train}}{V_{\text{so}} + V_b} = \frac{1800}{330 + 30} = \frac{180}{36} = 5 \text{ sec}$$

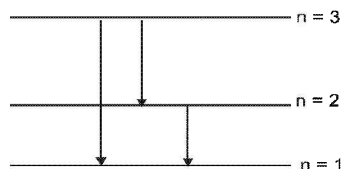
42. $\frac{Q_{\text{in}}}{W} = \frac{1}{\frac{T_H}{T_L} - 1} = \frac{T_L}{T_H - T_L} = \frac{270}{30} = 9$

$$W = \frac{Q_{\text{in}}}{9} = \frac{1260}{9} \text{ KJ/min} = \frac{1260}{9 \times 60} \text{ KJ/sec.}$$

$$= \frac{21}{9} = \frac{7}{3} \text{ KJ/sec} = 2.33 \text{ KJ/sec}$$

43. $R_n = 3 a_0$

$$L = mvr = \frac{3h}{2\pi} \quad [\text{as } n = 3, z = 3]$$



$$\frac{1}{\lambda} = R_z^2 \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

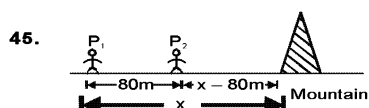
$$\frac{1}{\lambda_{3 \rightarrow 1}} = R9 \left[\frac{1}{1} - \frac{1}{9} \right] = 9R \frac{8}{9}$$

$$\frac{1}{\lambda_{2 \rightarrow 1}} = R9 \left[\frac{1}{1} - \frac{1}{4} \right] = \frac{3}{4} 9R$$

$$\frac{1}{\lambda_{3 \rightarrow 2}} = R9 \left[\frac{1}{4} - \frac{1}{9} \right] = \frac{5}{36} 9R$$

44. For maximum x coordinate
 $0 = 3V_0 - 2a_1 t$

$$t = \frac{3V_0}{2a_1}$$



$$\frac{2x}{V} = \frac{60}{40} \Rightarrow x = \frac{3V}{4} \quad \dots(1)$$

$$\frac{2(x-80)}{V} = 1 \Rightarrow \text{Put } V \text{ from (1)}$$

$$\therefore 2x - 160 = 4x/3$$

$$2x/3 = 160$$

$$x = 240 \text{ m}$$

46. $I = 2x^2 - 12x + 27$

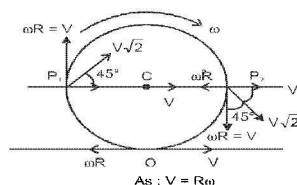
$$\frac{dI}{dx} = 4x - 12$$

$$\text{For minimum or maximum } \frac{dI}{dx} = 0 \Rightarrow x = 3$$

$$\frac{d^2I}{dx^2} = 4 \text{ Hence } I \text{ is minimum at } x = 3$$

$$I = 9 \text{ kg m}^2$$

47. Angle between velocity vectors and acceleration vector is either 45° at P_1 and 135° at P_2



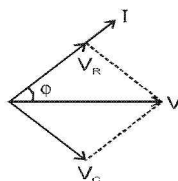
48. $V = \sqrt{V_R^2 + V_C^2}$

$$10 = \sqrt{(8)^2 + V_R^2}$$

$$100 - 64 = V_R^2$$

$$V_R = 6V$$

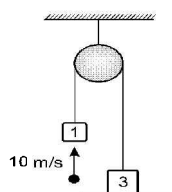
$$\tan \phi = \frac{8}{6} = \frac{4}{3}$$



49. $B_0 = \frac{E_0}{C} = \frac{0.5}{3 \times 10^8} \cong 1.66 \times 10^{-9}$

$$B_z = B_0 \cos \left[2\pi \times 10^8 \left\{ t - \frac{x}{c} \right\} \right]$$

50.



$$\text{Time after which 3 kg will be jerked into motion is } \frac{2 \times 10}{10} = 2 \text{ s}$$

$$\text{Velocity at that instant} = \frac{1 \times 10}{1+3} = \frac{10}{4} = 2.5 \text{ m/s}$$

$$\text{Retardations} = - \left[\frac{3-1}{3+1} \right] \times 10 = -5 \text{ m/s}^2$$

$$\text{Maximum height} = \frac{(2.5)^2}{2 \times 5} = 0.625 \text{ m}$$

$$\text{Time taken to reach that height} = \frac{2.5}{5} = 0.5 \text{ s}$$

$$J = 3(2.5 - 0)$$

51. $\beta = \frac{I_C}{I_B}$

$$I_B = \frac{I_C}{\beta} \Rightarrow I_B = \frac{5 \text{ mA}}{100} = 5 \times 10^{-5} \text{ A}$$

$$I_C = \frac{10 - V_{CE}}{1 \text{ k}\Omega} = 5 \text{ mA}$$

$$R_B = \frac{10 - V_{CE}}{I_B} = \frac{10 - 0}{5 \times 10^{-5}} = 2 \times 10^5 \Omega$$

52. $\phi = \frac{m}{2 \left(\frac{1}{4\pi G} \right)} \Rightarrow \phi = 2Gm\pi$

53. $A(t) = Ae^{-\lambda t}$

$$975 = 9750 e^{-\lambda(5)} \Rightarrow \frac{1}{10} = e^{-5\lambda}$$

$$\ln 10 = 5\lambda$$

$$\lambda = \frac{1}{5} 2.303 \log 10 = 0.461 \text{ per minute}$$

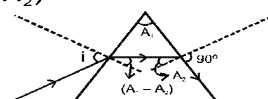
54. $dF = I_2 dx \frac{\mu_0 i_1}{2\pi x} \Rightarrow d\tau = \left(i_2 dx \frac{\mu_0 i_1}{2\pi x} \right) x$

$$Mg \frac{L}{2} = \frac{\mu_0 i_1 i_2}{2\pi} \int_0^L dx \Rightarrow i_2 = \frac{Mg\pi}{\mu_0 i_1}$$

55. $\mu = \frac{1}{\sin A_2} \Rightarrow \text{As } r = A_1 - A_2$

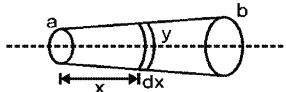
$$\therefore \sin i_{\min} = \frac{1}{\sin A_2} \sin(A_1 - A_2)$$

$$i_{\min} = \sin^{-1} \left[\frac{\sin(A_1 - A_2)}{\sin A_2} \right]$$



56. Applying Ampere's law along the upper circular part of the sieve, $\int \vec{B} \cdot d\vec{\ell} = \mu_0 i = \mu_0$ (charge on water falling per sec) $= \mu_0$ (vol of water falling per sec $\times$ conc. of Na^+ ions $\times$ charge on one ion) $= \mu_0 a v c e$

57.
$$Y = \frac{F}{\frac{\pi(y)^2}{\Delta \ell} dx}$$



$$\int \Delta \ell = \frac{F}{Y\pi} \int_0^L \frac{dx}{\left[\left(\frac{b-a}{L}\right)x + a\right]^2} = \frac{FL}{\pi abY}$$

58.
$$F_{\text{net}} = \sigma \pi D - P_{\text{ex}} \times \frac{\pi D^2}{4}$$

$$[P_{\text{ex}} = \left(\frac{\sigma}{r_1} + \frac{\sigma}{r_2}\right) = \frac{\sigma}{r_1} \quad (r_2 = \infty)]$$

$$F_{\text{net}} = \sigma \pi D - \frac{2\sigma}{D} \times \frac{\pi D^2}{4} = \frac{\sigma \pi D}{2}$$

59.
$$I_p = I_1 + I_2 + \sqrt{2I_1 I_2} \cos \phi$$

$\phi = 8\pi$ and $I_1 \neq I_2$ at the point given on screen so $I_p \neq 0$

60. Centre of gravity is point of application of gravitational force while centre of mass is a point about which mass moment of the object is zero.

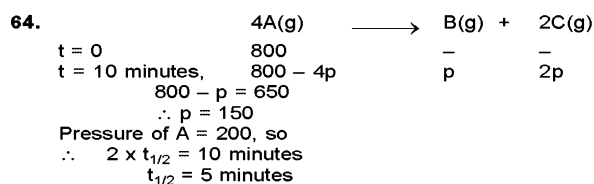
PART-C CHEMISTRY

61.
$$\frac{W_{\text{urea}}}{W_{\text{water}}} = \frac{0.2 \times 60}{0.8 \times 18} = \frac{5}{6}$$

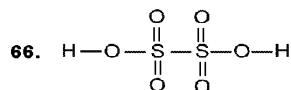
62.
$$\frac{\pi}{RT} = i \cdot C \quad i = \text{slope}$$

$$M_{\text{ABNORMAL}} = \frac{M_{\text{NORMAL}}}{i}$$

63. (1), (2), (3) are wrong statement.
(4) Spin quantum numbers of two electrons of 2s orbital are $+1/2$ and $-1/2$.



65. Cations are present at edge centers and body center through which this plane does not pass. So at this plane only anions are present.



In H_2SO_5 the oxidation number of O is +6 as two O are having peroxide bond.

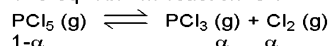
67.
$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -RT \ln K_{\text{eq}} = 0 \quad [K_{\text{eq}} = 1]$$

$$\Rightarrow T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{200 \times 10^3}{40} = 5000 \text{ K}$$

68. Given
$$\frac{r_{\text{O}_2}}{r_{\text{mix}}} = 2.084 = \sqrt{\frac{M_{\text{mix}}}{32}}$$

$$M_{\text{mix}} = 139 \text{ u}$$

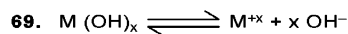
The equilibrium reaction is :



$$1-\alpha \quad \alpha \quad \alpha$$

$$M_{\text{mix}} = \frac{M_{\text{PCl}_5}}{1+\alpha} = \frac{208.5}{1+\alpha} = 139, \alpha = 0.5$$

$$K_p = \frac{0.5 \times 0.5 \times 2}{1-0.25} = \frac{2}{3}$$

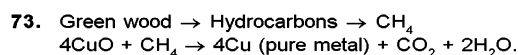
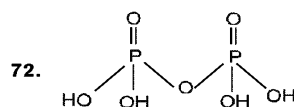


$$K_{\text{sp}} = S \cdot (XS)^x = X^x \cdot S^{(x+1)} = 4 \times 10^{-12} \quad \text{given } S = 10^{-4}$$

$$\therefore x = 2$$



71. (1) Due to chelate effect A is more thermodynamically stable.
(2) Cis isomer of $[\text{Co(en)}_2\text{Cl}_2]^+$ shows optical isomerism.
(3) $\text{Co}^{+3} = 3d^6 t_{2g}^4 e_g^2$
(4) (1) and (2) both are inner orbital complex.



74. Alkali metals and alkaline earth metals belong to I and II group of the periodic table with one and two valence electrons respectively in their 's' subshell. The properties of alkali metals and alkaline earth metals are the direct result of the number of valence electrons. As the size of atoms increase in the group, the ionisation energy decreases and electropositive character of alkali metals increases.
The carbonates of IA group metals are stable while thermal stability of carbonates of IIA group metals increase with increase of atomic number.

75. $E^\circ_{\text{cell}} = 0.76 \text{ volt}$
Applying nerst equation

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0591}{2} \log \frac{[\text{Zn}^{2+}][\text{P}_{\text{H}_2}]}{[\text{H}^+]^2}$$

$$0.169 = 0.76 - \frac{0.0591}{2} \log \frac{(0.1)(1)}{[\text{H}^+]^2}$$

$$\log \frac{0.1}{[\text{H}^+]^2} = \frac{2 \times 0.591}{0.0591}$$

$$\text{or } \log 0.1 - \log [\text{H}^+]^2 = 20$$

$$\text{or } 2\text{pH} = 20 - \log 0.1 \quad \text{pH} = \frac{21}{2} = 10.5$$

76. Interstitial compounds are chemically inert

77. Statement (1), (2), (3) are correct.

Statement (4) is incorrect.

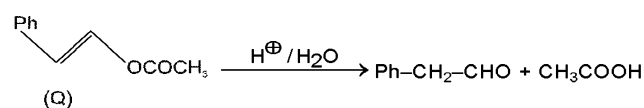
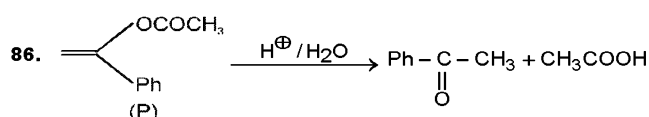
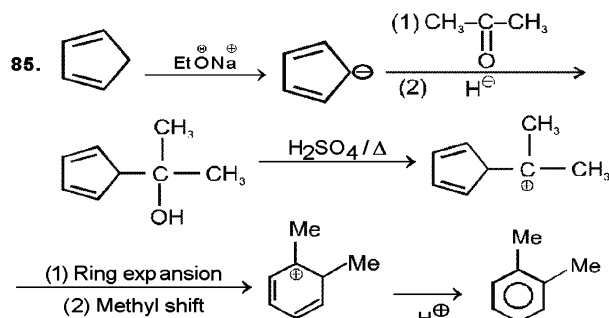
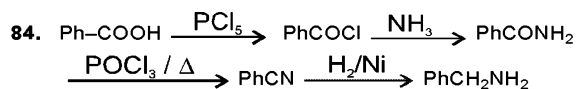
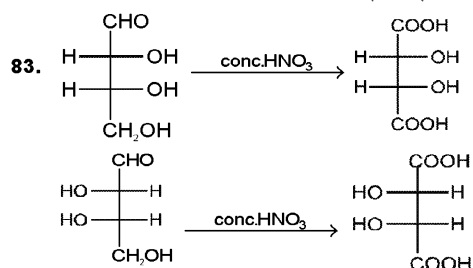
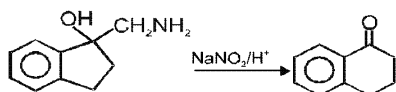
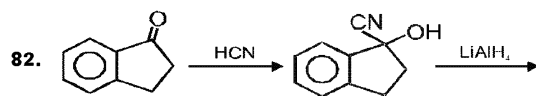
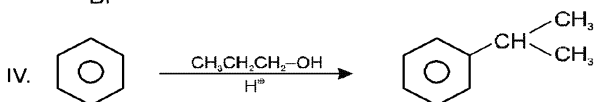
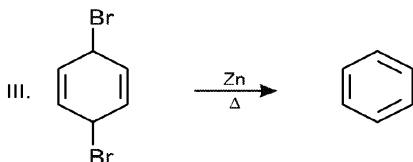
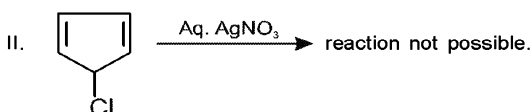
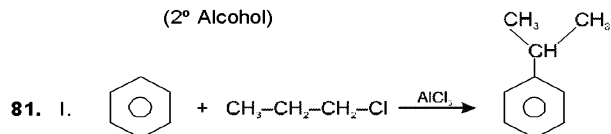
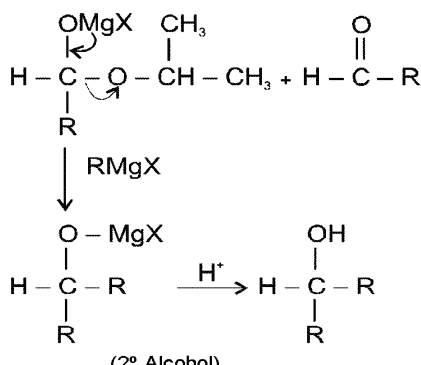
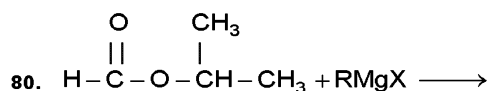
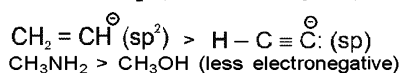
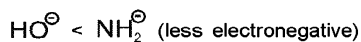
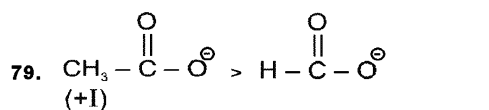
correct order of energies of molecular orbitals of O_2 is

$$E(\sigma 1s) < E(\sigma^* 1s) < E(\sigma 2s) < E(\sigma^* 2s) < E(\sigma 2p_z) < E(\pi 2p_x) = E(\pi 2p_y) < E(\pi^* 2p_x) = E(\pi^* 2p_y) < E(\sigma^* 2p_z)$$

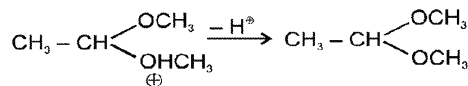
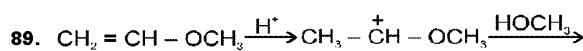
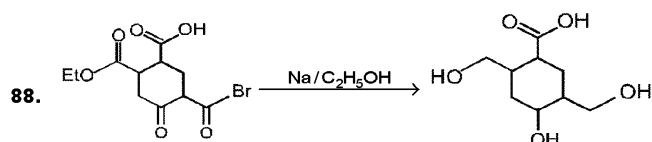
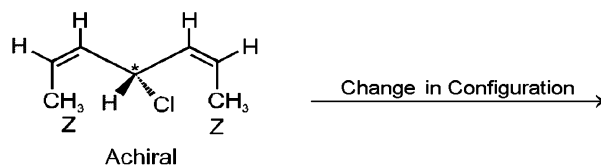
78. $\wedge_{\text{eq}} = 19.6 \text{ Scm}^2 \text{ eq}^{-1}$, $C = 0.01$

$$\alpha = \frac{\wedge_{\text{eq}}}{\wedge^\circ_{\text{eq}}} \Rightarrow \alpha = \frac{19.6}{392} = 5 \times 10^{-2}$$

$$[\text{H}^+] = C\alpha \Rightarrow \text{pH} = 3.3$$



87. Since both the side groups are identical, therefore the configuration of starred carbon depends on configuration of adjacent atoms.



90. (1) R has more priority over the S and $-\text{NH}_3^\oplus$ is more prior

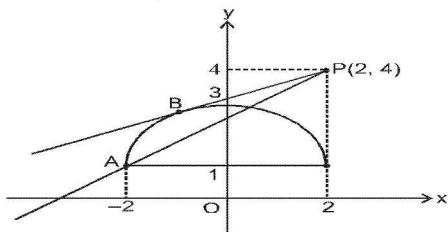
than $-\ddot{\text{N}}\text{H}_2$.

(2) Trans isomers do not possess plane of symmetry.
(3) anti conformation of ethylene glycol has centre of symmetry.
(4) The gauche form of meso tartaric acid do not have plane of symmetry and centre of symmetry so it is chiral.

MOCK TEST 3

PART - A MATHEMATICS

- Centre of the circle is given by $x = t^2 - 3t + 1$, $y = t^2 + 2t$
eliminating t , we get $x^2 - 2xy + y^2 - 12x - 13y + 11 = 0$ (the required locus)
- $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$
 $n(1 + x + x^2)^{n-1}(1 + 2x) = a_1 + 2a_2x + 3a_3x^2 + 4a_4x^3 + \dots$
Put $x = 1$, $n3^n = a_1 + 2a_2 + 3a_3 + 4a_4 + 5a_5 + 6a_6 + \dots$ (i)
Put $x = \omega$, $0 = a_1 + 2a_2\omega + 3a_3\omega^2 + 4a_4 + 5a_5\omega + 6a_6\omega^2 + \dots$ (ii)
Put $x = \omega^2$, $0 = a_1 + 2a_2\omega^2 + 3a_3\omega + 4a_4 + 5a_5\omega^2 + 6a_6\omega + \dots$ (iii)
Adding (i), (ii) and (iii) $n3^{n-1} = a_1 + 4a_4 + 7a_7 + 10a_{10} + \dots$
- Line passes through point (2, 4)
slope of PA = $\frac{3}{4}$



slope of PB = $\frac{5}{12}$ $\therefore k \in \left(\frac{5}{12}, \frac{3}{4}\right]$

- $x_1 + x_2 = (1 - \sin\theta)$, $x_1 x_2 = -\frac{1}{2} \cos^2\theta$
Now, $x_1^2 + x_2^2 = (x_1 + x_2)^2 - 2x_1x_2$
 $= (1 - \sin\theta)^2 + \cos^2\theta$
 $= 2 - 2\sin\theta$
 $\Rightarrow x_1^2 + x_2^2 \Big|_{\max} = 2 + 2 = 4.$

Required probability = ${}^{36}C_2 \left(\frac{1}{6}\right)^{34} \left(\frac{5}{6}\right)^2$

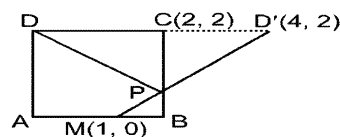
$= \frac{36 \cdot 35}{2} \cdot \frac{25}{6^{34} \times 36} = \frac{875}{2 \times 6^{34}}$

- A B C D E

Let AB = BC = CD = DE
The longest piece will be at least 3 times as long as the shortest piece if we cut between A and B or between D and E.

Probability = $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

- We use reflection principle



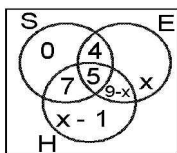
Let $D'(4, 2)$ be image of D in BC
 $\Rightarrow DP + PM = D'P + PM$
Now $D'P + PM$ will be least if P is collinear with M and D'
Equation of MD' is $3y = 2x - 2$
Equation of BC is $x = 2$

$\therefore P\left(2, \frac{2}{3}\right)$

$\therefore \text{minimum } DP + PM = \sqrt{1 + \frac{4}{9}} + \sqrt{4 + \frac{16}{9}}$

$= \frac{\sqrt{13}}{3} + \frac{2\sqrt{13}}{3} = \sqrt{13}$

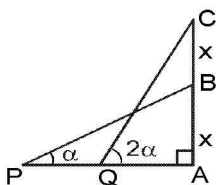
8. Let x be number of employs speaking only English



- $\Rightarrow$ numbers of employs speaking only English and Hindi will be $18 - 4 - 5 - x = 9 - x$
 number of employs speaking only Hindi will be $20 - 7 - 5 - (9 - x) = x - 1$
 $\Rightarrow 30 = 0 + 7 + 5 + 4 + x + (9 - x) + (x - 1)$
 $6 = x$

$$\text{required probability} = \frac{14}{30} = \frac{7}{15}$$

9. In $\triangle PAB$ $\tan \alpha = \frac{x}{AP}$



$$\text{In } \triangle QAC \tan 2\alpha = \frac{2x}{AQ}$$

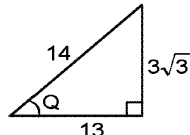
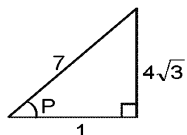
$$PQ = AP - AQ$$

$$= \frac{x}{\tan \alpha} - \frac{2x}{\tan 2\alpha} = x \tan \alpha$$

$$\Rightarrow AP \cdot PQ = \frac{x}{\tan \alpha} \cdot x \tan \alpha = x^2$$

$$\Rightarrow AP, x, PQ \text{ are in G.P.}$$

$$10. \cos(P - Q) = \frac{1}{7} \cdot \frac{13}{14} + \frac{4\sqrt{3}}{7} \cdot \frac{3\sqrt{3}}{14}$$



$$= \frac{13 + 36}{7 \cdot 14} = \frac{49}{7 \cdot 14} = \frac{1}{2}$$

$$P - Q = 60^\circ$$

11. Path of man is an ellipse. Let $a > b$
 $2a = 10$, $2ae = 8$, $b^2 = a^2(1 - e^2)$
 $a = 5$, $e = 4/5$, $b = 3$
 area = $\pi ab = 15\pi$

$$12. x^2 - b_n x + \frac{10^n - 1}{9} = n$$

$$\Rightarrow x^2 - b_n x + \frac{10^n - 9n - 1}{9} = 0$$

$$\text{Product of roots } \alpha\beta = \frac{10^n - 9n - 1}{9}$$

$$= \frac{1}{9} [{}^nC_0 9^n + {}^nC_1 9^{n-1} + {}^nC_2 9^{n-2} + \dots + {}^nC_{n-2} 9^2]$$

Which is divisible by 9

Hence these equation can have 9 as a common root.

But 9 is not the root for all the equations as $\alpha\beta$ has other divisors also for example $n = 4$

$\alpha\beta$ has other divisors also for example $n = 4$

$$\Rightarrow \alpha\beta = {}^4C_0 9^3 + {}^4C_1 9^2 + {}^4C_2 9$$

$$= 9 \times 123$$

One equation has 123 as a root.

$$13. \tan^2 \frac{\pi}{16} + \tan^2 \frac{7\pi}{16} + \tan^2 \frac{2\pi}{16} + \tan^2 \frac{6\pi}{16} + \tan^2 \frac{3\pi}{16} + \tan^2 \frac{5\pi}{16} + 1$$

$$= \tan^2 \frac{\pi}{16} + \cot^2 \frac{\pi}{16} + \tan^2 \frac{2\pi}{16} + \cot^2 \frac{2\pi}{16} + \tan^2 \frac{3\pi}{16} + \cot^2 \frac{3\pi}{16} + 1$$

$$= \left(\tan \frac{\pi}{16} - \frac{1}{\tan \frac{\pi}{16}} \right)^2 + \left(\tan \frac{2\pi}{16} - \frac{1}{\tan \frac{2\pi}{16}} \right)^2 + \left(\tan \frac{3\pi}{16} - \frac{1}{\tan \frac{3\pi}{16}} \right)^2 + 1 + 6$$

$$= \frac{4}{\tan^2 \frac{\pi}{8}} + \frac{4}{\tan^2 \frac{2\pi}{8}} + \frac{4}{\tan^2 \frac{3\pi}{8}} + 7 = \frac{4}{\tan^2 \frac{\pi}{8}} + 4 \tan^2 \frac{\pi}{8} + 11$$

$$= 4 \left(\tan \frac{\pi}{8} - \frac{1}{\tan \frac{\pi}{8}} \right)^2 + 8 + 11 = 4 \times \frac{4}{\tan^2 \frac{\pi}{4}} + 19 = 35$$

$$14. f_1 = \{(1, 2), (2, 3), (3, 4)\}$$

4 has no image under f_1 hence not function

$$f_2 = \{(1, 4), (2, 3), (2, 4), (3, 3), (3, 4), (4, 4), (4, 1), (4, 2), (4, 3)\}$$

2 has more than one image under f_2 hence not a function

$$f_3 = \{(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)\}$$

1 has more than one image under f_3 hence not a function

$$f_4 = \{(1, 4), (2, 3), (3, 2), (4, 1)\}$$
 it is function

$$15. \sim(p \vee \sim q) \equiv \sim p \wedge \sim(\sim q) \equiv \sim p \wedge q$$

$$16. |z - 2 - 2i| = |z - 6 - 6i| \Rightarrow a + b = 8$$

$$\Rightarrow A = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$$

$$\arg(z - 4 - 4i) = \pi/4 \Rightarrow a = b \text{ and } a, b > 4 \Rightarrow B = \{(5, 5), (6, 6)\}$$

$$17. \text{For } n = 1, a = \ell = s \Rightarrow (d + 2a)^2 = 8da \Rightarrow (d - 2a)^2 = 0$$

$$18. q - p = pq$$

$$q - p - pq + 1 = 1$$

$$(q + 1) - p(1 + q) = 1$$

$$(1 + q)(1 - p) = 1$$

$$\ell n(1 + q) = -\ell n(1 - p)$$

$$q - \frac{q^2}{2} + \frac{q^3}{3} - \frac{q^4}{4} + \dots \infty = p + \frac{p^2}{2} + \frac{p^3}{3} + \frac{p^4}{4} + \dots \infty$$

$$19. 1 \text{ white} + 2 \text{ non-white} \Rightarrow {}^3C_1 \times {}^8C_2 = 84$$

$$2 \text{ white} + 1 \text{ non-white} \Rightarrow {}^3C_2 \times {}^8C_1 = 24$$

$$3 \text{ white} \Rightarrow {}^3C_3 \times {}^8C_0 = 1$$

$$\therefore 109 \text{ ways}$$

$$20. \Rightarrow \frac{\sin 8\alpha}{8 \sin \alpha} = \frac{1}{8}$$

$$8\alpha = n\pi + (-1)^n \alpha$$

$$n = 2m \quad 7\alpha = 2m\pi \Rightarrow \alpha = \frac{2m\pi}{7} : m = 1,$$

$$\Rightarrow \alpha = \frac{2\pi}{7}$$

$$n = (2m + 1) \quad 8\alpha = (2m + 1)\pi - \alpha \Rightarrow \alpha = (2m + 1) \frac{\pi}{9}$$

$$m = 0, 1 \quad \alpha = \frac{\pi}{9}, \frac{2\pi}{9}$$

$\therefore$ Number of acute angles are 3

$$\begin{aligned} 21. \quad 11^n - 1 &= (10 + 1)^n - 1 \\ &= {}^nC_0 10^n + {}^nC_1 10^{n-1} + \dots + {}^nC_{n-2} 10 + 1 - 1 \\ &= 10^2 (\text{some integer}) + 10n \\ &\Rightarrow n = 10 \end{aligned}$$

$$\begin{aligned} 22. \quad \text{We have } \frac{1+x^4+x^8+\dots+x^{4(p-1)}}{1+x+x^2+\dots+x^{p-1}} &= \frac{1-x^{4p}}{1-x^4} \cdot \frac{1-x}{1-x^p} \\ &= \frac{(1+x^p)(1+x^{2p})}{(1+x)(1+x^2)} \Rightarrow \quad P \text{ is odd} \\ \text{also sum of the roots of } f(x) &\Rightarrow \quad P+6 \text{ is odd} \\ &\Rightarrow \quad 2 \text{ is a root} \\ &\Rightarrow \quad p+q+2 = -2 \end{aligned}$$

$$\begin{aligned} 23. \quad ax + by + c &= 0 \\ a + c &= 2b \Rightarrow \quad a - 2b + c = 0 \\ &\quad x = 1, y = -2 \\ &\quad (1, -2) = (\alpha, \beta) \\ (x-1)^2 + (y+2)^2 &= \gamma \Rightarrow \quad x^2 + y^2 - 2x + 4y + 5 - \gamma = 0 \\ \text{it is orthogonal to } x^2 + y^2 - 4x - 4y - 1 &= 0 \\ \Rightarrow \quad 4 - 8 = 5 - \gamma - 1 \\ &\quad \gamma = 8 \\ \alpha + \beta + \gamma &= 1 - 2 + 8 = 7 \end{aligned}$$

$$24. \quad (\sin \theta + \cos \theta) \tan \theta = |\tan \theta| \text{ clearly } \theta = 0, \pi, 2\pi \text{ are solutions}$$

Case - I if $\tan \theta > 0$, then $\sin \theta + \cos \theta = 1$

$$\therefore \sin \left(\theta + \frac{\pi}{4} \right) = \sin \frac{\pi}{4}$$

$$\text{i.e. } \theta + \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{4} \quad \text{gives no solution}$$

Case - II if $\tan \theta < 0$
 $\sin \theta + \cos \theta = -1$

$$\therefore \sin \left(\theta + \frac{\pi}{4} \right) = -\sin \frac{\pi}{4}$$

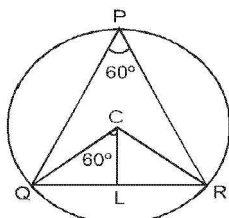
$$\therefore \theta + \frac{\pi}{4} = n\pi + (-1)^n \left(-\frac{\pi}{4} \right) \text{ gives no solution}$$

$\therefore$ Number of solutions is 3.

$$\begin{aligned} 25. \quad \text{Given Circle is } x^2 + y^2 + 2gx + 2fy + c &= 0 \quad \dots\dots\dots(1) \\ \text{Let } C \text{ be its centre and } PQR \text{ be an equilateral triangle inscribed} \\ \text{in the circle then } C \equiv (-g, -f) \text{ and radius of the circle } CQ &= \sqrt{g^2 + f^2 - c} \end{aligned}$$

$$\text{From } \triangle QLC, QL = CQ \sin 60^\circ = \frac{\sqrt{3}}{2} \sqrt{g^2 + f^2 - c}$$

$$\triangle QLC \quad QL = CQ \sin 60^\circ = \frac{\sqrt{3}}{2} \sqrt{g^2 + f^2 - c}$$



$$\therefore QR = 2QL = \sqrt{3} \cdot \sqrt{g^2 + f^2 - c}$$

$$\begin{aligned} \text{Now area of } \triangle PQR &= \frac{\sqrt{3}}{4} QR^2 = \frac{\sqrt{3}}{4} \cdot 3(g^2 + f^2 - c) \\ &= \frac{3\sqrt{3}}{4} (g^2 + f^2 - c) \end{aligned}$$

$$\begin{aligned} 26. \quad &20a^s, 0b^s \\ &a \ a \ \dots \ a \quad 1 \text{ ways} \\ &19 \ a^s, 1 \ b^s \quad 20 \text{ ways} \\ &18 \ a^s, 2 \ b^s \quad 19 \text{ ways} + 3 - 2 = 20 \\ &\vdots \\ &11 \ a^s, 9 \ b^s \quad 12 \text{ ways} + 10 - 2 = 20 \\ &10 \ a^s, 10 \ b^s \quad 11 \text{ ways} + 11 - 2 = 20 \\ &9 \ a^s, 11 \ b^s \quad 10 \text{ ways} + 12 - 2 = 20 \\ &\vdots \\ &1 \ a^s, 19 \ b^s \quad 20 \text{ ways} \\ &0 \ a^s, 20 \ b^s \quad 1 \text{ ways} \\ \text{required} &= 1 + 19 \times 20 + 1 = 382 \end{aligned}$$

$$\begin{aligned} 27. \quad A &= \frac{\pi}{2} \Rightarrow \quad b^2 + c^2 = a^2 \\ &\quad (b+c)^2 - 2bc = a^2 \\ (b+c)^2 - a^2 &= 2bc \\ &= 4 \Delta \\ (b+c-a)(b+c+a) &= 4(a+b+c) \\ \Rightarrow \quad b+c-a &= 4 \end{aligned}$$

$$\begin{aligned} 28. \quad \hat{\mu} &= \frac{0 + x_2 + x_3 + \dots + x_{n-1} + (x_1 + x_n)}{n} = \mu \\ \hat{\sigma}^2 &= \frac{1}{n} \sum y_i^2 - \hat{\mu}^2 \\ &= \frac{1}{n} [0^2 + x_2^2 + x_3^2 + \dots + (x_{n-1})^2 + (x_1 + x_n)^2] - \mu^2 \\ \hat{\sigma}^2 - \sigma^2 &= \frac{2x_1 x_n}{n} > 0 \\ \text{so } \hat{\sigma} &> \sigma \end{aligned}$$

$$\begin{aligned} 29. \quad \sin^6 \theta + \cos^6 \theta &\leq 1 \\ \Rightarrow (\sin^2 \theta + \cos^2 \theta)^3 - 3\cos^2 \theta \sin^2 \theta (\sin^2 \theta + \cos^2 \theta) &\leq 1 \\ \Rightarrow 1 - 3\sin^2 \theta \cos^2 \theta &\leq 1 \\ \Rightarrow 3\sin^2 \theta \cos^2 \theta &\geq 0 \\ \Rightarrow \frac{3}{4} (2\sin \theta \cos \theta)^2 &\geq 0 \\ \Rightarrow \frac{3}{4} (\sin 2\theta)^2 &\geq 0 \end{aligned}$$

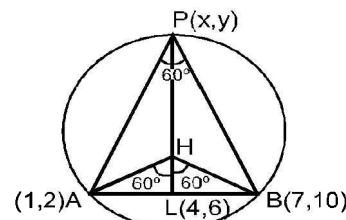
Statement 2 is not necessarily true.

$$30. \quad \text{Since base AB of the triangle is fixed, therefore for maximum area of } \triangle PAB, \text{ height of the triangle should be maximum. This is possible only when P lies on the perpendicular bisector of AB.}$$

$$\text{Equation of perpendicular bisector of AB is } y-6 = -\frac{1-7}{2-10} (x-4)$$

$$\text{or } y-6 = -\frac{3}{4} (x-4)$$

$$\text{or } 4y-24 = -3x+12$$



$$\text{or } 3x + 4y - 36 = 0$$

PART - B

PHYSICS

31. The increase in pressure in liquid is given by buoyant force divided by the area of bottom surface.

$$\Delta P = \frac{\rho_L V g}{A} = 1 \text{ N/m}^2$$

32. from graph

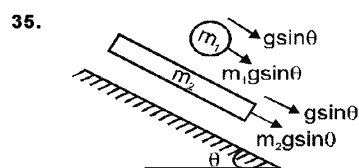
$$V = \left(-\frac{2700}{0.6} \right) s + 3600$$

$$2v \frac{dv}{ds} = \left(-\frac{2700}{0.6} \right), v \frac{dv}{ds} = -2250$$

33. $a_c = \frac{|v|^2}{R} = \frac{(20)^2}{400} = \pi \text{ m/s}^2$

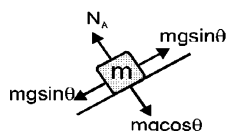
34. $\frac{m\ell^2}{3} = \frac{m\ell^2}{12} + m \left(\left(x - \frac{\ell}{2} \right)^2 + y^2 \right)$

$$\left(x - \frac{\ell}{2} \right)^2 + y^2 = \frac{\ell^2}{3} - \frac{\ell^2}{12} = \frac{3\ell^2}{12} = \frac{\ell^2}{4}$$



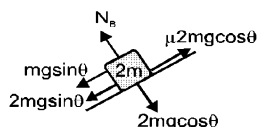
Neither there is relative motion nor there is tendency of relative motion.

36. Just after cutting the string,
F.B.D of A



$$\Rightarrow a_A = 0$$

F.B.D of B



$$a_B = \frac{1}{2m} (3mg \sin \theta - 2\mu mg \cos \theta)$$

$$= 2.6 \text{ m/s}^2 \text{ down the incline.}$$

37. $A' = Ae^{-bt/2m}$

$$\frac{A}{e} = Ae^{-bt_0/2m}$$

$$\frac{bt_0}{2m} = 1$$

$$t_0 = \frac{2m}{b} = \frac{2 \times 150}{50} = 6s$$

$$E' = \frac{1}{2} KA^2 e^{-bt/m} = \frac{1}{2} KA^2 e^{-2}$$

$$\% \text{ Loss} = \frac{E - E'}{E} \times 100 = \frac{1 - e^{-2}}{1} \times 100$$

$$= \frac{e^2 - 1}{e^2} \times 100$$

38. Apply Bernoulli's theorem between free surface and orifice

$$p_0 + \frac{1}{2} \rho V^2 + \rho gh = p_0 + \frac{1}{2} \rho v^2$$

$$A > a$$

$$\text{So, } V < v$$

$$v = \sqrt{2gh}$$

39. Pitch = $\frac{1}{2} \text{ mm}$

$$\text{least count} = \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}$$

$$\text{Reading} = \left[2 \times \frac{1}{2} + 20 \times 0.01 - (-30 \times 0.01) \right] = 1.5 \text{ mm}$$

40. Let the gravitational potential energy be zero when spring is at its natural length. Further let H be the height of the water column at natural length of spring. Now if x is the small extension in the spring.

$$\frac{1}{2} kx^2 + (\rho s x) g \left(\frac{H}{2} + \frac{x}{2} \right) + \frac{1}{2} mv^2 = C$$

$$\Rightarrow \frac{kx dx}{dt} + \rho s g x \frac{dx}{dt} + \rho s g \frac{H}{2} \frac{dx}{dt} + mv \frac{d^2 x}{dt^2} = 0$$

$$\Rightarrow \omega = \sqrt{\frac{k + \rho s g}{m}}$$

41. Velocity of bike $V_b = 30 \text{ m/s}$
Let velocity of car be V_c

$$f_{\text{obs}} = f_{\text{actual}} \times \frac{V_{\text{so}} + V_b}{V_{\text{so}} - V_c}$$

$$120 \text{ Hz} = 100 \text{ Hz} \times \frac{330 + 30}{330 - V_c}$$

$$\Rightarrow V_c = 30 \text{ m/s}$$

Distance between initial and final pulse of horn

$$= V_{\text{so}} \times \Delta t - V_c \Delta t$$

$$(\Delta t \text{ is duration of emission of sound})$$

$$= (330 - 30) \times 6 = 1800 \text{ m}$$

This wave train travels with 330 m/s and cross bike rider moving with velocity of 30 m/s towards it.

$$\text{time taken} = \frac{\text{length of wave train}}{V_{so} + V_b} = \frac{1800}{330 + 30} = \frac{180}{36} = 5 \text{ sec}$$

$$42. \quad \frac{Q_{in}}{W} = \frac{1}{\frac{T_H}{T_L} - 1} = \frac{T_L}{T_H - T_L} = \frac{270}{30} = 9$$

$$W = \frac{Q_{in}}{9} = \frac{1260}{9} \text{ KJ/min} = \frac{1260}{9 \times 60} \text{ KJ/sec.}$$

$$= \frac{21}{9} = \frac{7}{3} \text{ KJ/sec}$$

$$= 2.33 \text{ KJ/sec}$$

$$43. \quad \therefore x\hat{i} + y\hat{j} + z\hat{k} = \hat{i} + 4t^2\hat{j} + t\hat{k}$$

$$\therefore x = 1$$

$$y = 4t^2$$

$$z = t$$

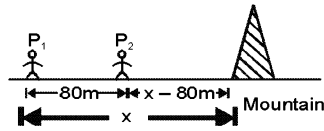
$$\text{Hence } y = 4z^2 \quad (\text{parabola})$$

44. For maximum x coordinate

$$0 = 3V_0 - 2a_1 t$$

$$t = \frac{3V_0}{2a_1}$$

45.



$$\frac{2x}{V} = \frac{60}{40} \Rightarrow x = \frac{3V}{4} \quad \dots (1)$$

$$\frac{2(x - 80)}{V} = 1 \Rightarrow \text{Put } V \text{ from (1)}$$

$$\therefore 2x - 160 = 4x/3$$

$$2x/3 = 160$$

$$x = 240 \text{ m}$$

$$46. \quad I = 2x^2 - 12x + 27$$

$$\frac{dI}{dx} = 4x - 12$$

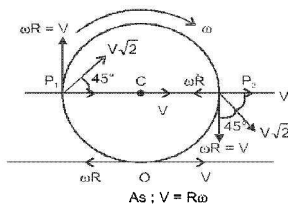
For minimum or maximum

$$\frac{dI}{dx} = 0 \Rightarrow x = 3$$

$$\frac{d^2I}{dx^2} = 4 \text{ Hence } I \text{ is minimum at } x = 3$$

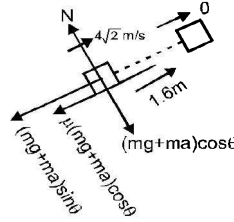
$$I = 9 \text{ kg m}^2$$

47. Angle between velocity vectors and acceleration vector is either 45° at P_1 and 135° at P_2

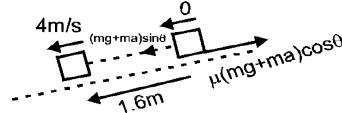


$$\text{As } V = R\omega$$

48.



$$0 = (4\sqrt{2})^2 - 2(g + a)(\sin\theta + \mu\cos\theta) \times 1.6$$



$$4^2 = 0^2 + 2(g + a)(\sin\theta - \mu\cos\theta) \times 1.6$$

$$\text{Solving, } a = 2.5 \text{ m/s}^2$$

$$\mu = 0.25.$$

49. As $k_1 x_1 = k_2 x_2$

$$25x_1 = 10x_2 \Rightarrow x_2 = 2.5 x_1$$

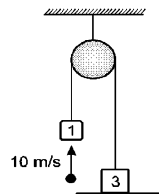
$$\text{and } x_1 + x_2 = 0.5$$

$$\therefore x_1 = \frac{1}{7} \quad \text{and} \quad x_2 = \frac{5}{14}$$

$$\text{Hence their stretched lengths are } x_1 + 0.5 = \frac{9}{14} \text{ met.}$$

$$\text{and } x_2 + 1 = \frac{19}{14} \text{ met}$$

50.



Time after which 3 kg will be jerked into motion is

$$\frac{2 \times 10}{10} = 2 \text{ s}$$

$$\text{Velocity at that instant} = \frac{1 \times 10}{1 + 3} = \frac{10}{4} = 2.5 \text{ m/s}$$

$$\text{Retardations} = -\left[\frac{3-1}{3+1}\right] \times 10 = -5 \text{ m/s}^2$$

$$\text{Maximum height} = \frac{(2.5)^2}{2 \times 5} = 0.625 \text{ m}$$

$$\text{Time taken to reach that height} = \frac{2.5}{5} = 0.5 \text{ s}$$

$$J = 3(2.5 - 0)$$

51. At A & B

$$n = \frac{P_0(2V_0)}{R(2T_0)} = \frac{P_B(3V_0)}{R(2T_0)}$$

$$P_B = \frac{2P_0}{3} = P_C \text{ (BC is isobaric process)}$$

$$\text{Work done from A to B} = nR(2T_0) \ln\left(\frac{3}{2}\right) = 2P_0V_0 \ln\left(\frac{3}{2}\right)$$

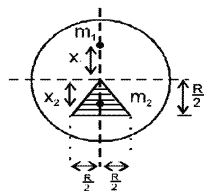
$$52. K = \frac{1}{2}mv^2$$

$$\frac{dK}{dt} = mv \times \frac{dv}{dt} = m \frac{dv}{dt} v = mav, \quad mav = nv$$

$$ma = n, \quad a = \frac{n}{m}$$

$$53. \text{Mass of remaining portion} = m_1 = \left(\pi R^2 - \frac{R^2}{4} \right) \sigma$$

$$\text{Mass of triangular portion} = m_2 = \frac{R^2}{4} \sigma$$



$$m_1 x_1 = m_2 x_2, \quad m_1 x_1 = m_2 \times \frac{R}{3}, \quad x_1 = \frac{R}{3(4\pi - 1)}$$

$$54. R = \frac{u_2^2}{g(1 + \sin \alpha)} \Rightarrow u_2^2 = Rg(1 + \sin \alpha) \quad \dots(1)$$

$$2R \cos \alpha = \frac{u_3^2}{g} \Rightarrow u_3^2 = 2Rg \cos \alpha \quad \dots(2)$$

$$\text{Now, } u_1 u_2 = Rg \sqrt{1 - \sin^2 \alpha} = Rg \cos \alpha = \frac{u_3^2}{2}$$

$$\therefore u_3^2 = 2u_1 u_2$$



$$V_1 \cos 60^\circ = \frac{3V_0}{2} \cos 30^\circ \Rightarrow V_1 = \frac{3\sqrt{3}}{2} V_0$$

$$56. a_1 = 1 \text{ m/s}^2 \quad v = t + 2$$

$$\text{at } t = 2 \text{ sec.} \quad a(2) = \sqrt{1^2 + \left\{ \frac{(2+2)^2}{R} \right\}^2}$$

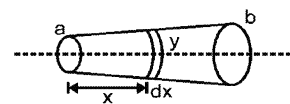
$$a(5) = \sqrt{1^2 + \left\{ \frac{(5+2)^2}{R} \right\}^2}$$

$$2\sqrt{1 + \frac{16^2}{R^2}} = \sqrt{1 + \frac{49^2}{R^2}}$$

$$4 + \frac{(4)16^2}{R^2} = 1 + \frac{49^2}{R^2}$$

$$3 = \frac{49^2}{R^2} - 4 \frac{16^2}{R^2} \Rightarrow R = 3\sqrt{51} \text{ m.}$$

$$57. Y = \frac{F}{\frac{\pi(y)^2}{\Delta \ell} dx}$$



$$\int \Delta \ell = \frac{F}{Y\pi} \int_0^L \frac{dx}{\left[\left(\frac{b-a}{L} \right) x + a \right]^2} = \frac{FL}{\pi abY}$$

$$58. F_{\text{net}} = \sigma \pi D - P_{\text{ex}} \times \frac{\pi D^2}{4}$$

$$[P_{\text{ex}} = \left(\frac{\sigma}{r_1} + \frac{\sigma}{r_2} \right) = \frac{\sigma}{r_1} \quad (r_2 = \infty)]$$

$$F_{\text{net}} = \sigma \pi D - \frac{2\sigma}{D} \times \frac{\pi D^2}{4} = \frac{\sigma \pi D}{2}$$

59. Statement-1 is False, Statement-2 is True

60. Centre of gravity is point of application of gravitational force while centre of mass is a point about which mass moment of the object is zero.

PART-C CHEMISTRY

$$61. \frac{W_{\text{urea}}}{W_{\text{water}}} = \frac{0.2 \times 60}{0.8 \times 18} = \frac{5}{6}$$

62. (1), (2), (3) are wrong statement.
(4) Spin quantum numbers of two electrons of 2s orbital are +1/2 and -1/2.

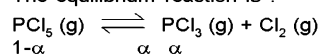
$$63. \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -RT \ln K_{\text{eq}} = 0 [K_{\text{eq}} = 1]$$

$$\Rightarrow T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{200 \times 10^3}{40 \times 10^3} = 5 \text{ K}$$

$$64. \text{Given } \frac{r_{\text{O}_2}}{r_{\text{mix}}} = 2.084 = \sqrt{\frac{M_{\text{mix}}}{32}}$$

$$M_{\text{mix}} = 139 \text{ u}$$

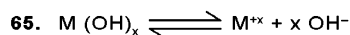
The equilibrium reaction is :



$$1-\alpha \quad \alpha \quad \alpha$$

$$M_{\text{mix}} = \frac{M_{\text{PCl}_5}}{1+\alpha} = \frac{208.5}{1+\alpha} = 139, \quad \alpha = 0.5$$

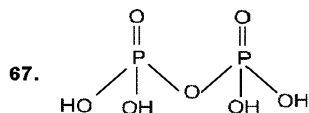
$$K_p = \frac{0.5 \times 0.5 \times 2}{1-0.25} = \frac{2}{3}$$



$$K_{\text{sp}} = S \cdot (XS)^x = X^x \cdot S^{(x+1)} = 4 \times 10^{-12} \text{ given } S = 10^{-4}$$

$$\therefore x = 2$$





68. Alkali metals and alkaline earth metals belong to I and II group of the periodic table with one and two valence electrons respectively in their 's' subshell. The properties of alkali metals and alkaline earth metals are the direct result of the number of valence electrons. As the size of atoms increase in the group, the ionisation energy decreases and electropositive character of alkali metals increases. The carbonates of IA group metals are stable while thermal stability of carbonates of IIA group metals increase with increase of atomic number.

69. Statement (1), (2), (3) are correct. Statement (4) is incorrect.

correct order of energies of molecular orbitals of O_2 is $E(\sigma_{1s}) < E(\sigma^*_{1s}) < E(\sigma_{2s}) < E(\sigma^*_{2s}) < E(\sigma_{2p_z}) < E(\pi_{2p_x}) = E(\pi_{2p_y}) < E(\pi^*_{2p_x}) = E(\pi^*_{2p_y}) < E(\sigma^*_{2p_z})$

71. $XeF_5^- \rightarrow 72^\circ$
 $XeF_4 \rightarrow 90^\circ$
 $XeF_2 \rightarrow 180^\circ$

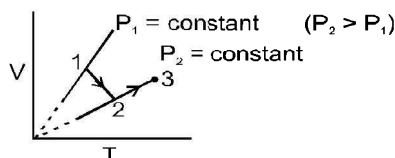
72. Since system is conducting frictionless
Hence $T_A = T_B$
and $P_A = P_B$
Since volume is different hence
 $n_B = 3n_A$

73. $Cu_2O(s) + \frac{1}{2} O_2(g) \rightleftharpoons 2CuO(s)$

$\Delta G^0_{\text{reaction}} = [2 \times (-30.4)] - [-34.98] = -25.82 \text{ kcal}$
and $-25.82 \times 10^3 = 2.303 \times 2 \times 298 \log K$
 $\therefore K \approx 10^{19}$, a very high value, hence reaction will be almost complete with a trace of Cu_2O .

74. Slope of graph = $n - 1$
As n increases, slope increases, graph moves up

75. V-T curve for a given mass of the gas is given



76. $HA + Ba(OH)_2 \longrightarrow BaA_2 + H_2O$ (in titration)

$A^- + H_2O \rightleftharpoons HA + OH^-$

For titration

$$N_{\text{acid}} V_{\text{acid}} = N_{\text{base}} V_{\text{base}}$$

$$\frac{1}{4} \times 40 = \frac{1}{6} \times V$$

$$V = 60 \text{ mL}$$

In resulting solution

$$[A^-] = \frac{\frac{1}{4} \times 40}{100} = 0.1 \text{ M}$$

$$K_h = \frac{K_w}{K_a} = \frac{10^{-14}}{2 \times 10^{-12}} = 5 \times 10^{-3}$$

$$\text{or } K_h = \frac{0.1 h^2}{(1-h)} = 5 \times 10^{-3} \text{ (solve quadratic equation to get 'h',)}$$

as we can't write $1 - h \approx 1$ since h is > 0.1

$$h = 0.2$$

$$[OH^-] = 0.1 \times 0.2 = 2 \times 10^{-2} \text{ M}$$

Therefore, $[H^+] = 5 \times 10^{-13} \text{ M}$

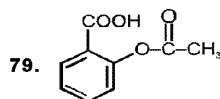
77. (1) The double chain silicates can be drawn in which two simple chains are joined together by shared oxygen. Such compounds are also known as amphiboles.

(2) If two oxygen atoms per tetrahedron are shared to form closed rings such that the structure with general formula $(SiO_3^{2-})_n$ or $(SiO_3)_n^{2n-}$ is obtained, the silicates containing these anions are called cyclic silicates.

(3) Orthosilicates contain discrete $[SiO_4]^{4-}$ units i.e., there is no sharing of corners with one another.

78. S_1 and S_2 are correct statements.

S_3 : GeO_2 is acidic while SnO_2 is amphoteric in nature.



Aspirin (2-Ethanoxybenzoic acid)

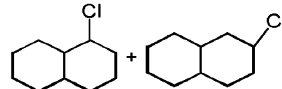
80. Aromatic carbocation are very stable here (D) is most stable since it is aromatic carbocation.

81. The group $-Cl$ has $(-I > +ve)$ effect, it will have higher dipole moment in which it operates its only $-I$ effect.

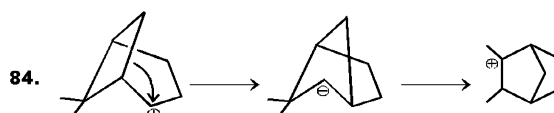
82. The electron density in benzene ring is proportional to the effect of substituent. In structure II due to SIR effect $-NR_2$ does not operate its $+m$ effect properly, hence it has least electron density in the aromatic ring.

83. (1) gives

(2) gives



(4) gives



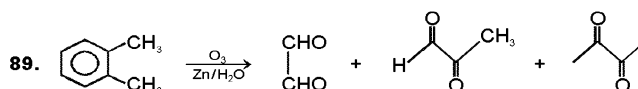
85. For aromaticity the species must have $(4n + 2)\pi e^-$ in cyclic conjugation in cyclic planar species which is not possible in structure B which has only $4\pi e^-$ which are not in cyclic conjugation.

86. $-OMe$, $-O-C(=O)-Me$, $-NMe_2$ are $+m$ groups.

87. For showing Iodoform test the compound should be carbonyl

having $-C(=O)-CH_3$ group.

88. The compound has 3 acidic hydrogen which on reaction with Na metal liberate 1.5 moles of H_2 gas and two functional groups $-SO_3H$ and $-COOH$ will liberates 2 moles of CO_2 gas with $NaHCO_3$.



90. $-CMe_3$, $-SO_3^-Na^+$, $-COO^-Na^+$ all are $+I$ effect groups.

MOCK TEST 4 **PART- A**

MATHEMATICS

1. (3)

$\because |A| \neq 0 \Rightarrow A^{-1}$ का अस्तित्व होगा
 दिया है $ABC = 0$
 $\Rightarrow A^{-1}(ABC) = A^{-1} \cdot 0$
 $\Rightarrow BC = 0$
 माना $|C| \neq 0 \Rightarrow C^{-1}$ का अस्तित्व होगा
 $\Rightarrow (BC)C^{-1} = OC^{-1} = 0$
 $B = 0$ जोकि सम्भव नहीं
 अतः $|C| = 0$.

2. (1) $\vec{a} \cdot \vec{b} < 0$ तथा $\vec{a} = \lambda \vec{c}$

i.e $2x^2\hat{i} + 4x\hat{j} + \hat{k} = \lambda\hat{i} + 20\lambda\hat{j} + 50\lambda\hat{k}$

$\therefore 2x^2 = \lambda, 4x = 20\lambda, 1 = 50\lambda \Rightarrow x = \frac{1}{10}$

तथा $\vec{a} \cdot \vec{b} = 14x^2 - 8x + x$

$= 14x^2 - 7x$ ऋणात्मक है, जब $x = \frac{1}{10}$

3. (1)

चूंकि $n = 9$, माध्यिका पद $= \left(\frac{9+1}{2}\right)^{th} = 5^{वाँ}$ पद

अब प्रत्येक प्रेक्षण को 2 से बढ़ाया गया है।
 इसलिए माध्यिका, 5 प्रेक्षण है जो कि 2 से बढ़ा दिया है।

4. (2)

यहाँ 49 गुणनखण्ड है इसलिए x^{48} का गुणांक

$= -2 - 4 - 6 - \dots - 98$

$= -\frac{49}{2}(2 + 98) = -2450$

5. (4)

4 युग्म को चुने जाने के तरीके ${}^5C_4 = 5$ तरीके। अब चुने हुए प्रत्येक युग्म से ठीक एक जूता चुनने के तरीके $= {}^2C_1 \times {}^2C_1 \times {}^2C_1 \times {}^2C_1 = 16$ तरीके
 अभीष्ट क्रमचय $= 5 \times 16 = 80$

6. (3)

$|(adjA^{-1})| = |A^{-1}|^2 = \frac{1}{|A|^2}$

$|(adjA^{-1})^{-1}| = \frac{1}{|adjA^{-1}|} = |A|^2 = 2^2 = 4$

7. (2)

$\vec{r} = (\vec{b} + \lambda \vec{c})$

$= (1 + \lambda)\hat{i} + (2 + \lambda)\hat{j} - (1 + 2\lambda)\hat{k}$

$= \vec{r} \cdot \vec{a} = \vec{b} \cdot \vec{a} + \lambda \vec{a} \cdot \vec{c}$

$\lambda = -3$ or $\lambda = 1$

$\therefore \vec{r} = -2\hat{i} - \hat{j} + 5\hat{k}$ or $\vec{r} = 2\hat{i} + 3\hat{j} - 3\hat{k}$

8. (1)

On solving, we get the value of $x = 0$

9. (4)

2 को 2 तरीके से लिया जा सकता है (2^0 या 2^2)

3 को 3 तरीके से लिया जा सकता है (3^0 या 3^2 या 3^4)

इसलिए 5 को 4 तरीके से लिया जा सकता है (5^0 या 5^2 या 5^4 या 5^6)

तथा 7 को 5 से तरीके से लिया जा सकता है (7^0 या 7^2 या 7^4 या 7^6 या 7^8)

अतः कुल भाजक जो पूर्ण वर्ग है $= 2 \cdot 3 \cdot 4 \cdot 5 = 120$

10. (3)

रेखाएं समतलीय होगी जबकि

$$\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ \ell_1 & m_1 & n_1 \\ \ell_2 & m_2 & n_2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & -1 & -1 \\ 1 & 1 & -k \\ k & 2 & 1 \end{vmatrix} = 0$$

$\Rightarrow k^2 + 3k = 0$

$\Rightarrow k = 0$ or -3

11. (3)

समीकरण निकाय का अशून्य हल है तो

$$\begin{vmatrix} 1 & 2a & a \\ 1 & 3b & b \\ 1 & 4c & c \end{vmatrix} = 0$$

Applying $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$

$$\begin{vmatrix} 1 & 2a & a \\ 0 & 3b - 2a & b - a \\ 0 & 4c - 2a & c - a \end{vmatrix} = 0$$

$\Rightarrow (3b - 2a)(c - a) - (4c - 2a)(b - a) = 0$

$\Rightarrow 3bc - 3ba - 2ac + 2a^2 = 4bc - 2ab - 4ac + 2a^2$

$\Rightarrow 4ac - 2ac = 4bc - 2ab - 3bc + 3ab$

$\Rightarrow 2ac = bc + ab$

abc को विभाजित करने पर

$$\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$$

अतः a, b, c हरात्मक श्रेढी में है।

12. (4)

$$\text{चूँकि } \vec{a} \times \vec{b} + \vec{c} = \vec{0}$$

$$\Rightarrow \vec{a} \times (\vec{a} \times \vec{b}) + \vec{a} \times \vec{c} = \vec{0}$$

$$\Rightarrow (\vec{a} \cdot \vec{b})\vec{a} - (\vec{a} \cdot \vec{a})\vec{b} + \vec{a} \times \vec{c} = \vec{0}$$

$$\text{चूँकि } \vec{a} \times \vec{c} = -2\hat{i} - \hat{j} - \hat{k}$$

$$\Rightarrow 3(\hat{j} - \hat{k}) - 2\vec{b} - 2\hat{i} - \hat{j} - \hat{k} = \vec{0}$$

$$\Rightarrow \vec{b} = -\hat{i} + \hat{j} - 2\hat{k}$$

अतः सही विकल्प (4) है।

13. (1)

$$P(A \cap B) = \frac{1}{4} \Rightarrow P(\overline{A \cap B}) = \frac{3}{4} \Rightarrow P(\overline{A} \cup \overline{B}) = \frac{3}{4}$$

$$\therefore P(\overline{B}) = P(\overline{A} \cup \overline{B}) + P(\overline{A} \cap \overline{B}) - P(\overline{A})$$

$$= \frac{3}{4} + \frac{1}{6} - \frac{1}{4} = \frac{2}{3}$$

$$\text{पुनः } P(A) \cdot P(B) = \frac{3}{4} \cdot \frac{1}{3} = \frac{1}{4} = P(A \cap B)$$

$\therefore$ A और B स्वतंत्र घटनाएँ हैं।

14. (2)

कम से कम एक कालांश की पुनरावृत्ति 5C_1 तरीके से है।

अब 6 विषय को $\frac{6!}{2!}$ तरीके से व्यवस्थित किया जा सकता है।

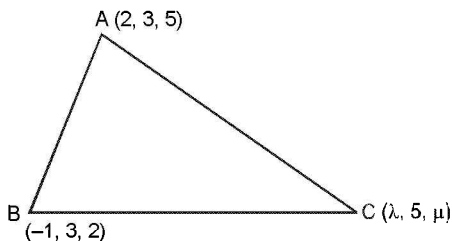
$$\text{अतः कुल क्रमचय } 5 \times \frac{6!}{2!} = 1800 \text{ है।}$$

15. (3)

$$\text{BC का मध्य बिन्दु } \left(\frac{\lambda-1}{2}, 4, \frac{2+\mu}{2} \right) \text{ है।}$$

A से गुजरने वाली माधिका के दिक अनुपात है।

$$\frac{\lambda-1}{2} - 2, 4-3, \frac{2+\mu}{2} - 5, \text{ i.e. } \frac{\lambda-5}{2}, 1, \frac{\mu-8}{2}$$



माधिका अक्षों के साथ बराबर झुकी हुई है तथा दिक अनुपात बराबर है इसलिए

$$\frac{\lambda-5}{2} = 1 = \frac{\mu-8}{2} \Rightarrow \lambda = 7, \mu = 10$$

16. (2)

$$\text{चूँकि mean } np = \frac{7}{3}, \text{ variance } npq = \frac{14}{9}, \text{ therefore}$$

$$\text{अतः } \frac{7}{3}q = \frac{14}{9} \Rightarrow q = \frac{2}{3} \Rightarrow p = \frac{1}{3}$$

$$\text{इस प्रकार } n\left(\frac{1}{3}\right) = \frac{7}{3} \Rightarrow n = 7$$

$$\text{अब } P(X = 6 \text{ or } 7) = P(X = 6) + P(X = 7) \\ = {}^7C_6 p^6 q^1 + {}^7C_7 p^7 q^0$$

$$= 7\left(\frac{1}{3}\right)^6 \left(\frac{2}{3}\right) + \left(\frac{1}{3}\right)^7 = \frac{5}{729}$$

17. (4)

रेखाएँ $\vec{r} \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = 0$ और $\vec{r} \cdot (3\hat{i} + 3\hat{j} + \hat{k}) = 0$ के समान्तर होंगी

$$(3\hat{i} + 3\hat{j} + \hat{k}) \times (\hat{i} + 2\hat{j} + 3\hat{k}), \text{ i.e. } (7\hat{i} - 8\hat{j} + 3\hat{k})$$

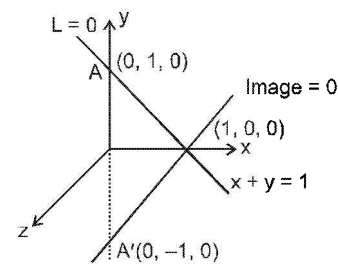
यदि θ अभीष्ट कोण है, तब

$$\cos \theta = \frac{7}{\sqrt{49 + 64 + 9}} = \frac{7}{\sqrt{122}}$$

18. (1)

इसलिए अभीष्ट रेखा

$$= \frac{x-1}{1} = \frac{y-0}{1} = \frac{z-0}{0}$$



$$\Rightarrow x-1=y=\frac{z}{0}$$

$$\text{i.e. } x-y=1, z=0$$

19. (1)

$$({}^7C_0 + {}^7C_1) + ({}^7C_1 + {}^7C_2) + \dots + ({}^7C_6 + {}^7C_7) \\ = {}^8C_1 + {}^8C_2 + \dots + {}^8C_7 = {}^8C_0 + {}^8C_1 + {}^8C_2 + \dots + {}^8C_7 + {}^8C_8 \\ - ({}^8C_0 + {}^8C_8) = 2^8 - 2$$

20. (1)

$$1! + 2! + 3! + 4! = 1 + 2 + 6 + 24 = 33 \text{ और } 15, n! \text{ को}$$

$n \geq 5$ ($n \in \mathbb{N}$) के लिए विभाजित करता है

अभीष्ट शेषफल 3, उस शेषफल के बराबर है जबकि 33 को 15 से विभाजित किया जाता है।

21. (4)

$$[\vec{a} - 2\vec{b} \quad \vec{b} - 3\vec{c} \quad \vec{c} - 4\vec{a}]$$

$$= ((\vec{a} - 2\vec{b}) \times (\vec{b} - 3\vec{c})) \cdot (\vec{c} - 4\vec{a})$$

$$= (\vec{a} \times \vec{b} - 3\vec{a} \times \vec{c} + 6\vec{b} \times \vec{c}) \cdot (\vec{c} - 4\vec{a})$$

$$= (\vec{a} \times \vec{b}) \cdot \vec{c} - 24(\vec{b} \times \vec{c}) \cdot \vec{a}$$

$$= -23 |\vec{a}| |\vec{b}| |\vec{c}| \sin 90^\circ$$

$$= -23 (1) (3) (5)$$

$$= -69 \times 5 = -345$$

22. (1)

$$(A + I)^3 = A^3 + 3A^2 + 3A + I = I$$

$$\therefore (A + I)(A^2 + 2A + I) = I$$

$$\therefore A^2 + 2A + I = (A + I)^{-1}$$

23. (1)

अभीष्ट प्रायिकता = $1 - P$ (जबकि P कोई भी पत्ता पान का इक्का नहीं हों) की प्रायिकता होगी)

$$= 1 - \frac{51}{52} \cdot \frac{51}{52} = \frac{(52)^2 - (51)^2}{52 \cdot 52} = \frac{(52-51)(52+51)}{2704}$$

$$= \frac{103}{2704}$$

24. (1)

$$\text{Let } \vec{a} = \hat{i} + \hat{j} + 2\hat{k}$$

$$\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$$

$$\vec{c} = \hat{i} + \hat{j} + \hat{k}$$

$$\vec{r} = \pm \frac{\vec{c} \times (\vec{a} \times \vec{b})}{|\vec{c} \times (\vec{a} \times \vec{b})|}$$

$$\text{Here } \vec{c} \times (\vec{a} \times \vec{b}) = (\vec{b} \cdot \vec{c})\vec{a} - (\vec{c} \cdot \vec{a})\vec{b}$$

$$\text{Hence } \vec{r} = \pm \left(\frac{\hat{k} - \hat{j}}{\sqrt{2}} \right)$$

25. (4)

$$\text{माध्य} = \bar{X} = \frac{X_1 + X_2 + \dots + X_{100}}{100} = 55$$

यदि प्रत्येक आँकड़े को 2 से बढ़ा दिया जाये तब

$$\bar{X}_{\text{new}} = \frac{(X_1 + X_2 + \dots + X_{100}) + 2 \times 100}{100} = \bar{X} + 2$$

$$\Rightarrow \bar{X}_{\text{new}} = 57$$

$$\text{प्रसरण} = \frac{\sum (x - \bar{x})^2}{n} = 16$$

प्रत्येक आँकड़े में 2 से वृद्धि करने पर प्रसरण का मान परिवर्तित नहीं होगा।

$$\therefore \text{गया प्रसरण} = 16$$

$$\Rightarrow \text{मानक विचलन} = \sqrt{\text{variance}} = 4$$

$$\therefore \text{माध्यों का और मानक विचलन का योग} = 57 + 4 = 61$$

26. (1)

माना x_w, x_R, x_B सफेद गेंदों लाल गेंदों और नीली गेंदों की संख्या है।

$$\text{हम जानते हैं } x_w + x_R + x_B = 10$$

अभीष्ट तरीके = number of non negative integral solutions of

$$x_w + x_R + x_B = 10 \text{ which are}$$

$${}^{3+10-1}C_{10} = {}^{12}C_{10} = {}^{12}C_2$$

27. (1)

माना कि समतल $\frac{x}{\lambda} + \frac{y}{\mu} + \frac{z}{\nu} = 1$ है। चूंकि यह बिन्दु

(a, b, c) से गुजरता है।

$$\therefore \frac{a}{\lambda} + \frac{b}{\mu} + \frac{c}{\nu} = 1$$

अतः बिन्दु (λ, μ, ν) का बिन्दुपथ $\frac{a}{x} + \frac{b}{y} + \frac{c}{z} = 1$ है।

28. (4)

29. (2)

30. (1)

$$P(A) = P(A/B) = \frac{P(A \cap B)}{P(B)} \Rightarrow P(A \cap B) = P(A) \cdot P(B)$$

$\therefore$ A और B स्वतंत्र है

$\therefore$ कथन-2 सत्य है

$$P(A) = \frac{4}{52} = \frac{1}{13}, P(B) = \frac{1}{4}$$

यदि यह ज्ञात है कि निकाला गया पत्ता हुकुम है, तो इसके इक्का होने की घटना A/B है।

$$P(A/B) = \frac{1}{13} = P(A)$$

$\therefore$ A और B स्वतंत्र है।

PART- B PHYSICS

31. (2)

Sol. गॉउस प्रमेय से

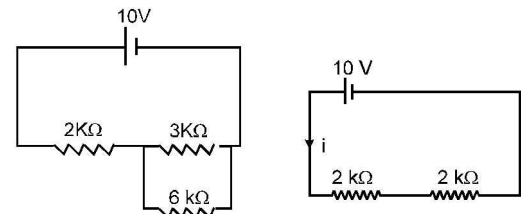
बेलन के अन्दर कुल आवेश = 0

$$\Rightarrow \int (2\pi r dr) h \rho = 0 \Rightarrow \int_0^R (ar + br^2) r dr = 0$$

$$\Rightarrow \frac{aR^3}{3} + \frac{bR^4}{4} = 0 \Rightarrow \frac{a}{b} = -\frac{3R}{4}$$

32. (2)

Sol. दिये गये परिपथ को चित्रानुसार व्यवस्थित करने पर



$$i = \frac{10}{4} = 2.5 \text{ mA}$$

$$\text{वोल्टमीटर का पाठ्यांक} = (2.5 \text{ mA}) (2 \text{ K}\Omega) = 5 \text{ V}$$

33. (3)

Sol. जब ब्लॉक स्थिर है

$$mg \sin \theta = f$$

$$\frac{f}{W} = \sin \theta$$

गति के दौरान

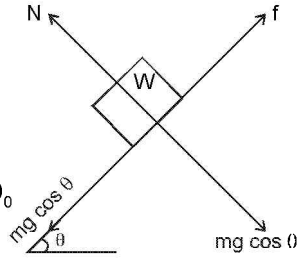
$$mg \sin \theta_0 = \mu_s mg \cos \theta_0$$

$$\tan \theta_0 = \mu_s$$

जब ब्लॉक फिसलता है

$$f = \mu_k mg \cos \theta$$

$$\frac{f}{W} = \mu_k \cos \theta$$

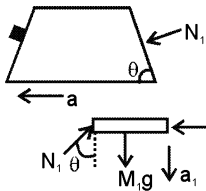


34. (3)

Sol. माना M_1 छड़ का द्रव्यमान है

$$M_1 g - N_1 \cos \theta = M_1 a_1 \quad \dots(i)$$

$$N_1 \sin \theta = (M + M_1) a \quad \dots(ii)$$



$$a = g \tan \theta \quad \dots(iii)$$

A_1 तथा A में सम्बन्ध

$$a_1 = a \tan \theta$$

इन समीकरणों को हल करने पर $M_1 = 3M = 3\text{kg}$

35. (1)

$$\text{Sol. } h = \frac{1}{2} g t^2$$

$$\text{प्रथम गेंद के लिए } 125 = \frac{1}{2} \times 10 \times t^2$$

$$t = 5\text{s}$$

$$\text{द्वितीय गेंद के लिए } 125 = u \times 3 + \frac{1}{2} \times 10 \times 9$$

$$u = \frac{80}{3} = 26.66 \text{ m/s}$$

36. (4)

Sol. बिन्दु C पर पहुँचने की शर्त

$$\tan 45^\circ = \frac{v_y}{v_x}, v_y = v_x$$

$$(v_R + v_M \cos \theta) = v_M \sin \theta$$

$$1 + 5 \cos \theta = 5 \sin \theta$$

$$1 + 5 \cos \theta = 5 \sqrt{1 - \cos^2 \theta}$$

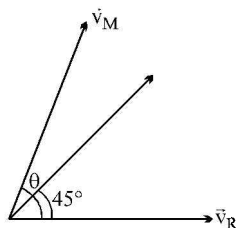
वर्ग करने पर

$$1 + 25 \cos^2 \theta + 10 \cos \theta = 25 - 25 \cos^2 \theta$$

$$50 \cos^2 \theta + 10 \cos \theta - 24 = 0$$

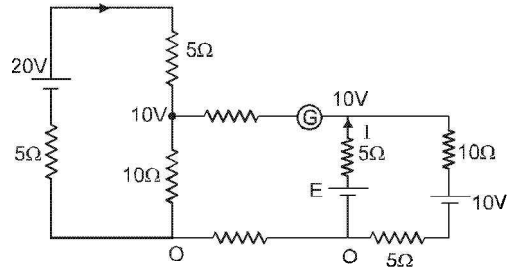
हल करने पर

$$\theta = 53^\circ$$



37. (1)

$$\text{Sol. } \frac{E - 10}{20} = I$$



$$10 = E - \frac{E - 10}{20} \times 5$$

$$40 = 4E - E + 10$$

$$30 = 3E$$

$$E = 10 \text{ V.}$$

38. (1)

$$\text{Sol. आवर्धन } M = \frac{F}{f}$$

लेंसो के मध्य दूरी

$$d = F + f$$

$$\text{दिया गया है } \frac{F}{f} = 9 \text{ i.e., } F = 9f \quad \dots(1)$$

$$\text{तथा } F + f = 20 \quad \dots(2)$$

समीकरण (1) तथा समीकरण (2) से

$$9f + f = 20 \Rightarrow 10f = 20 \Rightarrow f = \frac{20}{10} = 2\text{cm}$$

$$\therefore F = 9f = 9 \times 2 = 18 \text{ cm}$$

$$\therefore F = 18 \text{ cm, } f = 2 \text{ cm}$$

39. (1)

Sol. आम के सापेक्ष पत्थर की गति आम की ओर है।

अतः, यह आम से टकरायेगा।

40. (4)

Sol. $R = 10\text{m}$

$H = 25\text{m}$

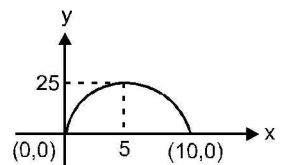
$$\frac{u^2 \sin^2 \theta}{2g} = 25$$

$$u \sin \theta = 10\sqrt{5}$$

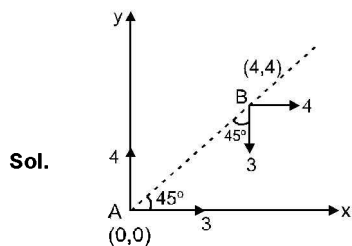
$$\frac{dy}{dx} (0,0) \text{ पर}$$

$$\tan \theta = 10$$

$$\sin \theta = \frac{10}{\sqrt{101}}$$



41. (4)



Sol.

$$\begin{aligned} V_{BA} &= V_B - V_A \\ &= [4\hat{i} - 3\hat{j}] - [3\hat{i} + 4\hat{j}] = \hat{i} - 7\hat{j} \\ V_{app} &= 4 \cos 45^\circ + 3 \cos 45^\circ + 3 \cos 45^\circ - 4 \cos 45^\circ \\ &= 6 \cos 45^\circ \\ &= 3\sqrt{2} \text{ m/s} \end{aligned}$$

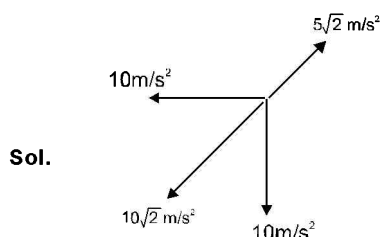
42. (1)

Sol. $\frac{u^2}{2g} = 10\text{m}$

$$\frac{2u}{g} = 4 \text{ sec}$$

हल करने पर $g = 5 \text{ m/s}^2$.

43. (4)



Sol.

व्यक्ति के सापेक्ष

$$\begin{aligned} S &= ut + \frac{1}{2} at^2 \\ &= 5\sqrt{2} \times 1 + \frac{1}{2} (-10\sqrt{2}) 1^2 = 0. \end{aligned}$$

44. (3)

Sol. $m = \frac{f}{f+u}$

$$m_1 = \frac{f}{f-x_1}$$

$$m_2 = \frac{f}{f-x_2} \quad m_1 = -m_2$$

$$2f = (x_1 + x_2)$$

$$f = \frac{x_1 + x_2}{2}$$

45. (3)

Sol. $i + e = A + \delta$

$$A = (i + e) - \delta = (37^\circ + 42^\circ) - 40^\circ = 39^\circ.$$

46. (1)

Sol. रजतित दर्पण की फोकस दूरी

$$\frac{1}{f} = \frac{1}{f_m} - \frac{2}{f_l}$$

$$\frac{1}{f} = \frac{1}{\infty} - \frac{2}{f_l}$$

$$f = -\frac{f_l}{2}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{-60} + \frac{1}{-30} = \frac{1}{f}$$

$$f = -20 \text{ cm}$$

बिना रजतित लैन्स की फोकस दूरी 40 cm है।

अतः $\frac{1}{v} + \frac{1}{60} = \frac{1}{40}$

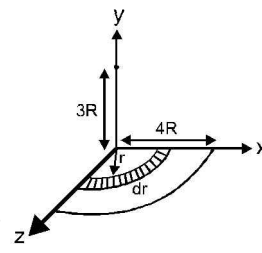
$$\frac{1}{v} = \frac{1}{40} - \frac{1}{60} = \frac{3-2}{120} = +\frac{1}{120} \Rightarrow v = 120 \text{ cm}.$$

47. (3)

48. (5)

Sol. (1) $v = \int_0^{4R} \frac{K \frac{\pi}{2} r dr \sigma}{\sqrt{r^2 + (3R)^2}} = \frac{\sigma R}{4 \epsilon_0}$

(2) $v = \int_0^{4R} \frac{4r K \frac{\pi}{2} r dr \sigma}{r} = \frac{\sigma R}{2 \epsilon_0}$



49. (4)

Sol. q_1 के समीप विभव $+\infty$ है, अतः $q_1 > 0$ है। q_2 के समीप विभव $-\infty$ है, अतः $q_2 < 0$ है। $x_2 < x_1$ पर विभव शून्य है अतः $|q_1| > |q_2|$ है।

50. (5)

Sol. z-अक्ष पर मूलबिन्दु से r दूरी पर विद्युत क्षेत्र है

$$E = \frac{1}{4\pi\epsilon_0} \left(\frac{Qr}{(r^2 + R^2)^{3/2}} - \frac{\sqrt{8}Qr}{(r^2 + 4R^2)^{3/2}} \right) = 0$$

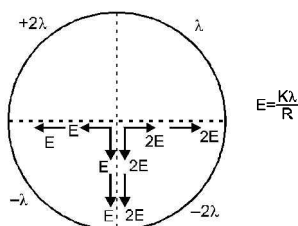
हल करने पर हम प्राप्त करते हैं $r = \sqrt{2}R$ Ans.

51. (3)

Sol. $E = \sqrt{\left(\frac{2k\lambda}{R} \frac{1}{\sqrt{2}} \right)^2 + \left(\frac{2k(2\lambda)}{R} \frac{1}{\sqrt{2}} \right)^2}$

$$= 2\sqrt{10} \frac{k\lambda}{R} = 5.4 \times 10^4 \text{ N/C}$$

वैकल्पिक हल:



$$E_{\text{net}} = 2\sqrt{10} \frac{K\lambda}{R} = 2\sqrt{10} \times \frac{9 \times 10^9 \times 3 \times 10^{-6}}{\sqrt{10}} = 54 \times 10^3 \text{ N/C.}$$

52. (1)

Sol. $mg \sin \theta = ma \cos \theta$

$$\Rightarrow a = g \tan \theta \Rightarrow \tan \theta = \frac{1}{2}$$

$$N = mg \cos \theta + ma \sin \theta = 5\sqrt{5} \text{ N}$$

53. (4)

Sol. $E = \frac{KQx}{(x^2 + R^2)^{3/2}}$, E का मान $x = \frac{R}{\sqrt{2}}$ के लिए अधिकतम है।

54. (3)

Sol. संधारित्र पर प्रारम्भिक आवेश $= KC_0 E = 20 \mu\text{C}$

पट्टिका को हटाने के पश्चात् आवेश $= C_0 E = 5 \mu\text{C}$

बैटरी से प्रवाहित आवेश की मात्रा $= KC_0 E - C_0 E = C_0 E (K - 1) = 15 \mu\text{C}$

बैटरी द्वारा अवशोषित ऊर्जा $= C_0 E (K - 1)$, $E_0 = C_0 E^2 (K - 1) = 75 \mu\text{J}$

संधारित्र में संग्रहित प्रारम्भिक ऊर्जा $= 1/2 KC_0 E^2 = 50 \mu\text{J}$

संधारित्र में संग्रहित अन्तिम ऊर्जा $= 1/2 C_0 E^2 = 12.5 \mu\text{J}$

संधारित्र में ऊर्जा में कमी $= 1/2 C_0 E^2 (K - 1) = 37.5 \mu\text{J}$

बाह्य कारक द्वारा किया गया कार्य $\frac{1}{2} E^2 C_0 (K - 1) = 37.5 \mu\text{J}$

55. (2)

56. (2)

Sol. τ अर्थात् RC समय नियतांक वह समय है जिसके पश्चात् संधारित्र प्रारम्भिक आवेश का 0.67 भाग ह्रास कर देता है।

57. (2)

Sol. $U_s = 2U_h + \text{अन्तः ऊर्जा}$

58. (2)

Sol. 1- सत्य 2-असत्य

59. (2)

Sol. अयताकार कौंच की पट्टिका दो प्रिज्मों के संयोजन की तरह व्यवहार करती है तथा यह कुल विचलन शून्य उत्पन्न करती है।

60. (4)

PART-C CHEMISTRY

61. CsBr कोर लम्बाई 4.3 Å के साथ b.c.c.

Sol. (1)

$$r_+ + r_- = \frac{\sqrt{3} a}{2} = \frac{\sqrt{3} \times 4.3}{2} = 3.72 \text{ Å}$$

62. समान आकार के गोलों से बनी hcp (ABAB...)

Sol. (1)

$$\text{अतः ccp में एकक कोष्ठिका आयतन} = \left(\frac{4a}{\sqrt{2}} \right)^3$$

$$\text{अतः प्रति गोले/परमाणु का आयतन} = \frac{1}{4} \times \frac{64a^3}{2\sqrt{2}} = 5.66a^3.$$

63. दो प्रथम कोटि की अभिक्रियाओं की अर्द्ध-आयु 3 : 2

Sol. (4)

25% पूर्ण होने में लगा समय

$$C_t = C_0 e^{-k_1 t_1} \Rightarrow \frac{3}{4} = e^{-k_1 t_1}$$

$$\text{जैसे } \ln(3/4) = -k_1 t_1 \text{ so } t_1 = \frac{1}{k_1} \ln \left(\frac{4}{3} \right).$$

$$75\% \text{ पूर्ण होने में लगा समय } t_2 = 2 \cdot \frac{\ln 2}{k_2}$$

$$\text{अतः आवश्यक अनुपात } \frac{t_1}{t_2} = \frac{\ln(4/3)k_2}{k_1 \cdot 2\ln 2} = \frac{3}{2} \times \frac{(\ln 4 - \ln 3)}{\ln 4} = 0.3 : 1$$

64. किसी निश्चित अभिक्रिया के लिए दर नियतांक

Sol. (2)

$$\frac{k_1}{k_0} = (TC)^{t-0/10}$$

$$\log \text{ लेने पर } \log_e k_1 - \log_e k_0 = \frac{t}{10} \log_e (TC)$$

$$\Rightarrow \ln k_1 = \ln k_0 + \left(\frac{\ln (TC)}{10} \right) t$$

$$\text{तुलना इंगित करती है } \ln (TC) = \ln 3 \Rightarrow TC = 3$$

65. निम्न में से कौनसा असत्य है.....

Sol. (1)

अधिशोषण के दो प्रकार की जानकारी से।

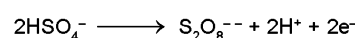
66. As_2S_3 सोल को स्कंदित करने में निम्न

Sol. (1)

हार्डी-शुल्जे नियम के अनुसार

67. HSO_4^- आयनों के विलयन के विद्युत अपघटन

Sol. (1)



अतः आवश्यक दर = 1 mole/hr

= 2 mole of e^- /hr

$$= \frac{2 \times 96500 \text{ C}}{3600 \text{ sec}} = \frac{2 \times 965}{36} \text{ A} \approx 53.6 \text{ A}$$

$$\text{अतः आवश्यक धारा} = \frac{4}{3} \times 53.6 \text{ A} = 71.4 \text{ A}$$

68. यदि $HA + NaOH \longrightarrow NaA + H_2O$

Sol. (2)

HA प्रबल अम्ल होगा, अतः इसका विलयन न्यून pH रखता है।

69. बेरियम आयन, CN^- व Co^{2+} एक आयनिक

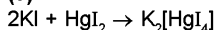
Sol. (4)

$$\alpha = \frac{i-1}{x-1} = \frac{3}{4} = \frac{4-1}{x-1} \quad \therefore x = 5 \quad \therefore$$

$Ba_3[Co(CN)_5]_2$ संकुल है।

70. 1000 g जल में 0.2 मोल KI ($\alpha = 100\%$) का

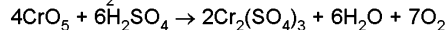
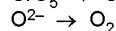
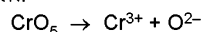
Sol. (3)



71. अभिक्रिया $CrO_5 + H_2SO_4 \rightarrow Cr_2(SO_4)_3 + H_2O + O_2$

Sol. (4)

संतुलित



72. अभिक्रिया $2MnO_4^- + 5H_2O_2 + 6H^+ \rightarrow 2Mn^{2+}$

Sol. (2)

H_2O_2 के ऑक्सीकरण के दौरान O-O बंध नहीं टूटता है।

73. 20 ml H_2O_2 को तनु H_2SO_4 के साथ अम्लीकरण

Sol. (1)

$$30 \times \frac{1}{12} = 20 \times N'$$

$$N' = \frac{30}{12 \times 20} = \frac{1}{8}$$

$$\therefore \text{सामर्थ्य} = N' \times \text{तुल्यांकी द्रव्यमान} = \frac{1}{8} \times 17 = 2.12 \text{ g/L.}$$

74. 80% भार से 100ml HCl विलयन

Sol. (3)

विलेय का द्रव्यमान तनुकरण के पहले व बाद में समान रहता है।

$$100 \times 1.5 \times 0.8 = (100 + V) \times 1 \times 0.4$$

$$V = 200 \text{ ml}$$

75. STP पर आर्द्रश गैस SO_x का 16 g, 5.6 लीटर

Sol. (2)

$$\text{गैस का अणुभार} = \frac{16 \times 22.4}{5.6} = 64 \text{ g}$$

$$32 + 16x = 64$$

$$x = 2$$

76. यदि $\frac{P_C V_C}{RT_C} = \frac{3}{8}$ एवं $\frac{P_r V_r}{T_r} = 2.4$, हो तो

Sol. (4)

$$\frac{P_r V_r}{T_r} = \frac{P}{T_c} \cdot \frac{V_m}{V_c} \cdot \frac{T_c}{T} = \frac{PV_m}{RT} \cdot \frac{RT_C}{P_C V_C}$$

$$\frac{PV_m}{RT} = 2.4 \times \left(\frac{3}{8}\right)$$

$$\text{so } V_m = \frac{0.0821 \times 200}{8.21} \times 0.9$$

$$= 1.8 \text{ L}$$

अतः दो मोल का आयतन = 3.6L.

77. यदि हवा के 600 ml का तापमान, नियत दाब पर

Sol. (3)

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\text{अतः } V_2 = \frac{T_2}{T_1} \cdot V_1 = \frac{320}{300} \times 600 \text{ ml} = 640 \text{ ml}$$

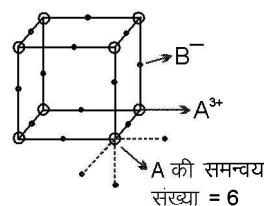
$$\text{इसलिए वृद्धि} = (640 - 600) \text{ ml} = 40 \text{ ml}$$

78. एक क्रिस्टल जिसका सूत्र AB_3 है

Sol. (3)

$$\frac{\text{धनायन की समन्वय संख्या}}{\text{ऋणायन की समन्वय संख्या}} = \frac{\text{धनायन पर आवेश}}{\text{ऋणायन पर आवेश}} = \frac{3}{1} = 3$$

अब A की समन्वय संख्या जो कि प्रदर्शित है इसलिए B की समन्वय संख्या 2 होगी।



79. निम्न समान्तर श्रृंखला अभिक्रिया के लिए

Sol. (4)

$$\frac{2k_1}{3k_2} = \frac{[B]}{[C]} = \frac{4}{9}$$

$$[B] + [C] = 2M$$

$$\frac{8}{13}M, \frac{18}{13}M$$

80. जब एक ग्राफ को $\log x/m$ तथा $\log p$ के मध्य खींचा

Sol. (2)

$$\log \frac{x}{m} = \log k + \frac{1}{n} \log P$$

$$\frac{1}{n} = \tan 45^\circ \quad \log k = 0.3010$$

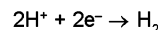
$$n = 1 \quad k = 2$$

$$\frac{x}{m} = 2 \times (0.3)^1$$

$$x = 0.6$$

81. हाइड्रोजन अर्द्ध-सेल का अपचयन विभव

Sol. (3)



$$\therefore RP = 0.0 - \frac{0.059}{2} \log \frac{P_{H_2}}{[H^+]^2}$$

82. 106.2 g इथाइलीन ग्लाइकोल का 1 मोलल

Sol. (2)

1 kg जल में विलेय का द्रव्यमान = 62 g

$$\therefore 106.2 \text{ ग्राम विलयन में विलेय का द्रव्यमान} = \frac{62}{1062} \times 106.2 = 6.2 \text{ g}$$

$$3.72 = 1.86 \times \frac{6.2 \times 1000}{62 \times w}$$

$$w = 50 \text{ g}$$

$$\therefore \text{पृथक बर्फ का द्रव्यमान} = 100 - 50 = 50 \text{ g}$$

83. अम्लीय माध्यम में FeSO_4 और $\text{Fe}_2(\text{SO}_4)_3$ के मिश्रण

Sol. (4)

FeSO_4 के मिली तुल्यांक = KMnO_4 के मिली तुल्यांक

$$\text{FeSO}_4 \text{ के तुल्यांक} = \frac{100 \times 2 \times 5}{1000} = 1$$

FeSO_4 के मोल = 1

$$\therefore \text{FeSO}_4 \text{ का मोल प्रभाज} = \frac{1}{3}$$

84. साधारण नमक जो समुद्र से प्राप्त होता है

Sol. (3)

NaCl का द्रव्यमान = $10 \times 0.96 = 9.6 \text{ g}$

$$\text{NaCl का मोल} = \frac{9.6}{58.5}$$

$$\text{अणुओं की संख्या} = \frac{9.6}{58.5} \times 6.023 \times 10^{23} \approx 10^{23}$$

85. एक HCl(g) , $\text{PCl}_3(\text{g})$ तथा $\text{PCl}_5(\text{s})$

Sol. (4)

यदि प्रत्येक में N परमाणु है तो Cl -परमाणु के मोल = $\frac{N}{N_A}$

$\text{HCl} : \text{PCl}_3 : \text{PCl}_5$

$$\frac{N}{N_A} : \frac{1}{3} \frac{N}{N_A} : \frac{1}{5} \frac{N}{N_A}$$

15 : 5 : 3

86. 27°C पर ओजोन तथा ऑक्सीजन की rms

Sol. (3)

$$\frac{V_{\text{O}_3, \text{rms}}}{V_{\text{O}_2, \text{rms}}} = \sqrt{\frac{M_{\text{O}_2}}{M_{\text{O}_3}}} = \sqrt{\frac{32}{48}} = \sqrt{\frac{2}{3}}$$

87. समान मात्रा (द्रव्यमान) के मेथेन

Sol. (4)

$$E_1 = \frac{3}{2} \times \frac{M}{16} RT_1$$

$$E_2 = \frac{3}{2} \times \frac{M}{30} RT_2 \quad \Rightarrow \quad \frac{E_1}{E_2} = \frac{30}{16} \frac{T_1}{T_2}$$

$$\Rightarrow \frac{3}{1} = \frac{30}{16} \cdot \frac{T_1}{T_2} \quad \text{or} \quad \frac{T_1}{T_2} = \frac{8}{5}$$

88. निम्नलिखित में कौनसा वक्तव्य

Sol. (3)

SRP मान के अनुसार क्लोरीन आयोडीन आयन को ऑक्सीकृत कर सकता है

89. अम्लीय विलयन में जबकि Fe^{3+} स्थायी

Sol. (4)

Mn^{3+} जल को O_2 में ऑक्सीकृत करता है जबकि Fe^{3+} ऑक्सीकृत नहीं करता अतः यह स्थायी है।

90. ऐनिलीन से प्राप्त सोडियम संगलन निष्कर्ष, आयरन

Sol. (1)

प्रुशियन ब्लू = $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$

PREP JEE

MOCK TEST-5

PREPARE JEE EXAMINATION WITH US

DETAILED SOLUTION

PART-A MATHEMATICS

1. If $A = \sin^{-1} \dots\dots\dots$
Sol. (1)

$$\text{Let } \sin^{-1} \left(\sqrt{\frac{13}{17}} \right) = \theta \Rightarrow \sin \theta = \sqrt{\frac{13}{17}}$$
$$\Rightarrow \cot \theta = \frac{2}{\sqrt{13}} \Rightarrow A = \sin^{-1} \frac{2}{\sqrt{13}} = \tan^{-1} \frac{2}{3}$$

2. If $\lim_{h \rightarrow 0} \left[\frac{f(a+h)}{f(a)} \right] = c$ where.....
Sol. (1)

$$\text{Local minima } \Rightarrow f(a+h) > f(a)$$
$$\Rightarrow \frac{f(a+h)}{f(a)} > 1 \Rightarrow \lim_{h \rightarrow 0} \left[\frac{f(a+h)}{f(a)} \right] = 1$$
$$\text{Local maxima } \Rightarrow \frac{f(a+h)}{f(a)} < 1 \Rightarrow \lim_{h \rightarrow 0} \left[\frac{f(a+h)}{f(a)} \right] = 0$$

3. If A and B be positive
Sol. (1)

$$\sin 2B = \frac{3}{2} \sin 2A \text{ and } 3 \sin^2 A = 1 - 2 \sin^2 B = \cos 2B$$
$$\text{so, } \tan 2B = \frac{(3/2) \cdot \sin 2A}{3 \sin^2 A} = \cot A$$
$$\text{or } \tan A \cdot \tan 2B = 1$$
$$\Rightarrow A + 2B = \pi/2 \Rightarrow B = \pi/4 - A/2$$

4. If $\lim_{\theta \rightarrow 0} \left(\frac{1+a \cos \theta}{\theta^2} - \frac{b \sin \theta}{\theta^3} \right) \dots\dots\dots$

Sol. (1) $\lim_{\theta \rightarrow 0} \left(\frac{1+a \cos \theta}{\theta^2} - \frac{b \sin \theta}{\theta^3} \right) = 1$

$$\lim_{\theta \rightarrow 0} \frac{\left[\theta \left[1+a \left(1-\frac{\theta^2}{2} \right) \right] - b \left[\theta - \frac{\theta^3}{3!} \right] \right]}{\theta^3} = 1$$
$$\Rightarrow \lim_{\theta \rightarrow 0} \left(\frac{\theta(1+a-b) - \frac{a\theta^3}{2} + \frac{b\theta^3}{6}}{\theta^3} \right) = 1$$
$$\therefore 1+a-b=0$$
$$\text{and, } \frac{b}{6} - \frac{a}{2} = 1$$
$$\Rightarrow a = -5/2$$
$$\Rightarrow b = -3/2$$
$$\Rightarrow b+a = -4$$
$$\therefore \lambda = 4$$

5. The maximum values
Sol. (4)

$$y = \frac{\log_e x}{x}$$

$$\frac{dy}{dx} = -\frac{1}{x^2} \log_e x + \frac{1}{x} \cdot \frac{1}{x} = \frac{1}{x^2} (1 - \log_e x) = 0$$
$$\log_e x = 1 \text{ or } x = e$$
$$x < e \Rightarrow \log_e x < 1$$
$$x > e \Rightarrow \log_e x > 1$$

at $x = e$, $\frac{dy}{dx}$ changes sign from positive to negative and hence if is more at $x = e$ and its value $\frac{\log_e e}{e} = e^{-1}$.

6. If $f(x) = \frac{\sin(e^{x-2}-1)}{\ln(x-1)}$,
Sol. (3)

$$\lim_{x \rightarrow 2} \frac{\sin(e^{x-2}-1)}{(e^{x-2}-1)} \times \frac{(e^{x-2}-1)}{x-2} \times \frac{x-2}{\ln(1+x-2)} = 1$$

7. Let $f(x) = x^x$;
Sol. (2)

$$\text{we have } f(g(x)) = g(x)^{g(x)} = x$$
$$\text{also } g(f(x)) = x$$
$$\Rightarrow g'(f(x)) \cdot f'(x) = 1 \Rightarrow g'(f(x)) = \frac{1}{f'(x)}$$
$$\Rightarrow g'(f(x)) = \frac{1}{x^x \cdot (1 + \ln x)}$$
$$\Rightarrow g'(f(g(x))) = \frac{1}{(g(x))^{g(x)} \cdot (1 + \ln(g(x)))}$$
$$\Rightarrow g'(x) = \frac{1}{x(1 + \ln g(x))}$$

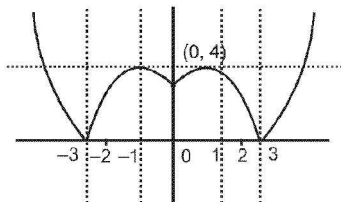
8. Rolle's Theorem holds
Sol. (2)

$$f(x) = x^3 + bx^2 + cx$$
$$f(1) = 1 + b + c$$
$$f(2) = 8 + 4b + 2c$$
$$\text{By Rolle's Theorem}$$
$$f(1) = f(2)$$
$$\Rightarrow 3b + c + 7 = 0 \dots\dots(i)$$
$$f'(x) = 3x^2 + 2bx + c$$
$$f'\left(\frac{4}{3}\right) = 0$$
$$\text{By Rolle's theorem}$$
$$\Rightarrow 8b + 3c + 16 = 0 \dots\dots(ii)$$
$$\text{By (i) \& (ii)}$$
$$\frac{b}{-5} = \frac{c}{8} = \frac{1}{1}$$
$$b = -5$$
$$c = 8.$$

9. A curve passes through
Sol. (1) $y' = x^2 - 2x$

$$y'(0^-) > 0, y'(0^+) < 0 \\ \Rightarrow \text{local maxima at } x = 0$$

10. If $f(x) = ||x|^2 - 2|x| - 3|$
Sol. (4)



11. If in a $\triangle ABC$, $a \cos^2 \frac{C}{2} + c \cos^2 \frac{A}{2} = \frac{3b}{2}$
Sol. (4)

$$a \cos^2 \left(\frac{C}{2} \right) + c \cos^2 \left(\frac{A}{2} \right) = \frac{3b}{2}$$

$$a \left(\frac{s(s-c)}{ab} \right) + c \left(\frac{s(s-a)}{bc} \right) = \frac{3b}{2}$$

$$\text{L.H.S.} \Rightarrow s \left[\frac{s-c}{b} + \frac{s-a}{b} \right]$$

$$\Rightarrow \frac{s}{b} [a + b + c - a - c] \Rightarrow \frac{a+b+c}{2} = \frac{3b}{2}$$

$$\Rightarrow a + c = 2b$$

12. If $f(\theta) = (\sin \theta + \operatorname{cosec} \theta)^2$
Sol. (3)

$$f(\theta) = 1 + 4 + \operatorname{cosec}^2 \theta + \sec^2 \theta$$

$$\Rightarrow 5 + \frac{1}{\sin^2 \theta \cos^2 \theta} \Rightarrow 5 + \frac{4}{\sin^2 2\theta}$$

$$\text{min. value} \Rightarrow 9$$

13. The set $(A \cup B \cup C)$
Sol. (1)

$$(A \cup B \cup C) \cap (A \cap B' \cap C')' \cap C'$$

$$= (A \cup B \cup C) \cap (A' \cup B \cup C) \cap C'$$

$$= [(A \cap A') \cup (B \cup C)] \cap C'$$

$$= (\phi \cup B \cup C) \cap C' = (B \cup C) \cap C'$$

$$= (B \cap C') \cup (C \cap C')$$

$$= (B \cap C') \cup \phi = B \cap C'$$

14. The value of $\cos^{-1}$
Sol. (44)

$$\therefore \sin^{-1} \left(\frac{\sqrt{2-\sqrt{3}}}{2} \right)$$

$$= \tan^{-1} \left(\frac{\sqrt{2-\sqrt{3}}}{\sqrt{2+\sqrt{3}}} \right) = \tan^{-1} (2 - \sqrt{3}) = \frac{\pi}{12}$$

$$= \frac{\pi}{12} + \frac{\pi}{6} + \frac{\pi}{4} = \frac{\pi}{2}$$

$$\Rightarrow \cot \frac{\pi}{2} = 0 \text{ and } \cos^{-1}(0) = \frac{\pi}{2}$$

15. If $x^{3/4(\log_3 x)^2 + \log_3 x - 5/4} = \sqrt{3}$
Sol. (4)

Taking logarithm,

$$\left\{ \frac{3}{4} (\log_3 x)^2 + \log_3 x - \frac{5}{4} \right\} \log_3 x = \log_3 \sqrt{3}$$

$$\text{or } \frac{3}{4} y^3 + y^2 - \frac{5}{4} y = \frac{1}{2}, \text{ (writing } y \text{ for } \log_3 x)$$

$$\text{or } 3y^3 + 4y^2 - 5y - 2 = 0 \text{ or } (y-1)(3y^2 + 7y + 2) = 0$$

$$\text{or } (y-1)(3y+1)(y+2) = 0$$

$$\therefore \log_3 x = 1, -\frac{1}{3}, -2 \Rightarrow x = 3, 3^{-1/3}, 3^{-2}.$$

16. $\lim_{x \rightarrow \infty} \frac{2 \cdot x^{1/2} + 3 \cdot x^{1/3} + 4 \cdot x^{1/4} + \dots + n \cdot x^{1/n}}{(3x-4)^{1/2} + (3x-4)^{1/3} + \dots + (3x-4)^{1/n}}$
Sol. (1)

Divide N^r and D^r by $x^{1/2}$

$$\lim_{x \rightarrow \infty} \frac{2 + 3x^{(1/3-1/2)} + \dots + n \cdot x^{(1/n-1/2)}}{\left(3 - \frac{4}{x}\right)^{1/2} + x^{1/3-1/2} \left(3 - \frac{4}{x}\right)^{1/3} + \dots + x^{1/n-1/2} \left(3 - \frac{4}{x}\right)^{1/n}}$$

$$= \frac{2}{\sqrt{3}}$$

17. A solution (x, y) of
Sol. (3)

$$\cos^2(\pi x) - \sin^2(\pi y) = 1/2$$

$$\Rightarrow \cos \pi(x+y) \cos \pi(x-y) = 1/2$$

$$\Rightarrow \cos \pi(x+y) \cos(\pi/3) = 1/2 \quad [\because x-y = 1/3]$$

$$\Rightarrow \cos \pi(x+y) = 1$$

$$\Rightarrow \pi(x+y) = 2n\pi \Rightarrow x+y = 2n$$

$$\text{Now } x+y = 2n \text{ and } x-y = 1/3$$

$$\Rightarrow x = n + 1/6, y = n - 1/6, (n \in \mathbb{I})$$

$$\therefore (x, y) = \left(n + \frac{1}{6}, n - \frac{1}{6} \right) \text{ which is satisfied by (3) for } n = 2.$$

18. Let $R = \{(3, 3), (6, 6), \dots\}$
Sol. (1)

$$(3, 3), (6, 6), (9, 9), (12, 12) \rightarrow R$$

$$(3, 6) \text{ and } (6, 12) \in R \Rightarrow R$$

hence reflexive and transitive both

19. If $f(x)$ is a differentiable
Sol. (1)

$$\frac{f(3) - f(1)}{3 - 1} = f'(c), \text{ for some } c \text{ such that } 1 < c < 3$$

$$f'(c) = \frac{f(3) - 2}{2} < 2$$

$$\therefore f(3) < 6$$

$$f(x) \Big|_{x \rightarrow 2^+} \rightarrow 14$$

Thus for $f(x)$ to be strictly monotonically increasing $a \in [4, 14]$.

22. If the thrice repeated.....

Sol. (4)

As $x = 1$ is a repeated root

$$\therefore f(1) = 0 \Rightarrow a + b + c = 0$$

$$f'(1) = 0 \Rightarrow 2a + b + 4 = 0$$

$$f''(1) = 0 \Rightarrow 3a + b + 6 = 0$$

$$\therefore a = -2, b = 0, c = 2.$$

After :

$$x^4 + ax^3 + bx^2 + cx - 1 = (x - 1)^3 (x + 1)$$

by comparing coefficient

$$\therefore a = -2, b = 0, c = 2.$$

23. The curves $x^3 - 3xy^2 = a$

Sol. (3)

The two curves are

$$x^3 - 3xy^2 = a \quad \text{.....(1)}$$

$$\text{and } 3x^2y - y^3 = b \quad \text{.....(2)}$$

Differentiating (1) w.r.t. x we get

$$3x^2 - 3y^2 - 6xy \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{x^2 - y^2}{2xy}$$

Differentiating (2) w.r.t. x we get

$$6xy + 3x^2 \frac{dy}{dx} - 3y^2 \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{2xy}{x^2 - y^2}$$

$$m_1 m_2 = \left(\frac{x^2 - y^2}{2xy} \right) \times \left(\frac{-2xy}{x^2 - y^2} \right) = -1$$

$\therefore$ The curves cut each other orthogonally

24. If $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 2$, then

Sol. (3)

$$\text{Since } \lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 2$$

$$\therefore \lim_{x \rightarrow 0} f(x) = 0$$

Since the given limit is a positive quantity

$f(x)$ approaches 0 through positive values

$$\therefore \lim_{x \rightarrow 0} [f(x)] = 0$$

25. If $\sin A + \sin B = \sqrt{3} (\cos B - \cos A)$

Sol. (1)

$$\sin A + \sin B = \sqrt{3} (\cos B - \cos A)$$

$$\sin A + \sqrt{3} \cos A = \sqrt{3} \cos B - \sin B$$

$$\Rightarrow \frac{1}{2} (\sin A + \sqrt{3} \cos A) = \frac{1}{2} (\sqrt{3} \cos B - \sin B)$$

$$\sin (A + 60^\circ) = \sin (60^\circ - B)$$

$$A + 60^\circ = 2n\pi + 60^\circ - B \quad \text{or} \quad A + 60^\circ = (2n + 1)\pi - 60^\circ + B$$

$$\therefore \sin 3A + \sin 3B = 0$$

26. Let $f: \mathbb{R} - \{1, 2, 3\} \rightarrow \mathbb{R}$

Sol. (2)

we put $f(x) = 0$

27. If $f(x) = 1 - x - x^3$

Sol. (1)

$$f(x) = 1 - x - x^3$$

$$f(f(x)) = 1 - f(x) - f^3(x)$$

$$f'(x) = -1 - 3x^2 \quad (\text{decreasing function})$$

$$\therefore f'(x) < 1 - 5x$$

$$1 - x - x^3 < 1 - 5x$$

$$x^3 - 4x > 0$$

$$x(x^2 - 4) > 0$$

$$28. \text{ If } f(x) = \begin{cases} ax + b & , \quad x \leq 2 \\ x^2 - 5x + 6 & , \quad 2 < x < 3 \\ px^2 + qx + 1 & , \quad x \geq 3 \end{cases}$$

Sol. (4)

$$\lim_{x \rightarrow 2^-} f(x) = 2a + b$$

$$\lim_{x \rightarrow 2^+} f(x) = 0$$

$$\therefore 2a + b = 0 \quad \text{..... (i)}$$

$$\lim_{x \rightarrow 3^-} f(x) = 0, \quad \lim_{x \rightarrow 3^+} f(x) = 9p + 3q + 1$$

$$\therefore 9p + 3q = -1 \quad \text{..... (ii)}$$

$$f'(x) = \begin{cases} a & , \quad x < 2 \\ 2x - 5 & , \quad 2 < x < 3 \\ 2px + q & , \quad x \geq 3 \end{cases}$$

$$\therefore f'(2^-) = a = -1 = f'(2^+) \quad \text{i.e. } a = -1 \quad \text{..... (iii)}$$

$$f'(3^-) = 1 = 6p + q = f'(3^+) \quad \text{i.e. } 6p + q = 1 \quad \text{..... (iv)}$$

from (i) and (iii), we get

$$a = -1, \quad b = 2$$

from (ii) and (iv), we get

$$p = \frac{4}{9}, \quad q = -\frac{5}{3}$$

$$a = -1, \quad b = 2$$

29. If $\alpha, \beta, \gamma \in \left(0, \frac{\pi}{2}\right)$

Sol. (1)

$$\sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$= \sin \alpha (\cos \beta \cos \gamma - 1) + \sin \beta (\cos \alpha \cos \gamma - 1) + \sin \gamma (\cos \alpha \cos \beta - 1) - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

$$\Rightarrow \sin(\alpha + \beta + \gamma) = \sin \alpha \cos \beta \cos \gamma + \sin \beta \cos \alpha \cos \gamma + \sin \gamma \cos \alpha \cos \beta - \sin \alpha \sin \beta \sin \gamma$$

30. If $f(x)$ is continuous,

Sol. (3)

$$\text{Here } h(x) = \frac{1}{1-x}; x \neq 1$$

$$\Rightarrow h(h(x)) = \frac{x-1}{x}; x \neq 0, 1$$

$$\therefore h(h(h(x))) = x; x \neq 0, 1$$

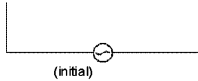
Also $g(x) \geq 0 \quad \forall x \in \mathbb{R}$

$$\Rightarrow f(x) \geq x \Rightarrow f(f(x)) \geq f(x) \geq x$$

$$\Rightarrow f(f(f(x))) \geq f(f(x)) \geq f(x) \geq x$$

$$\Rightarrow f(f(f(f(x)))) \geq f(f(f(x))) \geq f(f(x)) \geq f(x) \geq x$$

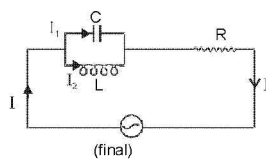
$$\Rightarrow f(f(f(f(f(x))))) - h(h(h(h(h(x)))) \geq 0 \text{ such that } x \neq 0, 1$$



In figure

$$I_1 = \frac{V_0}{X_C} \sin(\omega t + \pi/2)$$

$$\text{then } I_2 = \frac{V_0}{X_L} \sin(\omega t - \pi/2)$$



$$\text{and } I = I_1 + I_2 = 0$$

Ans. D

32. A conductor

Sol. (3)

$$B\ell Q = m\sqrt{2gh}$$

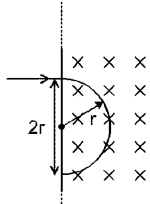
33. If a charged

Sol. (2)

Electromagnetic force will provide the necessary centripetal force.

$$eqv = \frac{mv^2}{r}$$

$$\Rightarrow r = \frac{mv}{qB} = \frac{v}{B\alpha} = \frac{(2\alpha d)(B)}{(B\alpha)} = 2d$$

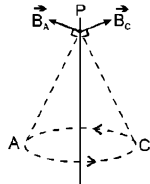


i.e. the electron will move out after travelling on a semicircular path of radius $r = 2d$. Hence (2)

34. A point charge is

Sol. (1)

The point charge moves in circle as shown in figure. The magnetic field vectors at a point P on axis of circle are $\vec{B}_A$ and $\vec{B}_C$ at the instants the point charge is at A and C respectively as shown in the figure.

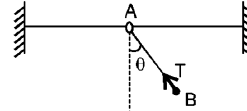


Hence as the particles rotates in circle, only magnitude of magnetic field remains constant at the point on axis P but its direction changes.

Sol. (2)

$$mgl \cos \theta = \frac{mv^2}{2}$$

$$T - mg \cos \theta = 2mg \cos \theta$$



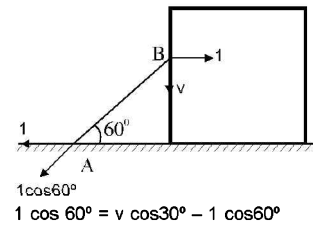
$$T = 3mg \cos \theta$$

$$3mg \cos \theta \sin \theta = \mu [mg + 3mg \cos^2 \theta]$$

$$\theta = 45^\circ$$

36. Rod AB is placed

Sol. (4)

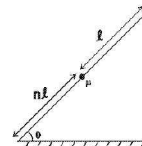


$$v = \frac{2}{\sqrt{3}}$$

$$v_{\text{net}} = \sqrt{\frac{7}{3}}$$

37. A particle starts

Sol. (3)



applying work-energy theorem from the top to bottom
 $(mg \sin \theta)l + (mg \sin \theta - \mu mg \cos \theta)n\ell = 0$

$$\sin \theta + n(\sin \theta - 3 \sin \theta) = 0$$

$$\sin \theta = 2n \sin \theta \Rightarrow n = \frac{1}{2}$$

38. If particle move

Sol. (1)

$$\vec{r} = x\hat{i} + y\hat{j}$$

$$\vec{v} = v_x\hat{i} + v_y\hat{j}$$

$$\text{Distance} = |\vec{v}| \Delta t = 10 \times \pi \times t$$

$$= 10 \times \frac{22}{7} \times \frac{7}{22} = 10 \text{ m.}$$

39. A rocket is

Ans. (4)

from (1) & (2)

$$\Rightarrow V_1 = \frac{V_0}{4} \text{ \& } V_2 = \frac{3V_0}{4}$$

When maximum extension occurs then angular speed with rest to centre for mass m_2 & m_3 are same. Using angular momentum conservation about centre of circle.

$$m \frac{3}{4} V_0 (2R) + 0 = mR^2\omega + m (2R)^2 \omega$$

$$\omega = \frac{3}{10} \frac{V_0}{R}$$

$$\text{So velocity of } m_2 = \frac{3}{5} V_0$$

$$\text{\& velocity of } m_3 = \frac{3}{10} V_0$$

When extension is maximum using energy conservation.

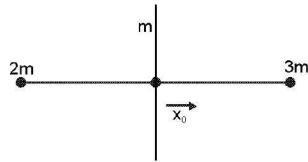
$$\frac{1}{2} m \left(\frac{3}{4} V_0 \right)^2 = \frac{1}{2} m \left(\frac{3}{5} V_0 \right)^2 + \frac{1}{2} m \left(\frac{3}{10} V_0 \right)^2$$

$$+ \frac{1}{2} kx_{\max}^2$$

$$\Rightarrow x_{\max} = \frac{3}{4} V_0 \sqrt{\frac{m}{5k}}$$

41. A frame made

Sol. (3)



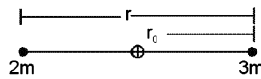
$$x_0 = \frac{2m \left(-\frac{\ell_0}{2} \right) + 3m \left(\frac{\ell_0}{2} \right)}{6m} = \frac{-m\ell_0 + \frac{3}{2}m\ell_0}{6m} = \frac{+\ell_0}{12}$$

42. In a binary star

Sol. (3)

$$3m \left(\frac{2\pi}{T} \right)^2 (r_0) = \frac{G(2m)(3m)}{r^2}$$

$$\text{and } r_0 = \frac{2}{5} r$$



$$T = 5\pi r_0 \sqrt{\frac{r_0}{2Gm}}$$

43. At $t = 0$ spring

Sol. (4)

At $t = 0$ system is in equilibrium position and perform SHM with

$$\text{time } t = 0, \frac{1}{2}, \frac{3}{2}, \dots$$

44. Initially A' and

Ans. (1)

45. An electromagnetic

Sol. (3)

$$\vec{B} = \frac{\vec{E}}{c} = \frac{1}{c} (E_1 \hat{i} + E_2 \hat{j}) \cos(kz - \omega t)$$

46. In the fraunhofer

Sol. (3)

$$\frac{1 \times \lambda_1 D}{d} = \frac{3 \times \lambda_2 D}{d}$$

$$\lambda_1 = 3\lambda_2$$

47. A polaroid is

Sol. (2)

$$I = I_0 \cos^2 \theta = I_0 \cos^2 45^\circ = \frac{I_0}{2}$$

48. In the given

Sol. (3)

At the instant of same energy, energy of capacitor is half of

maximum energy. So charge on capacitor is $\frac{q_{\max}}{\sqrt{2}}$

So, time required for this process is

$$\frac{T}{8} = \frac{1}{8} \frac{2\pi}{\omega} = \frac{\pi}{4\sqrt{LC}}$$

49. A car of mass

Sol. (2)

$$P = Fv$$

Engine power = 600 W = rate of increase in Potential energy + rate of increase in kinetic energy

$$= mgsin\theta v + m a v = 500 \times 10 \times \frac{1}{50} v + 500 \times 1 v$$

$$\therefore v = 1 \text{ m/s}$$

50. A small block

Sol. (3)

Apply Newton's laws on cart

$$10 = 2a$$

51. A man carries

Sol. (4)

$$N = P + w$$

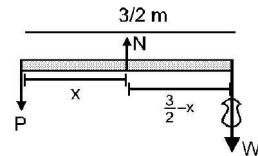
$$P = 2w$$

$$P_x = w \left(\frac{3}{2} - x \right)$$

$$2x = \frac{3}{2} - x$$

$$3x = \frac{3}{2}$$

$$x = \frac{1}{2}$$



Total charge flown through battery after reversing polarity is $2CV$
 Total energy given by battery + initial energy of capacitor
 = final energy of capacitor + Heat
 Heat = $2CV^2$.

53. An electric field

Sol. (2)

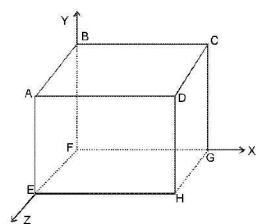
At face EFGH, $Y = 0$

$$\phi_{\text{out}} = 6 \text{ Nm}^2/\text{C}$$

At face ABCD, $Y = 1$

$$\phi_{\text{in}} = -9 \text{ Nm}^2/\text{C}$$

$$q_{\text{en}} = -3\epsilon_0$$



54. Two cells of

Sol. (4)

$$\epsilon_1 = 300 \alpha \quad \dots\dots\dots(i)$$

$$-\epsilon_2 + \epsilon_1 = 100 \alpha \quad \dots\dots\dots(ii)$$

where, α is the potential gradient

$$\therefore \frac{\epsilon_2}{\epsilon_1} = \frac{2}{3}$$

55. Two inductor

Ans. (2)

56. A uniform current

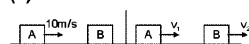
Sol. (2)

$$mgR = B_0 \pi R^2 i$$

$$i = \frac{mg}{\pi R B_0}$$

57. In the figure

Sol. (3)



$$m \times 10 = mv_1 + mv_2$$

$$\Rightarrow 10 = v_1 + v_2 \quad \dots(i)$$

$$\text{and } \frac{1}{2} \times 10 = v_2 - v_1 \quad \dots(ii)$$

From I and II

$$v_1 = \frac{5}{2} \text{ m/s}; v_2 = \frac{15}{2} \text{ m/s}$$

Distance between the two blocks

$$S = (-v_1 + v_2) \cdot t$$

$$= \left(-\frac{5}{2} + \frac{15}{2} \right) \times 5 = 25 \text{ m}$$

58. A particle is

Sol. (4)

Change in momentum = Impulse

$$\Delta \vec{P} = J_x \hat{i} + J_y \hat{j} + J_z \hat{k}$$

Sol. (1)

$$\text{Heat} = (i_{\text{rms}})^2 \cdot Rt$$

$$i_{\text{rms}} = \frac{i_0}{\sqrt{2}} = \frac{25}{Z\sqrt{2}}$$

$$\text{where, } Z = \sqrt{R^2 + (x_L - x_C)^2}$$

$$= \sqrt{4^2 + (7 - 4)^2} = 5 \Omega$$

$$\Rightarrow \text{Heat} = \left(\frac{25}{5\sqrt{2}} \right)^2 \cdot 4.80 = 4000 \text{ J and Amplitude of wattless}$$

$$\text{current} = i_0 \sin \phi$$

$$\text{where; } \phi = \tan^{-1} \left(\frac{x_L - x_C}{R} \right) = \tan^{-1} \left(\frac{7 - 4}{4} \right) = 37^\circ$$

$$\Rightarrow \text{Amplitude} = i_0 \sin 37^\circ = \frac{25}{5} \cdot \frac{3}{5} = 3 \text{ Ampere}$$

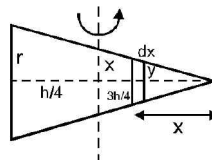
Hence (1)

60. Moment of inertia

Sol. (4)

$$I_c = \int_0^R \frac{1}{3} \frac{M}{\pi R^2 \cdot R} \cdot \pi x^2 \cdot dx \left[\frac{x^2}{3} + x^2 \right]$$

$$= \frac{3M}{R^3} \int_0^R \left(\frac{x^4}{4} + x^4 \right) dx$$

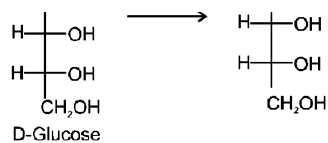


$$= \frac{3M}{R^3} \left(\frac{R^5}{20} + \frac{R^5}{5} \right) = \frac{3}{4} MR^2$$

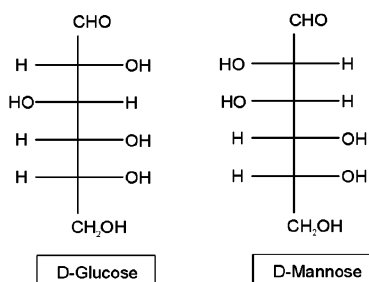
Applying parallel axis theorem,

$$\Rightarrow I_c = I_{\text{cm}} + \left(\frac{3R}{4} \right)^2 = \frac{3}{4} MR^2$$

$$I_{\text{cm}} = \frac{3MR^2}{16} \quad \text{Ans.}$$



62.



The above structures are related

Sol. (4)

All the above structures differs in configuration at one or two stereocentre.

63. RNA forms loop structure because

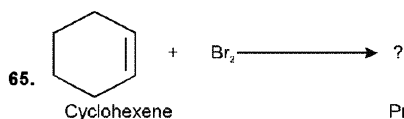
Sol. (2)

H-bond is formed is between complementary bases.

64. A protein attached to carbohydrate moiety

Sol. (4)

It is simple fact.



The correct name of the product

Sol. (3)

Addition of bromine is anti and the product is trans-1,2-dibromocyclohexane.

66. Which of the following statements is correct ?

(1) $-\text{NO}_2$ group activates the benzene

Sol. (2)

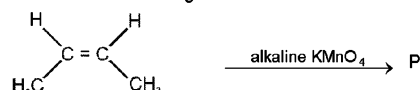
$-\text{NO}_2$ group is strong electron withdrawing due to $-M$ whereas $-\text{NH}_2$ group is strong electron donating group due to $+M$ effect.

67. The isoelectric point of an amino acid

Sol. (3)

At pI , the total(+) ion concentration is equal to total (-) ion concentration.

68. Which of the following statements is true



Sol. (1)

cis-2-Butene undergoes hydroxylation by syn addition forming meso product.

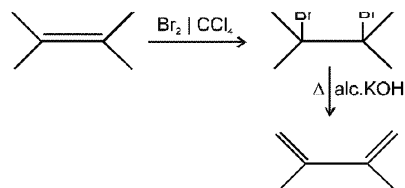
69. Which one of the following combinations is found in DNA?

Sol. (3)

Guanine and Cytosine are nitrogenous bases of DNA.

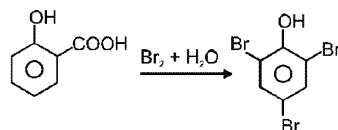
70. Compound 'X' on heating with Zn

Sol. (3)



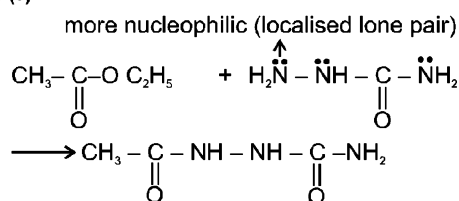
72. The compound which reacts with excess bromine.....

Sol. (3)



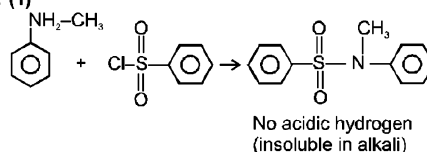
73. Ethyl acetate reacts with $\text{NH}_2\text{NHCONH}_2$

Sol. (3)



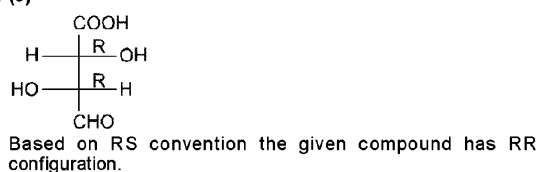
74. The compound X ($\text{C}_7\text{H}_9\text{N}$) reacts.....

Sol. (1)



75. The configurations of the carbon atoms C_2

Sol. (3)

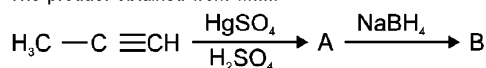


76. Major product of mononitration

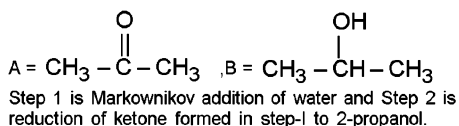
Sol. (4)

Nitration takes place at activated benzene ring.

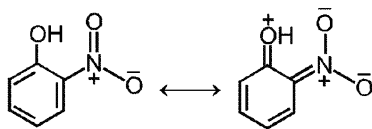
77. The product obtained from



Sol. (2)



High dipole moment of (I) is due to ionic structure of resonating form.



80. The order of acidities of the H-atoms.....

Sol. (1)

IV > II > I > III

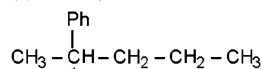
Structure IV is most acidic as the conjugate base is aromatic.

IV > II > I > III

81. The compound that is chiral.....

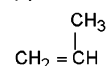
Sol. (3)

Only (3) has asymmetric carbon atom.



82. The monomer/s of the following polymer

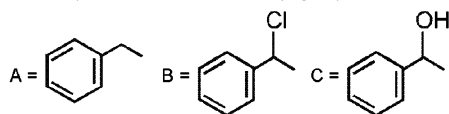
Sol. (2)



83. The product (3) of the following sequence

Sol. (4)

In our opinion there are no methyl group in above reaction



but official answer given A.

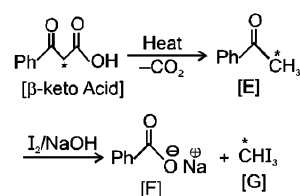
84. The order of reactivity of ammonia

Sol. (2)

Acid chloride is more electrophilic than alkyl halide, hence more reactive.

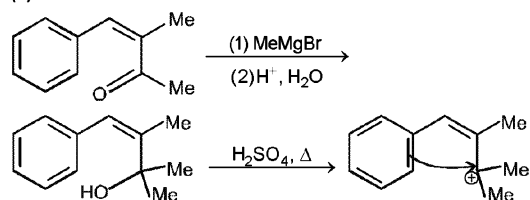
85. In the following reaction sequence, the correct

Sol. (3)

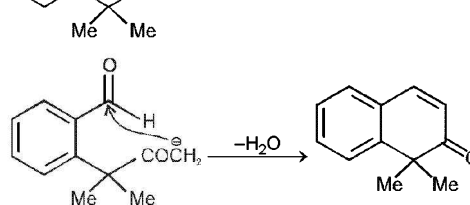


86. A carbonyl compound P, which.....

Sol. (2)

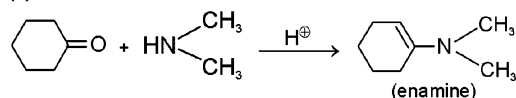


condensation

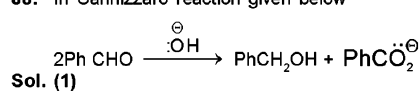


87. Reaction of cyclohexanone with dimethylamine.....

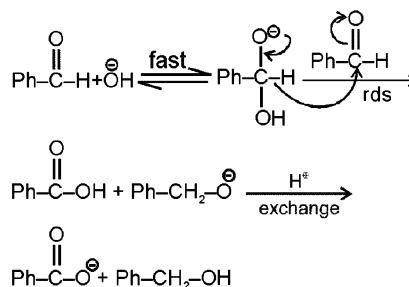
Sol. (3)



88. In Cannizzaro reaction given below

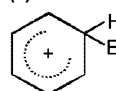


Sol. (1)



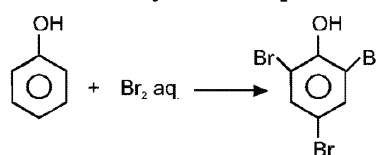
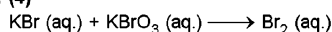
89. The electrophile, E⁺, attacks.....

Sol. (1)



90. Phenol is heated with a solution

Sol. (4)



2, 4, 6-tribromophenol

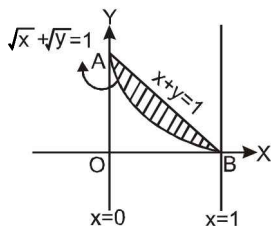
DETAILED SOLUTIONS

PART- A MATHEMATICS

1. The area bounded by the.....

Sol. (1)

The required area



$$= \text{area of } \triangle AOB - \int_0^1 (1 - \sqrt{x})^2 dx$$

$$= \frac{1}{2} - \left(x + \frac{x^2}{2} - 2 \frac{x^{3/2}}{3/2} \right)_0^1 = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$$

2. If $a, b, c, d \in \mathbb{R}$ then

Sol. (2)

The discriminants of the given equation are, $D_1 = a^2 + 12b$;

$D_2 = c^2 - 4b$ and $D_3 = d^2 - 8b$

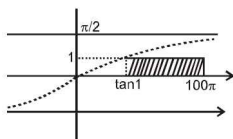
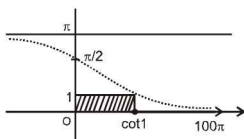
$\therefore D_1 + D_2 + D_3 = a^2 + c^2 + d^2 \geq 0$

$\Rightarrow$ At least one of D_1, D_2, D_3 is non-negative.

Hence, the equation has at least two real roots.

3. The value of $\int_0^{100\pi} (\cot^{-1} x) + [\tan^{-1} x] dx$

Sol. (2)



$$\int_0^{100\pi} (\cot^{-1} x) + [\tan^{-1} x] dx = \cot 1 + (100\pi - \tan 1)$$

$$= 100\pi + \frac{1 - \tan^2 1}{\tan 1} = 100\pi + 2 \cot 2$$

4. Let $f(x) = \int_{-2}^x e^{(1+t)^2} dt$

Sol. (1) $g(x) = \int_{-2}^{h(x)} e^{(1+t)^2} dt \Rightarrow g'(x) = h'(x) \cdot e^{(1+h(x))^2}$

$$\Rightarrow g'(2) = h'(2) \cdot e^{(1+h(2))^2}$$

$$\Rightarrow e^4 = 1 \cdot e^{(1+h(2))^2}$$

$$\Rightarrow 1 + h(2) = \pm 2$$

$$\Rightarrow h(2) = -3, 1$$

$$\Rightarrow \text{sum} = -2 \Rightarrow |\text{sum}| = 2$$

5. $\sum_{j=1}^{21} a_j = 693$, where

Sol. (2)

$$\sum_{j=1}^{21} a_j = 693 = \frac{21}{2} (a_1 + a_{21}), \text{ therefore } a_1 + a_{21} = 66$$

Now, $a_{11} = A.M. = 693/21 = 33$

Also, $a_2 + a_{20} = a_3 + a_{19} = \dots = a_9 + a_{13} = a_{10} + a_{12} = 66$

$$\therefore \sum_{i=0}^{10} a_{2i+1} = 5 \times (a_1 + a_{21}) + a_{11} = 5 \times 66 + 33 = 363.$$

6. $\int \frac{\sin 2x}{\sin 5x \sin 3x} dx$

Sol. (3)

$$\int \frac{\sin 2x}{\sin 5x \sin 3x} dx = \int \frac{\sin(5x - 3x)}{\sin 5x \sin 3x}$$

$$= \int \frac{\sin 5x \cos 3x - \cos 5x \sin 3x}{\sin 5x \sin 3x}$$

$$= \frac{1}{3} \ln \sin 3x - \frac{1}{5} \ln \sin 5x = c$$

7. The general solution.....

Sol. (3) $(x^5 y^4 + x^2) dy = (1 - x^5 y^5 - xy) dx$
 $x(xdy + ydx) - dx + x^5 y^4 (xdy + ydx) = 0$

$$\Rightarrow \frac{dx}{x} = (1 + x^4 y^4) d(xy)$$

$$\Rightarrow \ln |x| = xy + \frac{x^5 y^5}{5} + C$$

8. A spherical acetone

Sol. (1)

$$\frac{dv}{dt} = \lambda \cdot 4\pi r^2 \Rightarrow 3\pi r^2 \cdot \frac{dr}{dt} = \lambda \cdot 4\pi r^2$$

$$\Rightarrow \frac{dr}{dt} = \lambda \Rightarrow r = \lambda t + C$$

We have $r_0 = 3$ and $r = 2$ at $t = 1$

$$\Rightarrow C = 3, 2 = \lambda + 3 \Rightarrow r = 3 - t$$

9. If $x_1, x_2, x_3, \dots, x_{2008}$

Sol. (4)

Given $x_1, x_2, x_3, \dots, x_{2008}$ are in HP.

$$\therefore \frac{1}{x_1}, \frac{1}{x_2}, \frac{1}{x_3}, \dots, \frac{1}{x_{2007}}, \frac{1}{x_{2008}} \text{ are in AP.}$$

$$\text{Then, } \frac{1}{x_2} - \frac{1}{x_1} = \frac{1}{x_3} - \frac{1}{x_2} = \dots = \frac{1}{x_{2008}} - \frac{1}{x_{2007}} = D$$

$$\text{or } \frac{x_1 - x_2}{x_1 x_2} = \frac{x_2 - x_3}{x_2 x_3} = \dots = \frac{x_{2007} - x_{2008}}{x_{2007} x_{2008}} = D$$

$$= \frac{(x_1 - x_2) + (x_2 - x_3) + \dots + (x_{2007} - x_{2008})}{x_1 x_2 + \dots + x_2 x_3 + \dots + x_{2007} x_{2008}} = D$$

(by law of proportion)

$$\text{or } x_1 x_2 + x_2 x_3 + \dots + x_{2007} x_{2008} = \frac{x_1 - x_{2008}}{D}$$

$$\text{or } \sum_{i=1}^{2007} x_i x_{i+1} = \frac{x_1 - x_{2008}}{D} \quad \dots(i)$$

$$\text{Now, } \frac{1}{x_{2008}} = \frac{1}{x_1} + (2008 - 1) D$$

$$\text{or } \frac{x_1 - x_{2008}}{D} = 2007 x_1 x_{2008} \quad \dots(ii)$$

$\therefore$ From Eqs. (i) and (ii), we get

$$\sum_{i=1}^{2007} x_i x_{i+1} = 2007 x_1 x_{2008} \quad \therefore \lambda = 2007$$

$$10. \lim_{h \rightarrow 0} \frac{1}{h} \left[\int_a^{x+h} \sin^4 t \, dt - \int_a^x \sin^4 t \, dt \right] \dots\dots\dots$$

Sol. (1)

Use L' Hospital rule, we get

$$= \lim_{h \rightarrow 0} \frac{\sin^4(x+h) - 0}{1} = \sin^4 x.$$

11. If α, β be the roots of

Sol. (2)

$$\alpha + \beta = 1 \text{ and } \alpha\beta = -1$$

$$\text{AM of } A_{n-1} \text{ and } A_n = \frac{A_{n-1} + A_n}{2}$$

$$= \frac{\alpha^{n-1} + \beta^{n-1} + \alpha^n + \beta^n}{2}$$

$$= \frac{\alpha^{n-1}(1 + \alpha) + \beta^{n-1}(1 + \beta)}{2}$$

$$= \frac{\alpha^{n-1}(\alpha^2) + \beta^{n-1}(\beta^2)}{2} \quad (\because \alpha^2 = \alpha + 1 \text{ and } \beta^2 = \beta + 1)$$

$$= \frac{1}{2} (\alpha^{n+1} + \beta^{n+1}) = \frac{1}{2} A_{n+1}.$$

12. Solution of the differential

Sol. (1)

$$(2x - 10y^3) \frac{dy}{dx} + y = 0$$

$$\Rightarrow y \frac{dx}{dy} + 2x - 10y^3 = 0$$

$$\Rightarrow \frac{dx}{dy} + \frac{2}{y} x - 10y^2 = 0$$

$$\text{I.F. } e^{\int \frac{2}{y} dy} = y^2$$

$$\Rightarrow xy^2 = \int 10y^4 dy$$

$$\Rightarrow xy^2 = 2y^5 + c$$

$$\Rightarrow y^2(x - 2y^3) = c$$

13. If 2nd, 5th, and 9th terms

Sol. (3)

$a + d, a + 4d, a + 8d$ are in G.P.

$$(a + 4d)^2 = (a + d)(a + 8d)$$

$$a^2 + 8ad + 16d^2 = a^2 + 9ad + 8d^2$$

$$\Rightarrow 8d^2 = ad \quad \Rightarrow \frac{a}{d} = 8$$

14. If α, β are the roots

Sol. (2)

$$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$$

$$a\alpha + b = -a\beta$$

$$a\beta + b = -a\alpha$$

$$\therefore \frac{\alpha}{a\beta + b} + \frac{\beta}{a\alpha + b} = -\frac{2}{a}$$

15. Let $S_1, S_2, \dots, S_n$ be

Sol. (2)

We have length of a side of S_n = length of a diagonal of S_{n+1}

$$\Rightarrow \text{Length of a side of } S_n = \sqrt{2} \text{ (length of a side of } S_{n+1})$$

$$\Rightarrow \frac{\text{Length of a side of } S_{n+1}}{\text{Length of side of } S_n} = \frac{1}{\sqrt{2}} \text{ for all } n \geq 1$$

$$\Rightarrow \text{Sides of } S_1, S_2, \dots, S_n \text{ form a G.P. with common ratio } \frac{1}{\sqrt{2}}$$

and first term 10.

$$\therefore \text{Side of } S_n = 10 \left(\frac{1}{\sqrt{2}} \right)^{n-1} = \frac{10}{2^{(n-1)/2}}$$

$$\Rightarrow \text{Area of } S_n = (\text{side})^2$$

$$= \left(\frac{10}{2^{(n-1)/2}} \right)^2 = \frac{100}{2^{n-1}}. \text{ Now, area of } S_n < 1 \Rightarrow n - 1 \geq 7$$

16. The sum to n terms

Sol. (3)

Let S_n denote the sum to n terms of the given series. Then,

$S_n = 11 + 103 + 1005 + \dots$ to n terms

$$\Rightarrow S_n = (10 + 1) + (10^2 + 3) + (10^3 + 5) + \dots + \{10^n + (2n - 1)\}$$

$$\Rightarrow S_n = (10 + 10^2 + \dots + 10^n) + \{1 + 3 + 5 + \dots + (2n - 1)\}$$

$$\Rightarrow S_n = \frac{10(10^n - 1)}{(10 - 1)} + \frac{n}{2} (1 + 2n - 1) = \frac{10}{9} (10^n - 1) + n^2$$

17. The value of the integral

Sol. (1)

$$f(\theta) = \log \left(\frac{a - \sin \theta}{a + \sin \theta} \right) \text{ for which } f(-\theta) = \log \left(\frac{a + \sin \theta}{a - \sin \theta} \right)$$

$$= -\log \left(\frac{a - \sin \theta}{a + \sin \theta} \right) = -f(\theta)$$

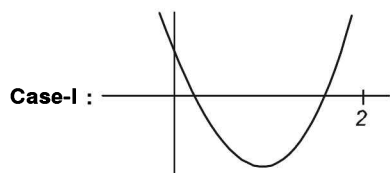
Hence, the integrand is an odd function. So, the given integral is zero.

18. The set of values of

Sol. (4)

$$4x^2 - 20px + (25p^2 + 15p - 66) = 0$$

$$x^2 - 5px + \frac{(25p^2 + 15p - 66)}{4} = 0$$



$$(i) D \geq 0$$

$$25p^2 - 4.1. \frac{(25p^2 + 15p - 66)}{4} \geq 0$$

$$-15p + 66 \geq 0$$

$$15p - 66 \leq 0$$

$$p \leq \frac{66}{15}$$

$$p \leq \frac{22}{5}$$

$$p \in \left(-\infty, \frac{22}{5} \right] \quad \dots(1)$$

$$(ii) f(2) > 0$$

$$16 - 40p + 25p^2 + 15p - 66 > 0$$

$$25p^2 - 25p - 50 > 0$$

$$p^2 - p - 2 > 0$$

$$p^2 - 2p + p - 2 > 0$$

$$p(p - 2) + 1(p - 2) > 0$$

$$(p - 2)(p + 1) > 0$$

$$p \in (-\infty, -1) \cup (2, \infty) \quad \dots(2)$$

$$(iii) \frac{-b}{2a} < 2$$

$$\frac{20p}{2.4} < 2 \Rightarrow p < \frac{16}{20} \Rightarrow p < \frac{4}{5}$$

$$p \in \left(-\infty, \frac{4}{5} \right) \quad \dots(3)$$

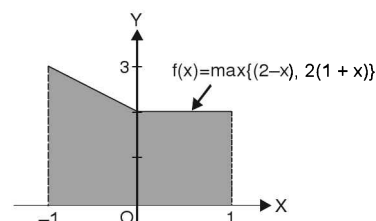
$$(1) \cap (2) \cap (3)$$

$$p \in (-\infty, -1).$$

19. Find the area covered.....

Sol. (4)

The graph of $(x) = \max \{2 - x, 2(1 + x)\}$ is shown as in the figure.



$$\text{therefore, } \int_{-1}^1 \max \{2 - x, 2, 1 + x\} dx$$

= Area of the shaded region

$$= \frac{1}{2} (2 + 3) \times 1 + 1 \times 2 = \frac{9}{2}$$

$$20. \text{ The value of } \int_0^{\pi/2} x \cot x dx \dots\dots\dots$$

Sol. (3)

Integrating by parts taking $\cot x$ as second function, given integral

$$I = [x \log \sin x]_0^{\pi/2} - \int_0^{\pi/2} \log \sin x dx$$

$$= 0 - \lim_{x \rightarrow 0} (x \log \sin x) - \int_0^{\pi/2} \log \sin x dx = \frac{1}{2} \pi \log 2$$

$$\text{as } \lim_{x \rightarrow 0} x \log \sin x = \lim_{x \rightarrow 0} \left(\frac{\log \sin x}{1/x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\cot x}{-1/x^2} \right) = \lim_{x \rightarrow 0} \left(\frac{-x^2}{\tan x} \right)$$

$$= \lim_{x \rightarrow 0} \left(-x, \frac{x}{\tan x} \right) = 0.1 = 0$$

21. The degree and order of

Sol. (1)

By repeated differentiation $y^2 = 4a(x + h)$

$$yy_1 = 2a \Rightarrow yy_2 + y_1^2 = 0$$

degree is 1 and order 2

22. The area bounded

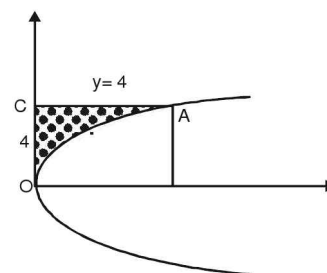
Sol. (2)

$y = 4$ meets the parabola $y^2 = x$ at A is (16, 4)

Required area = Area of rectangle OMAC - Area OMA

$$= 4 \times 16 - \int_0^{16} \sqrt{x} dx = 64 - \left[\frac{x^{3/2}}{3/2} \right]_0^{16}$$

$$= 64 - \frac{2}{3} (4)^3 = 64 - \frac{128}{3} = \frac{64}{3} \text{ sq. units}$$

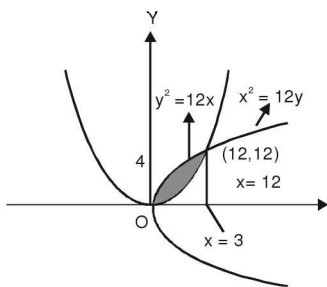


23. Thre ratio in which the

Sol. (1)

$$A_1 = \int_0^3 \sqrt{12x} dx - \int_0^3 \frac{x^2}{12} dx$$

$$= \sqrt{12} \left| \frac{2x^{3/2}}{3} \right|_0^3 - \frac{1}{36} |x^3|_0^3 = \frac{45}{4}$$



$$A_2 = \int_3^{12} \sqrt{12x} \, dx - \int_3^{12} \frac{x^2}{12} \, dx = \frac{147}{4} \text{ (on simplification)}$$

∴ Required ratio = 45 : 147, i.e., 15 : 49

24. The value of $\int_0^\infty f(x^n + x^{-n}) \dots\dots\dots$

Sol. (4)

Here limits and type of function suggest that there is something which is reciprocal to each other.

$$\text{Let, } t = 1/x \Rightarrow x = 1/t \Rightarrow dx = -\frac{1}{t^2} dt$$

Also when $x \rightarrow 0$, $t \rightarrow \infty$; $x \rightarrow \infty$, $t \rightarrow 0$

$$\Rightarrow I = \int_0^\infty f(x^n + x^{-n}) \ln x \frac{dx}{x}$$

$$= \int_\infty^0 f(t^{-n} + t^n) \ln \left(\frac{1}{t} \right) \frac{-dt}{\frac{1}{t}}$$

$$= - \int_0^\infty f(t^n + t^{-n}) \ln(t) \frac{dt}{t} = -I \Rightarrow 2I = 0 \Rightarrow I = 0$$

25. $\lim_{n \rightarrow \infty} \frac{1}{n} \dots\dots\dots$

Sol. (3)

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left(\sin^2 \frac{\pi}{2n} + \sin^2 \frac{2\pi}{2n} + \dots\dots + \sin^2 \frac{n\pi}{2n} \right)$$

$$= \int_0^1 \sin^2 \left(\frac{\pi}{2} x \right) dx = \int_0^1 \cos^2 \left(\frac{\pi}{2} x \right) dx$$

$$\therefore 2I = 1 \quad \therefore I = \frac{1}{2}$$

26. $\lim_{n \rightarrow \infty} \frac{1}{n^4} \dots\dots\dots$

Sol. (2)

$$\ln S = \frac{1}{n} \sum_{r=0}^{2n} \ln \frac{n^2 + r^2}{n^2} = \frac{1}{n} \sum_{r=0}^{2n} \ln \left(1 + \left(\frac{r}{n} \right)^2 \right)$$

$$= \int_0^2 \ln(1+x^2) \, dx = x \ln(1+x^2) \Big|_0^2 - \int_0^2 \frac{2x^2}{1+x^2} \, dx$$

$$= 2 \ln 5 - 2 \left(\int_0^2 1 \, dx - \int_0^2 \frac{dx}{1+x^2} \right) = 2 \ln 5 - 4 + 2 \tan^{-1} 2$$

$$\therefore S = e^{2 \ln 5 - 4 + 2 \tan^{-1} 2} = \frac{25}{e^4} e^{2 \tan^{-1} 2}$$

27. The solution of the

Sol. (3)

Put $y = vx$

Differential equation becomes

$$v + x \frac{dv}{dx} = \frac{x^2 + x^2v + v^2x^2}{x^2} = 1 + v + v^2$$

$$\int \frac{dv}{1+v^2} = \int \frac{dx}{x}$$

$$\tan^{-1} v = \ln x + c \Rightarrow \tan^{-1} \left(\frac{y}{x} \right) = \ln x + c$$

28. The solution of the

Sol. (3)

Put $y = vx$

Differential equation becomes

$$\int \frac{4(v^3 + 3v)}{v^4 + 6v^2 + 1} dv = -4 \int \frac{dx}{x}$$

$$\ln(v^4 + 6v^2 + 1) = -4 \ln x + \ln c$$

$$\Rightarrow x^4 + y^4 + 6x^2y^2 = c$$

29. If a, b be positive

Sol. (3)

$$\frac{a^{1/3} + b^{1/3}}{2} < \left(\frac{a+b}{2} \right)^{1/3} = \left(\frac{1}{2} \right)^{1/3} \quad (\because a + b = 1)$$

$$\therefore \frac{a^{1/3} + b^{1/3}}{2} < \frac{1}{2^{1/3}}$$

$$\Rightarrow A < \frac{2}{2^{1/3}} \quad \text{or} \quad A < 2^{2/3}$$

30. If x, y be positive real

Sol. (2)

$$\therefore \frac{(x^2)^{1/2} + (y^2)^{1/2}}{2} < \left(\frac{x^2 + y^2}{2} \right)^{1/2} = \left(\frac{8}{2} \right)^{1/2} = 2$$

$$\Rightarrow \frac{|x| + |y|}{2} \leq 2$$

$$\text{or } |x| + |y| \leq 4 \Rightarrow x + y \leq 4$$

PART- B

PHYSICS

31. In a Kundt's tube

Sol. (2)

$$\lambda_{\text{air}} = 2\Delta\ell$$

$$\lambda_{\text{gas}} = 2\Delta\ell'$$

$$\frac{V_{\text{gas}}}{V_{\text{air}}} = \frac{f\lambda_{\text{gas}}}{f\lambda_{\text{air}}} = \frac{\Delta\ell'}{\Delta\ell}$$

$$V_{\text{gas}} = \frac{1000}{3} \times \frac{3}{2} \frac{\Delta\ell}{\Delta\ell'} = 500 \text{ m/s}^{-1}$$

32. An N-P-N

Sol. (3)

The circuit arrangement is shown in figure. Collector current, I_C

$$= \frac{\text{Voltage drop across } R_L}{R_L} = \frac{1}{1000} = 10^{-3} \text{ amp}$$

$$\text{Now } V_{CE} = 9 - 1 = 8 \text{ volt}$$

$$\text{Current gain } \beta = \frac{I_C}{I_B}$$

$$\text{or } \frac{25}{26} = \frac{10^{-3}}{I_B}$$

$$\therefore I_B = 1.04 \times 10^{-3} \text{ amp}$$

$$\text{Voltage gain} = \beta \frac{R_L}{R_C}$$

$$= \frac{25}{26} \times \frac{1000}{200} = \frac{125}{26}$$

Power gain

$$= \beta^2 \frac{R_L}{R_C} = \left(\frac{25}{26}\right)^2 \times \frac{1000}{200} = \left(\frac{25}{26}\right)^2 \times 5 = 4.6$$

$$\text{Again } I_E = I_B + I_C = 1.04 \times 10^{-3} + 10^{-3} = 2.04 \times 10^{-3} \text{ A}$$

33. A piece of

Sol. (1)

Wein's displacement law

$$\lambda_{\text{max}} \propto \frac{1}{T}$$

34. In a common

Sol. (4)

$$A_v = \beta \frac{R_{\text{out}}}{R_{\text{in}}} \Rightarrow G = 25 \frac{R_{\text{out}}}{R_1} \quad \dots\dots(i)$$

$$G_m = \frac{\beta}{R_1} \Rightarrow R_1 = \frac{\beta}{G_m} = \frac{25}{0.03}$$

$$G = 25 \frac{R_{\text{out}}}{25} \times 0.03 \quad \dots\dots(ii)$$

$$G' = 20 \frac{R_{\text{out}}}{20} \times 0.02 \quad \dots\dots(iii)$$

$$G' = \frac{2}{3} G$$

35. If we study

Sol. (3)

Pressure change will be minimum at both ends

36. If photon of

Sol. (4)

$$P = \frac{E}{C} \quad \dots(i)$$

$$\lambda_p = \frac{hC}{E} \quad \dots(ii)$$

$$\lambda_e^2 = \frac{h}{2mE}$$

$$\frac{\lambda_p}{\lambda_e^2} = 2mC$$

$$\lambda_p \propto \lambda_e^2$$

37. The half life of

Sol. (1)

$$\begin{array}{ccc} X & \rightarrow & Y_0 \\ \text{at } t = 0 & & N_0 \\ \text{at } t = t & & N \end{array} \quad \begin{array}{c} 0 \\ N_0 - N \end{array}$$

$$\frac{N}{N_0 - N} = \frac{1}{7}$$

$$\frac{N}{N_0} = \frac{1}{8}$$

$$t = 3t_{1/2}$$

$$t = 3 \times 20 = 60 \text{ year}$$

38. During an adiabatic

Sol. (3)

$$P \propto T^3$$

$$P = KT^3$$

$$P = K \left(\frac{PV}{nR} \right)^3$$

$$P^2V^3 = \text{constant}$$

$$PV^{3/2} = \text{constant}$$

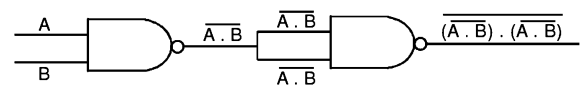
$$\gamma = \frac{C_p}{C_v} = \frac{3}{2}$$

39. Vernier constant

Ans. (1)

40. The output(X)

Sol. (2)



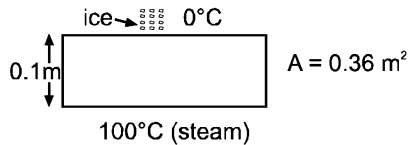
$$= \overline{A.B} = A.B$$

Alternate:

A	B	X
0	0	0
1	0	0
0	1	0
1	1	1

41. A slab of stone

Sol. (1)



Rate of heat given by steam = Rate of heat taken by ice

$$\frac{dQ}{dt} = \frac{KA(100-0)}{\ell} = m \frac{dL}{dt}$$

$$\frac{K \times 100 \times 0.36}{0.1} = \frac{4.8 \times 3.36 \times 10^5}{60 \times 60}$$

$$K = 1.24 \text{ J/m/s/}^\circ\text{C}$$

42. An ideal gas goes

Sol. (1)

Initial and final condition is same for all process

$$\Delta U_1 = \Delta U_2 = \Delta U_3$$

$$\Delta Q = \Delta U + \Delta W$$

Work done $\Delta W_1 > \Delta W_2 > \Delta W_3$ (Area of P.V. graph)

So $\Delta Q_1 > \Delta Q_2 > \Delta Q_3$

43. The equation of

Sol. (2)

$$y = 3 \sin \frac{\pi}{2} (50t - x)$$

$$y = 3 \sin \left(25\pi t - \frac{\pi}{2} x \right)$$

$$\text{Wave velocity } v = \frac{\omega}{k} = \frac{25\pi}{\pi/2} = 50 \text{ m/sec.}$$

$$v_p = \frac{\partial y}{\partial t} = 75\pi \cos \left(25\pi t - \frac{\pi}{2} x \right)$$

$$v_{p \text{ max}} = 75\pi$$

$$\text{then } \frac{v_{p \text{ max}}}{v} = \frac{75\pi}{50} = \frac{3\pi}{2}$$

44. The input resistance

Sol. (1)

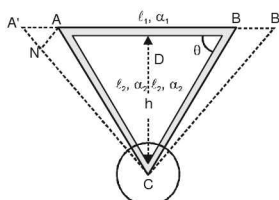
$$\text{Voltage gain} = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{I_{\text{out}}}{I_{\text{in}}} \times \frac{R_{\text{out}}}{R_{\text{in}}}$$

$$= \frac{2 \times 10^{-3}}{40 \times 10^{-6}} \times \frac{4 \times 10^3}{100}$$

$$= 2 \times 1000 = 2000$$

45. Figure shows isosceles

Sol. (1)



According to condition of the problem, height of the isosceles triangle ABC is unchanged. The dotted lines show configuration

after a temperature rise. Increase in length of rod AB,

$$\Delta \ell_1 = \ell_1 \alpha_1 \Delta T$$

Thus $AA' = 1/2 \ell_1 \alpha_1 \Delta T$

We draw a normal from A to A'C (the final length of AC). Increase in length of AC is A'N

$$A'N = \ell_2 \alpha_2 \Delta T$$

Considering increase in angle θ to be very small.

$$A'N \simeq AA' \cos \theta$$

$$\text{Where } \cos \theta = \frac{\ell_1}{2\ell_2}$$

$$\text{Thus, we have } \ell_2 \alpha_2 \Delta T = \left(\frac{1}{2} \ell_1 \alpha_1 \Delta T \right) \left(\frac{\ell_1}{2\ell_2} \right)$$

$$\text{Hence } \frac{\ell_1}{\ell_2} = 2 \sqrt{\frac{\alpha_2}{\alpha_1}}$$

46. The positron is

Sol. (4)

Cannot annihilate into a photon because momentum cannot be conserved.

The total momentum of the electron-positron pair is zero. Hence by momentum conservation, the momentum conservation, the momentum of a photon resulting from annihilate must also be zero. But this is impossible as the photon is stated to have non-zero momentum.

47. If refractive index

Sol. (4)

$$\mu = \frac{3}{2} = \frac{C}{V}$$

$$V = \frac{2C}{3} = \frac{1}{\sqrt{\epsilon_g \mu_g}} ; 2 \times 10^8 = \frac{1}{\sqrt{\epsilon_g \cdot \mu_g}}$$

$$4 \times 10^{16} = \frac{1}{3\epsilon_0 \cdot \mu_g} ; \mu_g = \frac{1}{3(4\epsilon_0) \times 10^{16}}$$

$$\mu_g = \frac{\pi}{3(4\pi\epsilon_0) \times 10^{16}} = \frac{\pi}{3} \times \frac{9 \times 10^9}{10^{16}}$$

$$= 3\pi \times 10^{9-16}$$

$$= 3\pi \times 10^{-7} = 9.42 \times 10^{-7}$$

48. Choose the

Sol. (3)

(1) $r + e = 1$ these means if r is large (good reflector) then e will be small (bad emitter)

(4) efficiency is equal to $1 - t_1/t_2$ (t_1 cannot be zero so, efficiency can not be 100%)

49. In a decay

Sol. (1)

$$\frac{N_0}{5} = N_0 e^{-\lambda t_1} \quad \dots\dots\dots(i)$$

$$\Rightarrow t_1 = \frac{2\ln 5}{\ln 2} = 2 \log_2 5 \quad (\text{use } \log_a b \cdot \log_b a = 1)$$

$$N_0 - \frac{9N_0}{10} = N_0 e^{-\lambda t_2} \quad \dots\dots\dots(ii)$$

$$\Rightarrow t_2 = \frac{2\ln 10}{\ln 2} = 2 \log_2 10 = \log_2 100$$

(i) / (ii) gives,

$$2 = e^{\lambda(t_2 - t_1)}$$

$$\ell n 2 = \lambda(t_2 - t_1)$$

From graphs $t_{1/2} = 2 \text{ sec.}$

So, $\lambda = \frac{\ell n 2}{t_{1/2}} = \frac{\ell n 2}{2}$

$$\ell n 2 = (t_2 - t_1)$$

$$t_2 - t_1 = 2.$$

50. Select correct

Ans. (4)

51. A source emit

Sol. (3)

$$\lambda' = \frac{V - V_s}{f} = \frac{332 - 32}{1000} = 0.3 \text{ m}$$

$$f' = f \frac{(V + V_0)}{V - V_s} = 1000 \times \frac{332 + 64}{332 - 32} = 1320 \text{ Hz}$$

$$\lambda'' = \frac{V - V_0}{f'} = 0.2 \text{ m.}$$

52. $p = \frac{a - t^2}{bx}$

Sol. (2)

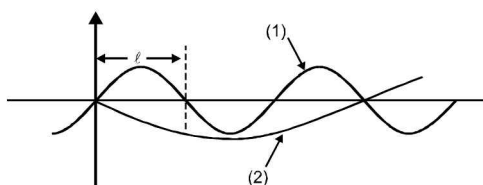
$$a = [T^2]$$

$$[p] = \frac{[T^2]}{[L]} ; \quad b = \frac{[T^2]}{[L] \left[\frac{MLT^{-2}}{L^2} \right]}$$

$$\frac{a}{b} = \frac{ML^2T^{-2}}{L^2} = MT^{-2}.$$

53. Two longitudinal

Sol. (4)



$$\frac{f_1}{f_2} = \frac{v/2\ell}{v/6\ell} = 3$$

$$\Rightarrow f_1 = 3f_2 = 3 \times 2 = 6 \text{ Hz } (\because f_2 = 2 \text{ Hz})$$

$$\Rightarrow \text{the difference in frequency of the two waves is } = f_1 - f_2 = 6 - 2 = 4 \text{ Hz}$$

54. Three samples

Sol. (1)

$$N_1\lambda_1 : N_2\lambda_2 : N_3\lambda_3 = 2:5:7$$

After two half life

$$\frac{N_1\lambda_1}{2^2} : \frac{N_2\lambda_2}{2^2} : \frac{N_3\lambda_3}{2^2} = 2:5:7$$

55. Density of the

Sol. (3)

For resonance with arm-1

$$f_0 = \frac{V}{4\ell_1}$$

$$\Rightarrow \ell_1 = 0.25 \text{ m}$$

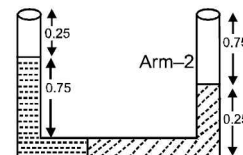
For resonance, with arm-2

$$f_0 = \frac{3v}{4\ell_2}$$

$$\ell_2 = 0.75 \text{ m}$$

$$\rho_0 + \rho_w g(0.75) = \rho_0 + \rho g(0.25)$$

$$\Rightarrow \rho = 3\rho_w.$$



56. Now we use a

Sol. (2)

For resonance appeared frequency for the tube should be still 300 Hz

$$f' = f_0 \left(\frac{V - V_0}{V - V_s} \right)$$

$$300 = 302 \left(\frac{300 - 0}{300 - (-V)} \right)$$

$V = 2 \text{ m/sec}$ away from the tube.

57. The modulation

Sol. (2)

$$\mu = \frac{A_m}{A_c} = \frac{20}{30}.$$

58. The lower

Sol. (4)

$$LSB = f_c - f_m = 12 \text{ MHz} - 12 \text{ kHz} = 11988000 \text{ Hz} = 11.988 \text{ MHz}$$

59. How many times

Ans. (2)

60. If power of sources

Sol. (1)

Phase difference at A is 9π (S_2 is lag in phase)

Phase difference at B is 3π (S_1 is lag in phase)

So, 6 times maximum sound can be observed from A to B.

$$\text{intensity at A } I_1 = \frac{7200}{4\pi(10)^2} = 18 \text{ watt/m}^2$$

$$\text{intensity at B } I_2 = \frac{7200}{4\pi(6)^2} = 50 \text{ watt/m}^2$$

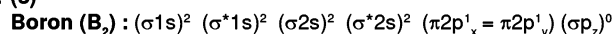
resultant intensity at B

$$= I_1 + I_2 + 2\sqrt{I_1 I_2} \cos 5\pi = (\sqrt{50} - \sqrt{18})^2.$$

PART-C CHEMISTRY

61. Which of the following

Sol. (3)

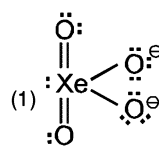


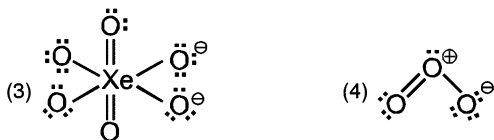
62. Unequal bond lengths

Sol. (3)

63. In which of the following species.....

Sol. (2)





64. Which of the following has

Sol. (3)

65. The oxide of an element

Sol. (1)

The element is phosphorus.

66. Which of the following statements

Sol. (3)

(1) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Br}_2$ and $[\text{Co}(\text{NH}_3)_4\text{Br}_2]\text{Cl} \cdot \text{H}_2\text{O}$; ionization as well as hydrate isomers.

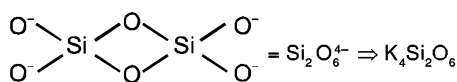
(2) $[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})](\text{NO}_3)_3$ and $[\text{Co}(\text{NH}_3)_5(\text{NO}_3)](\text{NO}_3)_2 \cdot \text{H}_2\text{O}$ hydrate isomers.

(3) $[\text{Pt}(\text{NH}_3)_4]^{2+} [\text{PtCl}_6]^{2-}$ and $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]^{2+} [\text{PtCl}_4]^{2-}$
Pt(II) Pt(IV) Pt(IV) Pt(II)

(4) $[\text{Co}(\text{NH}_3)_4(\text{NO}_2)\text{Cl}]\text{Cl}$; $[\text{Co}(\text{NH}_3)_4(\text{ONO})\text{Cl}]\text{Cl}$ linkage isomer and $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{NO}_2$ ionization isomer.

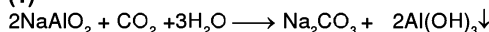
67. Usually a disilicate share

Sol. (2)



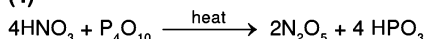
68. Bubbling CO_2 through

Sol. (1)



69. When conc. HNO_3

Sol. (4)



70. Which is not correctly

Sol. (1)

71. A substance X when

Sol. (2)

72. An aqueous solution

Sol. (1)

In excess of NH_4OH ppt of $\text{Zn}(\text{OH})_2$ will get dissolved.

73. (I) When copper

Sol. (2)

When copper ore is mixed with silica, in a reverberatory furnace copper matte is produced. The copper matte contains sulphides of copper (I) and iron (II).

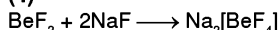
74. Among the following

Sol. (4)

On heating with concentrated NaOH solution in an inert atmosphere of CO_2 , white phosphorus gives PH_3 gas.

75. Which of the following

Sol. (4)



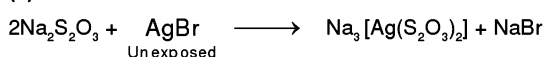
76. (I) V_2O_5 , Cr_2O_3

Sol. (3)

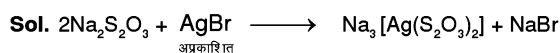
Interstitial compounds are chemically inert

77. $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$

Sol. (2)

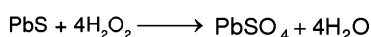
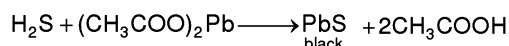


This property is used for fixing in photography.



78. H_2S reacts with lead acetate

Sol. (4)



79. (I) $[\text{MnCl}_6]^{3-}$, $[\text{FeF}_6]^{3-}$

Sol. (2)

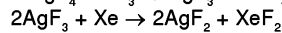
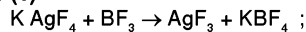
I : As all are weak field ligands therefore all will have same number of unpaired electrons as in central metal ion.

II : V.B.T. does not give any interpretation about the relative thermodynamic stabilities of various complexes. This is one of the limitation of V.B.T.

III : is correct statement.

80. When $[\text{K}]^+ [\text{AgF}_4]^-$

Sol. (1)



(red solid) (brown solid)

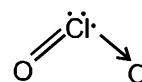
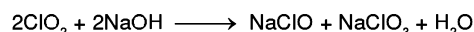
81. Which of the following ions

Sol. (1)

82. Which of the following

Sol. (2)

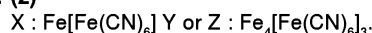
ClO_2 is powerful oxidising agent, also strong chlorinating agent. Its bleaching power is almost 30 times stronger than Cl_2 . In alkaline solution undergoes disproportionation.



unpaired electron that's why paramagnetic.

83. Correct observation

Sol. (2)

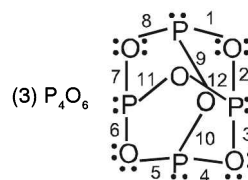
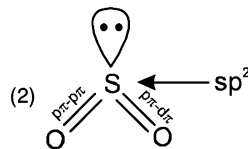


84. Which one is incorrect

Sol. (4)

(1) PH_5 does not exist as there is large difference in energies of s, p and d orbitals and hence it does not undergo sp^3d hybridisation.

In SCl_6 , smaller S cannot accommodate six larger Cl- ions. Fluorine can not expand its octet because it does not have vacant d-orbitals. So FCl_3 does not exist.



85. (I) $[\text{Co}(\text{EDTA})]^-$

Sol. (4)

(I), (II) and (III) are correct statements.

(IV) It is an example of only hydrate isomerism not ionisation isomerism because ionisation isomerism occurs owing to exchange of ions between coordination and ionisation spheres.

86. Identify the least stable

Sol. (1)

87. When haematite ore is burnt

Sol. (2)

88. Which of the following

Sol. (3)

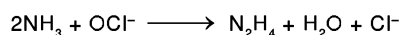
89. In which of the following complex

Sol. (3)

In $[\text{Fe}(\text{CN})_6]^{4-}$; Fe(II) is t_{2g}^6 , e_g^0 due to strong ligands.

90. Ammonia, on reaction

Sol. (3)



DETAILED SOLUTION

PART-A MATHEMATICS

1. Let number are x, y . माना संख्याएँ x, y हैं।

$$3\left(\frac{x+y}{2}\right) = a+b+c = 15$$

$$x+y=10$$

$$\frac{3}{2}\left(\frac{1}{x} + \frac{1}{y}\right) = \left(\frac{1}{p} + \frac{1}{q} + \frac{1}{r}\right) = \frac{5}{3}$$

$$\frac{1}{x} + \frac{1}{y} = \frac{10}{9}$$

$$x=9, \quad y=1$$

2. $(1-p)(1+3x+9x^2+27x^3+81x^4+243x^5) = 1-p^6$

$$\Rightarrow 1+3x+9x^2+27x^3+81x^4+243x^5 = \frac{1-p^6}{1-p}$$

$$\Rightarrow 1+3x+9x^2+27x^3+81x^4+243x^5$$

$$= 1+p+p^2+p^3+p^4+p^5$$

$$\text{Comparing, we get } p=3x \text{ or } p/x=3$$

$$\text{तुलना करने पर } p=3x \text{ या } p/x=3$$

3. $I = \int \frac{dx}{a \cos x + b \sin x}$

$$\text{Put } a = r \sin \theta, \quad b = r \cos \theta \quad \text{रखने पर}$$

$$r = \sqrt{a^2 + b^2}$$

$$\theta = \tan^{-1} \frac{a}{b}$$

$$I = \int \frac{1}{r} \operatorname{cosec}(x+\theta) dx$$

$$= \frac{1}{\sqrt{a^2+b^2}} \ln \left(\tan \left(\frac{x + \tan^{-1} \left(\frac{a}{b} \right)}{2} \right) \right) + C$$

5. $R = \{(x, y) : x^2 + y^2 = 1, x, y \in \mathbb{R}\}$

$$\text{letमाना } 3 \in \mathbb{R} \text{ and और } 3^2 + 3^2 = 18 \neq 1$$

$$\therefore (3, 3) \notin R \Rightarrow R \text{ is not reflexive}$$

$$R \text{ स्वतुल्य नहीं है।}$$

$$\text{letमाना } (x, y) \in R \therefore x^2 + y^2 = 1$$

$$\Rightarrow y^2 + x^2 = 1 \Rightarrow (y, x) \in R$$

$$\text{As } (x, y) \in R \Rightarrow (y, x) \in R$$

$$\Rightarrow R \text{ is symmetric } R \text{ सममित है।}$$

$$\text{Again पुनः } (0, 1), (1, 0) \in R$$

$$\therefore (a, b) \in R \text{ and और } (b, c) \in R$$

$$\text{but } (a, c) \notin R \text{ as } a=0=c \text{ so } x^2 + y^2 \neq 1$$

$$\text{परन्तु } (a, c) \notin R \text{ चूंकि } a=0=c \text{ इसलिए } x^2 + y^2 \neq 1$$

$$\therefore R \text{ is not transitive. } R \text{ संक्रामक नहीं है।}$$

6. Let 'A' be the top of hill and 'P' be a point on it's foot.

We have

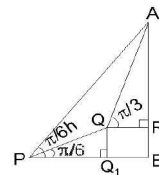
मानाकि बिन्दु 'A' पर्वत पर उच्चतम बिन्दु है तथा बिन्दु 'P' पर्वत के तल पर स्थित बिन्दु है।

जैसाकि

$$\angle APB = \frac{\pi}{4}, \angle QPB = \frac{\pi}{6}, \angle AQR = \frac{\pi}{3}, PQ = hm$$

In $\triangle APB$, में

$$PB = AB \cot \frac{\pi}{4} = AB$$



In $\triangle PQ_1Q$

$$QQ_1 = PQ \sin \frac{\pi}{6} = \frac{h}{2}$$

$$\Rightarrow \frac{h}{AB - \frac{h\sqrt{3}}{2}} = \sqrt{3}$$

$$AB = \frac{h}{(\sqrt{3}-1)} = \frac{h(\sqrt{3}+1)}{2} \text{ m}$$

7. Total arrangements of 5 papers are = $5! = 120$

पाँच परीक्षाओं को व्यवस्थित करने के कुल तरीके = $5! = 120$

When physics and chemistry come together = $4! = 48$

भौतिकी एवं रसायन शास्त्र की परीक्षा साथ साथ होने के तरीके = $4!$

$$2! = 48$$

Hence, required no. of ways are = $120 - 48 = 72$

अतः अभीष्ट तरीके = $120 - 48 = 72$

8. $\sim(\sim P \wedge Q) \wedge (P \vee Q) = [\sim(\sim p) \vee (\sim Q)] \wedge (P \vee Q)$

$$= [P \vee (\sim Q) \wedge (p \vee Q)]$$

$$= P \vee [(\sim Q) \wedge Q]$$

$$= P \vee F = P$$

9. The probability of the face number 1 to

$$\text{appear} = \frac{P(1)}{P(1)+P(2)} = \frac{0.1}{0.1+0.32} = \frac{5}{21}$$

$$\text{Hindi सतह 1 आने की प्रायिकता} = \frac{P(1)}{P(1)+P(2)} = \frac{0.1}{0.1+0.32}$$

$$= \frac{5}{21}$$

10. Since $\angle APB = \angle AQB = \frac{\pi}{2}$ so $y = mx + 8$ intersect the

circle whose diameter is AB.

Equation of circle is

$$x^2 + y^2 = 16$$

$$CD < 4$$

$$\Rightarrow \frac{8}{\sqrt{1+m^2}} < 4 \Rightarrow 1+m^2 > 4$$

$$\Rightarrow m \in (-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty)$$

If the line passing throw the point A (-4, 0), B (4, 0), then

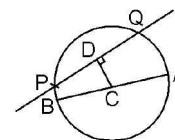
प्रतिच्छेद करती है जिसका

AB है।

वृत्त की समीकरण $x^2 + y^2 =$

$$CD < 4$$

व्यास



16 है।

$$\Rightarrow \frac{8}{\sqrt{1+m^2}} < 4$$

$$\Rightarrow 1+m^2 > 4 \Rightarrow m \in (-\infty, -\sqrt{3}) \cup (\sqrt{3}, \infty)$$

यदि रेखा बिन्दु A (-4, 0), B (4, 0) से गुजरता है, तो

$$\angle APB = \angle AQB = \frac{\pi}{2} \text{ कोण नहीं बनता है।}$$

$$\therefore m \neq \pm 2$$

11. For ellipse दीर्घवृत्त के लिए

$$8-a > 0 \text{ \& } a-2 > 0, 8-a \neq a-2$$

$$2 < a < 8, a \neq 5 \quad a \in (2, 8) - \{5\}$$

12. S_1 & S_2 : Obviously

$$S_3: y = f(x) \text{ and } y = f^{-1}(x) \text{ can intersect at points other than } y = x$$

$$\text{e.g. } y = -x + c \quad \text{or} \quad y = \sqrt{1-x^2}$$

Hindi S_1 तथा S_2 : स्पष्टतया

S_3 : $y = f(x)$ और $y = f^{-1}(x)$, $y = x$ के अलावा अन्य बिन्दुओं पर भी प्रतिच्छेद कर सकते हैं।

$$\text{उदाहरणार्थ } y = -x + c \quad \text{या} \quad y = \sqrt{1-x^2}$$

13. Normal to the required plane is

Perpendicular to $\vec{b}$ and $\vec{n}$

$\therefore$ Equation of the plane is समतल का समीकरण

$$(\vec{r} - \vec{a}) \cdot (\vec{n} \times \vec{b}) = 0$$

$$= \frac{1}{\sqrt{4+9+36}} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 6 \\ 2 & 1 & -1 \end{vmatrix}$$

$$= \frac{1}{\sqrt{49}} |-9\hat{i} + 14\hat{j} - 4\hat{k}| = \frac{\sqrt{293}}{7}$$

15. $f'(x) = 3x^2 - 6ax + 3(a^2 - 1) = 3(x - (a + 1))(x - (a - 1))$

so तथा $f'(x) = 0 \Rightarrow x = a + 1$ or $a - 1$

$\therefore a + 1 \in (-2, 4)$ if यदि $a \in (-3, 3)$

and एवं $a - 1 \in (-2, 4)$ if यदि $a \in (-1, 5)$

$\therefore a \in (-1, 3)$

16. $y = mx + c$ is normal to the parabola $y^2 = 4ax$ if

परवलय $y^2 = 4ax$ का अभिलम्ब $y = mx + c$ है यदि

$c = -2am - am^3$

$-\frac{k}{3} = -2 \times \frac{2}{3} - \frac{8}{27}$

$-\frac{k}{3} = -\frac{4}{3} - \frac{8}{27}$

$k = \frac{44}{9}$

17. Let माना $\log_a x = t \Rightarrow x = a^t$

$\therefore$ given limit दी गई सीमा $= \lim_{t \rightarrow \infty} \frac{\cot^{-1}\left(\frac{t}{a^{at}}\right)}{\sec^{-1}\left(a^{at} \cdot \frac{1}{t}\right)} = \frac{\pi/2}{\pi/2}$

$= 1$

18. It is a right Δ यह समकोण त्रिभुज है।

$\therefore$ point of intersection of perpendicular sides is the orthocentre.

लम्बवत् भुजाओं का प्रतिच्छेद बिन्दु ही लम्बकेन्द्र है।

$\Rightarrow -2\lambda - 10 = 0$

$\Rightarrow \lambda = -39$

20. $f(x) = y$ Domain प्रान्त $x \in (0, 1]$

$y' = \frac{1}{x} + \frac{1}{\sqrt{1-x^2}} > 0$

$\therefore f(x)$ is increasing वर्द्धमान है।

$\therefore$ range is परिसर $\left(-\infty, \frac{\pi}{2}\right]$.

21. $\frac{1}{2} \leq \sec^{-1} x < \frac{\pi}{2}$ or $\frac{\pi}{2} < \sec^{-1} x \leq \pi$

$\sec \frac{1}{2} \leq x < \infty$ or $-\infty < x \leq -1$

22. $\frac{dy}{dx} = \cos^{-1}(x^4) \cdot 2x - \cos^{-1}(x^2) \cdot 1$

$\left. \frac{dy}{dx} \right|_{x=\frac{1}{\sqrt[4]{2}}} = \left(\frac{\sqrt[4]{8}}{3} - \frac{1}{4} \right) \pi$

23. $a_1, a_2, \dots, a_5, a_6, a_7, \dots, a_{10}$

$\sum_{i=1}^{10} a_i = 12.5, \sum_{i=1}^6 a_i = 15, \sum_{i=6}^{10} a_i = 10$

$a_6 + 12.5 \times 10 = 90 + 50$

$a_6 = 15$

24. The given equation can be written as दी गई समीकरण को

निम्न प्रकार से लिखा जा सकता है

$\left(\frac{dy}{dx} - e^{-x} \right) \left(\frac{dy}{dx} - e^x \right) = 0$

$\Rightarrow \frac{dy}{dx} - e^{-x} = 0$ or या $\frac{dy}{dx} - e^x = 0$

$\Rightarrow dy - e^{-x} dx = 0$ or या $dy - e^x dx = 0$

$\Rightarrow y + e^{-x} = C$ or या $y - e^x = C$

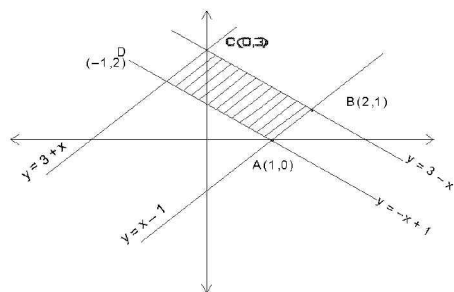
$$\Rightarrow a = 2 \cos 2, \frac{\pi}{6} = 1 \Rightarrow b = \frac{\sqrt{3}}{2} - \frac{\pi}{6}$$

$$26. y = |x-1| = \begin{cases} x-1 & x > 0 \\ -x+1 & x < 0 \end{cases}$$

$$y = 3 - |x| = \begin{cases} 3+x & x < 0 \\ 3-x & x > 0 \end{cases}$$

Solving $y = x-1$ and $y = 3-x$

$y = x-1$ तथा $y = 3-x$ को हल करने पर



$$\Rightarrow x-1 = 3-x$$

$$\Rightarrow x = 2 \text{ and तथा } y = 1$$

$$AB^2 = (2-1)^2 + (1-0)^2 = 2$$

$$AB = \sqrt{2}$$

Similarly इसी प्रकार, $BC = 2\sqrt{2}$

Area of rectangle ABCD = $AB \times BC$

आयत ABCD का क्षेत्रफल = $AB \times BC$

$$= \sqrt{2} \times 2\sqrt{2}$$

$$= 4 \text{ sq. units वर्ग इकाई}$$

$$27. {}^nC_0 - {}^nC_1 + {}^nC_2 - \dots + (-1)^r {}^nC_r = 28$$

$$\Rightarrow {}^nC_0 - ({}^{n-1}C_0 + {}^{n-1}C_1) + ({}^{n-1}C_1 + {}^{n-1}C_2) + \dots$$

$$+ (-1)^r ({}^{n-1}C_{r-1} + {}^{n-1}C_r) = 28$$

$$\Rightarrow (-1)^r {}^{n-1}C_r = 28$$

$$\Rightarrow {}^{n-1}C_r = 28 \quad (r \text{ सम होगा}) \quad (r \text{ must be even})$$

$$\Rightarrow {}^{n-1}C_r = {}^8C_2$$

$$\Rightarrow n-1 = 8, n = 9$$

Statement-2 Let $z = -2 + 0i$, then $\bar{z} = -2 - 0i$

$$\therefore \text{Arg}(z) + \text{Arg}(\bar{z}) = 2\pi \neq 0$$

$\therefore$ statement is wrong.

Hindi कथन-1 $\text{Arg}(2+3i) = \tan^{-1} \frac{3}{2}$

$$\text{Arg}(2-3i) = \tan^{-1} \left(-\frac{3}{2} \right)$$

$$\text{Arg}(2+3i) + \text{Arg}(2-3i) = 0$$

कथन-2 माना कि $z = -2 + 0i$ है, तो $\bar{z} = -2 - 0i$

$$\therefore \text{Arg}(z) + \text{Arg}(\bar{z}) = 2\pi \neq 0$$

$\therefore$ कथन असत्य है।

$$29. f(x) = x + \cos x$$

$$\therefore f'(x) = 1 - \sin x > 0 \quad \forall x \in \mathbb{R}, \text{ except at } x = 2n\pi + \frac{\pi}{2}$$

$$\text{and } f'(x) = 0 \text{ at } x = 2n\pi + \frac{\pi}{2}$$

$\therefore f(x)$ is strictly increasing.

Statement-2 is true but does not explain statement-1

$\therefore$ statement-2 gives $f'(x)$ may vanish at finite number of points

but in statement-1 $f'(x)$ vanishes at infinite number of points.

Hindi $f(x) = x + \cos x$

$$\therefore f'(x) = 1 - \sin x > 0 \quad x = 2n\pi + \frac{\pi}{2} \text{ के अलावा सभी } x \in \mathbb{R}$$

के लिए

$$\text{और } x = 2n\pi + \frac{\pi}{2} \text{ पर } f'(x) = 0$$

$\therefore f(x)$ निरन्तर वर्द्धमान है

कथन-2 सत्य है लेकिन कथन-1 को स्पष्ट नहीं करता है।

$\therefore$ कथन-2 कहता है कि $f'(x)$ कुछ परिमित बिन्दुओं पर शून्य हो सकता है लेकिन कथन-1 में $f'(x)$ अनन्त बिन्दुओं पर शून्य होता है।

PART-B PHYSICS

31. $N = mg \cos \theta$

32. $\frac{mg \cos \theta}{k}$

33. $y = A \sin \left(kx - \omega t + \frac{\pi}{2} \right), y = A \sin \left(kx + \omega t - \frac{\pi}{6} \right)$

35. frequency varies with time
आवर्ती समय के साथ परिवर्तित होती है।

37. $a \sin \theta = 2\lambda$

given दिया हुआ है $\lambda = 6 \times 10^{-7} \text{ m}$

$a = 24 \times 10^{-5} \times 10^{-2} \text{ m}$

$\sin \theta = \frac{2\lambda}{a} = \frac{2 \times 6 \times 10^{-7}}{24 \times 10^{-7}} = \frac{1}{2}$

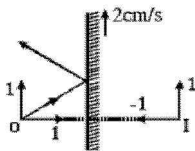
$\therefore \theta = 30^\circ$

38. $F_{\text{viscous}} = 6\pi\eta r v$. $F_d = -bv$ natural

frequency प्राकृतिक आवर्ती $= \sqrt{\frac{k}{m}}$

damping ratio मन्दन अनुपात $\zeta = \left(\frac{b}{2\sqrt{km}} \right)$

40.



Every reflected ray is going in $(-\hat{i})$ direction.

प्रत्येक परावर्तित किरण $(-\hat{i})$ दिशा में जाती है।

टक्कर के पश्चात् वस्तु विरामावस्था में आ जाती है तथा दर्पण u वेग से गति करता है।

Hence velocity of image $= 2\vec{u}$ (object is fixed & mirror is moving)

अतः प्रतिबिम्ब का वेग $= 2\vec{u}$ (वस्तु स्थिर है तथा दर्पण गतिशील है)

42. Number for yellow is 4, Number of is violet is 7.

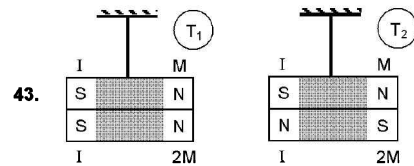
पीले के लिए संख्या 4 है, बैंगनी के लिए 7 है।

Brown colour gives multiplier 10^1 , Gold gives a tolerance of $\pm 5\%$

भूरा रंग 10^1 गुणज देता है, स्वर्ण $\pm 5\%$ सह्ययता (tolerance) देता है।

So resistance of resistor is $47 \times 10^1 \Omega \pm 5\%$ or $470 \pm 5\% \Omega$

अतः प्रतिरोधक का प्रतिरोध $47 \times 10^1 \Omega \pm 5\%$ या $470 \pm 5\% \Omega$



43.

$M_{\text{system}} = 2M + M = 3M$

$M_{\text{system}} = 2M - M = M$

$I_{\text{system}} = 2I$

$I_{\text{system}} = 2I$

$T = 2\pi \sqrt{\frac{I}{MB_H}}$ ($I_{\text{system}} \rightarrow$ same समान, $B_H \rightarrow$ same समान)

$T \propto \frac{1}{\sqrt{M}}$

$\frac{T_1}{T_2} = \sqrt{\frac{M_2}{M_1}} = \sqrt{\frac{M}{3M}} = \frac{1}{\sqrt{3}}$

44. $I = I_0 + I_0 + 2\sqrt{I_0 I_0} \cos \phi$

$\phi = \frac{2\pi}{\lambda} \Delta x = \frac{2\pi}{\lambda} \frac{\lambda}{4} = \frac{\pi}{2}$

so अतः $I = 2I_0$

$$47. \quad e = \frac{1}{2} B_0 \ell^2$$

$$i = \frac{e}{R}$$

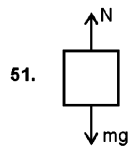
48. towards reference wire while loading & experimental wire while unloading
भार रखने के दौरान सन्दर्भ तार की तरफ तथा भार हटाने के दौरान प्रायोगिक तार की तरफ

49. $V' = V_0(1 + \gamma \Delta T)$, $A' = A_0(1 + 2\alpha \Delta T)$

$$\frac{\Delta \ell}{\ell} = \frac{\Delta V}{V} - \frac{\Delta A}{A}$$

50. Black bulb will radiate more energy to the air column causing more expansion of air column in limb x. So alcohol level in limb x will fall.

काला बल्ब अधिक ऊर्जा वायु स्तम्भ को विकिरित करता है, इसलिए वायु का स्तम्भ x में अधिक प्रसार होता है। अतः स्तम्भ -x में ऐल्कोहॉल का स्तर नीचे गिर जाता है।



$$N > mg$$

Newton's third law of action-reaction. (न्यूटन के तृतीय नियम के क्रिया प्रतिक्रिया से)

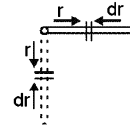
52. $\frac{KA(4-0)}{L-x} = \frac{3KA(0-(-2))}{x} \Rightarrow x = \frac{3L}{5}$

53. $MP = \frac{f_0}{f_e} = \frac{\beta}{\alpha} \Rightarrow \beta = \alpha \frac{f_0}{f_e} = 2 \times \frac{60}{5} = 24$

54. $v \frac{dv}{dx} = -9x \Rightarrow \int_6^v v dv = \int_0^2 -9x dx$

$$\Rightarrow \frac{v^2 - 6^2}{2} = -18 \quad \text{or} \quad v = 0$$

$$= mg \frac{\ell}{2}$$

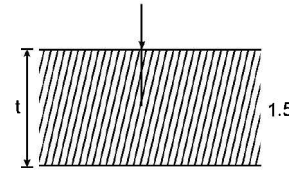


W_E = work done by electrostatic field स्थिर वैद्युत क्षेत्र द्वारा किया गया कार्य

$$= \int_{r=0}^{\ell} (\lambda dr) \cdot E \cdot r = \lambda E \frac{\ell^2}{2} = \frac{QE\ell}{2}$$

$$KE = mg \frac{\ell}{2} + \frac{QE\ell}{2}$$

58. Condition for constructive interference. सम्पौषी व्यतिकरण के लिए शर्त



$$2\mu t = (2n-1) \frac{\lambda}{2}; \quad 2 \times 1.5 \times 500 = (2n-1)$$

$$\therefore \lambda = \frac{3000}{(2n-1)}$$

Put रखने पर $n = 1, 2, 3, \dots$

59. If potential difference is increased, the electron will have more kinetic energy. As a result electron may be able to excited hydrogen atom to higher energy level. Hence statement I is true.

$$\lambda_d = \frac{h}{\sqrt{2mqv}} \propto \frac{1}{\sqrt{v}}. \text{ Hence statement-II is also true.}$$

विग्वान्तर बढ़ाने पर इलेक्ट्रॉन की गतिज ऊर्जा बढ़ेगी। परिणामस्वरूप इलेक्ट्रॉन हाइड्रोजन परमाणु को उच्च ऊर्जा स्तर पर उत्तेजित कर सकता है। अतः वक्तव्य I सही है।

$$\lambda_d \propto \frac{1}{\sqrt{v}}. \text{ अतः वक्तव्य -II भी सही है।}$$

इसलिये कुल त्वरण तथा कुल घर्षण बल केन्द्र की दिशा में नहीं होता है। इसलिये वक्तव्य-1 असत्य है।

PART-C CHEMISTRY

61. The movement towards anode shows that sol is negative. For coagulation of negative sol cation with higher charge is more effective.

एनोड की तरफ गतिशीलता दर्शाता है कि सॉल ऋणात्मक है। ऋणात्मक सॉल के स्कन्दन के लिए, उच्च आवेश युक्त धनायन अधिक प्रभावी होगा।

62. HCl and SO₂ are reducing agent can reduce MnO₄⁻ so CO₂ which is neither oxidising and nor reducing will provide only acidic medium. It can shift reaction in forward direction and reaction can go to completion.

Sol. HCl व SO₂ अपचायक अभिकर्मक है व MnO₄⁻ को अपचयित कर सकते हैं। CO₂ जो कि न तो ऑक्सीकारक है और न ही अपचायक है केवल अम्लीय माध्यम प्रदान करेगा। यह अभिक्रिया को अग्र दिशा में विस्थापित कर सकता है व अभिक्रिया को पूर्ण कर सकता है।

63. Since, the graph of t vs $(a-x)^{-1}$ is a straight line, it must be a second order reaction.

$$\therefore K = \frac{1}{t} \left[\frac{1}{(a-x)} - \frac{1}{a} \right]$$

$$\text{or } \frac{1}{a-x} = Kt + \frac{1}{a}$$

On comparing, slope

$$K = \tan\theta = 0.5 \text{ mol}^{-1} \text{ L min}^{-1}$$

$$OA = \frac{1}{a} = 2 \text{ L mol}^{-1}$$

$$\text{or } a = 0.5 \text{ mol L}^{-1}$$

$$\begin{aligned} \text{Rate} &= K(a)^2 = 0.5 \times (0.5)^2 \\ &= 0.125 \text{ mol L}^{-1} \text{ min}^{-1} \end{aligned}$$

Sol. चूँकि समय (t) व $(a-x)^{-1}$ के ग्राह्य आरेख एक सीधी रेखा है अतः यह द्वितीय कोटि अभिक्रिया होनी चाहिए।

$$K = \tan\theta = 0.5 \text{ mol}^{-1} \text{ L min}^{-1}$$

$$OA = \frac{1}{a} = 2 \text{ L mol}^{-1}$$

$$\text{या } a = 0.5 \text{ mol L}^{-1}$$

$$\text{दर} = K(a)^2 = 0.5 \times (0.5)^2$$

$$= 0.125 \text{ mol L}^{-1} \text{ min}^{-1}$$

$$64. P = m \times \frac{1}{V} + C$$

$$1 = m \times 1 + C$$

$$1.5 = m \times 2 + C$$

$$m = 0.5, C = 0.5$$

$$P = \frac{1}{2V} + \frac{1}{2}$$

$$\text{work done किया गया कार्य} = - \int PdV$$

$$= - \int_1^{0.5} \frac{1}{2} \left(\frac{1}{V} + 1 \right) dV = 0.6 \text{ Joule}$$

65. E°_{cell} सैल = 0.76 volt

Applying nernst equation नर्नस्ट समीकरण लागू करने पर

$$E_{\text{cell सैल}} = E^\circ_{\text{cell}} - \frac{0.0591}{2} \log \frac{[\text{Zn}^{2+}][\text{P}_{\text{H}_2}]}{[\text{H}^+]^2}$$

$$0.26 = 0.76 - \frac{0.0591}{2} \log \frac{(0.1)(1)}{[\text{H}^+]^2}$$

$$\log \frac{0.1}{[\text{H}^+]^2} = \frac{2 \times 0.50}{0.0591}$$

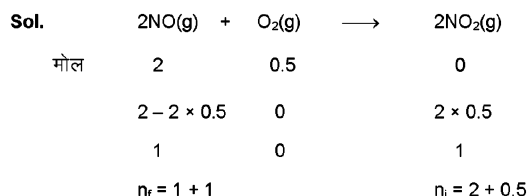
$$\text{या or } \log 0.1 - \log [\text{H}^+]^2 = 17$$

$$\text{या or } 2\text{pH} = 17 - \log 0.1$$

$$\text{pH} = \frac{18}{2} = 9$$

∴ change in pressure

$$\Delta p = \frac{\Delta nRT}{V} = 0.5 \times \frac{1}{12} \times \frac{300}{6.25} = 2 \text{ atm.}$$



∴ $\Delta n = (2.5 - 2) = 0.5$ मोल

∴ दाब में परिवर्तन

$$\Delta p = \frac{\Delta nRT}{V} = 0.5 \times \frac{1}{12} \times \frac{300}{6.25} = 2 \text{ atm.}$$

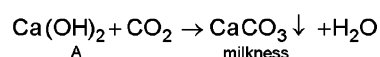
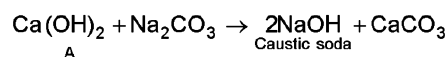
68. Vapour pressure of solution = P_A + P_B

$$= p_A^0 x_A + p_B^0 x_B = \frac{120 \times 2}{5} + \frac{180 \times 3}{5} = 156 \text{ mm Hg}$$

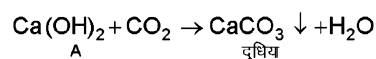
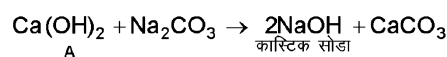
विलयन का वाष्प दाब = P_A + P_B

$$= p_A^0 x_A + p_B^0 x_B = \frac{120 \times 2}{5} + \frac{180 \times 3}{5} = 156 \text{ mm Hg}$$

69. $\text{Ca}(\text{HCO}_3)_2 + \text{Ca}(\text{OH})_2 \rightarrow 2\text{CaCO}_3 \downarrow + 2\text{H}_2\text{O}$
temp hardness



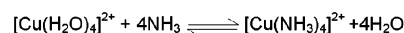
sol. $\text{Ca}(\text{HCO}_3)_2 + \text{Ca}(\text{OH})_2 \rightarrow 2\text{CaCO}_3 \downarrow + 2\text{H}_2\text{O}$
अस्थायी कठोरता



Time taken for this energy (इस ऊर्जा के लिए लगने वाला समय)

$$= \frac{213.6 \times 10^3 \text{ J}}{100 \text{ Js}^{-1}} = 2136 \text{ s} \approx 36 \text{ min.}$$

71. $[\text{Cu}(\text{H}_2\text{O})_4]^{2+} + 4\text{NH}_3 \rightleftharpoons [\text{Cu}(\text{NH}_3)_4]^{2+} + 4\text{H}_2\text{O}$ involves loss and gain of electrons. H₂O is coordinated to Cu by donating electrons (LHS). It is then removed by withdrawing electrons.



अभिक्रिया इलेक्ट्रॉन की प्राप्ति तथा हानि से सम्बन्धित है। दाता इलेक्ट्रॉन (LHS) द्वारा H₂O, Cu से उपसहसंयोजित रूप से बन्धित होता है। इसे फिर आकर्षी इलेक्ट्रॉन द्वारा हटाया जाता है।

72. Species	N ₂	O ₂	N ₂ ⁻	O ₂ ⁻
Bond-order	3	2	2.5	1.5
The O-O bond order in O ₂ ⁻ decrease				
स्पीसीज	N ₂	O ₂	N ₂ ⁻	O ₂ ⁻
बन्ध क्रम	3	2	2.5	1.5

O₂⁻ में O-O बन्ध क्रम घटता है।

73. Meq of A = Meq of B

$$\text{Meq of KMnO}_4 = 20 \times 0.5 = 10$$

$$\text{Meq of 50 ml of 0.1 M H}_2\text{C}_2\text{O}_4 = 50 \times 0.2 = 10$$

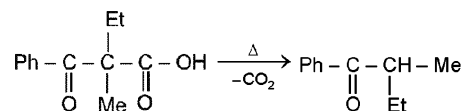
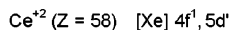
$$(0.1 \text{ M H}_2\text{C}_2\text{O}_4 = 0.2 \text{ N H}_2\text{C}_2\text{O}_4)$$

A का मिली तुल्यांक = B का मिली तुल्यांक

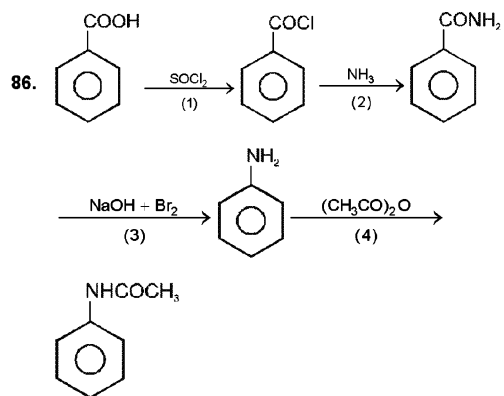
$$\text{KMnO}_4 \text{ के मिली तुल्यांक} = 20 \times 0.5 = 10$$

$$0.1 \text{ M H}_2\text{C}_2\text{O}_4 \text{ के } 50\text{ml का मिली तुल्यांक} = 50 \times 0.2 = 10$$

$$(0.1 \text{ M H}_2\text{C}_2\text{O}_4 = 0.2 \text{ N H}_2\text{C}_2\text{O}_4)$$

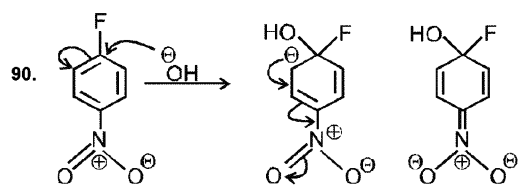
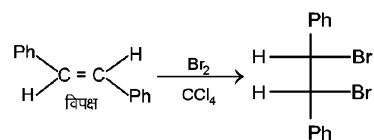
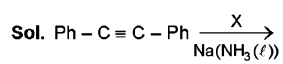
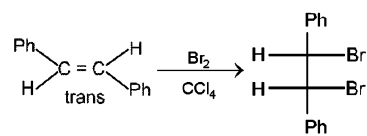
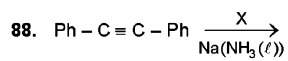


का स्थायित्व



87. D-Glucose and D-Mannose are epimers of each other.

D-ग्लूकोस और D-मैनोज एक दूसरे के एपीमर हैं।



DETAILED SOLUTION

PART-A MATHEMATICS

1. Let the numbers are a, ar, ar^2
 given $a + ar + ar^2 = 42$
 $\Rightarrow a(1 + r + r^2) = 42$
(i)
 also $2(ar + 2) = (ar^2 - 4) + (a + 2) \Rightarrow ar^2 - 2ar + a = 6$
(ii)
 on dividing (i) and (ii)
 $2r^2 - 5r + 2 = 0$
 $r = 2, \frac{1}{2}$
 and corresponding values of a are 6, 24 and hence numbers are 6, 12, 24

Hindi माना संख्याएँ a, ar, ar^2 हैं

दिया गया है $a + ar + ar^2 = 42$

$$\Rightarrow a(1 + r + r^2) = 42$$

.....(i)

$$\text{तथा } 2(ar + 2) = (ar^2 - 4) + (a + 2) \Rightarrow ar^2 - 2ar + a = 6$$

.....(ii)

(ii) में (i) का भाग लगाने पर

$$2r^2 - 5r + 2 = 0$$

$$r = 2, \frac{1}{2}$$

तथा a के संगत मान 6, 24 होंगे। तथा संख्याएँ 6, 12, 24 होंगी।

2. Let p, q, r be the three statements such that
 $p : x = 5, q : y = -2$ and $r : x - 2y = 9$
 Here given statement is $(p \wedge q) \rightarrow r$ and its contrapositive is

$$\sim r \rightarrow \sim(p \wedge q)$$

$$\text{i.e. } \sim r \rightarrow (\sim p \vee \sim q)$$

$$\text{i.e. if } x - 2y \neq 9 \text{ then } x \neq 5 \text{ or } y \neq -2$$

Hindi. माना p, q, r तीन कथन इस प्रकार हैं कि

$$p : x = 5, q : y = -2 \text{ and } r : x - 2y = 9$$

यहाँ दिया गया कथन $(p \wedge q) \rightarrow r$ है और इसका प्रतिपरिवर्तित $\sim r$

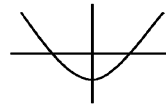
$$\rightarrow \sim(p \wedge q) \text{ है}$$

$$\text{अर्थात् } \sim r \rightarrow (\sim p \vee \sim q)$$

अर्थात् यदि $x - 2y \neq 9$ तब $x \neq 5$ या $y \neq -2$

3. $\text{tr}(A) = a + b + c = 10$
 There are three cases possible तीन स्थितियाँ संभव हैं।
 Case स्थिति-I $6 + 3 + 1 = 10$
 Case स्थिति-II $5 + 3 + 2 = 10$
 Case स्थिति -III $5 + 4 + 1 = 10$
 So total number of cases इसलिए कुल स्थितियों की संख्या $= 3 \times 3! \times 3!$

4.



root are opposite sign if $f(0) < 0$

मूल विपरीत चिन्ह के हैं यदि $f(0) < 0$

$$(a - 4)(a - 2) < 0$$

$$a \in (2, 4).$$

5. $n(C) = 20, n(B) = 50, n(C \cap B) = 10$
 Now अब, $n(C \cup B) = n(C) + n(B) - n(C \cap B)$
 $= 20 + 50 - 10 = 60.$
 Hence, required % of persons $= n(U) - n(C \cup B)$
 $= 100 - 60 = 40.$
 अतः व्यक्तियों का प्रतिशत $= n(U) - n(C \cup B) = 100 - 60 = 40.$

6. $240 = 2^4 \cdot 3^1 \cdot 5^1$
 We can choose divisors of the form $4n + 2$ ($n \geq 0$)
 $4n + 2$ ($n \geq 0$) रूप वाले भाजकों की संख्या
 $= 2 \cdot (3^0 + 3^1) \cdot (5^0 + 5^1)$
 Ways अभीष्ट तरीके $= 4$ since चूंकि $(2^1 \cdot 3^0 \cdot 5^0, 2^1 \cdot 3^0 \cdot 5^1, 2^1 \cdot 3^1 \cdot 5^0, 2^1 \cdot 3^1 \cdot 5^1)$

$$7. |z| = \frac{|3 + 4i|^{1/2} |12 - 5i|^{1/3}}{|12 + 5i|^{1/3}} = \frac{\sqrt{5} \times 13^{1/3}}{13^{1/3}} = \sqrt{5}$$

8. Total no. of ways = 2^n

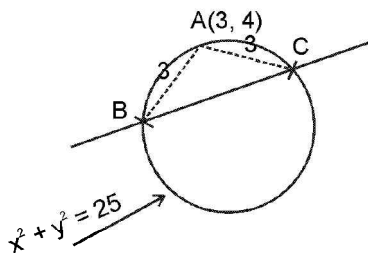
कुल तरीके = 2^n

If head comes odd times, then favourable ways = 2^{n-1}

यदि चित्त (head) विषम बार आये तो पक्ष में तरीके = 2^{n-1}

Required probability अभीष्ट प्रायिकता = $\frac{2^{n-1}}{2^n} = \frac{1}{2}$

9.



∴ eqⁿ of circle whose centre is at A(3, 4) and radius equal to 3 is

$$(x-3)^2 + (y-4)^2 = 9$$

..... (1)

$$x^2 + y^2 = 25$$

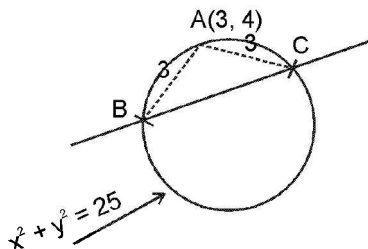
..... (2)

Now the eqⁿ of reqd. st. line will be the common chord of the two circle.

∴ eqⁿ of reqd. line is

$$6x + 8y - 41 = 0$$

Hindi



∴ उस वृत्त का समीकरण जिसका केन्द्र A(3, 4) एवं त्रिज्या 3 हो -

$$(x-3)^2 + (y-4)^2 = 9$$

..... (1)

$$x^2 + y^2 = 25$$

..... (2)

अभीष्ट सरल रेखा का समीकरण, दो वृत्तों की उभयनिष्ठ जीवा होगी।

∴ अभीष्ट रेखा का समीकरण

$$6x + 8y - 41 = 0$$

10. $\lim_{x \rightarrow 0} \frac{1 - \cos^3 x}{x \sin x \cdot \cos x}$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos x) \cdot (1 + \cos^2 x + \cos x)}{x^2 \cdot \left(\frac{\sin x}{x}\right) \cdot \cos x}$$

$$\lim_{x \rightarrow 0} \left(\frac{1 - \cos x}{x^2} \right) = \lim_{x \rightarrow 0} \frac{1}{\left(\frac{\sin x}{x}\right)}$$

$$\lim_{x \rightarrow 0} \left(\frac{1 + \cos^2 x + \cos x}{\cos x} \right)$$

$$= \left(\frac{1}{2} \right) \cdot (1) \cdot \left(\frac{3}{1} \right) = \frac{3}{2}$$

$$11. (x-2)^2 + (y-3)^2 = \left| \frac{3x-4y+7}{5} \right|^2$$

∴ focus is (2, 3) & directrix is $3x - 4y + 7 = 0$

latus vectum = $2 \times \perp$, distance from focus to directrix

$$= 2 \times \frac{1}{5} = 2/5$$

$$\text{Hindi } (x-2)^2 + (y-3)^2 = \left| \frac{3x-4y+7}{5} \right|^2$$

∴ नाभि (2, 3) एवं नियता $3x - 4y + 7 = 0$ है

नाभिलम्ब = $2 \times$ नाभि से नियता की लम्बवत् दूरी है

$$= 2 \times \frac{1}{5} = 2/5$$

$$12. g'(x) = \begin{cases} 2 - e^{x-1}, & 1 < x \leq 2 \\ x - e, & 2 < x \leq 3 \end{cases} = 0$$

$$\Rightarrow e^{x-1} = 2 \Rightarrow x = 1 + \ln 2$$

$$\text{and एवं } x - e = 0 \Rightarrow x = e$$

$$g''(x) = \begin{cases} -e^{x-1}, & 1 < x < 2 \\ 1, & 2 < x < 3 \end{cases}$$

At $x = 1 + \ln 2$, $g''(x) < 0 \Rightarrow x = 1 + \ln 2$ is the point of local maxima.

$x = 1 + \ln 2$, $g''(x) < 0 \Rightarrow x = 1 + \ln 2$ स्थानीय उच्चिष्ठ बिन्दु है।

At $x = e$, $g''(x) > 0 \Rightarrow x = e$ is the point of local minima.

$x = e$ पर, $g''(x) > 0 \Rightarrow x = e$ स्थानीय निम्निष्ठ बिन्दु है।

$$13. \begin{vmatrix} 1 & -c & -b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0.$$

$$\Rightarrow (1 - a^2) + c(-c - ab) - b(ac + b) = 0$$

$$\Rightarrow 1 - a^2 - c - abc - abc - b^2 = 0$$

$$a^2 + b^2 + c^2 + 2abc = 1$$

14. Domain प्रान्त : $x > 2$

$$f'(x) = \frac{2}{x-2} - 2x + 4$$

$$= \frac{2 - 2x(x-2) + 4x - 8}{x-2}$$

$$= \frac{-2x^2 + 8x - 6}{x-2}$$

$$= \frac{-2(x^2 - 4x + 3)}{x-2} = \frac{-2(x-1)(x-3)}{x-2}$$

$$f'(x) > 0 \Rightarrow x \in (2, 3)$$

15. $|\vec{a} + \vec{b} + \vec{c}| = 0$

$$|\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$\Rightarrow 9 + 1 + 16 + 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = 0$$

$$\Rightarrow (\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a}) = -13$$

16. $\therefore \lim_{x \rightarrow 3^-} \frac{\cos(2x-6) - 37}{2} = -18$

$$\text{and और } \lim_{x \rightarrow 3^-} \frac{x^2 |2x-6|}{x-3} = \lim_{x \rightarrow 3^-} \frac{2x^2 |x-3|}{(x-3)} = -18$$

$\therefore$ Using Sandwich theorem सेण्डविच प्रमेय से

$$\lim_{x \rightarrow 3^-} f(x) = -18$$

17. $|x|$ is derivable at all x in $(0, 2\pi]$

$(0, 2\pi]$ में सभी x पर $|x|$ अवकलनीय है।

$|\sin x|$ is non-derivable at $x = \pi$.

$x = \pi$ पर $|\sin x|$ पर अवकलनीय नहीं है।

$|x| + |\sin x|$ is non-derivable at $x = \pi$.

$x = \pi$ पर $|x| + |\sin x|$ अवकलनीय नहीं है।

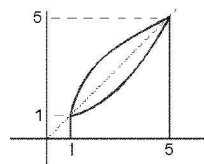
18. $\tan^{-1}(2\pi) + \tan^{-1}\left(\frac{1}{2\pi}\right) = \tan^{-1}2\pi + \cot^{-1}2\pi = \frac{\pi}{2}$

19. $\int_{\pi}^{5\pi/4} \frac{\sin 2x}{\cos^4 + \sin^4 x} dx = \int_{\pi}^{\pi/4} \frac{2 \sin x \cos x}{\cos^4 + \sin^4 x} dx$

$$= \int_0^{\pi/4} \frac{2 \tan x \cdot \sec^2 x}{1 + (\tan^2 x)^2} dx \quad \text{put } \tan^2 x = t \text{ रखने पर}$$

$$= \int_0^1 \frac{dt}{1+t^2} = \frac{\pi}{4}$$

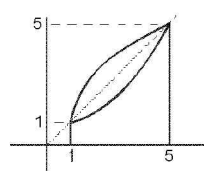
20. Area of trapezium = 12



area under $y = f(x)$ is 8 (given)

$\therefore$ area under $y = f^{-1}(x)$ is $12 + 12 - 8 = 16$

- Hindi. समलम्ब चतुर्भुज का क्षेत्रफल = 12



$y = f(x)$ का परिबद्ध क्षेत्रफल 8 है (दिया है)

$\therefore y = f^{-1}(x)$ परिबद्ध क्षेत्रफल $12 + 12 - 8 = 16$ है।

21. S.D. of $x_1, x_2, \dots, x_{18}$ का मानक विचलन

= S.D. of $x_1 - 8, x_2 - 8, \dots, x_{18} - 8$ का मानक विचलन

$$\sqrt{\frac{\sum (x_i - 8)^2}{n} - \left[\frac{\sum (x_i - 8)}{n} \right]^2} = \sqrt{\frac{45}{18} - \left(\frac{9}{18} \right)^2} = \frac{3}{2}$$

22. $\frac{dy}{dx} + \frac{1+y^2}{\sqrt{1-x^2}} = 0 \Rightarrow \frac{dy}{1+y^2} + \frac{dx}{\sqrt{1-x^2}} = 0$

$$\Rightarrow \tan^{-1} y + \sin^{-1} x = c$$

23. $y = x^2 - 2$

$$\frac{dy}{dx} = 2x$$

$$m_1 = 4$$

$$y = x^3 - 3x$$

$$\frac{dy}{dx} = 3x^2 - 3$$

$$m_2 = 9$$

$$\tan \theta = \left| \frac{4-9}{1+4 \cdot 9} \right| = \frac{5}{37}$$

$$24. {}^{n+2}C_2 = 36 \Rightarrow \frac{(n+1)(n+2)}{2} = 36$$

$$\Rightarrow n = 7$$

$$25. \sin x = \frac{-1 \pm \sqrt{1+48}}{12} = \frac{-1 \pm 7}{12} = -\frac{2}{3}, \frac{1}{2}$$

There is one solution in each of $\left(0, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, \pi\right),$

$$\left(\pi, \frac{3\pi}{2}\right), \dots, \left(5\pi, \frac{11\pi}{2}\right)$$

$\therefore$ There are 11 solutions

$$\text{Hindi. } \sin x = \frac{-1 \pm \sqrt{1+48}}{12} = \frac{-1 \pm 7}{12} = \frac{2}{3} - \frac{1}{2},$$

$$\left(0, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, \pi\right), \left(\pi, \frac{3\pi}{2}\right), \dots, \left(5\pi, \frac{11\pi}{2}\right) \text{ में से}$$

प्रत्येक में एक हल है।

$\therefore$ 11 हल हैं।

$$26. \int \frac{x^2+2}{x^4+4} dx = \int \frac{1+\frac{2}{x^2}}{\left(x-\frac{2}{x}\right)^2+4} dx = \frac{1}{2} \tan^{-1} \left(\frac{x-\frac{2}{x}}{2} \right) + c$$

$$27. y = |\cot \theta| = -\cot \theta$$

$$y' = \operatorname{cosec}^2 \theta \Rightarrow \left. \frac{dy}{d\theta} \right|_{\theta=\frac{3\pi}{4}} = 2$$

$$28. \text{Statement-1: } a = 3, b = 2, \text{ Slope of the line} = -\frac{2}{5}$$

$$\text{Slope of tangent line is } \frac{5}{2}$$

$$a^2m^2 - b^2 = 9 \cdot \frac{25}{4} - 4 > 0$$

$\therefore$ There are 2 tangents.

$\therefore$ Statement-1 is true.

Statement-2 is obviously true.

Hindi कथन-1: $a = 3, b = 2$, सरल रेखा की प्रवणता

$$= -\frac{2}{5}$$

$$\text{स्पर्श रेखा की प्रवणता} = \frac{5}{2}$$

$$a^2m^2 - b^2 = 9 \cdot \frac{25}{4} - 4 > 0$$

$\therefore$ 2 स्पर्श रेखाएं संभव हैं।

$\therefore$ कथन-1 सत्य है।

कथन-2 स्पष्टतया सत्य है।

29. Statement - 1 number of horizontal and vertical lines in a chessboard is 9.

$$\text{number of rectangles} = {}^9C_2 \cdot {}^9C_2.$$

Statement - 1 is false and Statement - 2 is true.

Hindi. कथन - 1 शतरंज के बोर्ड पर क्षैतिज एवं ऊर्ध्वाधर रेखाओं की संख्या 9 है।

$$\text{आयतों की संख्या} = {}^9C_2 \cdot {}^9C_2.$$

कथन - 1 असत्य है, वक्तव्य-2 सत्य है।

30. If $f(x)$ is increasing continuous function in $[\alpha, \beta]$, then its range is $[f(\alpha), f(\beta)]$ but for discontinuous function the statement is not true.

So D is correct.

Hindi यदि $f(x)$ अन्तराल $[\alpha, \beta]$ में एक सतत एवं वृद्धिमान फलन हो, तो इसका परिसर $[f(\alpha), f(\beta)]$ होता है लेकिन असतत फलन के लिए यह कथन सत्य नहीं है। अतः विकल्प (D) सही है।

PART-B PHYSICS

$$33. \therefore i_p + r_p = 90$$

$$\therefore r_p = 90 - i_p = 90 - 56.3 = 33.7$$

36. Vacancy in inner shell shows there is vacancy in outer shell but vacancy in outer shell doesn't mean vacancy in inner shell.

आन्तरिक कक्ष में रिक्ति प्रदर्शित करती है, बाह्य कोश में रिक्ति होगी परन्तु बाह्य कोश में रिक्ति होने का अर्थ यह नहीं है कि आन्तरिक कोश में भी रिक्ति उपस्थित हो।

$$38. \text{The earth's radius } R = 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$$

$$\text{पृथ्वी की त्रिज्या } R = 6400 \text{ km} = 6.4 \times 10^6 \text{ m}$$

Magnetization is the magnetic moment per unit volume.

Hence,

चुम्बकत्व प्रति इकाई आयतन में चुम्बकीय आघूर्ण द्वारा दिया जाता है अतः

$$I = \frac{M}{\frac{4\pi}{3}R^3} = \frac{8.0 \times 10^{22} \times 3}{4 \times \pi \times (6.4 \times 10^6)^3} = \frac{24.0 \times 10^4}{4 \times 262.1} = 729 \text{ Am}^{-1}$$

$$39. \text{path difference } (\Delta x) = d \sin \theta + d \sin \phi$$

$$\text{पथांतर } (\Delta x) = d \sin \theta + d \sin \phi$$

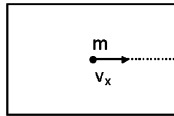
so phase difference = $k\Delta x$

अतः कलान्तर = $k\Delta x$

for constructive interference phase difference should be integral multiple of 2π .

समपोशी व्यतीकरण के लिए कलान्तर का मान 2π का पूर्ण गुणक होना चाहिए।

40. Momentum change/collision = $2mv_x$
संवेग परिवर्तन/ टक्कर = $2mv_x$



Also अतः $v^2 = v_x^2 + v_y^2 + v_z^2 \Rightarrow v = \frac{v^2}{3}$ [Since चूंकि $v_x^2 =$

$$v_y^2 = v_z^2]$$

$$\therefore v_x = \frac{v}{\sqrt{3}} \quad \therefore \text{momentum change/collision संवेग}$$

$$\text{परिवर्तन/ टक्कर} = 2m \frac{v}{\sqrt{3}}$$

$$\text{Let there be } n \text{ collision/sec area } p = 2m \frac{v}{\sqrt{3}} \times n \quad \therefore n =$$

$$\frac{p\sqrt{3}}{2mv}$$

माना इकाई क्षेत्रफल में प्रति सैकण्ड टक्करों की संख्या n हो तो $p =$

$$2m \frac{v}{\sqrt{3}} \times n \quad \therefore n = \frac{p\sqrt{3}}{2mv}$$

41. In rod छड़ में, $\frac{\lambda}{4} = \ell \Rightarrow \lambda = 4\ell \Rightarrow v_x = \lambda f = 4\ell f$

In air column वायु स्तम्भ में $v_a = 2\Delta \ell f$

$$\frac{v_r}{v_a} = \frac{2\ell}{\Delta \ell} \Rightarrow v_r = \frac{2 \times 1 \times 330}{0.6} = 1100 \text{ m/s}$$

42. $i_3 = i_1 + i_2$
 $= 5 \left(\frac{3}{5} \sin \omega t + \frac{4}{5} \cos \omega t \right)$
 $= 5 (\sin \omega t \cdot \cos 53^\circ + \cos \omega t \cdot \sin 53^\circ) = 5 \sin (\omega t + 53^\circ)$

43. $\phi = LI$
 $e = -\frac{d\phi}{dt} = -L \frac{dI}{dt}$

44. Least count अल्पतम माप = $\left(1 - \frac{49}{50} \right) 0.5^\circ = \frac{1^\circ}{100} = 0.6'$

45. $K_1 = \frac{Y_1 A}{L_1}, K_2 = \frac{Y_2 A}{L_2}$
 $K = \frac{K_1 K_2}{K_1 + K_2}$

46. After proportionality limit, strain rise is relatively more.
समानुपाती सीमा के पश्चात् विकृति सापेक्ष रूप से ज्यादा बढ़ती है।

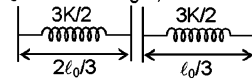
47. $\frac{dQ}{dt} = \sigma e A (T^4 - T_0^4) = ms \frac{d\theta}{dt}$

so $\frac{dQ}{dt}$ will be same 'e' is same but 's' is different so $\frac{d\theta}{dt}$ will be different.

इसलिए $\frac{dQ}{dt}$ समान होगा क्योंकि 'e' समान है परन्तु 's'

भिन्न-भिन्न होने के कारण $\frac{d\theta}{dt}$ भिन्न-भिन्न है।

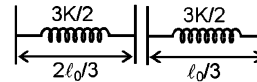
48. When a spring is cut into two parts each part has spring constant more than that of original spring. If k = spring constant & ℓ_0 = natural length, then for cut parts.



If they are stretched by same amount then work done in shorter part will be double than that in the case of longer part.

$\therefore$ Option (3) is correct

जब स्प्रिंग को काटा जाता है तो दोनों भागों का बल नियतांक वास्तविक स्प्रिंग के बल नियतांक से ज्यादा होता है। यदि k = स्प्रिंग नियतांक तथा ℓ_0 = सामान्य लम्बाई तो, काटे गये टुकड़ों के लिए।



यदि दोनों को समान मात्रा में विस्तारित किया जाये, तो छोटे भाग के लिए कार्य बड़े भाग तुलना में दोगुना होता है।

$\therefore$ विकल्प (3) सही है।

49. For lens $u = -\infty$ and $v = -100 \text{ cm}$

$$\therefore \frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \text{or} \quad \frac{1}{v} = \frac{1}{f} \Rightarrow f = v = -100 \text{ cm}$$

(concave) (अवतल)

$$\therefore \text{Power of lens (लेंस की शक्ति)} P = \frac{1}{f} = -\frac{1}{1} = -1 \text{ D}$$

51. For direction of motion (गति की दिशा के लिए)

$$\tan \theta_1 = \frac{10 \sin 30 - 10 \times 1}{10 \cos 30^\circ} = -\frac{1}{\sqrt{3}}$$

For displacement (विस्थापन के लिए)

$$\tan \theta_2 = \frac{10 \sin 30 - 10 \times \frac{1}{2} \times 1}{10 \cos \theta} = 0$$

$$\theta_1 = -30^\circ, \theta_2 = 0$$

$$\text{Angle (कोण)} = |\theta_1 - \theta_2| = 30^\circ$$

52. $\vec{L} = \vec{r} \times \vec{p} = M[(4\hat{i} - 2\hat{j}) \times 2t^2\hat{i}]$

$$\vec{L} = 8t^2 \hat{k}$$

$$\vec{\tau} = \frac{d\vec{L}}{dt} = 16t \hat{k}$$

$$\text{at } t = 2\text{S} \quad (t = 2\text{S पर}) ; \vec{\tau} = 32 \text{ N-m } (\hat{k})$$

53. The pressure due to surface tension = $\frac{4T}{R}$

पृष्ठ तनाव के कारण दाब = $\frac{4T}{R}$

The pressure due to electrostatic forces = $\frac{\sigma^2}{2\epsilon_0}$

स्थिर वैद्युत बलों के कारण दाब = $\frac{\sigma^2}{2\epsilon_0}$

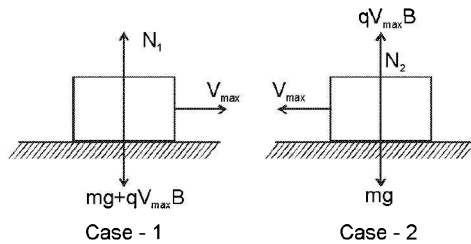
Just before the bubble bursts बुलबुले के फटने से ठीक पहले

$\frac{4T}{R} = \frac{\sigma^2}{2\epsilon_0}$ or अथवा $R = \frac{8T\epsilon_0}{\sigma^2}$

55. The velocity is maximum at mean position. Hence the magnetic force on block is maximum, at its mean position. The magnetic force on the block while it crosses the mean position towards right and left is as shown

माध्य स्थिति में वेग अधिकतम होता है। अतः माध्य स्थिति पर block पर लगने वाला चुम्बकीय बल भी अधिकतम होगा।

जब block दायी या बायीं दिशा में गति करता हुआ, माध्य स्थिति को पार करता है, तब इस पर लगने वाल चुम्बकीय बल चित्र में दिखाया गया है।



Hence normal reaction is maximum in case 1 & minimum in case 2. Hence correct option is (4).

अतः case : 1 में अभिलम्ब प्रतिक्रिया बल अधिकतम और case 2 में न्यूनतम है।

अतः विकल्प (4) सही है।

56. If he feels weightlessness then at the highest point, acceleration must be g.

यदि व्यक्ति उच्चतम बिन्दु पर भारहीनता अनुभव करता है, तो इसका त्वरण g होना चाहिए।

$\Rightarrow g = \omega^2 A \Rightarrow \omega = 2\pi f = \sqrt{2g} \Rightarrow f = \frac{\sqrt{2g}}{2\pi}$

57. Acute angle POR = 60° is given. According to the Biosawast law, the magnetic field at centre O is directly proportional to the current carrying segment. Hence,

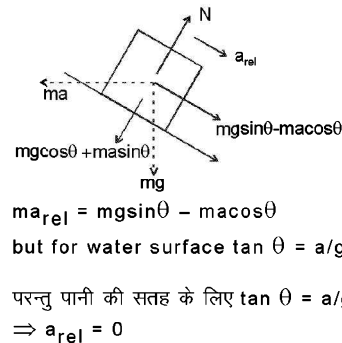
$\frac{B_1}{B_2} = \frac{\text{length of arc PQR}}{\text{length of arc PSR}}$
 $= \frac{\text{obtuse angle PSR}}{\text{acute angle PQR}} = \frac{300^\circ}{60^\circ} = 5$

न्यून कोण POR = 60° दिया हुआ है।

बायो सावर्ट के अनुसार केन्द्र O पर चुम्बकीय क्षेत्र धारावाही भाग के समानुपाती होता है। अतः,

$\frac{B_1}{B_2} = \frac{\text{PQR की लम्बाई}}{\text{PSR की लम्बाई}}$
 $= \frac{\text{अधिक कोण PSR}}{\text{न्यून कोण PQR}} = \frac{300^\circ}{60^\circ} = 5$

58.



$ma_{rel} = mg \sin \theta - ma \cos \theta$

but for water surface $\tan \theta = a/g$

परन्तु पानी की सतह के लिए $\tan \theta = a/g$

$\Rightarrow a_{rel} = 0$

59. Statement-1 is false, statement-2 is true
वक्तव्य-1 असत्य है, वक्तव्य-2 सत्य है।

60. Factual question. तथ्यपूर्ण प्रश्न

PART-C

CHEMISTRY

61. $\Delta H = \Sigma H_f^\circ (\text{products}) - \Sigma H_f^\circ (\text{reactants})$
 $= -110.5 + (-241.8) - (-393.5 + 0) = 41.2 \text{ KJ}$
 $\Delta H = \Sigma H_f^\circ (\text{उत्पाद}) - \Sigma H_f^\circ (\text{अभिकारक})$
 $= -110.5 + (-241.8) - (-393.5 + 0) = 41.2 \text{ KJ}$

62. Cyanide process is for gold (I-D); floatation process – pine oil (II-B); Electrolytic reduction – Al(III -C); Zone refining – Ge (IV - A).

गोल्ड के लिए सायनाइड प्रक्रम (I-D); झाग प्रक्रम-चीड का तेल

(II-B); Al(III -C) के लिए वैद्युत अपघटनी अपचयन; Ge (IV - A)

के लिए क्षेत्र परिशोधन।

63. $K_p = P_{H_2O}^2 = 1.086 \times 10^{-4} \text{ atm}^2$

$P_{H_2O} = 1.04 \times 10^{-2} \times 760 = 7.90 \text{ mm Hg}$

If the vapour pressure of water in air is less than 7.90 mm Hg, water of crystallization is given out.

यदि वायु में जल का वाष्प दाब 7.90 mm Hg से कम होगा तो क्रिस्टलन जल निकलेगा।

64. $2\text{XeF}_6 + \text{SiO}_2 \longrightarrow 2\text{XeOF}_4 + \text{SiF}_4$

65. Body diagonal contain two atoms at corners (A), one octahedral void (C) and two tetrahedral voids (D).

Sol. कोने के दो परमाणु (A), एक अष्टफलकीय छिद्र (C) व दो चतुष्फलकीय छिद्र (D) काय विकर्ण पर स्थित होंगे।

66. (I), (II) and (III) are correct statements.

(IV) It is an example of only hydrate isomerism not ionisation isomerism because ionisation isomerism occurs owing to exchange of ions between coordination and ionisation spheres.

Sol. (I), (II) तथा (III) सही हैं।

(IV) यह केवल हाइड्रेट समावयवता का उदाहरण है, आयनन समावयवता का नहीं, क्योंकि आयनन समावयवता समन्वय गोलक तथा आयनन गोलक के बीच आयनों के विनिमय से प्राप्त होती है।

67. The gas which shows higher value of 'a' will have higher value of critical temperature as vander Waal constant 'a' is directly proportional to critical temperature i.e., $T_c = \frac{8a}{27Rb}$.

Sol. गैस जिसके लिए वाण्डरवाल्स नियतांक 'a' अधिक होगा, उसके लिए क्रांतिक ताप भी अधिक होगा क्योंकि वाण्डरवाल्स नियतांक 'a' क्रांतिक ताप के समानुपाती होता है। अर्थात् $T_c = \frac{8a}{27Rb}$.

68. Let x gm of Zn deposit on 9 gm of Hg

$$\% \text{ of Zn in Amalgam} = \frac{m_{\text{Zn}} \times 100}{m_{\text{Zn}} + m_{\text{Hg}}} = \frac{x}{9 + x} \times 100 = 25$$

$$\therefore x = 3 \text{ gm}$$

$$\text{Eq. of Zn} = \frac{3 \times 2}{65.4}$$

$$\frac{n_{\text{Zn}}}{1} = \frac{n_{e^-}}{2} = \frac{Q}{2f} = \frac{it}{2f}$$

$$\text{Current } i = \frac{6}{65.4} \times \frac{96500}{1000} = 8.85 \text{ amp.}$$

Sol. माना 9 gm Hg पर Zn के x gm निक्षेपित होते हैं।

$$\text{अमलगम में Zn का \%} = \frac{m_{\text{Zn}} \times 100}{m_{\text{Zn}} + m_{\text{Hg}}} = \frac{x}{9 + x} \times 100 = 25$$

$$\therefore x = 3 \text{ gm}$$

$$\text{Zn का तुल्यांक} = \frac{3 \times 2}{65.4}$$

$$\frac{n_{\text{Zn}}}{1} = \frac{n_{e^-}}{2} = \frac{Q}{2f} = \frac{it}{2f}$$

$$\text{धारा } i = \frac{6}{65.4} \times \frac{96500}{1000} = 8.85 \text{ amp.}$$

$$69. \text{pH} = \text{p}K_1 + \log \frac{[\text{In}^-]}{[\text{HIn}]}$$

$$5 = \text{p}K_1 + \log \frac{75}{25}$$

$$\Rightarrow \text{p}K_1 = 4.523 \Rightarrow K_1 = 3 \times 10^{-5}$$

$$\text{pH} = 4.523 + \log \frac{10}{90} = 4.523 - 0.954 = 3.56$$

70. In solution $X_a = 0.3$ $\therefore X_b = 0.7$

In vapours $Y_a = 0.6$ $\therefore Y_b = 0.4$

using Dalton's law

विलयन में $X_a = 0.3$ $\therefore X_b = 0.7$

वाष्प में $Y_a = 0.6$ $\therefore Y_b = 0.4$

डॉल्टन नियम से

$$Y_a = \frac{P_a}{P_T}$$

$$\text{i.e. } Y_a = \frac{P_a^\circ \times X_a}{P_a^\circ \times X_a + P_b^\circ \times X_b}$$

$$\therefore Y_b = \frac{P_b^\circ \times X_b}{P_a^\circ \times X_a + P_b^\circ \times X_b}$$

$$\therefore \frac{Y_a}{Y_b} = \frac{P_a^\circ X_a}{P_b^\circ X_b}$$

$$\therefore \frac{P_a^\circ}{P_b^\circ} = \frac{Y_a}{Y_b} \times \frac{X_b}{X_a} = \frac{0.6}{0.4} \times \frac{0.7}{0.3} = 3.5$$

71. According to charge balance, number of Fe^{3+} ions replacing x

Fe^{+2} ions is equal to $\frac{2x}{3}$

$$\therefore \text{Vacancies of cations} = x - \frac{2x}{3} = \frac{x}{3}$$

$$\text{But } \frac{x}{3} = 1 - 0.94 = 0.06$$

$$\text{or } x = 0.06 \times 3 = 0.18 \text{ or } 18\%.$$

Sol. आवेश संतुलन के अनुसार x Fe^{+2} आयनों को प्रतिस्थापित करने वाले

Fe^{3+} आयनों की संख्या $\frac{2x}{3}$ के बराबर है।

$$\therefore \text{घनायनों की रिक्तियाँ} = x - \frac{2x}{3} = \frac{x}{3}$$

$$\text{लेकिन } \frac{x}{3} = 1 - 0.94 = 0.06$$

$$\text{या } x = 0.06 \times 3 = 0.18 \text{ या } 18\%.$$

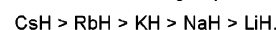
72. (1) This is because the heat from the flame excites the outermost orbital electron to a higher energy level.

When the excited electron comes back to the ground state, there is emission of radiation in the visible region.

(2) Here down the group the change in lattice energy is more than that of hydration energy.

(3) Because of low ionization energy and melting point.

(4) Reducing nature increases down the group as their stability decreases down the group

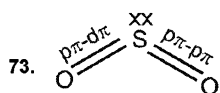
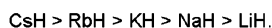


हल. (1) ऐसा इसलिए होता है क्योंकि ज्वाला द्वारा जो ऊष्मा दी जाती है वह बाह्यतम कोश इलेक्ट्रॉन को उत्तेजित कर उच्च ऊर्जा स्तर में स्थानान्तरित कर देती है। जब उत्तेजित इलेक्ट्रॉन आद्य अवस्था में पुनः वापिस आता है, जो दृश्य क्षेत्र में विकिरणों का उत्सर्जन करता है।

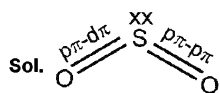
(2) यहाँ पर वर्ग में नीचे की ओर जालक ऊर्जा में परिवर्तन जलयोजन ऊर्जा की तुलना में अधिक होता है।

(3) निम्न आयनन ऊर्जा तथा गलनांक के कारण।

(4) वर्ग में नीचे की ओर अपचायक क्षमता बढ़ती है क्योंकि इनकी वर्ग में नीचे की ओर स्थायित्व घटता है।



SeF₄ has sp³d hybridisation and see-saw shape has sp³ hybridisation and bent shape.



SeF₄ का संकरण sp³d व नैज आकृति (see-saw shape) होती है।

I₃⁺ का संकरण sp³ व मुड़ी हुई आकृति (bent shape) होती है।

74. $mvr = \frac{nh}{2\pi} \quad \dots(1)$

$p = \frac{h}{\lambda} = mv \quad \dots(2)$

Placing the value of mv from Eq. (2) into Eq. (1) for 3rd orbit
3rd कक्ष के लिए समीकरण (2) से mv का मान समीकरण (1) में रखने पर

$$\frac{h}{\lambda} r_3 = \frac{3h}{2\pi}$$

$$\lambda = \frac{2\pi r_3}{3}$$

$$r_3 = n^2 r_0 = 9r_0$$

So इसलिए $\lambda = \frac{2\pi \cdot 9r_0}{3} = 6\pi r_0$

75. Normality of oxalic acid = $\frac{6.3 \times 1000}{63 \times 250} = 0.4N$

$$N_1 V_1 = N_2 V_2 \text{ or } 10 \times 0.4 = V \times 0.1 \text{ or } V = 40 \text{ ml.}$$

Sol. ऑक्सैलिक अम्ल की नार्मलता = $\frac{6.3 \times 1000}{63 \times 250} = 0.4N$

$$N_1 V_1 = N_2 V_2 \text{ or } 10 \times 0.4 = V \times 0.1 \text{ or } V = 40 \text{ ml.}$$

76. $[\text{NH}_3] = 0.3M \quad [\text{NH}_4^+] = 0.2M$
 $K_b = 1.8 \times 10^{-5}$

$$p^{OH} = pK_b + \log \frac{[\text{salt}]}{[\text{base}]} = 4.74 + \log \frac{0.2}{0.3}$$

$$= 4.74 + 0.3010 - 0.4771 = 4.56$$

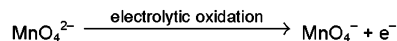
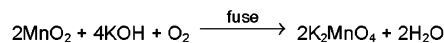
$$p^H = 14 - 4.56 = 9.436$$

77. $t = \frac{2.303}{k} \log \frac{a}{a-x}$

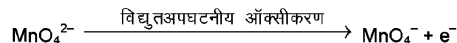
$$= \frac{2.303}{6} \log \frac{0.5}{0.05} = 0.384 \text{ min}$$

78. Fuse with KOH in presence of air or oxidising agents like KNO₃, KClO₃ etc.

वायु या ऑक्सीकारक जैसे KNO₃, KClO₃ इत्यादि की उपस्थिति में KOH के साथ संगलित करना।

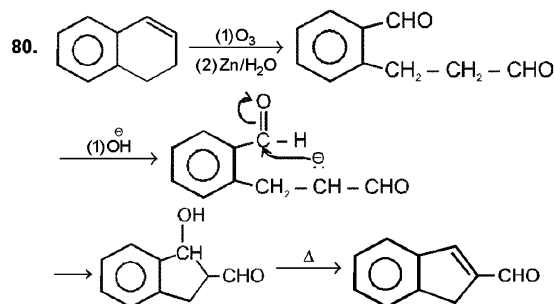


Sol. $2\text{MnO}_2 + 4\text{KOH} + \text{O}_2 \xrightarrow{\text{संगलन}} 2\text{K}_2\text{MnO}_4 + 2\text{H}_2\text{O}$;



79. All of these reaction gives CH₄.

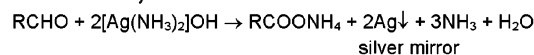
सभी अभिक्रियाओं में CH₄ बनता है।



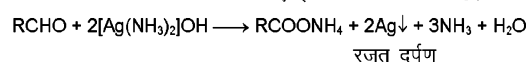
81. 2-acetoxybenzoic acid is IUPAC name of aspirin or acetyl salicylic acid

एस्पिरिन अर्थात् एसीटिल सैलिसिलिक अम्ल का IUPAC नाम 2-एसीटोक्सिबेन्जोइक अम्ल है।

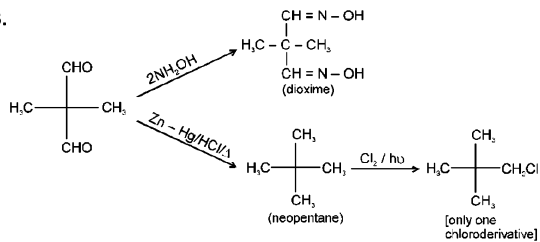
82. Tollen's reagent is a mild oxidising agent and oxidises only aldehydes to form silver mirror. It is used to differentiate between aldehydes and ketones



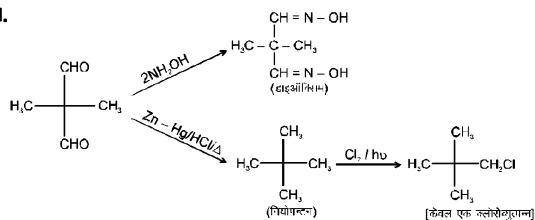
हल. टॉलेन अभिकर्मक एक मध्यम ऑक्सीकारी अभिकर्मक है, जो केवल एल्डिहाइड को ऑक्सीकृत कर रजत दर्पण बनाता है। एल्डिहाइड व कीटोन के मध्य अन्तर करने के लिए इसका उपयोग करते हैं।



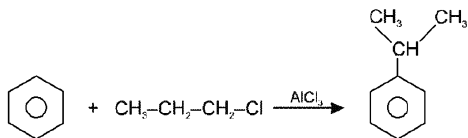
83.



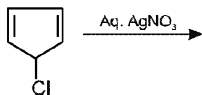
Sol.



84. I.

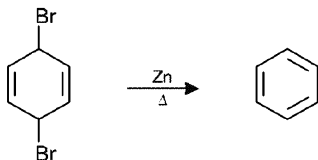


II.

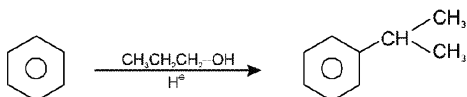


reaction not possible. (अभिक्रिया संभव नहीं है।)

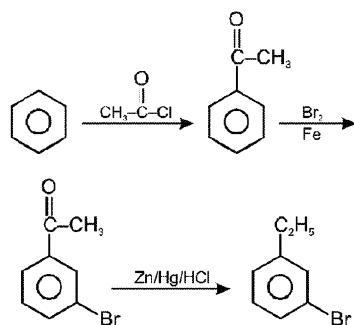
III.



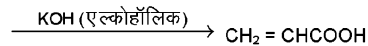
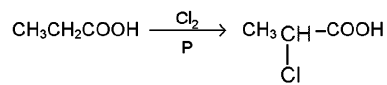
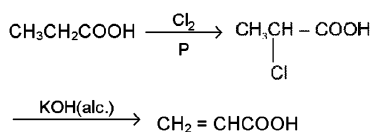
IV.



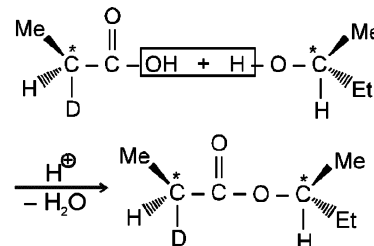
85.



86.



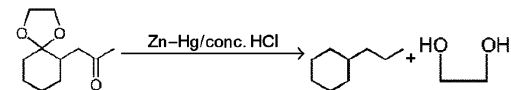
87.



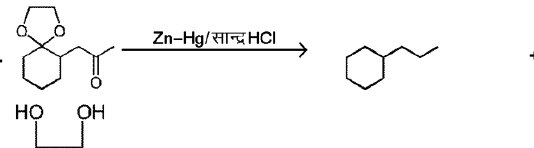
Since carbocation does not form any chiral carbon in this reactions hence racemic mixture can not form.

इस अभिक्रिया में किसी भी किरैल कार्बन पर कार्बधनायन का निर्माण नहीं होता। अतः रेसेमिक मिश्रण नहीं बन सकता।

88.



Sol.



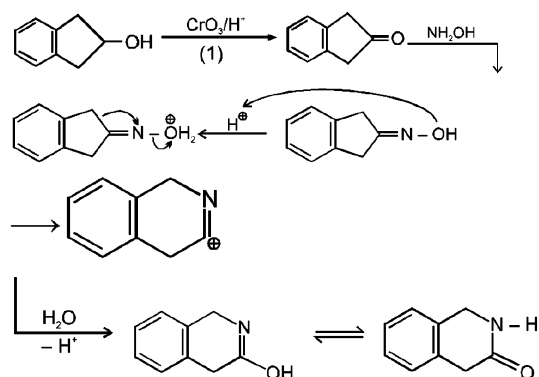
Ketal are easily hydrolysis into corresponding carbonyl compound only in acidic medium.

किटैल केवल अम्लीय माध्यम में संगत कार्बोनिल यौगिकों में आसानी से जल अपघटित हो जाते हैं।

89.

In salicylic acid, there is H-bonding in conjugate base.
सैलिसिलिक अम्ल के संयुग्मी क्षार में H-बन्धन होता है।

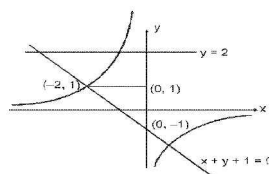
90.



DETAILED SOLUTION

Part-A- (Mathematics)

$$\begin{aligned}
 1. & (A \cup B \cup C) \cap (A \cap B' \cap C')' \cap C' \\
 &= (A \cup B \cup C) \cap (A' \cup B' \cup C) \cap C' \\
 &= [(A \cap A') \cup (B \cup C)] \cap C' = (\phi \cup (B \cup C)) \cap C' \\
 &= (\phi \cup (B \cup C)) \cap C' = (B \cup C) \cap C' \\
 &= (B \cap C') \cup (C \cap C') = (B \cap C') \cup \phi = B \cap C'
 \end{aligned}$$



$$2. a_1, a_2, \dots, a_5, a_6, a_7, \dots, a_{10}$$

$$\sum_{i=1}^{10} a_i = 12.5, \sum_{i=1}^6 a_i = 15, \sum_{i=6}^{10} a_i = 10$$

$$a_6 + 12.5 \times 10 = 90 + 50$$

$$a_6 = 15$$

3. p : it is a good watch (यह एक अच्छी घड़ी है)
 q : it is a Titan watch (यह एक टाइटन घड़ी है)
 $S_1 : p \rightarrow q$
 $S_2 : q$
 $S : p$

p	q	$p \rightarrow q$	S_1	S_2	S
T	F	F	F	F	T
F	T	T	T	T	F
T	T	T	T	T	T
F	F	T	T	F	F

As (क्योंकि) $S_1 : T, S_2 : T \Rightarrow S : T$ so invalid argument. (अतएव अनुपयुक्त कोणांक)

4. Point of intersection of $xy = -2$ and $x + y + 1 = 0$ is $(-2, 1)$
 $xy = -2$ और $x + y + 1 = 0$ का प्रतिच्छेद बिन्दु $(-2, 1)$ है

$$\text{Required area अभीष्ट क्षेत्रफल} = \frac{1}{2} (2) + \left| \int_1^2 -\frac{2}{y} dy \right|$$

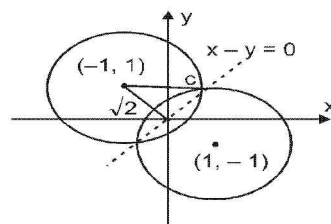
$$= 2 + \left| -2[\ln y]_1^2 \right|$$

$$= 2 + 2\ln 2$$

5. For real solutions वास्तविक मानों के लिए
 $1 \leq a^2 - 2a + 3 \leq 3$
 $\Rightarrow 1 \leq (a - 1)^2 + 2 \leq 3$
 $\Rightarrow -1 \leq (a - 1)^2 \leq 1$
 $\Rightarrow -1 \leq a - 1 \leq 1$
 $0 \leq a \leq 2.$

6. ${}^{2n}C_1, {}^{2n}C_2, {}^{2n}C_3$ are in A.P. सगान्तर श्रेणी में है।
 $\Rightarrow 2n^2 - 9n + 7 = 0$

7. Length of the common chord (समयनिष्ठ जीवा की लम्बाई) $= 2\ell$
 $= 2\sqrt{c^2 - 2}$



8. $z = \omega$
 $\Rightarrow (z - z^2 + 2z^3)(2 - z + z^2) = (\omega - \omega^2 + 2)(2 - \omega + \omega^2)$
 $= |2 - \omega + \omega^2|^2$
 $= |2 - i\sqrt{3}|^2 = 7$

9. For $x \neq -1$, we have

$$f(x) = 1 - 2x + 3x^2 - 4x^3 + \dots = (1+x)^{-2} = \left(\frac{1}{1+x}\right)^2$$

$$\lim_{h \rightarrow 0} f(-1-h) = \lim_{h \rightarrow 0} \frac{1}{(1-1-h)^2} \rightarrow \infty$$

$\Rightarrow f(x)$ is not continuous at $x = -1$

so, $f(x)$ is not derivable at $x = -1$

Hence $f(x)$ is neither continuous nor derivable at $x = -1$

Hindi. $x \neq -1$ के लिए

$$f(x) = 1 - 2x + 3x^2 - 4x^3 + \dots = (1+x)^{-2} = \left(\frac{1}{1+x}\right)^2$$

$$\lim_{h \rightarrow 0} f(-1-h) = \lim_{h \rightarrow 0} \frac{1}{(1-1-h)^2} \rightarrow \infty \Rightarrow f(x), x = -1 \text{ पर}$$

सतत् नहीं है।

$\therefore f(x), x = -1$ पर अवकलनीय नहीं है।

इसलिए $f(x), x = -1$ पर ना तो सतत् है और ना ही अवकलनीय।

10. Let माना $\lambda = \sqrt{x + \sqrt{x + \sqrt{x + \dots \infty}}}$

$$\Rightarrow \lambda^2 = x + \lambda$$

$$\lambda^2 - \lambda - x = 0$$

$$\lambda = \frac{1 \pm \sqrt{1+4x}}{2} = \frac{1 + \sqrt{1+4x}}{2}, \{\lambda \geq 0\}$$

$$I = \int_0^2 \frac{2dx}{1 + \sqrt{1+4x}}$$

$$\text{Let माना } 1 + 4x = t^2 \Rightarrow 4dx = 2t dt$$

$$I = \int_1^3 \frac{t dt}{1+t} = (t - \ln(1+t))_1^3 = 2 - \ln 2$$

$$2 - I = \ln 2 \Rightarrow e^{2-I} = 2$$

11. $\frac{dA}{dt} = 2\pi r \frac{dr}{dt} = 2 \times \frac{22}{7} \times 10 \times 3.5 = 220 \text{ sq.cm/sec}$

12. $\frac{y^2 dx + y dx - x dy}{y^2} = 0$

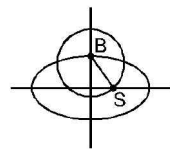
$$x dx + d\left(\frac{x}{y}\right) = 0$$

on integrating समाकलन करने पर $\frac{x^2}{2} + \frac{x}{y} + c = 0$

13. Centre $\equiv (0, 3)$ focus $S \equiv (\sqrt{7}, 0)$

$$\text{radius} = BS = 4$$

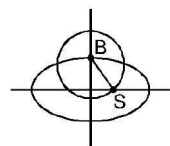
$$\therefore \text{its equation is } x^2 + (y-3)^2 = 4^2$$



Hindi केन्द्र $\equiv (0, 3)$ नाभि $S \equiv (\sqrt{7}, 0)$

$$\text{त्रिज्या} = BS = 4$$

$$\therefore \text{वृत्त का समीकरण } x^2 + (y-3)^2 = 4^2 \text{ है।}$$



14. $f(x) = \log_2 \left(\sin \left(x - \frac{\pi}{4} \right) + 3 \right)$

so domain of $f(x)$ is $x \in \mathbb{R}$

$$\text{Range} \rightarrow 2 \leq \sin \left(x - \frac{\pi}{4} \right) + 3 \leq 4$$

$$\text{so } 1 \leq \log_2 \left(\sin \left(x - \frac{\pi}{4} \right) + 3 \right) \leq 2$$

$$\text{Range} \in [1, 2]$$

Hindi $f(x) = \log_2 \left(\sin \left(x - \frac{\pi}{4} \right) + 3 \right)$

अतः $f(x)$ का प्रान्त $x \in \mathbb{R}$ है।

$$\text{परिसर} \rightarrow 2 \leq \sin \left(x - \frac{\pi}{4} \right) + 3 \leq 4$$

$$\text{अतः } 1 \leq \log_2 \left(\sin \left(x - \frac{\pi}{4} \right) + 3 \right) \leq 2$$

$$\text{परिसर} \in [1, 2]$$

$$15. \tan \beta = \frac{h}{x}$$

$$\tan \alpha = \frac{h}{d+x}$$

$$h = \frac{d}{\cot \alpha - \cot \beta}$$

16. Equation of asymptotes is अनन्तस्पर्शियों का समीकरण

$$y - 2 = \pm \frac{2}{3}(x - 1)$$

$$\Rightarrow 3y - 2x = 4 \text{ and } \text{और } 3y + 2x = 8$$

as given line is parallel to one of the asymptotes

दी गई रेखा इनमें से एक अनन्तस्पर्शी के समान्तर है।

so it will cut at one point only इसलिए यह केवल एक बिन्दु पर प्रतिच्छेद करेगी।

$$17. I = \int x^3(x^6 + x^3 + 1)(2x^6 + 3x^3 + 6)^{1/3} dx$$

$$= \int x^2(x^6 + x^3 + 1)(2x^9 + 3x^6 + 6x^3)^{1/3} dx$$

$$\text{Let माना } 2x^9 + 3x^6 + 6x^3 = t$$

$$18(x^9 + x^6 + x^3) dx = dt$$

$$I = \frac{1}{18} \int t^{1/3} dt = \frac{1}{18} \cdot \frac{t^{4/3}}{4/3} + c = \frac{1}{24} (2x^9 + 3x^6 + 6x^3)^{4/3} + c$$

$$18. \sum_{k=1}^{100} \tan^{-1} \left(\frac{2}{1+4k^2-1} \right)$$

$$= \sum_{k=1}^{100} \tan^{-1} \left(\frac{2}{1+(2k-1)(2k+1)} \right)$$

$$= \sum_{k=1}^{100} \tan^{-1} \left(\frac{(2k+1) - (2k-1)}{1+(2k-1)(2k+1)} \right)$$

$$= \sum_{k=1}^{100} \tan^{-1} ((2k+1) - \tan^{-1}(2k-1))$$

$$= (\tan^{-1}(3) - \tan^{-1}(1)) + (\tan^{-1}(5) - \tan^{-1}(3)) + \dots + (\tan^{-1}(201) - \tan^{-1}(199))$$

$$= \tan^{-1}(201) - \tan^{-1}(1)$$

$$= \tan^{-1} \left(\frac{201-1}{1+201} \right) = \tan^{-1} \left(\frac{200}{202} \right) = \tan^{-1} \left(\frac{100}{101} \right)$$

$$2p - q = 200 - 101 = 99$$

$$19. \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{\sin x}{x - \sin x}} = e^{\lim_{x \rightarrow 0} \frac{\sin x}{x - \sin x} \left(\frac{\sin x}{x} - 1 \right)} = e^{-1} = \frac{1}{e}$$

$$20. B = CAC^{-1}$$

$$B^2 = (CAC^{-1})(CAC^{-1}) = CA^2C^{-1}$$

$$B^3 = B^2B = (CA^2C^{-1})(CAC^{-1})$$

$$= CA^2(C^{-1}C)AC^{-1}$$

$$= CA^2 \cdot AC^{-1}$$

$$= CA^3C^{-1}$$

21. Let the equation of any normal be $y = -tx + 2t + t^3$

Since it passes through the points (15, 12)

$$\therefore 12 = -15t + 2t + t^3 \text{ ie } t^3 - 13t - 12 = 0$$

$$\therefore t = -1, -3, 4$$

$$\therefore \text{The points are } (1, -2), (9, -6), (16, 8)$$

$$\therefore \text{Centroid is } \left(\frac{26}{3}, 0 \right)$$

Hindi माना किसी अभिलम्ब का समीकरण $y = -tx + 2t + t^3$

यह बिन्दु (15, 12) से गुजरता है -

$$\therefore 12 = -15t + 2t + t^3 \text{ ie } t^3 - 13t - 12 = 0$$

$$\therefore t = -1, -3, 4$$

$\therefore$ परवलय पर तीन बिन्दु (1, -2), (9, -6), (16, 8) है, अतः केन्द्रक

$$\left(\frac{26}{3}, 0 \right)$$

$$22. {}^3C_2 \left(\frac{2}{3} \right)^2 \left(\frac{1}{3} \right) \left(\frac{3}{4} \right)^2 + {}^3C_2 \left(\frac{2}{3} \right)^3 \left(\frac{3}{4} \right)^2 \frac{1}{4}$$

$$= {}^3C_2 \left(\frac{2}{3} \right)^2 \left(\frac{3}{4} \right)^2 \cdot \left(\frac{1}{3} + \frac{2 \cdot 1}{3 \cdot 4} \right) = 3 \cdot \frac{4}{9} \cdot \frac{9}{16} \cdot \frac{1}{3} \left(\frac{3}{2} \right) = \frac{3}{8}$$

$$23. \log_2(a+b) + \log_2(c+d) \geq 4$$

$$\Rightarrow \log_2((a+b)(c+d)) \geq 4$$

$$\Rightarrow (a+b)(c+d) \geq 16$$

$$AM \geq GM$$

$$\therefore (a+b) + (c+d) \geq 2\sqrt{(a+b)(c+d)} \geq 8$$

$$24. C = 60^\circ$$

$$\text{Hence अतः } c^2 = a^2 + b^2 - ab$$

$$\Rightarrow \frac{a+b}{\sqrt{a^2-ab+b^2}} = \frac{a+b}{c} = 2 \cos \left(\frac{A-B}{2} \right)$$

25. Clearly the lines given by $3x^2 - 8xy - 3y^2 = 0$ are perpendicular to each other.

$\therefore$ the triangle must be right angled.

Equation of the angle bisectors of the given lines is

$$\frac{x^2 - y^2}{6} = \frac{xy}{-4}$$

and $x = 2y$ does not satisfy it.

$\therefore$ the triangle is not isosceles.

Hindi स्पष्टतया दी गयी रेखाये $3x^2 - 8xy - 3y^2 = 0$ एक दूसरे के लम्बवत् है।

$\therefore$ त्रिभुज समकोण होगा।

दी गयी रेखाओं के कोण समद्विभाजक का समीकरण $\frac{x^2 - y^2}{6} = \frac{xy}{-4}$

तथा $x = 2y$ इसे सतुष्ट नहीं करती है।

$\therefore$ त्रिभुज समद्विबाहु नहीं है।

26. $x + y + z = 10$

$$1 \leq x, y, z, \leq 6$$

Coefficient of $x^{10} : (x^1 + x^2 + x^3 + x^4 + x^5 + x^6)^3$

$$\text{coefficient of } x^7 : \left(\frac{1 - x^6}{1 - x} \right)^3$$

$$: (1 - {}^3C_1 x^6 + \dots) (1 - x)^{-3}$$

$$: (1 - {}^3C_1 x^6 + \dots) (1 + {}^3C_1 x + {}^4C_2 x^2)$$

$$\text{Coefficient of } x^7 = {}^9C_7 - {}^3C_1 \cdot {}^3C_1$$

$$= 36 - 3 \times 3$$

$$= 27$$

Hindi. $x + y + z = 10$

$$1 \leq x, y, z, \leq 6$$

x^{10} का गुणांक $: (x^1 + x^2 + x^3 + x^4 + x^5 + x^6)^3$

$$x^7 \text{ का गुणांक } : \left(\frac{1 - x^6}{1 - x} \right)^3$$

$$: (1 - {}^3C_1 x^6 + \dots) (1 - x)^{-3}$$

$$: (1 - {}^3C_1 x^6 + \dots) (1 + {}^3C_1 x + {}^4C_2 x^2)$$

$$x^7 \text{ का गुणांक } = {}^9C_7 - {}^3C_1 \cdot {}^3C_1$$

$$= 36 - 3 \times 3$$

$$= 27$$

27. The equation of any plane containing the given line is $(x + y + 2z - 3) + \lambda(2x + 3y + 4z - 4) = 0$

$$\Rightarrow (1 + 2\lambda)x + (1 + 3\lambda)y + (2 + 4\lambda)z - (3 + 4\lambda) = 0$$

..... (i)

If this plane is parallel to z-axis then

$$(1 + 2\lambda)(0) + (1 + 3\lambda)(0) + (2 + 4\lambda)(1) = 0$$

$$\Rightarrow \lambda = -\frac{1}{2}$$

$$\lambda = -\frac{1}{2} \text{ put in (1) then}$$

$$y - 2 = 0$$

..... (ii)

S.D. = distance of any point say $(0,0,1)$ on z-axis from plane (ii) =

$$\frac{2}{\sqrt{(1)^2}} = 2$$

Hindi समीकरण जबकि कोई समतल दी गई रेखाओं को रखता है $(x + y + 2z - 3) + \lambda(2x + 3y + 4z - 4) = 0$

$$\Rightarrow (1 + 2\lambda)x + (1 + 3\lambda)y + (2 + 4\lambda)z - (3 + 4\lambda) = 0$$

..... (i)

यदि यह समतल z-अक्ष के समान्तर है, तो

$$(1 + 2\lambda)(0) + (1 + 3\lambda)(0) + (2 + 4\lambda)(1) = 0$$

$$\Rightarrow \lambda = -\frac{1}{2}; \quad \lambda = -\frac{1}{2} \text{ (1) में रखने पर}$$

$$y - 2 = 0$$

..... (ii)

लघुतम दूरी = z-अक्ष पर किसी बिन्दु $(0,0,1)$ से समतल (ii) की दूरी =

$$\frac{2}{\sqrt{(1)^2}} = 2$$

28. **Statement-1 :** $\frac{d}{dx} (\tan^{-1}(\sec x + \tan x))$

$$= \frac{d}{dx} \left(\tan^{-1} \left(\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right) \right) = \frac{d}{dx} \left(\frac{\pi}{4} + \frac{x}{2} \right) = \frac{1}{2}$$

$$= \frac{d}{dx} \left(\cot^{-1} \left(\frac{1 + \cos x}{\sin x} \right) \right) = \frac{d}{dx} \left(\cot^{-1} \left(\cot \frac{x}{2} \right) \right) = \frac{1}{2}$$

$\therefore$ statement is false

29. Let $y = \frac{x^2 + 34x - 71}{x^2 + 2x - 7}$ then

$$x^2(y - 1) + 2x(y - 17) - (7y - 71) = 0$$

Since x is real, therefore above equation has real roots.

So, Discriminant ≥ 0

$$\Rightarrow 4(y - 17)^2 + 4(y - 1)(7y - 71) \geq 0$$

$$\Rightarrow y^2 - 14y + 45 \geq 0$$

$$\Rightarrow (y - 5)(y - 9) \geq 0$$

$$\Rightarrow y \in (-\infty, 5] \cup [9, \infty)$$

Hence the value of given expression can't lie in $(5, 9)$

$\Rightarrow$ Statement-1 is true

And Statement-2 is obviously true and

Statement-2 is the correct explanation of Statement-1.

हल माना $y = \frac{x^2 + 34x - 71}{x^2 + 2x - 7}$ तो

$$x^2(y - 1) + 2x(y - 17) - (7y - 71) = 0$$

चूँकि x वास्तविक है अतः उपरोक्त समीकरण के मूल वास्तविक होंगे

इसलिये, विविक्तकर ≥ 0

$$\Rightarrow 4(y - 17)^2 + 4(y - 1)(7y - 71) \geq 0$$

$$\Rightarrow y^2 - 14y + 45 \geq 0$$

$$\Rightarrow (y - 5)(y - 9) \geq 0$$

$$\Rightarrow y \in (-\infty, 5] \cup [9, \infty)$$

अतः दिये गए व्यंजक का मान (5, 9) में विद्यमान नहीं हो सकता।

$\Rightarrow$ कथन-1 सत्य है

तथा कथन-2 स्पष्टतया सत्य है

तथा कथन-2, कथन-1 की सही व्याख्या करता है।

30. $\cot \theta - \tan \theta = 2$

$$\frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} = 2$$

$$\frac{\cos^2 \theta - \sin^2 \theta}{2 \cos \theta \sin \theta} = 1$$

$$\cot 2\theta = 1$$

$$\Rightarrow 2\theta = n\pi + \pi/4$$

$$\theta = (4n + 1)\pi/8 \Rightarrow \text{Statement-1 is true (कथन-1 सत्य है)}$$

Statement-2 is true

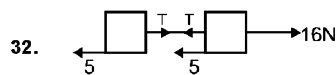
कथन-2 सत्य है

Part-B (Physics)

31. $\frac{1}{2} mv^2 = ct$

$$\frac{m}{2} 2v \frac{dv}{dt} = c \Rightarrow m \frac{dv}{dt} = \frac{c}{v}$$

$$F \propto \frac{1}{v}$$



$$16 - 10 = 10a$$

$$a = 0.6$$

$$T - 5 = 5a$$

$$T - 5 = 5 \times 0.6 = 3$$

$$T = 8.$$

33. (i) $V = \frac{V_0}{T/4} t$ $V = \frac{4V_0}{T} t$

The average value of voltage in one complete cycle will be zero in the given case.

दी गई स्थिति में एक पूर्ण चक्र में वोल्टेज का औसत मान शून्य होगा।

34. $\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2 = \left(\frac{\sqrt{4} + \sqrt{1}}{\sqrt{4} - \sqrt{1}} \right)^2 = 9$

$$\text{दृश्यता} = \frac{I_{\max}}{I_{\min}} = \frac{\left(\frac{I_{\max}}{I_{\min}} - 1 \right)}{\left(\frac{I_{\max}}{I_{\min}} + 1 \right)} = \frac{9 - 1}{9 + 1} = \frac{4}{5}$$

35. Linear momentum $\Rightarrow mv \propto \frac{1}{n}$

angular momentum $\Rightarrow mvr \propto n$

$\therefore$ product of linear momentum and angular momentum $\propto n^0$

रेखीय संवेग $\Rightarrow mv \propto \frac{1}{n}$

कोणीय संवेग $\Rightarrow mvr \propto n$

$\therefore$ कोणीय व रेखीय संवेग का गुणनफल $\propto n^0$

36. $\Delta E \propto Z^2$

i.e. greatest for Li^{++} and minimum for H.

अर्थात् Li^{++} के लिए अधिकतम व H के लिए न्यूनतम है।

38. Emf between a point and center is given by $\frac{1}{2} B \omega R^2$.

Therefore emf between two points will be

$$\frac{1}{2} B \omega (R_1^2 - R_2^2) \text{ where } R_1 \text{ and } R_2 \text{ are the radial distances of}$$

the two points from the center. Since point C is farthest from the center and the point E nearest, the emf will be maximum between these points.

बिन्दु तथा केन्द्र के मध्य विद्युत वाहक बल $\frac{1}{2} B \omega R^2$

अतः दो बिन्दुओं के मध्य विद्युत वाहक बल $\frac{1}{2} B \omega (R_1^2 - R_2^2)$ होगा

जहाँ R_1 तथा R_2 केन्द्र से दोनों बिन्दुओं की त्रिज्यीय दूरी है। चूँकि बिन्दु C, केन्द्र से अधिकतम दूरी पर है तथा बिन्दु E निकटतम दूरी पर है। इन बिन्दुओं के मध्य विद्युत वाहक बल अधिकतम होगा।

39. The electric field due to both straight wires shall cancel at common centre O. The electric field due to larger and smaller semi circular rings at O be E and E' respectively.

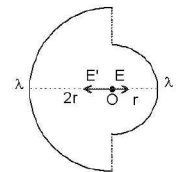
सीधे तारों के कारण उभयनिष्ठ केन्द्र O पर विद्युत क्षेत्र निरस्त हो जाती है। बड़ी और छोटी अर्द्धवृत्ताकार वलयों के कारण O पर विद्युत क्षेत्र क्रमशः E और E' है।

$$E = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{2r}; E' = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{r}$$

$\therefore$ Magnitude of electric field at O is

$\therefore$ O पर विद्युत क्षेत्र का परिमाण

$$= E' - E = \frac{1}{4\pi\epsilon_0} \left(\frac{2\lambda}{r} - \frac{\lambda}{r} \right) = \frac{1}{4\pi\epsilon_0} \frac{\lambda}{r}$$



40. The acceleration due to gravity at a distance x ($x < R$) from centre of earth (of radius R) is

(त्रिज्या R की) पृथ्वी के केन्द्र से x ($x < R$) दूरी पर गुरुत्वीय त्वरण है—

$$g(x) = g \frac{x}{R} \quad ; \quad \therefore g\left(\frac{R}{2}\right) = \frac{g}{2}$$

41. (1) Current will be same for each section

प्रत्येक काट क्षेत्र पर धारा समान होगी

(2) $E \propto V_d$, so अतः $E_B > E_A > E_C$

$$J \propto E \quad \Rightarrow \quad E \propto \frac{1}{A}$$

$$(3) \phi = \vec{E} \cdot \vec{A} = \frac{J}{\sigma} \cdot \vec{A}$$

So, अतः $\phi = i$

as 'i' is same for each section, so ϕ is also same

चूँकि 'i' प्रत्येक काट क्षेत्र पर समान है, अतः ϕ भी समान होगा।

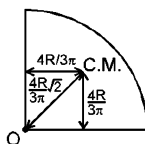
(4) current flows from high potential to low potential so $V_A > V_B > V_C$.

धारा उच्च विभव से निम्न विभव की ओर प्रवाहित होती है अतः $V_A > V_B > V_C$

42. M.I. about 'O' is $\frac{MR^2}{2}$

M.I. जड़त्व आघूर्ण 'O' के सापेक्ष $\frac{MR^2}{2}$

By parallel-axis theorem : $\frac{MR^2}{2}$



समान्तर अक्ष प्रमेय से : $\frac{MR^2}{2}$

$$= I_{cm} + M\left(\frac{4R}{3\pi} \cdot \sqrt{2}\right)^2 \Rightarrow I_{cm} = \frac{MR^2}{2} - M\left(\sqrt{2} \cdot \frac{4R}{3\pi}\right)^2$$

43. Restoring torque is given by, प्रत्यानयन आघूर्ण

$$(C) KX \cdot \left(\frac{\ell}{2}\right) = I\alpha$$

$$K\left(\frac{\ell}{2}\theta\right) \cdot \left(\frac{\ell}{2}\right) = \frac{m\ell^2}{12} \cdot (\alpha) \Rightarrow \alpha = \frac{3K}{m}\theta \Rightarrow \omega = \sqrt{\frac{3K}{m}}$$

44. Energy Density = $\frac{1}{2}$ stress $\times$ strain = $\frac{1}{2}$ Y (strain)² = 2880 J/m³

$$\text{ऊर्जा घनत्व} = \frac{1}{2} \text{ प्रतिबल} \times \text{विकृति} = \frac{1}{2} Y (\text{विकृति})^2 = 2880 \text{ J/m}^3$$

45. When two drops of radius r each combine to form a big drop, the radius of big drop will be given by

जब r त्रिज्या की दो गेंदे संयुक्त होकर बड़ी बून्द बनाये तो बड़ी बून्द की त्रिज्या होगी।

$$\frac{4}{3}\pi R^3 = \frac{4\pi}{3}r^2 + \frac{4\pi}{3}r^3$$

$$\text{or } R^3 = 2r^3 \quad \text{or } R = 2^{1/3}r$$

$$\text{Now } \frac{V_R}{V_r} = \left(\frac{R}{r}\right)^2 = 2^{2/3} = 4^{1/3}$$

$$\therefore V_R = 5 \times 4^{1/3} \text{ cm}^3/\text{s}$$

46. $v_A = v_B$ (since area is uniform)

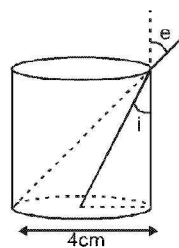
$v_A = v_B$ (क्योंकि क्षेत्रफल एकसमान है।)

From Bernoulli's principle

बरनौली समीकरण से

$$\frac{v_A^2}{2} + gh + \frac{p_A}{\rho} = \frac{v_B^2}{2} + 0 + \frac{p_B}{\rho} \Rightarrow p_A < p_B$$

- 47.



$$\sqrt{\frac{5}{2}} \sin i = 1 \sin e$$

$$\Rightarrow \sqrt{\frac{5}{2}} \frac{2 \text{ cm}}{\sqrt{4 \text{ cm}^2 + h^2}} = 1 \times \frac{4 \text{ cm}}{\sqrt{16 \text{ cm}^2 + h^2}}$$

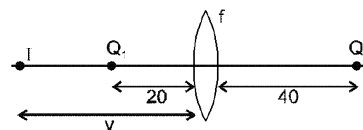
$$\Rightarrow h = 4 \text{ cm}$$

48. for object O_1 O_1 वस्तु के लिए

$$\frac{1}{v} + \frac{1}{20} = \frac{1}{f} \quad \dots (1)$$

for object O_2 O_2 वस्तु के लिए

$$\frac{1}{v} - \frac{1}{40} = -\frac{1}{f} \quad \dots (2)$$



from equation 1 and 2 we get

समीकरण 1 व 2 से

$$f = 80/3$$

49. Spring constant (स्प्रिंग नियतांक) $K = \frac{6.4}{0.1} = 64 \text{ N/m}$.

Now (अब) $T = 2\pi\sqrt{\frac{m}{k}}$ or $\frac{\pi}{4} = 2\pi\sqrt{\frac{m}{64}} \therefore m = 1 \text{ kg}$

50. $T = 2\pi\sqrt{\ell/g}$

$$\frac{\Delta T}{T} = \frac{1}{2} \frac{\Delta \ell}{\ell} = \frac{1}{2} \alpha \Delta t$$

51. $V = KT + C$

$$P = \frac{nRT}{V} \Rightarrow P = \frac{nRT}{KT + C};$$

$$\frac{dP}{dT} = \frac{nRC}{(KT + C)^2}$$

As $C < 0$ by diagram

चित्र से $C < 0$

$$\Rightarrow \frac{dP}{dT} < 0 \text{ for all } T \Rightarrow P \text{ continuously decreases.}$$

$$\Rightarrow \frac{dP}{dT} < 0 \text{ सभी } T \text{ पर } \Rightarrow P \text{ लगातार घटेगा।}$$

52. As the container is insulated $\Delta Q = 0$

चूँकि बर्तन ऊष्मारोधी है $\Delta Q = 0$

The expansion of gas is in vacuum $W = 0$

गैस का प्रसार निर्वात में है $W = 0$

$\therefore$ From first law प्रथम नियम से ,

$$\Delta Q = W + \Delta U$$

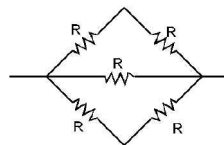
$$\Rightarrow \Delta U = 0$$

53. $\frac{dQ}{dt} = \frac{KA\Delta T}{2\ell} = \frac{\Delta T}{\frac{2\ell}{KA}} = \frac{10}{120} \text{ J/sec.}$

New rate ऊष्मा प्रवाह की दर $\frac{dQ}{dt} = \frac{\Delta T}{\frac{2\ell}{KA}} = \frac{40}{120} \text{ J/sec.}$

So time taken is अतः लगा समय

$$t = \frac{20}{40} \times 120 \text{ sec.} = 60 \text{ sec.}$$



54. $\lambda = 2\ell = 3\text{m}$

Equation of standing wave

$$y = 2A \sin kx \cos \omega t$$

$y = A$ as amplitude is $2A$.

$$A = 2A \sin kx$$

$$\frac{2\pi}{\lambda} x = \frac{\pi}{6} \Rightarrow x_1 = \frac{1}{4} \text{ m}$$

$$\text{and } \frac{2\pi}{\lambda} \cdot x = \frac{5\pi}{6} \Rightarrow x_2 = 1.25 \text{ m} \Rightarrow x_2 - x_1 = 1 \text{ m}$$

$$\lambda = 2\ell = 3\text{m}$$

अप्रगामी तरंग का समीकरण ($x = 0$ पर निस्पन्द लेते हुए)

$$y = 2A \sin kx \cos \omega t$$

$y = A$ चूँकि आयाम $2A$ हैं

$$A = 2A \sin kx$$

$$\frac{2\pi}{\lambda} x = \frac{\pi}{6} \Rightarrow x_1 = \frac{1}{4} \text{ m}$$

$$\text{और } \frac{2\pi}{\lambda} \cdot x = \frac{5\pi}{6} \Rightarrow x_2 = 1.25 \text{ m} \Rightarrow x_2 - x_1 = 1 \text{ m}$$

55. Initial Velocity = $V = u$ (say)

प्रारम्भिक वेग = $V = u$ (say)

Velocity at time $t = v$

t समय पर वेग = v

$$\frac{dv}{dt} = -\frac{v^2}{4\pi R} \Rightarrow -\int \frac{dv}{v^2} = \int \frac{dt}{4\pi R}$$

$$\frac{1}{u} = 0 + C \therefore \frac{1}{v} = \frac{1}{4\pi R} t + \frac{1}{u}$$

$$= \frac{ut + 4\pi R}{4\pi Ru} \Rightarrow \int \frac{dt}{ut + 4\pi R} = \int \frac{ds}{4\pi Ru}$$

$$\frac{1}{u} \cdot \log_e (ut + 4\pi R) = \frac{1}{4\pi Ru} s + C$$

$$\frac{1}{u} \cdot \log_e (0 + 4\pi R) = 0 + C$$

$$\log_e \left(\frac{ut + 4\pi R}{4\pi R} \right) = \frac{S}{4\pi R}; \text{ For } t = \frac{8\pi R}{u}$$

$$\log_e \left(\frac{u}{4\pi R} \times \frac{8\pi R}{u} + \frac{4\pi R}{4\pi R} \right) = \frac{S}{4\pi R}$$

$$\therefore S = (4\pi R) \log_e 3 = 4\pi R \text{ (about लगभग)}$$

$$\therefore \text{ Number of revolution} = 2$$

$$\therefore \text{ कुल घूर्णन} = 2$$

56. $\ell_1 = 52 + 1 = 53 \text{ cm}$

$\ell_2 = 48 + 2 = 50 \text{ cm}$

$$\frac{\ell_1}{\ell_2} = \frac{x}{R} \Rightarrow \frac{53}{50} = \frac{x}{10}$$

$x = 10.6 \Omega$

57. जब when $A = 1$, $B = 0$, $C = 1$ then तब $Y = 1$

60. If the rate at which molecules of same mass having same rms velocity striking a wall decreases, then the rate at which momentum is imparted to the wall decreases. This results in lowering of pressure. Hence statement-2 is correct.

In statement-1 the rms velocity of gas remains same on increasing the volume of container by piston, since the given process is isothermal. Now the piston is at a greater distance from opposite wall and hence time taken by gas molecules from near the opposite wall to reach the piston will be more. Thus rate of molecules striking the piston decreases. Hence statement-1 is correct and statement-2 is correct explanation.

यदि वह दर जिस पर समान द्रव्यमान तथा समान वर्गमाध्यमूल वेग के अणु दीवार पर टकराते हैं, घटती है तो वह दर जिस पर संवेग दीवार पर दिया जाता है, घट जाती है। परिणामस्वरूप दाब घट जाता है। इसलिये वक्तव्य-2 सत्य है।

वक्तव्य-1 में गैस का वर्गमाध्यमूल वेग पिस्टन द्वारा बर्तन का आयतन बढ़ाने पर समान रहता है चूंकि दी गई प्रक्रिया समतापी है। अब पिस्टन विपरीत दीवार से अधिक दूरी पर होगा तथा इसलिये गैस अणुओं द्वारा विपरीत दीवार के पास से पिस्टन तक पहुँचने में लिया गया समय अधिक होगा। इसलिये पिस्टन पर अणुओं के टकराने की दर घट जाती है। इसलिये वक्तव्य-1 सही है तथा वक्तव्य-2 सही स्पष्टीकरण है।

Part-C (Chemistry)

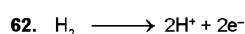
61. $\log \frac{K_2}{K_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$

$$\log 2 = \frac{E_a}{2.303 \times 8.314} \left[\frac{1}{293} - \frac{1}{308} \right]$$

$$0.3010 = \frac{E_a}{2.303 \times 8.314} \times \frac{15}{293 \times 308}$$

$= 34673 \text{ J mole}^{-1}$

$= 34.7 \text{ kJ mole}^{-1}$



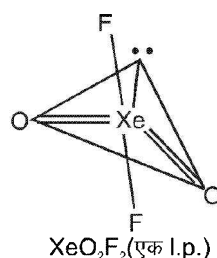
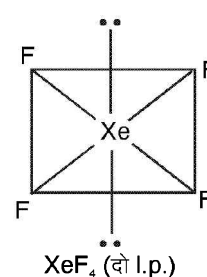
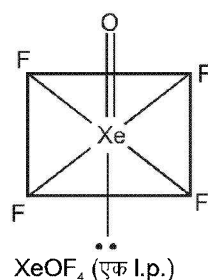
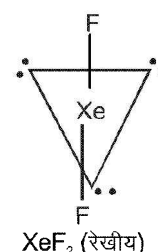
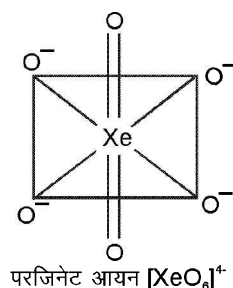
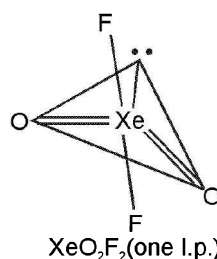
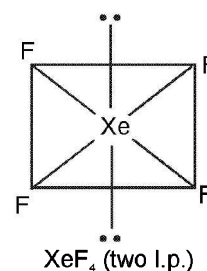
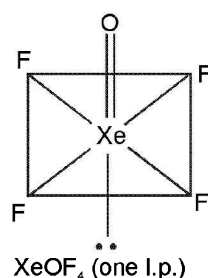
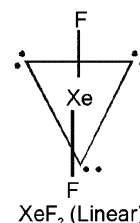
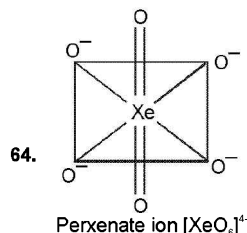
1 atm 10^{-10}

$$E_{\text{H}_2/\text{H}^+} = 0 - \frac{0.059}{2} \log \frac{(10^{-10})^2}{1}$$

$E_{\text{H}_2/\text{H}^+} = +0.59 \text{ V}$

63. Structure of BN is similar with graphite (Known as inorganic graphite.)

हल. BN की संरचना ग्रेफाइट के समान है। (इसे अकार्बनिक ग्रेफाइट कहा जाता है)



65. At Boyle temperature gas behaves as ideal gas ($Z = 1$) in a range of low pressure but at high pressure $Z > 1$.

बॉयल ताप पर दाब की निम्न दाब परास में गैस आदर्श गैस ($Z = 1$) जैसा व्यवहार करती है परन्तु उच्च दाब पर $Z > 1$ होता है।

66. $\text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons \text{PCl}_5(\text{g})$ $K_p = 1/5 \text{ atm}$

Initial pr. P_0 P_0

प्रारम्भिक दाब

at equilibrium $P_0 - 0.75P_0$ $P_0 - 0.75P_0$ $0.75P_0$

साम्य पर

$$\frac{1}{5} = \frac{(0.75 P_0)}{(0.25 P_0)(0.25 P_0)}$$

$$0.25 P_0 = 15$$

$$P_{\text{Total}} = 1.25 P_0 = 75 \text{ atm}$$

68. $\text{MnO}_2 + \text{H}_2 \longrightarrow \text{Mn}_x\text{O}_y + \text{H}_2\text{O}$

17.4 g 12.6 g

Applying POAC for Mn (Mn के लिए POAC प्रयुक्त कर)

$$1 \times \frac{17.4}{87} = x \times \frac{12.6}{55x + 16y}$$

$$0.2 = \frac{12.6}{55 + 16\left(\frac{y}{x}\right)}$$

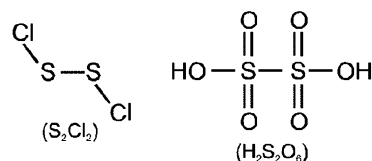
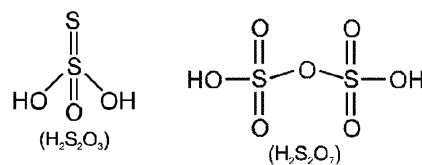
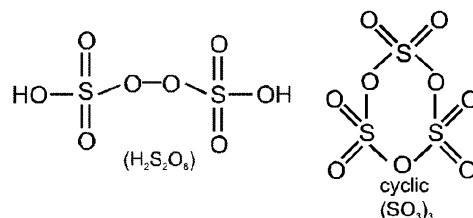
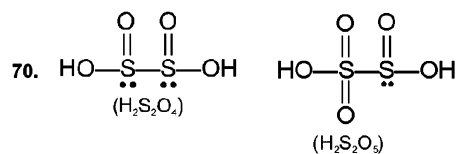
$$11 + 3.2\left(\frac{y}{x}\right) = 12.6$$

$$\frac{y}{x} = \frac{1}{2}$$

69. At equivalence point $[\text{BCl}] = 0.01 \text{ M}$

तुल्याक बिन्दु $[\text{BCl}]$ पर $= 0.01 \text{ M}$

$$[\text{H}^+] = \sqrt{\frac{K_w}{K_b} \cdot C} = 10^{-5} \text{ M}$$

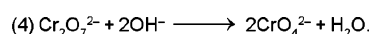
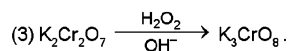
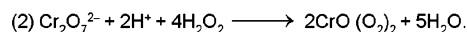
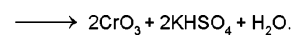


71. scc : $a = 2r$; bcc : $\sqrt{3}a = 4r$; fcc : $\sqrt{2}a = 4r$

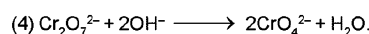
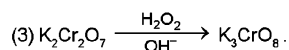
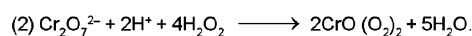
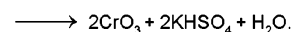
72. As temperature increases, physical adsorption decreases because physical adsorption is exothermic and reversible process.

हल. जैसे ही ताप बढ़ता है, भौतिक अधिशोषण घट जाता है, क्योंकि भौतिक अधिशोषण एक ऊष्माक्षेपी तथा उत्क्रमणीय प्रक्रिया होती है।

73. (1) $\text{K}_2\text{Cr}_2\text{O}_7 + 2\text{H}_2\text{SO}_4$ (concentrated & cold)



हल. (1) $\text{K}_2\text{Cr}_2\text{O}_7 + 2\text{H}_2\text{SO}_4$ (सान्द्र तथा ठण्डा)



74. $\Delta T_f = K_f m$

$$\Delta T_f = 2.8 = 1.86 \times \frac{a}{0.5}$$

$$\therefore \text{mass of } C_2H_6O_2 = a \times 62$$

$$\therefore C_2H_6O_2 \text{ का द्रव्यमान} = a \times 62$$

$$= \frac{2.8 \times 0.5}{1.86} \times 62 \approx 46.6 \text{ g.}$$

75. $\Delta G_{\text{system}} = -T\Delta S_{\text{total}}$

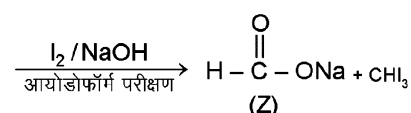
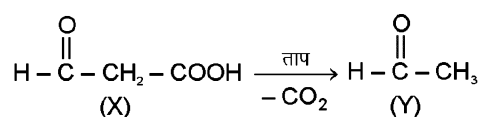
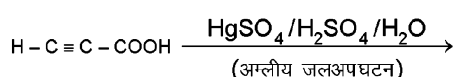
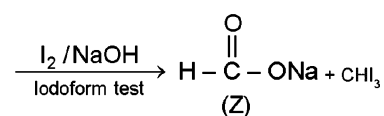
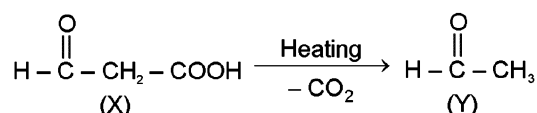
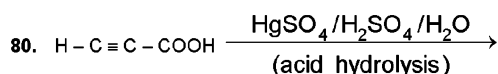
$$\Delta G_{\text{तन्त्र}} = -T\Delta S_{\text{कुल}}$$

76. $Na_2[Cr(edta)]$ is correct representation.

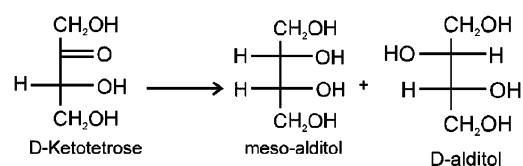
सही प्रदर्शन, $Na_2[Cr(edta)]$ है।

77. Freezing is exothermic process. The heat released increases the entropy of surrounding.

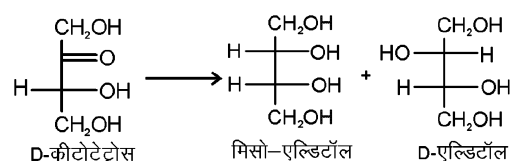
जमना ऊष्माक्षेपी अभिक्रिया होती है। उत्सर्जित ऊष्मा परिवेश की एन्ट्रॉपी में वृद्धि करती है।



82. The product is an optically active mixture :

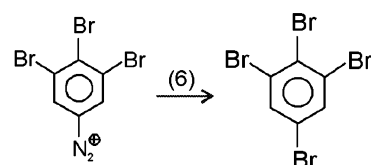
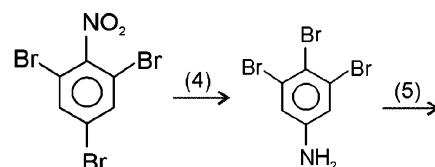
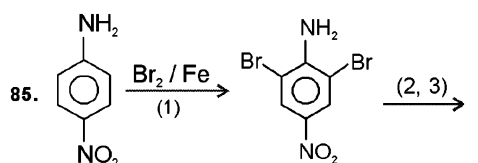
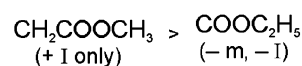
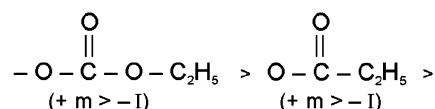


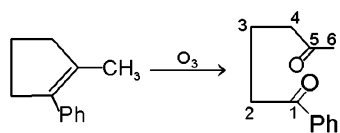
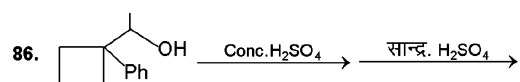
उत्पाद एक प्रकाशिक सक्रिय (ध्रुवण घूर्णक) मिश्रण है :



84. The electron releasing effect of the following groups follows the order :

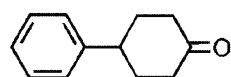
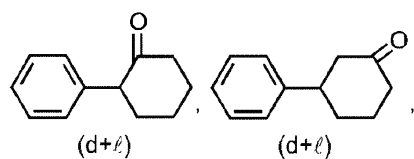
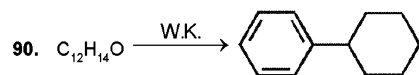
निम्न समूहों का इलेक्ट्रॉन देने का प्रभाव का क्रम निम्न है।





89. LiAlH_4 does not reduce the $\text{C}=\text{C}$ bonds.

यह $\text{C}=\text{C}$ को अपचयित नहीं करता है।



5 isomeric reactants are possible. (5 सगावयवी क्रियाकारक सम्भव है।)

DETAILED SOLUTION

Part-A- (Mathematics)

1. $I = \int_0^{\pi/2} \frac{1+2\cos x}{(2+\cos x)^2} dx$

$$= \int_0^{\pi/2} \frac{\operatorname{cosec}^2 x + 2 \operatorname{cosec} x \cot x}{(2 \operatorname{cosec} x + \cot x)^2} dx$$

$$= \left[\frac{1}{2 \operatorname{cosec} x + \cot x} \right]_0^{\pi/2} = \frac{1}{2}$$

2. $\frac{f(x)+g(x)}{2} = a$

$\Rightarrow f(x) + g(x) = 2a \Rightarrow h(x) = 2a$ which is many one into. जो कि बहुएक अन्तःक्षेपी है।

3. $f(x) = \sin \frac{2x}{3} + \cos 4x + |\tan 3x| + \operatorname{sgn}(x^2 + 4x + 15)$

period of $\sin \frac{2x}{3} = 3\pi$

$\cos 4x = \pi/2$

$|\tan 3x| = \pi/3$

while, $\operatorname{sgn}(x^2 + 4x + 15) = 1$ (a constant function)

$\therefore$ fundamental period of $f(x) = 3\pi$

हल $f(x) = \sin \frac{2x}{3} + \cos 4x + |\tan 3x| + \operatorname{sgn}(x^2 + 4x + 15)$

$\sin \frac{2x}{3} = 3\pi$

$\cos 4x = \pi/2$

$|\tan 3x| = \pi/3$

जबकि, $\operatorname{sgn}(x^2 + 4x + 15) = 1$ (एक अचर फलन है)

$\therefore$ मूलभूत आवर्तकाल $f(x) = 3\pi$

4. $\therefore \omega = \omega^4 = \omega^7 = \omega^{10} = \dots = \omega^{28}$
 $\omega^2 = \omega^5 = \omega^8 = \omega^{11} = \dots = \omega^{29}$
 $\omega^3 = \omega^6 = \omega^9 = \dots = \omega^{30} = 1$
 $\therefore (1 + \omega)(1 + \omega^2)(1 + \omega^3)(1 + \omega^4) \dots (1 + \omega^{30})$
 $= \{(1 + \omega)(1 + \omega^2)(1 + \omega^3)\}^{10}$
 $= \{(-\omega^2)(-\omega)(2)\}^{10}$
 $= 2^{10}$

5. $f'(x) = \frac{x^4}{4} - \frac{x^3}{3} = \frac{x^3}{12}(3x - 4)$

$f''(x) = x^3 - x^2 = x^2(x - 1)$

for $x < 0$, $f'(x) > 0$ and $x > 0$, $f'(x) < 0$

Hence $x = 0$ is a point of maxima

also $f''(x) = 0$ where $x = 1$ but at $x = 1$ we have an inflection point.

If $x = \frac{4}{3}$ is point of minima

Hindi $f'(x) = \frac{x^4}{4} - \frac{x^3}{3} = \frac{x^3}{12}(3x - 4)$

$f''(x) = x^3 - x^2 = x^2(x - 1)$

$x < 0$ के लिए, $f'(x) > 0$ तथा $x > 0$, $f'(x) < 0$

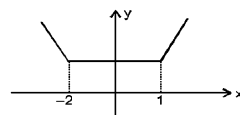
अतः $x = 0$ उच्चिष्ठ बिन्दु है।

तथा $f''(x) = 0$ जहाँ $x = 1$ परन्तु $x = 1$ पर नतिपरिवर्तन बिन्दु है।

यदि $x = \frac{4}{3}$ स्थानीय निम्निष्ठ बिन्दु है।

6. $f(x) = |x + 2| + |x - 1| = \begin{cases} -2x - 1, & x < -2 \\ 3, & -2 \leq x \leq 1 \\ 2x + 1, & x > 1 \end{cases}$

graph of $f(x)$ is $(f(x))$ का आलेख है)



7. $y = x^4$

$\therefore \frac{dy}{dx} = 4x^3 \quad \therefore \frac{dy}{dx} \Big|_{\text{at}(1,1)} = 4$

$\therefore$ equation of tangents at $(1, 1)$ is $y - 1 = 4(x - 1)$ i.e. $4x - y = 3$ (i)

$\frac{dy}{dx} \Big|_{\text{at}(-1,1)} = -4$

$\therefore$ Slope of normal is $\frac{1}{4}$

$\therefore$ Equation of the normal is $y - 1 = \frac{1}{4}(x + 1)$

i.e. $x - 4y = -5$ (ii)

$\therefore$ point of intersection is $\left(\frac{17}{15}, \frac{23}{15}\right)$

Hindi. $y = x^4$

$$\therefore \frac{dy}{dx} = 4x^3 \quad \therefore \left. \frac{dy}{dx} \right|_{\text{at}(1,1)} = 4$$

$\therefore$ (1, 1) पर स्पर्श रेखा का समीकरण $y - 1 = 4(x - 1)$ है।

i.e. $4x - y = 3$ (i)

$$\left. \frac{dy}{dx} \right|_{\text{at}(-1,1)} = -4$$

$\therefore$ अभिलम्ब की प्रवणता $\frac{1}{4}$ है।

$\therefore$ अभिलम्ब का समीकरण $y - 1 = \frac{1}{4}(x + 1)$ है।

i.e. $x - 4y = -5$ (ii)

$\therefore$ प्रतिच्छेदन का बिन्दु $\left(\frac{17}{15}, \frac{23}{15}\right)$ है।

8. $x(1 + y^2) dx + y(1 + x^2) dy = 0$

$$\Rightarrow x(1 + y^2) dx = -y(1 + x^2) dy$$

$$\Rightarrow \int \frac{x}{1+x^2} dx = - \int \frac{y}{1+y^2} dy$$

$$\Rightarrow \frac{1}{2} \ln(1 + x^2) = -\frac{1}{2} \ln(1 + y^2) + c$$

$$\Rightarrow \frac{1}{2} \ln(1 + x^2)(1 + y^2) = c$$

$$\Rightarrow \ln(1 + x^2)(1 + y^2) = 2c = \text{a constant (अचर)} = c_1$$

$$\Rightarrow (1 + x^2)(1 + y^2) = k$$

9. general term $= (1 + x + 2x^2)^4 C_r (3x^2)^{4-r} \left(\frac{-1}{3x^2}\right)^r = (1 + x + 2x^2)^4 C_r$

$$3^{4-r} \left(\frac{-1}{3}\right)^r x^{8-4r}$$

$$= {}^4C_r 3^{4-r} \left(\frac{-1}{3}\right)^r x^{8-4r} + {}^4C_r 3^{4-r} \left(\frac{-1}{3}\right)^r x^{8-4r} + {}^4C_r 3^{4-r} 2 \left(\frac{-1}{3}\right)^r$$

$$x^{10-4r}$$

For independent term of x

$$8 - 4r = 0$$

$$\Rightarrow r = 2$$

$$\text{and } 9 - 4r = 0$$

$$r = \frac{9}{4} \text{ Not possible}$$

$$\text{and } 10 - 4r = 0$$

$$r = \frac{5}{2} \text{ Not possible}$$

$$\text{term} = {}^4C_2 \cdot 3^2 \times \frac{1}{3^2} = 6$$

Hindi. व्यापक पद $= (1 + x + 2x^2)^4 C_r (3x^2)^{4-r} \left(\frac{-1}{3x^2}\right)^r = (1 + x + 2x^2)^4$

$${}^4C_r 3^{4-r} \left(\frac{-1}{3}\right)^r x^{8-4r}$$

$$= {}^4C_r 3^{4-r} \left(\frac{-1}{3}\right)^r x^{8-4r} + {}^4C_r 3^{4-r} \left(\frac{-1}{3}\right)^r x^{8-4r} + {}^4C_r 3^{4-r} 2 \left(\frac{-1}{3}\right)^r$$

$$x^{10-4r}$$

x से स्वतंत्र पद के लिए

$$8 - 4r = 0$$

$$\Rightarrow r = 2$$

$$\text{और } 9 - 4r = 0$$

$$r = \frac{9}{4} \text{ अंशभव}$$

$$\text{और } 10 - 4r = 0$$

$$r = \frac{5}{2} \text{ अंशभव}$$

$$\text{term} = {}^4C_2 \cdot 3^2 \times \frac{1}{3^2} = 6$$

10. Let k_i be the number of blanks between the i th and $(i + 1)$ th letters.

$$\therefore k_i \geq 3, k_1 + k_2 + k_3 + k_4 = 15, 1 \leq i \leq 4$$

$\therefore$ No. of possible solutions

$$= \text{coeff. of } x^{15} \text{ in } (x^3 + x^4 + \dots)^4$$

$$= \text{coeff. of } x^{15} \text{ in } \left(\frac{x^3}{1-x}\right)^4$$

$$= \text{coeff. of } x^{15} \text{ in } x^{12} (1-x)^{-4}$$

$$= \text{coeff. of } x^3 \text{ in } (1 + 4x + 10x^2 + 20x^3 + \dots) = 20$$

Also, no. of arrangements of 5 letters $= 5! = 120$

$$\therefore \text{Total no. of ways} = 20 \times 120 = 2400.$$

Hindi. माना i वें एवं $(i + 1)$ वें अक्षरों के बीच खाली जगहों की संख्या k_i

$$\therefore k_i \geq 3, k_1 + k_2 + k_3 + k_4 = 15, 1 \leq i \leq 4$$

$$\therefore \text{संभव हलों की संख्या} = (x^3 + x^4 + \dots)^4 \text{ में } x^{15} \text{ का गुणांक}$$

$$= \left(\frac{x^3}{1-x}\right)^4 \text{ में } x^{15} \text{ का गुणांक}$$

$$= x^{12} (1-x)^{-4} \text{ में } x^{12} \text{ का गुणांक}$$

$$= (1 + 4x + 10x^2 + 20x^3 + \dots) \text{ में } x^3 \text{ का गुणांक} = 20$$

तथा 5 अक्षरों की व्यवस्थाओं की संख्या $= 5! = 120$

$$\therefore \text{कुल तरीके} = 20 \times 120 = 2400.$$

11. Probability that problem is not solved by 1st $= 1 - \frac{1}{2} = \frac{1}{2}$

$$\text{Probability that problem is not solved by 2nd} = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\text{Probability that problem is not solved by 3rd} = 1 - \frac{1}{4} = \frac{3}{4}$$

$\therefore$ Probability that problem is not solved by any one of the

$$\text{three} = \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{4}$$

$$\text{Hence the required Probability } 1 - \frac{1}{4} = \frac{3}{4}$$

Hindi. समस्या को प्रथम के द्वारा हल नहीं करने की प्रायिकता = $1 - \frac{1}{2} = \frac{1}{2}$

समस्या को द्वितीय के द्वारा हल नहीं करने की प्रायिकता =

$$1 - \frac{1}{3} = \frac{2}{3}$$

समस्या को तृतीय के द्वारा हल नहीं करने की प्रायिकता =

$$1 - \frac{1}{4} = \frac{3}{4}$$

∴ तीनों में से किसी एक के द्वारा हल नहीं करने की प्रायिकता =

$$\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{4}$$

अभिष्ट प्रायिकता $1 - \frac{1}{4} = \frac{3}{4}$

$$12. T_n = \frac{(2n+1)}{1^2 + 2^2 + \dots + n^2}$$

$$\Rightarrow \frac{(2n+1)}{\left(\frac{n(n+1)(2n+1)}{6}\right)} \Rightarrow \frac{6}{n(n+1)}$$

$$\Rightarrow 6 \left(\frac{1}{n} - \frac{1}{n+1} \right)$$

$$\therefore S_n = 6 \left(1 - \frac{1}{n+1} \right) \Rightarrow \left(\frac{6n}{n+1} \right)$$

$$13. \ell n(x^2 + x + k) \in (-\infty, \infty)$$

$$\Rightarrow x^2 + x + k \in (0, \infty)$$

$$\Rightarrow x^2 + x + k > 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow D < 0$$

$$\Rightarrow 1 - 4k < 0$$

$$\Rightarrow k > \frac{1}{4}$$

$$14. \text{ We have } 12x^2 - 20xy + 7y^2 = 0$$

$$\Rightarrow (2x - y)(6x - 7y) = 0$$

So, the equations of three sides of the triangle are त्रिभुज के तीनों

गुजाओं का समीकरण है

$$2x - y = 0, 6x - 7y = 0, 2x - 3y + 4 = 0$$

The coordinates of vertices of the triangle are (0, 0), (7, 6) and (1, 2)

त्रिभुज के शीर्षों के निर्देशांक (0, 0), (7, 6) तथा (1, 2) है।

[Solving the above three equations, pairwise]

∴ Centroid of the triangle is त्रिभुज का केन्द्रक

$$\left(\frac{0+7+1}{3}, \frac{0+6+2}{3} \right) \text{ i.e. } \left(\frac{8}{3}, \frac{8}{3} \right)$$

$$15. \text{ Area of the required triangle is } \frac{r S_1^{3/2}}{r^2 + S_1}$$

$$\text{क्षेत्रफल} = \frac{r S_1^{3/2}}{r^2 + S_1}$$

$$= \frac{5 \times 16^{3/2}}{25 + 16} = \frac{320}{41} \text{ sq. units}$$

16. Equation of the plane OQR is $r = \lambda \vec{r}_2 + \mu \vec{r}_3$, i.e. $r(\vec{r}_2 \times \vec{r}_3) = 0$
OQR की समीकरण है

$$\text{So the distance of P from the plane OQR is } \frac{|\vec{r}_1 \cdot (\vec{r}_2 \times \vec{r}_3)|}{|\vec{r}_2 \times \vec{r}_3|}$$

अतः P की समतल OQR से दूरी है

$$\text{Since (चूँकि) } \vec{r}_2 \times \vec{r}_3 = -10 \hat{i} + 10 \hat{j} - 5 \hat{k},$$

$$\text{so (अतः) } |\vec{r}_2 \times \vec{r}_3| = 15$$

$$\text{and the perpendicular distance} = \frac{|-30 - 20 + 5|}{15} = 3$$

तथा लम्बवत् दूरी है

17. Let P be the point $(at^2, 2at)$, then mid point of SP is

$$\left(\frac{a(t^2 + 1)}{2}, at \right)$$

$$\text{Its locus is given by } x = \frac{a(t^2 + 1)}{2}, y = at$$

$$\text{i.e. } y^2 = 2a \left(x - \frac{a}{2} \right)$$

$$\text{Its vertex is } \left(\frac{a}{2}, 0 \right), \text{ directrix is y-axis, latus rectum} = 2a$$

and focus is $(a, 0)$

Hindi माना $P(at^2, 2at)$ है, तब SP का मध्य बिन्दु $\left(\frac{a(t^2 + 1)}{2}, at \right)$ है

$$\text{इसका बिन्दुपथ दिया गया है } x = \frac{a(t^2 + 1)}{2}, y = at$$

$$\text{i.e. } y^2 = 2a \left(x - \frac{a}{2} \right)$$

$$\text{इसका शीर्ष } \left(\frac{a}{2}, 0 \right), \text{ नियता y-अक्ष है नाभिलम्ब} = 2a \text{ और नाभि}$$

$(a, 0)$ है।

18. ∴ Mean of $x_1, x_2, \dots, x_n$ is $\bar{X}$.

$x_1, x_2, \dots, x_n$ का माध्य $\bar{X}$ है

$$\therefore \bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

Let $\bar{y}$ be the mean of observations $x_i + 2i, i = 1, 2, 3, \dots, n$. Then

माना $x_i + 2i, i = 1, 2, 3, \dots, n$ का माध्य $\bar{y}$ है, तो

$$\bar{y} = \frac{(x_1 + 2) + (x_2 + 2.2) + (x_3 + 2.3) + \dots + (x_n + 2.n)}{n}$$

$$= \frac{(x_1 + x_2 + \dots + x_n) + 2(1 + 2 + \dots + n)}{n}$$

$$= \frac{x_1 + x_2 + \dots + x_n}{n} + \frac{2n(n+1)}{2.n}$$

$$\Rightarrow \bar{y} = \bar{X} + (n+1)$$

19. $\sim(p \rightarrow q) \equiv p \wedge \sim q$
 where p : it rains
 q : I shall go to school
 जहाँ p : बारिश होती है
 q : मैं स्कूल जाऊँगा

20. In an equilateral triangle एक समबाहु त्रिभुज में $r = \frac{R}{2}$

Also exradius बहिर्त्रिज्या $r_1 = 4R \sin \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2} = 4R \frac{1}{2}$

$$\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} = \frac{3}{2} R$$

$\Rightarrow r, R, r_1$ are in A.P. समान्तर श्रेणी में है।

21. $\cos^6 \alpha + \sin^6 \alpha = (\cos^2 \alpha + \sin^2 \alpha)(\cos^4 \alpha + \sin^4 \alpha - \sin^2 \alpha \cos^2 \alpha)$
 $= 1 - 3 \sin^2 \alpha \cos^2 \alpha = 1 - \frac{3}{4} \sin^2 2\alpha$

$$\therefore 1 - \frac{3}{4} \sin^2 2\alpha + k \sin^2 2\alpha = 1$$

$$\Rightarrow k = 3/4$$

22. Since, R and R' are not disjoint, there is at least one ordered pair, say (a, b) in $R \cap R'$.
 But $(a, b) \in R \cap R' \Rightarrow (a, b) \in R$ and $(a, b) \in R'$.
 Since, R and R' are symmetric relations, we get
 $(b, a) \in R$
 and $(b, a) \in R'$
 and consequently $(b, a) \in R \cap R'$.
 Similarly, any other ordered pair $(c, d) \in R \cap R'$, then we must also have $(d, c) \in R \cap R'$.
 Hence, $R \cap R'$ is symmetric relation.

हल चूंकि, R तथा R' विंशधीत है अतः $(a, b) \in R \cap R'$.
 लेकिन $(a, b) \in R \cap R' \Rightarrow (a, b) \in R$ तथा $(a, b) \in R'$.
 चूंकि R व R' सममित है अतः
 $(b, a) \in R$
 तथा $(b, a) \in R'$
 $\Rightarrow (b, a) \in R \cap R'$.
 अतः $R \cap R'$ सममित है।

23. $\lim_{\theta \rightarrow 0} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right)$

We, know (हम जानते हैं)

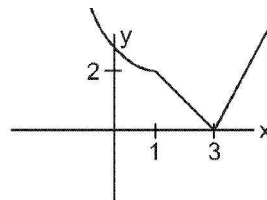
$$\frac{\sin \theta}{\theta} < 1, \quad \text{while (जबकि)} \quad \frac{\tan \theta}{\theta} > 1$$

thus, (अतः)

$$\lim_{\theta \rightarrow 0} \left[\frac{n \sin \theta}{\theta} \right] = n - 1, \quad \lim_{\theta \rightarrow 0} \left[\frac{n \tan \theta}{\theta} \right] = n$$

$$\therefore \lim_{\theta \rightarrow 0} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right) = 2n - 1$$

24. $f(x)$ at $x = 3$ is given by $|x - 3|$.
 $|x - 3|$ is continuous at $x = 3$ but not differentiable.
 $x = 3$ पर $f(x) = |x - 3|$.
 $|x - 3|$, $x = 3$ पर सतत् है किंतु अवकलनीय नहीं



25. Let $\sin^{-1} \left(\sqrt{\frac{13}{17}} \right) = \theta \Rightarrow \sin \theta = \sqrt{\frac{13}{17}}$

$$\Rightarrow \cot \theta = \frac{2}{\sqrt{13}} \Rightarrow A = \sin^{-1} \frac{2}{\sqrt{13}} = \tan^{-1} \frac{2}{3}$$

Hindi. माना $\sin^{-1} \left(\sqrt{\frac{13}{17}} \right) = \theta \Rightarrow \sin \theta = \sqrt{\frac{13}{17}}$

$$\Rightarrow \cot \theta = \frac{2}{\sqrt{13}} \Rightarrow A = \sin^{-1} \frac{2}{\sqrt{13}} = \tan^{-1} \frac{2}{3}$$

26. $f''(x) = -f(x)$, $f'(x) = g(x)$
 $\therefore g'(x) = f''(x) = -f(x)$
 $h(x) = (f(x))^2 + (g(x))^2$
 $\therefore h'(x) = 2f(x) \cdot f'(x) + 2g(x) \cdot g'(x)$
 $= 2f(x) \cdot g(x) + 2g(x) \cdot (-f(x)) = 0$
 $\therefore h(x)$ must be a constant function, $\forall x \in \mathbb{R}$ ($\therefore h(x)$ एक नियतांक फलन होना चाहिये)
 $\therefore h(10) = h(5) = 9$.

27. (1) $y^2 = 25m$

$$y = x + 5 \Rightarrow \text{touches } c = \frac{a}{m} \text{ स्पर्श करती है।}$$

$$5 = \frac{25/4}{1}$$

$$(2) \frac{x^2}{9} + \frac{y^2}{4} = 1$$

touches if स्पर्श करती है यदि $c^2 = a^2 m^2 + b^2$

$$\Rightarrow 25 = 9 \cdot 1 + 4$$

$$(3) \text{ If यदि } c^2 = a^2 m^2 - b^2$$

$$= 29(1) - 4 = 25$$

$$(4) c^2 = a^2(m^2 + 1) = 25(1 + 1) = 50$$

28. $10 - a > 0$ and $4 - a > 0$ if $a < 4$.

Hindi $10 - a > 0$ तथा $4 - a > 0$ यदि $a < 4$

29. (1)

$$\text{If यदि } A = \begin{bmatrix} 0 & a & -b \\ -a & 0 & c \\ b & -c & 0 \end{bmatrix}$$

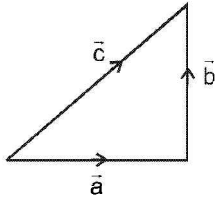
The system is निकाय

$$(I - A)X = 0 \Rightarrow X = 0$$

Since $I - A$ is non singular. चूंकि $I - A$ व्युत्क्रमणीय है।

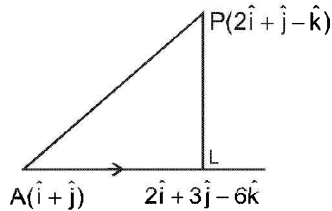
30. STATEMENT-2 $\frac{|\vec{c} \times \vec{a}|}{|\vec{a}|} = |\vec{c}| \sin \alpha = |\vec{b}|$

∴ STATEMENT-2 is true



STATEMENT-1 $\vec{AP} = \hat{i} - \hat{k}$

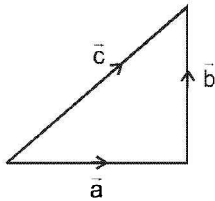
∴ $PL = \left| (\hat{i} - \hat{k}) \times \frac{2\hat{i} + 3\hat{j} - 6\hat{k}}{7} \right| = \frac{\sqrt{34}}{7}$



∴ STATEMENT-1 is false

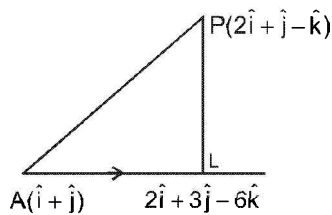
कथन-2 $\frac{|\vec{c} \times \vec{a}|}{|\vec{a}|} = |\vec{c}| \sin \alpha = |\vec{b}|$

∴ कथन-2 सत्य है।



कथन-1 $\vec{AP} = \hat{i} - \hat{k}$

∴ $PL = \left| (\hat{i} - \hat{k}) \times \frac{2\hat{i} + 3\hat{j} - 6\hat{k}}{7} \right| = \frac{\sqrt{34}}{7}$

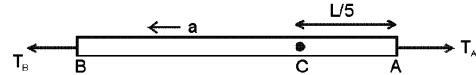


∴ कथन-1 असत्य है।

Part-B (Physics)

31. The F.B.D. of section of rope between A and B having acceleration a towards left is

रस्सी का त्वरण 'a' बायीं तरफ है तथा A व B के मध्य रस्सी के भाग का F.B.D.



Applying Newtons second law on section AB and section AC of rope we get

रस्सी के AB भाग व AC भाग पर न्यूटन के द्वितीय नियम लगाने पर

$$T_B - T_A = Ma \text{ and } T_C - T_A = \frac{M}{5} a$$

Solving हल करने पर $T_C = \frac{T_B + 4T_A}{5}$

32. The circuit will have inductive nature if $\omega > \frac{1}{\sqrt{LC}}$

$$\left(\omega L > \frac{1}{\sqrt{LC}} \right).$$

Hence A is false. Also if circuit has inductive nature the current will lag behind voltage. Hence D is also false.

If $\omega = \frac{1}{\sqrt{LC}}$ $\left(\omega L = \frac{1}{\omega C} \right)$ the circuit will have resistance nature. Hence B is false

$$\text{Power factor } \cos \phi = \frac{R}{\sqrt{R^2 + \left(\omega L + \frac{1}{\omega C} \right)^2}}$$

$$= 1 \text{ if } \omega L = \frac{1}{\omega C}. \text{ Hence C is true.}$$

Sol. यदि $\omega > \frac{1}{\sqrt{LC}}$ $\left(\omega L > \frac{1}{\omega C} \right)$ है तो परिपथ प्रेरणीक प्रकृति का होगा।
अतः A असत्य है। यदि परिपथ प्रेरणीक प्रकृति का है तो धारा वोल्टेज से पीछे होगी। अतः D भी असत्य है।

यदि $\omega = \frac{1}{\sqrt{LC}}$ $\left(\omega L = \frac{1}{\omega C} \right)$ है तो परिपथ प्रतिरोध प्रकृति का होगा। अतः B असत्य है।

$$\text{शक्ति गुणांक } \cos \phi = \frac{R}{\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C} \right)^2}} = 1 \text{ if } \omega L = \frac{1}{\omega C}.$$

अतः C सत्य है।

33. The distance between n th bright fringe and $(n + 1)^{\text{th}}$ dark fringe is

$$\text{equal to half fringe width} = \frac{\lambda D}{2d}$$

n वीं दीप्त फ्रिज तथा $(n + 1)$ वीं अदीप्त फ्रिज बीच की दूरी फ्रिज चौड़ाई

$$\text{के आधे के बराबर होती है} = \frac{\lambda D}{2d}$$

34. $T_{\alpha} = \frac{(Q - \Delta E)Y}{Y + 4} = \frac{(6.5 - 1)216}{216 + 4} = 5.40 \text{ MeV.}$

35. Production of neutrino along with the positron in β^+ -decay helps to explain the continuous spectrum
 β^+ -क्षय प्रक्रिया में पोजीट्रॉन के साथ न्यूट्रिनो का उत्पादन सतत स्पेक्ट्रम की व्याख्या करने में सहायता करता है।
 (D) इनमें से कोई नहीं।

36. $B = \frac{\mu_0 I}{4\pi \cdot \frac{R}{\sqrt{2}}} (\sin 90^\circ + \sin 135^\circ) = \frac{\mu_0 I}{4\pi R} (\sqrt{2} - 1)$

37. $L = \frac{\mu_0 N^2 \pi r^2}{\ell}$

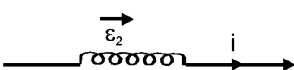
length of wire = $N 2\pi r$ = Constant (= C, suppose)

तार की लम्बाई = $N 2\pi r$ = अचर (= C, माना)

$$\therefore L = \mu_0 \left(\frac{C}{2\pi r} \right)^2 \frac{\pi r^2}{\ell} \quad \therefore L \propto \frac{1}{\ell}$$

$\therefore$ Self inductance will become $2L$.

$\therefore$ इसलिए स्वप्रेरकत्व $2L$ होगा

38. 

current is decreasing and धारा कम हो रही है तथा

$$L = \frac{17}{25kA} = 680 \mu H.$$

39. Potential difference विभवान्तर

$$V_1 - V_2 = \left(\frac{1}{4\pi \epsilon_0} \right) 'Q' \left(\frac{1}{R} - \frac{1}{2R} \right)$$

It will be doubled. दुगुना हो जाएगा।

40. $\frac{g \left(1 - \frac{2h}{R_e} \right) - g}{g} = -0.1 \times \frac{1}{100}$

$$\Rightarrow -\frac{2h}{R_e} = -\frac{1}{1000}$$

$$\Rightarrow h = \frac{R_e}{2000} = \frac{6400}{2000} \text{ km} = 3.2 \text{ km.}$$

41. $I = \frac{4}{10 \times 10^3} = 4 \times 10^{-4}$

but परन्तु

$$4 \times 10^{-4} = \frac{120}{x + 10^4}; \quad 4 \times 10^{-4}x + 4 = 120$$

$$x = \frac{116}{4} \times 10^4; \quad x = 290 \text{ k}\Omega$$

42. $E_2 = 2IR$

$$I' = \frac{E_2 - E_1}{R} = 2I - \frac{E_1}{R}$$

If यदि $E_1 < IR \Rightarrow I' > I$

If यदि $IR < E_1 < 2IR \Rightarrow I' < I$.

If यदि $E_1 = 2IR \Rightarrow I' = 0$.

43. Kinetic energy can become zero only

गतिज ऊर्जा शून्य हो सकती है।

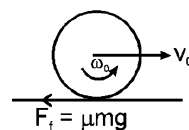
for the case shown in figure ;

चित्र में दी गई स्थिति के लिए चित्रानुसार

Torque equation :

बलाघूर्ण की समीकरण

$$(\mu mg) \cdot R = \frac{MR^2}{2} \cdot \alpha \Rightarrow \alpha = \frac{2\mu g}{R}$$



Therefore इसलिए, $t = \frac{\omega_0 R}{\alpha} = \frac{\omega_0 R}{2\mu g}$ (1)

For translational motion : $t = \frac{v_0}{\mu g}$ (2)

स्थानान्तरित गति के लिए : $t = \frac{v_0}{\mu g}$ (2)

From (1) & (2) : $\frac{\omega_0 R}{2\mu g} = \frac{v_0}{\mu g}$

समीकरण (1) तथा (2) से : $\frac{\omega_0 R}{2\mu g} = \frac{v_0}{\mu g}$

$$\Rightarrow \omega_0 = \frac{2v_0}{R} = \frac{2(10)}{0.2} = 100 \text{ rad/sec.} \quad \text{Ans.}$$

44. Acceleration = $\frac{\Sigma F}{M} = \frac{F}{M}$ ($\because$ friction force is zero)

त्वरण = $\frac{\Sigma F}{M} = \frac{F}{M}$ ($\because$ घर्षण बल शून्य है)

45. Strain developed :

$$\epsilon = \alpha \Delta T = (12 \times 10^{-6}) (50) = 6 \times 10^{-4}$$

Strain will be **negative**, as the rod is in a compressed state.

विकृति आयेगी -

$$\epsilon = \alpha \Delta T = (12 \times 10^{-6}) (50) = 6 \times 10^{-4}$$

विकृति ऋणात्मक होगी क्योंकि छड़ संकुचित अवस्था में हैं

46. Viscous force (श्यान बल) = $6 \pi \eta r v$

$$= 6\pi \times 18 \times 10^{-5} \times 0.03 \times 100$$

$$= 101.73 \times 10^{-4} \text{ dyne.}$$

47. $P_{\text{atm}} + \rho \cdot g \cdot 2 + 1/2 \rho \cdot V_1^2 = P_{\text{atm}} + 1/2 \cdot \rho V_2^2$

$$4g + V_1^2 = V_2^2$$

$$A_1 V_1 = A_2 V_2$$

$$4g + V_1^2 = 64 V_1^2$$

$$63V_1^2 = 40$$

$$V_1 = \sqrt{\frac{40}{63}} \text{ m/sec}$$

48. Given दिया गया है $i = 60^\circ$

$$A = \delta = e$$

$$\delta = i + e - A \Rightarrow \delta = i \quad (\because e = A)$$

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\frac{A}{2}}$$

Here angle of deviation is minimum ($\because i = e$)

$$\mu = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin\frac{A}{2}}$$

यहाँ न्यूनतम विचलन के लिए ($\because i = e$)

$$\mu = \frac{\sin\left(\frac{60^\circ + 60^\circ}{2}\right)}{\sin(60^\circ/2)}$$

49. $\frac{1}{f} = (1.5 - 1) \left(\frac{1}{-60} - \frac{1}{-30} \right)$

$$f = 120 \text{ cm}$$

$$y\text{-coordinate of focus} = -f\theta$$

$$\text{फोकस का } y\text{-निर्देशांक} = -f\theta$$

$$= -120 \times \frac{\pi}{180} \times 6^\circ$$

$$= -4\pi \text{ cm.}$$

50. $\frac{\Delta T}{T} \times 100 = \frac{1}{2} \frac{\Delta \ell}{\ell} \times 100$ is not valid as $\Delta \ell$ is not small.

$$\frac{\Delta T}{T} \times 100 = \frac{1}{2} \frac{\Delta \ell}{\ell} \times 100 \text{ यह सही नहीं है क्योंकि } \Delta \ell \text{ छोटा नहीं है।}$$

है।

$$T_1 = 2\pi \sqrt{\frac{\ell}{g}} \quad T_2 = 2\pi \sqrt{\frac{2\ell}{g}} \quad \% \text{ change परिवर्तन}$$

$$= \frac{T_2 - T_1}{T_1} \times 100 = (\sqrt{2} - 1) \times 100 = 41.4.$$

51. Law of equipartition of energy ऊर्जा के समवितरण नियम से

$$\langle KE_R \rangle = 2 \cdot \frac{1}{2} kT = E$$

$$\therefore \langle KE_T \rangle = 3 \cdot \frac{1}{2} kT$$

$$\langle KE_T \rangle = \frac{3E}{2}$$

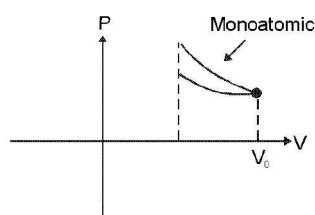
52. As (चुकि) $p \propto V^2$

$$T \propto V^3$$

i.e. if temperature increases, volume also increases hence w.d. will be positive.

अर्थात् यदि ताप बढ़ता है तो आयतन भी बढ़ता है इसलिए किया गया कार्य धनात्मक होगा।

- 53.



एक परमाणुक गैस के लिए दाब अधिक होगा इसलिए ताप भी उसका ज्यादा होगा।

54. $k = \frac{2\pi}{\lambda} = 0.025 \pi \Rightarrow \frac{\lambda}{2} = \frac{1}{0.025} = 40 \text{ cm.}$

The shortest possible length will be $L_{\min} = \frac{\lambda}{2} = 40 \text{ cm.}$

न्यूनतम संभव लम्बाई $L_{\min} = \frac{\lambda}{2} = 40 \text{ cm.}$

$$55. a_{\text{Net}} = \sqrt{a_t^2 + a_c^2}$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta$$

$$\therefore \omega_0 = 0$$

$$\text{so } \omega^2 = 2\alpha\theta$$

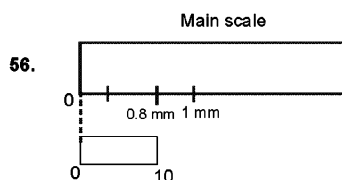
$$\omega^2 R = 2 (\alpha R \theta)$$

$$a_c = \omega^2 R = 2a_t \theta$$

$$1 = \sqrt{0.36 + (1.2 \times \theta)^2}$$

$$\Rightarrow 1 - 0.36 = (1.2 \theta)^2 \Rightarrow \frac{0.8}{1.2} = \theta \Rightarrow$$

$$\theta = \frac{2}{3} \text{ radian}$$



$$20 \text{ VSD} = 16 \text{ MSD}$$

$$1 \text{ VSD} = 0.8 \text{ MSD}$$

$$\text{Least count अल्पातमांक} = \text{MSD} - \text{VSD}$$

$$= 1 \text{ mm} - 0.8 \text{ mm}$$

$$= 0.2 \text{ mm}$$

$$57. \text{ Voltage gain वोल्टता लब्धि} = \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{I_{\text{out}}}{I_{\text{in}}} \times \frac{R_{\text{out}}}{R_{\text{in}}}$$

$$= \frac{2 \times 10^{-3}}{40 \times 10^{-6}} \times \frac{4 \times 10^3}{100}$$

$$= 2 \times 1000 = 2000$$

58. Since magnetic field is in vertical direction and Needle is free to rotate in horizontal plane only so magnetic force can not rotate the needle in horizontal plane so needle can stay in any position.

चुंकि चुम्बकीय क्षेत्र उर्ध्वाधर दिशा में इस कारण चुम्बकीय सूई क्षैतिज तल में घुमने के लिए स्वतंत्र है, इस कारण चुम्बकीय क्षेत्र क्षैतिज तल में चुम्बकीय सूई को नहीं घुमा पायेगा साथ ही चुम्बकीय सूई क्षैतिज तल में किसी भी दिशा में ठहर सकती है।

59. Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

वक्तव्य-1 सत्य है, वक्तव्य-2 सत्य है; वक्तव्य-2 वक्तव्य -1 का सही स्पष्टीकरण है।

60. Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1

वक्तव्य-1 सत्य है, वक्तव्य-2 सत्य है; वक्तव्य-2 वक्तव्य -1 का सही स्पष्टीकरण है।

Part-C (Chemistry)



As number of ions will decrease ($2\text{KI} \rightarrow 4 \text{ ions}$, $\text{K}_2\text{HgI}_4 \rightarrow 3 \text{ ions}$)

जैसे ही आयनों की संख्या में कमी होगी



$$\therefore \Delta T_f \downarrow F_p \uparrow$$

$$62. 30 \times \frac{1}{12} = 20 \times N'$$

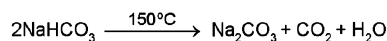
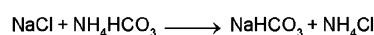
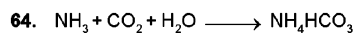
$$N' = \frac{30}{12 \times 20} = \frac{1}{8}$$

$$\therefore \text{strength} = N' \times \text{equivalent mass} = \frac{1}{8} \times 17 = 2.12 \text{ g/L.}$$

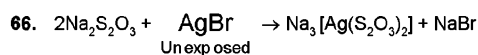
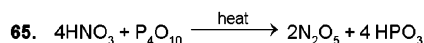
$$\therefore \text{सामर्थ्य} = N' \times \text{तुल्यांकी द्रव्यमान} = \frac{1}{8} \times 17 = 2.12 \text{ g/L.}$$

63. Cation will have more value of ionisation energy than corresponding atom.

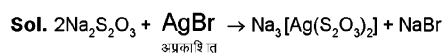
धनायन में संगत परमाणु की तुलना में आयनन ऊर्जा का मान अधिक होगा।



(byproduct) (सह उत्पाद).



This property is used for fixing in photography.



यह गुण फोटोग्राफी में स्थायीकरण के लिए प्रयुक्त होता है।

$$67. \alpha = \frac{i-1}{x-1} = \frac{3}{4} = \frac{4-1}{x-1}$$

$$\therefore x = 5 \quad \therefore \text{Complex is } \text{Ba}_3[\text{Co(CN)}_6]_2$$

$$\alpha = \frac{i-1}{x-1} = \frac{3}{4} = \frac{4-1}{x-1}$$

$$\therefore x = 5 \quad \therefore \text{Ba}_3[\text{Co(CN)}_6]_2 \text{ संकुल है।}$$

68. Let x gm of Zn deposit on 9 gm of Hg

$$\% \text{ of Zn in Amalgam} = \frac{m_{\text{Zn}} \times 100}{m_{\text{Zn}} + m_{\text{Hg}}}$$

$$= \frac{x}{9+x} \times 100 = 25$$

$$\therefore x = 3 \text{ gm}$$

$$\text{Eq. of Zn} = \frac{3 \times 2}{65.4}$$

$$\frac{n_{\text{Zn}}}{1} = \frac{n_{e^-}}{2} = \frac{Q}{2f} = \frac{it}{2f}$$

$$\text{Current } i = \frac{6}{65.4} \times \frac{96500}{1000} = 8.85 \text{ amp.}$$

Sol. माना 9 gm Hg पर Zn के x gm निक्षेपित होते हैं।

$$\text{अमलगम में Zn का \%} = \frac{m_{\text{Zn}} \times 100}{m_{\text{Zn}} + m_{\text{Hg}}} = \frac{x}{9+x} \times 100 = 25$$

$$\therefore x = 3 \text{ gm}$$

$$\text{Zn का तुल्यांक} = \frac{3 \times 2}{65.4}$$

$$\frac{n_{\text{Zn}}}{1} = \frac{n_{e^-}}{2} = \frac{Q}{2f} = \frac{it}{2f}$$

$$\text{धारा } i = \frac{6}{65.4} \times \frac{96500}{1000} = 8.85 \text{ amp.}$$

69. $\frac{V}{T} = \text{constant (नियत)}$

$$\log V = \log T + \log k \quad \text{where } K = \frac{nR}{P}$$

for 2 graph (2 आरेख के लिए)

$$\log \frac{nR}{P} = 0 = \log 1$$

$$\frac{nR}{P} = 1 \quad (R = P) \text{ or } n = 1 \text{ mol}$$

for 1 graph (1 आरेख के लिए)

$$\log \frac{nR}{P} = \log 3 \text{ or } n = 3 \text{ mol}$$

for 3 graph (3 आरेख के लिए)

$$\log \frac{nR}{P} = -\log 2 = \log \frac{1}{2}$$

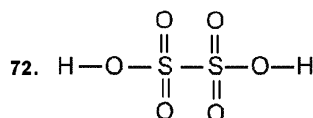
$$n = \frac{1}{2} \text{ mol}$$

70. HOMO can acts as Lewis base.

$$71. Q_c = \frac{\left(\frac{3}{10}\right)^4 \left(\frac{3}{10}\right)}{\left(\frac{2}{10}\right) \left(\frac{4}{10}\right)^2} = \frac{243}{32} \times 10^{-2} = 7.59 \times 10^{-2} > K_c$$

so, reaction will proceed in backward direction.

इसलिए, अभिक्रिया पश्च दिशा में जायेगी।



In H_2SO_5 the oxidation number of O is +6 as two O are having peroxide bond.

H_2SO_5 में O का ऑक्सीकरण अंक +6 है क्योंकि दो O परॉक्साइड बंध से बंधित हैं।

$$73. [\text{NH}_3] = 0.3 \text{ M} \quad [\text{NH}_4^+] = 0.2 \text{ M}$$

$$K_b = 1.8 \times 10^{-5}$$

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{salt}]}{[\text{base}]} = 4.74 + \log \frac{0.2}{0.3}$$

$$= 4.74 + 0.3010 - 0.4771 = 4.56$$

$$\text{pH} = 14 - 4.56 = 9.436$$

$$74. \text{ so, volume of unit cell in ccp} = \left(\frac{4a}{\sqrt{2}}\right)^3$$

$$\text{so volume per spheric/atom} = \frac{1}{4} \times \frac{64a^3}{2\sqrt{2}} = 5.66a^3.$$

$$\text{Sol. अतः ccp में एकक कोष्ठिका आयतन} = \left(\frac{4a}{\sqrt{2}}\right)^3$$

$$\text{अतः प्रति गोले/परमाणु का आयतन} = \frac{1}{4} \times \frac{64a^3}{2\sqrt{2}} = 5.66a^3.$$

$$75. K = A e^{-\frac{E_a}{RT}}$$

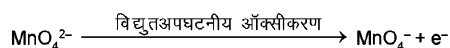
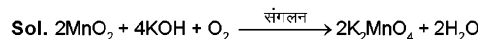
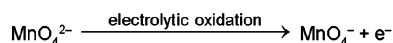
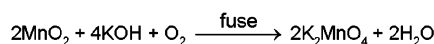
$$\log K = \log A - \frac{E_a}{2.303 R} \times \frac{1}{T}$$

$$\text{Slope ढाल} = \tan \theta = \left[-\frac{1}{2.303}\right] = -\frac{E_a}{2.303 R}$$

$$\text{so अतः } E_a = R = 2 \text{ cal/mol}$$

76. Fuse with KOH in presence of air or oxidising agents like KNO_3 , KClO_3 etc.

वायु या ऑक्सीकारक जैसे KNO_3 , KClO_3 इत्यादि की उपस्थिति में KOH के साथ संगलित करना।



77. According Hardy-Schulze rule as As_2S_3 is a negative sol.

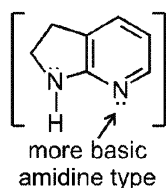
हार्डी-शुल्जे नियम के अनुसार क्योंकि As_2S_3 एक ऋणात्मक सॉल है।

78. Relatively HA will be stronger acid, so its solution will have lower pH.

तुलनात्मक रूप में HA प्रबल अम्ल होगा, अतः इसका विलयन न्यून pH रखता है।

79. (1) is feasible
(2) is feasible.

(3) is not feasible.

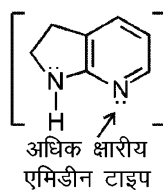


(4) is feasible by product in Wurtz reaction.

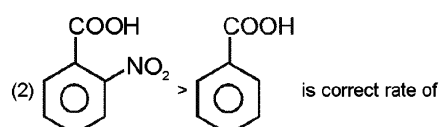
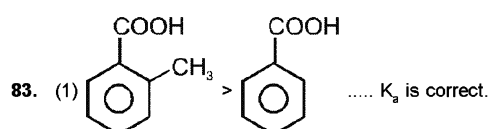
(1) सम्भव है।

(2) सम्भव है।

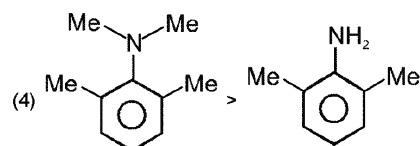
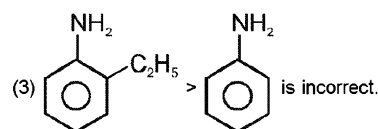
(3) सम्भव नहीं है।



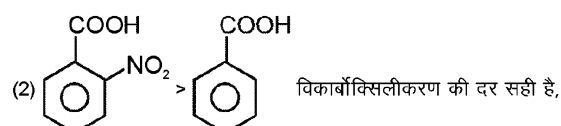
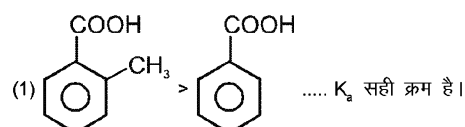
(4) बुद्ध अतिक्रिया में यह उत्पाद सम्भव है।



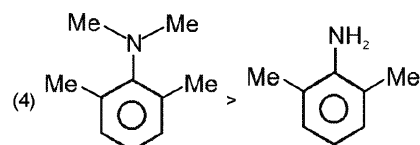
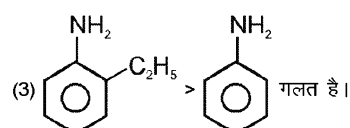
decarboxylation intermediate is $\text{C}^\ominus$ (more stable).



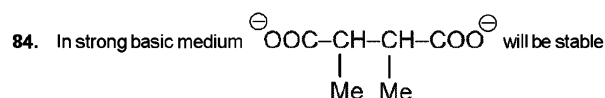
..... K_b is correct. Due to SIR



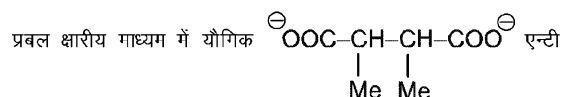
मध्यवर्ती $\text{C}^\ominus$ है (अधिक स्थायी)



..... K_b सही हैं SIR के कारण



in its anti form to minimise e-repulsion.



रूप में स्थायी होता है। जिससे की इलेक्ट्रॉन्स के मध्य प्रतिकर्षण न्यूनतम हो जाये।

85. $\text{I} > \text{II} > \text{IV} > \text{III}$

I $\rightarrow$ three resonating structure (तीन अनुनादी संरचना)

II $\rightarrow$ two resonating structure (दो अनुनादी संरचना)

III $\rightarrow$ 2° alcohol (2° एल्कोहॉल)

IV $\rightarrow$ 1° alcohol (1° एल्कोहॉल)

